SMC-BVS: State of the Project

Method, empirical results, theorem targets, work done and open problems

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Agent brief (read this page first)

What this is. The single starting point for the SMC-BVS theory loop: the method as specified and implemented (§2), every empirical result (§3), the approved paper-1 theorem targets with exact statements and per-clause status (§4), the work done and the routes that failed (§6), and what is open (§7). It consolidates the archive of the theory campaign thread "New BVS Headquarters" (2026-11-02 to 04), abbreviated `NBH/` = `Theory_work/Old_Projects/New_BVS_Headquarters/`, and the repository results. All citations refer to repo commit `21ebc28`.

Where things stand (one paragraph each).

- **Goal.** Objectives in order: (1) accuracy in recovering the true posterior, (2) parallel scalability over particles and islands (not over variables), (3) BVS selection accuracy; target venue Biometrika with Lean certificates for key results (§1). Thesis: "In split-and-merge, correlation breaks correctness. In local-to-global SMC, correlation costs only sequential depth and particles, never correctness" [pos 1044].
- **Method.** SMC-BVS v3.1 = the v3 study method unchanged as a statistical method (g-prior with $g = n$, BB$(1, p)$ size prior, cap $K = \min(p, \lfloor n/2 \rfloor)$, correlation-clustered exact block start π_0, tempering bridge, islands, unbiased $\hat{Z}$, HPM = best model ever scored) plus a parallel island layer, optional stacking, LS0, the $\hat{Z}$-weighted PIP, a general-h accumulator, a certificate card and NA outputs [NBH/START_HERE.md:16-25]. Not every v3.1 component is in the frozen production code; see the divergence boxes in §2.
- **Empirics.** At $n = 200$: low $p \in \{20, 30\}$ (exact posterior by enumeration) and high $p \in \{500, 1000\}$ (2,400 fits; 1,829 exact recoveries, 565 posterior misses, 0 search misses, 6 stopped runs) [NBH/START_HERE.md:54], [Docs/HQ/STATUS.md:23-66]. Read §3 for what "0 search misses" does and does not identify.
- **Theory.** The paper-1 list (main cards M1–M8 = `TH01`–`TH08`; supplement cards `TH09`–`TH20`) is APPROVED (F.B., 2026-11-07) as the current targets. Approval does *not* change proof status: most clauses are candidate proofs ("PROVED (candidate)", "PROVED (by this seat, unchecked)", "PROOF SKETCH"), checked per clause by the same model family (SF), a different family (XF) or Lean; nothing is adopted as final [NBH/START_HERE.md:42-45]; the approval is recorded in `sources/approval_2026-11-07.md`. Lean: the library has 101 declarations with 0 `sorry` [NBH/branch_snapshots/L-LEAN/STATUS.md:3], formalising abstract finite cores (T21 inequality, T07, T11(iii), Theorem A; T12 core only partially, with `hstep` an explicit hypothesis); **no main card is Lean-certified as a whole** [NBH/03_results_and_conclusions.md:22]. External human review of the proofs has not happened; all checks are AI-based (two model families plus Lean for finite cores) [NBH/03_results_and_conclusions.md:340].
- **Main open edges.** p ≫ n versions (candidate repair API-PXNGAP, not cross-checked); N1 for general h (binding proposal §13.1); §5.3 constants only useful for $\varepsilon \lesssim n^{-1/2}/p$; the Song–Liang

separation is not proved (numerical only); asymptotic onsets near $n \approx 2 \cdot 10^6$ versus the study's $n = 200$ [NBH/START_HERE.md:46-51]; the sampler-level results about the true model are stated only on explicit design families (M1(i) on families with osc $d \le D_* = O_P(1)$, under the T03♯ hypotheses and stacking R'_n, exhibited on W; M4(f) and M5(b1-LH) on LH, as joint statements with the output event, with no delivery claim; LH-iid only in an unreviewed candidate), and the exact-posterior result T03♯ needs fixed-design hypotheses; none covers the study's own generator (§4, §7).

Rules for agents of the loop.

R1. Never upgrade a status. Copy status words verbatim; a result becomes stronger only through a recorded proof plus the check level the acceptance criteria demand (§7).

R2. Every claim you add cites a source (`\src{path:line}` or `\pos{N}`) or a new file you created; new mathematics goes in a new file with its own proof and status ("PROVED (candidate)" until checked).

R3. Respect Felipe's ground rules GR1–GR8 (§1.2); losing a binding goal (§5.3 weight stability, §13.1 general-h accuracy) without an approved replacement is escalated to Felipe, never silently dropped.

R4. Before attacking a problem, read its history (§6, the "do not retry" table) and its obligations (§5.4); many natural routes are already refuted.

R5. Keep the regime of every result explicit: any X / fixed design under T03♯ hypotheses / LH / W / LH-iid / study design; asymptotic vs finite-n; "jointly with output".

How to navigate. Every result has a stable ID (`TH05`, `OB01`, `RF101`, ...); grep the `.tex` sources or the machine-readable registry `registry/*.tsv` (columns: id, native name, kind, status verbatim, checks, regime, dependencies, open items, sources). Appendix B renders the registry; Appendix A defines every symbol; Appendix D defines every status word; Appendix C indexes files. The authoritative mathematical texts are the branch snapshots `NBH/branch_snapshots/<BRANCH>/`; the verbatim thread is `NBH/transcript/`.

How to update this document. Edit the registry row first, then the section; run `scripts/build.sh` (LuaLaTeX, then `scripts/check.py`: citations resolve, statuses match the archive registry, IDs defined once, no dropped glyphs); bump `\sopversion` in `preamble.tex`; add a changelog entry (Appendix E) listing every ID promoted, demoted, added or removed, with the evidence. The review rubric for each new version is `review/RUBRIC.md`.

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1 Problem, objectives and thesis

1.1 The problem the project addresses

The project starts from Felipe Bedoya's proposal *Local-to-Global Bayesian Computation* [NBH/inputs/Local_to_Global_draft.txt:1-16]. Its principle is

local posterior computation ⟶ merge local structures ⟶ correct toward the global posterior,

with the correction step essential: "Local posterior samples are useful computational building blocks, but they are not the target posterior distribution" [NBH/inputs/Local_to_Global_draft.txt:38-43]. Bayesian variable selection (BVS) is the controlled test bed, because there the merge is easy but uncertainty is hard: two correlated predictors X_j, X_ℓ placed in different blocks can each look important locally, while the full posterior puts its mass on $S = \{j\}$ or $S = \{\ell\}$ and little on $S = \{j, \ell\}$; "naive merging can destroy joint posterior uncertainty" [NBH/inputs/Local_to_Global_draft.txt:168-186]. The proposal's message is "easy merge ⇏ valid posterior uncertainty" [NBH/inputs/Local_to_Global_draft.txt:211-218], and its Risk 4 is that "local proposals miss important posterior regions" [NBH/inputs/Local_to_Global_draft.txt:410-413]; the campaign's "masked regime" is exactly this risk [pos 1028].

The proposal names the theory a paper needs (§5.1 exactness under Metropolis–Hastings correction; §5.2 SMC consistency; §5.3 weight stability for variable selection under weak cross-block dependence; §5.4 approximation error in KL, TV, PIP, joint-PIP and credible-interval terms) [NBH/inputs/Local_to_Global_draft.txt:253-296], and adds in §13.1: "We should prove a theorem where our estimates of posterior expectation of $h(\theta)$ converge to the theoretical ground truth" [NBH/inputs/Local_to_Global_draft.txt:499-501]. These goals are binding on the theorem list (ground rule 7 below); §1.4 maps each to the cards.

1.2 Felipe's objectives and ground rules

The ground rules Felipe set on 2026-11-03 are binding on every agent [NBH/branch_snapshots/MEET-v31/DOSSIER.md:9-18]. In summary (the source is verbatim; read it before changing the method):

GR1 Baseline and burden of proof. The method stays close to the v3 study method (composite likelihood, SMC structure, HPM rule) unless there are "irrefutable reasons" [NBH/branch_snapshots/MEET-v31/DOSSIER.md:11].

GR2 Real issues only. Changes must address real issues such as "theorems breaking down or parallel computing orientation" [NBH/branch_snapshots/MEET-v31/DOSSIER.md:12].

GR3 Provable or nothing. "a fancy heuristic and hypothetically gives me perfect accuracy but cannot be formally proved is as good as nothing to me" [NBH/branch_snapshots/MEET-v31/DOSSIER.md:13].

GR4 What parallel means. "Parallel computing is the final goal but it is not the main driver, the final methodology must be parallelizable enough so you could scale it to p values that you could not otherwise in sequential MCMC." "We are not trying to parallelize over variables." The parallel axis is particles and islands: stack CPUs/GPUs, compute a large total number of particles [NBH/branch_snapshots/MEET-v31/DOSSIER.md:14], [pos 1113].

GR5 Local-to-global is foundational; dropping it needs "a very serious, irrefutable argument" [NBH/branch_snapshots/MEET-v31/DOSSIER.md:15].

GR6 Objectives, in order: (1) accuracy in recovering the true posterior; (2) parallel scalability; (3) BVS selection accuracy. "It must work well under correlated designs, unlike Song & Liang (2015) split-and-merge" [NBH/branch_snapshots/MEET-v31/DOSSIER.md:16].

GR7 Theory standard. The theory must reach every goal of the proposal (§5.1–5.4, §13.1) and extract "until the last drop" the provable information the method has, as Song & Liang (2015) did for theirs [NBH/branch_snapshots/MEET-v31/DOSSIER.md:17].

GR8 Re-runs. If the final method is close to v3, the 2,400-fit study is not re-run now; "it is re-run at the end only if needed"; empirical checks use at most 10–20 seeds [NBH/branch_snapshots/MEET-v31/DOSSIER.md:18].

The ordering of objectives came from Felipe's messages of 2026-11-03: precision of BVS selection "is not the main target", the primary target is "precision in terms of recovering the true posterior", and the project was born to "pursue parallel computing", unlike Song (2015) it "needs to do well also in correlated data scenarios" [pos 1020] [NBH/transcript/part_05_958-1107.md:321]. Felipe then clarified that the current theoretical goals are kept, and stated the long-range aim: "the big picture goal is basically to retire MCMC for BVS, the next gen samplers must be parallel and with the same degree of theoretical guarantees" [pos 1030] [NBH/transcript/part_05_958-1107.md:374]. Paper 1 is BVS; a second paper "will deal with trees and not variable selection, but it is the same logic" [pos 1020].

1.3 Thesis, scope and venue

The paper-1 thesis, written by the old HQ and accepted by Felipe [pos 1044] [NBH/START_HERE.md:9]:

In split-and-merge, correlation breaks correctness. In local-to-global SMC, correlation costs only sequential depth and particles, never correctness.

In the method this reads: *local step* = exact enumeration of correlation-clustered blocks, all in parallel; *merge* = the product start π_0 (used only as a starting point, never as the answer); *correction* = the tempering bridge to the exact posterior; *width* = particles and independent islands whose unbiased $\hat{Z}$ lets them be combined exactly [NBH/transcript/part_05_958-1107.md:450-456]. The thesis is scoped by the theorems themselves: the thesis remark TH22 is a remark only, and the results do not prove "that the posterior itself remains statistically consistent under arbitrary correlation" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1283]; see §4 for the regime map.

Venue and standard. Felipe's strategy (2026-11-03): "build something rigorous enough to make it to Biometrika. Down to Lean certificates. Then we will analyze what more challenging results can we go to" [pos 1015] [NBH/transcript/part_05_958-1107.md:296]. Paper scope on parallelism: Felipe decided that paper 1 claims *island-level* depth only [NBH/branch_snapshots/MEET-v31/MINUTES.md:135].

1.4 How the proposal's goals map to the approved theorem list

The theorem meeting's coverage matrix records, for each binding proposal goal, the cards that deliver it [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:14-23]. The cards are in §4; proof status is per clause and is *not* implied by this table.

Proposal goal	Delivered by (as recorded in the coverage matrix)
§5.1 exactness under MH correction	TH01 M1(a) (cited; validity); TH12 S-KERNEL; TH10 S-TARGET; numerical fidelity = FW16
§5.2 SMC consistency	TH01 M1(a),(f),(i): cited fixed-data limits, finite-R' bound, asymptotic with R'_n; TH17 S-POOL
§5.3 weight stability (binding)	TH02 M2 (a)(b)(c) with scale disclosure; TH03 M3 (d_σ); TH07 M7; TH06 M6 (Lemma KL); TH20 S-LEAK; FW22
§5.4 PIP and joint-PIP error	TH01 M1 (e),(f),(g),(h); TH04 M4(b)
§5.4 TV and KL	TH04 M4(a),(b),(d),(g); TH02 M2(a); TH06 M6; TH13 S-IDENT (C15-TV, C15-KL); masked final-estimate TV/KL = FW2 (not binding)
§5.4 credible-interval coverage	TH13 S-IDENT C15-CI (posterior-content error); TH01 M1(g); frequentist coverage = FW1
§5.4 masked accuracy	TH04 M4(f) composed with TH05 M5 (b1-LH)
§13.1 $E_\pi[h] \to$ truth (binding)	TH01 M1 (b),(e),(i); unbounded functionals = FW27; safeguard: if N1 (OB01) fails, a narrower theorem ships and §13.1 escalates

For agents — objectives

Judge every proposed change or new theorem against GR1–GR8 in this order: does it serve objective (1) posterior accuracy, then (2) parallel scalability over particles/islands, then (3) selection accuracy; is it provable (GR3); does it keep the method close to v3.1 (GR1) and the 2,400-fit study valid (GR8)? A result that only improves selection accuracy is the lowest priority. The binding proposal goals are §5.3 and §13.1; losing either without an approved replacement must be escalated to Felipe, never silently dropped [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:16], [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:23].

2 The method: SMC-BVS v3.1

This section states the method exactly as specified (method v3.1 = METHODOLOGY_v3 plus the MEET-v31 diff) and as implemented in the frozen production code used for every study fit (code v2.2.1). Version numbers of the method (v3, v3.1) and of the code (v2.1, v2.2, v2.2.1) are different series [README.md:5-9]. The authoritative method text is `METHOD_v3.1_FINAL.md`, written as a diff against `METHODOLOGY_v3.md` ("M" below) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:1-9]. Definition IDs DF01–DF25 are anchored in Appendix A; method-meeting decisions MD01–MD13 are defined in §2.13.

Plain reading — the method in one paragraph

Given X ($n \times p$) and y, SMC-BVS computes the posterior over subsets S of predictors under a unit-information g-prior with a unit ridge, a vague inverse-gamma noise prior and a beta-binomial BB$(1, p)$ size prior capped at $\lfloor n/2 \rfloor$ variables. When the capped model space is small it enumerates it exactly. Otherwise it starts from an exactly sampleable product-of-blocks approximation π_0 built from small correlation blocks, each enumerated exactly, and tempers from π_0 to the posterior with an SMC bridge; a calibration phase builds the tempering schedule and a fresh final run with frozen settings produces the reported HPM, PIPs and evidence [Docs/Methodology/METHODOLOGY_v3.md:25-33]. v3.1 is "v3 unchanged as a statistical method, plus a list of additions" (parallel layer, optional stacking, LS0, reporting changes, wording fixes) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:26-41].

UNCERTAIN — approval state of the method text

`METHOD_v3.1_FINAL.md` still says: "**Status: FINAL from MEET-v31 (chair), for HQ approval.** Not yet approved by HQ or Felipe." [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:5]. The later plain summary says "Approved by HQ after the method meeting (MEET-v31), two independent reviews (Sol 6.1: a proof check and an audit), and the N9 kernel check." [NBH/branch_snapshots/MEET-v31/FOR_FELIPE_FINAL_METHOD.md:3]; HQ wrote "**The final method (v3.1) is approved.**" [pos 1584]. Felipe approved the four method-process decisions [pos 1304] [NBH/START_HERE.md:16]; no source records Felipe approving the v3.1 method text itself. Treat v3.1 as HQ-approved; the header line of the method file is stale.

2.1 Inputs, preprocessing, statuses

Input (M §2): numeric $X \in \mathbb{R}^{n \times p}$, $y \in \mathbb{R}^n$, an integer seed $s \in [1, 2^{31} - 1]$, and the job's resource allocation (memory, wall time; default 8 GiB) [Docs/Methodology/METHODOLOGY_v3.md:37-38]. v3.1 rewrites Requirement (1) as: "Runnable from X, y and one integer seed, with no statistical tuning beyond published defaults. The job's allocation (memory, wall time) is a declared (c) input that decides delivery only. Every estimate on $\mathcal{O}$ is a function of (X, y, seed, binary)." (item e8) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:179].

Preprocessing (DF04; M §4) [Docs/Methodology/METHODOLOGY_v3.md:80-93]:

(1) stop with `invalid_input` if any entry is not finite, or $n < 3$, or $p < 1$ (v3; v3.1 item e3 proposes to admit $n = 2$, see below);

(2) drop column j only if all its entries are *exactly* equal; $p :=$ number kept; if $p = 0$ return $\hat{S} = \emptyset$ with status `no_predictors`;

(3) if all entries of y are exactly equal return $\hat{S} = \emptyset$, PIPs and evidence unavailable, status `y_constant`;

(4) standardise stably: divide each column by its maximum absolute value, compute the mean, centre, divide by the sample SD (denominator $n - 1$), so $\|x_j\|^2 = n - 1$; same for y, recording $s_y^2 = \|Cy\|^2/(n-1)$, $C = I - \mathbf{1}\mathbf{1}^\top/n$;

(5) report connected components of $\{|r_{jl}| \geq 1 - 10^{-10}\}$ (duplicates are kept);

(6) resource preflight (M §8.6).

These rules make the target invariant to affine maps of y ($a \neq 0$) and of each column, and to column permutation up to index-based tie-breaks [Docs/Methodology/METHODOLOGY_v3.md:92-93]. Code: [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_preprocess.R:97-132] (rejects $n < 3$ at line 106; the near-constant "anchored two-pass" route at $\sqrt{\varepsilon_{\text{mach}}}$ is at line 47).

Statuses (exactly one): `exact`, `completed`, `schedule_limit`, `numerical_failure`, `resource_limit`, `y_constant`, `no_predictors`, `invalid_input` [Docs/Methodology/METHODOLOGY_v3.md:48-52]; v3.1 adds `degenerate_n` (e3) and `unsupported_size` (e7) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:80]. Item e3: "Reject n = 0. n = 1 → status `degenerate_n` [...]. **n = 2 is admitted**: K = 1, B = 1, E = p + 1, so the exact branch runs." The $n = 2$ output statement is "seat-derived, unchecked", and "current preprocessing still rejects n < 3 and raw p = 0" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:182].

Flags: `escalated`, `checks_passed_calibration`, `checks_passed_final`, `forced_pilot_steps`, `grid_unconverged`, `local_limit`, `low_empirical_ESS` (any final stage below $0.5N$), `near_tie_truncated`, `local_certified`, `global_certified` [Docs/Methodology/METHODOLOGY_v3.md:54-58]; v3.1 adds `ls0_improved` and documents `global_certified` as "a heuristic numerical flag" (e5) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:81] [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:184].

2.2 Target posterior: likelihood, priors, score, evidence

Definitions DF01 (model, likelihood, g-prior, σ^2 prior), DF02 (size prior, cap), DF03 (score, evidence, posterior). v3.1 keeps the target "**unchanged** (Felipe's model)" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:84].

Statement (verbatim from source) Docs/Methodology/METHODOLOGY_v3.md:97-108

On the standardised data (X, y), for $S \subseteq \{1, \dots, p\}$ with $k = |S| \leq K$, $G_S = X_S^\top X_S$, $z_S = X_S^\top y$:

$$y \mid \beta_S, \sigma^2, S \sim N_n(X_S\beta_S, \sigma^2 I_n) \qquad \text{(no intercept; } y \text{ is centred)}$$

$$\beta_S \mid \sigma^2, S \sim N_k\big(0,\, n\sigma^2(G_S + I_k)^{-1}\big) \qquad g = n \text{ (unit information)},\ \tau = 1 \text{ (one pseudo-observation)}$$

$$\sigma^2 \sim \mathrm{IG}(a_0, b_0), \quad a_0 = b_0 = 0.01$$

$$\varpi(S) = w(k)\,1\{k \leq K\}/W_K, \quad w(k) = B(1{+}k, 2p{-}k)/B(1,p) = \tfrac{1}{2}\big/\tbinom{2p-1}{k}, \quad W_K = \textstyle\sum_{j \leq K} \binom{p}{j} w(j)$$

$$H_S = (n+1)G_S + I_k, \qquad Q_S = y^\top y - n\, z_S^\top H_S^{-1} z_S,$$

$$\log m(S) = \log C_n - \tfrac{1}{2}\big[\log\det H_S - \log\det(G_S + I_k)\big] - (a_0 + n/2)\log(2b_0 + Q_S),$$

$$C_n = (2b_0)^{a_0}\,\Gamma(a_0{+}n/2)\big/\big(\pi^{n/2}\,\Gamma(a_0)\big), \qquad \pi(S \mid y) = \varpi(S)\,m(S)/Z, \qquad Z = \textstyle\sum_{|S| \leq K} \varpi(S)\,m(S).$$

The cap is $K = \min(p, \lfloor n/2 \rfloor)$ and the model space is $\Omega = \{S : |S| \le K\}$ [Docs/Methodology/METHODOLOGY_v3.md:65] [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:64]. The shape is $\nu = a_0 + n/2$ and the theory cards write $\bar{Q} = 2b_0 + y^\top y$ [NBH/notation_and_conventions.md:142].

The score F. Throughout the method and the theory, the *score* of a model is the log unnormalised posterior

$$F(S) = \log \varpi(S) + \log m(S), \qquad e^{F(S)} = \varpi(S)\, m(S), \qquad Z = \textstyle\sum_{S \in \Omega} e^{F(S)},$$

("the strict maximum of log ϖ(S) + log m(S)" [Docs/Methodology/METHODOLOGY_v3.md:263]; "$Z_{\mathcal{T}} = \sum_{S \in \mathcal{T}} e^{F(S)} = \sum_{\mathcal{T}} \varpi(S)\, m(S)$" [NBH/notation_and_conventions.md:91]; code `F = log varpi(|S|) + log m(S)` [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_engine.h:89-98]). So F includes the prior; the marginal likelihood alone is $m(S)$. The Bayes factor against the empty model is $\mathrm{BF}(S) = m(S)/m(\emptyset)$ [Docs/Methodology/METHODOLOGY_v3.md:176]. The final theory cards write $m(U) = c_n[\det H_U / \det(G_U + I)]^{-1/2}(.02 + Q_U)^{-\nu}$ and note "The common positive c_n cancels from posterior ratios" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:65].

Reasons recorded for the target (accepted by Felipe): $g = n$ is unit information and $\tau = 1$ keeps every $m(S)$ finite for collinear X_S; IG(0.01, 0.01) is "proper, as the project proofs require"; BB$(1, p)$ "depends on p only; prior mean size p/(p+1) ≈ 1; each extra variable costs log((2p−k−1)/(k+1)) nats"; the cap "keeps near-interpolating models out of the space; the ½ is a declared restriction, not needed for propriety" [Docs/Methodology/METHODOLOGY_v3.md:111-119]. The v3 claim that the IG effect is negligible was narrowed in v3.1: its pairwise effect "relative to the gap [...] is not uniformly negligible [...]; negligibility needs residual-scale assumptions; not negligible at tiny n" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:299].

Reduction statement (consensus of the v3 consultants; "to be proved, V1"): with $\theta = s_y \beta$ and $v = s_y^2 \sigma^2$, the posterior is that of the contrast regression of $U^\top y^r$ on $U^\top X$ with $n - 1$ Gaussian errors, slab $\theta_S \mid v, S \sim N(0, nv(G_S + I)^{-1})$ and $v \sim \mathrm{IG}(0.51, 0.01 s_y^2)$; "This is not a claim that the working evidence equals a raw-data marginal likelihood" [Docs/Methodology/METHODOLOGY_v3.md:121-127].

Code check (prior/score). The scorer sets $K = \min(p, \lfloor n/2 \rfloor)$, $a_0 = b_0 = 0.01$, $\nu = a_0 + n/2$, $\log C_n$, $\log w(k) = -\log 2 - \log \binom{2p-1}{k}$, $\log W_K$ by log-sum-exp [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_score.h:39-72], forms $H = (n+1)G_S + I$ and $A = G_S + I$ by Cholesky and computes Q by subtraction with recomputation by the non-negative form below $10^{-8} y^\top y$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_score.h:175-224]; Cholesky, refactor, augmented QR, then `NumericalFailure` [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_score.h:227-238]. This matches M §5 and §6.13 [Docs/Methodology/METHODOLOGY_v3.md:286-289]. No divergence found.

2.3 Budgets and the exact branch

Symbol	Definition (M §3)	study ($n = 200$; $p = 500$ / 1000)
K	size cap $\min(p, \lfloor n/2 \rfloor)$	100 / 100
B	block size $\min(\lfloor \sqrt{n} \rfloor, \lfloor \sqrt{p} \rfloor, B^*)$, $B^* = 16$	14 / 14
ℓ	$\lceil \log_2(n + p + 1) \rceil$ (local-search and rebuild budget)	10 / 11
L	$\max(4, \lceil \log_2(n+1) \rceil)$ (burn-in rounds at $\varphi = 1$)	8 / 8
N_0	$\max(2500, n, p)$ final particles per island	2500 / 2500
H_0	$\lceil 512 N_0 / 2500 \rceil$ calibration particles (pilot and each diagnostic island)	512 / 512
T_0	$\max(1000, \ell^2)$ stage ceiling	1000 / 1000
R	4 islands (final and diagnostic)	4 / 4
tier e	$e \in \{0, 1\}$; $(N_e, H_e, T_e) = 4^e (N_0, H_0, T_0)$	

Source [Docs/Methodology/METHODOLOGY_v3.md:62-76]; "These are **declared resource budgets**, not accuracy guarantees." v3.1 keeps the values and relabels classes: $B^* = 16$ is "a declared **(c) budget for**

exhaustive tables" (e1) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:180]; "the ℓ^2 term exceeds 1,000 exactly when n + p ≥ 2^{31}" (w4) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:195]. Code: [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_preprocess.R:19-33] (exact integer $\lfloor\sqrt{\cdot}\rfloor$ and $\lceil\log_2 \cdot\rceil$; matches).

Exact branch (M §6.2): if $E = \sum_{k\le K}\binom{p}{k} \le (5H_0 + 4N_0)\,p$ (saturating integer arithmetic), score every model with $|S| \le K$ in a streaming pass (running best in the strict HPM order; log-sum-exp for Z and PIP numerators); status `exact` [Docs/Methodology/METHODOLOGY_v3.md:156-160]. "Because the threshold grows with $N_0 \ge n$, every fixed-p problem is eventually enumerated as n grows." v3.1 corrects "all p ≤ 3,000 at n ≤ 5" to "all p at n ≤ 5" (w2) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:193]. Code: 128-bit saturating count [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:97-110]; enumeration [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:526-570]. On this branch `global_certified` is set iff $F(\text{HPM}) - \eta(\text{HPM}) > \max_{S\neq\text{HPM}}[F(S) + \eta(S)]$ with the scorer's heuristic error expression η [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_exact.R:3-17]; v3.1 (e5) labels this "a **heuristic numerical margin**" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:184]. The 2,400-fit study used the SMC branch, where the flag is always FALSE [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:147].

2.4 The block start π_0

Definitions DF05 (blocks and tables), DF06 (start). v3.1 keeps blocks, tables and start "**unchanged**"; the table stage runs in parallel (#4) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:87].

Statement (verbatim from source) Docs/Methodology/METHODOLOGY_v3.md:164-177

- **Clustering:** complete linkage on the distance $1 - |r_{jl}|$ computed from X; ties between candidate merges broken lexicographically on the sorted original-index lists.
- **Cut:** top-down from the root; any node with more than B members is split into its two children (visited smallest index first) until every retained cluster has at most B members.
- **Packing:** next-fit in dendrogram leaf order, whole clusters only (a retained cluster is never split), into blocks $C_1, \ldots, C_J$ of at most B members. $J \ge \lceil p/B\rceil$ (packing leaves gaps).
- **Tables:** for each block, every sub-model $U \subseteq C_b$ is scored against the full y; $r_b(U) = m(U)/m(\emptyset)$ (table exponent 1). Since $B \le K$ for $n \ge 3$, no sub-model exceeds the cap. Every table score counts for the HPM. Table scores are not inserted in the score cache.
- **Start:** $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b)$, $\pi_0 = \gamma_0/\bar{Z}_0$. $\bar{Z}_0$ is computed exactly by convolving the block size profiles $e_b(j) = \sum_{|U|=j} r_b(U)$ up to K with the size prior (valid because ϖ depends on S only through $|S|$). Exact iid draws: draw the total size, then block sizes backwards, then each block's sub-model within its size class.
- **Discrepancy:** $d(S) = \log \text{BF}(S) - \sum_b \log r_b(S \cap C_b)$, $\text{BF}(S) = m(S)/m(\emptyset)$; bridge $\gamma_\varphi(S) = \gamma_0(S)e^{\varphi d(S)}$, $\varphi \in [0,1]$; $\gamma_1 \propto$ the posterior; $Z = m(\emptyset)\sum_S \gamma_1(S)$.

Block-size rule and no correlation cut. $B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, B^*)$ is "Felipe's rule; B ≤ K and B ≤ p" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:307]. w5: "The dendrogram is cut by node size only (≤ B). There is no correlation-height threshold." [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:196]. e6 records the study's actual block counts: "J = 46–60 (p = 500) and 95–120 (p = 1,000) versus proxies 36 and 72" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:185]. The theory's partition hypothesis **SIZE** $g_{\max} \le B < 2g_{\min}$ is "a hypothesis of one partition theorem [...], not a method parameter. The study's g = 5 is the theorem obligation PART-pairing" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:180] (obligation OB08; card TH11).

Exact sampling (`draw_pi0`) and $\bar{Z}_0$ in code. Tables score every $U \subseteq C_b$ with $|U| \le K$ against the full y [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_start.h:73-98]; size

profiles $\log e_b(j)$, partial convolutions $\log F_b(k)$ truncated at K, $\log \bar{Z}_0 = \log \sum_k \varpi(k) F_J(k)$, and the backward conditional $P(k_b = j \mid r) = e_b(j) F_{b-1}(r-j)/F_b(r)$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_start.h:100-151]; the draw uses one uniform for the total size, one per block $b = J, \ldots, 1$ while the remainder is positive, and one per non-empty block for the sub-model [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_start.h:153-176]. Blocks: [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_blocks.h:39-144] (complete linkage with the union-lexicographic tie rule; left child = child with smaller minimum index; next-fit packing). All match M §6.3. Exactness of `draw_pi0` and $\bar{Z}_0$ is part of N9 (OB09), PASSED [NBH/branch_snapshots/MEET-v31/MINUTES.md:150] (card TH12; clause TH01.a).

Start mode $\tilde{S}_0$. $\tilde{S}_0 = \arg\max \gamma_0$, "computed exactly by the block DP, with the size-then-lexicographic tie rule" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:150]. The header function `start_mode` computes per-block per-size maxima, a max-plus convolution up to K, picks the size by strict improvement ("ties keep the smaller size") and enumerates all optimal choices keeping the lexicographically smallest set, capped at 10^6 sets with a truncation flag [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_start.h:178-240]. e4: "On truncation, use a deterministic tie-aware DP that returns the prescribed smallest-size, lexicographically smallest maximiser without enumerating ties" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:150]; "The invariant 'DP output = the prescribed choice' is an OPEN obligation (N3)" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:183] (OB03). In the study, "0/2,400 tie truncations; all 2,400 had a single tie" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:150].

Dilution (cite, do not derive). The tables score each block's sub-models against the *full* y with the noise variance integrated out per sub-model, so a block's table does not see the other blocks' signal except as noise. O-DILUTE: "Seen from a truth, the v3 table's own-σ^2 integration *dilutes*; seen from a null, it *calibrates* the leakage" [NBH/branch_snapshots/O-DILUTE/THEORY.md:114-115]; mechanism "σ^2-dilution of the truths' table log-odds" with "About 7 of the 14 truths are invisible at n = 200 for every c" [NBH/03_results_and_conclusions.md:269]. The exact identity $-\log \pi_0(S_0) = V + C - \log A(|S_0|)$ is card M3(d) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:206] (TH03). The common plug-in σ^2 alternative was rejected (it "floods the start") [NBH/03b_reasoning_and_traces.md:83-88].

Item 1 (start ablation). The start is kept under Outcome 2: "the start is presented as the exactly computable, theorem-bearing local step (S1/S2) plus the parallel table stage, with **no depth or speed claim**" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:118] (MD01).

2.5 Bridge, neighbour graph and move kernel

Definitions DF07 (discrepancy and bridge), DF08 (moves and round). The bridge family is $\gamma_\varphi = \gamma_0 e^{\varphi d}$ (above); the theory cards write the normalised family $q_t \propto \gamma_0 e^{td}$ [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:67] and the registrar's canonical notation is $q_\varphi := \gamma_\varphi / \bar{Z}_\varphi$ [NBH/notation_and_conventions.md:55].

Neighbour graph (M §6.4): $r_n = \sqrt{2 \log p/n}$; for each j the partners $l \neq j$ with $|r_{jl}| > r_n$, always including j's single largest-$|r|$ partner, keeping at most the B largest (ties by original index); then the symmetric union $N(j)$ (degrees may exceed B); no edges when $p = 1$ [Docs/Methodology/METHODOLOGY_v3.md:181-183]. w3: when $r_n \geq 1$ (i.e. $p > e^{n/2}$) "the graph is the symmetric 1-nearest-neighbour graph" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:194]. Code matches [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_blocks.h:146-173].

Statement (verbatim from source) Docs/Methodology/METHODOLOGY_v3.md:187-201

A round is the list of $m = p + B + 2$ labelled elementary steps — the p Gibbs coordinate updates, one uniform swap, B neighbour swaps, one independence proposal — applied **in a uniformly random order drawn independently of the particle's state, afresh for every particle and every round**.

- **Gibbs coordinate j:** draw j's inclusion from its conditional under γ_φ, with an addition at $|S| = K$ excluded. Record the conditional inclusion probability o_j immediately before the draw (used for PIPs in measured rounds).
- **Uniform swap:** skipped (self-loop) if $|S| \in \{0, p\}$; delete j uniformly from S, add l uniformly from outside; accept with $1 \wedge \gamma_\varphi(S')/\gamma_\varphi(S)$.
- **Neighbour swap:** skipped if $S = \emptyset$; delete j uniformly from S; if $N(j) \setminus S = \emptyset$ reject; else add l uniformly from $N(j) \setminus S$; accept with $1 \wedge [\gamma_\varphi(S')/\gamma_\varphi(S)] \cdot |N(j) \setminus S|/|N(l) \setminus S'|$.
- **Independence proposal:** $S' \sim \pi_0$ (exact draw), accepted with $1 \wedge e^{\varphi[d(S')-d(S)]}$. Every proposal is scored, accepted or not.

Each step is reversible with respect to γ_φ; the composition in a random order whose law is invariant under reversal is self-adjoint (checked by all three consultants; V5). An individual realised order is not.

Invariance status. N9 ("kernel reversibility and self-adjointness, exact start sampler") was a closure gate of the method meeting; "**N9 PASSED** — all seven sub-claims PROVED (candidate, author-only), with exact numerical verification on the production engine headers and failing negative controls" [NBH/branch_snapshots/MEET-v31/MINUTES.md:150]. The sub-claim count is UNCERTAIN: other records give 7 [pos 1550] and 8 [pos 1593] (see the uncertain boxes under TH12 and OB09). See OB09, TH12 (S-KERNEL) and TH01.a for the statement and its exact status words. v3.1 keeps the round unchanged; "Changing it is WORKHORSE" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:89].

Code check (kernels). [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_engine.h:111-215]: Gibbs computes $o_j = \text{plogis}(\log\gamma_\varphi(S \cup j) - \log\gamma_\varphi(S \setminus j))$, sets $o_j = 0$ for an infeasible addition at $|S| = K$, records o_j before the draw, and counts both configurations as scored ("SR"); uniform swap self-loops at $|S| \in \{0, p\}$; neighbour swap self-loops at $S = \emptyset$, rejects without scoring when $N(j) \setminus S = \emptyset$, and includes the Hastings factor $|N(j) \setminus S|/|N(l) \setminus S'|$; the independence step draws $S' \sim \pi_0$ and accepts with $\exp\{\varphi(d(S') - d(S))\} \wedge 1$; the order is a Fisher–Yates permutation of the m labels drawn afresh per particle and round. All match M §6.5. The exact uniform-consumption order is documented in [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:24-40].

2.6 Resampling, pilot schedule, candidate grid and calibration

Definitions DF09 (resampling, pilot, ESS 0.90), DF10 (candidate set, grid, rebuild loop), DF11 (calibration record, tiers, final runs).

Statement (verbatim from source) Docs/Methodology/METHODOLOGY_v3.md:205-227

- **Resampling:** fresh multinomial resampling after every reweighting, including the terminal stage ($O(N)$ per stage with a linear-time categorical sampler).
- **Pilot:** H_e particles from π_0; at each stage the next temperature is the largest φ' with population ESS of the weights $e^{(\varphi'-\varphi)d}$ at least $0.90 \cdot H_e$ (weight $\text{CV}^2 \le 1/9$), found by 60 bisection halvings; exploratory floor $1/T_e$ (forced steps counted as `forced_pilot_steps`); ends exactly at $\varphi = 1$; then L burn-in rounds, 4 measured rounds and a local search.
- **V:** the union of the top $20p$ scored models by $\log\gamma_0$, the top $20p$ scored models by $\log\gamma_1$ (the target), and $\emptyset$; deduplicated; list ties lexicographic; frozen during each construction.
- **Restricted log-normaliser:** $F_V(t) = \log\sum_{S\in V}\gamma_0(S)e^{t\,d(S)}$.
- **Restricted ESS** between temperatures $u < v$: $\rho_V(u,v) = \exp\{2F_V(v) - F_V(u) - F_V(2v - u)\}$, the population ESS fraction of the weights $e^{(v-u)d}$ under the restricted law $q_u(\cdot \mid V)$ (evaluating F_V beyond 1 is a finite-sum moment, not a production temperature).
- **Construction:** start at $u = 0$ with the pilot temperatures as targets. For the next pilot tem-

perature b: if $\rho_V(u,b) \geq 0.90$ append b; otherwise bisect on $(u,b]$ with u fixed, at most 60 halvings, and append the largest tested endpoint that passes; continue from the appended point. Validate strict numerical progress and the inequality at every appended point. Non-progress, failed validation, or more than T_e intervals → the construction fails.

- **Rebuild loop:** after each set of diagnostic islands, update V and rebuild; stop when the rebuilt grid equals the previous one; at most ℓ iterations [...]. If not converged: flag `grid_unconverged` (a calibration trigger); at tier 1 the last grid is used and the flag is reported. If a construction fails: trigger escalation at tier 0; status `schedule_limit` at tier 1.

[...] = the lead's ruling that convergence is judged on the grid, not on V, with Fable-T3's dissent (judge on V) [Docs/Methodology/METHODOLOGY_v3.md:224-225]. Purpose and limits: the grid inserts temperatures "exactly where models already scored would make the bridge jump"; it "controls restricted one-step weight dispersion only; it does not control full-space ESS, particle occupancy, mixing, or modes never scored" [Docs/Methodology/METHODOLOGY_v3.md:228-236].

Calibration and escalation (M §6.8–6.9): calibration at tier e runs the pilot and, per rebuild iteration, 4 diagnostic islands of H_e particles on the current grid, each followed by L burn-in rounds, 4 measured rounds and a local search; streams 0 and 1–4 (tier 0), 5 and 6–9 (tier 1); score leaders are kept across tiers [Docs/Methodology/METHODOLOGY_v3.md:240-242]. Triggers (any one ⇒ at most one escalation to tier 1, decided before any final randomness): E1 (some diagnostic island's best score more than 1 nat below the overall best), E2 (SD of the 4 diagnostic log $\hat{Z}$ exceeds 1 nat), E3 (a local search improved the best), forced pilot steps, `grid_unconverged`, a failed grid construction, `local_limit`; after the final run E1–E3 and the empirical-ESS check are recomputed as flags only [Docs/Methodology/METHODOLOGY_v3.md:249-254]. v3.1 keeps these "as calibration triggers, which define the law of $\mathcal{C}$" and labels them heuristics (MD09) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:93].

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:170

Calibration contract. "The calibration output $\mathcal{C}$ (grid, budgets N_e, tier, R', V-independent settings) is frozen before any final randomness. Validity of the final estimates (unbiased $\hat{Z}_r$, PIP identities) holds conditionally on **any** frozen $\mathcal{C}$, whatever happened in calibration. The grid satisfies Theorem G's cap. There are $\leq \ell$ rebuilds per tier ($\leq 2\ell$ in total with escalation, `v3_calib.R:85, 141--149`). The restricted ESS controls one-step weight dispersion on V only. The statuses `schedule_limit`, `grid_unconverged` and `escalated` are distinguished. Rebuild convergence, $P(S_0 \in V)$ and $P($`schedule_limit`$)$ are **OPEN theorem obligations, not inherent limitations**. Rebuilds, tier, T and convergence are tabulated POST HOC from saved fits."

Code check (calibration). Pilot next temperature (ESS $\geq 0.90H$, 60 halvings, floor $1/T_e$, snap within 10^{-12} of 1) and grid construction (targets = pilot temperatures, 60 halvings, validation, T_e cap) [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_calib.h:25-98]; rebuild loop of at most ℓ iterations with convergence by grid equality [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_calib.R:84-119]; at most one escalation, re-preflight before tier 1, freeze of (grid, tier, N, L, inherited leaders and HPM) [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_calib.R:135-175]. The frozen record in code also carries the *inherited* calibration stores (top store and both leader stores) and the calibration HPM with its provenance, which the final run offers to its stores [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:731-743] [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:443-466]. Matches M §6.6–6.9.

2.7 Final islands, streams, stacking and the parallel layer

Definitions DF11, DF12 (islands and stacking). **Final run** (M §6.8): 4 islands $\times N_e$ particles from fresh iid π_0 draws, the frozen grid, multinomial resampling, one move round per stage, then L burn-in rounds

and 4 measured rounds at $\varphi = 1$, then a local search; streams 10–13, independent of all calibration randomness; no forced steps [Docs/Methodology/METHODOLOGY_v3.md:243-245]. "Final observations never revise the schedule and never select replacement estimates." [Docs/Methodology/METHODOLOGY_v3.md:151-152]. In code, each stream is a xoshiro256** generator seeded by SplitMix64 from $x = (\text{seed} \ll 32) + \text{stream}$; uniforms are $(\text{next} \gg 11)\, 2^{-53}$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:12-23].

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:142

Optional stacking $R' = 4k$ at N_e.

- "$R' = 4$ by default. Optionally $R' = 4k$, at **unchanged** N_e, declared before final randomness (before stream 10)."
- "Final islands use streams 10, … , $9 + R'$. Islands 1–4 keep streams 10–13. N_e **is never split** across more islands."
- "With $R' > 4$, an **extra local search runs from the islands-1–4 centre** c_4, besides the search from the expanded centre. The post-search scored set is then nested and HPM$(R') \geq$ HPM(4) (Cor W-R with the extra search, PC1 claim 6)."
- "Two records are kept. The **expanded record** covers all R' islands, pooled $\hat{Z}$ over R' and the expanded store; it carries the run's reported flags. The **baseline record** covers islands 1–4, with its own baseline HPM, the search from c_4 and its own E1–E3, low-ESS and `local_certified`. It is kept whenever baseline reproducibility is claimed."
- "Each record has its own certificate card (#13)."

AMENDMENT_HQ_02 adds the optional configuration $R'_n = 4\lceil \log_2(n + p + 1) \rceil$ (default stays $R' = 4$); the rate $4\lceil (n + p)^6/4 \rceil$ is "a sufficient rate inside theorem M1 only, not a method configuration" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:425] [NBH/branch_snapshots/MEET-v31/AMENDMENT_HQ_02.md:9-12]. An adaptive R' is not allowed ("R′ must be declared in advance") [NBH/branch_snapshots/METH-AUDIT/REPORT.md:214-215]. Splitting N_e into many islands (C8) was rejected [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:274].

Parallel execution layer (MD04) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:131-136]: (a) the R' final islands run as concurrent *process* workers with their own score cache and ledger, merged by Lemma S before the final local search; "Output is bitwise-identical to serial execution (Prop W). 'Bitwise' is scoped to one binary (`Makevars.v3`: `-ffp-contract=off -march=x86-64-v2`), single-thread BLAS and process workers"; (b) the 4 diagnostic islands per rebuild run concurrently — "'Bitwise' may **not** be claimed for (b) until W-cal has been executed"; (c) local-search scan candidates may be scored in parallel with a max-reduction that keeps the lexicographic tie rule; (h) block tables are scored in parallel, with no change of floating-point summation order under a bitwise claim; workers are processes (threads would need private counters). (a) "has been executed bitwise on smoke plus 3/3 study designs (CONC)" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:46]; the speed figure 1.9× "is a 2-core measurement" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:56].

GUARD-W (#6): "Any worker stop, LS0 stop or guard stop ⇒ no output. The fit returns the corresponding unavailable status with all estimate fields NA (#15), never a partial estimate. […] A completed LS0 that ends on its move cap is not a stop: it returns output with `local_limit`, and L1 = 'n/a' (#10). Lemma RO (a guard that reads no uniforms is stop-only) stays as recorded (XF VERIFIED). That the v3.1 worker, merge and LS0 code satisfy GUARD-W is a **code-audit obligation** (§6), not a mathematical gap." [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:136].

Depth and cost (#19, MD06): "With island parallelism, the particle-round contribution to depth is $N_e \cdot (T+L+4) \cdot (p+B+2)$ **sequential elementary-step slots** per island at tier e" and "'Depth free of N, H, R' is **not** claimed for v3.1"; the single-tier calibration/final trajectory ratio $(1+4I)H_0/(4N_0) = 0.0512 + 0.2048 \cdot I$ is "**seat-derived, unchecked**" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:171]. Local-search cost bound: scan size $p + k^*(p - k^*)$ with $k^* = \min(K, \lfloor p/2 \rfloor)$, times $(\ell + 1)$ scans [NBH/branch_

snapshots/MEET-v31/METHOD_v3.1_FINAL.md:99]. Declared v3.2 path (not shipped): C6 per-particle counter-based streams, C7 Gram-free scorer, C11 faster exact linkage [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:259-265].

2.8 Outputs

2.8.1 Evidence $\hat{Z}$

Definition DF13.

Statement (verbatim from source) Docs/Methodology/METHODOLOGY_v3.md:279-282

Final island r: $\hat{Z}_r = m(\emptyset)\,\bar{Z}_0 \prod_t (1/N_e) \sum_i \exp\{(\varphi_t - \varphi_{t-1})\, d(S_i^{(t-1)})\}$, with W_K inside γ_0 exactly once. Pooled $\hat{Z} = (1/4)\sum_r \hat{Z}_r$ via log-sum-exp; 95% lower bound $0.05 \cdot \hat{Z}$ (Markov; one-sided, not an accuracy interval). Reported on the full working scale and relative to $m(\emptyset)$. Given the frozen calibration, each $\hat{Z}_r$ is unbiased (existing project results P6, P7); $\log \hat{Z}$ is not.

v3.1 pools over the R' islands ("pooled $\hat{Z}$ over R'") [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:142], adds $Z_{\mathcal{T}} = \sum_{S \in \mathcal{T}} e^{F(S)}$ with "$\hat{Z} < Z_{\mathcal{T}}$ implies $\hat{Z} < Z$" (#14) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:161], and the card bound $\max\{Z_{\mathcal{T}}, 0.05\hat{Z}\}$ which "keeps 95% coverage on the output event" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:173]. Paper sentence (#17): "The probability that a run returns a card with a false claim is at most 5%." [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:169]. Unbiasedness given $\mathcal{C}$ is clause TH01.a. Code: $\log \hat{Z}_r = \log m(\emptyset) + \log \bar{Z}_0 + \sum_t$ log-increment, increment $=$ LSE$(\Delta\varphi\, d_i) - \log N$, computed before resampling [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_engine.h:300-323]; pooled $\log \hat{Z}$, $\log(0.05) + \log \hat{Z}$, and the value relative to $m(\emptyset)$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:117-124]. Matches.

2.8.2 PIPs and MPMs

Definition DF14. In each island, the RB-PIP averages the recorded Gibbs conditionals o_j over particles and the 4 measured rounds [Docs/Methodology/METHODOLOGY_v3.md:271-272].

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:158

- "**Primary:** $\widehat{\text{PIP}}_j = \sum_r \hat{Z}_r \widehat{\text{PIP}}_{rj} / \sum_r \hat{Z}_r$ over the R' final islands."
- "**Secondary (named):** the equal-weight mean over islands."
- "Report $\text{MPM}_{\hat{Z}} = \{j : \text{primary } \widehat{\text{PIP}}_j > \frac{1}{2}\}$ and MPM_{eq}. Either may exceed K and is reported as computed."
- "Report $R_{\text{eff}} = (\sum_r W_r)^2 / \sum_r W_r^2$ with $W_r \propto \hat{Z}_r$."
- "Neither normalised estimate is generally unbiased at a finite budget."
- "The R-scaling guarantee is stated for the primary PIP; the equal-weight PIP keeps its single-island bias for every R (Prop P(i))."

This *swaps* v3's pooling, where the equal-weight mean was primary and the $\hat{Z}$-weighted one secondary [Docs/Methodology/METHODOLOGY_v3.md:272-275] (MD08). Code (v2.2.1) also computes *plain* PIPs: per island the fraction of particles containing j at the end of each of the 4 measured rounds [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_engine.h:350-371], pooled with equal weights as `pip_plain`, `mpm_plain` [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:167-169]. These are the plain measured-endpoint averages $H_r(h)$ of §2.8.4 for $h = 1\{j \in S\}$, equal-weight pooled (this identification is an editorial reading of the two definitions, not a source statement).

2.8.3 HPM, local search, LS0

Definitions DF15 (HPM and local search), DF16 (LS0).

Statement (verbatim from source) Docs/Methodology/METHODOLOGY_v3.md:258-267

- **Local search:** from the current best scored model (the centre), scan every feasible flip and swap (additions beyond K excluded) with the centre frozen during the scan; move to the best strict improvement (lexicographic ties); at most ℓ moves, then one further scan; stop earlier when a scan finds no improvement. Every evaluated model updates the global best. If the last scan still improves, set `local_limit` (no restart). `local_certified` only when the returned model is a scanned centre with no improving neighbour. The search never alters particles.
- **HPM:** the strict maximum of $\log \varpi(S) + \log m(S)$ over every model ever scored (tables, pilot, calibration, final rounds, rejected proposals, independence proposals, local searches), kept as a running best that survives cache eviction. Only exactly equal scores invoke the tie rule: smallest $|S|$, then lexicographically smallest index set. A near-tie list (models within 10^{-9} nat of the best, drawn from the stored top $40p + 1$) is reported; it never reorders, and `near_tie_truncated` is set if the list may be incomplete.

Equivalently the HPM is the $\succ$-maximum of the scored set $\mathcal{S}$ for the strict total order $S \succ S' \iff (F(S), -|S|, \cdot)$ is larger, the last component being "lexicographically smaller index set first"; code: `hpm_better` [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_util.h:120-126], running best updated on every fresh score event [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_ledger.h:83-115], table scores ingested first [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:402-417]. Lexicographic comparison is on sorted index lists with a proper prefix first [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_util.h:117-119]. The final local search starts from the global best [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:481]; within a scan the best neighbour is chosen by F, exact ties by plain lexicographic order (not size first), and a move needs a strict F increase [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_engine.h:377-425]. v3.1 adds an *informational* heuristic near-tie list at $\max(10^{-9}, 2\eta(\hat{S}))$ (e2) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:181]. Stores: top $40p + 1$ by F in HPM order; leader stores of $20p$ by $\log \gamma_0$ and by $\log \gamma_1$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_ledger.h:73-81].

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:150

LS0: local search from the start mode.

- "After the final local search (and after the extra search from c_4 when $R' > 4$), run one local search with the §6.10 rules from the start mode $\tilde{S}_0 = \arg\max \gamma_0$. $\tilde{S}_0$ is computed exactly by the block DP, with the size-then-lexicographic tie rule."
- "LS0 reads no uniforms. Every model it scores joins the all-scored set and the stores."
- "Models first scored by LS0 have provenance `ls0`. The flag `ls0_improved` is set if LS0 raised the best score."
- "**An LS0 improvement does not fire E3.** E3 refers only to the final local search from the HPM centre."
- "An operational LS0 interruption means no complete output (GUARD-W, #6). A completed LS0 capped at `local_limit` still returns output (#10)."
- "Start-mode ties: keep the enumeration fast path (cap 10^6, `cpp/v3_start.h:183`). On truncation, use a deterministic tie-aware DP that returns the prescribed smallest-size, lexicographically smallest maximiser without enumerating ties (e4, #29)."

#9: the bounded top store after LS0 is "$\mathcal{T}_{\text{new}} = \text{top}_c(\mathcal{T}_{\text{old}} \cup \mathcal{S}_{\text{added}})$, not a superset of $\mathcal{T}_{\text{old}}$"; "stores nested" and "bitwise fit" are struck [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:151]. #10: `local_certified` and card item L1 are evaluated on the finally returned $\hat{S}$, from whichever search ended on it [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:152]. LS0 cost: "The runtime overhead is under 1 s per fit"; in the O-LS0 test export it changed the HPM in 0 of 2,394 fits [NBH/branch_snapshots/O-LS0/REPORT.md:380] [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:48]. Rejected: an HPM–MPM disagreement flag ("No provable power") [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:277].

2.8.4 General-h accumulator, $\hat{f}$, cross-block proxy

Definitions DF18, DF19, DF20.

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:159

Declaration. "$\boldsymbol{\mathcal{H}}$ is a finite list of bounded model-space functions. The default list is fixed from (X, y): model-size indicators, and joint inclusion on neighbour-graph edges ($\leq 2pB$ pairs). A user may also declare h measurable in $(X, y, \mathcal{C})$, fixed before stream 10. An h defined from other calibration information (streams 0–9) is covered only if that information is included in $\boldsymbol{\mathcal{C}}$ before it is frozen (enlarge $\boldsymbol{\mathcal{C}}$ when necessary); measurability in streams 0–9 need not imply measurability in the compressed $\boldsymbol{\mathcal{C}}$. No final-run selection." **Accumulation.** "$H_r(h) = (4N_e)^{-1} \sum_{q=1..4} \sum_i h(S_{rqi})$, taken after each complete measured round ($\varphi = 1$, equal weights after multinomial resampling). Report $\hat{\mu}(h) = \sum_r \hat{Z}_r H_r(h) / \sum_r \hat{Z}_r$ and the equal-weight version." **Not allowed.** "Products of marginal RB-PIPs are not joint estimates. v3.1 ships plain measured-round particle averages for joint h ($c_h = 0$). Rao–Blackwellisation is used only for the marginal PIPs. No RB joint estimator ships." **Archive (optional).** "The measured-round populations as a key → count multiset. An h chosen adaptively after the run does not inherit the fixed-h guarantee." **Continuous parameters.** "Use $\bar{h}(S) = E[h \mid S, X, y]$." **Descriptive only.** "The measured-round frequency of the returned HPM is **not** in $\boldsymbol{\mathcal{H}}$. It may be printed as a descriptive diagnostic with no theorem, next to $e^{F(\hat{S})}/\hat{Z}$." Placement in code: `cpp/v3_engine.h:351--359`. The identity for general h is a PROOF SKETCH (N1).

AMENDMENT_HQ_02 adds d to the default $\boldsymbol{\mathcal{H}}$ ("raw d is recorded", theorems use $(d - \min_\Omega d)/D$), the estimate $\hat{J} = \sum_r \hat{Z}_r B_r / \sum_r \hat{Z}_r - (1/R') \sum_r A_r$ of the Jeffreys divergence $J = E_\pi d - E_{\pi_0} d$ with $A_r = N^{-1} \sum_i d(S_{ri}^{(0)})$ (initial start particles) and $B_r = (4N)^{-1} \sum_{k=1}^{4} \sum_i d(S_{ri}^{(k)})$ (measured endpoints), labelled "Post hoc Monte Carlo estimate of Jeffreys divergence; not a certificate; floating-point error unvalidated" [NBH/branch_snapshots/MEET-v31/AMENDMENT_HQ_02_DEFINITIONS_from_MEET-THM.md:16-41]; and the cross-block proxy $\hat{\varepsilon}_\times = \max_{b \neq c,\, j \in C_b,\, k \in C_c} |X_j^\top X_k|/(n-1)$ (`cross_block_rmax`) with companions $\hat{\kappa} = \min_b \lambda_{\min}(G_{C_b}/(n-1))$, $\hat{\delta} = (K-1)\hat{\varepsilon}_\times/\hat{\kappa}$, $\hat{u} = \sum_b \max_{U \subseteq C_b} x_b(U)$, $\hat{r} = \hat{\delta}\hat{u}/[(1-\hat{\delta})(1-\hat{u})]$, $\hat{\eta} = K \log(1/(1-\hat{\delta})) + \nu \log(1/(1-\hat{r}))$ on the stated domain, else "n/a" [NBH/branch_snapshots/MEET-v31/AMENDMENT_HQ_02_DEFINITIONS_from_MEET-THM.md:43-58]. "Reporting fields 1–3 are specified, not yet coded" [NBH/branch_snapshots/MEET-v31/AMENDMENT_HQ_02.md:15].

2.8.5 Certificate card

Definition DF17; decision MD10. The card is POST HOC, "computed after the run and never feeds back into calibration". Abridged items below keep every quantifier; full text [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:212-255].

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:214-253

Header. Every line is stated in exact arithmetic. Until SC(η_F) is established by validated arithmetic and VAL-FP holds at production size, every computed number carries the label "numerical implementation of an exact-arithmetic certificate", with η printed. L1, L2 and R run under one declared

certificate budget; on exhaustion the item reports "unresolved". A failed or unfinished certificate never changes the fit's status. Under stacking there is one card per reported record.
G1 (deterministic). $\mathcal{T}$ is the deduplicated reported store and $Z_{\mathcal{T}} = \sum_{S\in\mathcal{T}} e^{F(S)}$. Then: $Z_{\mathcal{T}} \le Z$; $\pi(S) \le e^{F(S)}/Z_{\mathcal{T}}$ for every S; $\pi(A) \le \sum_A e^F/Z_{\mathcal{T}}$ for every $A \subseteq \mathcal{T}$; $\pi(S)/\pi(\hat{S}) \le e^{-g}$ on $\mathcal{T}\setminus\{\hat{S}\}$, with $g = F(\hat{S}) - \max_{\mathcal{T}\setminus\{\hat{S}\}} F$; $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$.
G2 (one-sided, output-level). Simultaneously for all S: $\pi(S) \le \min\{1, e^{F(S)}/\max(Z_{\mathcal{T}}, 0.05\hat{Z})\}$; $\pi(\Omega\setminus\mathcal{T}) \ge 1 - Z_{\mathcal{T}}/(0.05\hat{Z})$. This is mass **outside the reported store** (never scored, or scored and evicted). It is **not** a bound on never-scored mass. G2 fails only on $\{Z < 0.05\hat{Z}\}$, and P(run returns output and G2 is false) $\le 0.05 \cdot P(\text{SMC}\cap\mathcal{A}) \le 0.05$. This is a probability **joint with output, never conditional on completion**. [...] If $Z_{\mathcal{T}} \ge 0.05\hat{Z}$, G2 holds surely.
L1 (local). Evaluated on the finally returned $\hat{S}$ [...]. If `local_certified`, then $F(S) \le F(\hat{S})$ for every feasible flip or swap. Report g_{add} ($+\infty$ if $|\hat{S}| = K$), g_{del} ($+\infty$ if $\hat{S} = \emptyset$), g_{swap} ($+\infty$ if there are no swaps), $\rho_{\text{loc}} = \pi(\hat{S} \mid \{\hat{S}\} \cup \mathcal{N}(\hat{S}))$.
L2 (subsets). $g_{\text{sub}} = F(\hat{S}) - \max_{S\subsetneq\hat{S}} F(S)$ and $\sum_{S\subsetneq\hat{S}} \pi(S)/\pi(\hat{S})$. With $\max\emptyset = -\infty$: when $\hat{S} = \emptyset$, $g_{\text{sub}} = +\infty$ and the relative subset mass is 0.
L3 (best-of-scored). If $\mathcal{T}$ = top-$(40p+1)$ of the realised scored set $\mathcal{S}$ and $|\mathcal{T}| \ge 2$, then $\pi(S) \le e^{-g}\pi(\hat{S})$ for all $S \in \mathcal{S}\setminus\{\hat{S}\}$, with $g = F(\hat{S}) - F(\text{top}[2])$. This holds under LS0 because $\mathcal{T}_{\text{new}} = \text{top}_c(\mathcal{T}_{\text{old}} \cup \mathcal{S}_{\text{added}})$.
R (optional). A declared region rule $U(X, y, \hat{S}) \ni \hat{S}$ [...]. B&B on $[\emptyset, U] \cap \Omega$, dropping leaves with $|L| > K$, gives (a)–(c) as in PC2. "Unresolved" on budget exhaustion. In floating point, (a) needs $\tilde{\Phi} \le \tilde{F}(\hat{S}) - 2\eta_F$.
Card-level statement. "The probability that a run returns a card with a false G2, L or R claim is at most 5%", in exact arithmetic. G1, L and R are true surely. [...] **Excluded:** C2; C5; the "unscored mass" wording; the B&B cost remark; any rigorous reading of `global_certified`.

Cuts [...]: G2's premise list (PC1 claim 2 (i)–(iv), premise (i) = Lemma RO plus GUARD-W, a code-audit obligation); L1's LS0 cap/interruption sentence (as #10); R's printed node budget and τ; the floating-point version of the card-level statement (needs validated SC and VAL-FP). See the source span.

2.8.6 Output convention, output event, unavailable outputs

Definition DF21; decision MD12.

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:168

Output convention. "Every estimate on the output event $\mathcal{O}$ is a function of (X, y, seed, binary). The allocation and hardware decide **delivery only** (stop-only guard, Lemma RO). Guarantees are stated for the potential (uninterrupted) run and as **P{$\mathcal{O}$, ·}** (joint with output), never as P(· | $\mathcal{O}$). P(no output) is an operational frequency that depends on the allocation and hardware, and is reported with them (V13)."

#15: "On every return without complete output (`resource_limit`, `schedule_limit`, `numerical_failure`, GUARD-W stops, LS0 stop, degenerate statuses), `fit$log_z` and **every estimate field** ($\hat{Z}_r$, $\hat{Z}$, bounds, PIPs, MPMs, $\hat{\mu}(h)$, card) are NA by exact name. [...] This fixes the partial-match hazard, where `fit$log_z` silently returned `log_zbar0`" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:162]. On `schedule_limit` and `numerical_failure` the v3 rule is "the best scored model is returned", labelled incomplete [Docs/Methodology/METHODOLOGY_v3.md:49-51].

2.9 The settings table and the (n, p) rules

Definition DF25; decision MD13. Ground rule 9 requires every setting to be "(a) modelling convention with a stated reason; (b) explicit function of (n, p) or X; (c) declared resource budget; (d) numeri-

cal tolerance; (e) arbitrary", and "**No setting is class (e) in the method's law**" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:287-294]; "No separate p ≪ n / p ≫ n rule is needed: K, B, N_0, the exact-branch rule and r_n switch regime by formula" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:210]. The constants paragraph: "These values are conventions, not derived optima; changing them is not automatically law-preserving." [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:208]. The table below transcribes all 49 rows of METHOD §5 (all six columns; markdown converted to LaTeX text) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:296-346]; row k is source line $297+k$, shown in parentheses in the first column. "Numerical boundaries quoted from seats are seat-derived unless marked [chair-checked]." Code pointers inside the rows (e.g. `v3_core.cpp:142`) are METHOD's own and refer to the v2.1 tree (see the citation-version divergence in §2.12).

# (line)	setting (value; source)	class	reason	behaviour as n, p grow	action
1 (298)	g = n, τ = 1 (M §5; `v3_score.h:175`)	(a), (b)	unit information; ridge keeps m(S) finite for collinear X_S	p-free; τ matters more for p ≫ n (collinear sub-models)	keep (Felipe)
2 (299)	IG(0.01, 0.01) (M §5)	(a)	proper, as the proofs require	pairwise effect ≤ $n \cdot b_0 \lvert 1/Q_1 - 1/Q_2 \rvert$ on standardised data; relative to the gap it is not uniformly negligible (two equal-size, nearly collinear singleton models give a relative rate effect → 0.0142; review check, SOL1 #43); negligibility needs residual-scale assumptions; not negligible at tiny n	keep; no universal "negligible" claim (R2_C)
3 (300)	BB(1, p), w(k) = ½/C(2p−1, k), W_K explicit (M §5)	(a), (b)	sparse, multiplicity-aware; the project theory is for it	defined for every finite p; dense growing truths are not its regime	keep
4 (301)	Cap K = min(p, ⌊n/2⌋) (M §3, §5)	(a), (b)	keeps near-interpolating models out	void for p ≤ n/2; binds for p ≳ n/2; K = 1 at n = 2, 3; $s_0 > K$ unrecoverable by definition	keep
5 (302)	Input domain: n ≥ 2, p ≥ 1 (v3.1, proposed; current code rejects n < 3, `v3_preprocess.R:103–107`)	(b)	standardisation and the target are defined at n = 2	n = 2 → exact branch (K = B = 1, E = p + 1); n = 1 → `degenerate_n`; n = 0 rejected	**e3** (§2 #25)
6 (303)	Exact-constant column drop; `y_constant`; `no_predictors` (M §4)	(a)	removes undefined scales	regime-free	keep
7 (304)	Standardisation: max-abs, centre, SD (n−1) (M §4)	(a), (b)	affine and permutation invariance	regime-free	keep
8 (305)	Near-constant anchored two-pass route at $\sqrt{\varepsilon_{\text{mach}}}$ (`v3_preprocess.R:47`)	(d)	cancellation protection, not screening	regime-free	keep; document (R2_C)
9 (306)	Duplicate report $\lvert r \rvert \geq 1 - 10^{-10}$ (M §4 rule 5; `v3_core.cpp:722`)	(d)	reporting only; duplicates are kept	regime-free	keep
10 (307)	B = min(⌊√n⌋, ⌊√p⌋, B*) (M §3)	(a), (b)	Felipe's rule; B ≤ K and B ≤ p	√p binds for p < n, √n for p > n; B = 1 at n = 2, 3	keep
11 (308)	**B* = 16** (M §3, §8.7; `v3_core.cpp:142`)	**(c)**	exhaustive-table budget: ≤ 4,096p scores and 65,536p bytes	binds once n, p ≥ 289; theorem SIZE condition: B = 16 covers groups 9..16	**e1** (§2 #23)
12 (309)	$\ell = \lceil \log_2(n + p + 1) \rceil$ (M §3)	(b), (c)	local-search move cap and rebuild cap	10–11 at the study, 20 at n + p ≈ 10^6; not guaranteed enough for growing support (obligation)	keep
13 (310)	L = max(4, $\lceil \log_2(n + 1) \rceil$) (M §3)	(b), (c)	burn-in at φ = 1; floor binds for n ≤ 15	grows with n, not with p; "L logarithmic versus t_{mix} possibly polynomial" is V10 OPEN	keep
14 (311)	N_0 = max(2,500, n, p) (M §3)	(b), (c)	floor: the 2,500 is a declared budget. "Per-island RB-PIP MC SD ≤ 0.01" is an iid-sampling benchmark only ($1/(2\sqrt{N})$); resampling creates dependence, so it is not a guarantee. Grows with dimension	floor binds at the study; N_0 = max(n, p) above 2,500 ⇒ Θ(p·max(n, p)) move steps per stage, Θ(p^2) only when n = O(p); score-size costs extra (preflight reports it)	keep; C6 iff a particle-depth claim or any demonstrated fit with p > 2,500 (§3)
15 (312)	$H_0 = \lceil 512 \cdot N_0/2{,}500 \rceil$ (M §3)	(b), (c)	declared calibration/final ratio 0.2048	grows linearly; single-tier calibration/final = 0.0512 + 0.2048·I ≤ ≈ 2.1 at I = ℓ = 10 (seat-derived; ≤ ℓ rebuilds per tier)	keep; print the formula (§2 #19)
16 (313)	T_0 = max(1,000, ℓ^2) (M §3)	(b), (c)	finite stage ceiling; exceeding it ⇒ `schedule_limit`	the ℓ^2 term binds only for n + p ≥ 2^{31}; ideal-grid T* = O(log log p); as-coded T_n OPEN (Conjecture P)	keep; **w4** (§2 #33)

# (line)	setting (value; source)	class	reason	behaviour as n, p grow	action
17 (314)	Tiers e ∈ {0, 1}, factor 4, ≤ 1 escalation (M §3, §6.9)	(c)	bounded escalation, decided before final randomness	≤ ≈ 16× bridge work; no unbounded rescue; beyond → `schedule_limit`	keep
18 (315)	R = 4 final islands; optional R′ = 4k at N_e, declared before stream 10 (M §3; §2 #7)	(c)	replication and parallel width	fixed 4 gives no growing-dimension accuracy; stacking at N_e lowers the variance of the averaged evidence exactly ∝ 1/R′; for normalised PIPs and general h, 1/R′ is an asymptotic (delta-method) rate under moment conditions, not finite-R monotonicity	keep; **w7**
19 (316)	4 diagnostic islands per rebuild (M §6.8)	(c)	E2 is an SD over 4 values	regime-free	keep
20 (317)	4 measured rounds (M §6.11)	(c)	RB averaging after burn-in	constant; correlated rounds are not 4 replications	keep; general-h accumulator (§2 #12)
21 (318)	Exact branch $\Sigma_{k\le K}\, C(p, k) \le (5H_0 + 4N_0)\cdot p$ (M §6.2)	(b), (c)	enumerate when it costs about one SMC stage of all populations	fixed p is eventually exact as n grows ($N_0 \ge n$); all p at n ≤ 5 [chair-checked]; n = 10: p ≤ 36; n ≥ 22: every p ≤ 17 is exact (a uniform sufficient range; the maximum for n ≤ 2,500, while larger n can admit more, e.g. n = 3,000, p = 18; review check, SOL1 #33); growing p ≪ n need not be exact	keep; **w2**
22 (319)	Clustering: complete linkage on 1 − |r|, size-only cut, next-fit packing, lexicographic ties (M §6.3)	(b), (c)	deterministic X-dependent partition under the table budget	no correlation cut; $O(p^2)$ memory, $O(p^2)$–$O(p^3)$ time; packing can merge separated groups	keep; **w5**; C11 (§3)
23 (320)	Tables (all subsets per block, exponent 1, full y) and start π_0 (M §6.3)	(a), (b)	exactly computable composite step, corrected at φ = 1	constant B bounds each table; the number of blocks grows with p	keep; item 1 = Outcome 2: no depth or speed claim for the start
24 (321)	Neighbour graph: $r_n = \sqrt{(2 \log p / n)}$, strict exceedance, forced best partner, ≤ B outgoing, symmetric union (M §6.4)	(b), (c)	chance-correlation scale of p columns; proposal heuristic	$r_n \to 0$ when log p = o(n); $r_n \ge 1$ iff $p \ge e^{n/2}$ (equality impossible for integer n, p) ⇒ 1-NN graph for p > 1; degrees may exceed B	keep; **w1**, **w3**
25 (322)	`cut_group` diagnostic at $\rho = r_n$ (`v3_core.cpp:760–770`)	(b)	descriptive only	can give large components or none	keep, descriptive
26 (323)	Move round m = p + B + 2, fresh random order; multinomial resampling each stage (M §6.5–6.6)	(b), (c)	reversible, self-adjoint kernel; fixed mutation allocation	Θ(p) sequential steps per particle per round: inherent to the kernel (I1)	keep; changing it is WORKHORSE
27 (324)	ESS threshold 0.90, pilot and grid (M §6.6–6.7; `v3_calib.h:31, 79`)	**(c)**, with B's caveat	one-step dispersion budget (weight $CV^2 \le 1/9$). Caveat: "defines the law of $\boldsymbol{C}$; carried by the theorems as a hypothesis (Theorem G needs the same threshold for pilot and grid)"	dimensionless; does not control accumulated error or omitted mass	keep; S3 R3-4 records (c) over B's (a)
28 (325)	60 bisection halvings; endpoint snap 10^{-12} (`v3_calib.h:34–43, 82–94`)	(d)	2^{-60} below double resolution on [0, 1]	at very sharp crossings, progress may not resolve; failure is reported	keep
29 (326)	Pilot floor $1/T_e$ (M §6.6)	(b), (c)	exploratory progress within the ceiling; forced steps counted and trigger escalation	shrinks as T_e grows	keep
30 (327)	V = top 20p by γ_0 ∪ top 20p by γ_1 ∪ {∅}, |V| ≤ 40p + 1 (M §6.7)	(b), (c)	O(p) = size of a 1-flip neighbourhood; caps grid cost $60\cdot\|V\|\cdot T_e$	the retained fraction of model space vanishes; omitted mass not controlled	keep

# (line)	setting (value; source)	class	reason	behaviour as n, p grow	action
31 (328)	Grid convergence = grid equality; ≤ ℓ rebuilds (M §6.7)	stopping rule (b), a deterministic rule (reclassified from the lead's (a), R2_B row 20, by HQ amendment 01; values unchanged); cap (b)/(c)	a flat posterior keeps V changing by MC noise	convergence and P($S_0 \in V$) OPEN (obligations)	keep; contract (§2 #18)
32 (329)	E1 > 1 nat, E2 SD > 1 nat, E3, forced steps, grid failure/non-convergence, `local_limit` (M §6.9)	(c) diagnostic criteria (S3 R3-4)	odds factor e; triggers define the law of $\mathcal{C}$	scale-free; no positive power as dimension grows	keep; labelled heuristics
33 (330)	Low-ESS flag: final stage < 0.5N (M §2)	(c), (b)	report flag	regime-free	keep
34 (331)	Top store 40p + 1; leader stores 20p (M §6.10)	(b), (c)	HPM survives eviction; near-ties, Z_$\mathcal{T}$ and the card read it	linear in p; may truncate many near-equivalent models; restricted summaries labelled restricted	keep
35 (332)	Score cache 500p, clear when full (M §8.6)	(b), (c)	a hit returns identical bits; no effect on the law	key storage up to O(pK)	keep
36 (333)	Near-tie window 10^{-9} nat (M §6.10)	(d)	informational; the HPM uses exact equality	can fall below scorer error at large n; η depends on k, Q and conditioning, not on n alone, so a numerical crossing point needs a specified design and model (η ≈ $6{\cdot}10^{-10}$ at n = 200; seat-derived)	keep + **e2** heuristic list (§2 #24)
37 (334)	HPM tie rule: exact equality, then smallest |S|, then lexicographic (M §6.10)	(a)	argmax convention	scale-free	keep
38 (335)	MPM threshold ½; α = 0.05 for the Markov bound (M §6.11–6.12)	(a)	Hamming-loss rule; declared confidence	dimension-free; MPM may exceed K	keep
39 (336)	Q recompute below $10^{-8}{\cdot}y^Ty$; Cholesky → refactor → augmented QR → `numerical_failure` (M §6.13)	(d)	stable evaluation of one score	conditioning worsens with n, K; failure explicit	keep
40 (337)	Scorer error expression η (64ε…), used by `global_certified` (`v3_score.h:240`)	(d) as an indicator; (e) only if read as rigorous	the source calls it heuristic	no uniform validity established	**e5** (§2 #27)
41 (338)	LS0 start-mode tie enumeration cap 10^6 (`v3_start.h:183`)	(d)/(c) fast path	DP argmax exact; only the tie choice can truncate	ties can proliferate with duplicates or symmetry	**e4** (§2 #26)
42 (339)	Certificate budget (scores, B&B nodes), region rule U, τ (card)	(c) declared certificate-region settings	PC2: Seat C's correction 6 upheld	"unresolved" on exhaustion; L2 needs $2^{\|\hat{S}\|} - 1$ scores	new in v3.1 (§2.1)
43 (340)	Default $\mathcal{H}$: size indicators, neighbour-edge pairs ≤ 2pB (§2 #12)	(b)	fixed from (X, y) via the graph	linear in pB	new in v3.1
44 (341)	Allocation (default 8 GiB, 24 h), preflight, stop-only wall guard (M §2, §8.6)	**(c) declared input**	decides delivery only (Lemma RO)	a fixed allocation cannot support unrestricted growth (V13, by definition); the setup component alone, $8(3np + 3p^2)$ bytes, gives $p \le [-n + \sqrt{(n^2 + 4{\cdot}8{\cdot}2^{30}/24)}]/2$ at 8 GiB (18,818 at n = 200; 11,398 at n = 20,000; review check, SOL1 #39); this excludes all other allocations and workers and is not a feasible-fit threshold (seat-derived)	**e8**, **e6** (§2 #22, #28)

# (line)	setting (value; source)	class	reason	behaviour as n, p grow	action
45 (342)	Seed ∈ [1, 2^{31} − 1]; stream map 0–9 (calibration), 10..9+R′ (final); 53-bit uniforms; integer dimensions	(c), (d)	reproducibility conventions; "bitwise" is per binary and BLAS	integer ranges can be exceeded at huge sizes	**e7** (§2 #29); IND-T for 10..9+R′ (§6)
46 (343)	Workers: processes; worker count (§2 #1–#5)	(c)	delivery only; bitwise to serial	regime-free	new in v3.1
47 (344)	Archive/export limits: V snapshot all if p ≤ 25, else ≤ 10^4; R Gram copy only for p ≤ 2,000 (`v3_calib.R:15-20`; `v3_setup.R:9`)	(c)	storage choices; full V still builds the grid	truncation grows with p; never reconstruct a "full" V from a snapshot	keep; expose truncation
48 (345)	Other operational constants: exact-branch timing (200 scores, ≥ 64 columns), duplicate-check chunk 512, compiler-lock waits	(c)	profiling and workspace	hardware-dependent; outside the theory	keep
49 (346)	Gram-based scorer; O(p^2) correlations and distances (M §8.1)	(c) implementation of the exact target	—	the dominant practical storage limit of v3.1; integer-dimension, stream and allocation ranges and checked-arithmetic rejection are also hard implementation limits (§2 #29, row 45)	v3.2: C7, C11 (§3)

Per-regime summary (tiny n; fixed p, $n \to \infty$; growing $p \ll n$; $p \approx n$; $p \le n^A$; $p \le e^{Cn^\alpha}$; joint growth) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:350-377] and the adopted theory-scope sentence: "The p ≤ n^A scope of the selection theorems comes from the target theory (Felipe's prior and the proof technique), not from a sampler choice. [...] Whether our prior admits it is OPEN, not impossible." [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:381]. Method-induced restrictions: the cap excludes $s_0 > K$; fixed B limits whole-group recovery (as a single block) to $g \le B$; bounded stores can omit posterior mass; logarithmic ℓ and L may be short for some proof routes; $N_0 \ge p$ and full-coordinate rounds impose real computational growth [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:383-388].

2.10 The pipeline end to end

Assembled from M §6.1 and §7 [Docs/Methodology/METHODOLOGY_v3.md:133-149] [Docs/Methodology/METHODOLOGY_v3.md:293-306] with the v3.1 additions [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:85]. Steps marked [v3.1] are not in the frozen code v2.2.1 (§2.12).

SMC-BVS v3.1(X, y, s) — pipeline

1. **Checks and preprocessing** (§2.1); statuses `invalid_input`, `no_predictors`, `y_constant` [v3.1: `degenerate_n`, `unsupported_size`; $n = 2$ admitted]. Preflight → maybe `resource_limit`.
2. **Budgets** $K, B, \ell, L, N_0, H_0, T_0$ from (n, p). If $\sum_{k \le K} \binom{p}{k} \le (5H_0 + 4N_0)p$: **enumerate** (exact Z, PIPs, HPM in the strict order, near ties), status `exact`; report and stop.
3. **Setup**: Gram, correlations; complete-linkage blocks $C_1..C_J$ (size cut B, next-fit packing); exact tables $r_b(U)$ for all $U \subseteq C_b$ (all table scores enter the HPM ledger) [v3.1: tables in parallel]; size profiles, $\bar{Z}_0$ by convolution, sampler of π_0; neighbour graph $N(\cdot)$.
4. **Calibration**, tier $e = 0$ (streams 0 and 1–4): pilot (H_e particles from π_0; adaptive ESS-0.90 schedule, floor $1/T_e$; multinomial resampling and one round per stage; L burn-in + 4 measured rounds; local search) → V, preliminary grid → repeat ≤ ℓ times: 4 diagnostic islands (H_e each) on the grid [v3.1: concurrent] → local search → update V → rebuild grid; stop when the grid repeats → triggers E1–E3, forced steps, grid failure/non-convergence, `local_limit`.
5. If any trigger fired at $e = 0$: re-preflight, then **one** escalation to tier 1 (streams 5 and 6–9), keeping score leaders. If grid construction fails at $e = 1$: `schedule_limit` (best scored model returned).
6. **Freeze** $\mathcal{C}$: grid, tier, N_e, L, inherited leaders/HPM [v3.1: and R', declared before stream 10; default $R' = 4$; optional $R' = 4k$ or $R'_n = 4\lceil \log_2(n + p + 1) \rceil$].
7. **Final islands** $r = 1..R'$ on streams 10..9 + R' [v3.1: concurrent process workers, merged by Lemma S]: N_e fresh iid π_0 draws; for $t = 1..T$: weights $e^{(\varphi_t - \varphi_{t-1})d}$, log-increment, multinomial resampling, one

random-order round at φ_t; then L burn-in and 4 measured rounds at $\varphi = 1$ with RB recording of o_j [v3.1: $H_r(h)$ for $h \in \mathcal{H}$; d means for $\hat{f}$].

8. **Final local search** from the global best (all scored models, all phases) [v3.1: scans in parallel; if $R' > 4$ also from c_4]; then [v3.1] **LS0** from $\tilde{S}_0$.
9. **Report**: HPM $\hat{S}$ (strict order F, then $|S|$, then lex) with provenance; near ties; $\hat{Z}_r$, pooled $\hat{Z}$, $0.05\hat{Z}$; PIPs [v3.1: $\hat{Z}$-weighted primary, equal-weight secondary, both MPMs, R_{eff}; $\hat{\mu}(h)$; $\hat{f}$, $\hat{\varepsilon}_\times$ and companions]; flags; diagnostics [v3.1: POST HOC certificate card per record; on any return without complete output every estimate field NA].

2.11 v3 → v3.1: what changed and what did not

"**Does the 2,400-fit study need a re-run? No.** The study stands as **v3.1 at R′ = 4, with POST HOC additions**"; its saved sampler estimates are reusable with stated reporting limitations, and "Full v3.1 records are not yet produced" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:45]. Unchanged (M §§): target (§5); blocks, tables, composite start (§6.3); neighbour graph (§6.4); move round (§6.5); resampling (§6.6); calibration law (§6.6–6.9); budgets (§3); HPM decision (§6.10); evidence estimator (§6.12) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:26-34]. The full per-section map is [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:74-106]. Change classes: ADDITIVE (outputs on completed fits unchanged or nested), POST HOC (computable from saved fits), LAW-PRESERVING (same law, different seed-to-uniform map), WORKHORSE (changes the method's law) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:16-20]. "No change is WORKHORSE." [NBH/branch_snapshots/MEET-v31/MINUTES.md:28]. The table summarises METHOD §2 rows (names shortened; see source lines).

#	change	class	effect on the 2,400 fits (source line)
0	composite block start kept; item 1 = Outcome 2 (no depth/speed claim)	none	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:118]
1	(a) concurrent final islands, Prop W	ADDITIVE (bitwise)	none; executed bitwise smoke + 3/3 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:131]
2	(b) concurrent diagnostic islands, Prop W-cal	ADDITIVE (bitwise) claimed from code reading only	none; W-cal not executed [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:132]
3	(c) parallel local-search scans	ADDITIVE	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:133]
4	(h) parallel block-table stage	ADDITIVE (bitwise)	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:134]
5	process workers; counter rule	ADDITIVE (engineering)	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:135]
6	GUARD-W semantics	ADDITIVE (spec)	none; the 6 `resource_limit` fits stay stops [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:136]
7	optional stacking $R' = 4k$ at N_e; c_4 search; two records	ADDITIVE (nested)	none; study is $R' = 4$ [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:142]
8	LS0 from the start mode	ADDITIVE	0 of 2,394 HPMs changed (test export); merged replay pending [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:150]
9	D3c wording for LS0 (store = top$_c$, not superset)	spec text	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:151]
10	certificate attaches to the returned $\hat{S}$	ADDITIVE (spec)	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:152]

#	change	class	effect on the 2,400 fits (source line)
11	$\hat{Z}$-weighted PIP primary; equal-weight secondary; both MPMs; R_{eff}	POST HOC	no refit; primary needs full rescoring (3 designs done: MPMs never differ, $\max\|\Delta\text{PIP}\| \le 1.3 \cdot 10^{-3}$) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:158]
12	general-h accumulator	ADDITIVE	not recoverable: no joint-summary claims [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:159]
13	certificate card	POST HOC	computable where the top store was saved [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:160]
14	$Z_{\mathcal{T}}$ and the underestimation certificate	POST HOC	as #13 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:161]
15	unavailable output = explicit NA	ADDITIVE (interface)	only the 6 stopped fits' records change [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:162]
16–21	output convention; card sentence; calibration contract; depth/cost; memory wording; §9 limitation wordings	spec text	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:168-173]
22–29	item-9 e8, e1, e2, e3, e4, e5, e6, e7	spec text / POST HOC / ADDITIVE	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:179-186]
30–36	wording fixes w1–w7	spec text	none [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:192-198]

Not claimed / not proved — what is not claimable for the study

Quoting METHOD §0 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:52-61]: joint posterior summaries ("No joint-summary claims for the 2,400 study fits"); the $\hat{Z}$-weighted (primary) PIPs and MPM on the full study until full rescoring ("So far only 3 designs have been rescored"; the equal-weight PIPs remain valid as the **secondary**); any stacking ($R' > 4$) result; a 4-core speed-up ("The 1.9× figure is a 2-core measurement"); "Bitwise" for (b) until W-cal runs; "Depth free of N, H, R"; "LS0 as 'run by the production core' until the merged-LS0 replay runs"; the certificate card wherever the top store was not saved; "Any depth or speed claim for the block start".

Rejected changes (with reasons): backfit start (WORKHORSE, no exact sampler with known normaliser, no theorem); C8 split N_e; C9 shorter/shared calibration; C10 parallel-in-particle kernel; HPM–MPM flag; truth-based gate for the primary PIP; RB joint estimator; PSIS-$\hat{k}$/$\hat{R}$ flags; feeding certificates back [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:269-281]. Engineering checks that gate claims (W-cal, merged-LS0 replay, GUARD-W audit, IND-T, VAL-FP, preflight, checked arithmetic, full PIP rescoring, calibration timing) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:392-407].

2.12 Implementation status (frozen code v2.2.1 vs spec)

All study fits used the frozen tree `Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/`; "v2.2.1 = v2.2 core, byte-identical, plus one merged study layer" and "No numerical path changed" [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/CHANGES_v2_2_1.md:3-25]; "The v2.2.1 rerun reproduces the 2026-10-23 fits (`Results/SMC_BVS_v3_production/`) exactly." [README.md:88]. Paths below are relative to that tree's `core/`. "Spec only" = specified, no code in the repo; "prototype" = code on a `theory/*` branch (the archive snapshots hold only .md/.tex/.lean).

component	where (code location, verified)	status
preprocessing, statuses, budgets	`v3_preprocess.R:19-33, 97-132`	implemented (v3 rules; $n < 3$ rejected)
$n = 2$, `degenerate_n`, `unsupported_size` (e3, e7)	—	spec only

component	where (code location, verified)	status
prior, score $m(S)$, $F(S)$	`cpp/v3_score.h:39-72, 175-238`; `cpp/v3_engine.h:89-98`	implemented, matches
exact branch	`cpp/v3_core.cpp:97-110, 526-570`; `v3_exact.R`	implemented
blocks, cut, packing; neighbour graph	`cpp/v3_blocks.h:39-173`	implemented, matches
tables, $\bar{Z}_0$, `draw_pi0`	`cpp/v3_start.h:73-176`	implemented (serial tables)
start mode $\tilde{S}_0$ (DP, cap 10^6)	`cpp/v3_start.h:178-240` (header only; production removed its Rcpp export)	present, unused in production; e4 tie-aware DP spec only
move kernels, random-order round	`cpp/v3_engine.h:111-215`	implemented; N9 PASSED
resampling, island loop, RB recording	`cpp/v3_engine.h:231-375`	implemented
pilot rule, V, grid, rebuild, escalation, freeze	`cpp/v3_calib.h:25-98`; `v3_calib.R:28-175`	implemented
final run, 4 islands, streams 10–13	`cpp/v3_core.cpp:421-520`; `v3_run.R:84-90`	implemented, serial, $R = 4$ fixed
(a) concurrent final islands	METH-AUDIT harness on branch `theory/METH-AUDIT`	prototype; executed bitwise 3/3
(b) concurrent diagnostic islands; (c) parallel scans; (h) parallel tables	—	spec only (W-cal not executed)
stacking R', c_4 search, two records	—	spec only
LS0, `ls0_improved`, provenance `ls0`	`ls0_core.R`, `ls0_exports.cpp` on branch `theory/O-LS0`	prototype (test export); not merged
HPM order, stores, near ties (10^{-9})	`cpp/v3_util.h:120-126`; `cpp/v3_ledger.h:32-115`; `v3_run.R:136-140`	implemented, matches
local search, `local_certified`	`cpp/v3_engine.h:377-425`; `v3_run.R:141-147`	implemented
e2 heuristic near-tie list, $\eta(\hat{S})$	(`error_bound` exists, `cpp/v3_score.h:240-246`)	spec only
$\hat{Z}_r$, $\hat{Z}$, $0.05\hat{Z}$	`cpp/v3_engine.h:300-323`; `v3_run.R:117-124`	implemented
equal-weight RB-PIP and MPM	`v3_run.R:126-130` (named `pip_primary`)	implemented (v3.1 *secondary*)
plain PIP	`cpp/v3_engine.h:350-371`; `v3_run.R:167-169`	implemented (v2.2 addition)
$\hat{Z}$-weighted PIP, MPM$_{\hat{Z}}$, R_{eff}	`pip_rescore.R` (MEET-v31 evidence, POST HOC)	POST HOC script; 3 designs rescored
general-h (joint), $\hat{J}$, $\hat{\varepsilon}_\times$	—	spec only
$Z_{\mathcal{T}}$, certificate card	—	spec only (POST HOC)
E1–E3, low-ESS flags; calibration triggers	`v3_run.R:148-156`; `v3_calib.R:58-118`	implemented
NA estimate fields	—	spec only
stop-only runtime guard	`cpp/v3_engine.h:336-339`; `v3_fit.R:76-81`	implemented (serial core); v3.1 worker/LS0 audit open
preflight	`v3_fit.R:16-46`	implemented: SMC memory only

Location sources: LS0 prototype [NBH/branch_snapshots/O-LS0/REPORT.md:69] [NBH/branch_snapshots/O-LS0/REPORT.md:95-97]; METH-AUDIT harness [NBH/04_code_and_files_index.md:366]; `pip_rescore.R` [NBH/04_code_and_files_index.md:369]; production removals [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/README.md:15-19].

DIVERGENCE — spec v3.1 vs code: primary PIP

v3.1 makes the $\hat{Z}$-weighted PIP primary (#11, MD08). The frozen core computes only the equal-weight mean and calls it `pip_primary` ("equal-weight mean over islands (M §6.11)") [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:129]; production removed "secondary PIP computation and outputs on both branches" [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/README.md:16], and the study adapter stops if a secondary PIP is returned [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/study/adapter.R:229]. METH-AUDIT flagged it: "the $\hat{Z}$-weighted (secondary) PIP of METHODOLOGY §6.11 is not computed by the core" [NBH/branch_snapshots/METH-AUDIT/REPORT.md:232]. Consequence: every study field or table named "primary PIP" is the v3.1 *secondary* PIP. The v3.1 primary exists only via POST HOC rescoring, done on 3 designs [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:54].

DIVERGENCE — decision vs code: LS0 not in the production core

MD03 decides "LS0 in the shipped core". In fact "It is not yet in the production core, and the integrated replay is pending" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:48]; METH-AUDIT: "**LS0 is not in the production core**" [NBH/branch_snapshots/METH-AUDIT/REPORT.md:235]. The O-LS0 check ran LS0 as a test export on the production core, and "No production SMC fits were rerun with LS0 integrated (`ls0_enable_integrated` exists in `ls0_core.R` but was not used)" [NBH/branch_snapshots/O-LS0/REPORT.md:69]. The start-mode function is in the header but production removed "the start-mode Rcpp export" [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/README.md:15]. Hence OP-LS0 is a premise, not a code fact, and theorems "about the shipped method" that use LS0 (e.g. TH05 (b1-LH)) are about the spec; merged-LS0 replay = engineering check 2 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:399].

DIVERGENCE — spec v3.1 vs code: islands, streams, parallel layer

Code fixes `islands = 4L` [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:88] and seeds final islands on streams $9+r, r = 1..4$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:468-478], serially; the stream map is 0–13 [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:8-11]. No R', no c_4 search, no baseline/expanded records, no concurrency: "The production core is serial throughout" [NBH/branch_snapshots/METH-AUDIT/REPORT.md:172]. Prop W was executed with the METH-AUDIT harness, not the core; W-cal "is code reading only" [NBH/branch_snapshots/METH-AUDIT/REPORT.md:216].

DIVERGENCE — spec v3.1 vs code: unavailable outputs, statuses, guard

#15 requires every estimate field NA by exact name on any return without complete output. The core sets `fit$log_Zbar0` after setup [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_fit.R:146] and writes `log_Z` only on completion [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:121]; on a stop, R's partial matching makes `fit$log_Z` return `log_Zbar0` — the hazard #15 names [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:162]. Statuses `degenerate_n` and `unsupported_size` do not exist; $n < 3$ is `invalid_input` [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_preprocess.R:106]. Also v3 §2 defines `resource_limit` as "preflight predicts the memory or time allocation will be exceeded" [Docs/Methodology/METHODOLOGY_v3.md:51-52], while the code also returns it from a mid-run stop-only guard with partial, labelled results [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/README.md:19]; the 6 study stops are such wall-time stops [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:64]. v3.1's GUARD-W wording covers this (guard stop ⇒ no output).

DIVERGENCE — spec v3 vs code: preflight

M §8.6: "estimate peak memory [...] and wall time [...]; if either exceeds the job allocation, return `resource_limit`" [Docs/Methodology/METHODOLOGY_v3.md:321-324]. Code: "Lead rulings [...] SMC preflight is memory only" [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_fit.R:16], with block count proxy $\lceil p/B \rceil$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_fit.R:28] — the e6 issue ("use J ≤ p blocks (not ⌈p/B⌉)") [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:185].

DIVERGENCE — reporting fields not coded

General-h for joint h (sizes, neighbour-edge pairs, d), $\hat{J}$, $\hat{\varepsilon}_\times$ and companions, $Z_{\mathcal{T}}$ and the certificate card, R_{eff}, $\text{MPM}_{\hat{Z}}$, the e2 heuristic near-tie list: no code in v2.2.1 (searched the tree); "Reporting fields 1–3 are specified, not yet coded" [NBH/branch_snapshots/MEET-v31/AMENDMENT_HQ_02.md:15]. The only measured-round particle averages recorded are the marginal plain PIPs [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_engine.h:359-361]; "The measured-round populations were not saved" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:53].

DIVERGENCE — citation versions: METHOD's code line numbers

METHOD_v3.1_FINAL cites code lines that match the v2.1 tree `Scripts/SMC_BVS/core/`, not v2.2.1: e.g. "B* = 16 (M §3, §8.7; `v3_core.cpp:142`)" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:308] is `B_MAX` at line 142 of [Scripts/SMC_BVS/core/cpp/v3_core.cpp:142] but at line 170 of [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/cpp/v3_core.cpp:170]; `timings$calibration` and `timings$final` are at `v3_fit.R:124, 136` in v2.1 [Scripts/SMC_BVS/core/v3_fit.R:124] but at lines 158 and 170 in v2.2.1 [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_fit.R:158]. The header-file citations (`v3_score.h`, `v3_start.h`, `v3_calib.h`, `v3_preprocess.R`) agree in both trees where checked. Not a behavioural divergence; agents must resolve METHOD's `file:line` pointers against `Scripts/SMC_BVS/core/`.

Checked and found consistent (no divergence): prior and score; cap K in scorer, Gibbs step and local search; block construction and tables; start convolution and sampler; neighbour graph; all four elementary steps and the random order; multinomial resampling; pilot and grid rules; rebuild loop and single escalation; evidence estimator; HPM order and running best over tables, calibration (inherited) and final phases; local-search rules; E1–E3 definitions.

2.13 Method decisions MD01–MD13 (MEET-v31, 2026-11-03)

Decision text = MINUTES §2 "Decision" column (verbatim); reason = its "Reason" column (verbatim unless marked); status = `ID_REGISTRY` status column. Meeting: chair Opus 5.5 (non-voting), seats A, B (Fable 5.1), C (gpt-6-astra) [NBH/branch_snapshots/MEET-v31/MINUTES.md:5-11]. Felipe's ground rules (DOSSIER §1) are summarised in [NBH/notation_and_conventions.md:11]; objectives: (1) accuracy in recovering the true posterior, (2) parallel scalability, (3) BVS selection accuracy.

MD01 MEET-v31 decision 1 decided (MEET-v31 MINUTES §2); closed in: R1 (all three); R2 rule; R3-6 procedure; result §3

Decision. **Block (local-to-global) start kept.** The backfit start is rejected for paper 1. Arm D is v2 evidence only. [NBH/branch_snapshots/MEET-v31/MINUTES.md:36]

Reason. "Ground rules 1, 5 and 7: no irrefutable case for dropping it; it carries the S1/S2 theorems. Backfit is WORKHORSE (full re-run) with no theorem, and no exact sampler with a known normaliser was exhibited for it [...], so $\hat{Z}$ unbiasedness is not available. The earlier 'headline gain capped at 6/2,400' rationale is withdrawn [...]." Felipe: "I'm open to dropping it but I need a very serious, irrefutable argument" [pos 1132]; ablation result Outcome 2 [NBH/03b_reasoning_and_traces.md:161-166]. METHOD §4 cites ground rules "1, 5, 8"; HQ: "both sets are correct in context" [NBH/branch_

snapshots/MEET-v31/AMENDMENT_HQ_01.md:141].

MD02 MEET-v31 decision 2 decided (MEET-v31 MINUTES §2); closed in: R1 (all three)

Decision. **HPM stays the only decision rule.** MPMs of both PIPs are reported as descriptions. No HPM-vs-MPM disagreement flag. [NBH/branch_snapshots/MEET-v31/MINUTES.md:37]

Reason. "HPM is the 0–1 Bayes rule (T07), and the finite-n spine T38(0)/N200 is about it. A disagreement flag has no provable power (ground rule 3)." HQ: "HPM stays primary, because it's the rule that is easiest to prove for an SMC output" [NBH/03b_reasoning_and_traces.md:168-172].

MD03 MEET-v31 decision 3 decided (MEET-v31 MINUTES §2); closed in: R2 D3c

Decision. **LS0 in the shipped core,** after the final local search. Own provenance and `ls0_improved` flag. Does not fire E3. [NBH/branch_snapshots/MEET-v31/MINUTES.md:38]

Reason. "Makes T38(b1-LH) a theorem about the shipped method. ADDITIVE." Evidence: "O-LS0: 0/2,394 HPM changes". Earlier HQ advice: "It closes the main theorem's biggest hole", "It costs almost nothing"; caveat "most T38 premises fail at n=200 […], so LS0's value is asymptotic" [NBH/03b_reasoning_and_traces.md:68-75]. See the LS0 divergence above.

MD04 MEET-v31 decision 4 decided (MEET-v31 MINUTES §2); closed in: R1–R2 (D3e, D3f)

Decision. **Parallel layer** (each item ADDITIVE / bitwise unless stated): (a) concurrent final islands; (b) concurrent diagnostic islands; (c) parallel scans with lexicographic reduction; (h) the block-table stage in parallel; process workers (threads need private counters); GUARD-W "any worker / LS0 / guard stop ⇒ no output" (an LS0 stop is an operational interruption; a completed LS0 capped at `local_limit` still returns output, with L1 = n/a); (g) estimate fields NA on unavailable output. [NBH/branch_snapshots/MEET-v31/MINUTES.md:39]

Reason. "Real issue: the parallel orientation (ground rule 2). The axis is islands and scans, not variables (ground rule 4)." HQ: "The real problem is in the code, not the method" (production code serial everywhere) [NBH/03b_reasoning_and_traces.md:153-159].

MD05 MEET-v31 decision 5 decided (MEET-v31 MINUTES §2); closed in: R2 D3a, D3b

Decision. **Stacking R′ = 4k at N_e,** declared before stream 10 (streams 10..9+R′). N_e is never split. Extra local search from the R = 4 centre. Baseline and expanded records each carry their own flags and card. [NBH/branch_snapshots/MEET-v31/MINUTES.md:40]

Reason. "Keeps stacking ADDITIVE (nested)." Evidence: "PC1 claim 6: without the extra search the post-search HPM is **not** monotone in R′ (COUNTEREXAMPLE). No listed theorem needs monotonicity; the extra search costs ≤ ℓ + 1 scans".

MD06 MEET-v31 decision 6 decided (MEET-v31 MINUTES §2); closed in: R2 D3d, D4; C6 trigger: HQ (AMENDMENT_HQ_01)

Decision. **Depth and cost are stated at the implemented (island) level.** "Free of N, H, R" is removed. Calibration is counted. C6 (per-particle streams) and C7 (Gram-free scorer) are the declared v3.2 path. C6 is implemented and law-checked iff paper 1 claims particle-parallel depth **or** demonstrates any fit with p > 2,500. […] Felipe decided (2026-11-03) that paper 1 claims island-level depth only, and the study has p ≤ 1,000, so neither trigger currently fires. [NBH/branch_snapshots/MEET-v31/MINUTES.md:41]

Reason. "Honest scaling claim." Evidence: calibration took 0.45×, 0.62× and 1.76× the serial final run on S1, S5 and S4. […] = the trigger's provenance (union of seat positions, HQ disposition), quoted in full in the registry row.

MD07 MEET-v31 decision 7 decided (MEET-v31 MINUTES §2); closed in: R1–R2 (all three)

Decision. **Calibration kept unchanged, with a contract:** $\mathcal{C}$ frozen before final randomness; validity holds for any frozen $\mathcal{C}$; Theorem G cap; ≤ ℓ rebuilds per tier (≤ 2ℓ with escalation); restricted ESS on V only; statuses distinguished. Shortening it (C9) is rejected. [NBH/branch_snapshots/MEET-v31/MINUTES.md:42]

Reason. "Validity does not depend on calibration converging. The missing theorems concern efficiency and delivery (theorem obligations)." Shortening calibration "weakens three existing theorems" [NBH/

03b_reasoning_and_traces.md:157].

MD08 MEET-v31 decision 8 decided (MEET-v31 MINUTES §2); closed in: R2 D5 (all three)

Decision. **Ẑ-weighted PIP is primary,** equal-weight is secondary, and both MPMs and R_eff are reported. POST HOC. [NBH/branch_snapshots/MEET-v31/MINUTES.md:43]

Reason. "The only PIP with a consistency-in-R theorem (Prop P(ii), T31(i)). Equal weighting keeps the single-island bias for every R. The truth-based gate was withdrawn: closeness to the binary truth is objective 3, not objective 1." Evidence: 3-design rescoring (MPMs never differ, max|ΔPIP| ≤ $1.3{\cdot}10^{-3}$) [NBH/03b_reasoning_and_traces.md:174-176].

MD09 MEET-v31 decision 9 decided (MEET-v31 MINUTES §2); closed in: R1 (all three)

Decision. **Flags E1–E3, ESS and grid flags kept** as calibration triggers (they define the law of $\mathcal{C}$) and as labelled heuristics. No PSIS/R̂ flags. The certificate card never feeds back into calibration. [NBH/branch_snapshots/MEET-v31/MINUTES.md:44]

Reason. "Removing them is WORKHORSE. Their theory is negative (V12, E1c)."

MD10 MEET-v31 decision 10 decided (MEET-v31 MINUTES §2); closed in: R2 D2; R3-1

Decision. **Certificate card (POST HOC):** PC2's minimal card. G1 [...]; G2 [...], joint with output, ≥ 95%; L1–L3 [...]; R optional (box). One declared certificate budget, with "unresolved" on exhaustion. L1 is evaluated on the returned Ŝ (after LS0). Each number carries the label "numerical implementation of an exact-arithmetic certificate" until validated arithmetic (SC) and VAL-FP exist; a long-double replay alone does not remove it. [NBH/branch_snapshots/MEET-v31/MINUTES.md:45] ([...] = item contents, given in full in §2.8.5.)

Reason. "The only diagnostics with a provable positive property (ground rule 3; HQ objection 8)."

MD11 MEET-v31 decision 11 decided (MEET-v31 MINUTES §2); closed in: R2 D7; R3-3

Decision. **General-h estimator:** H_r(h) after each complete measured round. Ẑ-weighted and equal-weight pools are reported. The default $\mathcal{H}$ is model size and pairwise inclusion on neighbour-graph edges. h must be fixed from (X, y), or from (X, y, $\mathcal{C}$) before stream 10. Plain averages for joint h; Rao–Blackwellisation for marginal PIPs only. HPM terminal frequency is descriptive only. ADDITIVE. [NBH/branch_snapshots/MEET-v31/MINUTES.md:46]

Reason. "Proposal §13.1 needs joint posterior summaries. The marginal RB product is not a joint estimate." The identity is a PROOF SKETCH (obligation N1, OB01).

MD12 MEET-v31 decision 12 decided (MEET-v31 MINUTES §2); closed in: R2 item 9 e8

Decision. **Output convention:** every estimate on the output event $\mathcal{O}$ is a function of (X, y, seed, binary). The allocation and hardware decide only whether output is delivered (stop-only guard, Lemma RO). Guarantees are stated for the potential run and as $P\{\mathcal{O}, \cdot\}$, never conditional on completion. [NBH/branch_snapshots/MEET-v31/MINUTES.md:47]

Reason. "Ground rule 9 (`resource_limit` vs 'runnable from X, y and one seed')."

MD13 MEET-v31 decision 13 decided (MEET-v31 MINUTES §2); closed in: R2–R3-4

Decision. **Item 9: generality in (n, p).** No setting is class (e) in the method's law. Fixes e1–e8 and wordings w1–w7 are adopted (list in §4). No separate $p \ll n$ / $p \gg n$ rule is needed. [NBH/branch_snapshots/MEET-v31/MINUTES.md:48]

Reason. "Ground rule 9; METHODOLOGY_v3 Requirement (4)." Felipe's rule [pos 1214]: every setting a modelling convention with a reason, an explicit function of (n, p) or X, or a declared budget [NBH/03b_reasoning_and_traces.md:183-185].

Closure: "**open method questions: none**" after N9 passed [NBH/branch_snapshots/MEET-v31/MINUTES.md:150]. Binding-goal rule: if an obligation tied to a binding proposal goal fails (N1 / general h, §13.1; S4 weight stability, §5.3; N9, §5.1), "the binding goal is not met by withdrawal: HQ escalates to Felipe for a method reconsideration or an explicit scope change" [NBH/branch_snapshots/MEET-v31/MINUTES.md:161]. Growth rates in (n, p) may be added to budgets, "limited to growth rates only, with no other method

change" (approved [pos 1304]) [NBH/notation_and_conventions.md:404].

For agents — what a theorist must keep in mind about the method

1. **Object vs output.** The posterior π, Z and the exact HPM/MPM are properties of the target (§2.2); the method outputs $\hat{S}$ (best of the *scored* set $\mathcal{S}$), $\hat{Z}$, pooled PIPs. Statements about the method's output are about the *potential (uninterrupted) run* and are transferred to delivery only as $P\{\mathcal{O} \cap \cdot\}$, never as $P(\cdot \mid \mathcal{O})$ (MD12, DF21); Lemma RO "does not imply $P(B \mid \mathcal{O}, \mathcal{C}) \leq \alpha$" and CEX-guard is a recorded COUNTEREXAMPLE [NBH/notation_and_conventions.md:433] (see TH15).
2. **Spec vs code.** v3.1 items LS0, stacking, concurrency, $\hat{Z}$-weighted primary PIP, general h, card, NA fields are spec; the study's numbers come from v2.2.1 (§2.12). OP-LS0, GUARD-W, IND-T for $R' > 4$ are *premises*. Theorems "about the shipped method" with LS0 are about the spec until the merged-LS0 replay.
3. **Which "primary".** In code and study tables `pip_primary` = equal-weight RB-PIP = v3.1 *secondary*.
4. **Score.** $F(S) = \log \varpi(S) + \log m(S)$ includes the capped prior; HPM ties use $(F, -|S|, \text{lex})$; the local-search scan breaks exact ties by lex only. Notation F clashes with V9's $F(S)$ and with a Frobenius F_S (Appendix A).
5. **Calibration is part of the law.** Triggers, ESS 0.90, grid equality and the inherited leaders define the law of $\mathcal{C}$; validity is conditional on *any* frozen $\mathcal{C}$; convergence/delivery facts are OPEN obligations (OB05).
6. **Randomness.** One serial stream per pilot/island; final streams ≥ 10 are fresh. IND-T is recorded as "IND-T (PROVED, XF VERIFIED*, A-REV1 only)", while "IND-T for streams 10..9+R': code audit, not done" [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:160] [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:231]. The independence step draws a fresh exact π_0 sample at every visit.
7. **Fixed budgets.** $R = 4$, 4 measured rounds, floors 2,500/1,000, ESS 0.90, ×4 tier are declared (c) budgets; growth rates may be added only as explicit functions of (n, p); anything else is a method change (WORKHORSE needs Felipe).
8. **Do not** claim depth/speed for the block start, depth free of N, H, R, joint summaries or primary PIPs for the 2,400 study fits, or "bitwise" for concurrent calibration.

3 Empirical results

This section reports every empirical result of the project: the New Campaign study (2026-10-28 to 2026-11-03) at low p (exact posterior available) and at $p \gg n$ (truth only), and the numerical companions run during the theory campaign ("New BVS Headquarters", 2026-11-02 to 04). It adds no analysis: every number is either generated from a machine-readable file in `Results/` by `scripts/make_tables.py` (tables marked "Generated"), or transcribed with its source. The empirical phase is closed: "the New Campaign's empirical phase is COMPLETE (HQ D94)" [Docs/HQ/STATUS.md:3]. Key facts get stable IDs `EMP01–EMP27` (registry rows in `registry/empirics.tsv`); §3.5 maps them to the theory IDs they inform.

Plain reading — how to read the numbers

A fit "recovers" when its selected set equals the true support S_0 exactly; failed or stopped fits count as misses [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:3]. At low p the SMC output is compared with the exact posterior of the *same* model (full enumeration of all 2^p models); at $p \gg n$ "the true Z, exact HPM and exact PIPs are not computed" (D22) [Docs/HQ/DECISIONS.md:34], so only

selection against the truth is reported [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:24]. The reports' verdict scale (support / contradict / silent) is about what one $n = 200$ design can show; "failure of a finite-n *premise* of a conditional statement is reported as such, and is **not** a contradiction of the implication" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:14-17].

3.1 Study design

Data generator (both regimes). $n = 200$; X is drawn as $N(0, \Sigma_s)$ rows ("X first via chol, then noise"), column-standardised (`scale`, sd with $n-1$), $\varepsilon \sim N(0,1)$, and $Y_c = c\,X\beta + \varepsilon$ for the stored scales; only $c \in \{1, 0.5\}$ (`c100`, `c050`) are fitted [Scripts/Data_generators/gen_prod_data.R:28-40] [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:13-17] [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:70]. The scenario covariances are built on **groups of 5 consecutive predictors** (`group <- ceiling(seq_len(p)/5)`), $H = \mathbf{1}\{\text{same group}\}$ [Scripts/Data_generators/gen_dev_data.R:18-30]:

scenario	Σ_s (verbatim construction, `gen_dev_data.R:23-29`)
S1	I
S2	$0.5^{\lvert j-k\rvert}$ (AR(1))
S3	$0.7H + 0.3I$ (equicorrelation 0.7 within each group of 5)
S4	$0.85H + 0.15I$ (equicorrelation 0.85 within groups)
S5	$0.3\,\mathbf{1}\mathbf{1}^\top + 0.4H + 0.3I$ (0.7 within, 0.3 between groups)
S6	within group b: $\rho_b = 0.4 + 0.1 \cdot ((b-1) \bmod 6)$, diagonal 1, 0 between groups

Truth. One signal per group of 5, placed by `truth(p)`: $k = 0, \ldots, s_0 - 1$, block $= 1 +$ step $\cdot\, k$, active index $= 5(\text{block} - 1) + \text{pos}[k \bmod 6]$ with pos $= (3, 2, 4, 1, 5, 3)$, $\beta = \text{pattern}[k \bmod 6]$ with pattern $= (1.4, -1.0, 1.8, -1.2, 0.8, -1.6)$; $s_0 = 14$ with step 7 ($p = 500$) and 13 ($p = 1000$) [Scripts/Data_generators/gen_dev_data.R:33-42]. At low p the same rule with step 1 gives $s_0 = 4$ at $p = 20$ (active 3, 7, 14, 16; $\beta = 1.4, -1.0, 1.8, -1.2$) and $s_0 = 5$ at $p = 30$ (active 3, 7, 14, 16, 25; $\beta = 1.4, -1.0, 1.8, -1.2, 0.8$) [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:21-26]. Nominal SNR $c^2\beta'\Sigma_s\beta$ at low p ranges from 5.65 to 8.28 at $c = 1$ and 1.41 to 2.07 at $c = 0.5$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:39-48]; the smallest eigenvalue of Σ_s is 0.15 in S4 and 0.10 in S6 at $p = 30$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:32-35].

For agents — the study design is not a theory family

The study generator has groups of size $g = 5$. The theory's look-alike exhibit LH uses $g = 9$ ("SIZE: $g_{\max} \le B < 2g_{\min}$; holds on LH for $n \ge 9^6$ with $g = 9$") [NBH/notation_and_conventions.md:434], and "The study generator ($g = 5$, $2g \le B$) is the obligation PART-pairing (N8)" [NBH/notation_and_conventions.md:259], recorded as "OPEN; Prop Q PROOF SKETCH; W7-PART CE2 COUNTEREXAMPLE" [NBH/05_open_questions_and_next_steps.md:94]. The archive states it plainly: "the $g = 5$ study is not the LH exhibit" [NBH/03_results_and_conclusions.md:74]. No theorem about LH or LH-iid therefore applies to these data as stated; the data can only be consistent or inconsistent with observable consequences.

Design summary.

item	low p (PROD-L)	high p (PROD-H)
n, p	200; $p \in \{20, 30\}$	200; $p \in \{500, 1000\}$
scenarios, replicates, c	S1–S6, 100, $c \in \{1, 0.5\}$	S1–S6, 100, $c \in \{1, 0.5\}$
datasets / cases	1,200 / 2,400 [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:15]	1,200 / 2,400 fits [NBH/03_results_and_conclusions.md:157]
data seed	base $+10000s + r$; base 1,210,000 ($p = 20$), 1,310,000 ($p = 30$) [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:54-59]	base $+10000s + r$; base 1,410,000 ($p = 500$), 1,510,000 ($p = 1000$) [Scripts/Data_generators/gen_prod_data.R:3-5]
method seed	data seed + 4,000,000; repeat m: $+1{,}000m$ [Scripts/Campaigns/2026-10-28_New_Campaign/code/lowp/LOWP_SPEC.md:63-65]	data seed + 4,000,000 [Scripts/Data_generators/gen_prod_data.R:5]
SMC-BVS fits	2,400 primary + 12,000 repeats (5 extra seeds per case); exact enumeration of 1,200 datasets [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:3]	2,400 (run `20261030T0934Z_prodH`, frozen v2.2.1; reproduces the 2026-10-23 fits) [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:6-10]
benchmarks	8 methods × 2,400 cases = 19,200 records [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:176]	8 methods (aLasso rerun 2,400 fits, D86) [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:13-15]
reference	exact posterior of the same model	truth only (D22)

Benchmarks and rules. F1 lasso, F2 elastic net, F3 adaptive lasso, F4 MCP, F5 ABESS, A1 SuSiE, A2 spike-and-slab lasso (SSL), A3 horseshoe. Headline rules: F1–F3 `ebic1_K37` with CV `cv_min` "as a disclosed sensitivity per D35/D42"; MCP and ABESS `ebic1_K37`; SuSiE `pip05`, SSL `ssl_final`, horseshoe `ci95` [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:121]. `ebic1_K37` is EBIC($\gamma = 1$) over candidates with $k \leq K$, "$K = \lfloor n/\log n \rfloor = 37$" [Scripts/Campaigns/2026-10-28_New_Campaign/code/bench_v2/SPEC.md:38]; at $p \leq 30$, $K = \min(\lfloor n/\log n \rfloor, p) = p$, so K37 and cap rows coincide [Scripts/Campaigns/2026-10-28_New_Campaign/code/bench_lowp/ADAPTATIONS.md:8]. At $p \gg n$ the MCP/ABESS primary is "EBIC($\gamma = 1$) at $c = 1$, and EBIC($\gamma = 1$) with the size bound $K = 37$ at $c = 0.5$ (pre-registered)" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:30]. The adaptive lasso uses the lasso `cv_min` pilot (it replaced a ridge pilot, D42 item 2) [Docs/HQ/DECISIONS.md:109].

UNCERTAIN — version label of the $p \gg n$ fits

The same 2,400 fits are called "the 2026-10-23 v2.1 production fits" [Docs/Records/DATA_INDEX.md:11], "the existing 2,400 SMC-BVS v3 fits" [Results/2026-10-28_New_Campaign/H_existing_20261030/HEADLINE_H.md:5], and the "v3 study" that "stands as v3.1 with 4 islands" [NBH/03_results_and_conclusions.md:155]; the rerun used "frozen code v2.2.1" whose core is "byte-identical to v2.2" [Scripts/Campaigns/2026-10-28_New_Campaign/README.md:16]. The labels mix code versions (v2.x) and method versions (v3, v3.1); no source states the mapping in one place.

3.2 Low p: SMC-BVS against the exact posterior

PROD-L: "1,200 datasets (p ∈ {20, 30}, S1–S6, 100 replicates) × c ∈ {1, 0.5} = 2,400 cases. The exact posterior comes from full enumeration of all 2^p models. Truth was used only in scoring. Reconciliation: enumeration 1200/1,200, SMC primary 2400/2,400, SMC repeats 12000/12,000. Failed: 0. Missing: 0." [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:3]. Review: "Terra tie-outs PASS, Sol ACCEPT (D75, D94)" [Docs/HQ/STATUS.md:24].

DIVERGENCE — which PIP the study reports

Every SMC-BVS PIP in this section (low and high p) is pooled over islands with *equal weights* by the frozen v2.2.1 core: the RB PIP is `pip_primary`, the equal-weight mean of the island RB PIPs [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:126-129], and the plain PIP is `pip_plain`, the equal-weight mean of the island plain PIPs [Scripts/Campaigns/2026-10-28_New_Campaign/code/core/v2_2_1/core/v3_run.R:167-168] (see §2.8). Equal-weight pooling is the v3.1 *secondary* PIP; the v3.1 primary ($\hat{Z}$-weighted) exists only through POST HOC rescoring, done on 3 designs [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:54]. See the divergence box "primary PIP" in §2.12. Theorems stated for the $\hat{Z}$-weighted PIP are therefore not directly tested by these tables.

- **EMP01** **HPM is exact; 0 search misses.** "SMC HPM = exact HPM in 2,400/2,400 primary fits and 12,000/12,000 repeat fits. HPM = truth 92.3%. Every miss is a miss of the exact posterior itself; there are 0 search misses" [Docs/HQ/STATUS.md:29]. Miss classes: 2,216 hits, 0 search misses, 184 posterior misses, "F(S_0) ≤ F(HPM)" in every miss, no ties (closest gap −0.0017, median −1.10) [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:101].
- **EMP02** **Evidence $\hat{Z}$.** Over the 2,400 primary fits $e = \log \hat{Z} - \log Z$ has "bias -0.0002, RMSE 0.012, median |e| 0.007, and 90th-percentile |e| 0.020", and "the 0.05·$\hat{Z}$ lower bound covered Z in 100% of fits (0 violations)" [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:7]. Over all 14,400 fits the mean relative error $\hat{Z}/Z - 1$ is −0.00011 with SE 0.00010 ("within 1.1 SE of 0"; "cancellation across cases cannot be excluded"), and the lower bound "never fails, in 14,400/14,400 fits" [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:150-154].
- **EMP03** **PIPs and MPM.** Mean per-fit max |PIP − $\text{PIP}^{\text{exact}}$| is 0.0005 (RB, 90th pct 0.0014) and 0.0020 (plain, 90th pct 0.0033); "The MPM equals the exact MPM in 99.9% of cases (RB) and 99.8% (plain). The RB PIPs are about 4× closer." [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:9].
- **EMP04** **Seed stability.** "all 6 seeds gave the same HPM in **100%** of cases (2,400/2,400)"; the sd of log $\hat{Z}$ across seeds has mean 0.011 (90th pct 0.018, max 0.030), "similar to the RMSE against the exact value (0.012), consistent with seed variability dominating the error"; the 12,000 extra-seed fits "are as accurate as the primaries" [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:150-153]. The spread uses primary + 5 extra seeds, "a slight widening of the brief's 'across the 5 extra seeds' definition" [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:148].

Accuracy by cell (Q1–Q3; Generated):

p	c	cell	bias e	RMSE e	p90 $\lvert e \rvert$	HPM= S_0 %	RB err	plain err	MPM_{RB}=exact %	MPM_{pl}=exact %
20	1	S1	-0.0001	0.005	0.007	96.0	0.0001	0.0017	100.0	100.0
20	1	S2	-0.0011	0.010	0.016	98.0	0.0002	0.0017	100.0	100.0
20	1	S3	0.0005	0.016	0.027	97.0	0.0002	0.0017	100.0	100.0
20	1	S4	-0.0021	0.017	0.029	95.0	0.0002	0.0018	100.0	100.0
20	1	S5	-0.0015	0.014	0.024	95.0	0.0002	0.0017	100.0	100.0
20	1	S6	0.0008	0.015	0.026	95.0	0.0001	0.0018	100.0	100.0
20	1	**pooled**	-0.0006	0.014	0.024	96.0	0.0002	0.0017	100.0	100.0
20	0.5	S1	0.0004	0.005	0.008	96.0	0.0001	0.0019	100.0	100.0
20	0.5	S2	0.0004	0.010	0.016	97.0	0.0002	0.0018	100.0	100.0
20	0.5	S3	-0.0012	0.014	0.025	96.0	0.0004	0.0019	100.0	99.0
20	0.5	S4	0.0021	0.015	0.025	87.0	0.0017	0.0025	99.0	99.0
20	0.5	S5	0.0030	0.014	0.021	92.0	0.0005	0.0018	100.0	99.0
20	0.5	S6	-0.0020	0.013	0.020	94.0	0.0003	0.0017	100.0	100.0
20	0.5	**pooled**	0.0005	0.012	0.022	93.7	0.0005	0.0019	99.8	99.5
30	1	S1	0.0007	0.006	0.012	95.0	0.0001	0.0018	100.0	100.0
30	1	S2	-0.0010	0.010	0.016	95.0	0.0002	0.0018	100.0	100.0
30	1	S3	-0.0008	0.009	0.015	97.0	0.0002	0.0017	100.0	100.0
30	1	S4	0.0005	0.011	0.019	89.0	0.0004	0.0019	100.0	100.0
30	1	S5	-0.0014	0.015	0.024	97.0	0.0002	0.0017	100.0	100.0
30	1	S6	-0.0003	0.010	0.015	98.0	0.0002	0.0017	100.0	100.0

p	c	cell	bias e	RMSE e	p90 $\lvert e\rvert$	HPM= S_0 %	RB err	plain err	MPM_{RB}=exact %	MPM_{pl}=exact %
30	1	**pooled**	-0.0004	0.010	0.017	95.2	0.0002	0.0018	100.0	100.0
30	0.5	S1	-0.0015	0.008	0.014	94.0	0.0001	0.0020	100.0	100.0
30	0.5	S2	0.0016	0.010	0.017	91.0	0.0004	0.0020	100.0	100.0
30	0.5	S3	-0.0011	0.010	0.016	94.0	0.0011	0.0022	100.0	100.0
30	0.5	S4	0.0006	0.012	0.020	61.0	0.0030	0.0036	99.0	99.0
30	0.5	S5	-0.0000	0.014	0.023	86.0	0.0010	0.0022	99.0	100.0
30	0.5	S6	-0.0018	0.011	0.017	81.0	0.0018	0.0026	100.0	100.0
30	0.5	**pooled**	-0.0004	0.011	0.018	84.5	0.0012	0.0024	99.7	99.8
all	all	**all 2,400**	-0.0002	0.012	0.020	92.3	0.0005	0.0020	99.9	99.8

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/L_20261102/tables/Q1_logZ.csv]; [Results/2026-10-28_New_Campaign/L_20261102/tables/Q2_hpm.csv]; [Results/2026-10-28_New_Campaign/L_20261102/tables/Q3_pip.csv]. $e = \log \hat{Z} - \log Z$ over the 2,400 primary fits; "err" = mean over fits of the per-fit $\max_j |\text{PIP}_j - \text{PIP}_j^{\text{exact}}|$ (its 90th percentiles and the median $|e|$ are further columns of the source files). In *every* row the SMC HPM equals the exact HPM in 100% of cases, the $0.05\hat{Z}$ lower bound covers Z in 100%, and SMC HPM = S_0 equals exact HPM = S_0 (all checked by the script; columns omitted). Pooled and all-2,400 rows are the source files' own `all` rows.

Seed stability and repeat-fit accuracy (Generated):

p	c	fits	mean sd log $\hat{Z}$	p90	max	same HPM, all 6 seeds %	max # distinct HPMs	max sd PIP (RB)	p90	(plain)	p90
20	1	3,600	0.0119	0.0195	0.0303	100	1	0.0002	0.0004	0.0015	0.0024
20	0.5	3,600	0.0114	0.0184	0.0303	100	1	0.0005	0.0015	0.0017	0.0027
30	1	3,600	0.0098	0.0164	0.0305	100	1	0.0002	0.0005	0.0016	0.0025
30	0.5	3,600	0.0100	0.0161	0.0277	100	1	0.0012	0.0033	0.0021	0.0037
all	all	14,400	0.0108	0.0178	0.0305	100	1	0.0005	0.0016	0.0017	0.0029

Accuracy of the 12,000 extra-seed fits against the exact posterior:

p	c	extra-seed fits	RMSE e	p90 $\lvert e\rvert$	LB cov. %	HPM=exact %	RB err	plain err	MPM_{RB}=ex. %	MPM_{pl}=ex. %
20	0.5	3,000	0.0128	0.0211	100	100	0.0005	0.0019	99.83	99.73
20	1	3,000	0.0130	0.0219	100	100	0.0002	0.0017	100.00	100.00
30	0.5	3,000	0.0108	0.0178	100	100	0.0012	0.0024	99.77	99.70
30	1	3,000	0.0110	0.0182	100	100	0.0002	0.0018	100.00	99.93
all	all	12,000	0.0119	0.0199	100	100	0.0005	0.0020	99.90	99.84

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/L_20261102/tables/stability.csv]; [Results/2026-10-28_New_Campaign/L_20261102/tables/repeat_fit_accuracy.csv]. Upper table: spread over the 6 fits of each case (primary + 5 extra seeds; the source's `all` rows); "max sd PIP" = mean over cases of the max over variables of the sd of the PIP across seeds.

Recovery against benchmarks. **EMP05** Pooled exact recovery /2,400: SSL 2,251; SMC-BVS HPM (= exact HPM) 2,216; SuSiE 2,183; adaptive lasso 2,066; MCP 2,061; ABESS 2,051; horseshoe 2,028; lasso 1,852; elastic net 1,692 [Docs/HQ/STATUS.md:33-45]. The exact HPM equals the truth in 92.3%, rising with signal: 96.0% / 93.7% ($p = 20$, $c = 1$ / 0.5) and 95.2% / 84.5% ($p = 30$) [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:73-78]; the hardest cell is S4 at $p = 30$, $c = 0.5$ [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:10].

method	rule	p=20, c=1	p=20, c=0.5	p=30, c=1	p=30, c=0.5	all /2,400
SMC-BVS HPM	`hpm`	576	562	571	507	**2,216**
SMC-BVS MPM (RB)	`pip05`	576	552	571	491	**2,190**
SMC-BVS MPM (plain)	`mpm_plain`	576	551	571	490	**2,188**
Exact-posterior HPM	`hpm`	576	562	571	507	**2,216**
Exact-posterior MPM	`mpm`	576	553	571	490	**2,190**
Lasso	`ebic1_K37`	541	482	478	351	**1,852**
Elastic net	`ebic1_K37`	502	449	416	325	**1,692**
Adaptive lasso	`ebic1_K37`	534	521	531	480	**2,066**
MCP	`ebic1_K37`	534	517	532	478	**2,061**
ABESS	`ebic1_K37`	534	516	529	472	**2,051**
SuSiE	`pip05`	575	550	570	488	**2,183**
SSL	`ssl_final`	590	574	586	501	**2,251**
Horseshoe	`ci95`	573	520	568	367	**2,028**
Lasso (CV sensitivity)	`cv_min`	5	5	1	1	**12**
Elastic net (CV sensitivity)	`cv_min`	2	4	1	1	**8**
Adaptive lasso (CV sensitivity)	`cv_min`	156	120	127	77	**480**

Generated by `scripts/make_tables.py` from [`Results/2026-10-28_New_Campaign/L_20261102/tables/Q4_recovery.csv`]. Exact recovery (selected set = S_0) out of 600 per $p \times c$. Benchmarks under their headline rule; CV rows are disclosed sensitivities (D42). Not run: 0; failed: 0 in every cell (checked).

Per-scenario counts, with the passing-only SMC counts (two flagged primary fits, both hits, are not credited: "p = 20, c = 1 has 575/600 and S4 94/100; p = 30, c = 1 has 570/600 and S3 96/100") [`Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:144`]:

method (rule as above)	p	c = 1: S1–S6, /600							c = 0.5: S1–S6, /600						
SMC-BVS HPM	20	96	98	97	95	95	95	**576**	96	97	96	87	92	94	**562**
	30	95	95	97	89	97	98	**571**	94	91	94	61	86	81	**507**
SMC-BVS MPM (RB)	20	96	98	97	95	94	96	**576**	96	98	94	78	92	94	**552**
	30	94	95	97	90	97	98	**571**	93	91	90	53	85	79	**491**
SMC-BVS MPM (plain)	20	96	98	97	95	94	96	**576**	96	98	94	78	91	94	**551**
	30	94	95	97	90	97	98	**571**	93	91	90	53	84	79	**490**
Exact-posterior HPM	20	96	98	97	95	95	95	**576**	96	97	96	87	92	94	**562**
	30	95	95	97	89	97	98	**571**	94	91	94	61	86	81	**507**
Exact-posterior MPM	20	96	98	97	95	94	96	**576**	96	98	94	79	92	94	**553**
	30	94	95	97	90	97	98	**571**	93	91	90	53	84	79	**490**
Lasso	20	89	93	94	75	93	97	**541**	89	92	80	52	79	90	**482**
	30	90	92	84	45	88	79	**478**	89	83	52	13	56	58	**351**
Elastic net	20	89	93	92	39	91	98	**502**	89	92	71	37	75	85	**449**
	30	90	95	68	13	78	72	**416**	89	80	43	9	49	55	**325**
Adaptive lasso	20	88	91	88	91	84	92	**534**	88	91	86	80	84	92	**521**
	30	89	86	93	80	89	94	**531**	89	85	90	55	81	80	**480**
MCP	20	88	91	88	91	84	92	**534**	88	91	86	77	84	91	**517**
	30	89	88	93	78	89	95	**532**	89	86	90	49	82	82	**478**
ABESS	20	88	91	88	91	84	92	**534**	88	91	87	75	84	91	**516**
	30	89	87	92	77	89	95	**529**	89	85	86	50	83	79	**472**
SuSiE	20	91	90	99	99	99	97	**575**	90	91	98	80	95	96	**550**
	30	85	90	100	98	99	98	**570**	83	89	94	55	88	79	**488**
SSL	20	99	100	99	97	97	98	**590**	99	100	98	85	95	97	**574**
	30	100	98	98	93	99	98	**586**	96	93	89	58	86	79	**501**
Horseshoe	20	94	99	95	96	95	94	**573**	95	99	89	57	89	91	**520**
	30	94	95	97	86	98	98	**568**	93	88	62	15	63	46	**367**
Lasso (CV sensitivity)	20	0	1	0	0	3	1	**5**	0	1	0	0	3	1	**5**
	30	0	0	0	0	0	1	**1**	0	0	0	0	0	1	**1**
Elastic net (CV sensitivity)	20	0	1	0	0	1	0	**2**	0	1	0	0	3	0	**4**
	30	0	0	0	0	0	1	**1**	0	0	0	0	0	1	**1**
Adaptive lasso (CV sensitivity)	20	12	28	22	41	30	23	**156**	11	25	17	27	22	18	**120**
	30	9	16	25	28	24	25	**127**	7	11	14	16	12	17	**77**

Generated by `scripts/make_tables.py` from [`Results/2026-10-28_New_Campaign/L_20261102/tables/Q4_recovery.csv`]. Counts out of 100 per scenario. Passing-only SMC HPM counts (the 2 flagged primary fits, failed final check with low ESS, not credited): $p = 20$: 575 ($c = 1$), 562 ($c = 0.5$); $p = 30$: 570, 507 (column `n_exact_passing`).

Low-p benchmark disclosures ("Disclosed, not audited (D94)" [`Docs/HQ/STATUS.md:47-51`]):

- "SSL > exact HPM is organic (the exact posterior is our model's own, not an oracle)" [Docs/HQ/STATUS.md:48]; SSL is above the exact HPM in three of the four $p \times c$ cells (not at $p = 30$, $c = 0.5$: 501 vs 507) [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:265]; the $p = 30$, $c = 0.5$ counts [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:243-258].
- Lasso and EN "are ≈ 0 under CV (as audited in D42)" [Docs/HQ/STATUS.md:49]: 12 and 8 of 2,400 [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:266].
- The horseshoe (`ci95`) "drops at p = 30, c = 0.5" (367/600), "Plausibly the conservative credible-interval rule under weak signals; disclose and leave for the paper phase" [Docs/HQ/DECISIONS.md:299].
- ABESS "is RSS-optimal in 2,375/2,400 fits" [Docs/HQ/STATUS.md:51]; replacing the 25 suboptimal supports by the exact best subset of the same size would change its count by +7 [Results/2026-10-28_New_Campaign/L_20261102/HEADLINE_L.md:142]. Generated:

p	c	fits	(a)	(b)	(c)	(d)	(e)	(f)	(g)
20	0.5	600	5	3	0	0.0122	523	2,052	12,000
20	1	600	1	0	0	0.0122	522	2,007	12,000
30	0.5	600	18	6	2	0.0274	592	5,902	18,000
30	1	600	1	0	0	0.0004	592	5,751	18,000
all	all	2400	25	9	2	0.0274	2229	15,712	60,000

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/L_20261102/abess_check/abess_size_summary.csv]. ABESS exact size-k check (D54.3): headline `ebic1_K37` support vs the exact best subset of the same size by RSS. (a) headline supports not RSS-optimal; (b) the best subset would recover S_0 and ABESS did not; (c) ABESS recovered and the best subset would not; (d) max relative RSS gap of headline supports; (e) fits with any suboptimal path size; (f) suboptimal (fit, size) pairs on the stored path; (g) all (fit, size) pairs.

3.3 High p ($p \gg n$): selection against the truth

Review: "Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)" [Docs/HQ/STATUS.md:54]. The truth has 14 variables [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:3].

- **EMP06 Reproduction.** "The v2.2.1 rerun reproduces the 2026-10-23 fits exactly (2,400/2,400, 11 fields)." [Docs/HQ/STATUS.md:56]; "The 6 stopped fits match too, including their unavailable (empty) fields. Every available estimate reproduces exactly, with max |diff| = 0" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:7-9]. Generated:

field	cases	identical	max |diff|
`fit$hpm`	2400	2400	0.0
`fit$hpm_F`	2400	2400	0.0
`fit$log_Z`	2400	2400	0.0
`fit$log_Z_r`	2400	2400	0.0
`fit$mpm`	2400	2400	0.0
`fit$pip_primary`	2400	2400	0.0
`fit$pip_r`	2400	2400	0.0
`native_status`	2400	2400	–
`record$hpm`	2400	2400	0.0
`record$mpm`	2400	2400	0.0
`record$pip`	2400	2400	0.0

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/repro/repro_summary.csv]; [Results/2026-10-28_New_Campaign/H2_20261102/repro/repro_mismatches.tsv]. v2.2.1 rerun (run `20261030T0934Z_prodH`) vs the 2026-10-23 records; the mismatch list has a header only.

- **EMP07 Exact recovery /600 (HPM).** 559 / 560 at $c = 1$ and 370 / 340 at $c = 0.5$ ($p = 500$ / 1000) [Docs/HQ/STATUS.md:57-62]; shares 93.2%, 93.3%, 61.7%, 56.7%, with "Monte Carlo standard error of each pooled share …about 1–2 percentage points" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:116-121].

- **EMP08 MPM rows.** "RB and plain MPM agree in 2,391 of 2,394 completed fits"; each disagreement "involves a single variable whose two PIPs straddle 0.5, within 0.003 of each other"; "The largest |RB PIP − plain PIP| over all variables and fits is 0.0067" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:123-127]. This is internal agreement only, not accuracy (D22) [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:194-200].

Exact recovery, $c = 1$ (Generated):

method / rule	role	$p = 500$: S1–S6, /600							$p = 1000$: S1–S6, /600						
SMC-BVS HPM	primary	95	96	94	91	89	94	**559**	95	95	97	93	84	96	**560**
SMC-BVS MPM (RB)	secondary	92	97	93	93	90	94	**559**	94	95	97	94	85	96	**561**
SMC-BVS MPM (plain)	secondary, new	92	97	93	93	90	94	**559**	94	95	97	94	85	96	**561**
SuSiE	primary	99	98	97	95	98	98	**585**	100	96	98	95	94	99	**582**
SuSiE-refine	companion (R4), post-results arm	99	98	97	95	98	98	**585**	100	96	98	95	93	99	**581**
Spike-and-slab lasso	primary	99	99	97	84	93	97	**569**	100	96	98	92	97	99	**582**
Horseshoe (95% CI)	primary	92	98	100	87	98	99	**574**	91	95	100	91	98	100	**575**
MCP `ebic1`	pre-declared primary	39	69	77	44	78	70	**377**	18	28	60	32	49	48	**235**
MCP `ebic1_K37`	post hoc	69	79	77	44	78	70	**417**	66	74	80	32	69	65	**386**
ABESS `ebic1`	pre-declared primary	66	77	71	26	76	49	**365**	40	63	76	24	66	54	**323**
ABESS `ebic1_K37`	post hoc	68	77	71	26	76	49	**367**	63	72	76	24	67	54	**356**
Lasso `ebic1_K37`	headline (D35/D42)	67	29	2	0	9	4	**111**	48	28	4	0	5	4	**89**
Elastic net `ebic1_K37`	headline (D35/D42)	48	11	0	0	2	0	**61**	38	12	0	0	0	0	**50**
Adaptive lasso `ebic1_K37`	headline (D42/D86), lasso pilot	73	89	85	82	82	82	**493**	82	85	91	83	77	94	**512**

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/headline_h2_long.csv]. Exact recovery against the truth, $c = 1$; count out of 100 per scenario, pooled out of 600; failed fits count as misses.

Exact recovery, $c = 0.5$ (Generated):

method / rule	role	$p = 500$: S1–S6, /600							$p = 1000$: S1–S6, /600						
SMC-BVS HPM	primary	79	84	64	25	59	59	**370**	71	75	67	35	31	61	**340**
SMC-BVS MPM (RB)	secondary	74	83	63	18	53	47	**338**	70	76	63	24	32	56	**321**
SMC-BVS MPM (plain)	secondary, new	74	83	63	18	53	47	**338**	70	76	62	24	32	56	**320**
SuSiE	primary	65	70	34	9	3	39	**220**	49	57	47	21	0	31	**205**
SuSiE-refine	companion (R4), pre-registered	66	71	43	17	6	46	**249**	49	57	49	22	3	32	**212**
Spike-and-slab lasso	primary	80	84	53	9	42	35	**303**	67	64	43	15	28	38	**255**
Horseshoe (95% CI)	primary	74	69	18	0	10	14	**185**	60	50	11	0	11	4	**136**
MCP `ebic1_K37`	pre-registered primary	62	71	42	1	45	19	**240**	49	59	45	1	20	18	**192**
MCP `ebic1`	pre-declared, non-primary	38	58	42	1	45	19	**203**	13	26	34	1	18	10	**102**
ABESS `ebic1_K37`	pre-registered primary	67	72	34	4	48	31	**256**	54	64	42	3	23	28	**214**
ABESS `ebic1`	pre-declared, non-primary	65	72	34	4	48	31	**254**	36	55	42	3	22	28	**186**
Lasso `ebic1_K37`	headline (D35/D42)	17	5	0	0	1	1	**24**	11	3	1	0	0	0	**15**
Elastic net `ebic1_K37`	headline (D35/D42)	11	3	0	0	0	0	**14**	8	2	0	0	0	0	**10**
Adaptive lasso `ebic1_K37`	headline (D42/D86), lasso pilot	64	69	34	11	38	37	**253**	59	53	32	7	14	32	**197**

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/headline_h2_long.csv]. Exact recovery against the truth, $c = 0.5$; count out of 100 per scenario, pooled out of 600; failed fits count as misses.

Lasso-family CV rules and supplementary rows (Generated):

rule	c=1, p500	c=1, p1000	c=0.5, p500	c=0.5, p1000
Lasso `cv_min`	0	0	0	0
Lasso `cv_1se`	0	0	0	0
Elastic net `cv_min`	0	0	0	0
Elastic net `cv_1se`	0	0	0	0
Adaptive lasso `cv_min`	0	0	0	0
Adaptive lasso `cv_1se`	7	1	2	0

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/headline_h2_long.csv]. Lasso-family CV rules (registered primary before D42; `cv_1se` companion R2), pooled /600.

rule	c=1, p500	c=1, p1000	c=0.5, p500	c=0.5, p1000
SSL-RS (suppl.)	593	592	32	32
SCAD `ebic1` (S1 arm)	454	417	95	55
SuSiE-L100 (S1 arm)	585	582	221	205
Adaptive lasso, ridge pilot `ebic1_K37` (superseded)	247	96	36	12
Adaptive lasso, ridge pilot `cv_min` (superseded)	0	0	0	0
Adaptive lasso, ridge pilot `cv_1se` (superseded)	9	0	0	0

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H_existing_20261030/tables/all_rules_recovery_long.csv]. Supplementary and superseded rows, pooled exact recovery /600 (not part of the 9-method ranking).

Rankings. EMP09

- "c = 0.5: SMC-BVS is first or tied-first in 11 of 12 cells, ahead of the best benchmark (SSL) by 67 datasets at p500 and 85 at p1000." [Docs/HQ/STATUS.md:65]. The exception is p500 S1 (79 vs SSL 80) [Results/2026-10-28_New_Campaign/H_existing_20261030/HEADLINE_H.md:88]. Paired against SSL, only SMC-BVS recovers in 106 / 117 datasets vs only SSL in 39 / 32 [Results/2026-10-28_New_Campaign/H_existing_20261030/HEADLINE_H.md:90-95].
- "c = 1: SMC-BVS ranks 4th of 9, behind SuSiE, SSL and the horseshoe. It beats the adaptive lasso (rerun with the lasso pilot: 493 / 512 at c = 1) in 24/24 cells." [Docs/HQ/STATUS.md:66]; "Almost every miss adds variables to the truth" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:133]. The HPM also beats MCP and ABESS `ebic1_K37` in all 24 cells [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:136].

Generated (per-cell comparison; paired comparisons):

c	p	cell	SMC HPM	SMC MPM	best benchmark (pre-F3-rerun rows)	best	HPM−best	HPM rank /9	aLasso rerun
1	500	1	95	92	SuSiE,Spike-and-slab lasso	99	-4	3	73
1	500	2	96	97	Spike-and-slab lasso	99	-3	4	89
1	500	3	94	93	Horseshoe	100	-6	4	85
1	500	4	91	93	SuSiE	95	-4	2	82
1	500	5	89	90	SuSiE,Horseshoe	98	-9	4	82
1	500	6	94	94	Horseshoe	99	-5	4	82
1	500	pooled	559	559	SuSiE	585	-26	4	493
1	1000	1	95	94	SuSiE,Spike-and-slab lasso	100	-5	3	82
1	1000	2	95	95	SuSiE,Spike-and-slab lasso	96	-1	3	85
1	1000	3	97	97	Horseshoe	100	-3	4	91
1	1000	4	93	94	SuSiE	95	-2	2	83
1	1000	5	84	85	Horseshoe	98	-14	4	77
1	1000	6	96	96	Horseshoe	100	-4	4	94
1	1000	pooled	560	561	SuSiE,Spike-and-slab lasso	582	-22	4	512
0.5	500	1	79	74	Spike-and-slab lasso	80	-1	2	64
0.5	500	2	84	83	Spike-and-slab lasso	84	0	1	69
0.5	500	3	64	63	Spike-and-slab lasso	53	11	1	34
0.5	500	4	25	18	SuSiE,Spike-and-slab lasso	9	16	1	11
0.5	500	5	59	53	ABESS (ebic1_K37)	48	11	1	38
0.5	500	6	59	47	SuSiE	39	20	1	37
0.5	500	pooled	370	338	Spike-and-slab lasso	303	67	1	253
0.5	1000	1	71	70	Spike-and-slab lasso	67	4	1	59
0.5	1000	2	75	76	Spike-and-slab lasso,ABESS (ebic1_K37)	64	11	1	53
0.5	1000	3	67	63	SuSiE	47	20	1	32
0.5	1000	4	35	24	SuSiE	21	14	1	7
0.5	1000	5	31	32	Spike-and-slab lasso	28	3	1	14
0.5	1000	6	61	56	Spike-and-slab lasso	38	23	1	32
0.5	1000	pooled	340	321	Spike-and-slab lasso	255	85	1	197

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H_existing_20261030/tables/smc_vs_best_benchmark.csv]; [Results/2026-10-28_New_Campaign/H2_20261102/tables/headline_h2_long.csv]. Columns 1–9 are the 2026-10-30 comparison (benchmark primaries; adaptive lasso then with the ridge pilot `cv_min`); the last column is the D86 adaptive-lasso rerun (`ebic1_K37`, lasso pilot) from `headline_h2_long.csv`, added so the reader can see where it would change the "best" column.

c	p	comparator	both	only SMC HPM	only comparator	neither
1	500	SuSiE	558	1	27	14
1	1000	SuSiE	557	3	25	15
1	500	Spike-and-slab lasso	543	16	26	15
1	1000	Spike-and-slab lasso	557	3	25	15
1	500	Horseshoe	541	18	33	8
1	1000	Horseshoe	549	11	26	14
0.5	500	SuSiE	192	178	28	202
0.5	1000	SuSiE	181	159	24	236
0.5	500	Spike-and-slab lasso	264	106	39	191
0.5	1000	Spike-and-slab lasso	223	117	32	228
0.5	500	Horseshoe	167	203	18	212
0.5	1000	Horseshoe	124	216	12	248

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H_existing_20261030/tables/paired_wins_losses.csv]. Paired exact recovery on the same 600 datasets per $p \times c$ (SMC-BVS HPM vs the three Bayesian benchmarks).

UNCERTAIN — one "best benchmark" cell is superseded by the aLasso rerun

The per-cell comparison file predates the D86 adaptive-lasso rerun; at $p = 500$, S4, $c = 0.5$ it lists SuSiE/SSL with 9 as best [Results/2026-10-28_New_Campaign/H_existing_20261030/tables/smc_vs_best_benchmark.csv:19], while the rerun aLasso `ebic1_K37` has 11 there [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:76]. The HPM (25) is still first, so "first or tied-first in 11 of 12" is unaffected, and HEADLINE_H states that "every paired comparison …stands unchanged" [Results/

2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:131]; but the file's HPM-minus-best value in that cell (16) is computed against the pre-rerun best (9). Separately, the archive's comparison table [NBH/03_results_and_conclusions.md:171-179] reproduces the pre-D42 rows of [Results/SMC_BVS_v3_production/HEADLINE_TABLES.md:5-33] (ABESS ebic1 254/186 at $c = 0.5$, CV lasso family 0); the final headline rows are those of HEADLINE_H (ABESS primary at $c = 0.5$ is ebic1_K37: 256/214).

High-p benchmark disclosures.

- Lasso-family headline ebic1_K37, CV as sensitivity (D42): "The CV rules' 0/2,400 is method behaviour: CV-min over-selects at p ≫ n by 30–80 false positives"; EBIC-K37 "was a pre-registered companion at c = 0.5 and post hoc at c = 1" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:89-92]; D42 item 6 [Docs/HQ/DECISIONS.md:113].
- Elastic-net α: "CV chose α* = 0.9, the top edge of the frozen grid …in **2,399 of 2,400 fits**"; "Whether a wider grid would change its EBIC rows is untested" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:98-100] (Generated below).
- Adaptive lasso cv_min chose the last λ of the grid in 804 of 2,400 fits (D55); it is a sensitivity row only [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:103-110]. The old ridge-pilot rows are superseded (ebic1_K37 247 → 493, 96 → 512, 36 → 253, 12 → 197) [Docs/HQ/DECISIONS.md:262].
- "The benchmark code is AI-written. Its frequentist reproduction check (G5) is partial. ABESS fails the independent same-size search check." [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:33-35]; the Bayesian benchmarks pass G5 [Results/2026-10-28_New_Campaign/H_existing_20261030/HEADLINE_H.md:156].

fits	α^* = 0.1	0.25	0.5	0.75	0.9 (grid edge)
2400	0	0	1	0	2399

Generated by scripts/make_tables.py from [Results/2026-10-28_New_Campaign/H2_20261102/tables/f2_alpha_disclosure.csv]. Elastic-net CV choice of α over the frozen grid (PREREG R3).

Accounting and stops. **EMP10** "2,400 fits = 2,394 completed + 6 resource_limit + 0 missing"; the 6 stops are "all in S5, one at p500 c = 1, three at p1000 c = 1 and two at p1000 c = 0.5"; each hit the runtime guard at its allocation (6,300 s at $p = 500$; 24,000 s at $p = 1000$); "The same-seed retry at doubled allocation was not run (Felipe, D62). The 6 count as misses in every table"; in 3 of the 4 $c = 1$ stops the support saved at the stop equals the truth, "It is not counted" [Results/2026-10-28_New_Campaign/H2_20261102/HEADLINE_H.md:140-145]. D62: "If the remaining 6 are not done then feel free to kill them, we have enough." [Docs/HQ/DECISIONS.md:191]. Maximum effect of completing them: "at most **6** additional recoveries overall (0.25 percentage points of 2,400)"; "The observed retry effect is unknown" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:166-173]. Generated (reconciliation; diagnostic flags, informational only per D13):

Low p (PROD-L):

piece	expected	scored	failed	missing	balances
enum (exact, datasets)	1200	1200	0	0	True
smc primary (cases)	2400	2400	0	0	True
smcrep (fits, 5 per case)	12000	12000	0	0	True

High p (2026-10-23 fits; reproduced fit for fit by the v2.2.1 rerun):

p	c	fits	completed	resource_limit	missing
500	0.5	600	600	0	0
500	1	600	599	1	0
1000	0.5	600	598	2	0
1000	1	600	597	3	0

Generated by scripts/make_tables.py from [Results/2026-10-28_New_Campaign/L_20261102/tables/reconciliation.csv]; [Results/2026-10-28_New_Campaign/H_existing_20261030/tables/reconciliation.csv].

p	c	fits	diag_flag	rate %	escalated	resource_limit	flagged recovered	unflagged rate %
500	1	600	34	5.7	57	1	34/34	92.8
500	0.5	600	6	1.0	6	0	4/6	61.6
1000	1	600	56	9.3	112	3	48/56	94.1
1000	0.5	600	14	2.3	33	2	6/14	57.0

Generated by scripts/make_tables.py from [Results/2026-10-28_New_Campaign/H_existing_20261030/tables/smc_diag_flag_rates.csv]; [Results/2026-10-28_New_Campaign/H_existing_20261030/tables/smc_recovery_given_flag.csv]. diag_flag is informational only (D13).

3.4 The miss decomposition and what is identified

Low p (exact reference). Every one of the 184 misses is a posterior miss ($F(S_0) \leq F(\text{HPM})$, strict in every case); 0 search misses; the truth was scored in 2,400/2,400 fits; the scored models carry exact posterior mass with median 0.99995 and minimum 0.9980 [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:95-103]. "In these fits, every observed HPM truth-recovery error is a posterior error; the search contributes none. This concerns HPM recovery only, not $\hat{Z}$ or PIP estimation error." [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:119]. Generated:

p	c	hit	search miss	post. miss	sup.	same-size	sub.	mixed	P(wrong)	E[mass off S_0]	T21 bound	cov. median	cov. min	S_0 scored
20	1	576	0	24	24	0	0	0	0.040	0.331	0.663	0.99999	0.9999	600
20	0.5	562	0	38	21	17	0	0	0.063	0.367	0.735	0.99999	0.9998	600
30	1	571	0	29	27	2	0	0	0.048	0.383	0.766	0.99988	0.9988	600
30	0.5	507	0	93	22	59	10	2	0.155	0.488	0.977	0.99983	0.9980	600
all	all	2216	0	184	94	78	10	2	0.077	0.393	0.785	0.99995	0.9980	2400

Generated by scripts/make_tables.py from [Results/2026-10-28_New_Campaign/L_20261102/evidence/miss_class.csv]; [Results/2026-10-28_New_Campaign/L_20261102/evidence/t21.csv]; [Results/2026-10-28_New_Campaign/L_20261102/evidence/coverage.csv]. Miss class against the exact posterior: search miss $\Leftrightarrow$ $F(S_0) > F(\widehat{\text{HPM}})$; post. miss $\Leftrightarrow$ $F(S_0) \leq F(\widehat{\text{HPM}})$, HPM $\neq S_0$; shape columns split the posterior misses. T21 bound = P(no output) + 2 E[mass off S_0] + P(never scored); P(no output) = 0 and P(never scored) = 0 and the bound holds in every row (checked; columns omitted). cov. = exact posterior mass of the models the SMC scored.

High p (score-based, no exact reference). EMP11 The operational decomposition is: no output (6 resource_limit); "detected search miss: $F(S_0) > F(\text{HPM})$"; "completed miss with a scored competitor above the truth: $F(S_0) \leq F(\text{HPM})$, with HPM $\neq S_0$" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:131-134]. Counts: 1,829 hits, 565 completed misses, 0 detected search misses, 6 no output; "There are no exact ties: the smallest $F(\text{HPM}) - F(S_0)$ is 0.0049 nats, 23 of 565 are within 0.1 nats, and 39% are within 1 nat." [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:136-143]. Generated:

p	c	hit	completed miss (competitor above S_0)	detected search miss*	no output
500	1	559	40	0	1
1000	1	560	37	0	3
500	0.5	370	230	0	0
1000	0.5	340	258	0	2
all		1829	565	0	6

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_miss_class.csv]. *The CSV has columns hit / no_output / posterior_miss per scenario (100 each); the detected-search-miss column is 600− (the other three), i.e. fits with $F(S_0) > F(\widehat{\text{HPM}})$; it is 0 everywhere.

Not claimed / not proved — what the $p \gg n$ decomposition identifies (repo reports, after Sol R1)

"**Shown:** each of the 565 completed misses has a scored competitor above S_0. Even scoring S_0 could not have made that report a hit; the error would remain. **Not identified:** whether S_0 was ever scored in those 565 fits. So T21's discovery term $P(U^c, S_0 \notin \mathcal{S})$ is not identified: its count among completed fits lies anywhere between 0 and 565. **Why:** the per-model truth trace and the scored mass are unavailable at p > 64, because the registry runs in HLL mode (`truth_scored`, `log_sum_exp_F` and `first_scored` are NA/NULL in all 2,400 records). HPM provenance certifies that S_0 was scored only in the 1,829 hits. **Not evaluated:** the posterior-mass expectation $2E_0[1 - \pi(S_0)]$." [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:147-150]. Verdict: "Consistent with T21's decomposition"; "T21's probability bound: untested (silent)" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:153-154]. The status board repeats it: "At high p, 0 detected, but whether the truth was ever scored is unidentified for the 565 completed misses, and the T21 probability bound is silent there" [Docs/HQ/STATUS.md:75]. This wording was imposed by Sol R1 finding 1 [Results/2026-10-28_New_Campaign/H2_20261102/review/SOL_REVIEW_H2_R1.md:11-13].

UNCERTAIN — was S_0 scored in the 565 completed misses?

EMP12 **Repo side (above):** not identified; between 0 and 565 [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:148]. **Archive side:** the theory-campaign data clerk S-ELS read the per-fit top store and reports: "**S_0 was scored by the fit in all 2,394 completed fits**: it is in the 20,001-model top store, and the store's $F(S_0)$ equals the offline $F(S_0)$ to 1e-12. The event E_LS (S_0 not scored before the search and better than the best scored model) is **empty in the data (0 of 2,394)**." [NBH/branch_snapshots/S-ELS/REPORT.md:10]; its own limit: "The store holds the top 20,001 scored models, so that check covers S_0 only because S_0 scored high enough to be kept, which it always was." [NBH/branch_snapshots/S-ELS/REPORT.md:103]. In 434 of the 541 misses whose final local-search centre lies in $\mathcal{R}_2 \setminus \{S_0\}$, "$S_0$ was a one-move neighbour: the single scan scored it, and it was worse than S_c" [NBH/branch_snapshots/S-ELS/REPORT.md:74]. The archive adopts S-ELS [NBH/03_results_and_conclusions.md:168] (registry CJ222, CJ223) and so does API-N200 ("S_0 scored in 2,394 of 2,394 completed fits, 0 search misses (S-ELS); exact 95% upper bound 0.125% per fit") [NBH/branch_snapshots/API-N200/REPORT.md:12]; CJ247 recommends "use S-ELS for counts". **No source reconciles the two statements explicitly**: the repo reports reasoned from the registry fields (NA in HLL mode), S-ELS from the top-store field. A further detail is unreconciled: S-ELS calls the store "20,001-model" at both p, while the v3.1 settings list "top store $40p+1$" [NBH/02_theory.md:648] (= 20,001 only at $p = 500$). Until this is settled, theory must not rely on "S_0 scored in every completed fit" at $p \gg n$ without citing S-ELS and this conflict.

Shapes of the completed misses. EMP13 "At c = 1 these misses are almost all 'truth + one extra variable' (median FP = 1, FN = 0), with small gaps"; at $c = 0.5$ "all shapes occur, and equal-size wrong supports are the largest class" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:185-186]; "207 of the 565 completed misses (37%) are equal-size wrong supports, where the likelihood ranks a wrong model of the true size above S_0. No size-only prior can repair these" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:190]. S-ELS: at $c = 0.5$, 201 of 488 are strict same-size misses, 161 of them single swaps, and S4 supplies half [NBH/branch_snapshots/S-ELS/REPORT.md:92]. Low p: 94 supersets, 78 same-size swaps, 10 subsets, 2 mixed among 184 [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:132-136]. Generated:

p	c	shape of HPM vs S_0	n	median gap	min gap	max gap	med. FP	med. FN
500	0.5	mixed_size	25	2.843	0.7828	9.05	2	2
500	0.5	same_size_swap	106	1.539	0.0233	7.65	1	1
500	0.5	subset	49	1.136	0.0049	4.51	0	1
500	0.5	superset	50	0.926	0.0694	5.00	1	0
500	1	mixed_size	1	1.221	1.2209	1.22	2	1
500	1	same_size_swap	3	3.883	2.8120	6.21	1	1
500	1	superset	36	0.597	0.0862	3.33	1	0
1000	0.5	mixed_size	32	2.603	0.4540	7.88	2	2
1000	0.5	same_size_swap	95	1.599	0.0321	7.58	1	1
1000	0.5	subset	84	1.576	0.0745	8.39	0	1
1000	0.5	superset	47	0.676	0.0273	3.44	1	0
1000	1	same_size_swap	3	0.598	0.5665	2.64	1	1
1000	1	superset	34	0.895	0.0085	3.67	1	0

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_posterior_miss_shape.csv]. Shape of the 565 completed misses; gap = $F(\widehat{\mathrm{HPM}}) - F(S_0)$ in nats; "same_size_swap" = equal-size wrong support.

Final local search. EMP14 "The production local search ran and scanned in every completed fit, but made no improving move (0 moves in 2,394 fits). In the 1,829 hits, the centre was already S_0. In 541 completed misses the centre lay in $\mathcal{R}_2 \setminus \{S_0\}$. Each of these has F(centre) ≥ F(S_0), so no improving path to S_0 exists." [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:94-96]; S-ELS: "every search ended `no_improvement`, `local_certified = TRUE`, `local_limit = FALSE`"; the 541 have median gap −1.33 (range −9.05 to −0.0049) [NBH/branch_snapshots/S-ELS/REPORT.md:9-12]. Generated:

p	c	final local-search centre	n	returned S_0	HPM = S_0	median moves
500	0.5	centre = S_0	370	370	370	0
500	0.5	centre in $\mathcal{R}_0 \setminus S_0$	49	0	0	0
500	0.5	centre in $\mathcal{R}_1 \setminus \mathcal{R}_0$	143	0	0	0
500	0.5	centre in $\mathcal{R}_2 \setminus \mathcal{R}_1$	26	0	0	0
500	0.5	centre outside $\mathcal{R}_2$	12	0	0	0
500	1	centre = S_0	559	559	559	0
500	1	centre in $\mathcal{R}_1 \setminus \mathcal{R}_0$	37	0	0	0
500	1	centre in $\mathcal{R}_2 \setminus \mathcal{R}_1$	3	0	0	0
1000	0.5	centre = S_0	340	340	340	0
1000	0.5	centre in $\mathcal{R}_0 \setminus S_0$	84	0	0	0
1000	0.5	centre in $\mathcal{R}_1 \setminus \mathcal{R}_0$	131	0	0	0
1000	0.5	centre in $\mathcal{R}_2 \setminus \mathcal{R}_1$	31	0	0	0
1000	0.5	centre outside $\mathcal{R}_2$	12	0	0	0
1000	1	centre = S_0	560	560	560	0
1000	1	centre in $\mathcal{R}_1 \setminus \mathcal{R}_0$	35	0	0	0
1000	1	centre in $\mathcal{R}_2 \setminus \mathcal{R}_1$	2	0	0	0

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_local_search.csv]. Final local search in the 2,394 completed fits (rows with $n = 0$ omitted).

3.5 Theory-companion empirics

The campaign measured, after the fit, quantities mapped to the pre-campaign theorems by `THEORY_EXTRACTION_STRATEGY.md` (D23a) [Docs/HQ/DECISIONS.md:115-117]. Pre-campaign labels and archive IDs: T21 = TH115, T22 = TH116, T27 = TH119, T12 = TH110, T07 = TH106, locked T03 = TH123; refined T37 (i)/(ii)/(iii) = TH199/TH201/TH204; T38 (0)/(a)/(b1) = TH281/TH283/TH284 [NBH/ID_REGISTRY.md:254-267] [NBH/ID_REGISTRY.md:347-352] [NBH/ID_REGISTRY.md:429-432]. The final paper-1 cards that carry these themes are TH03 (M3 · Theorem START), TH05 (M5 · Theorem HPM), TH08 (M8 · Proposition D) and TH09 (S-T, T03♯) APPROVED (F.B., 2026-11-07). The pre-campaign label T26 ("Thm 6") has no row in the archive ID registry.

3.5.1 Start mass and masking (T22, T27; TH03, TH116, TH119)

EMP15 High p. "$\pi_0(S_0)$ is tiny. Median $\log_{10}\pi_0(S_0)$ by cell is −12.4 to −25.6; the largest over all 2,400 cases is 10^−5.6. The iid intensity Q_f·$\pi_0(S_0)$ is below 1 in **2,393 of the 2,394** completed fits"; the exception (`p0500_S2_r034_c100`, intensity ≈ 1.09) first scored the truth in the pilot; the cell median falls by 1.9–4.1 orders of magnitude from $p = 500$ to 1000; pooled near-truth medians $\pi_0(\mathcal{R}_0) = 0.20$, $\pi_0(\mathcal{R}_1) = 0.56$, $\pi_0(\mathcal{R}_2) = 0.81$ [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:42-47]. Verdicts: masking "qualitatively consistent" with T27; T27(i)'s joint law, T22(iii), T38(a), and the premise $\pi_0(\mathcal{R}_d) \geq b_n$: silent [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:58-61]. Recovery does not track $\pi_0(\mathcal{R}_0)$ at $c = 1$ [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:55]. Generated:

p	c	cell	med. $\log_{10}\pi_0(S_0)$	max	max $\log_{10} Q_f\pi_0(S_0)$	# $Q_f\pi_0(S_0) \geq 1$	med. $\pi_0(\mathcal{R}_0)$	$\pi_0(\mathcal{R}_1)$	$\pi_0(\mathcal{R}_2)$	min $\pi_0(\mathcal{R}_2)$	HPM = S_0
500	1	S1	-12.4	-7.9	-2.27	0	0.466	0.807	0.946	0.76	95
500	1	S2	-12.7	-5.6	0.04	1	0.253	0.607	0.840	0.259	96
500	1	S3	-13.2	-6.7	-1.14	0	0.105	0.390	0.690	0.266	94
500	1	S4	-13.5	-8.9	-3.24	0	0.005	0.059	0.250	0.0137	91
500	1	S5	-19.6	-11.9	-6.29	0	0.324	0.672	0.883	$5.0 \cdot 10^{-6}$	89
500	1	S6	-12.8	-9.4	-3.80	0	0.056	0.334	0.654	0.0419	94
1000	1	S1	-14.5	-6.9	-1.33	0	0.495	0.827	0.952	0.559	95
1000	1	S2	-14.6	-8.9	-3.23	0	0.351	0.702	0.893	0.431	95
1000	1	S3	-15.1	-8.7	-3.02	0	0.166	0.483	0.757	0.152	97
1000	1	S4	-16.3	-11.0	-5.41	0	0.009	0.096	0.303	0.0105	93
1000	1	S5	-23.4	-17.4	-11.76	0	0.285	0.648	0.860	$1.4 \cdot 10^{-12}$	84
1000	1	S6	-15.0	-7.5	-1.92	0	0.083	0.414	0.728	0.124	96
500	0.5	S1	-14.0	-9.5	-3.81	0	0.518	0.843	0.960	0.782	79
500	0.5	S2	-14.3	-7.6	-1.97	0	0.314	0.687	0.889	0.222	84
500	0.5	S3	-15.1	-8.0	-2.41	0	0.137	0.458	0.744	0.366	64
500	0.5	S4	-15.4	-10.3	-4.69	0	0.008	0.084	0.284	0.034	25
500	0.5	S5	-21.6	-13.9	-8.25	0	0.434	0.777	0.930	$1.2 \cdot 10^{-3}$	59
500	0.5	S6	-14.8	-10.7	-5.00	0	0.082	0.403	0.699	0.0422	59
1000	0.5	S1	-16.8	-8.5	-2.85	0	0.548	0.861	0.965	0.731	71
1000	0.5	S2	-16.7	-10.8	-5.25	0	0.407	0.755	0.921	0.53	75
1000	0.5	S3	-17.1	-10.5	-4.86	0	0.175	0.541	0.800	0.148	67
1000	0.5	S4	-18.5	-12.4	-6.75	0	0.020	0.143	0.401	0.0237	35
1000	0.5	S5	-25.6	-20.4	-14.59	0	0.458	0.799	0.945	$1.5 \cdot 10^{-15}$	31
1000	0.5	S6	-17.3	-9.5	-3.84	0	0.119	0.447	0.773	0.267	61

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_start_mass.csv]. $Q_f = 4N_e(T + L + 5)$ (expected number of final-run iid start draws equal to S_0 is $Q_f\pi_0(S_0)$); a descriptive bin, not T22(iii)'s hypotheses.

c	$\pi_0(\mathcal{R}_0) \in$ (-0.001, 0.25]	$\pi_0(\mathcal{R}_0) \in$ (0.25, 0.5]	$\pi_0(\mathcal{R}_0) \in$ (0.5, 0.75]	$\pi_0(\mathcal{R}_0) \in$ (0.75, 0.9]	$\pi_0(\mathcal{R}_0) \in$ (0.9, 1.0]
1	660/704 (93.8%)	317/342 (92.7%)	137/147 (93.2%)	4/6 (66.7%)	1/1 (100.0%)
0.5	325/617 (52.7%)	218/333 (65.5%)	152/224 (67.9%)	15/25 (60.0%)	0/1 (0.0%)

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_recovery_by_pi0R0.csv]. HPM = S_0 / fits (rate) by bin of the near-truth start mass.

Low p. $\pi_0(S_0)$ "spans 38 orders of magnitude"; median $5.5 \cdot 10^{-3}$, 53% of cases below 0.01; S1 is favourable, S3–S6 at $p = 20$ strongly masked [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:50-54]. "In every class and cell the exact HPM has the same rate, so the SMC loses nothing relative to the exact posterior, even with a tiny start mass" [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:65]. Verdict: support for the masked-regime claim at low p; silent on rates [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:68-69]. Generated:

p	c	med. $\pi_0(S_0)$	% < 0.01	med. $\pi_0(\mathcal{R}_0)$	$\pi_0(\mathcal{R}_1)$	$\pi_0(\mathcal{R}_2)$	init	Gibbs	nswap	uswap	indep	med. s	p90 s	med. distinct
20	1	$1.4 \cdot 10^{-10}$	75.2	$6.3 \cdot 10^{-10}$	$2.3 \cdot 10^{-4}$	0.212	165	434	0	1	0	2.84	3.94	32,704
20	0.5	$5.3 \cdot 10^{-7}$	71.5	$4.0 \cdot 10^{-6}$	0.0134	0.41	191	395	1	13	0	2.61	3.65	34,122
30	1	0.0848	29.2	0.28	0.766	0.953	485	99	10	6	0	2.70	5.02	71,480
30	0.5	0.0317	36.3	0.326	0.766	0.952	463	110	20	6	1	2.77	4.44	84,700
all	all	$5.5 \cdot 10^{-3}$	53.0	0.0749	0.558	0.873	1304	1038	31	26	1	2.74	4.11	49,008

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/L_20261102/evidence/pi0.csv]; [Results/2026-10-28_New_Campaign/L_20261102/evidence/truth_first.csv]; [Results/2026-10-28_New_Campaign/L_20261102/evidence/cost.csv]. Start mass from the exact block DP; "init ...indep": mechanism that first *scored* S_0 (always in the pilot phase; first scoring ≠ visit/retention); cost = runtime per fit and distinct models scored; no fit escalated beyond tier 0 (checked).

p	c	$\pi_0(S_0) < 0.01$		0.01–0.1		0.1–0.5		≥ 0.5	
		cases	HPM= S_0 %	cases	%	cases	%	cases	%
20	1	451	95.8	25	96.0	23	95.7	101	97.0
20	0.5	429	92.5	35	94.3	54	98.1	82	96.3
30	1	175	93.7	143	93.7	182	97.3	100	96.0
30	0.5	218	76.6	180	83.9	151	92.1	51	98.0
all	all	1273	91.1	383	89.3	410	95.4	334	96.7

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/L_20261102/evidence/recovery_by_pi0.csv]. SMC HPM = S_0 rate by start-mass class; the exact-HPM rate is identical in every class and cell (checked).

3.5.2 Truth dominance and greedy repair (T37; TH199, TH05)

EMP16 **Low p.** The literal T37 condition "**fails in every case**, 2,400/2,400 at both d"; 97.8% of clause-(a)/(b) violations at $d = 1$ are explained by a null-out/truth-in swap [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:15-19]. The "toward S_0" wording holds strictly in 63.0% ($d = 1$) and 53.6% ($d = 2$) of cases; every case where it holds is a hit (1,511 at $d = 1$), it fails in all 184 posterior misses, and it also fails in 705 hits at $d = 1$ [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:21-35]. Verdict: contradict "the applicability of the literal T37 condition"; "A conditional theorem is not refuted merely because its premise fails"; the toward form is "a candidate refinement ...It needs its own assumptions and proof" [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:39-44]. Generated:

p	c	d	literal violations	truth-in swap share	toward holds %	toward: n	hit	post. miss	not toward: n	hit	post. miss
20	1	1	106,790	0.996	92.3	554	554	0	46	22	24
20	1	2	1,104,201	0.996	83.5	501	501	0	99	75	24
20	0.5	1	114,547	0.985	74.8	449	449	0	151	113	38
20	0.5	2	1,147,727	0.986	63.5	381	381	0	219	181	38
30	1	1	369,824	0.992	57.7	346	346	0	254	225	29
30	1	2	5,684,860	0.996	46.2	277	277	0	323	294	29
30	0.5	1	400,904	0.958	27.0	162	162	0	438	345	93
30	0.5	2	5,842,712	0.974	21.2	127	127	0	473	380	93
all	all	1	992,065	0.978	63.0	1511	1511	0	889	705	184
all	all	2	13,779,500	0.986	53.6	1286	1286	0	1114	930	184

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/L_20261102/evidence/td.csv]; [Results/2026-10-28_New_Campaign/L_20261102/evidence/td_vs_outcome.csv]. T37 truth-dominance audit TD(d) with the full null pool, $m = 0$; "literal" = T37 wording, "toward" = the "toward S_0" variant. "truth-in swap share" = share of clause (a)/(b) violations explained by a null-out/truth-in swap. The literal wording holds in 0% of cases and the audit is complete in every row (checked).

High p. TD(0, 0) was audited on a fixed sample of 196 cases (replicates 1, 25, 50, 75, 100 per cell plus every $c = 1$ HPM miss; 18.7 core-h); "The finite-n TD(0, 0) premise holds strictly in 0 of the 196 audited cases"; clause (a) fails at a median 77% of the 16,383 proper subsets; in the 4 diagnosed cases the competing move is mostly a truth-for-truth swap (lateral: $|S \Delta S_0|$ unchanged); clause (c) holds in 92/92 sampled hits and fails in 97/100 sampled misses [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:66-93]. Verdicts: "T37(i) ...**not contradicted**", (ii)–(iii) silent, "Repair by improving moves: not exercised" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:99-105]. Generated:

p	c	cell	sample	n	complete	TD holds strict	S_0 strict local max	med. viol. share	med. clause-(a) min margin	med. margin at S_0	HPM = S_0
500	0.5	S1	fixed	5	5	0	5	0.873	-7.83	1.77	5
500	0.5	S2	fixed	5	5	0	4	0.874	-8.45	1.59	4
500	0.5	S3	fixed	5	5	0	3	0.883	-8.45	0.69	3

p	c	cell	sample	n	com-plete	TD holds strict	S_0 strict local max	med. viol. share	med. clause-(a) min margin	med. margin at S_0	HPM = S_0
500	0.5	S4	fixed	5	5	0	2	0.804	-12.14	-1.19	2
500	0.5	S5	fixed	5	5	0	4	0.905	-8.86	0.48	4
500	0.5	S6	fixed	5	5	0	2	0.895	-8.33	-1.12	2
500	1	S1	c100_miss	5	5	0	0	0.707	-8.47	-0.44	0
500	1	S1	fixed	5	5	0	5	0.688	-7.67	2.27	5
500	1	S2	c100_miss	4	4	0	0	0.652	-7.66	-0.55	0
500	1	S2	fixed	5	5	0	5	0.714	-8.41	2.09	5
500	1	S3	c100_miss	5	5	0	0	0.720	-7.67	-0.71	0
500	1	S3	fixed	5	5	0	4	0.685	-8.40	2.06	4
500	1	S4	c100_miss	7	7	0	0	0.705	-9.95	-2.81	0
500	1	S4	fixed	5	5	0	3	0.697	-11.93	2.28	3
500	1	S5	c100_miss	11	11	0	2	0.733	-8.77	-0.78	0
500	1	S5	fixed	5	5	0	5	0.753	-8.49	2.88	5
500	1	S6	c100_miss	4	4	0	0	0.739	-9.34	-0.62	0
500	1	S6	fixed	5	5	0	3	0.708	-8.50	0.72	3
1000	0.5	S1	fixed	5	5	0	4	0.901	-8.95	1.26	3
1000	0.5	S2	fixed	5	5	0	5	0.878	-8.43	1.65	5
1000	0.5	S3	fixed	5	5	0	3	0.901	-8.80	0.14	3
1000	0.5	S4	fixed	5	5	0	2	0.930	-10.99	-0.17	2
1000	0.5	S5	fixed	5	5	0	0	0.962	-9.48	-2.53	0
1000	0.5	S6	fixed	5	5	0	4	0.869	-9.23	1.49	4
1000	1	S1	c100_miss	5	5	0	0	0.700	-8.67	-0.71	0
1000	1	S1	fixed	5	5	0	5	0.748	-8.89	2.17	5
1000	1	S2	c100_miss	5	5	0	0	0.671	-9.17	-1.71	0
1000	1	S2	fixed	5	5	0	5	0.720	-8.34	2.30	5
1000	1	S3	c100_miss	3	3	0	1	0.787	-8.46	-1.65	0
1000	1	S3	fixed	5	5	0	5	0.699	-8.57	2.34	5
1000	1	S4	c100_miss	7	7	0	0	0.759	-10.28	-0.60	0
1000	1	S4	fixed	5	5	0	5	0.763	-10.73	2.50	5
1000	1	S5	c100_miss	16	16	0	2	0.773	-9.48	-0.78	0
1000	1	S5	fixed	5	5	0	5	0.821	-9.24	1.06	5
1000	1	S6	c100_miss	4	4	0	0	0.776	-8.87	-0.83	0
1000	1	S6	fixed	5	5	0	5	0.691	-8.73	3.01	5

Generated by `scripts/make_tables.py` from [`Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_td0.csv`]. TD$(0,0)$ audit on the fixed sample (replicates 1, 25, 50, 75, 100 per cell, plus every $c = 1$ HPM miss); unweighted description of an enriched sample, not population estimates.

3.5.3 Discovery of the truth (T38; TH05, TH281, TH284)

EMP17 **High p.** The reported HPM was first scored in "the pilot (calibration) SMC in all 2,394 completed fits"; for the 1,829 hits by Gibbs flips (1,724), neighbour swaps (104) and a uniform swap (1); "Neither final-run iid draws (§1) nor local-search moves (§2) were the source" [`Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:110-113`]. Verdict: "motivates a pilot-discovery / T24-type route for T38"; "The formal premises of T38(b2) are unverified: silent"; "Pilot provenance does not identify a terminal-round repair trajectory" [`Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:123-127`]. **Low p:** the truth was always first scored in the pilot (initial draw 1,304, Gibbs flip 1,038, `nswap` 31, `uswap` 26, `indep` 1 of 2,400); "First *scoring* a model does not mean a particle visited or retained it" [`Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:105-115`]; this "neither tests the hypothesised local-search repair mechanism nor rules out a later role for local search" [`Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:37`]. Generated (high-p provenance and score ledger; registry in HLL mode):

fits	phase that first scored the reported HPM	Gibbs flip	nswap	uswap
completed miss	pilot	521	44	0
hit (HPM = S_0)	pilot	1724	104	1

Final-run fresh-score ledger (pooled over completed fits; registry in HLL mode in 2400 fits; `first_scored` present in 0, `log_sum_exp_F` non-NA in 0):

phase	share %	mechanism	share %
table	0.041	table	0.041
final_init	0.002	init_draw	0.002
final_bridge	72.609	gibbs_flip	97.934
burn_in	18.230	uswap	0.167
measured	9.115	nswap	1.746
local	0.003	indep	0.107
		local_scan	0.003

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_hpm_provenance.csv]; [Results/2026-10-28_New_Campaign/H2_20261102/tables/evidence_numbers.json].

3.5.4 Cost (T38 cost corollary; TH08)

EMP18 When p doubles (500 → 1000; medians): "×2^1.07 fresh scores, ×2^1.5 distinct models (HLL estimate), ×2^1.4 runtime"; escalation to tier 1 rises from 9.5% to 18.7% ($c = 1$) and 1.0% to 5.5% ($c = 0.5$); verdict: "descriptively consistent with polynomial growth …not a test of the asymptotic cost corollary" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:115-126]. Low p: median 2.7 s per fit, "essentially flat from p = 20 to p = 30"; distinct models scored about 3% of 2^{20} and 0.007% of 2^{30}; no escalation; "two values of p cannot establish a rate" [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:80-88]. Generated:

p	c	fits	HPM = S_0	escalated	med. runtime s	p90 s	med. fresh scores	med. distinct (HLL)	distinct after cal.
500	0.5	600	370	6	607	826	$1.57 \cdot 10^8$	$1.31 \cdot 10^8$	$6.78 \cdot 10^7$
500	1	600	559	57	565	964	$1.39 \cdot 10^8$	$1.12 \cdot 10^8$	$5.83 \cdot 10^7$
1000	0.5	600	340	33	1,616	2,369	$3.28 \cdot 10^8$	$3.70 \cdot 10^8$	$2.16 \cdot 10^8$
1000	1	600	560	112	1,550	3,839	$2.94 \cdot 10^8$	$3.24 \cdot 10^8$	$1.88 \cdot 10^8$

Exponent $\log_2$*(median at* $p = 1000$ */ median at* $p = 500$*):*

c	fresh scores	distinct (HLL)	runtime
1	1.078	1.526	1.456
0.5	1.065	1.496	1.412

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_cost.csv]; [Results/2026-10-28_New_Campaign/H2_20261102/tables/evidence_numbers.json].

3.5.5 Evidence and PIP estimators (T12 = TH110; T26)

Low p: support for T12 ("the pooled mean relative error is compatible with zero") and for T26(i)–(ii) "as finite-setting accuracy evidence at n = 200, p ≤ 30"; "Silent on consistency and rates: one n, two values of p, one computational configuration" [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:162-165] (**EMP02**, **EMP03**). The final-card form of $E[\hat{Z}_r \mid \mathcal{C}] = Z$ is in **TH01.a** and **TH16** [NBH/ID_REGISTRY.md:31] [NBH/ID_REGISTRY.md:96]. High p: silent (D22); RB vs plain agreement only. Generated:

p	c	completed	MPM_{RB} =MPM_{pl}	$\text{MPM}_{RB} = S_0$	$\text{MPM}_{pl} = S_0$	med. max|ΔPIP|	p99	max	med. island sd log $\hat{Z}$
500	0.5	600	599	338	338	0.0022	0.0051	0.0067	0.063
500	1	599	599	559	559	0.0016	0.0046	0.0064	0.146
1000	0.5	598	596	321	320	0.0022	0.0051	0.0062	0.076
1000	1	597	597	561	561	0.0015	0.0046	0.0054	0.251

Generated by `scripts/make_tables.py` from [Results/2026-10-28_New_Campaign/H2_20261102/tables/ev_pip_rb_plain.csv]. Internal agreement of the RB and plain PIP estimators only (no exactness claim at $p \gg n$, D22).

3.5.6 Theory-campaign numerical runs (archive §2)

These were run by theory-campaign agents, mostly post hoc on the PROD-H fits or on small exact designs. Their raw outputs are mostly not in the repo (branch snapshots keep reports and some result files); the numbers below are **transcribed from reports; raw data not in repo** unless stated.

- **EMP19 O-LS0** (CJ226, CJ228, CJ178). LS0 run post hoc on all 2,400 fits: "LS0's best-scored model **never beats the production HPM**: 0 / 2,394. The smallest margin is 0.002 nats"; exact recovery "1,119 / 1,200 at c = 1" and "710 / 1,200 at c = 0.5", "gain 0, harm 0"; LS0 alone scores S_0 in 1,087 ($c = 1$) and 796 ($c = 0.5$) fits; median cost 0.17 s / 0.37 s, "under 0.1% of a production fit" [NBH/branch_snapshots/O-LS0/REPORT.md:13-37]. T38(b1) premises mostly fail at $n = 200$: "(M1) $\pi_0(\tilde{S}_0) > ½$ in 18 of 2,400 fits" [NBH/03_results_and_conclusions.md:190]. Caveat: post hoc; "LS0 exists only in the O-LS0 test export"; the integrated LS0 check is an "OPEN implementation check" [NBH/03_results_and_conclusions.md:191].
- **EMP20 API-N200** (TH334, Theorem N200; CJ14, CJ238). Predicted wrong = observed $P(U) + \sum \hat{P}_0(L \mid X)$ vs observed wrong: "81.8 vs 81 at c = 1 and 476 vs 490 at c = 0.5; 558 vs 571 over 2,400 fits; RMS z = 1.08 over the 24 cells, max |z| = 2.5 (p = 1000, S5, c = 0.5)" [NBH/branch_snapshots/API-N200/REPORT.md:13]; pooled table [NBH/branch_snapshots/API-N200/REPORT.md:42-48]; "The prediction is a lower bound plus a measured term". Exact enumeration at $n = 40$, $p = 12$: $\Pi - P(L)$ is 0.002–0.083 across the nine design/c rows [NBH/branch_snapshots/API-N200/REPORT.md:58-70] (result file `API-N200/results/` holds only `n200_table.md`). "An upper bound on Π(X) at finite n is OPEN" [NBH/03_results_and_conclusions.md:208].
- **EMP21 API-RESID / common σ^2** (CJ162, RF150). Common $\hat{\sigma}^2$: "None of the three feasible plug-ins gives one favourable fit (0/2,400 each)"; residualised (3 sweeps): median $\log_{10} \pi_0(S_0)$ $-15.65 \rightarrow -1.07$, "1,888/2,400 fits are favourable"; fixed-point: favourable 2,203/2,400; status "**unproven candidate**" [NBH/03_results_and_conclusions.md:212-215].
- **EMP22 Concurrency** (TH394, TH395). 3/3 designs bitwise (358 → 189 s, 672 → 352 s, 413 → 219 s) at $R = 4$ concurrent islands; Prop W-cal "is not executed" [NBH/03_results_and_conclusions.md:222-229].
- **PIP rescoring** (chair helper). The equal-weight vs $\hat{Z}$-weighted MPM comparison "cannot be computed for the 2,394 completed fits from this machine"; on 3 designs max|ΔPIP| up to $1.29 \cdot 10^{-3}$ (S5 r002), MPMs differ in 0 rows; "It says nothing about the rate of MPM disagreement in the study" [NBH/03_results_and_conclusions.md:233-235].
- **EMP23 ABL-START / ABL-CONF** (CJ166, CJ164, CJ165). ABL-START: "**Outcome 2. Keep the block start, reframed. The paper makes no depth or speed claim for the block start.**"; the prior start (arm C) was certified to underestimate Z on 10/12 designs [NBH/03_results_and_conclusions.md:239-241]. ABL-CONF (118 fits, rule pre-registered): "Claim 2 (accuracy …): LICENSED", median log $\hat{Z}$ error of the prior start −13.4 (S1), −10.8 (S4), −5.7 (S5) nats; "Claim 1 (cost …): INCONCLUSIVE" (S5 holds: block start cheaper in 10/10 pairs, sign test $p = 0.001$, median wall 1,116 s vs 1,639 s, accuracy equivalent; S4: singleton factors did not deliver within the 75-min cap in 10/10 while the block start delivered in 9/10, "a delivery result, not an equal-accuracy result"; S1: no cost advantage, B cheaper in 8/10 pairs, $p = 0.055$; under the pre-registered rule "licensed iff it holds on every design", Claim 1 is not licensed). For Claim 2 the prior start "escalated to tier 1 in 7, 9 and 8 of 10 fits and cost 2–4× the block start's wall time" [NBH/branch_snapshots/ABL-CONF/REPORT.md:9-14]; reference table [NBH/branch_snapshots/ABL-CONF/REPORT.md:18-22] (only `REPORT_TABLES.md` and `DEVIATIONS.md` of `results/final/` are in the repo).
- **EMP24 Calibration cost formula** (TH08.b, OB15). "calibration/final ratio predicted 0.46/0.67/1.69 vs observed 0.45/0.62/1.76 on 3 designs"; status "seat-derived, unchecked" [NBH/03_results_and_conclusions.md:258-259].
- **EMP25 LIT-PAR** (CE213). On an LH-like design ($p = 16$), "the exact full posterior recovers S_0 in 20/20 datasets. SaM recovers S_0 in **5/318** random splits that separate the linked pair" [NBH/

03_results_and_conclusions.md:263]. Scope: the LIT-PAR construction, not the LH family; its masked variable x_4 is a singleton linked only to hub x_0 (group size 1, not the g-prior's g) [NBH/ID_REGISTRY.md:1309]; for group size 1 the isolation probability under a uniform equipartition is 0.53 at $K_n = 2$ versus 0.05 for group size 4 (CE214) [NBH/ID_REGISTRY.md:1310].

- **EMP26 Dilution** (LM212, TH177, TH158). "On 480 regenerated study fits, V carries 83% of −log $\pi_0(S_0)$, C carries 11% and the size term 6%"; "About 7 of the 14 truths are invisible at n = 200 for every c"; per-cell medians of $\pi_0(S_0)$ $10^{-11.1}$ to $10^{-25.6}$ [NBH/03_results_and_conclusions.md:268-269].
- **EMP27 O-CERT / O-VCERT** (LM204, TH330). "0 violations in 476,160 exhaustive (S, S+j) pairs"; "0 Φ-violations and 0 M-violations over all 285,712 boxes on 48 designs" [NBH/03_results_and_conclusions.md:274].
- **API-PXNGAP §4** (LH-iid, exact small-scale; TH05). LS0 "scores S_0 in 56/60 overall and in 35/36 at n ≥ 200"; (L) failures 11/60; "Everything is consistent with an asymptotic theorem whose finite-n reach starts around n ≈ 200 for c ≤ 2 on this family" [NBH/branch_snapshots/API-PXNGAP/REPORT.md:79-111].

UNCERTAIN — conflicts recorded inside the archive's empirical section

(i) Favourable count of the unmodified start: "API-RESID table 'v3 (method) … favourable (≥ 10^{-4}) 0' of 2,400; O-DILUTE §1 'Favourable fits: 0 of 480; study: 1 of 2,394'. Not reconciled." [NBH/03_results_and_conclusions.md:216]. (The "1 of 2,394" matches EVIDENCE_H's different criterion $Q_f\pi_0(S_0) \geq 1$ [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:44].) (ii) Concurrency: "the task statement says '32 islands within 2e-4'. The file shows R = 32 (N = 312) max ΔPIP ≤ 2e-4 only on S1; it is up to 2.3e-3 on S4 and 7e-4 on S5" [NBH/03_results_and_conclusions.md:228]. (iii) Single thresholds for "favourable from n": "590–730, 800–1,600, 1,050–3,200" [NBH/03_results_and_conclusions.md:270]. (iv) Stopped fits "6 vs 4" across reports; O-SWAP's "4 no-output fits" is superseded by S-ELS's 6 (CJ223, CJ247).

Summary map (fact → theory).

fact	content	informs	source verdict (verbatim)
EMP01	low-p HPM = exact HPM 2,400/2,400; 0 search misses	TH05, TH115	"Support" (T21; bound loose) [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:118]
EMP02, EMP03	$\hat{Z}$ and PIP accuracy vs exact	TH110, TH01.a, TH16	"Support" T12; T26 finite-setting [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:163-164]
EMP11, EMP12	1,829 / 565 / 0 / 6 decomposition	TH115, TH05, TH334, LIM07	"Consistent with T21's decomposition" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:26]
EMP13	207/565 equal-size; supersets at $c = 1$	TH106, TH09, TH123	"T03: silent", "T07: support / relevant" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:28]
EMP14	local search never moved	TH199, TH05, CJ224	"Repair by improving moves: not exercised" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:105]
EMP15	masking: $Q_f\pi_0(S_0) < 1$ in 2,393/2,394	TH03, TH116, TH119	"qualitatively consistent with T27" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:23]
EMP16	TD fails (low p 2,400/2,400; high p 196/196)	TH199, TH201, TH204	"T37(i) … is not contradicted" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:24]
EMP17	first discovery in the pilot	TH281, TH284, TH218	"T38(b2)'s formal premises: unverified (silent)" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:25]
EMP18	cost exponents 1.07/1.5/1.4	TH08	"descriptively consistent with polynomial scaling" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:25]
EMP10	6 stops, P5 not run	TH05 (output event $\mathcal{O}$)	"Small observed completion gap" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:27]

3.6 Reproducibility and review trail

Every headline table was checked by an independent recomputation from raw records ("tie-out", worker Terra 6.1, with planted canaries) and reviewed by Sol 6.1. Bulk data are not in Git; they are on Felipe's PC (OneDrive "NOTS Data/New_Campaign/") with sha256 manifests [Docs/Records/DATA_INDEX.md:9]; per-case/per-fit tables cited by the reports (e.g. `case_theory_H.csv`, `smc_h2_per_fit.csv`, `ev_stops.csv`) are kept out of the repo [Docs/Records/DATA_INDEX.md:28-41].

analysis	tie-out	review / decision
Low p, SMC vs exact	PASS: 38/38 genuine cells, 5/5 canaries [Results/2026-10-28_New_Campaign/L_20261102/tieout/TIEOUT_VERDICT.md:8]	Sol R1 and R2 "ACCEPT WITH FIXES" [Results/2026-10-28_New_Campaign/L_20261102/review/SOL_REVIEW_R1.md:1] [Results/2026-10-28_New_Campaign/L_20261102/review/SOL_REVIEW_R2.md:1]; D75 [Docs/HQ/DECISIONS.md:227]
Low p, benchmarks (pass C)	PASS: 21/21 genuine, 4/4 canaries [Results/2026-10-28_New_Campaign/L_20261102/tieout_bench/TIEOUT_VERDICT.md:11]	Sol "ACCEPT" [Results/2026-10-28_New_Campaign/L_20261102/review/SOL_REVIEW_PASSC.md:3]; D94 [Docs/HQ/DECISIONS.md:294]
High p, existing fits	44 targeted checks, 5 canaries flagged, "The other 39 cells: all match" [Results/2026-10-28_New_Campaign/H_existing_20261030/tieout/TIEOUT_VERDICT.md:4-13]	Sol "ACCEPT WITH FIXES" [Results/2026-10-28_New_Campaign/H_existing_20261030/review/SOL_REVIEW_HEADLINE_H.md:1]
High p, H2 rerun	PASS: 46/46 genuine, 5/5 canaries [Results/2026-10-28_New_Campaign/H2_20261102/tieout/TIEOUT_VERDICT.md:26-27]	Sol R1 and R2 "ACCEPT WITH FIXES" [Results/2026-10-28_New_Campaign/H2_20261102/review/SOL_REVIEW_H2_R1.md:3] [Results/2026-10-28_New_Campaign/H2_20261102/review/SOL_RECHECK_H2_R2.md:3]; D78 [Docs/HQ/DECISIONS.md:232]
High p, aLasso rerun (F3)	PASS: 41/41 genuine, 3 canaries [Results/2026-10-28_New_Campaign/H2_20261102/tieout/F3_TIEOUT_VERDICT.md:10]	Sol "ACCEPT" [Results/2026-10-28_New_Campaign/H2_20261102/review/SOL_CHECK_F3.md:3]; D87 [Docs/HQ/DECISIONS.md:261]
Theory-campaign runs	none of the above kind	per-ID status in the registry (e.g. TH334 "author-only (no SF, no XF)")

Frozen code: `Scripts/Campaigns/2026-10-28_New_Campaign/` is "a byte-exact mirror of the campaign's code folders"; "Frozen means frozen" [Scripts/Campaigns/2026-10-28_New_Campaign/README.md:3] [Scripts/Campaigns/2026-10-28_New_Campaign/README.md:46]. All tables in this section regenerate with `python scripts/make_tables.py`, which also re-checks the reports' quoted totals (e.g. 2,216, 2,251, 559/560/370/340, 1,829/565/0/6, 207/565, 2,391/2,394) and exits non-zero on any mismatch.

3.7 What the empirics do not show

Not claimed / not proved — empirical scope

- **One n.** Every study fit has $n = 200$; consistency, rates and asymptotic statements are "silent" in every evidence report [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:87] [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:189]. Two values of p per regime "cannot establish a rate" [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:88].
- **One generator, not a theory family.** Groups of 5; not LH ($g = 9$) or LH-iid (§3.1).
- **No accuracy at $p \gg n$.** No $\hat{Z}$ or PIP accuracy is claimed at $p \gg n$ (D22) [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:8]; no head-to-head $\hat{Z}$/PIP comparison against other Bayesian methods exists [NBH/03_results_and_conclusions.md:181].
- **Discovery at $p \gg n$.** Whether S_0 was scored in the 565 completed misses is unidentified in the repo reports and asserted by S-ELS (EMP12); T21's probability bound is untested; $2E_0[1-\pi(S_0)]$ was not evaluated at $p \gg n$.

- **Repair.** The final local search never made an improving move at $p \gg n$; repair "by improving moves: not exercised". At low p first-score attribution "neither tests the hypothesised local-search repair mechanism nor rules out a later role".
- **TD at $p \gg n$** covers $d = 0$ only, on a 196-case sample enriched for misses; the move-type diagnosis covers 4 cases [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:231-233].
- **Stops.** The P5 retry was not run, so "The observed retry effect is unknown".
- **Benchmarks.** AI-written code, partial G5; ABESS not RSS-optimal; EN α at the grid edge; benchmark comparisons are "a comparison of *targets* as much as of *algorithms*" (F-REFEREE) [NBH/03_results_and_conclusions.md:181].
- **Theory-campaign runs** are mostly author-only, post hoc, or on 3–12 designs; LS0 is not in the production core.

For agents — empirical facts a theorist must explain or must not contradict

1. At low p the SMC HPM equals the exact HPM in 14,400/14,400 fits and there are 0 search misses (EMP01); any theorem whose finite-n reading predicts search failures on such designs must explain this.
2. At $n = 200$ **every observed wrong report among completed fits is a posterior (target) error**: 184/184 at low p (exact), and at $p \gg n$ each of the 565 completed misses has a scored competitor strictly above S_0 (min gap 0.0049 nats); the other 6 high-p errors are stopped fits with no output (EMP10) (EMP11;LIM07 "true by definition of the fixed target"). These observed competitor defeats establish target-mode misses *for the existing fits* only.

   UNCERTAIN — withdrawn inference: "a start change can add at most the 6 stopped fits"

   One archive summary still states: "no start or search change can fix them (API-RESID: a start change can add at most the 6 stopped fits, 0.25%)" [NBH/03_results_and_conclusions.md:183]. This inference was withdrawn at the method meeting: "The argument that the backfit start could gain at most 6 of 2,400 fits doesn't follow" [pos 1480] [NBH/03b_reasoning_and_traces.md:92] (see MD01 and reversal H.18 in §6). What stands is the definitional fact LIM07: under the fixed target, no search can make the HPM equal S_0 when $F(S_0) < F(\hat{S})$ [NBH/branch_snapshots/MEET-v31/MINUTES.md:92]. Do not cite the 6-fit cap.
3. Do not claim "S_0 was scored in every completed fit" at $p \gg n$ without the EMP12 conflict.
4. The start is masked almost everywhere ($Q_f \pi_0(S_0) < 1$ in 2,393/2,394; median $\log_{10} \pi_0(S_0)$ down to -25.6), yet at high p the reported HPM was first scored in the pilot in all 2,394 completed fits (for the 1,829 hits mostly by Gibbs flips, 1,724, and neighbour swaps, 104), and at low p the truth was always first scored in the pilot (initial draw 1,304, Gibbs flip 1,038 of 2,400) (EMP15, EMP17). At $n = 200$ "The rigorous a-priori bounds (T21 routes) are **vacuous**" [NBH/branch_snapshots/API-N200/REPORT.md:12]; EVIDENCE_H: "This motivates a pilot-discovery / T24-type route for T38" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:123].
5. Literal truth dominance fails in every checked case (low p 2,400/2,400; high p TD(0,0) 196/196, a $d = 0$ check on a miss-enriched sample), via truth-in swaps ($d \geq 1$) and lateral truth-for-truth swaps ($d = 0$) (EMP16); a usable repair premise "must tolerate lateral ($|S \Delta S_0|$-preserving) moves" [Results/2026-10-28_New_Campaign/H2_20261102/EVIDENCE_H.md:102].
6. Error shapes: supersets dominate at $c = 1$; equal-size wrong supports (37% of completed misses) and subsets at $c = 0.5$ (EMP13).
7. The N200 prediction (observed $P(U)$ plus the one-move term $\sum \hat{P}_0(L \mid X)$, "a lower bound

plus a measured term") is 558 vs 571 observed wrong over 2,400 fits (EMP20); an upper bound on $\Pi(X)$ at finite n is open.

8. RB PIPs are about 4× more accurate than plain PIPs at low p (EMP03); $\hat{Z}$'s pooled relative bias is within 1.1 SE of 0.
9. Cost from $p = 500$ to 1000 grows like $2^{1.07}$–$2^{1.5}$ in work (EMP18); two points, no rate.
10. SSL beats the exact HPM of SMC-BVS's own model at low p (2,251 vs 2,216; EMP05): "The exact posterior is that of the SMC-BVS model, not an oracle, so another prior can recover more often" [Results/2026-10-28_New_Campaign/L_20261102/EVIDENCE_L.md:265].

4 Theory: the approved paper-1 targets

This section is the ground truth for the approved paper-1 theorem list: the exact statements of the eight main cards (TH01–TH08, native M1–M8) and of the cited target-consistency card TH09 (S-T, T03♯), their exact proof statuses, what is still open for each, and how they depend on each other. The supplement cards TH10–TH20, obligations (OB), future work (CJ), drops (DR), limitations (LIM) and cited lemmas (CL) are written up in Section 5 and are only referenced here.

For agents — how to use this section

Every formal statement below sits in a *Statement (verbatim from source)* box and was transcribed mechanically from [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md] ("TLF"): only LaTeX formatting changed (Unicode sub/superscripts and accents set in math mode, Markdown tables set as labelled paragraphs, clause tags in the style of `TH01.a` inserted at the start of each clause). Status words are copied byte-for-byte from [NBH/ID_REGISTRY.md:23-89]. Before you try to prove, repair, demote or add anything: (1) read the card's *Status*, *Not claimed / not proved* box and *Open items*; (2) read the shared premises in §4.2, since most cards are stated *under named implementation premises* that are not model assumptions; (3) use the regime map in §4.4 to see which design family a result actually covers. A "PROVED" without "-LOCKED" is a candidate; "(by this seat, unchecked)" and "PROOF SKETCH" mean no checked proof exists [NBH/03_results_and_conclusions.md:23].

4.1 Overview

Thesis and objectives. The list serves the paper-1 thesis stated by HQ at [pos 1044]: "*In split-and-merge, correlation breaks correctness. In local-to-global SMC, correlation costs only sequential depth and particles, never correctness.*" [NBH/transcript/part_05_958-1107.md:455-456]. The project lead's motivation is parallel scalability with full guarantees under correlated designs, unlike Song–Liang (2015) split-and-merge [pos 1020]; the long-run aim is "to retire MCMC for BVS, the next gen samplers must be parallel and with the same degree of theoretical guarantees" [pos 1030] [NBH/transcript/part_05_958-1107.md:318-383]. The theorem meeting worked to three objectives, in order: "(1) accuracy in recovering the true posterior; (2) parallel scalability [...]; (3) BVS selection accuracy. It must work under **correlated designs**, unlike Song & Liang (2015) split-and-merge" [NBH/branch_snapshots/MEET-THM/DOSSIER.md:19]. The thesis sentence itself is *not* a theorem: it ships as a scoped remark after the main theorems, TH22 (Thesis CD, K96), in Seat C's wording "there is no retained non-corollary thesis sentence" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:26,1349]; see §4.4 for its scope sentence.

Approval. APPROVED (F.B., 2026-11-07): Felipe Bedoya approved the paper-1 list (M1–M8 = TH01–TH08, supplement cards TH09–TH20) on 2026-11-07 as the current targets (recorded after the frozen source commit) [Theory_work/State_of_Project/sources/approval_2026-11-07.md:3-5]. At the end of the archived thread the list was approved by HQ (with one change, growth rate (ii), RL06) but not yet by Felipe

[NBH/02_theory.md:66-67]; the 2026-11-07 approval does not change any proof status. The four decisions HQ put to Felipe (approve the list; Song–Liang separation as future work; Prop D (M8) in the main text; HQ's readings of proposal goals §5.3/§5.4) are in [NBH/branch_snapshots/MEET-THM/FOR_FELIPE_THEOREM_LIST.md:45-55].

Map of the list. One line per card. "Guarantee" paraphrases the card's own Guarantee row or HQ's plain-language table (main cards) or the registry's one-line description (supplement cards); statuses are verbatim *fragments* of the registry status (the full status is in each card or in Section 5).

ID	Native name	Guarantee (one line)	Placement	Status (verbatim fragment)
TH01	M1 · Theorem V/H	Any admissible data, any correlation, under the named implementation premises (EX-A, EX-S, CAL, IND-T, …): unbiased island identities given a frozen calibration ($\hat{Z}$ and PIP cited; general h via N1, a proof sketch); island-wise error of reported bounded summaries $\leq 2\delta_1 = 2(1 - \pi(S_0))$ in $\hat{Z}$-weighted expectation, given N1 (PROOF SKETCH; otherwise marginal PIPs only, per the Fallback row); fixed data: $\hat{\mu}(h) \to \pi(h)$ a.s. as $R' \to \infty$ (cited); $\hat{\mu}(h) \to h(S_0)$ in probability jointly with output only asymptotically, under the T03♯ hypotheses, $p \leq n^A$, $\operatorname{osc} d \leq D_* = O_P(1)$ and $R'_n = 4\lceil \log_2(n + p + 1) \rceil$ (M1(i), by this seat, unchecked).	main	N1: PROOF SKETCH; […] (h), (i): PROVED (by this seat, unchecked). — see card
TH02	M2 · Theorem WS (§5.3 package)	Inside the matrix domain H-DOM, start–posterior gap and bridge quantities controlled by two data-computable terms through bounds continuous in ε ((b7)); the integer stage count is a step function that can jump ((b8)); (b5)/(b6) need the grid premises (general D_* route a proof sketch); one-shot reading false (asymptotic obstructions (c1), (c2) under their stated hypotheses).	main	Seat C's D_* decomposition and actual-grid moment: PROOF SKETCH (NEW-COMP) — see card
TH03	M3 · Theorem START	Exact start-mass decomposition; under G on LH (exact Gram), start mass at truth $p^{-2}n^{-1}e^{O_P(1)} \to 0$ while $\pi(S_0) \to 1$ (T27(i) CONDITIONAL, Thm5-LH SAME-FAMILY-ONLY).	main	Final assembly: PROOF SKETCH, unchecked — see card

ID	Native name	Guarantee (one line)	Placement	Status (verbatim fragment)
TH04	M4 · Theorem $Z\mathcal{T}$ + certificate card	Deterministic store identities, and two-sided brackets when S_0 is stored; 95% certificate jointly with output under Lemma RO + GUARD-W (audit open); on LH under (b1-LH), $P(\mathcal{O} \cap \{TV(\pi_{\mathcal{T}}, \pi) > \varepsilon\}) \to 0$ (not $P(\mathcal{O}) \to 1$).	main	(f) assembly: PROOF SKETCH (seat-assembled; W2PC1/W2PC2 TRUE-AS-WRITTEN as an assembly) — see card
TH05	M5 · Theorem HPM	Wrong reported HPM ⇒ no output, posterior miss or never-scored truth; on LH with LS0, $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$, hence $\liminf P(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup P(U_n)$; delivery ($P(\mathcal{O}) \to 1$) is not claimed.	main	N3 "OPEN" [...] Assembly: PROOF SKETCH (by this seat, unchecked). — see card
TH06	M6 · Theorem SEP + Lemma KL	Under G on LH, iid start-only discovery misses S_0 w.p. $\to 1$ if $Q_n/(p^2 n) \to 0$ and $\to 0$ if $Q_n/(p^2 n) \to \infty$; one-shot evidence correction has a corresponding second-moment (L^2) obstruction, no matching upper bound.	main	KL: PROOF SKETCH (NEW-SYNTH) [...] Final assembly: PROOF SKETCH, unchecked. — see card
TH07	M7 · Theorem STG	Coded-grid stage laws: what ESS 0.9 certifies, stage-count bound, ideal-grid crossing $\log_2 A_0 + O(1)$.	main	ConjP: OPEN; [...]; (c'): PROOF SKETCH (Seat C, unchecked), W2PC2 TRUE-AS-WRITTEN — see card
TH08	M8 · Proposition D	Island-level work/depth/width in referee units, given the execution DAG as specified (audit open); polynomial work for $p \leq n^A$ under the shipped budgets and any permitted polynomial R'; no speed-up claim.	main	assembly PROOF SKETCH (by this seat, unchecked) — see card
TH09	S-T · Theorem 1, T03♯ (K93, K04)	Asymptotic: $1 - \pi(S_0) = O_P(\sqrt{\log n / n})$ under fixed-design hypotheses (fixed s_0, $p \leq n^A$, G, R, SE, BM-∞ or BM($c > c_0$)); exact HPM/MPM consistent.	supp. (cited in main)	T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2). [...] T01–T03: PROVED-LOCKED (wave 3; external review queued), SAME-FAMILY-ONLY
TH10	S-TARGET	Target propriety, invariance, envelopes, exact branch, odds certificate.	supp.	Remaining assembly: PROOF SKETCH, unchecked. — see §5
TH11	S-BLOCKS	Separation keeps groups; SIZE gives blocks = groups; block counts.	supp.	Counting/graph assembly: PROOF SKETCH, unchecked. PART-pairing remains OPEN, excluded. — see §5
TH12	S-KERNEL	N9 kernel facts, exact start sampler, minorisation, tie-aware DP (K09).	supp.	N9K: PROVED (candidate) [...] N3: OPEN [...] C10 (K09): PROOF SKETCH, unchecked — see §5
TH13	S-IDENT	Restricted-ESS coverage identity, terminal-law TV/KL calculus, credible-set clause C15-CI.	supp.	C15-TV, C15-KL, C15-CI: PROOF SKETCH, unchecked (assembly) — see §5
TH14	S-CERT	Local-search and certificate semantics, B&B soundness, perturbation lemma.	supp.	N7 L1: CORRECT WITH AMENDMENT [...] — see §5

ID	Native name	Guarantee (one line)	Placement	Status (verbatim fragment)
TH15	S-OUT	Output-event validity (stop-only guard), worker equivalence, stacking.	supp.	GUARD-W: not audited [...] — see §5
TH16	S-Z	Evidence variance identities and bounds; masked tails.	supp.	T26 overall CONJECTURE [...] — see §5
TH17	S-POOL	Pooling rules; recording-marginal equal-weight bias comparison P1; R_{eff}.	supp.	MA-CorV: PROOF SKETCH [...] — see §5
TH18	S-LIM	Limitations stated as theorems (counterexample constructions).	supp.	COUNTEREXAMPLE (constructions); INP I5 PROVED [...] — see §5
TH19	S-REP	Run-card reporting corollary: $\hat{f}$ and $\hat{\varepsilon}_{\times}$ with companions.	supp.	Reporting assembly: PROOF SKETCH, unchecked. N1: PROOF SKETCH. [...] — see §5
TH20	S-LEAK	Leakage clauses (F), (S) as a supplementary corollary to M2.	supp.	the m lower bound: PROVED (by this seat, unchecked) [...] — see §5
TH21	S-STGI	Dissolved: K30 = TH07.d; K29 → CJ23.	—	DISSOLVED (MINUTES §2.2, corrected)
TH22	Thesis CD (K96)	Scoped remark after the main theorems.	remark	remark, not a theorem

Sources: [NBH/ID_REGISTRY.md:23-102]; [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:15-46]; [NBH/branch_snapshots/MEET-THM/FOR_FELIPE_THEOREM_LIST.md:16-43].

How to read a card. Each main card below has: a header box (ID, native name, approval, the full registry status, K-IDs, authoring seat and source span, HQ charter and wave, binding goal); the chair's integration notes (TLF clerk text; where a note changes the reading of a card, the note governs [NBH/02_theory.md:63]); the formal statement in one or more verbatim boxes, with each clause row of [NBH/ID_REGISTRY.md] tagged where the clause begins; a list of clause statuses; the remaining card fields (Guarantee, Serves, Status, Verdict rows, Self-checked flag, Novelty, Route/key lemma, Lean core, Dependencies, Risk, Fallback + stop rule, IC3 design, Numerical companion), verbatim; then this document's plain reading, the *not claimed / not proved* box and the open items (obligations OB, future work CJ, and the per-theorem HQ charter's deliverables, verbatim). The status vocabulary (PROVED-LOCKED, PROVED (candidate), PROVED-UNDER, PROOF SKETCH, SF/XF, VERIFIED*, TRUE-AS-WRITTEN, SAME-FAMILY-ONLY, AUTHOR-ONLY, ...) is defined in the glossary, Appendix D. One distinction matters throughout and is quoted here:

Statement (verbatim from source) NBH/02_theory.md:690

What "checked" means in these cards. (i) Clause verdict rows such as "W1PC1:109–110 TRUE" or "W2PC2 TRUE-AS-WRITTEN" are statement-sanity checks (counterexample search, non-vacuity), "not a proof" (check-wave briefs); they never upgrade a status. (ii) "SF"/"XF" in a status are proof checks of the ingredient proofs recorded in the registry. (iii) "(by this seat, unchecked)" is the only status a seat may give its own argument.

Binding safeguards (MINUTES §2.3 §2.7): "§13.1: if N1's general record identity fails, the narrower theorem ships and §13.1 escalates. §5.3: if both the D_* route and the correlated-family route fail, it escalates. Also: loss of the masked statement without an approved replacement escalates (C §S5)"; and "N2(b), N2(c), N2(e), N14(c), the N3 DP invariant, and the integrated streams/LS0/GUARD-W/execution-DAG audits are **named implementation premises in the charters**" [NBH/branch_snapshots/MEET-THM/MINUTES.md:89-90]. The per-theorem HQs run in four waves: 1 =

M1, M3, LP-FOUND; 2 = M2, M7, M6, LP-IDENT; 3 = M5, M4, M8, LP-AUDIT; 4 = LP-REST, LP-REP [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:46]. At the end of the archive no per-theorem HQ had been launched [NBH/02_theory.md:68].

4.2 Shared definitions and premises

The cards cannot be read without these two blocks. The first (Seat C's shared definitions, R3_C) fixes the data, score, prior, start/bridge, the sampling model G, the design families **LH** and **W**, and the computational contracts; "Used by M3, M6 and S-REP (and by the other cards where they cite LH, W, G, EX-A)" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:56]. The second (Seat D's implementation premises, R3_D) names the premises carried by M5 and Prop D (M8) and, by name, by M1, M2, M4 [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:80].

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:62-74

Shared definition. Exact scope

D0: data/target. $n \geq 3, p \geq 1$; nonconstant columns and response, centered and standardized. $K = \min(p, \lfloor n/2 \rfloor)$, $\Omega = \{S : |S| \leq K\}$. Use the shipped $g = n, \tau = 1, a_0 = b_0 = .01$ score and normalized capped BB$(1, p)$ prior.

Score. $G_U = X_U^\top X_U$, $z_U = X_U^\top y$, $H_U = (n+1)G_U + I$, $Q_U = y^\top y - n z_U^\top H_U^{-1} z_U$, $\nu = .01 + n/2$. $m(U) = c_n[\det H_U / \det(G_U + I)]^{-1/2}(.02 + Q_U)^{-\nu}$. The common positive c_n cancels from posterior ratios.

Prior. $w(k) = [2\binom{2p-1}{k}]^{-1}$, $W_K = \sum_{k \leq K} \binom{p}{k} w(k)$, $\varpi(S) = w(|S|)/W_K$.

Start/bridge. Actual blocks C_b, $r_b(U) = m(U)/m(\emptyset) > 0$. $\gamma_0(S) = \varpi(S) \prod_b r_b(S_b)$, $\pi_0 = \gamma_0/\bar{Z}_0$, $\pi = \varpi m/Z$, $d = \log BF - \sum_b \log r_b$, $q_t \propto \gamma_0 e^{td}$.

G: sampling model. Fixed raw design; $y^r = \alpha \mathbf{1} + X^r \beta + \epsilon$, $\epsilon \sim N(0, \sigma^2 I)$, fixed $\sigma^2 > 0$. Standardize afterwards. Standardized errors are not assumed independently Gaussian.

LH: design. $p = 9\lfloor n^{1/3}/9 \rfloor$, groups of nine, within-group correlation 0.4. Two disjoint cross-group links have correlation 0.2; remaining cross-group correlations vanish. Each linked pair has coefficients $(1, -0.2)$; others zero; $\sigma^2 = 1$. Centered Gram: $X^\top X = (n-1)\Sigma$.

LH: non-vacuity. $\lambda_{\min}(\Sigma) \geq 0.4$, $p < n$, support size four. For $n \geq 9^6$, $9 \leq B \leq 16 < 18$; separation and SIZE give actual blocks equal groups. Two true coordinates have zero population marginal covariance with the response. W1P2 "Shared · LH family"; W2P2 C5/C11.

W: reporting exhibit. Same p, groups of nine; within correlation $1/4$, cross correlation $\varepsilon_n = 1/(np) > 0$. Two coefficients $n^{-1/4}$ in different groups; $\sigma^2 = 1$. Also retain the checked stronger exhibit $\varepsilon_n = n^{-1/2}/p$. Actual blocks equal groups for $n \geq 9^6$; $D_* = O_P(1)$. W2P2 §1, IC3/scale rows.

EX-A. Exact mathematical scores, arithmetic, sums and comparisons. This is an idealisation, not a validated floating-point property.

MC contract. Exact iid start draws, multinomial resampling, actual invariant move rounds, independent ideal uniforms; successful full calibration $\mathcal{C}$ frozen before final randomness; independent final islands; common N; $R' = 4k$ declared in advance. These are computational premises.

Output contract. Potential uninterrupted final run first. GUARD-W identifies delivered statistical records with potential records on $\mathcal{O}$. Its integrated audit remains open. Bounds transfer by intersection with $\mathcal{O}$, never by conditioning on delivery.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:86

Shared implementation premises (named once, used by both cards): **EX-A** exact scores and arithmetic; **EX-S** fresh independent ideal uniforms (H-U); **CAL** a successful frozen calibration record

$\mathcal{C}$ on streams 0–9; **IND-T** final streams 10..9+R′ independent of $\mathcal{C}$ and of each other, R′ = 4k declared before stream 10; **GUARD-W** stop-only guard (operational interruption ⇒ no output); **SR** the method's stop/return rule as in the T38 hypotheses; **OP-LS0** LS0 integrated as specified (METHOD §2 #8); **e4** emitted LS0 centre = the prescribed smallest-size, lexicographically smallest start-mode maximiser (DP invariant N3 OPEN); **complete scans** no operational interruption of any search. $\mathcal{O}$ is the output event; statements are $P\{\mathcal{O} \cap \cdot\}$; $D_n := P(\mathcal{O} \cap \{S_0 \notin \mathcal{S}^+\})$ in potential-run notation (the overbar is the potential run, not set complement; W2PC1:122).

Notes on the shared objects.

- **Premises are not model conditions.** EX-A (exact arithmetic) is "an idealisation, not a validated floating-point property" and the MC contract items are "computational premises" (row EX-A and MC contract above). Every card number of M4 is labelled "numerical" until SC + VAL-FP exist (N16, OB16, CJ16) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:248].
- **LH is parametrised differently by different cards.** The shared row fixes groups of nine, within correlation 0.4, two cross-group links 0.2, coefficients $(1, -0.2)$, $p = 9\lfloor n^{1/3}/9 \rfloor$, exact Gram $X^\top X = (n-1)\Sigma$. M4(f) restates the same family with $n \geq 9^6$ and SIZE $9 \leq B \leq 16 < 18$ [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:247]. M5 states (b1-LH) for a general LH with "c linked pairs, group size g, within-correlation ρ, link ψ with $|\psi| < \rho$, fixed β" and SIZE $g_{\max} \leq B < 2g_{\min}$, and its no-LS0 variant (b2-LH) on STAB-LH ($g = 9$, $c = 2$, $\rho = 0.4$, $\psi = 0.2$) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:292]. LH-iid (random Gaussian rows with Σ_{LH}) is a *different* family used only by the unreviewed candidate TH161 (§4.4).
- **The raw discrepancy d in the declared list.** d is a default member of the declared list $\mathcal{H}$ (TH01 note 2); because $\mathcal{H}$ is $[0,1]$-valued, "**raw d is recorded**, and theorems (M1) apply to the globally range-scaled $(d - \min_\Omega d)/D$, with $D = \max_\Omega d - \min_\Omega d > 0$. If $D = 0$, then $\hat{J} = J = 0$" [NBH/branch_snapshots/MEET-THM/AMENDMENT_HQ_02_DEFINITIONS.md:41] (final-audit correction #10, [NBH/branch_snapshots/MEET-THM/CLOSE_AUDIT.md:18]).
- **The matrix domain H-DOM** and the constants $a, \delta_A, \delta_H, r, C_\sigma, b_X, D_*$ are defined inside card M2 (TH02, row H-DOM; registry DF24).

4.3 The main cards M1–M8 and the cited card S-T

4.3.1 TH01 · M1 · Theorem V/H (validity of v3.1 and accuracy of reported bounded functionals)

TH01 · M1 · Theorem V/H

APPROVED (F.B., 2026-11-07) Placement: main text, wave 1 (HQ charter `HQ_CHARTERS/M1.md`).
Status (registry, verbatim): T12, T31(i): PROVED-LOCKED; Prop2(a,b,d,e): PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: GAP (A-REV4)); N9K: PROVED (candidate), code-checked; CHK_INHERITED: VERIFIED WITH AMENDMENT; N1: PROOF SKETCH; A1new: PROOF SKETCH (NEW-SYNTH); FMP-M1: PROVED (SF: none; XF: VERIFIED (A-REV9)); PC1c3: PROVED (PC1 claim 3), 'E[W_r $\hat{h}_r$] = π(h) is still a premise'; MA-PropP: PROVED (elementary), author-only; IND-T: PROVED (SF: none; XF: VERIFIED* (A-REV1)); LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)); CEX-guard: COUNTEREXAMPLE; N6(a): PROVED (potential run; output form CORRECT WITH AMENDMENT); T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)); (h), (i): PROVED (by this seat, unchecked).
K-IDs (MINUTES §2.1, as in the registry): K46, K32, K86, K35, K36, K50, K51, K52, K48, K62; cited K20, K21, K08, K47 (the card's own header lists a different set; integration note 1).
Authoring seat / card source: Seat A (Fable 5.1), `R3_A.md:112–137`, clerk copy TLF lines 100–129.
Binding goal: proposal §13.1 ("$E_\pi[h]$ → truth"), with the safeguard that if N1 fails the narrower

theorem ships and §13.1 escalates; also §5.1, §5.2 [NBH/02_theory.md:719-730].

TLF clerk text (verbatim). Chair's integration note.

1. K-ID header: the card's own header line lists "K46, K32, K86, K85, K36, K78, K51, K52, K48, K62; cited K20, K21, K08, K35, K47"; MINUTES §2.1 lists K46, K32, K86, K35, K36, K50, K51, K52, K48, K62; cited K20, K21, K08, K47. The registry (REGISTRY_PAPER1.md) follows MINUTES. The card body is unchanged.

2. Growth rates (MINUTES §2.3 §2.6, confirmed by all four): clause (i) is the first stacking configuration, $R'_n = 4\lceil \log_2(n + p + 1) \rceil$, for a fixed finite set of declared summaries on families with $D_* = O_P(1)$; the second optional configuration $R'_n = 4\lceil (n + p)^6/4 \rceil$ for simultaneous control over $\leq (n + p)^2$ declared summaries is added to the statement of (i). ℓ, L and N need no edit; shipped $R' = 4$ carries only the fixed-R statements. d is a default member of the declared list $\mathcal{H}$ (A, B; AMENDMENT_HQ_02).

3. Binding §13.1: if N1's general record identity fails, the narrower theorem ships and §13.1 escalates (MINUTES §2.3 §2.7; the card's own fallback agrees). K63 (worker equivalence) and K86 (IND-T) are named premises here and proved in S-OUT (R3-4).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:109-111

Model/design hypotheses. Admissible standardised (X, y), $n \geq 3$; $h \in \mathcal{H}$ with $0 \leq h \leq 1$, measurable in (X, y, $\mathcal{C}$), declared before stream 10 (M9). For (i) only: T03#'s hypotheses (R1, G, R, SE_λ, SE_w, BM-∞ or BM(c)) on a family with osc $d \leq D_* = O_P(1)$ (M2(a)), $p \leq n^A$.

Implementation premises (named). EX-A exact arithmetic; H-U/EX-S exact iid uniforms and exact start sampler (N9); CAL: $\mathcal{C}$ frozen on streams 0–9; IND-T: streams 10..9+R′ independent (audited for $R' = 4$; extension audit pending, S-OUT); $R' = 4k$ declared before stream 10; multinomial resampling; GUARD-W and Lemma RO for (d); M9 accumulator; stacking configuration R'_n for (i).

Statement.

TH01.a (a) Validity [cited]: each elementary move is γ_φ-reversible, the uniformly random-order round q_φ-invariant and self-adjoint under N9's sufficient conditions, draw_pi0 and $\bar{Z}_0$ exact; each island is a Feynman–Kac model, so $E[\hat{Z}_r \mid \mathcal{C}] = Z$ and $E[\hat{Z}_r \, PI\hat{P}_{rj} \mid \mathcal{C}] = Z \cdot PIP_j$ with the RB record before the Gibbs draw (weighted identity; no terminal stationarity claimed).

TH01.b (b) Lemma N1: $E[\hat{Z}_r \, H_r(h) \mid \mathcal{C}] = Z \cdot \pi(h)$, H_r the plain measured-endpoint average.

TH01.c (c) Contract: (a),(b) hold conditionally on any successful frozen $\mathcal{C}$; any R′ declared before stream 10 keeps them; adaptive R′ or conditioning on completion **need not** preserve them (CEX-guard).

TH01.d (d) Output event: $P(\mathcal{O} \cap \{0.05\hat{Z} > Z\}) \leq 0.05 \cdot P(SMC \cap \mathcal{A}) \leq 0.05$, never conditional on $\mathcal{O}$.

TH01.e (e) Islandwise, every regime and frozen grid: $E[W_r \cdot |H_r(h) - h(S_0)| \mid \mathcal{C}] \leq 2\delta_1$, $\delta_1 = 1 - \pi(S_0)$.

TH01.f (f) Finite R′, $M_2 = E[{W_r}^2 \mid \mathcal{C}]$: $P(|\hat{\mu}(h) - \pi(h)| > \varepsilon \mid \mathcal{C}) \leq \min\{1, 4(M_2 - 1)/R' + 4M_2/(R'\varepsilon^2)\}$; $P(|\hat{\mu}(h) - h(S_0)| > t \mid \mathcal{C}) \leq \exp\{-R'(1-s)^2/(2M_2)\} + 2\delta_1/(st)$; vacuous at $R' = 4$, informative under stacking; fixed data: $\hat{\mu}(h) \to \pi(h)$ a.s. as $R' \to \infty$ [cited].

TH01.g (g) Declared classes: joint inclusions $h_A = 1\{A \subseteq S\}$ on $\leq 2pB$ declared pairs (plain averages); interval probabilities $h_I(S) = P(\beta_j \in I \mid S, X, y)$ for a finite declared list, union bound; posterior-content error only, no quantile claim (HQ Q2).

TH01.h (h) Decision transfer: exact PIPs $\geq \delta$ from ½ and uniform error $< \delta \Rightarrow$ reported MPM = exact MPM.

TH01.i (i) Asymptotic, with $R'_n = 4\lceil \log_2(n+p+1) \rceil$: $\hat{\mu}(h) \to h(S_0)$ in probability, jointly with output; liminf $P(\mathcal{O} \cap \text{correct}) = 1 - \text{limsup } P(U_n)$.

Clause rows and their registry statuses. TH01.a (M1(a) Validity [cited]): T12, T31(i): PROVED-LOCKED; Prop2(a,b,d,e): PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: GAP (A-REV4)); N9K: PROVED (candidate), code-checked;

CHK_INHERITED: VERIFIED WITH AMENDMENT

TH01.b (M1(b) Lemma N1): N1: PROOF SKETCH

TH01.c (M1(c) Contract): CEX-guard: COUNTEREXAMPLE; W1PC1:127 AMBIGUOUS → ”need not” (W2PC1:92 REPAIRED)

TH01.d (M1(d) Output event): N6(a): PROVED (potential run; output form CORRECT WITH AMENDMENT); LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)); GUARD-W audit open

TH01.e (M1(e) Islandwise bound): PC1c3: PROVED (PC1 claim 3), ’E[W_r $\hat{h}_r$] = π(h) is still a premise’ (card status row); verdict rows W1PC1:118,120; W2PC1:93 TRUE

TH01.f (M1(f) Finite-R′ bound): same as TH01.e (PC1c3: PROVED, N1 a premise); fixed data $\hat{\mu}(h) \to \pi(h)$ a.s. [cited]

TH01.g (M1(g) Declared classes): verdicts W1PC1:263 (interval claim narrowed); same status block as TH01.e

TH01.h (M1(h) Decision transfer): (h), (i): PROVED (by this seat, unchecked)

TH01.i (M1(i) Asymptotic with R'_n): (h), (i): PROVED (by this seat, unchecked); T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)); second rate: see U1

Remaining card fields (verbatim, TLF 105, 112–128). K-IDs: K46, K32, K86, K85, K36, K78, K51, K52, K48, K62; cited K20, K21, K08, K35, K47.

Verdict rows per clause. (a) W1PC1:109–110 TRUE; W2PC1:92 TRUE; CHK:94, 140–142 (N9 amendments adopted); RB wording per W2PC1:106. (b) W1PC1:110, W2PC1:92 TRUE; N1 PROOF SKETCH. (c) W1PC1:127 AMBIGUOUS → ”need not” (W2PC1:92 REPAIRED). (d) W1PC1:126 TRUE. (e),(f),(g),(h) W1PC1:118, 120; W2PC1:93 TRUE (interval claim narrowed, W1PC1:263); W1PC1:240 ($0 \le h \le 1$); (h) W1PC1:302 TRUE. (i) W2PC1:94 TRUE on C’s family; liminf form W1PC1:170.

Guarantee. Whatever the correlation, every island’s unnormalised outputs are exactly unbiased given any frozen calibration, every reported bounded summary is right on the islands that carry evidence weight, and pooling enough declared islands by $\hat{Z}$ makes it right.

Serves. Objective (1); proposal §5.1, §5.2, §5.4 (PIP, joint-PIP, posterior-content error), §13.1 (binding); Song–Liang Thm 3 (reported PIPs), UQ of merged output; INV rows 4, 6, 8, 10.

Status (verbatim). T12, T31(i): PROVED-LOCKED; Prop2(a,b,d,e): PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: GAP (A-REV4)); N9K: PROVED (candidate), code-checked; CHK_INHERITED: VERIFIED WITH AMENDMENT; N1: PROOF SKETCH; A1new: PROOF SKETCH (NEW-SYNTH); FMP-M1: PROVED (SF: none; XF: VERIFIED (A-REV9)); PC1c3: PROVED (PC1 claim 3), ’E[W_r $\hat{h}_r$] = π(h) is still a premise’; MA-PropP: PROVED (elementary), author-only; IND-T: PROVED (SF: none; XF: VERIFIED* (A-REV1)); LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)); CEX-guard: COUNTEREXAMPLE; N6(a): PROVED (potential run; output form CORRECT WITH AMENDMENT); T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)); (h), (i): PROVED (by this seat, unchecked).

Fidelity to v3.1. (a)–(h) as shipped (LS0 alters no particle; any frozen grid; $\hat{Z}$-weighted primary). (i) uses the optional stacking configuration (AMENDMENT_HQ_02), not the default four islands.

Self-checked flag. N1 unchecked (PROOF SKETCH, numerically TRUE-AS-WRITTEN); MA-PropP author-only; N9 candidate plus cross-family CHK.

Novelty. (a), (f) fixed-data limits KNOWN—CITE: Del Moral 2004; Del Moral–Doucet–Jasra 2006; Whiteley–Lee–Heine 2016; Dai et al. 2022 (LIT-SMC rows 1, 21–22). (d) Markov KNOWN—CITE Grosse et al. 2015; the output-event discipline ”genuinely non-standard” (LIT-SMC). (b), (e), (g), (h): NEW-COMP (M9 mechanism, measured-record identity). Checked against LIT-SMC/LIT-PAR digests only; citation checker to confirm volumes.

Route / key lemma. N1: finite-state resample–move recursion; the state before each measured record is Zπ-weighted because the prefix of a uniformly random order is state-independent (terminal-round prefix identity, N9 §3). (e): $|H_r - h(S_0)| \le$ fraction of measured particles off S_0, then N1 on $1\{S \neq S_0\}$. (f): Chebyshev with the denominator event ≥ ½ (PC1 claim 3); Maurer/Hoeffding. (i): $M_2 \le e^{D_*}$ (M2(b)) and $R'_n \to \infty$. Key lemma: N1.

Difficulty. M (N1 one page; the rest assembly).

Lean core. T12Core `unbiased_of_step` with `hstep` discharged for multinomial resampling + an invariant kernel; detailed balance of the four kernels (finite matrices); (e) deterministic step; (f) finite inequality on abstract nonnegative weights; (h) finite; CEX-guard as a two-point example.

Dependencies. S-KERNEL (N9), S-OUT (Lemma RO, GUARD-W, IND-T extension), S-TARGET (propriety), M2(b)(iv) and S-T (T03♯) for (i); feeds S-POOL, S-REP, M4(d), M8.

Risk. N1 fails (binding §13.1); GUARD-W or IND-T extension audit fails (then (c),(d) stay potential-run statements).

Fallback + stop rule. Fallback: (a), (c), (d) plus (e)–(i) for marginal PIPs only (locked RB identity; no general h). Stop: one HQ wave on N1; if not closed, ship the fallback and **escalate §13.1 to HQ** (never dropped).

IC3 design, exhibited. (a)–(h): any admissible X, including duplicate columns (W1PC1 D0/E1 finite enumeration, W1PC1:69, 110). (i): C's family W (within 1/4, $\varepsilon_n = 1/(np) > 0$, two $n^{-1/4}$ signals, blocks = groups for $n \geq 9^6$), $D_* = O_P(1)$, checked with the method's own partition (W2PC2:53–58); also at $\varepsilon = n^{-1/2}/p$.

Numerical companion (≤ 20 seeds). N9-KERNEL CHECK (DB ≤ $2.3 \cdot 10^{-12}$); PC1 E1 two-state exact enumeration (exists). For the HQ: exact enumeration on $p \leq 8$ of $E[\hat{Z}_r\, H_r(h)]$ against $Z \cdot \pi(h)$ and of $E[W_r | H_r - h(S_0)|]$ against $2\delta_1$ on a coded-grid replica, 10 seeds.

Split note (if R3-1 (ii) were chosen instead): Proposition V+ = clauses (a)–(d) with the same premises; Theorem H = (e)–(i) with "Setting V+". Nothing else changes.

UNCERTAIN — U1: the second growth rate in clause (i)

Integration note 2 says the second configuration $R'_n = 4\lceil (n+p)^6/4 \rceil$ "for simultaneous control over ≤ $(n+p)^2$ declared summaries is added to the statement of (i)" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:96]; the verbatim card clause (i) prints only $R'_n = 4\lceil \log_2(n+p+1) \rceil$ [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:111]; HQ keeps configuration (ii) "**only as a sufficient rate inside the theorem** (M1's simultaneous-control clause: "for any $R'_n \geq 4\lceil (n+p)^6/4 \rceil$"), not as a runnable configuration of the method" [NBH/branch_snapshots/MEET-v31/AMENDMENT_HQ_02.md:12]. The exact final wording of clause (i) with the second rate is not printed in any snapshot [NBH/02_theory.md:2781].

Plain reading

For any admissible data set (any correlation) and any successful calibration frozen before the final islands, under the named implementation premises (EX-A, EX-S/H-U with N9's sufficient conditions, CAL, IND-T, R' declared before stream 10, multinomial resampling), each island's unnormalised outputs are unbiased: $E[\hat{Z}_r \mid \mathcal{C}] = Z$ and the Rao–Blackwellised PIP identity hold (clause (a), cited); the same identity for a general bounded summary h is Lemma N1 (clause (b), a proof sketch). On each island, the evidence-weighted error of a reported bounded summary against its value at the true model is at most $2\delta_1 = 2(1 - \pi(S_0))$ in expectation (e), given N1 (the route applies N1 to $1\{S \neq S_0\}$; if N1 fails only marginal PIPs remain, per the Fallback row). The finite-R' tail bounds (f) are vacuous at the shipped $R' = 4$ and informative only under stacking. Convergence of $\hat{\mu}(h)$ to $h(S_0)$ in probability, jointly with output (i), is asymptotic and needs the optional stacking configuration $R'_n = 4\lceil \log_2(n+p+1) \rceil$ and T03♯'s hypotheses with $p \leq n^A$ on a family with osc $d \leq D_* = O_P(1)$; it is "PROVED (by this seat, unchecked)". Output statements are always of the form $P(\mathcal{O} \cap \cdot)$, never conditional on delivery.

Not claimed / not proved

- Lemma N1 (general h) has no checked proof: "N1: PROOF SKETCH" (numerically TRUE-AS-WRITTEN); PC1 claim 3 takes "$E[W_r\, \hat{h}_r] = \pi(h)$" as a premise (card status row). If N1 fails, (e)–(i) ship for marginal PIPs only and §13.1 escalates (Fallback row).
- Adaptive R' or conditioning on completion "**need not**" preserve (a),(b) (clause (c); CEX-guard COUNTEREXAMPLE). (c),(d) are output-form statements only if the GUARD-W and IND-T-extension audits pass; otherwise they stay potential-run statements (Risk row).
- No quantile claim for interval probabilities (g); posterior-content error only (HQ Q2, RL02); frequentist coverage is CJ01; unbounded functionals are CJ27.

- The shipped $R' = 4$ carries only the fixed-R statements; (i) uses the optional stacking configuration, not the default four islands (Fidelity row; integration note 2).
- (h), (i) are "PROVED (by this seat, unchecked)"; MA-PropP is author-only; N9 is a candidate with a cross-family CHK (Self-checked row).

Open items for this card. Obligations: OB01 (N1, the crux; binding §13.1), OB02 (N2(e) GUARD-W stop-only coupling), OB14 (N14(c) IND-T extension for $R' > 4$; K84/N14(a) extra search), OB04 (finite-R' accuracy), OB05 (calibration contract), OB09 (N9, closed with amendments). Future work touching the card: CJ01, CJ16 (validated arithmetic), CJ27. Named implementation premises and audits [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:92]. HQ charter deliverables (verbatim):

Charter M1 · Theorem V/H.

• Goal: the §13.1 and §5.4 accuracy theorem for every reported bounded summary, with validity as clause (a).

• Statement: §S2 card M1, clauses (a)–(i).

• First lemma: N1 (measured-endpoint identity $\mathrm{E}[\hat{Z}_r\, H_r(h) \mid \mathcal{C}] = Z \cdot \pi(h)$) via the terminal-round prefix identity; then (e) as its corollary on $1\{S \neq S_0\}$.

• Lean target: T12Core `unbiased_of_step` with `hstep` for multinomial resampling + an invariant kernel; detailed balance of the four kernels; the finite inequality (f).

• Independent check (cross-family): N1 proof to PC1 (Codex Sol); the Lean file to the registrar; a $p \leq 8$ coded-grid replica of $\mathrm{E}[\hat{Z}_r\, H_r(h)]$ (10 seeds) by PC2.

• Fallback and stop: marginal PIPs only (locked identity) for (e)–(i); one HQ wave on N1.

• Escalation evidence: N1 refuted or unclosed after one wave (→ §13.1 unmet, HQ); GUARD-W or IND-T extension audit failing (→ (c),(d) revert to potential-run statements and N2 reopens).

4.3.2 TH02 · M2 · Theorem WS, the §5.3 package

TH02 · M2 · Theorem WS (the §5.3 package): (a) discrepancy; (b) bridge-form stability with Q5 content; (c) Proposition ONE-SHOT

APPROVED (F.B., 2026-11-07) Placement: main text, wave 2 (HQ charter `HQ_CHARTERS/M2.md`).
Status (registry, verbatim): OWS-WS3, OWS-WS4: PROVED (SF: none; XF: none) + CHK "VERIFIED (PC1, cross-family, line by line)". OWS-WS6d, OWS-WS7: PROVED (SF: none; XF: none) + CHK "VERIFIED WITH AMENDMENT". OWS-WS1: PROVED, AUTHOR-ONLY. FD-W1: PROVED, XF VERIFIED (A-REV4). AREV9-OWSTAB: "VERIFIED under H-ORTH exactly as stated" (not used beyond H-ORTH). Seat C's D_* decomposition and actual-grid moment: PROOF SKETCH (NEW-COMP); W1PC2/W2PC2 TRUE-AS-WRITTEN (statement sanity, not a proof). (b2)–(b4), (b6) assemblies: PROVED (by Seats A/B/D, unchecked), W2PC1 TRUE-AS-WRITTEN. C8 pilot extension: PROOF SKETCH, unchecked (W2PC2 PLAUSIBLE); T26BUD-G: PROVED, deterministic, AUTHOR-ONLY. (b7): PROVED (by this seat, unchecked). T03: PROVED-LOCKED; T27(i): PROVED-LOCKED, CONDITIONAL. OWS-WS6e: CONJECTURE; OWS-WS8d: OPEN (neither used).
K-IDs: K102, K10, K11, K25, K53; K103/K104 as used; K27 cited for the count.
Authoring seat / card source: Seat B (Fable 5.1), `R3_B.md:151–187` (contains Seat C's D_* decomposition and actual-grid moment).
Binding goal: proposal §5.3 (HQ Q1 (a)(b)(c) with Q5's two conditions, RL01, RL05); "Every fallback keeps (a), a stage-count clause and (c)" [NBH/02_theory.md:780-791].

TLF clerk text (verbatim). **Chair's integration note.** No change.

(B's leakage clauses (F), (S) are already outside this card; they are the supplement S-LEAK, R3-5. The §5.3 escalation safeguard of MINUTES §2.3 §2.7 matches the card's fallback row.)

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:147-163

Setting (model/design; everything computed from (X, y) and the built blocks). Standardised X, centred nonconstant y, $n \geq 3$, $p \geq 1$; shipped target ($g = n$, IG(a_0, b_0), capped BB(1,p), $K = \min(p, \lfloor n/2 \rfloor)$); blocks $C_1..C_J$ as built by the method from X; tables r_b from the same (X, y); $\Omega = \{|S| \leq K\}$; $d = \log BF - \Sigma_b \log r_b$; $q_t \propto \gamma_0 e^{td}$; $G = X^T X$, G_D its block-diagonal part, $E = G - G_D$, $A_D = G_D + I$, $H_D = (n+1)G_D + I$, $z = X^T y$, $\bar{Q} = 2b_0 + y^T y$, $\nu = a_0 + n/2$ (C:50, 63).

H-DOM (matrix domain). $a := n z^T H_D^{-1} z/\bar{Q} < 1$; $\delta_A := \|A_D^{-1/2} E A_D^{-1/2}\| < 1$; $\delta_H := \|(n+1)H_D^{-1/2} E H_D^{-1/2}\| < 1$; $r := \delta_H a/[(1-\delta_H)(1-a)] < 1$. $C_\sigma := \nu a^2/[2(1-a)]$; $b_X := (K/2)[-\log(1-\delta_A) - \log(1-\delta_H)] - \nu \log(1-r)$; $D_* := C_\sigma + 2b_X$; x_b, $d_\sigma(S) := \nu[\Sigma_b \log(1-x_b) - \log(1-\Sigma_b x_b)]$ (C:64–65).

Implementation premises (named, not model conditions). EX-A exact arithmetic; EX-S/H-U exact start sampler and ideal uniforms (for (b3), (b5)); CAL: a successful calibration record $\mathcal{C}$ frozen on streams 0–9; $R' = 4k$ declared before stream 10; kernels as coded under N9's sufficient conditions; for (b5)–(b6): ESS .9 for pilot and grid, 2^{-60} bisection, 10^{-12} snap/validation, a successful grid (no ceiling/snap failure), $2^{-60} \leq h/2$. No output clause, so GUARD-W is not used.

TH02.a1 **(a1) Decomposition.** Under H-DOM, for every $S \in \Omega$: $d(S) = d_\sigma(S) + d_X(S)$ with $0 \leq d_\sigma(S) \leq C_\sigma$ and $|d_X(S)| \leq b_X$; hence $\operatorname{osc}_\Omega d \leq D_*$. The σ^2-coupling term C_σ survives $E = 0$. [W2PC2 C2(a)·split TRUE-AS-WRITTEN, 104 domain designs; W1PC2 D14·deterministic bound; W2PC1 B.R3(a) TRUE-AS-WRITTEN]

TH02.a2 **(a2) Divergences.** Under H-DOM: $KL(\pi\|\pi_0) \leq D_*$, $KL(\pi_0\|\pi) \leq D_*$, $J := KL(\pi\|\pi_0) + KL(\pi_0\|\pi) = E_\pi d - E_{\pi_0} d \leq D_*$; $TV(\pi, \pi_0) \leq \tanh(\operatorname{osc} d/2)$; $1 + \chi^2(\pi\|\pi_0) \leq e^{\operatorname{osc} d}$. [W2PC2 C2(a)·divergences; W1PC1 §16 KL/TV rows TRUE-AS-WRITTEN]

TH02.a3 **(a3) Block-orthogonal case and per-model η-form.** If $E = 0$: $d = d_\sigma \geq 0$, zero iff at most one block has positive fit fraction; with known σ^2 (tables recomputed under known precision) $d \equiv 0$. For S with $A_S \succ 0$ and $\varepsilon_S < 1 - \bar{x}_S$: $|d_X(S)| \leq \eta_S := \nu\varepsilon_S\bar{x}_S/(1-\varepsilon_S-\bar{x}_S) + F_S/(4(1-\varepsilon_S))$. [CHK §3 WS3 VERIFIED; FD-W1; W1PC1 §5 zero case; CHK §4 WS4 VERIFIED; W2PC1 D.R3(a)]

TH02.b1 **(b1) Per-stage χ^2.** Under H-DOM, for $0 \leq u < u+\Delta \leq 1$: $\chi^2(q_{u+\Delta}\|q_u) \leq e^{\Delta D_*} - 1$. [W2PC2 C2(b)·increments TRUE-AS-WRITTEN; W1PC2 D14·one-shot/increment]

TH02.b2 **(b2) Four-normaliser transfer (same data, fixed partition).** With ${q^\sigma}_t \propto \gamma_0 e^{td_\sigma}$: $e^{-4\nu b_X} \leq [1+\chi^2(q_v\|q_u)]/[1+\chi^2(q^\sigma{}_v\|q^\sigma{}_u)] \leq e^{4\nu b_X}$, $0 \leq u < v \leq 1$. On any nonempty $W \subseteq \Omega$ with $\eta_W := \max_W \eta_S$: the same with $e^{\pm 4\nu\eta_W}$, and $TV({q_v}^W, {q_v}^{\sigma,W}) \leq 1 - e^{-2\nu\eta_W}$. [W2PC2 C2(b)·continuity; W1PC1 B.R3(b)(1)–(4); CHK §4 WS4(e)]

TH02.b3 **(b3) First stage.** Particles iid π_0: $1 + N_e \cdot \operatorname{Var}(\hat{Z}_{\text{stage }1}/(\bar{Z}_{\varphi_1}/\bar{Z}_0)) = 1 + \chi^2(q_{\varphi_1}\|\pi_0)$ exactly, and this quantity is within $e^{\pm 4\varphi_1\eta}$ of its block-orthogonal value. Transfer of $1 + N_e$ Var, never of Var. [W2PC1 B.R3(b)(1)–(4); A.R3(b)(i),(ii),(v); W2PC2 C2·variance transfer]

TH02.b4 **(b4) Doeblin constants.** $M_\varphi := \max_S q_\varphi(S)/\pi_0(S) \leq e^{2\varphi\eta} M^\sigma{}_\varphi$, so the coded round minorises q_φ with constant $\geq e^{-2\varphi\eta}/M^\sigma{}_\varphi$. [W2PC1 B.R3(b)(1)–(4); D.R3(b)(i),(ii),(iv)]

TH02.b5 **(b5) Pathwise island moment.** On every successful coded grid, with the kernels as coded: $W_r := \hat{Z}_r/Z \in [e^{-\operatorname{osc} d}, e^{\operatorname{osc} d}] \subseteq [e^{-D_*}, e^{D_*}]$ pathwise and $E[W_r \mid \mathcal{C}] = 1$, hence $M_2 := E[{W_r}^2 \mid \mathcal{C}] \leq e^{D_*}$. An upper bound only; no comparison to the block-orthogonal M_2. [W2PC2 C2(b)·weights; W1PC2 D14·actual-grid moment (1,600 islands, poor V included); W2PC1 B.R3(b)(5),(6); SYNTH §3 A.R3(b)(iv)]

TH02.b6 **(b6) Stage count: the sandwich.** For exact-frontier greedy ESS-.9 grids on a fixed common nonempty V with common pilot targets, same-data block-diagonal baseline d_σ on the fixed partition, and $.9e^{4\eta} < 1$: $T_{bd}(.9e^{-4\eta}) \leq T \leq T_{bd}(.9e^{4\eta})$; for the coded grid add the 2^{-60} bisection and 10^{-12} snap/validation qualifications. [W2PC1 B.R3(b)(1)–(4); A.R3(b)(iii)·sandwich; D.R3(b)(iii)·sandwich; E-pilot 16 cases, 0 violations]

TH02.b6p **(b6′) Local constancy, integer bound, envelope.** If the baseline is not at a critical threshold, $T = T_{bd}(.9)$ for all small ε. Under the pilot premises (global range bound D_*, $2^{-60} \leq$

h/2, h = $2c_\rho/D_*$, $c_\rho = \sqrt{\log(10/9)}$): $P \le \lceil 2/h \rceil + 1$ and $T \le 2\lceil D_*/c_\rho \rceil + 2 \le 4 + 2D_*/c_\rho$, a continuous envelope; $D_* = 0 \Rightarrow T = 1$. Checked fallback: $T \le P + \lceil D_*/c_\rho \rceil + 1$. [W2PC1 B.R3(b) continuity verdict (local constancy); W2PC2 C2(b)·stage count, C8·pilot extension, 600 adversarial runs]

TH02.b7 (b7) Q5 condition (1): continuity and Lipschitz statements in ε. On the fixed-partition, same-data family $E = \varepsilon E_1$ (E_1 a fixed cross-block pattern): $\delta_A(\varepsilon) = \varepsilon\delta_A(1) =: c_A\varepsilon$ and $\delta_H(\varepsilon) = c_H\varepsilon$ exactly. Hence on $[0, \varepsilon_1]$ with $c_A\varepsilon_1, c_H\varepsilon_1 < 1$ and $r(\varepsilon_1) < 1$, the functions $b_X(\varepsilon)$, $D_*(\varepsilon)$, $e^{\pm 4\nu b_X(\varepsilon)}$, $e^{\Delta D_*(\varepsilon)} - 1$, $e^{D_*(\varepsilon)}$ and the Doeblin factor are continuous with $b_X(0) = 0$ and Lipschitz with constant $L(\varepsilon_1) := (K/2)[c_A/(1-c_A\varepsilon_1) + c_H/(1-c_H\varepsilon_1)] + \nu c_H a/[(1-a)(1-c_H\varepsilon_1)^2(1-r(\varepsilon_1))]$ (for b_X; $e^{\cdot}$ factors by the chain rule). Per-stage χ^2: $|\log(1+\chi^2(q_v\|q_u))(\varepsilon) - (\cdot)(\varepsilon')| \le 4\nu\cdot\sup_S|d(S;\varepsilon) - d(S;\varepsilon')| \le 4\nu[b_X(\varepsilon) + b_X(\varepsilon')]$, so at the baseline the modulus is $4\nu L\varepsilon$. Status: PROVED (by this seat, unchecked), elementary calculus on the checked formulas; local check above, 0 violations.

TH02.b8 (b8) Q5 condition (2), verbatim in the theorem. "T(ε) is a step function of ε. It can jump at critical thresholds: PC1 exhibits an actual design (n = 40, p = 2, B = 1) with one-sided baseline counts {2, 4} and T = 3 at $\varepsilon = 10^{-5}$. 'Stable stage count' means the sandwich (b6), and nothing more." [W2PC1 D.R3(b)(iii)·jump corollary; E-jump]

TH02.c1 (c1) Zero-dependence obstruction (WS6(d) with CHK amendment). Under G (Gaussian noise, fixed $\sigma^2 > 0$), ORTH ($X^TX = (n-1)I$, $p \le n-1$), fixed nonempty S_0 with fixed nonzero coefficients, the shipped score/prior/state space, partition fixed in X, truth-free null count $m \to \infty$ with $\log m = O(\log n)$, $m \ge p/2$ and $p^{(\alpha-1)\bar{R}^2}/[\sqrt{n}(\log p)^{(1+(\alpha-1)\bar{R}^2)/2}] \to \infty$: $D_\alpha(\pi\|\pi_0) \to \infty$ in probability; for $\alpha = 2$, $\chi^2(\pi\|\pi_0) \to \infty$ and $ESS_\infty \to 0$, while in the bounded-block setting (shipped $B \le 16$) or assuming $\pi_0(S_0) \to 1$, $TV(\pi, \pi_0) \to 0$. Instance: $p = n-1$, one coefficient 2, $\sigma^2 = 1$. [CHK §1, §6 VERIFIED WITH AMENDMENT; W2PC1 B.R3(c)]

TH02.c2 (c2) Masked obstruction (WS7 with CHK amendments). If $\pi_0(S_0) = p^{-c}n^{-c/2}e^{O_P(1)}$ (T27(i)'s conclusion; H3 and the joint-ratio law are non-design premises, instantiated on LH) and $\pi(S_0) \to 1$: $ESS_\infty \le \pi_0(S_0)/\pi(S_0)^2 \to 0$, $KL(\pi\|\pi_0) \ge \pi(S_0)\log(1/\pi_0(S_0)) - \log 2 \to \infty$, $TV \to 1$; the initial population holds S_0 with probability $1 - (1-\pi_0(S_0))^N \le N\pi_0(S_0)$, later stages only "≤"; restricted-grid lower bounds with the set-mass ($\log \pi_0(V) = o_P(c \log p + (c/2)\log n)$) and singleton-endpoint amendments. Neither (c1) nor (c2) refutes the multistage sampler or (b). [CHK §2, §6; W2PC1 B.R3(c)]

Clause rows and their registry statuses. TH02.a1 (M2(a1) Decomposition): Seat C's D_* decomposition: PROOF SKETCH (NEW-COMP); W1PC2/W2PC2 TRUE-AS-WRITTEN (statement sanity, not a proof)

TH02.a2 (M2(a2) Divergences): as a1; W1PC1 §16 KL/TV rows TRUE-AS-WRITTEN

TH02.a3 (M2(a3) Block-orthogonal case and per-model η-form): OWS-WS3, OWS-WS4: PROVED (SF: none; XF: none) + CHK "VERIFIED (PC1, cross-family, line by line)"; FD-W1: PROVED, XF VERIFIED (A-REV4)

TH02.b1 (M2(b1) Per-stage χ^2): W2PC2 C2(b)·increments TRUE-AS-WRITTEN (statement sanity)

TH02.b2 (M2(b2) Four-normaliser transfer): (b2)–(b4), (b6) assemblies: PROVED (by Seats A/B/D, unchecked), W2PC1 TRUE-AS-WRITTEN

TH02.b3 (M2(b3) First stage): as b2

TH02.b4 (M2(b4) Doeblin constants): as b2

TH02.b5 (M2(b5) Pathwise island moment): Seat C's D_* decomposition and actual-grid moment: PROOF SKETCH (NEW-COMP); W1PC2/W2PC2 TRUE-AS-WRITTEN (statement sanity, not a proof)

TH02.b6 (M2(b6) Stage count: the sandwich): (b2)–(b4), (b6) assemblies: PROVED (by Seats A/B/D, unchecked), W2PC1 TRUE-AS-WRITTEN

TH02.b6p (M2(b6′) Local constancy, integer bound, envelope (b6′)): C8 pilot extension: PROOF SKETCH, unchecked (W2PC2 PLAUSIBLE); T26BUD-G: PROVED, deterministic, AUTHOR-ONLY

TH02.b7 (M2(b7) Q5 condition (1): continuity/Lipschitz in ε): PROVED (by this seat, unchecked), elementary calculus on the checked formulas

TH02.b8 (M2(b8) Q5 condition (2): step function statement): stated verbatim in the theorem (card row (b8)); required by HQ Q5 condition (2); no proof status needed

TH02.c1 (M2(c1) Zero-dependence obstruction (WS6(d) with CHK amendment)): OWS-WS6d: PROVED (SF: none; XF: none) + CHK ”VERIFIED WITH AMENDMENT”

TH02.c2 (M2(c2) Masked obstruction (WS7 with CHK amendments)): OWS-WS7: PROVED (SF: none; XF: none) + CHK ”VERIFIED WITH AMENDMENT”; T27(i): PROVED-LOCKED, CONDITIONAL

Remaining card fields (verbatim, TLF 143, 164–179). Card M2 · Theorem WS, the §5.3 package · K102, K10, K11, K25, K53; K103/K104 as used; K27 cited for the count; part (c) = Proposition ONE-SHOT (OWS-WS6(d), OWS-WS7)

Scale disclosure (not a theorem). Useful (O(1)) constants need signal energy $O(\sqrt{n})$ ($\nu a^2 = O(1)$) and $\varepsilon \lesssim n^{-1/2}/p$ on the exhibited family; at fixed ε, D_* grows ($534 \to 6.45\cdot10^4$ on W at ε = .01) and the domain fails. η-form constants are O(1) iff $\varepsilon = O(1/(n\bar{x}))$. A sharp necessity statement is FUTURE_WORK (all three seat forms COUNTEREXAMPLE; C’s is a range obstruction only). [W2PC2 C2·ε-scale; W2PC1 B.R3·scale; CHK §4 scale remark]

Removed from my R2 card. Prop NEC (W2PC1 B.R3 NEC COUNTEREXAMPLE); ”every constant in (b1)–(b5), including T, tends to its BO value” (replaced by (b6′), (b8)); the q_2 tilt-tail route (W1PC1 IMPLAUSIBLE); the m-form of (S) and (F) (to the leakage corollary, R3-5).

Guarantee (one sentence). On the method’s own partition, the gap between the block-product start and the posterior, and every bridge quantity (per-stage χ^2, island weight moments, Doeblin constants, and the stage count through its sandwich and envelope), is controlled by two data-computable terms, the σ^2-coupling term and the cross-block term, continuously in the cross-block correlation; and the one-shot importance-weight reading of §5.3 is false even at zero correlation.

Objective / goal / Song–Liang. Objective (1); objective (2) through (b5) (island count via $M_2 \le e^{D_*}$) and (b6′) (depth envelope); proposal §5.3 (binding; HQ Q1 (a)(b)(c), Q5); Song–Liang Thm 4/Cor 2 as ”specific start-distortion analysis, compared not exceeded” (d_σ quantifies the block-start distortion).

Status (verbatim, with qualifiers). OWS-WS3, OWS-WS4: PROVED (SF: none; XF: none) + CHK ”VERIFIED (PC1, cross-family, line by line)”. OWS-WS6d, OWS-WS7: PROVED (SF: none; XF: none) + CHK ”VERIFIED WITH AMENDMENT”. OWS-WS1: PROVED, AUTHOR-ONLY. FD-W1: PROVED, XF VERIFIED (A-REV4). AREV9-OWSTAB: ”VERIFIED under H-ORTH exactly as stated” (not used beyond H-ORTH). Seat C’s D_* decomposition and actual-grid moment: PROOF SKETCH (NEW-COMP); W1PC2/W2PC2 TRUE-AS-WRITTEN (statement sanity, not a proof). (b2)–(b4), (b6) assemblies: PROVED (by Seats A/B/D, unchecked), W2PC1 TRUE-AS-WRITTEN. C8 pilot extension: PROOF SKETCH, unchecked (W2PC2 PLAUSIBLE); T26BUD-G: PROVED, deterministic, AUTHOR-ONLY. (b7): PROVED (by this seat, unchecked). T03: PROVED-LOCKED; T27(i): PROVED-LOCKED, CONDITIONAL. OWS-WS6e: CONJECTURE; OWS-WS8d: OPEN (neither used).

Fidelity to v3.1 as shipped. Partition: the method’s own (verified computed on W). Kernels: as coded (b5). Grid: coded rule for (b5), (b6′); (b6)’s exact-frontier comparison is a labelled comparison object. (c) concerns the start/target laws, not the sampler. LS0 n/a; pooling n/a; stacking n/a.

Self-checked flag. Mixed: WS3/4/6d/7 cross-family checked (CHK); C’s D_* route PC2-checked (cross-family to C) but no filed proof; (b7) seat-only; A/B/D’s (b) assemblies statement-checked by PC1 only.

Novelty, citations checked. NEW-COMP. Nearest predecessor OWS-WS4 (internal; PC2 novelty note). Generic χ^2 precedent: Frazier et al. 2023, Assumption 6(iv) (via THM-INV §4b; not re-fetched by me, to the citation checker). Cérou–Del Moral–Guyader 2011 Thms 1.4–1.5 (variance theorems; not re-sold). Chatterjee–Diaconis 2018 Thms 1.1–1.3 (exp(KL) scale behind (c2)). Štefankovič–Vempala–Vigoda 2009 (greedy-schedule optimality behind (b6); LIT-SMC #9). Gelman–Meng 1998 (J identity). YWJ/ZYVR not touched.

Proof route and key lemma. (a1)–(a2): PC2 Appendix A’s four steps (exact split; $d_D \in [0, \nu x^2/(2(1-x))]$; eigenvalues of I + M in $[1-\delta, 1+\delta]$; resolvent bound $|t - t_D| \le \delta_H t_D/(1-\delta_H)$). **Key lemma:** restriction monotonicity of $z^T H^{-1} z$ and of relative operator norms. (b2): WS1(a) cumulant identity and $|\log Z_t - \log Z^{\sigma}{}_t| \le |t| b_X$. (b5): product of stage averages in $[e^{\Delta_t\, d_{\min}}, e^{\Delta_t\, d_{\max}}]$, $\Sigma\Delta_t = 1$, E W = 1 (T12/N9). (b6): feasibility inclusion plus constrained greedy optimality (Theorem G). (b6′): empirical ESS monotone in Δ (convex log-MGF), Popoviciu, bisection invariant. (b7): homogeneity and chain rule.

Difficulty. (a) M; (b1)–(b5), (b7) S; (b6)–(b6′) M (the pilot interface); (c) done (cited).

Lean core. Finite-space: WS1(a) cumulant identity; the four-normaliser bound given $\sup|d - d'| \le \eta$; the product-of-averages pathwise bound; the scalar log inequalities defining b_X and the Lipschitz algebra of (b7). Not Lean: (c) and the probability-in-the-design clauses.

Dependencies (card IDs). M3 (d_σ, WS3, FD-W1; T27(i) masking law for (c2)); M7 (grid rule, Theorem G count, K25 coverage identity for $V \neq \Omega$); M1 (T12/N9 E $W_r = 1$ for (b5)); S-KERNEL (N9 amendments); S-T (T03 concentration in (c1)); M6 (Lemma KL in (c2)); S-BLOCKS (blocks = groups on W); S-REP ($\hat{\eta}$, $\hat{\varepsilon}_\times$, $\hat{J}$ report the constants).

Risk. D_* route is a PROOF SKETCH: if the general matrix route stalls, prove on W directly (C's fallback). The useful scale is perturbative ($\varepsilon \lesssim n^{-1/2}/p$): disclosed. Pilot-extension envelope unchecked as a proof: the checked fallback count stands. CAL/EX-S premises.

Fallback statement and stop rule. If (a1) (D_*) fails: ship (a3) η-form (VERIFIED) with (b2)–(b4) in η_W form, (b5) with osc d, (b6) sandwich, (c); the scale shrinks to $\varepsilon = O(1/(n\bar{x}))$. If (b6′) fails: ship (b1)–(b6) plus the checked count $T \le P + \lceil D_*/c_\rho \rceil + 1$ and (b8). Every fallback keeps (a), a stage-count clause and (c) (PC1: omitting the count would not meet Q1(b)). If both the D_* route and the direct W-family route fail: escalate to HQ as unmet binding §5.3; never dropped (§2.7). Stop: one check wave per half; one HQ cycle for the pilot-extension lemma.

IC3 design, exhibited, where checked. W (SMC-dispatched): $p = 9\lfloor n^{1/3}/9 \rfloor$, groups of 9 at ¼, every cross correlation $\varepsilon_n = 1/(np) > 0$, two $n^{-1/4}$ signals in two groups, $\sigma^2 = 1$, $n \ge 9^6$; own partition computed, blocks = groups for $n \in \{9^6, 10^6, 2{\cdot}10^7, 10^8\}$, H-DOM 10/10 seeds per n, D_* median 1.08 → 1.04, $b_X \approx 5{\cdot}10^{-4}$ to $3{\cdot}10^{-5}$ (W2PC2 §0 item 3, `w_family.py`). **W at $\varepsilon = n^{-1/2}/p$**: $b_X \approx 0.35$, $D_* \approx 1.73$–1.90, blocks = groups, domain 10/10 (W2PC2 §0 item 3). Component checks (exact branch dispatches): Dε, n = 40, p = 8, B = 2 (W1PC1 "Dε"; my R2 check 1, 40/40; W2PC1 B.R3·scale); A's n = 64, p = 16, B = 4. (c1): CHK §6 example $X^TX = (n-1)I$, p = n−1, coefficient 2. (c2): LH (W1PC2 shared LH).

Numerical companion (≤ 20 seeds). Existing: W2PC2 `r3_clauses.py` (104 domain designs, 0 violations), `w_family.py` (10 seeds per n), `c8_pilot.py`, W1PC2 `smc_coded.py` (1,600 islands), W1PC1 E-pilot (16 cases), my R2 check 1 (10 seeds × 4 ε), today's Lipschitz check (5 seeds, 75 pairs). For the HQ: ≤ 20 seeds on W at $n = 9^6$, reporting (a, δ_A, δ_H, r, b_X, D_*), the sandwich at $\varepsilon \in \{1/(np), n^{-1/2}/p\}$, M_2 on four coded islands, and $L(\varepsilon_1)$ against the observed b_X increments.

Plain reading

On the method's own partition and inside the matrix domain H-DOM (every constant computable from (X, y) and the built blocks), the log-discrepancy d between posterior and block-product start splits into a σ^2-coupling part bounded by C_σ (present even with orthogonal blocks) and a cross-block part bounded by b_X, so $\mathrm{osc}_\Omega\, d \le D_*$ (a1). This one constant then bounds KL/TV/χ^2 between start and posterior (a2), every per-stage χ^2 of the bridge (b1), the island-weight second moment $M_2 \le e^{D_*}$ (b5), Doeblin constants (b4) and the stage count (sandwich (b6), envelope (b6′)); along a one-parameter family $E = \varepsilon E_1$ these bounds are continuous and Lipschitz in ε (b7), while the integer stage count itself is a step function of ε (b8). Part (c): the one-shot reading of §5.3 is false, already at zero cross-block correlation (c1: under G, ORTH and the stated null-count growth) and on masked designs (c2: conditional on T27(i)'s start-mass law and $\pi(S_0) \to 1$, instantiated on LH). Parts (a),(b) are fixed-data statements under H-DOM and the named implementation premises ((b5) on a successful coded grid, (b6) for exact-frontier greedy grids with $.9e^{4\eta} < 1$); (c) is asymptotic.

Not claimed / not proved

- The general D_* route (a1) and the actual-grid moment (b5) are "PROOF SKETCH (NEW-COMP)"; W1PC2/W2PC2 TRUE-AS-WRITTEN are "statement sanity, not a proof". Fallback: prove on W directly, or ship the η-form (a3) (Fallback row).
- (b6′)'s pilot-extension envelope is "PROOF SKETCH, unchecked"; the checked fallback count $T \le P + \lceil D_*/c_\rho \rceil + 1$ stands.
- Useful (O(1)) constants need $\varepsilon \lesssim n^{-1/2}/p$ on W; at fixed ε, D_* grows and the domain fails (Scale disclosure). A sharp necessity statement is CJ22 (FW22); all three seat forms of Prop NEC were COUNTEREXAMPLE.
- (b5) is an upper bound only, with no comparison to the block-orthogonal M_2; masked M_2 / two-sided masked evidence precision is CJ24.

- (c1), (c2) do not refute the multistage sampler or (b) (row (c2)). OWS-WS6e (CONJECTURE) and OWS-WS8d (OPEN) are not used.

Open items for this card. The D_* decomposition proof (first lemma of the charter), the pilot-extension lemma behind (b6′), CJ22, CJ24; reporting of the constants via TH19 (S-REP). Named premises: CAL/EX-S, EX-A; "No output clause, so GUARD-W is not used" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M2.md:95]. HQ charter deliverables (verbatim):

Charter M2 · Theorem WS (§5.3 package).

• Goal: ship HQ Q1 (a), (b), (c) with Q5's two conditions, on the method's own partition, with every constant data-computable and reported (S-REP).

• Statement: card M2 above, rows (a1)–(c2) plus (b7), (b8).

• First lemma to attack: C's D_* decomposition (a1) via the restriction-monotonicity lemma (PC2 Appendix A steps 2–4), written as a filed proof; in parallel the pilot-progress lemma behind (b6′) (ESS monotone in Δ, Popoviciu, bisection invariant).

• Lean target: WS1(a) cumulant identity; four-normaliser bound under $\sup|d - d'| \le \eta$; pathwise product-of-averages bound; the scalar log inequalities of b_X and $L(\varepsilon_1)$.

• Independent-check plan (cross-family): (a1) proof by an Astra-family prover, checked by Opus/Fable; (b2)–(b7) proof by a Claude-family prover, checked by gpt-6.1-sol; (c) already CHK-verified, reread only for wording. Numerical: ≤ 20 seeds on W as in the card.

• Fallback and stop: as in the card; one check wave per half; one HQ cycle for the pilot-extension lemma, then the checked count fallback.

• Escalation evidence: a counterexample to (a1) on the matrix domain *and* failure of the direct proof on W; or a design in the domain with a successful coded grid violating (b5). Either reopens §5.3 at HQ as unmet (never dropped).

4.3.3 TH03 · M3 · Theorem START

TH03 · M3 · Theorem START

APPROVED (F.B., 2026-11-07) Placement: main text, wave 1 (HQ charter HQ_CHARTERS/M3.md).
Status (registry, verbatim): Status: deterministic: T22(i)(ii)(iv): **PROVED-LOCKED**. DIL-E: **PROVED; SF: none; XF: VERIFIED (A-REV9)**. DIL-E1: **PROVED; SF: none; XF: VERIFIED* (A-REV9)**. Status: statistical: T27(i): **PROVED-LOCKED, CONDITIONAL**. Thm3/4-EXP: **PROVED; SF: VERIFIED (W3-VEXP); XF: VERIFIED (A-REV2)**. Thm5-LH: **PROVED; SF: VERIFIED (W3-VEXP); XF: none; SAME-FAMILY-ONLY**. Status: coupling/assembly: OWS-WS3: **PROVED; SF: none; XF: none; AUTHOR-ONLY** in the inherited ledger; CHK separately records **VERIFIED (PC1, cross-family, line by line)**. Final assembly: **PROOF SKETCH, unchecked**.
K-IDs: K15, K17, K18, K10 (K16 supplementary conditional lemma).
Authoring seat / card source: Seat C (gpt-6-astra), R3_C.md:144–177.
Binding goal: none rests on M3 alone; escalation evidence: "A failed LH premise affecting M5/M6 or the retained masked requirement; any failure leaving Q1(c) without its independently checked obstruction" [NBH/02_theory.md:847-858].

TLF clerk text (verbatim). **Chair's integration note.** No change. (K16 stays a supplementary conditional lemma, as the card states.)

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:198-212

Hypotheses. D0 and EX-A. Deterministic clauses allow any fixed partition with $|C_b| \le K$ and the stated positive tables. Score-coupling clauses require production tables. The masked clause

additionally uses G on the exhibited LH family.

Implementation premises. Tables, size cap and prior are those specified above. No calibration, random-number, LS0, pooling or delivery premise is needed for the mathematical start-law claims. Actual execution fidelity remains distinct from these identities.

AGG definitions I. For feasible reference S°, $s = |S^\circ|$, $U_b^* = S^\circ \cap C_b$, define $a_{bj} = \sum_{U \supseteq U_b^*, |U \setminus U_b^*| = j} r_b(U)/r_b(U_b^*)$, $D_b = \max_{j \geq 1} (j! a_{bj})^{1/j}$, empty maximum zero.

AGG definitions II. Set $q = (p - s)^{-1} \sum_b D_b$ for $s < p$, otherwise zero. Set $e = 2p \max_{b,U:\, a = |U_b^* \setminus U| > 0} [r_b(U)/r_b(U \cup U_b^*)]^{1/a}$, empty maximum zero.

TH03.a **M3(a): deterministic mass.** If $q < 1$, $\pi_0(S^\circ) \geq (1 - q)^{s+1}/(1 + e)^s$. For feasible S_0, $\pi_0(S_0)/\pi_0(S^\circ) = [w(|S_0|)/w(s)] \prod_b r_b(S_{0b})/r_b(U_b^*)$. **Evidence:** W1P1 §5, A.C8(a–e)/B.D4/D.M6; source T22(i),(ii).

TH03.b **M3(b): bridge mass.** For feasible S_0, $\inf_{0 \leq t \leq 1} q_t(S_0) \geq \min\{\pi_0(S_0), \pi(S_0)\}$. This is a finite-law mass bound, not a particle-discovery assertion. **Evidence:** same W1P1 §5 row; T22(iv).

TH03.c **M3(c): exact factorization.** $B_b = \sum_{U \subseteq C_b} r_b(U)$, $q_b(U) = r_b(U)/B_b$, $A(k) = \varpi(k) \prod_b B_b/\bar{Z}_0$. Then $\pi_0(S) = A(|S|) \prod_b q_b(S_b)$. **Evidence:** W1P2 cluster 5, "D2 · exact factorisation"; W2P2 §2, C5.

Dilution definitions. For nonempty S_{0b}, let $\lambda_b = \log r_b(S_{0b})$, $\eta_b = [B_b - 1 - r_b(S_{0b})]/[1 + r_b(S_{0b})]$. For empty S_{0b}, let $\eta_b = B_b - 1$. Production tables have $r_b(\emptyset) = 1$, so $\eta_b \geq 0$.

TH03.d **M3(d): dilution.** $V = \sum_{b: S_{0b} \neq \emptyset} \log(1 + e^{-\lambda_b})$, $C = \sum_b \log(1 + \eta_b)$. Then $-\log \pi_0(S_0) = V + C - \log A(|S_0|)$. **Evidence:** W1P1 §5, dilution clause; DIL-E/E1, here at tilt $t = 1$. No sign claim for $-\log A$.

Coupling definitions. Let $\bar{Q} = .02 + y^\top y$, $x_b = n z_{S_b}^\top H_{S_b}^{-1} z_{S_b}/\bar{Q}$, $x = \sum_b x_b$. Define $d_\sigma(S) = \nu[\sum_b \log(1 - x_b) - \log(1 - x)]$ when $x < 1$, and $d_X = d - d_\sigma$.

TH03.e **M3(e): shared variance.** Under block orthogonality, $x < 1$, $d_X = 0$, and $0 \leq d_\sigma \leq \nu \sum_{b<c} x_b x_c/(1-x)$. Equality $d_\sigma = 0$ holds exactly when at most one block has positive fit fraction. **Evidence:** CHK §3, WS3(b),(c); W2P1 D.M6.

Correlated extension interface. On M2's matrix domain, the same split satisfies $0 \leq d_\sigma \leq C_\sigma$, $|d_X| \leq b_X$. The quantitative perturbation theorem belongs to M2. **Evidence:** W2P2 §1, "C2(a) · split".

TH03.f **M3(f): masked law.** Under G on LH, $\log \pi_0(S_0) + 2 \log p + \log n = O_P(1)$, whereas $\pi(S_0) \to 1$. **Evidence:** W1P2 cluster 5, "D2 · LH masking law"; W2P2 §2, C5. Posterior concentration uses S-T.

Conditional-law safeguards. T27(i) retains a fixed feasible completion chain, jointly two-sided tight successive log-table ratios and a normalized reference-mass lower bound. LH supplies the application; cancellation alone does not discharge these premises on another design.

TH03.K16 **K16 placement.** Cite T22(iii), **PROVED-LOCKED CONDITIONAL**, with G, R, within-block SE, EXP′, VIS, NL_η, $\mathrm{NL}_{\mathrm{avg}}(\kappa)$ retained. Its favourable rate is a separate regime. It is not an additional unconditional assertion in M3.

Statement (verbatim from source) Docs/Theory/proofs/T27i_T20C_cancellation_budget/STATEMENT.md:26-49

T27(i): sequential cancellation mass law (TH119; locked pre-campaign statement; the input of M3(f) and M2(c2)).
Every premise. H1. **Fixed feasible completion chain.** Fix integers $s \geq 0$ and $c \geq 1$. There is a specified data-measurable chain $S^{(0)} \subset \cdots \subset S^{(c)} = S_0$ with $|S^{(0)}| = s$ and $S^{(i+1)} = S^{(i)} \cup \{l_i\}$, $l_i \notin S^{(i)}$, for $i = 0, \ldots, c - 1$. Every state is feasible and $|S_0| = s + c \leq K$ eventually. Deterministic chains are included. Let b_i be the block of l_i and $T_i = S^{(i)} \cap C_{b_i}$.

H2. **Joint sequential ratio law, assumed statistical input.** For this specified chain,

$$E_{n,i} = \log\left[r_{b_i}(T_i \cup \{l_i\})/r_{b_i}(T_i)\right] + \frac{1}{2}\log n, \qquad \max_{0\le i<c} |E_{n,i}| = O_{P_0}(1).$$

This is two-sided log tightness jointly at the successive states. A data-selected chain must itself satisfy this premise; no uniform guarantee over all selection rules is asserted. The remainders need not be independent. Cancellation at an initial table state alone does not assert H2.
H3. **Normalized reference-mass lower bound, assumed statistical input.** A deterministic constant $c_0 \in (0, 1]$ satisfies $P_0\{\pi_0(S^{(0)}) \ge c_0\} \to 1$. This premise includes the full capped composite normalizer. A lower bound on an unnormalized table product does not replace it.
H4. **Retained operational assumptions.** EX-A: exact scores, convolutions, probabilities, comparisons. EX-S: independent ideal uniform streams with the method's stream separation. CAL: calibration uses streams 0-9 only, final streams 10-13 are fresh and independent of calibration, and escalation precedes final randomness. Successful final execution: the final run completes its frozen schedule. These are retained as the theorem file's operational setting; the mass implication uses the exact start definition and H1-H3 and is independent of final-run randomness. No probability-of-completion assertion is added. In particular no stochastic-order statement conditioned on completion or diagnostics is claimed.
The scaling p asymptotic to n^a is retained, although the mass implication itself only needs $p \ge s+c$ and fixed s, c. G/R, Gaussianity, eigenvalue or beta-min assumptions are not silently used to derive H2 or H3.
Exact conclusion.

$$\log \pi_0(S_0) + c \log p + (c/2) \log n = O_{P_0}(1),$$

equivalently $\pi_0(S_0) = p^{-c} n^{-c/2} \exp\{O_{P_0}(1)\}$. Here $O_{P_0}(1)$ means tightness: for each $\epsilon > 0$ a finite M and N exist such that $P_0\{|Z_n| > M\} < \epsilon$ for $n \ge N$. The positive mass multiplier and its reciprocal are both tight.
Empty-chain extension: if $c = 0$, $S^{(0)} = S_0$, H2 is vacuous (empty maximum zero), and the conclusion is $\log \pi_0(S_0) = O_{P_0}(1)$, with empty products one and empty sums zero.

Status of the locked statement: "H2 and H3 are statistical inputs, not proved" [Docs/Theory/proofs/LOCKED_PROOFS.md:45], [Docs/Theory/proofs/LOCKED_PROOFS.md:93]; proof file `Docs/Theory/proofs/T27i_T20C_cancellation_budget/PROOF_v3.tex`.

Clause rows and their registry statuses. TH03.a (M3(a) deterministic mass): T22(i)(ii)(iv): PROVED-LOCKED

TH03.b (M3(b) bridge mass): T22(iv): PROVED-LOCKED

TH03.c (M3(c) exact factorization): Final assembly: PROOF SKETCH, unchecked

TH03.d (M3(d) dilution): DIL-E: PROVED; SF: none; XF: VERIFIED (A-REV9). DIL-E1: PROVED; SF: none; XF: VERIFIED* (A-REV9)

TH03.e (M3(e) shared variance): OWS-WS3: PROVED; SF: none; XF: none; AUTHOR-ONLY in the inherited ledger; CHK separately records VERIFIED (PC1, cross-family, line by line)

TH03.f (M3(f) masked law): T27(i): PROVED-LOCKED, CONDITIONAL. Thm3/4-EXP: PROVED; SF: VERIFIED (W3-VEXP); XF: VERIFIED (A-REV2). Thm5-LH: PROVED; SF: VERIFIED (W3-VEXP); XF: none; SAME-FAMILY-ONLY

TH03.K16 (K16 placement (supplementary conditional lemma)): T22(iii), PROVED-LOCKED CONDITIONAL

Remaining card fields (verbatim, TLF 197, 213–226). **IDs. M3: K15, K17, K18, K10**; K16 is an attributed supplementary conditional lemma.

Guarantee. The shipped start has an exact mass decomposition and can assign vanishing mass to truth on a correlated family whose full posterior concentrates on truth.

Goals. Objectives (1), (3); proposal §5.3 discrepancy and §5.4 start approximation; Song–Liang Thm 4 comparison only, plus statistical context for Thm S. No identification of π_0 with Song–Liang's final output.

Status: deterministic. T22(i)(ii)(iv): **PROVED-LOCKED**. DIL-E: **PROVED; SF: none; XF: VERIFIED (A-REV9)**. DIL-E1: **PROVED; SF: none; XF: VERIFIED* (A-REV9)**.

Status: statistical. T27(i): **PROVED-LOCKED, CONDITIONAL**. Thm3/4-EXP: **PROVED; SF: VERIFIED (W3-VEXP); XF: VERIFIED (A-REV2)**. Thm5-LH: **PROVED; SF: VERIFIED (W3-VEXP); XF: none; SAME-FAMILY-ONLY**.

Status: coupling/assembly. OWS-WS3: **PROVED; SF: none; XF: none; AUTHOR-ONLY** in the inherited ledger; CHK separately records **VERIFIED (PC1, cross-family, line by line)**. Final assembly: **PROOF SKETCH, unchecked**.

Fidelity/self-check flag. Actual v3.1 target, tables, cap and partition. No altered start or known-variance substitution. Not self-check-only at the ingredient level; Thm5-LH's qualification and the absence of a full assembly proof check remain explicit.

Novelty/citation. NEW-COMP, centered on this start's masking and shared-variance distortion. Block computation is established precedent: Papaspiliopoulos–Rossell (2017), §§3.1–3.2 (https://pmc.ncbi.nlm.nih.gov/articles/PMC5975653/). DIL-E/E1 and OWS-WS3 are the internal predecessors.

Proof route/key lemma. Import T22 and DIL identities; match production normalization. For LH, verify actual partition equality, then import the joint ratio/reference-mass application of Thm5-LH and S-T. Shared-variance algebra is CHK WS3.

Difficulty/Lean core. Moderate assembly; substantial qualification audit. Lean: finite products, prior ratios, logarithmic dilution identity, endpoint Hölder inequality and scalar shared-variance bounds. Gaussian asymptotics remain cited prose.

Dependencies. S-TARGET; S-BLOCKS; S-T for $\pi(S_0) \to 1$. The M2 perturbation clause is a cross-reference, not a prerequisite for the BO or LH clauses.

Risk. Losing normalized reference mass; treating the capped start as independent blocks; confusing sample and population cancellation; exporting exact-Gram results to adaptive random packing.

Fallback/stop rule. Retain deterministic mass, factorization, dilution and BO coupling if the LH application fails its full check. After one proof-and-repair cycle, move the failed statistical extension to FUTURE_WORK. Notify M5/M6 and HQ of any resulting loss of required masked accuracy or Q1(c) coverage.

IC3 exhibit/check. LH above supplies genuine cross-block correlation and the masked conclusion; W1P2 shared-family/cluster-5 rows and W2P2 C5/C11 verify the stated scope. BO components allow correlated columns within blocks; CHK §3 checks them.

Numerical companion. None newly run. Proposed companion: at most ten Gaussian seeds; compute exact start mass by block convolution on LH and compare its log scaling. Use small enumerable component cases for factorization; label forced/component dispatch explicitly.

Plain reading

The start π_0 (prior times block tables, normalised) factorises exactly into a size factor $A(|S|)$ times independent within-block laws q_b (c); for a feasible reference model with $q < 1$ its mass has a deterministic lower bound (a), and along the tempering bridge the mass of a feasible S_0 never drops below $\min\{\pi_0(S_0), \pi(S_0)\}$ (b). The start's deficit at the truth decomposes exactly into a "dilution" part V, a competitor part C and a size term (d). With orthogonal blocks the start/posterior discrepancy is a nonnegative shared-variance term (e). On the explicit correlated family LH (exact Gram), under the sampling model G, $\log \pi_0(S_0) + 2\log p + \log n = O_P(1)$, i.e. the start puts mass $p^{-2}n^{-1}e^{O_P(1)}$ on S_0, while $\pi(S_0) \to 1$ (via S-T) (f): the start is "masked". The statistical inputs are T27(i) (PROVED-LOCKED, CONDITIONAL) and Thm5-LH (SAME-FAMILY-ONLY); the final assembly is a proof sketch.

Not claimed / not proved

- Final assembly: "PROOF SKETCH, unchecked"; Thm5-LH (the LH input) is SAME-FAMILY-ONLY; OWS-WS3 is AUTHOR-ONLY in the inherited ledger (CHK separately VERIFIED).
- The masking law (f) is for LH with exact Gram; "exporting exact-Gram results to adaptive

random packing" is a named risk; the study generator (groups of 5) is not LH (CJ25, OB08); masking as an inherent limitation is scoped to its design class (LIM03).

- K16 (T22(iii)) is a separate favourable-rate regime, "PROVED-LOCKED CONDITIONAL", and "not an additional unconditional assertion in M3".
- No sign claim for $-\log A$ (row (d)); the M2 perturbation clause is a cross-reference, not part of M3. Start-mode consistency under orthogonal blocks is CJ03.

Open items for this card. Full assembly check; LH normalized reference mass and joint sequential-ratio premises (first lemma); CJ03, CJ25. HQ charter deliverables (verbatim):

M3 charter. Instruction

Goal. Deliver exact start mass/decomposition and the explicit correlated masking law with production fidelity.

Statement. Prove §S2 M3(a–f), preserving the separate conditional K16 citation.

First lemma. Match production $\gamma_0, \bar{Z}_0$ to L-DIL; then audit LH's normalized reference mass and joint sequential ratio premises.

Lean target. Finite factorization, prior odds, dilution identity and endpoint mass inequality.

Independent check. Opus-family checker reviews Astra's assembly, especially cap normalization, actual partition, standardization and Thm5-LH's SAME-FAMILY-ONLY input. Check Gaussian inputs against original statements, not simulations alone.

Fallback/stop. One complete proof-and-repair cycle; retain deterministic/BO clauses if the statistical application fails. Record the failed extension as FUTURE_WORK.

Escalation evidence. A failed LH premise affecting M5/M6 or the retained masked requirement; any failure leaving Q1(c) without its independently checked obstruction.

Numerical companion. At most ten seeds; exact convolution and small enumeration as specified in M3.

4.3.4 TH04 · M4 · Theorem Z$\mathcal{T}$ and the certificate card, with the masked-regime statement

TH04 · M4 · Theorem Z$\mathcal{T}$ and the certificate card, with the masked-regime statement (HQ Q3)

APPROVED (F.B., 2026-11-07) Placement: main text, wave 3 (HQ charter HQ_CHARTERS/M4.md).
Status (registry, verbatim): MZ-P3: "PROVED (P3 top store, deterministic); P3' conditional on T03 + discovery"; "candidate: not reviewed, not adopted"; AUTHOR-ONLY. N13: PROVED (PC1 claim 5). PC1c5: (a) PROVED, ½ sharp. N17: PROVED (PC1 claim 1; PC2 C1(c)). PC1c2: PROVED for the potential run; N6(a) output form CORRECT WITH AMENDMENT; N6(b) PROVED modulo premise (i) (GUARD-W not audited). LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)). CEX-guard: COUNTEREXAMPLE (XF VERIFIED*). OCERT-C1: XF ERROR as quantified (O-VCERT R1 wording used). T38-b1-LH: PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8)). Thm5-LH: PROVED (SF: VERIFIED (W3-VEXP); XF: none), SAME-FAMILY-ONLY. T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)). (f) assembly: PROOF SKETCH (seat-assembled; W2PC1/W2PC2 TRUE-AS-WRITTEN as an assembly).
K-IDs: K42, K43, K45, K44; K36, K78 (output-event clauses); K95 cited from S-TARGET.
Authoring seat / card source: Seat B (Fable 5.1), R3_B.md:189–216 (Seat D's "M4(e)" = final (f)).
Binding goal: proposal §5.4 masked-regime accuracy (HQ Q3, RL03: "at least one masked-regime accuracy statement that paper 1 actually proves"; it is clause (f)); loss without approved replacement escalates [NBH/02_theory.md:910-921].

TLF clerk text (verbatim). 1. K95 (the odds certificate, $\eta < 1$) lives in S-TARGET (R3-4, unanimous); M4 cites it. No clause of this card changes.

2. The masked statement is clause (f) of this card (Seat D's lettering "M4(e)" in R3_D is the same statement). Per MINUTES §2.3 §2.7 and R3_C §S5, loss of the masked statement without an approved replacement **escalates**; the card's fallback row says HQ "is told … (never silently)", which is to be read as an escalation.

3. N2(e) GUARD-W, the LS0 retention rule/OP-LS0 integration and the N3 DP invariant are named implementation premises of this card (MINUTES §2.3, "Audit obligations").

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:247-255

Setting (model/design). Finite Ω, shipped target $\pi(S) = \varpi(S)m(S)/Z$; $\mathcal{T} \subseteq \Omega$ the nonempty deduplicated reported store, $Z_{\mathcal{T}} := \Sigma_{\mathcal{T}}\, \varpi m$, $\pi_{\mathcal{T}} := \pi(\cdot|\mathcal{T})$; $\delta_1 := 1 - \pi(S_0)$; h: $\Omega \to [0, 1]$ (or bounded, scaled by $\|h\|_\infty$); $\mathcal{S}$ the set of scored models; $\mathcal{T}_{\text{new}} = \text{top}_c(\mathcal{T}_{\text{old}} \cup \mathcal{S}_{\text{added}})$, $c \geq 1$ (the LS0 union-then-top-c rule, METHOD #9). For (f): the LH family (W1PC2 shared LH): $p = 9\lfloor n^{1/3}/9 \rfloor$, groups of 9 at .4, two disjoint hub links .2, coefficients (1, −.2), $\sigma^2 = 1$, exact Gram, $n \geq 9^6$, SIZE $9 \leq B \leq 16 < 18$; G; T03♯'s hypotheses hold on LH (Thm5-LH, SAME-FAMILY-ONLY).

Implementation premises (named). EX-A (exact arithmetic; the card is "numerical" until SC + VAL-FP, N16); SR (all scored candidates enter the winner comparison; union-then-top-c storage); Lemma S and the inheritance merge (PC2 code facts); for (e): Lemma RO + GUARD-W (stop-only coupling, audit N2(e) open), CAL, EX-S, $\hat{Z} = Z$ on the exact branch, $E[\hat{Z}_r \mid \mathcal{C}] = Z$; for (f): OP-LS0 (completed scans), the e4 centre rule (tie-aware DP invariant N3 OPEN, carried as hypothesis), $\ell \geq \max\{\Phi(S^\circ) - 1, 0\}$ (= 1 on LH; shipped ℓ suffices).

TH04.a (a) Deterministic identities. $Z_{\mathcal{T}} \leq Z$; $\pi(S) \leq \varpi(S)m(S)/Z_{\mathcal{T}}$ for every S; $TV(\pi_{\mathcal{T}}, \pi) = 1 - Z_{\mathcal{T}}/Z$; $KL(\pi_{\mathcal{T}}\|\pi) = \log(Z/Z_{\mathcal{T}})$; $KL(\pi\|\pi_{\mathcal{T}}) = \infty$ iff $\pi(\mathcal{T}^c) > 0$; $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$. [W1PC1 §3 A.C7/B.D3(a–d) TRUE-AS-WRITTEN; W1PC1 §16 reverse KL repaired with $\pi(\mathcal{T}^c) > 0$; W1PC2 D5·TV certificate, D5·underestimation; W2PC1 A.M5, B.M5]

TH04.b (b) Brackets when the truth is stored. If $S_0 \in \mathcal{T}$: $1 - \delta_1 \leq Z_{\mathcal{T}}/Z \leq 1$; $TV(\pi_{\mathcal{T}}, \pi) \leq \delta_1$; $|PIP^{\mathcal{T}}{}_j - PIP_j| \leq \delta_1$ for all j; $|\pi_{\mathcal{T}}(h) - \pi(h)| \leq 2\delta_1\|h\|_\infty$; joint inclusions likewise. [W1PC1 §3 B.D3(b) COUNTEREXAMPLE repaired with $\|h\|_\infty$; W2PC1 B.M5 TRUE-AS-WRITTEN]

TH04.c (c) Retention (N13). If $S_0 \in \mathcal{S}$ and $\delta_1 < ½$ then $S_0 \in \mathcal{T}$ and $MPM^{\mathcal{T}} = S_0$; ½ is sharp; holds under LS0 with $\mathcal{T}_{\text{new}} = \text{top}_c(\mathcal{T}_{\text{old}} \cup \mathcal{S}_{\text{added}})$. Unqualified retention is false. [N13 PROVED (PC1 claim 5); W1PC1 §3; W2PC2 C4·retention TRUE-AS-WRITTEN; W2PC1 B.M5]

TH04.d (d) Analytic envelope (N17). $m(S) \leq U_m := C_n(2b_0 + y^T y/(n+1))^{-\nu}$ for every S; hence $Z \leq U := Z_{\mathcal{T}} + U_m[1 - \Sigma_{\mathcal{T}}\, \varpi]$ and $TV(\pi_{\mathcal{T}}, \pi) \leq 1 - Z_{\mathcal{T}}/U$, a computable bound (loose, reported as such). [N17 PROVED (PC1 claim 1; PC2 C1(c)); W1PC2 D5·envelope, 40 designs × 5 stores; W2PC1 A.M5]

TH04.e (e) Output-event certificate. Under Lemma RO + GUARD-W: $P(\mathcal{O} \cap \{\max(Z_{\mathcal{T}}, 0.05\hat{Z}) > Z\}) \leq 0.05 \cdot P(\text{SMC} \cap \mathcal{A}) \leq 0.05$; island-minimum bound $0.05^{1/R'} \min_r \hat{Z}_r$ at the same level; card level $P(\mathcal{O} \cap \{\text{some G2/L/R item false}\}) \leq 0.05$. Never conditional on completion: $P(B \mid \mathcal{O}, \mathcal{C}) \leq \alpha$ does not follow (CEX-guard). [W1PC1 §3 A.C3(a,b,c)/B.D3(e) TRUE-AS-WRITTEN; W1PC2 D5·LCB, D12; W2PC1 B.M5]

TH04.f (f) Masked-regime statement (HQ Q3). On LH under M5's hypotheses (b1-LH): for every $\varepsilon > 0$, $P(\mathcal{O} \cap \{TV(\pi_{\mathcal{T}}, \pi) > \varepsilon\}) \to 0$; $P(\mathcal{O} \cap \{\max_j |PIP^{\mathcal{T}}{}_j - PIP_j| > 1 - \pi(S_0)\}) \leq P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) + P(\pi(S_0) \leq ½) \to 0$; $P(\mathcal{O} \cap \{MPM^{\mathcal{T}} \neq S_0\}) \to 0$; and surely $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$. Reading: vanishing P(output and error); not conditional success, and not $P(\mathcal{O}) \to 1$. Non-vacuous on a masked family: $\pi_0(S_0) = p^{-2}n^{-1}e^{O_P(1)} \to 0$ while $\pi(S_0) \to 1$. [W2PC1 B.R4·Z$\mathcal{T}$/CARD TRUE-AS-WRITTEN; D.R4; W2PC2 §3 §R4 named guarantee TRUE-AS-WRITTEN (assembly)]

TH04.g (g) Definitional facts. For a proper store ($\pi(\mathcal{T}^c) > 0$), $KL(\pi\|\pi_{\mathcal{T}}) = \infty$; any positive-mass state absent from the records makes $KL(\pi\|\hat{\pi}) = \infty$, surely when $|\Omega| > 4N_eR'$. The masked final-estimate TV/KL bound is FW2. [W1PC1 §16; W2PC2 C15-KL]

Clause rows and their registry statuses. TH04.a (M4(a) Deterministic identities): MZ-P3: ”PROVED (P3 top store, deterministic); P3′ conditional on T03 + discovery”; ”candidate: not reviewed, not adopted”; AUTHOR-ONLY; W1PC1 §3 TRUE-AS-WRITTEN

TH04.b (M4(b) Brackets when the truth is stored): same as (a); W1PC1 §3 B.D3(b) COUNTEREXAMPLE repaired with $\|h\|_\infty$

TH04.c (M4(c) Retention (N13)): N13: PROVED (PC1 claim 5)

TH04.d (M4(d) Analytic envelope (N17)): N17: PROVED (PC1 claim 1; PC2 C1(c))

TH04.e (M4(e) Output-event certificate): PC1c2: PROVED for the potential run; N6(a) output form CORRECT WITH AMENDMENT; N6(b) PROVED modulo premise (i) (GUARD-W not audited); LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)); CEX-guard: COUNTEREXAMPLE (XF VERIFIED*)

TH04.f (M4(f) Masked-regime statement (HQ Q3)): (f) assembly: PROOF SKETCH (seat-assembled; W2PC1/W2PC2 TRUE-AS-WRITTEN as an assembly); T38-b1-LH: PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8)); Thm5-LH: PROVED (SF: VERIFIED (W3-VEXP); XF: none), SAME-FAMILY-ONLY

TH04.g (M4(g) Definitional facts): definitional

Remaining card fields (verbatim, TLF 243, 256–270). Card M4 · Theorem Z$\mathcal{T}$ and the certificate card, with the masked-regime statement · K42, K43, K45, K44; K36, K78 (output-event clauses); K95 cited from S-TARGET

Removed from my R2 card. The M_2 aside ”masking forces the actual M_2 to diverge” (W2PC1 B.R4 NEEDS-HYPOTHESIS).

Guarantee (one sentence). The run’s own stored record gives deterministic two-sided brackets on the evidence and on every reported probability whenever the truth was scored, a 95% certificate jointly with output, and, on a masked family where the start misses the truth polynomially, the reported restricted posterior is still within δ_1 of the exact one jointly with output.

Objective / goal / Song–Liang. Objective (1); proposal §5.4 (PIP/joint-PIP error, deterministic TV, HQ Q3’s required masked statement); §4.4/§8.2 diagnostics (what the certificate proves); Song–Liang Thm 3 (estimated PIPs via the restricted store) and Thm S (reported MPM/HPM on LH, additional to the exact-posterior counterpart).

Status (verbatim). MZ-P3: ”PROVED (P3 top store, deterministic); P3′ conditional on T03 + discovery”; ”candidate: not reviewed, not adopted”; AUTHOR-ONLY. N13: PROVED (PC1 claim 5). PC1c5: (a) PROVED, ½ sharp. N17: PROVED (PC1 claim 1; PC2 C1(c)). PC1c2: PROVED for the potential run; N6(a) output form CORRECT WITH AMENDMENT; N6(b) PROVED modulo premise (i) (GUARD-W not audited). LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)). CEX-guard: COUNTEREXAMPLE (XF VERIFIED*). OCERT-C1: XF ERROR as quantified (O-VCERT R1 wording used). T38-b1-LH: PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8)). Thm5-LH: PROVED (SF: VERIFIED (W3-VEXP); XF: none), SAME-FAMILY-ONLY. T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)). (f) assembly: PROOF SKETCH (seat-assembled; W2PC1/W2PC2 TRUE-AS-WRITTEN as an assembly).

Fidelity to v3.1 as shipped. LS0: yes (the union-then-top-c rule); coded grid: yes (no grid hypothesis); pooling: n/a; exact arithmetic premise, numerical label until N16; (f) uses v3.1 with LS0 as shipped, integration audit open.

Self-checked flag. MZ-P3 AUTHOR-ONLY; (f) seat-assembled; GUARD-W un-audited; N13, N17 PC1/PC2 checked.

Novelty, citations checked. NEW in kind (certificates on reported quantities), elementary; Markov lower bound KNOWN—CITE (Grosse–Ghahramani–Adams 2015); the restricted-measure TV identity is elementary (no citation claimed). (f) NEW-COMP (composition with T38-b1-LH), worded ”for the paper’s own target and start”; Hou–Wang–Atchadé arXiv:2407.20580: never ”first finite-n”; YWJ 2016 / ZYVR 2022 cited next to M5.

Proof route and key lemma. Finite sums and $\pi_{\mathcal{T}} = \pi/\pi(\mathcal{T})$ for (a), (b), (d); Lemma S plus the inheritance merge for (c); Markov + Lemma RO + GUARD-W for (e); event containment for (f): $\{\mathcal{O}, \mathrm{TV} > \varepsilon\} \subseteq \{\mathcal{O}, \hat{S}^+ \neq S_0\} \cup \{\pi(S_0) \le \max(½, 1-\varepsilon)\}$. **Key lemma:** N13 retention.

Difficulty. S.

Lean core. (a)–(d): finite inequalities on a finite Ω; (e): finite event algebra given Lemma RO; (f): the containment is finite; the limits are prose.

Dependencies (card IDs). M5 (b1-LH) for (f); S-T (T03♯, Thm5-LH) for $\pi(S_0) \to 1$; S-OUT (Lemma RO, GUARD-W) for (e); S-TARGET (N17) for (d); S-CERT (G2/L/R item semantics); M1 (T12 $E[\hat{Z}_r|\mathcal{C}] = Z$ for Markov); S-BLOCKS (SIZE on LH).

Risk. GUARD-W audit (N2(e)); N16 floating point; N3 (e4 DP) hypothesis in (f); SAME-FAMILY-ONLY on LH; MZ-P3 AUTHOR-ONLY.

Fallback statement and stop rule. (a)–(d) always ship (deterministic). If the GUARD-W audit fails: (e) in potential-run form only; the output form is withheld and CEX-guard disclosed (engineering fix, not a theorem fix). If LS0 integration (N3/OP-LS0) stays open: (f) ships under the named premises, as now. If (f) were lost altogether: the deterministic (b) plus $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$ remain and HQ is told that Q3's named masked statement has shrunk (never silently). Stop: none for (a)–(d); one HQ cycle for the GUARD-W audit; (f) follows M5's stop rule.

IC3 design, exhibited, where checked. (a)–(e): any design (W1PC1 D0/E2 finite stores incl. the h = (0, 100) repair; W1PC2 40 designs × 5 random stores incl. duplicate columns). (f): LH exact Gram as above, verified W1PC2 shared LH ($\lambda_{\min}(\Sigma) = .4212$, $\mathrm{cov}(x_{\mathrm{hub}}, y) = 0$ exactly, strict separation .6 < .8, LH-LAT margin $m_0 = .0325$, $B \in [9, 16]$ for $n \geq 9^6$): a genuinely masked family.

Numerical companion (≤ 20 seeds). Existing: W1PC2 `misc_checks.py` (D5U, D5TV); PC2's exact-rational card test (60 cases). For the HQ: ≤ 20 seeds on LH at $n = 10^4$ (p = 18) with forced SMC dispatch plus one run at $n = 9^6$, recording $S_0 \in \mathcal{S}$, $S_0 \in \mathcal{T}$, $1 - Z_{\mathcal{T}}/Z$, $\max_j |\mathrm{PIP}^{\mathcal{T}}{}_j - \mathrm{PIP}_j|$, $\mathrm{MPM}^{\mathcal{T}} = S_0$, and the $\hat{Z} < Z_{\mathcal{T}}$ flag.

Plain reading

From the run's own deduplicated store $\mathcal{T}$ of scored models, for any design (under EX-A; the card is "numerical" until SC + VAL-FP): the restricted evidence $Z_{\mathcal{T}}$ never exceeds Z and the restricted posterior's TV distance to π is exactly $1 - Z_{\mathcal{T}}/Z$ (a); if the true model is in the store, the evidence ratio, every store-restricted PIP $PIP_j^{\mathcal{T}}$ and every bounded store summary $\pi_{\mathcal{T}}(h)$ are within $\delta_1 = 1 - \pi(S_0)$ (times $2\|h\|_\infty$ for h) of their exact values (b); these are store quantities, not the particle-based reported PIPs; a scored truth with $\delta_1 < \frac{1}{2}$ is retained and is the store's MPM (c); a computable analytic envelope bounds Z (d); and, under Lemma RO + GUARD-W (stop-only audit open), jointly with output (not conditionally), the 95% certificate $\max(Z_{\mathcal{T}}, 0.05\hat{Z}) \leq Z$ fails with probability at most 0.05 (e). On LH under M5's (b1-LH) hypotheses and the named premises (OP-LS0, the e4 centre rule with N3 OPEN), the probability of "output and store far from π" tends to zero (f) — not conditional success, and not $P(\mathcal{O}) \to 1$; (f) is a proof sketch.

Not claimed / not proved

- (e) rests on Lemma RO + GUARD-W; the GUARD-W stop-only audit (N2(e)) is open; if it fails, (e) is a potential-run statement only. Never $P(B \mid \mathcal{O}, \mathcal{C}) \leq \alpha$ (CEX-guard).
- (f) assembly is "PROOF SKETCH (seat-assembled; W2PC1/W2PC2 TRUE-AS-WRITTEN as an assembly)"; it carries N3 (e4 DP invariant, OPEN) and OP-LS0 as hypotheses and the SAME-FAMILY-ONLY Thm5-LH; MZ-P3 is AUTHOR-ONLY.
- The card is "numerical" until SC + VAL-FP (N16, `OB16`, `CJ16`); the envelope (d) is "loose, reported as such".
- The masked final-estimate TV/KL bound is FW2 (`CJ02`); masked two-sided evidence precision is `CJ24`; the removed aside "masking forces the actual M_2 to diverge" was NEEDS-HYPOTHESIS. Unqualified retention is false; ½ is sharp (c).

Open items for this card. `OB02` (N2(e) GUARD-W), `OB03` (N3 DP invariant, via M5), `OB06` (N6 output-event statements), `OB07` (N7 card semantics; SC(η_F)/VAL-FP), `OB16`; proved obligations used: `OB13` (N13 retention), `OB17` (N17 envelope). Future work: `CJ02`, `CJ16`, `CJ24`. HQ charter deliverables (verbatim):

Charter M4 · Theorem $Z\mathcal{T}$ and the certificate card.

• Goal: deterministic two-sided brackets and a joint-output certificate from the stored record, and the masked statement (f) that HQ Q3 requires.

• Statement: card M4 rows (a)–(g).

• First lemma to attack: N13 retention under the coded LS0 rule $\mathcal{T}_{\text{new}} = \text{top}_c(\mathcal{T}_{\text{old}} \cup \mathcal{S}_{\text{added}})$, confirmed against the code (Lemma S, inheritance merge); then the GUARD-W stop-only coupling audit N2(e), on which (e) rests.

• Lean target: (a)–(d) on a finite Ω (store algebra, TV identity, envelope); the event containment of (f).

• Independent-check plan (cross-family): the Claude-authored assembly (f) and the card semantics checked by gpt-6.1-sol; the AUTHOR-ONLY MZ-P3 re-derived by the checker (it is one page). Numerical: ≤ 20 seeds on LH as in the card.

• Fallback and stop: as in the card; (a)–(d) need no stop rule; one HQ cycle for the GUARD-W audit; (f) follows M5.

• Escalation evidence: a path-dependent stop that reaches output (CEX-guard realised in v3.1 code) → engineering decision for HQ/Felipe, output-event clauses withheld meanwhile; a code retention rule differing from the stated union-then-top-c → N13's premise fails → HQ.

4.3.5 TH05 · M5 · Theorem HPM

TH05 · M5 · Theorem HPM (composition at every n; reported HPM with LS0 on LH; Theorem A)

APPROVED (F.B., 2026-11-07) Placement: main text, wave 3 (HQ charter `HQ_CHARTERS/M5.md`).
Status (registry, verbatim): Thm4-C1 "PROVED (SF: VERIFIED* (W5-LOCKBOOK); XF: none)"; T38-(0) "PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))"; N200 "PROVED (elementary, every n); Π(X) upper bound OPEN" (author-only); TheoremA "PROVED (SF: VERIFIED (W3-VT37R); XF: VERIFIED* (A-REV2))"; LS-score "PROVED (SF: VERIFIED (W5-VMASK); XF: VERIFIED* (A-REV8))"; T38-b1-LH "PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8))" (registry ⚑ CONFLICT on the iid sub-claim, excluded); Thm-b1 "PROVED-UNDER (SF: VERIFIED (W5-VMASK); XF: VERIFIED (A-REV8))"; T38-(a) "PROVED-UNDER (SF: VERIFIED* (inputs); XF: none)"; B2E-B2LH "PROVED-UNDER (SF: none; XF: VERIFIED* (A-REV7))"; STAB-cert(A7) "REFEREE (Astra-authored)"; PART-ThmA "PROVED (XF: VERIFIED* (A-REV9))"; N3 "OPEN"; Thm5-LH (input via M3) "PROVED (SF: VERIFIED (W3-VEXP); XF: none)" SAME-FAMILY-ONLY. Assembly: PROOF SKETCH (by this seat, unchecked).
Fidelity: v3.1 with LS0 (METHOD #8) for (0), (a), (b1-LH); any frozen grid; pooling not involved (scored-set maximum); (b2-LH) is v3 without LS0, labelled; (a-fav) is an iid-draw variant, labelled.
K-IDs: K72, K69, K65, K68; K63 hypothesis; K70 labelled no-LS0 variant; K82 remark.
Authoring seat / card source: Seat D (Fable 5.1, fresh-eyes), `R3_D.md:141–158`.
Binding goal: none rests on M5 alone, but M5 feeds M4(f) (binding masked statement); escalation evidence: "a counterexample to T38-b1-LH on LH; the integrated LS0 differing from the specified LS0; N3 failing outright" [NBH/02_theory.md:970-981].

TLF clerk text (verbatim; note 4 is the final-audit δ_1 correction and governs the reading of clause (0)). Chair's integration note.

1. K70 stays in this card as the labelled no-LS0 variant clause (b2-LH) (A, B, D: clause; C: supplementary attributed variant; MINUTES §2.1 follows the clause form). K63 is a hypothesis here, proved in S-OUT.

2. N2(c) (integrated LS0 replay), the N3 DP invariant (e4 centre), GUARD-W, N14(c) (IND-T extension for $R' > 4$) and K84/N14(a) (extra search at $R' > 4$) are named implementation premises (MINUTES §2.3, "Audit obligations").

3. If (b1-LH) falls, the §5.4 masked statement rests on M4's regime-free certificate; that loss escalates (MINUTES §2.3 §2.7).

4. **δ_1 correction (final audit #7).** In the deterministic clause, read $\delta_1 := 1 - \pi(S_0)$, pointwise, as in

M4: "if $S_0 \in \mathcal{S}^+$ and $1 - \pi(S_0) < ½$, then $\hat{S}^+ = S_0$". The probability bound keeps its expectation, written $2E_0[1 - \pi(S_0)] + D_n$. The card's single symbol "$\delta_1 := E_0[1 - \pi(S_0)]$" conflates the two; the HQ uses this reading.

For agents — δ_1 in M5

Read clause (0) as: deterministic part "if $S_0 \in \mathcal{S}^+$ and $1 - \pi(S_0) < \frac{1}{2}$, then $\hat{S}^+ = S_0$"; probability bound $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2E_0[1-\pi(S_0)] + D_n$. The verbatim card below still writes "$\delta_1 := E_0[1 - \pi(S_0)]$"; that symbol conflates the two (final audit #7, [NBH/branch_snapshots/MEET-THM/CLOSE_AUDIT.md:15]). The Lean lemma `t21_C1` already uses $\delta_1 = 1 - \pi(S_0)$ pointwise [NBH/branch_snapshots/L-LEAN/REPORT.md:34].

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:292

Final statement. Hypotheses (model/design): standardised X, the shipped prior and hyperparameters (ridge/BB(1,p)/cap, IG(a_0, b_0)); for (b1-LH) and (b2-LH): LH exact Gram with $p \leq n - 1$, c linked pairs, group size g, within-correlation ρ, link ψ with $|\psi| < \rho$, fixed β; the T03♯ hypotheses on LH (G, R, $SE_\lambda(j_0)$, $SE_w(m♯; g_0)$, BM) and SIZE $g_{max} \leq B < 2g_{min}$ (holds on LH for $n \geq 9^6$ with g = 9, W1PC2 D4).
Implementation premises: EX-A, EX-S, CAL, SR, GUARD-W, OP-LS0, e4, complete scans, $\ell \geq \tau(C)$ with a completed final scan; IND-T when $R' > 4$.
`TH05.0` **(0) Composition at every n, any X.** $\hat{S}^+$ the $\succ$-maximum of the all-scored set $\mathcal{S}^+$ under (F, −|S|, lex) (K63). $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2E_0\delta_1 + D_n$, $\delta_1 := E_0[1 - \pi(S_0)]$. If $S_0 \in \mathcal{S}^+$ and $\delta_1 < ½$ then $\hat{S}^+ = S_0$. N200 bracket: $P(U \cup \{\hat{S} \neq S_0\}) \leq P(U) + \Pi(X) + P(\mathcal{O}, S_0 \notin \mathcal{S})$, with the matching lower bracket, Π(X) measured, not bounded. Transfer to $R' > 4$ only with the extra search from c_4 (nested scored set).
`TH05.a` **(a) Theorem A / LS-score.** Under TD'_∞ on Lat(S°), local search from $C \in Lat(S°)$ with $\ell \geq \tau(C)$ and a completed final scan reaches and certifies S_0 within Φ(C) moves; LS0 scores S_0 within $\max\{\Phi(S°) - 1, 0\}$ moves (scoring, not certification).
`TH05.b1LH` **(b1-LH).** On LH with LS0: $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$; hence liminf $P(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 -$ limsup $P(U_n)$. No delivery claim (K82 remark).
`TH05.afav` **(a-fav) labelled variant:** under (F) with iid π_0 draws, $D_n \to 0$.
`TH05.b2LH` **(b2-LH) labelled variant (K70):** without LS0 on STAB-LH (g = 9, c = 2, ρ = 0.4, ψ = 0.2), any frozen grid, the same conclusion.

Clause rows and their registry statuses. `TH05.0` (M5(0) Composition at every n, any X): T38-(0) "PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))"; N200 "PROVED (elementary, every n); Π(X) upper bound OPEN" (author-only)

`TH05.a` (M5(a) Theorem A / LS-score): TheoremA "PROVED (SF: VERIFIED (W3-VT37R); XF: VERIFIED* (A-REV2))"; LS-score "PROVED (SF: VERIFIED (W5-VMASK); XF: VERIFIED* (A-REV8))"

`TH05.b1LH` (M5(b1-LH) HPM with LS0 on LH): T38-b1-LH "PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8))" (registry ⚑ CONFLICT on the iid sub-claim, excluded); Thm-b1 "PROVED-UNDER (SF: VERIFIED (W5-VMASK); XF: VERIFIED (A-REV8))"; N3 "OPEN"

`TH05.afav` (M5(a-fav) labelled variant): T38-(a) "PROVED-UNDER (SF: VERIFIED* (inputs); XF: none)"; labelled variant

`TH05.b2LH` (M5(b2-LH) labelled variant (K70)): B2E-B2LH "PROVED-UNDER (SF: none; XF: VERIFIED* (A-REV7))"; STAB-cert(A7) "REFEREE (Astra-authored)"

Remaining card fields (verbatim, TLF 288, 293–305). Card M5 · Theorem HPM (composition at every n; reported HPM with LS0 on LH; Theorem A) — K72, K69, K65, K68; K63 hypothesis; K70 labelled variant; K82 remark

Guarantee. A wrong reported model means no output, a posterior miss, or a never-scored truth; on an explicit

correlated family where the start misses the truth polynomially, the shipped method still reports it, jointly with output.

Serves. Objective (3); Song–Liang Thm S (”exact-HPM counterpart plus a separate reported-HPM result on LH”); THM-INV rows 11, 12; feeds M4(e) for §5.4’s masked statement (HQ Q3).

Status (verbatim). Thm4-C1 ”PROVED (SF: VERIFIED* (W5-LOCKBOOK); XF: none)”; T38-(0) ”PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))”; N200 ”PROVED (elementary, every n); Π(X) upper bound OPEN” (author-only); TheoremA ”PROVED (SF: VERIFIED (W3-VT37R); XF: VERIFIED* (A-REV2))”; LS-score ”PROVED (SF: VERIFIED (W5-VMASK); XF: VERIFIED* (A-REV8))”; T38-b1-LH ”PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8))” (registry ⚑ CONFLICT on the iid sub-claim, excluded); Thm-b1 ”PROVED-UNDER (SF: VERIFIED (W5-VMASK); XF: VERIFIED (A-REV8))”; T38-(a) ”PROVED-UNDER (SF: VERIFIED* (inputs); XF: none)”; B2E-B2LH ”PROVED-UNDER (SF: none; XF: VERIFIED* (A-REV7))”; STAB-cert(A7) ”REFEREE (Astra-authored)”; PART-ThmA ”PROVED (XF: VERIFIED* (A-REV9))”; N3 ”OPEN”; Thm5-LH (input via M3) ”PROVED (SF: VERIFIED (W3-VEXP); XF: none)” SAME-FAMILY-ONLY. Assembly: PROOF SKETCH (by this seat, unchecked). **Fidelity:** v3.1 with LS0 (METHOD #8) for (0), (a), (b1-LH); any frozen grid; pooling not involved (scored-set maximum); (b2-LH) is v3 without LS0, labelled; (a-fav) is an iid-draw variant, labelled.

Verdict rows used. (0): W1PC1 §7 TRUE-AS-WRITTEN (E1, 1,440 cases), W2PC1:122 TRUE-AS-WRITTEN; R′ transfer W1PC1 §7 TRUE. (a): W1PC1 §6 NEEDS-HYPOTHESIS repaired to $\ell \geq \tau(C)$ + final scan, W2PC1:122 REPAIRED; W2PC2:127 (scoring vs certification) TRUE-AS-WRITTEN. (b1-LH): W1PC1 §6 TRUE-AS-WRITTEN, W1PC2 D4 TRUE as cited, W2PC1:122, W2PC2:126; liminf form W1PC1 §6. SIZE: W1PC1 §6 (PART-CE2), W2PC2:158. Variants: W1PC1 §6 TRUE-AS-WRITTEN as labelled variants.

Self-checked flag. N200 author-only; T38-(a) XF none; Thm5-LH SAME-FAMILY-ONLY; STAB-cert Astra-authored, Claude-verified. Integrated LS0 / emitted-centre / GUARD-W audits open (W2PC2:39; W2PC1:122 GAP).

Novelty. NEW for this target and method (NEW-COMP). Checked: Yang–Wainwright–Jordan, AoS 44(6) 2016 Thm 2, Lemma 4 (LIT-MCMC #1); Zhou–Yang–Vats–Roberts–Rosenthal, JRSSB 84(5) 2022 Thm 1 (LIT-MCMC #2): write ”with their priors LH is covered by rapid mixing; ours is for the paper’s own target and start”. Hou–Wang–Atchadé arXiv:2407.20580 Thm 2 (LIT-STAT #19): never ”first finite-n”. Rossell 2022 Prop 1: the $2E_0\delta_1$ term is KNOWN—CITE. Barbieri–Berger 2004: HPM/MPM facts (K60, K61 cited lemmas).

Route / key lemma. (0) finite-set argument. (a) TD′ lattice descent (TheoremA) with the capped-scan interface. (b1-LH): LH centre law (M3) + LH-LAT margin $m_0 > 0$ + LS-score with the e4 centre + T03♯ (S-T) + retention (M4(b)). **Key lemma:** LS-score with the emitted centre equal to the prescribed start mode (N3). Difficulty: S (assembly); M for the N3 DP invariant.

Lean core. `T37Greedy.lean`, `T37Lattice.lean` (sorry-free); `T21HPM.lean` (`t21_C1` sorry-free). To add: the capped-scan certification interface ($\ell \geq \tau(C)$, completed scan ⇒ certified return) and (0) as a finite Boolean lemma with the output event. Cited probability: T03♯ (S-T), Thm5-LH masking law (M3).

Dependencies. Uses: S-T ($\delta_1 \to 0$), M3 (start-mode centre, $\pi_0(S_0)$ law), S-BLOCKS (SIZE ⇒ blocks = groups, K07/K06), S-CERT (K64 search semantics and scan bound), S-OUT (K77 Lemma RO, K78 GUARD-W, K84 stacking nesting, K63 order). Used by: M4(e), Prop D (search cost term), COVERAGE_MATRIX Thm S row.

Risk. N3 OPEN (DP invariant); integrated-LS0 replay (N2(c)) pending; Π(X) only measured (K94 FW); study generator N8 uncovered (the g = 5 study is not the LH exhibit).

Fallback + stop rule. Fallback: (0) + (a) + (b2-LH) ship unconditionally; (b1-LH) labelled ”conditional on N3” if the DP invariant is unproved after one HQ cycle. Stop: the merged-LS0 replay contradicting LS-score’s centre rule, or a counterexample to T38-b1-LH on LH ⇒ (b1-LH) withdrawn, objective (3) coverage reported as (b2-LH) only; the §5.4 masked statement then rests on M4(d) + M4(b) (no binding goal dropped).

IC3 design (exhibited). LH exact Gram, g = 9, c = 2, ρ = 0.4, ψ = 0.2, $n \geq 9^6$ (blocks = groups, SIZE, LH-LAT $m_0 = 0.0325 > 0$): W1PC2 D4 DESIGN-VERIFIED (`lh_family.py`), W2PC2:126 DESIGN-VERIFIED. (0): any X (W1PC1 §7 DESIGN-VERIFIED, Dε exact branch).

Numerical companion (≤ 20 seeds). Filed: 0/2,394 HPM changes by LS0 (O-LS0 test export). New: forced-dispatch LH replica (PC2’s `lh_family.py` + `smc_coded.py`) with LS0 at component size, ≤ 20 seeds, reporting the frequency of $\hat{S}^+ = S_0$, of $S_0 \in \mathcal{S}^+$, and of the D_n event; a component check, since production dispatches the exact branch at that size.

Plain reading

At every sample size and for any design, under the named implementation premises: if S_0 was scored ($S_0 \in \mathcal{S}^+$) and $1 - \pi(S_0) < \frac{1}{2}$, the reported HPM is S_0; so a wrong reported HPM on output requires a posterior miss ($\pi(S_0) \leq \frac{1}{2}$) or a true model that the run never scored, and $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2E_0[1 - \pi(S_0)] + D_n$ (0). Local search started from a centre $C \in \mathrm{Lat}(S^\circ)$, under the descent condition TD'_∞ on $\mathrm{Lat}(S^\circ)$, with $\ell \geq \tau(C)$ and a completed final scan, reaches and certifies S_0 within $\Phi(C)$ moves, and LS0 scores (does not certify) S_0 within $\max\{\Phi(S^\circ) - 1, 0\}$ moves (a). On the explicit correlated family LH, where the start misses the truth polynomially, under the T03♯ hypotheses, SIZE and the named premises (including the e4 centre rule, N3 OPEN), the method with LS0 as specified has $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$, hence $\liminf P(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup P(U_n)$; delivery ($P(\mathcal{O}) \to 1$) is not claimed (b1-LH). The labelled variants cover iid start draws under (F) ($D_n \to 0$, a-fav) and v3 without LS0 on STAB-LH (b2-LH).

Not claimed / not proved

- No delivery claim: $P(U_n) \to 0$ is FW13 (CJ13, K82 remark); the liminf statement is in terms of $\limsup P(U_n)$.
- (b1-LH) is about "v3 + LS0" with LS0 as specified: the e4 tie-DP invariant N3 is OPEN ("the only OPEN mathematical obligation in the chain", charter) and the integrated-LS0 replay N2(c) is pending; "LS0 exists only in the O-LS0 test export" [NBH/03_results_and_conclusions.md:191]. Fallback: (b1-LH) labelled "conditional on N3".
- T38-b1-LH carries a registry ⚑ CONFLICT on the iid sub-claim, which is excluded; (b1-LH-iid) is CJ10 (FW10).
- $\Pi(X)$ is measured, not bounded (CJ14); finite-n Gibbs discovery is CJ26; the $g = 5$ study generator is not the LH exhibit (CJ25, OB08); T38(b1) premises mostly fail at $n = 200$ [NBH/03_results_and_conclusions.md:190].
- Assembly "PROOF SKETCH (by this seat, unchecked)"; N200 author-only; T38-(a) XF none; Thm5-LH SAME-FAMILY-ONLY.

Open items for this card. OB03 (N3: DP invariant, first lemma), OB02 (N2(c) integrated LS0 replay; GUARD-W), OB11 (N11, ℓ versus moves; growing s_0 OPEN), OB14 (N14(a),(c) for $R' > 4$). Future work: CJ10, CJ13, CJ14, CJ19, CJ25, CJ26. Named premises [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:79]. HQ charter deliverables (verbatim):

Charter M5 · Theorem HPM.

• Goal: prove that the shipped method (v3.1 with LS0) reports S_0 jointly with output on an explicit correlated masked family, and the composition inequality at every n.

• Statement: §S2 card M5, clauses (0), (a), (b1-LH); variants (a-fav), (b2-LH) labelled.

• First lemma: the e4 tie-DP invariant (N3): the DP returns the prescribed smallest-size, lexicographically smallest start-mode maximiser (W1PC2 D9: TRUE on 400 random instances, DESIGN-ASSERTED). It is the only OPEN mathematical obligation in the chain. In parallel, the merged-LS0 replay (N2(c)) against LS-score's centre rule.

• Lean target: `T37Greedy/T37Lattice` (existing, sorry-free) plus the capped-scan certification interface; `T21HPM` plus (0) as a finite Boolean lemma with the output event.

• Independent-check plan: cross-family (Codex checker on this Claude-authored assembly); route Thm5-LH (SAME-FAMILY-ONLY) and T38-(a) (XF none) to a cross-family check inside this HQ; numerical: ≤ 20-seed forced-dispatch LH replica with LS0 (component check) plus the filed 0/2,394 export.

• Fallback and stop: as the card; (b1-LH) becomes "conditional on N3" after one unproductive HQ

cycle; (0) + (a) + (b2-LH) always ship.

• Escalation evidence: a counterexample to T38-b1-LH on LH; the integrated LS0 differing from the specified LS0; N3 failing outright. Each changes objective (3) coverage and the §5.4 masked statement (M4(e)), so HQ decides.

4.3.6 TH06 · M6 · Theorem SEP + Lemma KL

TH06 · M6 · Theorem SEP + Lemma KL

APPROVED (F.B., 2026-11-07) Placement: main text, wave 2 (HQ charter `HQ_CHARTERS/M6.md`).
Status (registry, verbatim): SEP-A/SEP-B: **PROVED; SF: none; XF: VERIFIED (A-REV9)**. KL: **PROOF SKETCH (NEW-SYNTH)**, **AUTHOR-ONLY** in IDX. Statement checks remain TRUE-AS-WRITTEN; they do not upgrade KL. Final assembly: **PROOF SKETCH, unchecked**.
K-IDs: K19, K106, K105.
Authoring seat / card source: Seat C (gpt-6-astra), `R3_C.md:179–206`.
Binding goal: none rests on M6 alone; it supplies M2(a)'s lower bound and M2(c2) (Lemma KL) [`NBH/02_theory.md:1021-1032`].

TLF clerk text (verbatim). Chair's integration note. No change.

Statement (verbatim from source) `NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:324-334`

Hypotheses. Finite Ω, strictly positive normalized π, π_0, feasible distinguished S_0. Write $a = \pi_0(S_0)$, $b = \pi(S_0)$. Statistical consequences use D0, EX-A and G on LH, with M3(f) and S-T.

Implementation premises. Comparison experiment: $Q \geq 1$ exact iid draws from π_0, conditionally on data. One-shot estimation uses exact weights. No production-grid, mutation, LS0 or delivery premise enters these comparisons.

`TH06.a` **M6(a): discovery.** Conditional on data, $P(S_0 \text{ absent from all } Q \text{ draws}) = (1-a)^Q \leq e^{-Qa}$. **Evidence:** W1P2 cluster 8, "D3 · miss probability"; W2P2 §2, C6.

`TH06.b` **M6(b): LH threshold.** Let $Q_n^\star = p^2 n$. Under LH, $Q_n/Q_n^\star \to 0$ gives miss probability tending to one; $Q_n/Q_n^\star \to \infty$ gives miss probability tending to zero, over data and comparison draws. **Evidence:** W2P2 §2, "C6 · Q* scale".

Tables qualification. On LH, truth spans several actual blocks, so no block table evaluates the full truth. Adding existing block-table evaluations to this start-only comparison does not change its discovery event. **Evidence:** W1P1 §8, SEP-A scope; checked LH partition.

`TH06.c` **M6(c): singleton second moment.** $1 + \chi^2(\pi\|\pi_0) \geq b^2/a$. On LH this yields an inverse-start-mass lower bound $p^2 n \exp\{O_P(1)\}$, with a two-sided tight positive multiplier. **Evidence:** W1P2 cluster 8, "D3 · χ^2 bound"; CHK §2(a),(b).

Evidence estimator. Let $f(S) = \varpi(S)m(S)$, $Z = \sum_S f(S)$, and $\widehat{Z}_Q^{IS} = Q^{-1}\sum_{i=1}^Q f(S_i)/\pi_0(S_i)$. Then $\mathrm{Var}(\widehat{Z}_Q^{IS} \mid X, y)/Z^2 = \chi^2(\pi\|\pi_0)/Q$. **Evidence:** W1P2 cluster 8, "D3 · IS variance".

Interpretation of variance. Consequently, below the LH discovery scale the one-shot evidence estimator's relative variance diverges in probability. This is an L^2 obstruction; large variance alone is not a high-probability error lower bound. No matching upper bound is asserted.

`TH06.KL` **Lemma KL.** $KL(\pi\|\pi_0) \geq b\log(1/a) - \log 2$. **Evidence:** W1P2 cluster 8, "D3 · KL bound"; CHK §2(a).

KL consequence. On LH, $KL(\pi\|\pi_0) \geq b\log Q_n^\star + O_P(1) - \log 2$, hence diverges in probability. The finite singleton bound is the retained theorem; no universal sample-complexity conclusion for arbitrary functionals is attached.

Scope boundary. This is not a lower bound for production SMC, start-plus-search, or every one-shot estimator. SEP-taska remains a **COUNTEREXAMPLE**. Ideal logarithmic crossing counts do

not imply logarithmic total work.

Clause rows and their registry statuses. TH06.a (M6(a) discovery): SEP-A/SEP-B: PROVED; SF: none; XF: VERIFIED (A-REV9)

TH06.b (M6(b) LH threshold): SEP-A/SEP-B: PROVED; SF: none; XF: VERIFIED (A-REV9)

TH06.c (M6(c) singleton second moment): SEP: PROVED; XF VERIFIED (A-REV9); CHK §2(a),(b)

TH06.KL (Lemma KL): KL: PROOF SKETCH (NEW-SYNTH), AUTHOR-ONLY in IDX. Statement checks remain TRUE-AS-WRITTEN; they do not upgrade KL

Remaining card fields (verbatim, TLF 323, 335–346). **IDs. M6: K19, K106, K105**. Generic finite-law inequalities are supporting lemmas; the LH consequence is the method-specific package.

Guarantee. The specified iid start-only comparison needs the LH inverse-mass scale for discovery, while one-shot evidence correction inherits a corresponding second-moment obstruction.

Goals. Objective (1), with motivation for objectives (2), (3); proposal §5.3 one-shot obstruction and §5.4 start KL. Song–Liang Thm 4 is contextual comparison only; no analytic SL/FDR separation.

Status. SEP-A/SEP-B: **PROVED; SF: none; XF: VERIFIED (A-REV9)**. KL: **PROOF SKETCH (NEW-SYNTH)**, **AUTHOR-ONLY** in IDX. Statement checks remain TRUE-AS-WRITTEN; they do not upgrade KL. Final assembly: **PROOF SKETCH, unchecked**.

Fidelity/self-check flag. Actual shipped π_0, π; explicitly different comparison algorithms. SEP ingredients have filed cross-family checks. KL and the final assembly lack a full proof certificate. SEP-A-iid and broader exponent claims are not imported.

Novelty/citation. NEW-COMP for the explicit LH start-mass mechanism. Chatterjee–Diaconis, Thms 1.1–1.3 (https://arxiv.org/html/1511.01437v3) supplies the established KL-scale context, with log-weight-tail and estimand qualifications. Finite coarsening and iid variance identities are not novelty claims.

Proof route/key lemma. M3 supplies two-sided tightness of $Q_n^\star a_n$; apply the exact miss formula. Retain the singleton term in the second-moment sum. Binary KL data processing and binary entropy $\leq \log 2$ prove Lemma KL.

Difficulty/Lean core. Routine after M3. Lean: iid miss probability, singleton second-moment inequality, binary coarsening/log-sum inequality and entropy bound. Probability tightness arguments remain prose unless separately formalized.

Dependencies. M3; S-T; S-TARGET. No dependence on M7, a production mixing theorem or Proposition D.

Risk. Replacing a lower bound by an asymptotic equality; confusing evidence variance with self-normalized variance; claiming probability failure from variance alone; claiming universal necessity of tempering.

Fallback/stop rule. Ship the exact discovery formula and finite singleton/KL/variance statements. If the LH application fails, withhold its scale claim until M3 is repaired. After one repair cycle, classify the extension FUTURE_WORK and notify M2; preserve CHK's independently verified orthogonal Q1(c) obstruction.

IC3 exhibit/check. LH, with actual blocks and $n \geq 9^6$. W1P2 shared-family/cluster-8 rows, W2P2 C6, and CHK §2 supply the checked instantiation and finite inequalities.

Numerical companion. None newly run. Proposed: at most ten LH data seeds; compute a_n by exact convolution and evaluate $(1-a_n)^Q$ analytically. Avoid simulating $p^2 n$ draws. Enumerate small correlated component cases for KL/second-moment inequalities.

Plain reading

For the comparison algorithm that only draws Q exact iid samples from the start π_0, the chance of never drawing S_0 is exactly $(1-a)^Q$ with $a = \pi_0(S_0)$ (a); under G on LH, with $Q_n^\star = p^2 n$, a number of draws of smaller order than $Q_n^\star$ misses S_0 with probability tending to one and a number of larger order finds it with probability tending to one (b). The one-shot importance-sampling evidence estimator has relative variance $\chi^2(\pi\|\pi_0)/Q$, and $1 + \chi^2 \geq b^2/a$ with $b = \pi(S_0)$ (c), so below the LH discovery scale its relative variance diverges in probability. Lemma KL (a proof sketch):

$KL(\pi\|\pi_0) \geq b\log(1/a) - \log 2$, which diverges in probability on LH.

Not claimed / not proved

- "This is not a lower bound for production SMC, start-plus-search, or every one-shot estimator. SEP-taska remains a **COUNTEREXAMPLE**" (Scope boundary row). Tempering is not proved necessary.
- Lemma KL: "PROOF SKETCH (NEW-SYNTH)", AUTHOR-ONLY; statement checks "do not upgrade KL". Final assembly: PROOF SKETCH, unchecked.
- The variance result is an L^2 obstruction only: "large variance alone is not a high-probability error lower bound. No matching upper bound is asserted."
- The LH application depends on M3 (and Thm5-LH, SAME-FAMILY-ONLY); SEP-A-iid and broader exponent claims are not imported.

Open items for this card. Lemma KL proof (binary coarsening, singleton second-moment term); two-sided tightness of $Q_n^\star a_n$ from M3; citation check of every $\exp(KL)$ sentence against Chatterjee–Diaconis (CL07). No OB/CJ item is assigned to M6 in the coverage matrix. HQ charter deliverables (verbatim):

M6 charter. Instruction

Goal. Quantify the start-only discovery obstruction and the specified one-shot evidence loss without universal algorithm claims.

Statement. Prove §S2 M6(a–c), the evidence variance identity and Lemma KL.

First lemma. Binary coarsening plus the singleton second-moment term; explicitly distinguish probability error from variance.

Lean target. Finite KL lower bound, entropy $\leq \log 2$, iid miss formula and variance of the exact importance-weight mean.

Independent check. Opus-family checker checks the Astra assembly, estimand definitions, two-sided tightness and conditioning. Citation check compares CD18 Thms 1.1–1.3 with every sentence invoking an $\exp(KL)$ scale.

Fallback/stop. One proof-and-repair cycle. Ship finite identities if the LH application cannot close; remove its scale claim and record the dependency failure.

Escalation evidence. Refutation of a claimed production-law instantiation, or failure of the checked Q1(c) obstruction relied upon by M2. Ordinary scope narrowing remains within meeting authority.

Numerical companion. At most ten data seeds; analytic miss probabilities and small finite-law enumeration, without enormous iid simulations.

4.3.7 TH07 · M7 · Theorem STG (stage laws as coded; coverage identity)

TH07 · M7 · Theorem STG (stage laws as coded; coverage identity)

APPROVED (F.B., 2026-11-07) Placement: main text, wave 2 (HQ charter `HQ_CHARTERS/M7.md`).
Status (registry, verbatim): STG-G, STG-L0, STG-L1: PROVED (SF: none; XF: none); T26BUD-G: PROVED, deterministic (AUTHOR-ONLY, W2PC2:155); STG-M: PROVED (SF: none; XF: VERIFIED* (A-REV9)), explicit decimal constants GAP (W2PC2 C14); STG-L: PROVED (SF: none; XF: none); ConjP: OPEN; STG-cexW2: COUNTEREXAMPLE; OWS-WS7(d): PROVED (SF: none; XF: none) + CHK VERIFIED WITH AMENDMENT; (b), (c): D/E proposals, TRUE-AS-WRITTEN (W1PC1:192–193); (c'): PROOF SKETCH (Seat C, unchecked), W2PC2 TRUE-AS-WRITTEN; STG-F: PROVED (SF: none; XF: none), cited only (K29 FW).

K-IDs: K26, K25, K27, K30, K33 (deterministic termination/count form); K28 cited.
Authoring seat / card source: Seat A (Fable 5.1), `R3_A.md:139–162`.
Binding goal: none rests on STG alone: M2(b)'s stage-count clause has its own checked fallback `[NBH/02_theory.md:1078-1089]`. Dissent: Seat D would place K33 in S-IDENT (DS01); K30 placement dissent (DS04).

TLF clerk text (verbatim). Chair's integration note.

1. K33 enters in its deterministic termination/count form only: the card's clause (a) (termination, $\rho_V \geq 0.9$ per appended point, $\leq \ell$ rebuilds per tier, $T \leq T_e$) and clause (c) (the count). Rebuild convergence N5(b), P($S_0 \in V$) and P(schedule_limit) → FUTURE_WORK (K31) in either case; Seat D's dissent (K33 in S-IDENT, R3_D §S1) is recorded in MINUTES §2.3.

2. K30 is clause (d), labelled ideal two-state grid, and is stated only here (S-STGI is dissolved). K29 is FUTURE_WORK FW23; its conditional implication may be cited as an attributed lemma.

3. The card's escalation statement: no binding goal rests on STG alone; M2(b) has its own checked fallback $T \leq P + \lceil D_*/c_\rho \rceil + 1$.

Statement (verbatim from source) `NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:369-371`

Model/design hypotheses. Admissible (X, y) with positive finite tables, so $\gamma_0 > 0$ on Ω; a fixed nonempty $V \subseteq \Omega$; $D_V := \mathrm{osc}_V\, d$; for (d) a two-state restriction with initial log-odds $-A_0$ and terminal log-odds bounded away from 0 on the positive side.

Implementation premises (named). EX-A; the coded grid rule (`v3_calib.h`: ESS threshold 0.9 for pilot and grid, P retained pilot targets, 60 halvings, 10^{-12} snap/validation, T_e cap) equal to its mathematical rule (Python translation checked, no compiled bitwise replay: W1PC1:61); bi-section resolution $\leq h_V/2$ ($2^{-60} \leq h_V/2$); "success" is a reported status, with ceiling/snap failures excluded.

Statement.
TH07.a (a) Monotonicity and construction: $\rho_V(u,v)$ is nondecreasing in u and nonincreasing in v; bisection accepts only intervals with $\rho_V \geq 0.9$; build_grid terminates, every appended point has $\rho_V \geq 0.9$, $\leq \ell$ rebuilds per tier; $T \leq T_e$ on success; at exact frontiers the greedy grid is shortest among refinements of the retained pilot targets, up to the 2^{-60} slack (no "exact/minimal").
TH07.b (b) Coverage identity: $\rho_{\mathrm{full}}(u,v) = \rho_V(u,v)\cdot a_u\, a_w/{a_v}^2$, $w = 2v-u$, $a_t = q_t(V)$; $a_u, a_w \geq 1-\eta \Rightarrow \rho_{\mathrm{full}} \geq 0.9(1-\eta)^2$.
TH07.c (c) Interval inequality and count: every accepted interval has $(v-u)^2\cdot \mathrm{Var}_{q_\xi|V}(d) \leq \log(10/9)$ for some $\xi \in [u, 2v-u]$; if $D_V > 0$ every increment $\leq h_V := 2c_\rho/D_V$ passes ($c_\rho = \sqrt{\log(10/9)}$); with resolution $\leq h_V/2$ and absent ceiling/snap failure, $T \leq P + \lceil 2/h_V \rceil + 1 = P + \lceil D_V/c_\rho \rceil + 1$; a uniform full-space grid ($W = \Omega$) with $\max\{1, \lceil \mathrm{osc}_\Omega\, d/(2c_\rho)\rceil\}$ stages has full ESS ≥ 0.9 (existence).
TH07.cp (c′) Pilot-count clause, cited from M2(b): under a global range bound D_* and $2^{-60} \leq h/2$, $P \leq \lceil 2/h \rceil + 1$, hence $T \leq 2\lceil D_*/c_\rho \rceil + 2 \leq 4 + 2D_*/c_\rho$; $D_* = 0 \Rightarrow T = 1$ (PROOF SKETCH, Seat C).
TH07.d (d) Ideal grid, labelled, not the coded grid: on the exact-frontier two-state ESS-0.9 grid a dominance crossing of gap A_0 costs $\log_2 A_0 + O(1)$ stages; separated pair-dominant crossings add; restricted ESS alone forces only $\log_2(9.5\ell_0)$ stages (WS7(d) with the set-mass and endpoint amendments).
TH07.e (e) [cited] $\sqrt{c}\cdot T^*(c) \to \Lambda_{\mathrm{len}} := \int_0^1 \mathrm{sd}_{q_\varphi}(d)\,d\varphi \leq \sqrt{J}$ as $c \to 0$.
TH07.f (f) The as-coded masked stage count T_n is Conjecture P, OPEN → FUTURE_WORK (K31); the shipped guarantee is $T \leq T_e$. Favourable O(1) (K29) → FUTURE_WORK.

Clause rows and their registry statuses. TH07.a (M7(a) Monotonicity and construction): STG-G, STG-L0, STG-L1: PROVED (SF: none; XF: none); W1PC1:190 TRUE; exact/minimal W1PC1:191 COUNTEREXAMPLE repaired
TH07.b (M7(b) Coverage identity): (b), (c): D/E proposals, TRUE-AS-WRITTEN (W1PC1:192–193)

TH07.c (M7(c) Interval inequality and count): W1PC1:194 COUNTEREXAMPLE repaired to P + ...; (b), (c) TRUE-AS-WRITTEN

TH07.cp (M7(c′) Pilot-count clause (cited from M2(b))): (c′): PROOF SKETCH (Seat C, unchecked), W2PC2 TRUE-AS-WRITTEN

TH07.d (M7(d) Ideal grid (K30), labelled, not the coded grid): STG-M: PROVED (SF: none; XF: VERIFIED* (A-REV9)), explicit decimal constants GAP (W2PC2 C14); OWS-WS7(d): PROVED (SF: none; XF: none) + CHK VERIFIED WITH AMENDMENT

TH07.e (M7(e) [cited]): KNOWN—CITE Syed et al. 2024 Thm 3, Gelman–Meng 1998

TH07.f (M7(f) as-coded masked T_n): ConjP: OPEN; STG-cexW2: COUNTEREXAMPLE; STG-F: PROVED (SF: none; XF: none), cited only

Remaining card fields (verbatim, TLF 365, 372–386). K-IDs: K26, K25, K27, K33, K30 (labelled ideal); K28 cited.

Verdict rows per clause. (a) W1PC1:190 TRUE; exact/minimal W1PC1:191 COUNTEREXAMPLE repaired; termination/rebuilds W1PC1:198 TRUE; W2PC1:99 TRUE. (b) W1PC1:192 TRUE; PC2 D7 to 10^{-8}. (c) interval W1PC1:193 TRUE; count W1PC1:194 COUNTEREXAMPLE repaired to P + ...; range/resolution W1PC1:195 NEEDS-HYPOTHESIS repaired; W = Ω W1PC1:140; zero range W2PC1:56 repaired max{1,·}. (c′) W2PC2:105, 156 TRUE-AS-WRITTEN (600 coded-pilot runs). (d) W1PC1:197 TRUE; CHK:91 (WS7 amendments). (e) W1PC1:257 (Λ_{len} renamed).

Guarantee. Every accepted stage of the coded grid is certified by an exact restricted-ESS inequality whose full-bridge meaning is the coverage factor; the stage count is bounded by the pilot targets plus the spread of d on V; and on the ideal grid correlation enters depth only through the crossing gap, as $\log_2 A_0$.

Serves. Objective (2) (island depth); proposal §4.4/§8.2 (what ESS 0.9 certifies); input to M2(b) (sandwich, envelope) and M8 (T); INV rows 5, 7 × C.

Status (verbatim). STG-G, STG-L0, STG-L1: PROVED (SF: none; XF: none); T26BUD-G: PROVED, deterministic (AUTHOR-ONLY, W2PC2:155); STG-M: PROVED (SF: none; XF: VERIFIED* (A-REV9)), explicit decimal constants GAP (W2PC2 C14); STG-L: PROVED (SF: none; XF: none); ConjP: OPEN; STG-cexW2: COUNTEREXAMPLE; OWS-WS7(d): PROVED (SF: none; XF: none) + CHK VERIFIED WITH AMENDMENT; (b), (c): D/E proposals, TRUE-AS-WRITTEN (W1PC1:192–193); (c′): PROOF SKETCH (Seat C, unchecked), W2PC2 TRUE-AS-WRITTEN; STG-F: PROVED (SF: none; XF: none), cited only (K29 FW).

Fidelity to v3.1. (a)–(c), (c′): the coded rule (translation-checked). (d), (e): ideal exact-frontier / fine grid, labelled comparison objects; the gap to the coded grid is K31 (FUTURE_WORK).

Self-checked flag. Yes for STG-G/L0/L1/L (author-only) and (b), (c) (statement sanity only); STG-M cross-family VERIFIED*; (c′) unchecked.

Novelty. (a)(i) greedy optimality KNOWN—CITE Štefankovič–Vempala–Vigoda 2009 (LIT-SMC row 9); (e) KNOWN—CITE Syed et al. 2024 Thm 3, Gelman–Meng 1998 (LIT-SMC row 15); (b), (c), (d) NEW, elementary (checked LIT-SMC digest; no published restricted-ESS coverage identity found).

Route / key lemma. Lemma RS ($\rho_V(u,v) = Z_V(u)Z_V(w)/Z_V(v)^2$, convex log-MGF ⇒ monotonicity); second-order mean value for log-sum-exp gives (c); Popoviciu Var ≤ ${D_V}^2/4$ gives the passing step; bisection invariant keeps the lower endpoint passing; two-point algebra for (d). Key lemma: Lemma RS.

Difficulty. S for (a)–(e); (c′) S–M.

Lean core. Lemma RS identity; Theorem G optimality among refinements of fixed targets; (b) identity; (c) finite inequality with the Popoviciu step; (d) two-state recurrence bound.

Dependencies. S-TARGET (Lemma 0: d finite); M3 (A_0 through $\pi_0(S_0)$ for (d)); M1(c) (frozen $\boldsymbol{C}$); S-IDENT (K25 lemma, K104 for (e)). Feeds M2(b), M8.

Risk. The resolution/snap premises; the (c′) pilot-progress lemma unchecked; no compiled replay of the coded rule.

Fallback + stop rule. (a), (b), (d), (e) with $T \le T_e$ and the interval inequality of (c); drop the counts if the resolution/progress lemma fails. Stop: one wave on (c)/(c′). No binding goal rests on STG alone: M2(b)'s stage-count clause has its own checked fallback $T \le P + \lceil {D_*}/{c_\rho} \rceil + 1$.

IC3 design, exhibited. (a)–(c): Dε (n = 40, p = 8, built pairs, cross 10^{-5}–10^{-4}; W1PC1:71) and my p = 16 exhibit (W2PC1:61), component checks (production dispatches exact); C's family W with the coded pilot replica, 600

adversarial runs (W2PC2:50, 156), SMC-dispatched for n ≥ 9^6. (d): the truth/start-mode two-state restriction of LH (c = 2), masked ($A_0 \to \infty$).

Numerical companion (≤ 20 seeds). O-STAGES numerics $\log_2 A_0 + 7.7 \pm 0.4$ (exists); PC2 `c8_pilot.py` (exists); E-pilot 16 cases (W2PC1:55). For the HQ: coded-grid replica on W at n = 9^6, 10 seeds, T against P + $\lceil D_V/c_\rho \rceil + 1$ and $2\lceil D_*/c_\rho \rceil + 2$.

Plain reading

For the grid rule as coded: each accepted tempering stage passes an exact restricted-ESS test ($\rho_V \geq 0.9$ on the candidate set V), the construction terminates, and on success $T \leq T_e$ (a). The restricted ESS converts into a full-bridge statement through the coverage factor $a_u a_w / a_v^2$ (b). With bisection resolution $\leq h_V/2$ and no ceiling/snap failure, the number of stages is at most the number of retained pilot targets plus $\lceil D_V/c_\rho \rceil + 1$, where D_V is the spread of d on V (c); with M2's global range bound and $2^{-60} \leq h/2$, $T \leq 4 + 2D_*/c_\rho$ (c′, a proof sketch). On an *ideal* exact-frontier two-state grid (not the coded one), a dominance crossing of gap A_0 costs $\log_2 A_0 + O(1)$ stages (d). The as-coded masked stage count is open (f).

Not claimed / not proved

- Conjecture P (as-coded masked T_n) is OPEN → FUTURE_WORK (K31, `CJ06`); favourable O(1) (K29) → `CJ23`. The shipped guarantee is only $T \leq T_e$.
- (d), (e) are about ideal exact-frontier or fine grids, "labelled comparison objects"; the gap to the coded grid is K31.
- The coded rule is checked against a Python translation only ("no compiled bitwise replay"); the resolution/snap premises and the unchecked (c′) pilot-progress lemma are risks.
- No "exact/minimal" claim: greedy optimality holds only among refinements of the retained pilot targets, up to the 2^{-60} slack. STG-G/L0/L1/L are author-only (SF none, XF none); explicit decimal constants of STG-M are GAP.
- Rebuild convergence N5(b), $P(S_0 \in V)$ and P(schedule_limit) are future work (`CJ06`, `OB05`).

Open items for this card. Lemma RS, then (b) and (c) proofs and the cross-family check; a compiled replay of the coded rule; (c′) pilot-progress lemma; `CJ06`, `CJ23`, `OB05`. HQ charter deliverables (verbatim):

Charter M7 · Theorem STG.

• Goal: the depth theorem of the coded grid (what ESS 0.9 certifies; how many stages; the ideal-grid crossing law).

• Statement: §S2 card M7, clauses (a)–(e), (c′) cited from M2.

• First lemma: Lemma RS ($\rho_V(u,v) = Z_V(u)Z_V(w)/Z_V(v)^2$, monotone by convexity of the log-MGF), then the coverage identity (b) and the interval inequality (c).

• Lean target: Lemma RS; Theorem G optimality among refinements of fixed pilot targets; (b) identity; (c) finite inequality.

• Independent check (cross-family): PC1 (Codex Sol) on the proofs of (b), (c) (both author-side Claude); PC2 (Opus) on (c′), Seat C's pilot-progress lemma; a compiled replay of the coded rule against the Python translation (registrar).

• Fallback and stop: (a), (b), (d), (e) with $T \leq T_e$; drop the counts of (c)/(c′) if the resolution/progress lemma fails; one wave.

• Escalation evidence: a coded-rule behaviour not captured by the translation (snap/validation) invalidating (a); none binding (M2(b)'s stage-count clause has its own checked fallback).

4.3.8 TH08 · M8 · Proposition D (work, depth and width at the island level)

TH08 · M8 · Proposition D (work, depth and width at the island level)

APPROVED (F.B., 2026-11-07) Placement: main text, wave 3 (HQ charter `HQ_CHARTERS/M8.md`); HQ recommended keeping it main (Seat D's alternative: supplement, main = 7) [NBH/branch_snapshots/MEET-THM/FOR_FELIPE_THEOREM_LIST.md:49].

Status (registry, verbatim): MA-PropD "PROVED (counting, given the code's structure)" AUTHOR-ONLY; T32(i) "PROVED (SF: VERIFIED (W7-SEC); XF: none)", displayed rate GAP (A-REV5), registry ⚑ CONFLICT marker carried; Cost-T38 "PROVED-UNDER (SF: VERIFIED (W3-VT38E); XF: GAP (A-REV5))"; calibration share "seat-derived, unchecked" (N15); assembly PROOF SKETCH (by this seat, unchecked). **Fidelity:** island level, v3.1 as shipped; "depth free of N, H, R" struck (MEET-v31 decision 6); K88's particle-parallel critical path is v3.2 and not claimed.

K-IDs: K87, K88, K89, K06, K64; K90 remark.

Authoring seat / card source: Seat D (Fable 5.1, fresh-eyes), `R3_D.md:160–177`.

Binding goal: none; placement rests on objective (2) [NBH/02_theory.md:1134-1145].

TLF clerk text (verbatim). Chair's integration note.

1. K90 is a remark (the card has it as a remark); the Song–Liang §6 comparison is in the same units with no speed claim (MINUTES §2.1).

2. Seat D's note for Felipe (R3_D §S5, MINUTES §2.1): if objective (2) is served by a sentence in the introduction citing the supplement, Prop D moves to the supplement and main = 7. Nothing else changes.

3. The execution-DAG audit (N15) and N14(c) are named implementation premises (MINUTES §2.3).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:407

Final statement. Units: sequential elementary-step slots (METHOD §2 #19); score evaluations and memory counted separately.

Hypotheses (design/budget, all explicit functions of (n, p)): the shipped budgets $N_0 = \max\{2{,}500, n, p\}$, $H_0 = \lceil 512N_0/2{,}500 \rceil$, $L = \max\{4, \lceil \log_2(n+1) \rceil\}$, $\ell = \lceil \log_2(n+p+1) \rceil$, $T \le T_e$ on success, $B = \min\{\lfloor \sqrt{n} \rfloor, \lfloor \sqrt{p} \rfloor, 16\}$, K; $m := p + B + 2$; $k^* := \min(K, \lfloor p/2 \rfloor)$; $R' = 4k$ declared; w island workers.

Implementation premise: the execution DAG as specified (METHOD §6.8, §8; AMENDMENT_HQ_01 #18, #19, #42); the integrated DAG audit is open.

`TH08.a` **(a) Final run.** Work = $R' \cdot N_e \cdot (T + L + 4) \cdot m$ slots; island depth = $N_e \cdot (T + L + 4) \cdot m$, in $\lceil R'/w \rceil$ waves; width R'. With $w \ge R'$ the depth does not depend on R'.

`TH08.b` **(b) Calibration.** Work = $\Sigma_i N_i(T_i + L_i + 4)m$ over the executed pilot, diagnostic and rebuild trajectories, each with its own grid; per tier at most $(1 + 4I_e)H_e$ trajectories enter work and $(1 + I_e)H_e$ sequential trajectory equivalents enter four-worker depth, $I_e \le \ell$ rebuilds per tier ($\le 2\ell$ with escalation).

`TH08.c` **(c) Searches.** Each local search (final, extra from c_4 when $R' > 4$, LS0) scores at most $(\ell + 1)[p + k^*(p - k^*)]$ models; each is counted separately.

`TH08.d` **(d) Polynomial work and scheduling.** For $p \le n^A$, under the budgets above and any permitted polynomial R', total work is polynomial in n; on w workers, time $\le W/w + D$ (Brent).

`TH08.e` **(e) Ceilings.** $\Theta(p)$ slots per round and $\Theta(p^2)$ Gram memory are not removed by islands or particles.

Remarks (not clauses): the SL §6 comparison in the same units states worker counts, n-vs-p scaling, T, score cost and the loss criterion, with no same-order, exact-output or speed claim; the single-tier calibration share $0.0512 + 0.2048 \cdot I$ is seat-derived, unchecked; on C's family M7 + M2(b) give $T \le 4 + 2D_*/c_\rho = O_P(1)$, so the island depth is $O_P(N_e(L + D_*/c_\rho)(p + B))$.

Clause rows and their registry statuses. TH08.a (M8(a) Final run): MA-PropD "PROVED (counting, given the code's structure)" AUTHOR-ONLY

TH08.b (M8(b) Calibration): "calibration share seat-derived, unchecked" (N15)

TH08.c (M8(c) Searches): T32(i) "PROVED (SF: VERIFIED (W7-SEC); XF: none)", displayed rate GAP (A-REV5)

TH08.d (M8(d) Polynomial work and scheduling): Cost-T38 "PROVED-UNDER (SF: VERIFIED (W3-VT38E); XF: GAP (A-REV5))"

TH08.e (M8(e) Ceilings): assembly PROOF SKETCH (by this seat, unchecked)

Remaining card fields (verbatim, TLF 403, 408–420). Card Prop D · Proposition D (work, depth and width at the island level) — K87, K88, K89, K06, K64; K90 remark

Guarantee. In a referee's units, islands add width without adding depth; the work is polynomial; accuracy grows with width through M1(b),(d′) at fixed depth; no speed-up over sequential MCMC is claimed.

Serves. Objective (2); Song–Liang §6 ("same-unit accounting; speed comparison not established"); THM-INV row 8 (work–depth), I1/I2/I9 ceilings.

Status (verbatim). MA-PropD "PROVED (counting, given the code's structure)" AUTHOR-ONLY; T32(i) "PROVED (SF: VERIFIED (W7-SEC); XF: none)", displayed rate GAP (A-REV5), registry ⚑ CONFLICT marker carried; Cost-T38 "PROVED-UNDER (SF: VERIFIED (W3-VT38E); XF: GAP (A-REV5))"; calibration share "seat-derived, unchecked" (N15); assembly PROOF SKETCH (by this seat, unchecked). **Fidelity:** island level, v3.1 as shipped; "depth free of N, H, R" struck (MEET-v31 decision 6); K88's particle-parallel critical path is v3.2 and not claimed.

Verdict rows used. (a): W1PC2 D6 "final run" TRUE as counting; W2PC2:133 TRUE-AS-WRITTEN (m = p + B + 2; waves); W2PC1:205 TRUE-AS-WRITTEN as corrected accounting. (b): W1PC2 D6 "calibration and LS" TRUE (amendments #18, #42); W2PC2:133; per-trajectory sums repair W1PC1 §13 AMBIGUOUS → REPAIRED (W2PC1:205). (c): W1PC2 D6 TRUE. (d): W1PC1 §13 "polynomial/Brent" TRUE-AS-WRITTEN. (e): W1PC1 §13 first row (memory, block count) DESIGN-VERIFIED on Dε. SL remark: W1PC1 §13 NEEDS-HYPOTHESIS, repaired to a remark (W2PC1:205).

Self-checked flag. MA-PropD author-only; T32(i) XF none; calibration share unchecked. Route to a cross-family code read in the charter.

Novelty. MODEST (accounting, TCS-standard). Checked: Brent 1974 (cited for W/w + D); Zhou–Yang–Vats–Roberts–Rosenthal 2022 §3 (LIT-MCMC #2): different kernel and per-step cost, no comparison made; Song–Liang 2015 §6 (`theory_campaign/inputs/`): $K_n = \sqrt{P_n}$, $O(P_n^{(1+\alpha)/2}) + O(nP_n)$, quoted in the remark only.

Route / key lemma. Counting over the execution DAG. **Key lemma:** the trajectory inventory (every executed trajectory with its own (N_i, T_i, L_i); every completed search), which is exactly the W1PC1 §13 repair. Difficulty: S (mathematics), M (code audit).

Lean core. The counting identity (sum of products over the inventory), the scan bound $p + k^*(p - k^*) \le p + p^2/4$, and Brent's inequality as finite lemmas; low value but certifiable. Nothing rests on probability theory.

Dependencies. Uses: M7 ($T \le T_e$; on C's family the M2(b) envelope), S-CERT (K64 search bound), S-OUT (K83 worker equivalence, K84 stacking nesting for the extra search). Used by: the CD remark, COVERAGE_MATRIX §6 row.

Risk. The DAG audit finds an uncounted serial component (e.g. a reduction across islands that is not trivially $O(R')$), or a calibration path that is not a trajectory sum; then (a)/(b) are restated, not dropped.

Fallback + stop rule. Fallback: W and D_{PEL} as symbolic sums without rates and without the calibration-share remark. Stop: simplification stops at any unaudited component; if the audit shows depth growing with R' at $w \ge R'$, clause (a)'s last sentence is withdrawn and HQ is told (objective (2) wording changes). No binding goal is involved; a $p > 2{,}500$ demonstration would fire the C6 trigger and go to HQ.

IC3 design (exhibited). Any admissible input. Checked on Dε (J = 4 blocks, $\Sigma 2^{|C_b|} = 16 = p2^B/B$; W1PC1 §13 DESIGN-VERIFIED) and on C's family W (W2PC2:133 DESIGN-VERIFIED, any input).

Numerical companion (≤ 20 seeds). Filed: calibration/final ratio predicted 0.46/0.67/1.69 vs observed 0.45/0.62/1.76 on 3 designs. New: the timing tabulation of AMENDMENT_HQ_01 §4 item 9

(`timings$calibration`, `timings$final`, rebuilds, tier, T) on ≤ 20 saved study fits, compared with (a)–(b) evaluated from the saved budgets; Owls ≤ 8 cores, read-only.

Plain reading

Counted in sequential elementary-step slots, with the execution DAG as specified (its audit is open): the final run costs $R' \cdot N_e \cdot (T+L+4) \cdot m$ slots of work, but each island is only $N_e(T+L+4)m$ deep and islands run in $\lceil R'/w \rceil$ waves on w workers, so with $w \geq R'$ workers depth does not grow with the number of islands (a). Calibration and every local search are counted too (b), (c). For $p \leq n^4$, under the shipped budgets and any permitted polynomial R', the total work is polynomial in n and scheduling obeys Brent's bound (d). Islands and particles do not remove the $\Theta(p)$ per-round sequential factor or the $\Theta(p^2)$ Gram memory (e). No speed-up over MCMC is claimed.

Not claimed / not proved

- No speed comparison: the Song–Liang §6 comparison is a remark "with no same-order, exact-output or speed claim"; accuracy-matched comparison and K91 are dropped/FW (DR05, CJ12).
- The execution-DAG audit (N15) is open; "depth free of N, H, R" was struck (MD06); particle-parallel depth (v3.2) is not claimed.
- The calibration share $0.0512 + 0.2048 \cdot I$ is "seat-derived, unchecked"; MA-PropD AUTHOR-ONLY; T32(i) displayed rate GAP (A-REV5); assembly PROOF SKETCH.
- In general only $T \leq T_e$ is available; $T = O_P(1)$ is stated only on C's family W (remark).

Open items for this card. OB15 (N15, DAG audit: the trajectory inventory, first lemma), OB14 (N14(c); K84/N14(a)); CJ09 (R vs N allocation), CJ12 (separation from single-chain MCMC). HQ charter deliverables (verbatim):

Charter Prop D · Proposition D.

• Goal: state objective (2) in a referee's units: island-level work, depth and width, polynomial work, depth independent of R′ at w ≥ R′ workers; no speed claim.

• Statement: §S2 card Prop D, clauses (a)–(e); remarks are not claims.

• First lemma: the trajectory inventory of the execution DAG (every executed pilot, diagnostic, rebuild and final trajectory with its own (N_i, T_i, L_i); every completed search), read from `v3_engine.h`, `v3_calib.h`, `v3_fit.R`; this is the W1PC1 §13 repair and the open audit (W2PC2:40).

• Lean target: the counting identity over the inventory, the scan bound, Brent's inequality; a finite certificate.

• Independent-check plan: cross-family code read (Codex on Claude accounting, or the reverse), reproducing (a)–(c) from the code paths; numerical: the AMENDMENT_HQ_01 §4 item 9 timing tabulation on ≤ 20 saved fits against the formulas, Owls ≤ 8 cores.

• Fallback and stop: symbolic sums without rates and without the calibration-share remark; stop simplification at any unaudited component; never state a comparator speed-up.

• Escalation evidence: the audit shows depth growing with R′ at w ≥ R′, or a non-trajectory calibration cost of the same order as the final run; either changes the objective (2) wording, which is Felipe's.

4.3.9 TH09 · S-T · Theorem 1, target consistency T03♯ (K93, K04)

TH09 · S-T · Theorem 1, target consistency T03♯ (K93, K04)

APPROVED (F.B., 2026-11-07) Placement: supplement card, cited in the main text; registered with LP-FOUND, no HQ work (wave 1, cited) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:32,438].
Status (registry, verbatim): T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2). T02(i♯), T02(ii♯): PROVED; SF/XF VERIFIED. T01–T03: PROVED-LOCKED (wave 3; external review queued), SAME-FAMILY-ONLY
Statement source: Seat C's full self-contained card C9 (R2_C); Seats A, B, D file it as "unchanged from R1" with scope notes; "Compatible; no choice needed" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:438].
Kind: the registry classifies TH09 as one of the supplement cards TH09–TH20 [NBH/ID_REGISTRY.md:4]. It is algorithm-free (a statement about the exact posterior π only).

TLF clerk text (verbatim). Decision note (MINUTES §2.2, `theory_campaign/MEET-THM/MINUTES.md:60`): content = Theorem 1, target consistency T03♯ (K93, K04). Notes and decisions = Unanimous since Round 2; cited in the main text

Registered here as a cited supplement; no HQ work (registered with LP-FOUND). A, B and D file the statement as "unchanged from R1" with scope notes; C filed the full self-contained statement (C9 with its hypotheses table). Compatible; no choice needed.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:448-451

R1, R. Fixed-design triangular array, fixed $s_0 \geq 1$, $p \leq n^A$ for fixed A, $s_0 \leq K$ eventually; G; $\mu = X\theta^*$, $\|\mu\|^2 \leq Mn$, and normalized truth Gram eigenvalue at least fixed $\phi_0^2 > 0$.

Tail cutoff. With per-model prior weight $w(k)$, $R_m = \sum_{k=\max(m+1,0)}^{K} \binom{p}{k} w(k)/w(s_0)$, and $m_n^\sharp = \min\{m \geq s_0 : R_m \leq n^{-(s_0+2)/2}\}$. It is deterministic and data-free.

Anchored eigenvalues. $j_0 = \lceil (2A+1)s_0 \rceil + 1$. $\mathrm{SE}_\lambda(j_0)$: every $S_0 \cup J$, $|J| \leq j_0$, has normalized minimum eigenvalue at least ϕ_0^2. $\mathrm{SE}_w(m_n^\sharp; g_0)$: every $S_0 \cup J$, $|J| \leq m_n^\sharp$, has unnormalized minimum eigenvalue at least fixed $g_0 > 0$.

Signal thresholds. BM-∞: $\theta_{\min}^{*2} n\phi_0^2/(\sigma^2 \log n) \to \infty$. Alternatively BM($c$) uses a fixed $c > c_0(s_0, A)$, with $c_0 = \max\{4C_*, 32[s_0(A+1)+2]\}$, C_* the locked T03 proof constant, exactly as FIX §1.1.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:462

Final statement. Under R1, G, R, $\mathrm{SE}_\lambda(j_0)$, $\mathrm{SE}_w(m_n^\sharp; g_0)$, and BM-∞ or BM($c > c_0$), $1 - \pi(S_0) = O_P(\sqrt{\log n/n})$; exact HPM and MPM are consistent. Without beta-min, for $J_n(c) = \{j \in S_0 : \theta_j^{*2} \geq c\sigma^2 \log n/(n\phi_0^2)\}$, $c > c_0$, $\pi(S \not\supseteq J_n(c)) = O_P(\sqrt{\log n/n})$, hence all corresponding exact PIPs converge to one. Empty $J_n(c)$ gives a vacuous inclusion statement.

Remaining card fields (verbatim, TLF 463–469) and the seats' one-line filings (TLF 478, 489, 498). Repairs. PC1/PC2 statistical clauses: no expectation rate inferred from O_P, no universal onset n_0, no random-design or growing-sparsity extension.

Status. T03♯: **PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2)**. T02(i♯), T02(ii♯): **PROVED; SF/XF VERIFIED**.

Placement. Supplement.

Route / key lemma. Cite FIX §1 and the filed exact-prior-tail/selection checks; no re-derivation.

Lean core. Finite prior-tail and posterior-to-decision inequalities; Gaussian asymptotics remain prose.

Fallback / stop. Fixed-p, uniformly nonsingular correlated design; narrow scope if hypotheses cannot be matched.

IC3. LH satisfies fixed sparsity, bounded energy, positive full Gram eigenvalue and BM-∞. Otherwise unchanged from R1 §D D8.

S1 · Theorem T (K93; K61 remark). Statement unchanged from R1 §D C15 with the scope sentence ($p \le n^A$ from prior and technique; e^{Cn^α} OPEN) and onset $n \gtrsim 2\cdot 10^6$. Status: T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)); T01–T03: PROVED-LOCKED (wave 3; external review queued), SAME-FAMILY-ONLY. Supplement; cited in the main setup. Rest unchanged.

Final statement unchanged from R1 §D D13 (R1 fixed s_0, $p \le n^A$; G, R, SE_λ, SE_w, BM-∞ or BM($c > c_0$); $1 - \pi(S_0) = O_P(\sqrt{(\log n/n)})$; HPM/MPM consistent; large-signal inclusion without beta-min). Status: T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)); T01–T03: PROVED-LOCKED (wave 3; external review queued) SAME-FAMILY-ONLY; onset $n \gtrsim 2\cdot 10^6$ disclosed. Repairs: none needed (PC1 §12 TRUE). Rest unchanged.

M12 Theorem 1 (T03♯) — K93, K04. Unchanged from R1 §D M12; supplement, cited in main text. | "PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2))"; T01-T03 "PROVED-LOCKED (wave 3; external review queued)"

Seat A's R1 base statement, to which the "scope sentence" above refers (verbatim):

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:417

Statement. R1 hypotheses (s_0 fixed, $p \le n^A$, G, R, SE, BM-∞ or BM(c)), fixed design: $1 - \pi(S_0) = O_P(\sqrt{(\log n/n)})$; HPM, MPM of π equal S_0 w.p. → 1; $P(S\dagger \ne S_0) \le 2E_0[1 - \pi(S_0)]$; large signals get PIP → 1 without beta-min; exact BB(1,p) tail T02(i♯). Scope sentence: $p \le n^A$ comes from the prior and proof technique; e^{Cn^α} OPEN.

Plain reading

For a fixed design sequence with fixed sparsity s_0, $p \le n^A$, Gaussian noise, bounded signal energy, sparse-eigenvalue conditions anchored at S_0 and a beta-min condition, the exact posterior puts mass $1 - O_P(\sqrt{\log n/n})$ on the true model, so the exact HPM and MPM are consistent; without beta-min, for a fixed $c > c_0$ every true coordinate with $\theta_j^{*2} \ge c\sigma^2 \log n/(n\phi_0^2)$ (the set $J_n(c)$) has exact PIP → 1, and $\pi(S \not\supseteq J_n(c)) = O_P(\sqrt{\log n/n})$; an empty $J_n(c)$ makes this vacuous. This is a statement about the target π, not about the sampler.

Not claimed / not proved

- "no expectation rate inferred from O_P, no universal onset n_0, no random-design or growing-sparsity extension" (Repairs row). $p \le n^A$ comes "from the prior and proof technique; e^{Cn^α} OPEN" (R1_A:417); exponential p is CJ18; growing support CJ19; Hellinger/predictive contraction CJ20.
- Asymptotic only, onset $n \gtrsim 2 \cdot 10^6$ (the binding condition $q_n \le \frac{1}{2}$ first holds at $n \approx 2.1 \cdot 10^6/2.5 \cdot 10^6$; the earlier "1.2e5" claim was an ERROR) [NBH/branch_snapshots/MEET-THM/R0_DIFF.md:207]. A finite-n bound on the posterior-miss probability $\Pi(X)$ is CJ14; at $n = 200$ the study has at least 565 strict posterior misses (LIM07).
- T01–T03 are PROVED-LOCKED but SAME-FAMILY-ONLY ("external review queued"). The registry records "beta-min constant ≥ 43× above executed signals" [NBH/03_results_and_conclusions.md:102].

Open items for this card. No HQ work is planned (cited card). Future work touching it: CJ14, CJ18, CJ19, CJ20; the candidate LH-iid extension inside TH161 (part (P), unreviewed) concerns this layer.

4.4 Consistency and regime map

This subsection answers, for the loop, three different questions that are easy to conflate: **(i)** is the *exact* posterior π concentrated on the true model S_0? **(ii)** does the *sampler's output* converge to π (or to $h(S_0)$)? **(iii)** is the *reported* HPM $\hat{S}^+$ equal to S_0? A result for one layer says nothing about another unless the card composes them explicitly (M1(i) composes (i)+(ii); M5 composes (i)+(iii); M4(f) composes (i)+(iii)+store retention).

Layer (i): the exact posterior. The only consistency theorem for π itself is TH09 (S-T, T03♯): under R1 (fixed-design triangular array, fixed s_0, $p \le n^A$), G, R, $\mathrm{SE}_\lambda(j_0)$, $\mathrm{SE}_w(m_n^\sharp; g_0)$ and BM-∞ or BM($c > c_0$), $1 - \pi(S_0) = O_P(\sqrt{\log n / n})$ and exact HPM/MPM are consistent (statement in the card below). It is algorithm-free [NBH/branch_snapshots/MEET-THM/R1_A.md:420], asymptotic, and its binding condition first holds at $n \approx 2.1 \cdot 10^6 / 2.5 \cdot 10^6$, hence the onset disclosure "n ≳ 2·10⁶" [NBH/branch_snapshots/MEET-THM/R0_DIFF.md:207]; [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:478,489]. On LH, $\pi(S_0) \to 1$ enters through Thm5-LH (status "PROVED; SF: VERIFIED (W3-VEXP); XF: none; SAME-FAMILY-ONLY") and M3(f) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:210,216]; S-T's own IC3 row states "LH satisfies fixed sparsity, bounded energy, positive full Gram eigenvalue and BM-∞" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:469]. The scoped thesis remark chosen by the chair (Seat C's wording, Chair's choice 5, [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1349]) says explicitly what is *not* proved:

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1280-1283

K96 disposition. Remark only. There is no retained non-corollary thesis sentence.

Fixed data. For each admissible finite input and successful frozen calibration, the idealized start, kernels and unnormalized identities target the specified posterior. Normalized finite-run posterior estimates are not generally unbiased or exact. Independent stacking gives fixed-data consistency for the stated records.

Asymptotic accuracy. Exact-target selection uses C9's model/design conditions. Growing-dimensional summary accuracy is established by the proposed C1/C2 assembly on W with explicit stacking. Masked output/store accuracy uses LH and LS0 with output/retention premises. These are separate scopes.

Correlation and cost. Dependence affects the start discrepancy, available bounds and search difficulty. The results do not prove that correlation only increases cost, that all correlated designs admit polynomial accuracy budgets, or that the posterior itself remains statistically consistent under arbitrary correlation.

Layer (ii): sampler output. Regime-free and fixed-data: M1(a) (cited validity and unbiasedness given a frozen $\mathcal{C}$), M1(b) (Lemma N1, PROOF SKETCH), M1(e) (islandwise $E[W_r | H_r(h) - h(S_0)| \mid \mathcal{C}] \le 2\delta_1$, given N1; route "(e): …then N1 on $1\{S \ne S_0\}$"), M1(f) (finite-R' tails, "vacuous at R′ = 4"; fixed data $\hat{\mu}(h) \to \pi(h)$ a.s. as $R' \to \infty$ [cited]) and M4(a)–(e) (store brackets, output-event certificate). Asymptotic: M1(i), $\hat{\mu}(h) \to h(S_0)$ in probability jointly with output, only with the optional stacking configuration $R'_n = 4\lceil \log_2(n + p + 1) \rceil$ and under T03♯'s hypotheses on a family with $\mathrm{osc}\, d \le D_* = O_P(1)$; exhibited on W [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:109,111,116,125]. Masked: M4(f) on LH bounds the *reported store* against π jointly with output [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:254]. Terminal-law TV/KL and the credible-set clause are in S-IDENT C15-TV/KL/CI (TH13, status "C15-TV, C15-KL, C15-CI: PROOF SKETCH, unchecked (assembly)" [NBH/ID_REGISTRY.md:93]). The TV/KL error of the final pooled estimate in the masked regime is CJ02 (FW2, FUTURE_WORK) and polynomial mixing of the $\varphi = 1$ round is CJ04 (FW4) [NBH/ID_REGISTRY.md:104,106].

Layer (iii): the reported HPM. M5(0) holds at every n for any X: $P(\mathcal{O} \cap \{\hat{S}^+ \ne S_0\}) \le 2E_0[1 - \pi(S_0)] + D_n$ (written with the δ_1 correction of the final audit), plus the N200 bracket; M5(a) is de-

terministic under TD'_∞ on Lat(S°), with $\ell \geq \tau(C)$ and a completed final scan. Convergence of the joint error $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\})$ to zero for the reported HPM is stated only on LH (no delivery claim): M5(b1-LH) with LS0 (premise N3 OPEN) and the labelled no-LS0 variant (b2-LH) on STAB-LH [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:282,292].

Regime	(i) exact π concentrates on S_0	(ii) sampler output → π / → $h(S_0)$	(iii) reported $\hat{S}^+ = S_0$
any X (fixed data, every n)	not covered (no regime-free consistency; T03♯ needs its hypotheses).	TH01.a (T12, T31(i): PROVED-LOCKED [...] see card); TH01.b (N1: PROOF SKETCH); TH01.e, TH01.f finite bounds (vacuous at $R' = 4$; fixed-data a.s. limit as $R' \to \infty$ cited); TH04.a–TH04.e store identities, brackets when S_0 is stored, and certificate (output form needs the GUARD-W audit).	TH05.0: bound $2E_0[1 - \pi(S_0)] + D_n$, not a limit (T38-(0) "PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))"); TH05.a under TD'_∞ (with $\ell \geq \tau(C)$, completed final scan); TH04.c retention (if S_0 scored and $\delta_1 < \frac{1}{2}$); TH01.h decision transfer ((h), (i): PROVED (by this seat, unchecked)).
fixed design under T03♯ hypotheses (R1: fixed s_0, $p \leq n^A$; G, R, SE_λ, SE_w, BM)	TH09: $1 - \pi(S_0) = O_P(\sqrt{\log n/n})$; exact HPM, MPM consistent (T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2)). Asymptotic; onset $n \gtrsim 2 \cdot 10^6$.	TH01.i only if in addition $\operatorname{osc} d \leq D_* = O_P(1)$ and the optional stacking $R'_n = 4\lceil \log_2(n + p + 1) \rceil$ ((h), (i): PROVED (by this seat, unchecked)); the shipped $R' = 4$ carries only fixed-R statements.	TH05.0: the posterior term $2E_0[1 - \pi(S_0)]$ is driven by S-T (M5 uses S-T for "$\delta_1 \to 0$"), but the never-scored term D_n is not controlled in general; TH05.afav (labelled iid-draw variant): $D_n \to 0$ under (F) (T38-(a) "PROVED-UNDER (SF: VERIFIED* (inputs); XF: none)").
LH (shared LH for M3(f), M4(f), M6: exact Gram, $p = 9\lfloor n^{1/3}/9 \rfloor$, $n \geq 9^6$; M5(b1-LH) is stated for a general LH with SIZE)	TH03.f: $\pi(S_0) \to 1$ with $\pi_0(S_0) \to 0$ via Thm5-LH (SAME-FAMILY-ONLY) and S-T.	TH04.f: $P(\mathcal{O} \cap \{TV(\pi_{\mathcal{T}}, \pi) > \varepsilon\}) \to 0$ for the reported *store* ((f) assembly: PROOF SKETCH); M1(i) is not exhibited on LH (its IC3 family is W); final pooled estimate TV/KL under masking = CJ02 (FUTURE_WORK).	TH05.b1LH with LS0: $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$ (T38-b1-LH "PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8))"; premise N3 "OPEN"); TH05.b2LH without LS0 on STAB-LH (B2E-B2LH "PROVED-UNDER (SF: none; XF: VERIFIED* (A-REV7))").
W (groups of nine at 1/4, cross $\varepsilon_n = 1/(np)$, two $n^{-1/4}$ signals)	No S-T row for W in TLF; T03♯'s hypotheses enter TH01.i as premises.	TH01.i (IC3 family; $D_* = O_P(1)$ checked numerically, median 1.08 → 1.04) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:125,178]; M2's constants (TH02) are exhibited on W.	not covered.
LH-iid, $18 \leq p \leq n^A$ (API-PXNGAP; *candidate, unreviewed, not on the list*)	part (P) of TH161: $1 - \pi(S_0) = O_P(\sqrt{\log n/n})$ under the joint law.	not covered ("masked PIP/evidence accuracy" explicitly not claimed).	TH161 (C): $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \leq 2E_0\delta_1 + D_n \to 0$, PROVED (candidate, API-PXNGAP) given T22(i),(ii), T03, CONC, W7 Thm B; no SF/XF review in the archive; related future work CJ10 (FW10, K71).

Regime	(i) exact π concentrates on S_0	(ii) sampler output $\to \pi$ / $\to h(S_0)$	(iii) reported $\hat{S}^+ = S_0$
study design ($n = 200$, $p \in \{500, 1000\}$, $s_0 = 14$, $B = 14$, groups of 5)	not covered: asymptotic results only, onset $n \gtrsim 2 \cdot 10^6$; at least 565 strict posterior misses among 2,394 completed fits (LIM07).	only the regime-free fixed-data statements (M1(a)–(h), M4(a)–(e)); asymptotic clauses do not apply.	not covered by (b1-LH): the study generator (groups of 5, $2g \le B$) is not LH (CJ25, FW25). Finite-n account: M5(0)/N200 bracket, predicted 558.1 vs 571 observed wrong; $\Pi(X)$ measured, not bounded (CJ14); $2E_0\delta_1 > 1$ at $n = 200$.

Sources for the table: [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:96,109-111,247-254,292,302,462-469]; [NBH/branch_snapshots/API-PXNGAP/REPORT.md:3-23,44-58]; [NBH/ID_REGISTRY.md:309]; [NBH/03_results_and_conclusions.md:157,183,190,195-206]; [NBH/branch_snapshots/MEET-THM/R0_DIFF.md:207].

Plain reading — the three layers in one paragraph

For any data set, under the named implementation premises (EX-A, exact uniforms, a frozen calibration, ...), the method's unnormalised island outputs are unbiased, and its stored record gives deterministic identities, two-sided brackets when S_0 is stored, and a computable (loose) envelope; that is the regime-free layer. Statements that the exact posterior is *right* (concentrates on S_0) are asymptotic, need fixed sparsity, $p \le n^A$, eigenvalue and beta-min conditions, and start to bite only around $n \approx 2 \cdot 10^6$. Asymptotic statements about the *reported* model hold only on LH designs: M5(b1-LH) is stated for a general LH (exact Gram, $p \le n - 1$, c linked pairs, group size g, within-correlation ρ, link $|\psi| < \rho$, fixed β) satisfying SIZE, under the T03♯ hypotheses on LH and the named implementation premises (N3 OPEN); the shared exhibit (groups of nine, $p \approx n^{1/3}$, $n \ge 9^6$) is one instance: $P(\mathcal{O} \cap \{\hat{S}^+ \ne S_0\}) \to 0$, hence $\liminf P(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup P(U_n)$; delivery ($P(\mathcal{O}) \to 1$) is not claimed, and nothing is conditional on delivery; an unreviewed candidate extends this to LH-iid with $p \le n^A$. None of the asymptotic statements covers the study's own design; there, M5(0)'s finite-n composition bracket (N200) is the account: its prediction, "a lower bound plus a measured term" ($\Pi(X)$ measured, not bounded), is 558.1 against 571 observed wrong [NBH/03_results_and_conclusions.md:204-206].

Not claimed / not proved — consistency

- That the posterior "remains statistically consistent under arbitrary correlation": explicitly not proved (scoped remark above, TLF:1283).
- Any expectation rate inferred from O_P, any universal onset n_0, any random-design or growing-sparsity extension of T03♯ [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:463]; exponential p (CJ18), growing support (CJ19), Hellinger/predictive contraction (CJ20).
- A finite-n upper bound on the posterior-miss probability $\Pi(X)$ (CJ14, FW14); delivery $P(U) \to 0$ (CJ13, FW13).
- Masked final-estimate TV/KL (CJ02); general production mixing (CJ04); as-coded masked stage count (CJ06).
- Reported-HPM consistency off LH, and on the study generator (groups of 5) in particular (CJ25; API-PXNGAP obstruction (O1) [NBH/branch_snapshots/API-PXNGAP/REPORT.md:17-20]).

4.5 Dependency graph

Edges are taken from the cards' *Dependencies* fields (and the M5/M1 "Lean core" fields for the Lean node); an arrow $X \to Y$ means "Y uses X". The consolidated seat edge lists are in [NBH/branch_snapshots/

MEET-THM/REGISTRY_PAPER1.md] (§dependencies) and in each charter's "Dependencies" section.

4.6 Lean coverage

No main card is Lean-certified as a whole [NBH/03_results_and_conclusions.md:22]. The Lean library SmcBvs (branch L-LEAN, Lean v4.35.0-rc2) formalises abstract finite versions of a few campaign-era results: "lake build is green: 0 errors, 0 warnings, 0 sorry. A per-declaration list (101 declarations) is in DECL_STATUS.md" [NBH/branch_snapshots/L-LEAN/STATUS.md:3]; "#print axioms shows only propext, Classical.choice and Quot.sound for all 56 audited declarations, and no sorryAx" [NBH/branch_snapshots/L-LEAN/REPORT.md:7]. Fidelity of the statements: Fable 5.1 GREEN; Astra round 1 RED on T37 coverage only, round 2 GREEN [NBH/branch_snapshots/L-LEAN/REPORT.md:9-12].

File	decls	sorry-free	What is formalised (abstract, finite)	Card link stated in the sources
Basic.lean	10	yes	finite posterior post, argmax facts, $\pi(S) > \frac{1}{2} \Rightarrow$ strict max	support for T21/T07
T21HPM.lean	8	yes	T21 (C1): hpm_eq_truth; t21_decomposition; t21_C1: $P(U \vee \hat{S} \neq S_0) \leq P(U) + 2E\delta_1 + P(U^c, S_0 \notin \mathcal{S})$, $\delta_1 = 1 - \pi(S_0)$	M5 Lean core: "T21HPM.lean (t21_C1 sorry-free)"; to add: (0) "as a finite Boolean lemma with the output event"
T37Greedy.lean	18	yes	Theorem A abstract core: Lemma 0, A1–A3, A4-PD for any best-improvement map	M5 Lean core (sorry-free); to add: "the capped-scan certification interface ($\ell \geq \tau$(C), completed scan ⇒ certified return)"
T37Lattice.lean	29	yes	Theorem A on the model lattice: move table, theoremA2_lattice_count, theoremA3_lattice_tau ($\ell \geq \tau$)	M5(a) (as above)
T07Bayes.lean	28	yes	T07: size-only ceiling; HPM minimises 0–1 loss, MPM minimises Hamming loss; reported-model clauses	not cited by a main card (campaign-era TH106)
T11Miss.lean	5	yes	T11(iii): $P(s \text{ never drawn in } Q \text{ draws}) = (1 - \pi_0\{s\})^Q \leq \exp(-Q\pi_0\{s\})$	same finite form as M6(a); reuse for TH06.a "not stated in the sources: **UNCERTAIN**" [NBH/03_results_and_conclusions.md:80]
T12Core.lean	3	yes, *partial*	T12(i.a): invariance_step, resample_move_step, unbiased_of_step; "**The resample–move recursion hstep is an explicit hypothesis**: the probabilistic island model is not formalised"	M1 Lean core: "T12Core unbiased_of_step with hstep discharged for multinomial resampling + an invariant kernel" (not done)

Sources: [NBH/branch_snapshots/L-LEAN/STATUS.md:5-11]; [NBH/branch_snapshots/L-LEAN/DECL_STATUS.md:3]; [NBH/branch_snapshots/L-LEAN/REPORT.md:19-92]; Lean-core rows [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:121,300]. Declaration counts per file are counted from DECL_STATUS.md. Every main card also names a planned "Lean core" (in its card fields above); none of these planned cores exists yet except the M5 parts listed in the table.

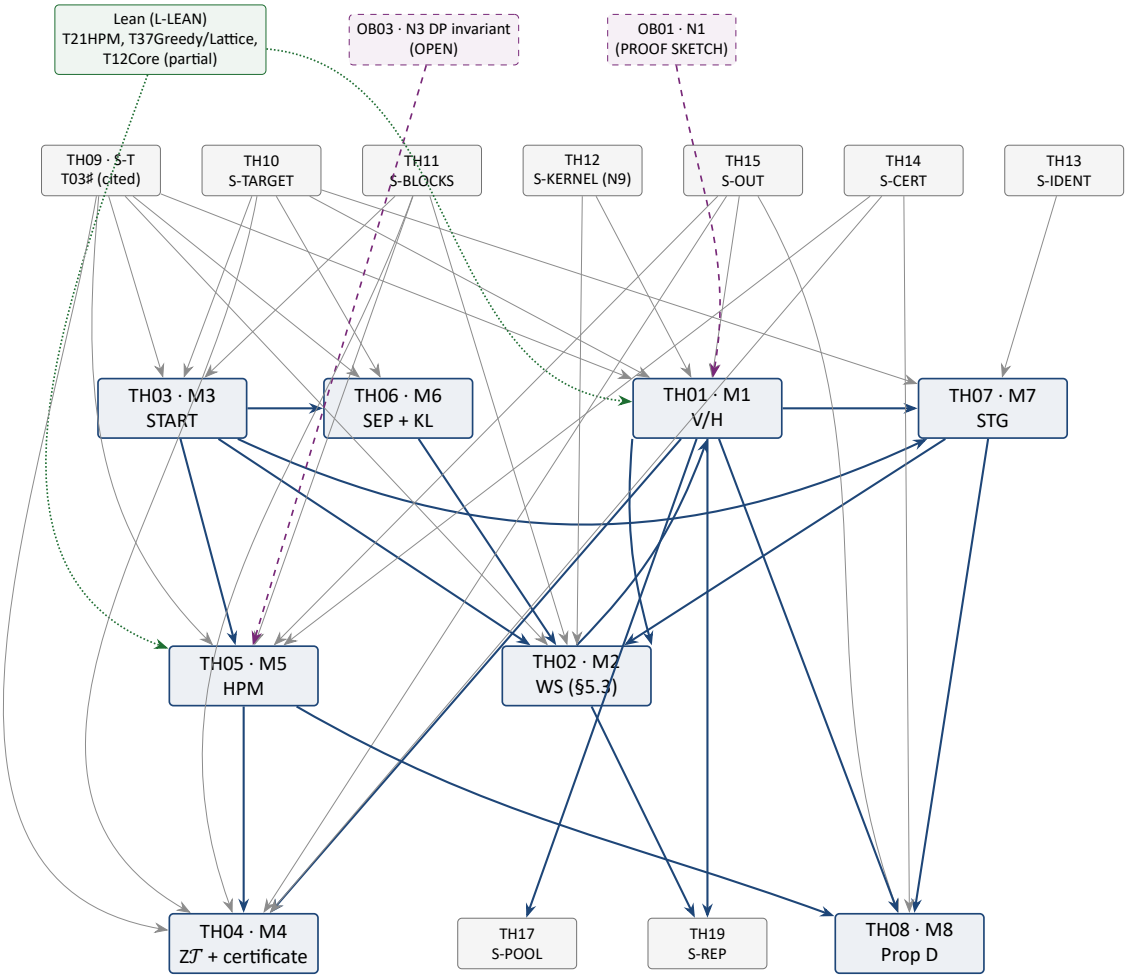


Figure 1: Dependencies of the main cards M1–M8 (blue boxes) on each other, on the key supplement cards (grey), on the two open obligations that carry the most weight (N1 behind M1(b) and the binding §13.1; N3 behind M5(b1-LH) and hence M4(f); dashed purple), and on the existing Lean library (dotted green). An arrow X → Y means "Y uses X". Clause-level meaning of the main-to-main edges: M1 → M2 is $E\,W_r = 1$ for M2(b5); M2 → M1 is M2(b) for M1(i); M1 → M7 is the frozen-$\mathcal{C}$ contract M1(c); M5 → M4 is (b1-LH) for M4(f); M5 → M8 is the search-cost term; Lean → M5 covers parts of (0) and (a) (`t21_C1`, Theorem A); Lean → M1 is the partial `T12Core` (hypothesis `hstep` not discharged). S-T edges are as the cards state them (for M1 only clause (i)). Not drawn: cited lemmas (CL), K-level edges, and the supplement packs' internal dependencies.

Sources: [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:122,175,222,266,301,342,382,416] (Dependencies rows); [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:121,300] (Lean-core rows).

5 Supplement, obligations and future work

This section holds everything of paper 1's theory that is not a main card (main cards TH01–TH09 are in the previous section): the eleven supplement cards TH10–TH20, the dissolved card TH21 and the thesis remark TH22; the seventeen proof obligations OB01–OB17; the 27 future-work items CJ01–CJ27; the six drops DR01–DR06; the nine inherent limitations LIM01–LIM09; and the 23 cited lemmas CL01–CL23. The exact status words come from part A of the ID registry [NBH/ID_REGISTRY.md:90-234].

Approval. APPROVED (F.B., 2026-11-07): Felipe Bedoya approved the paper-1 theorem list, including the supplement cards, on 2026-11-07. Approval fixes the cards as current targets. It changes no proof status.

How the supplement cards are stored in the source. THEOREM_LIST_FINAL.md (TLF) Part II gives, for each supplement card, the decision note from MINUTES §2.2 and then "the statement copied verbatim from the seat card(s) that carry it". Several seats (A, B, C, D) often filed the same card in different packaging. TLF keeps all of them and does not merge them [NBH/02_theory.md:1184]. MINUTES §2.2 says the statements are "as in the cited R2/R3 cards, with the SYNTH_R2 §3 repairs" [NBH/branch_snapshots/MEET-THM/MINUTES.md:56]. The chair's editorial choices fix wording in five places only: K29, K30, the S-Z variance clause, S-LEAK, and the Thesis CD wording [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1333-1349]. So each card below works like this:

- where TLF names a final card or the chair chose a wording, that text is transcribed in full and the other seat texts are transcribed after it;
- where TLF says the seat texts are complementary, every seat text is transcribed, because each one carries clauses the others lack;
- several seat texts read "Final statement as R1 §D … with: …" or "Rest unchanged". Their Round-1 base statements are not in TLF. Each card transcribes them from the R1 files under "Round-1 base statements". A base statement is superseded wherever the Round-2 text changes it.
- the only abridgements are Seat D's Round-1 base cards that D's Round-2 rows mark "unchanged". These are listed by span and not transcribed: R1_D S1 [NBH/branch_snapshots/MEET-THM/R1_D.md:376], S2 [NBH/branch_snapshots/MEET-THM/R1_D.md:391], S3 [NBH/branch_snapshots/MEET-THM/R1_D.md:406], S6 [NBH/branch_snapshots/MEET-THM/R1_D.md:446], S8 [NBH/branch_snapshots/MEET-THM/R1_D.md:465], S9 [NBH/branch_snapshots/MEET-THM/R1_D.md:475], S11 [NBH/branch_snapshots/MEET-THM/R1_D.md:493] and S12 [NBH/branch_snapshots/MEET-THM/R1_D.md:501]. Seats A, B and C cover the same content in the transcribed texts. R1_D S4, S5 and S7 are transcribed.

Who works on what. Supplement cards are worked by lemma-pack HQs:

- LP-FOUND, wave 1: TH10–TH12, with S-T (TH09) cited only;
- LP-IDENT, wave 2: TH13;
- LP-AUDIT, wave 3: TH14, TH15;
- LP-REST, wave 4: TH16–TH18;
- LP-REP, wave 4: TH19, TH20.

[NBH/branch_snapshots/MEET-THM/MINUTES.md:108-113]. Every pack charter carries the chair's generic fallback rule IC7, verbatim: "the card ships in the narrowest form its filed checks support. A clause whose proof or cross-family check fails after one repair cycle moves to FUTURE_WORK.md with the reason. It is never relabelled as a limitation, and no status is upgraded." The generic stop rule is "one proof attempt plus one cross-family check-and-repair cycle per card" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-FOUND.md:112-113].

Status vocabulary used below.

- SF = same-family check; XF = cross-family check. "VERIFIED*" means verified with a qualification.

- PC1 = the cross-family proof checker gpt-6.1-sol, run on Seats A, B and D. PC2 = Opus 5.5, run on Seat C.
- "TRUE-AS-WRITTEN" is a statement-sanity verdict, not a proof.

[NBH/02_theory.md:2794] [NBH/branch_snapshots/MEET-THM/MINUTES.md:22-26].

For agents — how to use this section

Each card below gives: the status words; the final statement(s) verbatim; repairs; dependencies and placement; a plain reading; and what is *not* claimed. Every "PROVED" in the obligation texts (§5.4) is a candidate: "not reviewed, not adopted" [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:9]. Do not upgrade any status word. Items in §5.5 (future work) and §5.6 (drops and limitations) record routes that were tried and why they were deferred or dropped. Read them before proposing a new result.

5.1 Index of the supplement cards

Dependencies are summarised from the lemma-pack charter edge lists ("$X \to Y$": Y uses X) [NBH/02_theory.md:1190-1203].

ID	Native	Pack (wave)	Uses / used by
TH10	S-TARGET	LP-FOUND (1)	→ M1 TH01, M2 TH02 (Lemma 0), M3 TH03, M4(d) TH04 (N17), M7 TH07, S-REP TH19, S-Z TH16, S-T TH09
TH11	S-BLOCKS	LP-FOUND (1)	→ M2 (IC3 partition), M3, M5 TH05, M8 TH08, M4(f) (SIZE on LH)
TH12	S-KERNEL	LP-FOUND (1)	→ M1(a), M3, M2(b4), S-IDENT TH13, S-REP; K09/N3 carried as hypothesis in M5 and M4(f)
TH13	S-IDENT	LP-IDENT (2)	← S-KERNEL (K103), M1, M2 (W rates), M7; → M2(b), M6 TH06, M7(e), S-REP
TH14	S-CERT	LP-AUDIT (3)	→ M4 (card semantics), M5 (Theorem A, LS-score, scan bound), M8 (search cost)
TH15	S-OUT	LP-AUDIT (3)	→ M1(c,d), M4(e), M5, M8, S-REP
TH16	S-Z	LP-REST (4)	← M2(a) (D), M3, M4 (envelope), S-TARGET
TH17	S-POOL	LP-REST (4)	← M1, M3
TH18	S-LIM	LP-REST (4)	← M4 (cited)
TH19	S-REP	LP-REP (4)	uses M1; M2 for informative W rates; M3 for discrepancy definitions; S-KERNEL for exact initialization; S-OUT for delivered-event transfer
TH20	S-LEAK	LP-REP (4)	← M2, M3, S-T
TH21	S-STGI	—	dissolved; K30 = M7(d) TH07.d; K29 = CJ23
TH22	Thesis CD (K96)	—	remark after the main theorems

5.2 Supplement cards TH10–TH20

TH10 · S-TARGET · Propriety, invariance, envelopes, exact branch, odds certificate (K01–K05, K76, K95)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): N17: PROVED. T01: PROVED-LOCKED (wave 3; external review queued); SAME-FAMILY-ONLY. Remaining assembly: PROOF SKETCH, unchecked. N12 eventual dispatch: seat-proved, unchecked; T26-exact-branch-paren: XF ERROR (corrected) [NBH/ID_REGISTRY.md:90]
Decision note (MINUTES §2.2). "K95 here (R3-4 unanimous)" [NBH/branch_snapshots/MEET-THM/MINUTES.md:62]. TLF: "K95 is here (R3-4, unanimous). Seat statements below are complementary (same content, different packaging)." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:508]. There is no chair's choice for this card, so all seat texts are transcribed.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:516-525

Seat C (R2_C), card C12 — K01/K02/K03/K04/K05/K76.
Final statement. For finite nonconstant standardized data, $n \geq 3, p \geq 1$, the shipped ridge/IG target is proper, including rank-deficient submodels. With $H_S = (n+1)G_S + I$, $Q_S = Y - nz_S' H_S^{-1} z_S$, $Q_S \geq Y/(n+1)$, and $1 \leq \det H_S / \det(G_S + I) \leq (n+1)^{|S|}$. Thus $m(S), Z \leq U_m = C_n(2b_0 + Y/(n+1))^{-\nu}$, where C_n is the shipped score's common prefactor.
Other clauses. Use the normalized capped prior once. Target probabilities are invariant under admissible affine rescaling and equivariant under column permutation. The contrast reduction uses IG shape $a_0 + 1/2$, rate $b_0 s_y^2$. Real-arithmetic exact enumeration returns exact summaries; every fixed-p sequence eventually enters that branch.
Repairs. PC2 status repair; target equivariance only. The pairwise IG perturbation coefficient is $2\nu b_0$, not universally nb_0.
Status. N17: **PROVED**. T01: **PROVED-LOCKED (wave 3; external review queued); SAME-FAMILY-ONLY**. Remaining assembly: **PROOF SKETCH, unchecked**.
Placement. Supplement.
Route / Lean. Checked spectral envelope and reduction; finite normalization/enumeration.
Fallback / stop. Proper target and exact-arithmetic enumeration; no machine-certification claim.
IC3. C10's duplicated, rank-deficient correlated design.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:534

Seat A (R2_A) S5. S5 · Proposition TARGET (K01–K05, K76). As R1 §D C1 with (ii) restricted to the target (affine invariance, permutation equivariance); algorithmic equivariance in law only for a tie-free partition and decision structure (PC1:301); K05 superset bound; K76 eventual exact dispatch (N12 seat-proved, unchecked); (iii) constant $2\nu b_0$ (PC2:216; PC1:300). Status: N17 PROVED; rest D/E proposals, TRUE-AS-WRITTEN (PC1:300); T01 reduction PROVED-LOCKED (wave 3; external review queued), SAME-FAMILY-ONLY. Supplement.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:543

Seat A (R2_A): K95 odds-certificate clause (fragment of A's Z$\mathcal{T}$ card, clause (f)).
(f) odds certificate: $\pi(S_0) = 1/(1+A(S_0))$; $A \leq \eta < 1 \Rightarrow$ HPM = MPM = S_0 (PC1:130).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:552-554

Seat B (R2_B) S7 · Proposition TARGET (K01, K02, K03, K04, K95, K76) · supplement.
Final statement as R1 §D D17 with: target affine-invariant and permutation-equivariant; algorithmic output need not be equivariant, even in law, under index ties (PC1 §20); $A(S_0) \leq \eta < 1 \Rightarrow$ HPM = MPM = S_0 (PC1 §20 NEEDS-HYPOTHESIS conceded); $2\nu b_0$ as the exact pairwise IG constant (PC2 cluster 20). Status: N17: PROVED (PC1 claim 1); T01 reduction: PROVED-LOCKED (wave 3; external review queued); N12 eventual dispatch: seat-proved, unchecked; T26-exact-branch-paren: XF ERROR (corrected). Rest unchanged.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:563-575

Seat D (R2_D) rows.
S1 target lemmas — K01–K04 (+K76, K95 via S3/S12): Target invariance and permutation equivariance; algorithmic equivariance in law only without partition/decision ties (cluster 20). [status: unchanged]
S3 decision rules — K62 moved to H′(f); K60, K61 cited; K95, K101: η < 1 stated (cluster 20). [status: unchanged]
S12 fixed p/tiny n — K76, K02: unchanged. [status: N12 "seat-proved, unchecked"]

Round-1 base statements. Seat A's "As R1 §D C1" and Seat B's "as R1 §D D17" point to the texts below. They are superseded wherever the Round-2 texts above say "with:" (equivariance restricted to the target; constant $2\nu b_0$).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:166

Seat A, R1 C1 · TARGET — Statement. For every (X, y) passing preprocessing (n ≥ 3, rank-deficient X_S allowed): (i) m(S) is finite and positive, Z finite, π proper, with envelopes $Q_S \geq y^\top y/(n+1)$, $\log \mathrm{BF}(S) = -\frac{1}{2}\log D_S - \nu \log(1-x_S)$, $0 \leq x_S < 1$ (OWS Lemma 0); (ii) π, Z/m(∅), PIPs are invariant under column/response affine maps and permutation-equivariant; per-seed output equivariant in law only; (iii) the working posterior equals the contrast-regression posterior with $\mathrm{IG}(a_0+\frac{1}{2}, b_0 s_y^2)$ noise; (iv) BB(1,p) sums to one, W_K divides once, size masses decay with ratio ≤ p/(2p−1); (v) $Z \leq C_n(2b_0 + y^\top y/(n+1))^{-\nu}$.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:442

Seat B, R1 D17 · Proposition TARGET — statement. $H_S \succ 0$ for rank-deficient X_S; $Q_S \geq y^\top y/(n+1)$; $-(|S|/2)\log(n+1) \leq \log \mathrm{BF}(S) \leq \nu \log[(2b_0+Y)/(2b_0+Y/(n+1))]$; affine invariance and permutation equivariance (ties break fixed-seed equivariance); contrast reduction with $\mathrm{IG}(a_0+\frac{1}{2}, b_0 s_y^2)$; $|\Delta \log \mathrm{BF}| \leq 2\nu b_0 |1/Q_1 - 1/Q_2|$ when $b_0 \to 0$; BB(1,p) identities; π(S₀) = 1/(1+A(S₀)) and A ≤ η ⇒ HPM = MPM = S₀; fixed p is eventually enumerated exactly.

Dependencies. Used by M1, M2 (Lemma 0), M3, M4(d) (N17), M7, S-REP, S-Z and S-T [NBH/02_theory.md:1193]. Obligations: OB17 (N17, PROVED) and OB12 (N12, eventual exact dispatch K76). K60/K61 (Barbieri–Berger) are cited as CL12.

Plain reading

The shipped target is a proper probability distribution on models for any finite non-constant standardised data with $n \geq 3$, including rank-deficient submodels. Every model's score is bounded by an explicit analytic envelope. Posterior probabilities do not change under admissible affine rescaling of the data, and they move with a permutation of the columns. For fixed p, the method eventually takes the exact-enumeration branch (N12: "seat-proved, unchecked"). If the posterior odds against the truth are at most $\eta < 1$, then HPM and MPM both equal S_0. All of this is about the *target*, for fixed data.

Not claimed / not proved

- Equivariance is claimed for the target only. The algorithm's output is "equivariant in law only for a tie-free partition and decision structure". Under index ties it "need not be equivariant, even in law" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:534].
- No machine-certification claim is made [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:524].
- Eventual exact dispatch (K76) is "seat-proved, unchecked".
- The remaining assembly is a PROOF SKETCH, unchecked.
- T01 is SAME-FAMILY-ONLY, with external review still queued.

TH11 · S-BLOCKS · Separation ⇒ groups kept; SIZE ⇒ equality (K07, K14, K06)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): PART-ThmA: PROVED; SF none; XF VERIFIED*. Counting/graph assembly: PROOF SKETCH, unchecked. PART-pairing remains OPEN, excluded. PART-CE2: COUNTEREXAMPLE [NBH/ID_REGISTRY.md:91]

Decision note. MINUTES gives no further decision ("—") [NBH/branch_snapshots/MEET-THM/MINUTES.md:68]. TLF: "Seats agree; C11 is the fullest. A files it inside S10 (BLOCKS part), D inside the S2 row." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:585]. C11 is transcribed first.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:593-601

Seat C (R2_C), card C11 — K07/K06/K22 (fullest).
Final statement. Suppose every within-group distance is strictly smaller than every between-group distance and every group has size at most B. Complete linkage forms each group before cross-group merging; cutting and whole-cluster packing preserve groups. If additionally $B < 2g_{\min}$, output blocks equal groups. For actual blocks, $\lceil p/B \rceil \le J \le 2\lceil p/B \rceil$, $\sum_b 2^{|C_b|} \le p2^B/B$; the symmetrized capped graph has at most pB edges and no isolated vertex when $p > 1$.
Repairs. PC2 Shared objects and cluster 4: replace unqualified eventual recovery by $n \ge 9^6$ for both nine-variable exhibits. PC1 cluster 6: explicit SIZE.
Status. PART-ThmA: **PROVED; SF none; XF VERIFIED***. Counting/graph assembly: **PROOF SKETCH, unchecked**. PART-pairing remains **OPEN**, excluded.
Placement. Supplement.
Route / Lean. Cite checked separation/packing result; finite partition, bin-count and graph-union arguments.
Fallback / stop. Group preservation and deterministic counts only; no size-five pairing law.
IC3. W and LH satisfy strict separation and $9 \le B \le 16 < 18$ for $n \ge 9^6$. Otherwise unchanged from R1 §D D10.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:610-614

Seat B (R2_B) S11 · Proposition BLOCKS (K07, K14, K06) · supplement.
Unchanged from R1 §D D21 (PC1 §19 B.D21 TRUE-AS-WRITTEN; preservation only, equality needs SIZE). Status: PART-ThmA: PROVED (SF: none; XF: VERIFIED* (A-REV9)); PART-pairing: OPEN; PART-CE2: COUNTEREXAMPLE.
Dropped from the list since R1: D10 (CD → remark), D20 (K94 → FUTURE_WORK).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:628

Seat D (R2_D) S2 row. S2 start machinery — K06, K07, K09, K14, K22: Graph "no isolated vertex" for p ≥ 2 (cluster 19); K07 preservation only, equality under SIZE. [status: unchanged]

Seat A's BLOCKS clauses are the last clauses of A's S10. That text is transcribed once, under TH13 [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:620].

Round-1 base statement. Seat B's "Unchanged from R1 §D D21" points to the text below. Seat C's "Otherwise unchanged from R1 §D D10" (IC3 row only) points to [NBH/branch_snapshots/MEET-THM/R1_C.md:297].

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:488

Seat B, R1 D21 · Proposition BLOCKS — statement. If every within-group distance is strictly below every between-group distance and groups have size ≤ B, complete linkage forms each group first and cut/packing never splits a group (so cross-block |r| ≤ max between-group |r|); every sub-model inside a block is scored at table time, so the HPM beats every block-local model and within-block masking of any order ≤ B costs nothing; study packing (2g ≤ B) is N8, OPEN.

Dependencies. Used by M2 (IC3 partition), M3, M5, M8 and M4(f) (SIZE on LH) [NBH/02_theory.md:1194]. Obligation OB08 (N8, PART-pairing) has its final placement here: "S-BLOCKS (preservation, SIZE equality); FW25" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:75]. See also CJ25.

Plain reading

Suppose the correlation clusters are strictly separated (every within-group distance is smaller than every between-group distance) and every group has size at most B. Then complete linkage, the size cut and the packing never split a group. If in addition every block is too small to hold two groups ($B < 2g_{\min}$), the blocks are exactly the groups. The deterministic counts of blocks, table entries and graph edges (no isolated vertex for $p > 1$) hold for any design; their assembly is a proof sketch.

Not claimed / not proved

- No pairing law when two groups fit in one block (PART-pairing OPEN). This includes the study generator ($g = 5, B = 14$).
- PART-CE2 is a counterexample: unequal groups are not merged.
- Equality of blocks and groups needs SIZE ($B < 2g_{\min}$). Without it only preservation holds.
- The counting and graph assembly is a PROOF SKETCH, unchecked.

[NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:597-600] [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:184-186]

TH12 · S-KERNEL · N9 kernel facts, graph (p ≥ 2), start sampler, minorisation (K20, K21, K08, K09, K22, K103)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): N9K: PROVED (candidate), N9-KERNEL §§0–3 VERIFIED WITH AMENDMENT (PC1, cross-family); T09: PROVED-LOCKED (wave 1); N3: OPEN; K22, K23, K103: PROVED (by this seat, unchecked), PC1 §19 TRUE-AS-WRITTEN. C10 (K09): PROOF SKETCH, unchecked [NBH/ID_REGISTRY.md:92]

Decision note. "N9 with CHK amendments. K09 is the tie-DP proposition (C10)" [NBH/branch_snapshots/MEET-THM/MINUTES.md:63]. TLF adds the following [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:638]:

- The CHK amendments are in `evidence/CHK_INHERITED/REPORT.md` §5 (VERIFIED WITH AMENDMENT).
- "K23 (R3-4): FUTURE_WORK (A, B, D); C dissents." The S-KERNEL HQ "may cite it as an attributed lemma without a theorem number".

- "The Dirichlet-form clause $\mathcal{E}_Q \geq \tfrac{1}{2}\mathcal{E}_{R_2}$ appears in seats A (S10) and B (S5) as filed; it is K23 and is not a numbered supplement claim."

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:646-648

Seat B (R2_B) S5 · Proposition KERNEL (K20, K21, K08, K09, K22, K23, K103) · supplement, cited lemma + Lean.
Final statement as R1 §D D16 with: reversibility under the sufficient conditions of N9 (a fixed order need not be self-adjoint); the graph clause for p ≥ 2 (PC1 §19); $Q_\varphi(S,\cdot) \geq \pi_\varphi / M_\varphi$ (E's constant $1/M_\varphi$, PC1 §19 TRUE; $1/(mM_\varphi)$ also valid) and $\|Q_\varphi^k(S,\cdot) - \pi_\varphi\|_{\mathrm{TV}} \leq (1 - 1/M_\varphi)^k$ with $M_\varphi \leq \exp\{\varphi \cdot \operatorname{osc} d\}$; $\mathcal{E}_Q \geq \frac{1}{2}\mathcal{E}_{R_2}$, $R_2 = (1/m)\Sigma P_j^2$; tie-aware DP returns the prescribed maximiser (N3 OPEN, hypothesis). Status: N9K: PROVED (candidate), N9-KERNEL §§0–3 VERIFIED WITH AMENDMENT (PC1, cross-family); T09: PROVED-LOCKED (wave 1); N3: OPEN; K22, K23, K103: PROVED (by this seat, unchecked), PC1 §19 TRUE-AS-WRITTEN. Rest unchanged.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:657-666

Seat C (R2_C), card C10 — K09 (tie-DP proposition).
Final statement. For positive finite block tables, size-only capped prior and exact comparisons, max-plus convolution gives the best start score at each size. Choose the smallest maximizing size. In ascending index order, include an index exactly when an optimal completion of that size consistent with the existing prefix remains possible. The result is the prescribed lexicographically smallest start maximizer.
Repairs. PC2 §0/D9: the new duplicate-column exhibit below puts the tie at the global start mode, not merely somewhere in a table.
Status. Proposed finite mathematical proposition: **PROOF SKETCH, unchecked**. Registry N3's integrated tie-DP invariant remains **OPEN**; it is not being listed as a proved implementation theorem.
Placement. Supplement.
Route / key lemma. Optimal-completion invariant and prefix induction.
Lean core. Finite constrained maximization and lexicographic order.
Fallback / stop. Unique-maximizer instances. If the prefix invariant fails, drop the tie proposition and retain the integration gap.
IC3. $n = 40, p = 20$; columns 1 and 2 identical; remaining columns mutually orthogonal and orthogonal to them. Standardized y has squared correlation $1/2$ with the duplicate direction and zero with other columns. Complete linkage keeps the duplicate pair together; adding zero-score columns decreases the start score. The two singleton duplicates are tied global modes; SMC dispatch holds.

Seat C filed a scalar sanity script for the C10 exhibit [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:675-679]. It prints log start scores relative to the empty model: 0, 7.858020937 and 4.91339373 for $k = 0, 1, 2$, with $2^{20} = 1048576$ models against a dispatch cutoff of 251200. The filed interpretation, verbatim: "Scalar sanity check of the exhibit, not a DP implementation audit or an exact floating-point certificate."

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:693

Seat A (R2_A) S6. S6 · Proposition PAR (K63, K83, K84, K86, K09). Unchanged from R1 §D S1 (Lemma S candidate; MA-PropW: PROVED (code reading; executed check); MA-Wcal: PROVED (code reading only; not executed); IND-T extension audit pending; tie-DP invariant N3 OPEN; PC2:198 TRUE-AS-WRITTEN on random instances, DESIGN-ASSERTED). Supplement.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:704-710

Seat D (R2_D) rows.
S2 start machinery — K06, K07, K09, K14, K22: Graph "no isolated vertex" for p ≥ 2 (cluster 19); K07 preservation only, equality under SIZE. [status: unchanged]
S8 Doeblin/calculus — K103, K104, K108: TV ≤ $(1 - 1/M_\varphi)^L$ with the shipped $L = \max(4, \lceil \log_2(n+1) \rceil) \to \infty$; useful iff $M_1 = O(1)$. [status: PC2 D15-TV TRUE]

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:719

Seat D (R2_D) validity statement (the N9 statement, H′(a)).
Statement. Under EX-A, EX-S, CAL, GUARD-W, R′ declared. (a) Validity: each Gibbs flip, uniform swap, neighbour swap and independence move is γ_φ-reversible, the uniformly random-order round is γ_φ-invariant and self-adjoint, draw_pi0 and $\bar{Z}_0$ are exact (N9, cited); conditionally on frozen $\boldsymbol{\mathcal{C}}$, $E[\hat{Z}_r] = Z$, $E[\hat{Z}_r \widehat{\mathrm{PIP}}_{rj}] = Z \cdot \mathrm{PIP}_j$ (locked), and $E[\hat{Z}_r H_r(h)] = Z \cdot \pi(h)$ for h ∈ $\boldsymbol{\mathcal{H}}$ measurable in (X, y, $\boldsymbol{\mathcal{C}}$) (N1). Potential-run statements; the intersection inequalities P($\mathcal{O}$ ∩ B) ≤ $P_{\mathrm{pot}}(B)$ transfer by Lemma RO + GUARD-W; expectations do not (D.M1(c) repair). (b) For 0 ≤ h ≤ 1, $M_2 = E[W_r^2 | \boldsymbol{\mathcal{C}}]$: $P(|\hat{\mu}(h) - \pi(h)| > \varepsilon \mid \boldsymbol{\mathcal{C}}) \le \min\{1, 4(M_2 - 1)/R' + 4M_2/(R'\varepsilon^2)\}$ (PC1c3). (c) Joint inclusions $h_A = 1\{A \subseteq S\}$ (plain averages, $c_h = 0$); predeclared $h_I(S) = P(\beta_j \in I \mid S, X, y)$ for a finite endpoint grid, with a union bound over the grid (cluster 16 repair: no "any interval"). (d′) Growth clause: on any family with osc d = $O_P(1)$ (model/design condition) and $R'(n,p) = 4\ell(n,p)$, $\ell = \lceil \log_2(n+p+1) \rceil$, under T03♯: $\hat{\mu}(h) \to h(S_0)$ jointly with output. (e) Fixed data: $\hat{\mu}(h) \to \pi(h)$ a.s. as R′ → ∞ (cited). (f) MPM margin: if every exact PIP is ≥ δ from ½ and the estimated PIPs err < δ, the reported MPM equals the exact MPM (K62).

TLF files this D text under S-KERNEL as "the N9 statement, H′(a)". Clauses (b)–(f) are main-card material; their final form is M1 TH01. Seat C's C15-TV (K103) and Seat A's S10 (minorisation, graph, Dirichlet form) are transcribed once, under TH13 [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:685] [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:696].

Round-1 base statement. Seat B's "as R1 §D D16" points to the text below. It is superseded where B's Round-2 text changes it: the minorisation constant is $1/M_\phi$, and "a fixed order need not be self-adjoint". Seat A's "Unchanged from R1 §D S1" (PAR) is transcribed under TH15. Seat D's S2 base is at [NBH/branch_snapshots/MEET-THM/R1_D.md:391] (not transcribed).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:426

Seat B, R1 D16 · Proposition KERNEL — statement. Each elementary move is γ_φ-reversible (cap and self-loops included); the uniformly random order is self-adjoint on $L^2(q_\varphi)$ (a realised order is not); draw_pi0 exact, $\bar{Z}_0$ exact; tie-aware DP returns the prescribed maximiser (N3 OPEN); the symmetrised capped graph has no isolated vertex and ≤ pB edges; $Q_\varphi(S, \cdot) \ge \pi_\varphi/(mM_\varphi)$ with $M_\varphi \le \exp\{\varphi(d_{\max} - E_{\pi_0} d)\}$, so $\|Q_\varphi^k(S, \cdot) - \pi_\varphi\|_{\mathrm{TV}} \le (1 - 1/(mM_\varphi))^k$; $\mathcal{E}_Q \ge \frac{1}{2}\mathcal{E}_{R_2}$.

K23 placement. REG-A lists "K23 … PROVED (by this seat, unchecked)" in this card's status words. The decision, however, is: K23 is FUTURE_WORK (CJ05). It may be cited as an attributed lemma without a theorem number (CL22; Seat C's dissent, recorded as DS02). The Dirichlet-form clause is therefore not a numbered claim of this card [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:638] [NBH/branch_snapshots/MEET-THM/MINUTES.md:79].

Dependencies. Used by M1(a), M3, M2(b4), S-IDENT and S-REP. K09/N3 is carried as a hypothesis in M5 and M4(f) [NBH/02_theory.md:1195]. Obligations:

- OB09 (N9, the closure gate; final placement "S-KERNEL (cited N9 with CHK amendments)") [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:76];

- OB03 (N3; final placement "Named premise (MINUTES §2.3) + K09 proposition C10 in S-KERNEL; not FW") [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:70].

Cited: CL01. Kernel mixing is deferred: CJ04, CJ05.

Plain reading

Each elementary move of the kernel (Gibbs flip, uniform swap, neighbour swap, independence move) leaves the tempered target invariant and is reversible, under N9's sufficient conditions. A uniformly random order of the moves is self-adjoint. The start sampler draws exactly from the block-product start. One full round is minorised by the target with constant $1/M_\phi$, so after k rounds the TV distance is at most $(1 - 1/M_\phi)^k$; this is useful only when $M_1 = O(1)$ (K103: "PROVED (by this seat, unchecked)"). N9's kernel facts are "PROVED (candidate)". Under exact comparisons, the max-plus DP picks the prescribed tie-broken start mode; that last clause is a proof sketch.

Not claimed / not proved

- A fixed (non-random) move order need not be self-adjoint.
- The minorisation bound is vacuous under masking: it is "useful iff $M_1 = O(1)$" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:710].
- The integrated tie-DP invariant (N3) is OPEN; C10 is "not being listed as a proved implementation theorem".
- No mixing-time result for the φ = 1 round (CJ04).
- The Dirichlet comparison K23 is not a numbered claim (CJ05).
- N9K is "PROVED (candidate)".

UNCERTAIN — number of N9 sub-claims

HQ reports "All 7 sub-claims are PROVED (candidate)" at [pos 1550] and "All 8 sub-claims are PROVED (candidate)" at [pos 1593]. MEET-v31 MINUTES §9 says "all seven sub-claims". The archive does not reconcile them [NBH/02_theory.md:2786].

TH13 · S-IDENT · Restricted-ESS coverage identity, TI/KL/TV calculus, C15-CI clause (K25 lemma, K104, K55, K103)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): C15-TV, C15-KL, C15-CI: PROOF SKETCH, unchecked (assembly); Independence-sampler foundation: KNOWN—CITE; S9: PROVED (by this seat, unchecked); OWS-WS1: PROVED (SF: none; XF: none) [NBH/ID_REGISTRY.md:93]

Decision note. "Seat B's 'missing' item 1 (R3_B §S4) adopted: HQ Q2's posterior-content reading needs a named clause" [NBH/branch_snapshots/MEET-THM/MINUTES.md:67]. TLF adds three points [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:729]:

- "C15-CI is named as a clause of this card … It is the HQ Q2 deliverable: posterior-content error, not frequentist coverage (FW1)."
- "K33 is not here (M7; D's dissent recorded in MINUTES §2.3)."
- "The minorisation constant is $1/M_\varphi$ in A, B, C and D (B notes $1/(mM_\varphi)$ is also valid); compatible."

Seat C's C15 package is the fullest text and is transcribed first. The other seats' texts follow.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:737-745

Seat C (R2_C), card C15-TV — K103.
Final statement. Under A, $M_\phi = \max_S q_\phi(S)/\pi_0(S)$. The complete production round minorizes q_ϕ by $1/M_\phi$, without a $1/m$ factor. Thus a terminal particle's marginal law after L terminal rounds has TV at most q^L, $q = 1 - 1/M_1$. For a predeclared partition into d_0 cells, $P(TV(\widehat{\pi}_{\text{cell}}, \pi_{\text{cell}}) > \epsilon \mid \mathcal{C}) \le \min\{1, 4(M_2 - 1)/R' + 4d_0M_2/(R'\epsilon^2)\}$.
Repairs. PC2 cluster 16 and §0 item 4: distinguish marginal law from realized empirical law; label default-4 bound vacuous. Full-state TV uses singleton cells and may be useless.
Status. Production-round/record assembly: **PROOF SKETCH, unchecked**. Independence-sampler foundation: **KNOWN—CITE**.
Placement. Supplement, one package with C15-KL/CI.
Route / Lean. Cite PC2's checked minorization/vector-ratio clauses; compose minorization through invariant kernels.
Fallback / stop. Coarsened TV and bounded-functional error; no useful masked full-TV claim.
IC3. W gives $M_1 \le e^{D_*} = O_P(1)$; §R6 supplies explicit L, R'.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:754-762

Seat C (R2_C), card C15-KL — K103/K104.
Final statement. Under C15-TV, $\lambda_L = (1 - q^L)\pi + q^L\nu_L$. Consequently $KL(\pi\|\lambda_L) \le -\log(1 - q^L)$ and $KL(\lambda_L\|\pi) \le q^L \log(1/\pi_{\min})$. Also $KL(\widehat{\pi}\|\pi) \le 4TV(\widehat{\pi}, \pi)^2/\pi_{\min}$, with the corresponding minimum cell mass for coarsenings. If any positive-posterior state is absent, $KL(\pi\|\widehat{\pi}) = \infty$; in particular this occurs surely when $|\Omega| > 4NR'$.
Repairs. PC2 cluster 16; HQ Q3: reverse empirical KL obstruction retained, useful masked final-TV/KL not claimed.
Status. PROOF SKETCH, unchecked assembly; elementary divergence foundations cited.
Placement. Supplement, C15 package.
Route / Lean. Cite checked mixture/log-sum clauses; finite support and convexity.
Fallback / stop. Fixed-data directional bounds. No forward-KL convergence rate without controlling $\pi_{\min}$; no silent smoothing.
IC3. Every finite admissible correlated design for inequalities; W for terminal reverse-KL convergence. Otherwise unchanged from R1 §F D15-KL.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:771-780

Seat C (R2_C), card C15-CI — K103 (the named posterior-content credible-set clause).
Final statement. Under C15-TV, for any possibly output-selected union C of the chosen cells and nominal content $1 - \alpha$, $|\pi(C) - (1 - \alpha)| \le TV(\widehat{\pi}_{\text{cell}}, \pi_{\text{cell}}) + |\widehat{\pi}(C) - (1 - \alpha)|$. Therefore the probability, jointly with $\mathcal{O}$ under GUARD-W, that the left side exceeds ϵ plus the observed nominal-content discrepancy is bounded by C15-TV's tail bound.
Available interval. Use model-size cells, $d_0 = K + 1$, and an interval selected from the existing size distribution. Discrete overshoot is retained explicitly; estimated content need not equal nominal content.
Repairs. HQ Q2 and PC1/PC2 cluster 16: absolute posterior-content deviation, not frequentist coverage. A pointwise fixed-h bound does not validate arbitrary selected sets.
Status. PROOF SKETCH, unchecked.
Placement. Supplement, C15 package.
Route / Lean. Simultaneous cell-law control and deterministic set-mass inequality.
Fallback / stop. Predeclared interval endpoints with simultaneous CDF bounds; unavailable historical records remain unavailable.
IC3. W with §R6's stacking rate and actual size-distribution records.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:789-795

Seat C (R2_C), agenda-7 rows.
C15 empirical TV/content. The same R'_n handles any predeclared coarsening with $d_0 \le s^2$, including model size. It does not make singleton full-state TV informative when $|\Omega|$ is exponential.
C15 terminal law. On W, $M_1 \le e^{D_*} = O_P(1)$; the existing logarithmic L gives terminal marginal TV and reverse-KL convergence in probability. Forward-KL convergence additionally needs control of $\log(1/\pi_{\min})$, which is not supplied.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:804-806

Seat B (R2_B) S9 · Proposition IDENT (K25, K104, K54, K55) · supplement.
Final statement as R1 §D D19 with $\Lambda_{\text{len}} := \int_0^1 \text{sd}_{q_\varphi}(d)\, d\varphi \le \sqrt{J}$ (the length, not the cumulant $\Lambda(1)$; PC1 §16 AMBIGUOUS). Status: PROVED (by this seat, unchecked); OWS-WS1: PROVED (SF: none; XF: none); PC1 §14, §16 TRUE-AS-WRITTEN. Rest unchanged.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:815

Seat A (R2_A) S10. S10 · Proposition KL/TV and BLOCKS (K104, K103, K07, K14, K22).
Exponential-family identities with the length $\Lambda_{\text{len}} := \int \text{sd}_{q_\varphi}(d)\, d\varphi \le \sqrt{J}$ (renamed; PC1:257); minorisation $Q_\varphi(S,\cdot) \ge \pi_\varphi / M_\varphi$ (constant $1/M_\varphi$, PC1:290; PC2:184), vacuous under masking; Dirichlet form $\mathcal{E}_Q \ge \frac{1}{2}\mathcal{E}_{R_2}$ with $R_2 = (1/m)\Sigma P_j^2$ (PC1:289); graph: ≤ pB edges, no isolated vertex for p ≥ 2 (PC1:291); strict-gap preservation (PART-ThmA) and the SIZE equality clause; every block sub-model is scored. Supplement. Status: D/E proposals TRUE-AS-WRITTEN at the cited rows; OWS-WS1 PROVED (no review).

Seat D's S8 row, which is identical to the one transcribed under TH12, is at [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:824]. The Dirichlet-form clause in Seat A's S10 is K23. It is not a numbered claim (see TH12, CJ05).
Round-1 base statements. Seat B's "as R1 §D D19" and Seat A's base S4 (KL/TV lemmas) are below. B's Round-2 text renames Λ to the length Λ_{len}.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:464

Seat B, R1 D19 · Proposition IDENT — statement. $\rho_{\text{full}}(u,v) = \rho_V(u,v) \cdot a_u a_w / a_v^2$, $a_t = q_t(V)$, w = 2v−u; $(\log Z_\varphi)' = Ed$, ″ = Var d; $KL(\pi\|\pi_0) + KL(\pi_0\|\pi) = \int_0^1 \text{Var}_{q_\varphi}(d)\, d\varphi =: J$; $\Lambda \le \sqrt{\text{J}}$; $R_{\text{eff}}/R \to Z^2/E[\hat{Z}^2|\mathcal{C}]$; sandwich variance estimator consistent as R → ∞; √N and √R CLTs [cited].

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:550

Seat A, R1 S4 · KL/TV lemmas — Statement. (a) π_φ is an exponential family in d: Λ′ = E d, Λ″ = Var d, $\log(Z/(m(\emptyset)\bar{Z}_0)) = \int_0^1 E_{\pi_\varphi} d$, $KL(\pi\|\pi_0) + KL(\pi_0\|\pi) = \int_0^1 \text{Var}_{\pi_\varphi}(d) = J$, J = 0 iff d constant, $\Lambda \le \sqrt{\text{J}}$; (b) if every log mass is perturbed by ≤ ε: $\tilde{Z}/Z \in [e^{-\varepsilon}, e^{\varepsilon}]$, TV($\tilde{\pi}$, π) ≤ tanh ε, a winner with gap > 2ε is unchanged; kernel errors accumulate by path coupling as $\Sigma\delta_t$.

Clause (b) of Seat A's S4 is the K108 perturbation lemma. It ships in S-CERT (TH14). Seat C's "R1 §F D15-KL" base is [NBH/branch_snapshots/MEET-THM/R1_C.md:389]; the related D15-TV and D15-CI base cards are [NBH/branch_snapshots/MEET-THM/R1_C.md:377] and [NBH/branch_snapshots/MEET-THM/R1_C.md:401].
Charter. LP-IDENT, wave 2. The binding requirement, verbatim: "S-IDENT's C15-CI is the HQ Q2 deliverable (posterior-content reading of §5.4 credible-interval coverage; the frequentist half is FW1): it must be proved or its loss reported, not silently dropped" [NBH/branch_snapshots/MEET-THM/

HQ_CHARTERS/LP-IDENT.md:57]. The charter names no audit obligation [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-IDENT.md:61].

Dependencies. Uses S-KERNEL TH12 (K103), M1, M2 (W rates) and M7. Used by M2(b), M6, M7(e) and S-REP TH19 [NBH/02_theory.md:1196]. Consumes OB01 (N1): "C: M1/L-N1; S-REP and S-IDENT consume it" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:68]. OB10 (N10) has its final placement as "FW4 (K24); S-IDENT minorisation/terminal-law TV ship" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:77].

Plain reading

Under the card's setting (a proof-sketch assembly, unchecked): one production round at temperature φ is minorised by the tempered target with constant $1/M_\phi$. So a terminal particle's *marginal* law is within q^L of π in TV, with $q = 1 - 1/M_1$. For any cell partition fixed in advance, the pooled empirical cell law has a finite-R' TV tail bound. That bound transfers, jointly with output under GUARD-W, to the posterior-content error of any union of cells, including a credible set chosen from the output, up to the observed nominal-content discrepancy. This is a statement about posterior content, not frequentist coverage. KL bounds follow in the reverse direction. Forward KL needs control of $\pi_{\min}$, which is not supplied. On the reporting family W, where $M_1 \le e^{D_*} = O_P(1)$, the logarithmic L gives terminal marginal TV and reverse-KL convergence in probability. Under masking the bounds are vacuous.

Not claimed / not proved

- No useful masked full-TV claim.
- The default $R' = 4$ bound is vacuous.
- Singleton full-state TV is uninformative when $|\Omega|$ is exponential.
- No forward-KL convergence rate.
- No frequentist coverage of credible intervals (CJ01).
- No masked final-estimate TV/KL (CJ02).
- No mixing-time result (CJ04).
- The assembly is a PROOF SKETCH, unchecked.

[NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:740-761] [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:789-795]

TH14 · S-CERT · Local search, certificate semantics, B&B soundness, perturbation lemma (K64, K73, K74, K05, K108)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): N7 L1: CORRECT WITH AMENDMENT; OCERT-LemmaM: PROVED (SF: VERIFIED (O-VCERT); XF: VERIFIED (A-REV9)); OCERT-C4: PROVED (XF: VERIFIED*) [NBH/ID_REGISTRY.md:94]

Decision note. Content: "Local search (weak local maximum, ties allowed), certificate semantics, B&B soundness, perturbation lemma (K64, K73 without the region-certification clause, K74, K05, K108)". "K73 per C's withdrawal (R3-8)" [NBH/branch_snapshots/MEET-THM/MINUTES.md:64]. TLF: "K107 (rounding bound) is FUTURE_WORK (N16 OPEN); K108 (deterministic perturbation lemma) stays. Seats A and B state the same card; D files it as the S9 row." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:934]. The seat texts are complementary: A adds the scan bound and K74. Both are transcribed.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:942-944

Seat B (R2_B) S8 · Proposition CERT (K64, K73, K05, K108) · supplement.
Final statement as R1 §D D18 with: `local_certified` = no strictly improving feasible flip or swap at the scanned centre (weak local maximum; ties allowed) (PC1 §17 COUNTEREXAMPLE conceded); B&B exact when it completes, "unresolved" otherwise, in exact arithmetic; log-mass perturbation $\varepsilon \Rightarrow \tilde{Z}/Z \in [e^{-\varepsilon}, e^{\varepsilon}]$, TV ≤ tanh ε, winners with gap > 2ε unchanged. The rounding bound (K107) is withdrawn to FUTURE_WORK (N16 OPEN; T33-errbound COUNTEREXAMPLE). Status: N7 L1: CORRECT WITH AMENDMENT; OCERT-LemmaM: PROVED (SF: VERIFIED (O-VCERT); XF: VERIFIED (A-REV9)); OCERT-C4: PROVED (XF: VERIFIED*). Rest unchanged.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:953

Seat A (R2_A) S8. S8 · Proposition CERT (K64, K73, K74, K05, K108). `local_certified` = no strictly improving feasible flip/swap neighbour at the scanned centre (weak local maximum; ties allowed; PC1:272); ≤ (ℓ+1) scans of ≤ p + k*(p−k*) models per search; card R (OCERT-C4 + Lemma M) exact on completion, "unresolved" on exhaustion, nothing outside U; perturbation lemma ($\tilde{Z}/Z \in e^{\pm\varepsilon}$, TV ≤ tanh ε, winner gap > 2ε); limitation CEX-localmax(A2), CEX2. K107 → FUTURE_WORK (N16 OPEN; T33-errbound COUNTEREXAMPLE). Status unchanged from R1 §D C16/S4. Supplement.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:962

Seat D (R2_D) S9 row. S9 search/certificates — K64, K73, K74, K05, K108: K107 rounding bound → FUTURE_WORK (cluster 17); K108 deterministic perturbation lemma stays. [status: N16 "OPEN"]

Round-1 base statements. These are superseded where Round 2 changes them. In particular, "strict … local maximum" became "weak local maximum; ties allowed" (PC1 §17 COUNTEREXAMPLE), and B's rounding clause was withdrawn (K107 = CJ16). Seat D's base S9 is at [NBH/branch_snapshots/MEET-THM/R1_D.md:475] (not transcribed). Seat A's "S4" half (K108) is transcribed under TH13.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:453

Seat B, R1 D18 · Proposition CERT — statement. Local search moves only on strict improvement, ≤ (ℓ+1) scans of ≤ p + k*(p−k*) models, `local_certified` = strict flip/swap local maximum at the scanned centre; B&B over [∅, U] with the Furnival–Wilson superset bound is exact when it completes, "unresolved" otherwise; score rounding error $\lesssim c \cdot k \cdot \kappa_2(H_S) \cdot u \cdot (\nu + k)$; if every log mass moves ≤ ε then $\tilde{Z}/Z \in [e^{-\varepsilon}, e^{\varepsilon}]$, TV ≤ tanh ε, winners with gap > 2ε unchanged.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:436

Seat A, R1 C16 · REPAIR and certificates — Statement. EX-A, completed scans, tie-broken greedy map, capped lattice. (a) Theorem A: under TD'_∞ on Lat(S°), the local search from C ∈ Lat(S°) with $\ell \geq \tau(C)$ reaches, scores and certifies S_0 in ≤ Φ(C) moves; LS-score: LS0 from the prescribed start mode scores S_0 within max{Φ(S°) − 1, 0} moves; Lemma M: margins ⇒ TD′ w.p. → 1 under SE; (b) Lemma S: the HPM is the ≻-maximum of the scored set, order-independent; top_c merges; with LS0, $\mathcal{T}_{\text{new}} = \text{top}_c(\mathcal{T}_{\text{old}} \cup \mathcal{S}_{\text{added}})$; (c) local search moves only on strict improvement, ≤ (ℓ+1) scans of ≤ p + k*(p−k*) models; local_certified means a strict flip/swap local maximum at the scanned centre; every block sub-model is scored, so the HPM beats every block-local model; (d) card R: B&B on [∅, U] with the leaps-and-bounds superset bound is exact when complete, "unresolved" on exhaustion, certifies nothing outside U; (e) limitation: a certified local maximum can lie far below the mode (CEX-localmax, CEX2).

Clause (a) of C16 (Theorem A, LS-score) is main-card material, M5 TH05.a.

Chair-specified fallback (IC7, close audit #8), verbatim: "the deterministic perturbation lemma K108 ($\tilde{Z}/Z \in [e^{-\varepsilon}, e^{\varepsilon}]$, TV ≤ tanh ε, and gap > 2ε preserves the winner). Also the exact semantics of the local-search flags (K64, a weak local maximum, with ties allowed). No B&B-soundness, rounding or region-certification claim. Stop rule: the generic rule." [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-AUDIT.md:101] [NBH/branch_snapshots/MEET-THM/CLOSE_AUDIT.md:16]. LP-AUDIT is the code-audit wave (wave 3). Its named audits are listed under OB02 [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-AUDIT.md:70].

Dependencies. Used by M4 TH04 (card semantics), M5 TH05 (Theorem A, LS-score, scan bound) and M8 TH08 (search cost) [NBH/02_theory.md:1197]. Obligations:

- OB07 (N7), final placement "S-CERT (K73 without region clause) + M4(e); N16 = FW16" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:74];
- OB11 (N11), final placement "M5 hypothesis (+ S-CERT)" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:78];
- OB16 (N16), final placement "FW16; K108 in S-CERT" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:83].

Cited: CL21 (Furnival–Wilson, K05). Deferred: CJ11 (NP-completeness), CJ16 (K107).

Plain reading

The flag `local_certified` means exactly this: no single flip or swap strictly improves the score at the scanned centre. That is a weak local maximum, and ties are allowed. Each search scans a bounded number of models. In exact arithmetic, the branch-and-bound card R is exact when it completes and says "unresolved" otherwise. If every log-mass is perturbed by at most ε, the evidence moves by a factor in $[e^{-\varepsilon}, e^{\varepsilon}]$, TV moves by at most tanh ε, and any winner with a gap above 2ε stays the winner.

Not claimed / not proved

- No region certification: K73's operational region-certification clause was withdrawn [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1673].
- No rounding-error bound (K107 = CJ16; T33-errbound COUNTEREXAMPLE). All certificates are exact-arithmetic statements, and production floating point is not validated (OB16).
- A certified weak local maximum can be far below the mode (CEX-localmax(A2); see TH18).
- No NP-completeness result (CJ11).
- If the full card fails, the fallback ships without any B&B-soundness claim.

TH15 · S-OUT · Output-event validity, worker equivalence, stacking (K77, K78, K83–K86, K63)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): LemmaRO: PROVED; SF VERIFIED; XF VERIFIED*. CEX-guard: COUNTEREXAMPLE; XF VERIFIED*. N6 potential-run: PROVED; output form: CORRECT WITH AMENDMENT. GUARD-W: not audited [NBH/ID_REGISTRY.md:95]

Decision note. "K63/K86 are also named as premises in M1/M5" [NBH/branch_snapshots/MEET-THM/MINUTES.md:69]. TLF: "Seat B has no S-OUT card in Round 2 (R3_B places K63/K86 in hypotheses …). Cards below are complementary." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:972].

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:980-988

Seat C (R2_C), card C13 — K77/K78.
Final statement. Under A and stop-only statistical coupling GUARD-W, a potential-run event B satisfying $P(B \mid \mathcal{C}) \le \alpha$ gives $P(\mathcal{O} \cap B \mid \mathcal{C}) \le \alpha$. It does not imply $P(B \mid \mathcal{O}, \mathcal{C}) \le \alpha$.
Repairs. PC2 cluster 3; preserve actual statistical-output scope and the unaudited integrated coupling.
Status. LemmaRO: **PROVED; SF VERIFIED; XF VERIFIED***. CEX-guard: **COUNTEREXAMPLE; XF VERIFIED***. N6 potential-run: **PROVED**; output form: **CORRECT WITH AMENDMENT**. GUARD-W: **not audited**.
Placement. Supplement.
Route / Lean. Cite filed coupling/event-containment result; finite probability core.
Fallback / stop. Potential-run theorem while the audit remains open. Never replace it with completion-conditioned validity.
IC3. Any admissible correlated input; unchanged from R1 §D D12 apart from explicit audit qualification.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:997

Seat D (R2_D) S4 row. S4 worker/stacking/output — K63, K83–K86, K77, K78: "Adaptive R′ need not preserve the bounds" (cluster 3). [status: unchanged]

Seat A's S6 (PAR: K63, K83, K84, K86, K09) is the same text transcribed under TH12 [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1006].

Round-1 base statements. TLF contains no full statement of the worker-equivalence and stacking clauses (K63, K83–K86). They live in the base cards transcribed here.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_D.md:424

Seat D, R1 S4 · Worker equivalence, stacking, output convention — Statement. Lemma S (top-c merge order-independent); Prop W bitwise (one binary, process workers; 3/3 designs); W-cal by code reading; Cor W-R(b) with the extra search; pooled bounds hold for any R′ declared before stream 10 and fail for adaptive R′; Lemma RO (stop-only guard); CEX-guard; IND-T stream separation.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:493

Seat A, R1 S1 · PAR — Statement. (a) Prop W / W-cal: concurrent islands, diagnostic islands and scans reproduce the serial run bitwise (one binary, single-thread BLAS, process workers, fixed reduction order); (b) Cor W-R: stacking R′ = 4k leaves islands 1–4 unchanged; with the extra search from c_4 the scored set nests and HPM(R′) ≥ HPM(4); without it, FALSE; (c) stream separation 10..9+R′ (IND-T extended; code audit); (d) tie-aware DP returns the prescribed start mode (N3 sub-obligation); (e) for fixed p the exact branch is eventually taken (N12).

Their obligation forms are OB02 (N2(a)–(e)) and OB14 (N14(a)–(d)), transcribed in §5.4. Seat C's C13 base, "R1 §D D12", is at [NBH/branch_snapshots/MEET-THM/R1_C.md:323].

Charter (LP-AUDIT). "No binding theorem of its own, but M1(c),(d), M4(e), M5 and M8 rest on S-OUT (Lemma RO, GUARD-W, IND-T extension, worker equivalence, stacking nesting). The M1 charter escalation evidence applies if the GUARD-W or IND-T extension audit fails." [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-AUDIT.md:66].

Dependencies. Used by M1(c,d), M4(e), M5, M8 and S-REP [NBH/02_theory.md:1198]. Obligations:

- OB06 (N6), final placement "M1(d), M4(e), S-OUT" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:73];
- OB14 (N14), final placement "S-OUT + M1(c)" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:81];
- OB02 (N2, the named implementation premises).

Cited: CL20 (Lemma RO, GUARD-W).

Plain reading

Fix a frozen calibration record, under the card's setting A. If a failure event has probability at most α for the *potential* run (the run as if it always completed), then the probability that the run both returns output and fails is still at most α. This needs the guard to be stop-only (GUARD-W). It does *not* say that a run that did return output fails with probability at most α.

Not claimed / not proved

- GUARD-W, the stop-only coupling of the integrated v3.1 code, is *not audited*. Until it is, the statement is a potential-run theorem.
- The probability is never conditional on completion (CEX-guard is a counterexample to that reading).
- Adaptive R' need not preserve the bounds.
- The IND-T extension to streams 10..9 + R' is a code audit, not done (OB14).

TH16 · S-Z · Evidence variance, masked tails (K37–K41, K53)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): OPW2-ThmA: PROVED (SF: none; XF: VERIFIED (A-REV9)); MZ-E1E2: PROVED (E1, E2), candidate: not reviewed, not adopted; FMP-Na: PROVED (SF: none; XF: GAP (A-REV9)); PC1c4: PROVED; T26-ThmA: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV4)); T26 overall CONJECTURE [NBH/ID_REGISTRY.md:96]

Decision note. "Terminal-off clause only in the exception-probability form" [NBH/branch_snapshots/MEET-THM/MINUTES.md:61]. The seats differ on the variance clause [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1016]. The chair resolves it: both bounds ship. The chair's choice is transcribed first, then the seat texts.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1339-1343

Chair's choice 3 (S-Z, variance). Both bounds ship, as two clauses:
– (vii-a) $\mathrm{Var}(\hat{Z}_r/Z \mid \mathcal{C}) \le (T+1)\cdot e^{2D}/N$, with D = $\mathrm{osc}_\Omega\, d$, the deterministic full-space range (A S4(vii), B S4(vi); wave-1 repair);
– (vii-b) $\mathrm{Var}(W_r \mid \mathcal{C}) \le e^{2\cdot \mathrm{osc}_\Omega\, d} - 1$ (D S6; it follows from the pathwise bound $W_r \in [e^{-\mathrm{osc}\, d}, e^{\mathrm{osc}\, d}]$ checked by PC2, W2_PC2 C2 rows).
Their minimum is the reported bound. (vii-b) is the checked fallback if (vii-a)'s 1/N route fails in the LP-REST HQ.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1024

Seat A (R2_A) S4. S4 · Proposition Z (K37–K41, K53). (i) $E[\hat{Z}_r \mid \boldsymbol{\mathcal{C}}] = Z$, $E[\log \hat{Z}_r] \le \log Z$, $P(0.05\hat{Z} > Z) \le 0.05$ (cited). (ii) exact variance identity (OPW2 Thm A, as a specialisation of CDG 2011 Thm 1.4 in kind); the exact-refresh product never substituted (OPW2-cex). (iii) masked lower tail: islands whose terminal population is off S_0 have $\hat{Z}_r \le \sqrt{\delta_1} Z$ **except on an event of probability ≤** $2\sqrt{\delta_1}$ (source form, PC1:247; divided by κ if conditioned on an off-event of probability κ); PC1 claim 4 wording mandatory. (iv) $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$. (vi) favourable $\hat{Z}/Z \to 1$ under (P1) (remark). (vii) $\mathrm{Var}(\hat{Z}_r/Z \mid \boldsymbol{\mathcal{C}}) \le (T+1)e^{2D}/N$ with $D := \mathrm{osc}_\Omega(d)$, deterministic, bounded by §R3(a) (PC1:249). Status unchanged from R1 §D C6 (FMP-Na: PROVED (XF: GAP (A-REV9)); MZ-E1E2: PROVED (E1, E2), candidate: not reviewed, not adopted; (vii) E proposal unchecked). Supplement.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1033-1035

Seat B (R2_B) S4 · Theorem Z, evidence variance (K39, K40, K38, K37, K41, K53) · supplement.
Final statement as R1 §D D12 with: (ii) in the exception-probability form, $P(\hat{Z}_r > \sqrt{\delta_1} Z$ on a terminal-off island) $\le 2\sqrt{\delta_1}$ (÷κ if conditioned on an off event of probability κ) (PC1 §15 COUNTEREXAMPLE conceded); the E1/E2 path-tail clause imported with its full hypotheses (A, x_m, τ_A, D_A, exceptions) or omitted; (vi) $\mathrm{Var}(\hat{Z}_r/Z \mid \boldsymbol{\mathcal{C}}) \le (T+1)e^{2D}/N$ with D a deterministic full-space range bound ($D = \mathrm{osc}\, d \le D_*$ under §R3(a)). Status: OPW2-ThmA: PROVED (SF: none; XF: VERIFIED (A-REV9)); MZ-E1E2: PROVED (E1, E2), candidate: not reviewed, not adopted; FMP-Na: PROVED (SF: none; XF: GAP (A-REV9)); PC1c4: PROVED; T26-ThmA: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV4)); T26 overall CONJECTURE. Novelty of (i) vs Cérou–Del Moral–Guyader Thm 1.4 unchecked (LIT-SMC §3A: "needs escalation"). Rest unchanged.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1044

Seat D (R2_D) S6 row. S6 evidence variance — K37–K41, K53: Terminal-off clause in exception-probability form; $\mathrm{Var}(W_r|\mathcal{C}) \le e^{2\cdot \mathrm{osc}_\Omega d} - 1$ (pathwise bound, PC2 D14) replaces the $(T+1)e^{2D}/N$ form; envelope per M3′(d). [status: OPW2-ThmA "PROVED (SF: none; XF: VERIFIED (A-REV9))"; MZ-E1E2 "PROVED (E1, E2)" candidate]

Round-1 base statements. These are superseded where Round 2 repairs them: the terminal-off tail is now in exception-probability form, and Var bounds follow chair's choice 3. Seat D's base S6 is at [NBH/branch_snapshots/MEET-THM/R1_D.md:446] (not transcribed). The E1/E2 path-tail clause ships "with its full hypotheses (A, x_m, τ_A, D_A, exceptions) or omitted". Its Round-1 form is B's clause (ii) below.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:261

Seat A, R1 C6 · Z — Statement. One island, setting C2. (i) $E[\hat{Z}_r \mid \mathcal{C}] = Z$, $E[\log \hat{Z}_r] \le \log Z$, $P(0.05\hat{Z} > Z) \le 0.05$; (ii) $N \cdot \mathrm{Var}(W_T \mid \mathcal{C}) = \Sigma_s(1-1/N)^{T-1-s}(E[W_s^2\mu_s(h_{s,T}^2)] - 1)$ for any invariant kernels; the ideal-refresh product $\prod(1+\chi_t^2/N) - 1$ is stated as an idealisation, never substituted (OPW2-cex); (iii) masked lower tail E1/E2 and N(a): $\hat{Z}_r \le \sqrt{\delta_1} Z$ on islands whose terminal population is off S_0; PC1 claim 4 wording: "underestimates with high probability; the complementary rare event carries compensating overestimates; Markov caps them"; (iv) $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$; (v) $Z \le C_n(2b_0 + y^\top y/(n+1))^{-\nu}$, gap data-dependent; (vi) favourable: $\hat{Z}/Z \to 1$ under (P1) (remark); (vii) crude universal bound $\mathrm{Var}(\hat{Z}_r/Z \mid \mathcal{C}) \le (T+1)e^{2D}/N$ (remark, E T19, unchecked).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:359

Seat B, R1 D12 · Theorem Z — statement. (i) exact variance identity $N \cdot \mathrm{Var}(W_T|\mathcal{C}) = \Sigma_s(1-1/N)^{T-1-s}(E[W_s^2\mu_s(h_{s,T}^2)] - 1)$; under exact refresh $\prod(1+\chi_t^2/N) - 1$ (idealisation, not the kernel: OPW2-cex). (ii) masked lower tail: $P(W > (T/\eta)[\pi_0(A^c)/\pi_0(x_m)]e^{-\varphi_{\tau_A} D_A}) \le \eta + N\pi_0(A)$; if the terminal population is off S_0 w.p. ≥ κ then $\hat{Z}_r \le \sqrt{\delta_1} Z$ there. (iii) $E[\log \hat{Z}_r] \le \log Z$. (iv) island-minimum bound $0.05^{1/R} \min_r \hat{Z}_r$ (independence). (v) favourable: $\hat{Z}/Z \to 1$ under (P1) (remark). (vi) design-free $\mathrm{Var}(\hat{Z}_r/Z|\mathcal{C}) \le (T+1)e^{2D}/N$ (crude).

Chair-specified fallback (LP-REST), verbatim: "The exact evidence identities (KNOWN—CITE); clause (vii-b), $\mathrm{Var}(W_r \mid \mathcal{C}) \le e^{2\cdot \mathrm{osc}\, d} - 1$, from the pathwise bound checked by PC2; the terminal-off clause in exception-probability form. Clause (vii-a), the 1/N form, is dropped if its route fails" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-REST.md:89].

Dependencies. Uses M2(a) (for D), M3, M4 (envelope) and S-TARGET TH10 [NBH/02_theory.md:1199]. Cited: CL08 (Cérou–Del Moral–Guyader). Deferred: CJ24 (useful two-sided masked evidence precision).

Plain reading

Each island's evidence estimate is unbiased for Z given the frozen calibration, and its variance has an exact identity. Under masking, an island whose final population misses the truth underestimates Z by a factor $\sqrt{\delta_1}$, except on an event of probability at most $2\sqrt{\delta_1}$. Two explicit variance bounds ship, and the reported bound is the smaller of the two. Both are finite-data, deterministic-range statements; but (vii-a), the $1/N$ form, is an "E proposal unchecked" and is dropped if its route fails, with (vii-b) as the checked fallback [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1343], [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-REST.md:89].

Not claimed / not proved

- T26 overall is a CONJECTURE.
- MZ-E1E2 is "candidate: not reviewed, not adopted".
- FMP-Na has an XF GAP.
- Clause (vii-a) was an "E proposal unchecked" (Seat A).
- The novelty of (i) relative to Cérou–Del Moral–Guyader Thm 1.4 is unchecked.
- The terminal-off tail holds only in exception-probability form: PC1 §15 is a counterexample to the unconditional form.
- No useful two-sided relative precision of $\hat{Z}$ under masking (CJ24; see also LIM05).
- Clause (vi), favourable $\hat{Z}/Z \to 1$, is a remark only.

TH17 · S-POOL · Pooling rules, recording-marginal equal-weight bias comparison P1, R_eff (K49, K50, K54, K56, K57-P1)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): MA-PropP: PROVED (elementary), author-only; Prop2(c): PROVED (XF VERIFIED); MZ-P1P2: P1 PROVED; MA-CorV: PROOF SKETCH; ADRAFT-ThmA Astra-only, cite Del Moral; OBG-LH-lock-general: XF ERROR [NBH/ID_REGISTRY.md:97]

Decision note. "General K57 accuracy → FW (B, C; R3-8)" [NBH/branch_snapshots/MEET-THM/MINUTES.md:66]. TLF: "C has no R2 card (cites). Seats are compatible; P1 is stated for the uniformly sampled recording-state / one-record marginal in all three." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1054]. The Round-2 texts below are deltas on Round-1 base cards, so the base statements are transcribed after them. Read the base text with the Round-2 changes applied.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1062-1064

Seat B (R2_B) S10 · Proposition PIP-eq (K49, K50, K57; K59 FUTURE_WORK) · supplement.
Final statement as R1 §D D14 with the TV clause on the uniformly sampled one-record marginal of the delivered population (PC1 §14 AMBIGUOUS) and the splitting clause labelled exact-refresh idealisation. Status: MA-PropP: PROVED (elementary), author-only; Prop2(c): PROVED (XF VERIFIED); MZ-P1P2: P1 PROVED; MA-CorV: PROOF SKETCH. Rest unchanged.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1073

Seat A (R2_A) S7. S7 · Proposition POOL (K49, K50, K54, K56, K57; K47, K55 cited). As R1 §D C4 (i), (iii)–(vi) and S2 P1, with (ii) moved to M2(b) and $0 \le h \le 1$ (PC1:240); P1 stated for the uniformly sampled recording-state marginal (PC1:237); "iff delivered" dropped. Status unchanged (MA-PropP: PROVED (elementary), author-only; Prop2(c): PROVED (XF VERIFIED); ADRAFT-ThmA Astra-only, cite Del Moral; MZ-P1P2: P1 PROVED; OBG-LH-lock-general: XF ERROR). Supplement.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1082-1088

Seat D (R2_D) S5 and S7 rows.
S5 pooling — K47 cited, K48–K50, K54, K56: unchanged. [status: unchanged]
S7 equal-weight P1 — K57: unchanged (bias ≤ TV of the one-record marginal, cluster 14). [status: MZ-P1P2 "P1 PROVED; P2 PROVED as a composition"]

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1097

Seat C (R3_C §S1), objection to unrestricted K57 accuracy (for the record). §2.2 S-POOL: Object to unrestricted K57 accuracy. Retain the checked recording-marginal bias comparison; general fixed-budget equal-weight accuracy remains FUTURE_WORK. W2P1 B.S10 verifies that narrower scope. [What would change it:] A checked estimator-level theorem for the actual records and budgets.

Round-1 base statements (superseded wherever the Round-2 texts above say "with:").

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:223

Seat A, R1 C4 · POOL — Statement. Setting C2, $W_r = \hat{Z}_r/Z$, $M = E[W_r^2 \mid \boldsymbol{\mathcal{C}}]$. (i) $\hat{Z}$-weighted $\hat{\mu}(h) = \Sigma_r \hat{Z}_r H_r(h)/\Sigma_r \hat{Z}_r \to \pi(h)$ a.s. as R → ∞ at fixed N; (ii) $P(|\hat{\mu} - \pi(h)| > \varepsilon \mid \boldsymbol{\mathcal{C}}) \leq \min\{1, 4(M-1)/R + 4M/(R\varepsilon^2)\}$; (iii) the equal-weight mean → $E[H_1(h) \mid \boldsymbol{\mathcal{C}}] \neq \pi(h)$ in general, and $E[\hat{\bar{Z}} \cdot \mathrm{PIP}_j^{\mathrm{eq}} \mid \boldsymbol{\mathcal{C}}] - Z \cdot \mathrm{PIP}_j = -(1-1/R)\mathrm{Cov}(\hat{Z}_1, \widehat{\mathrm{PIP}}_{1j} \mid \boldsymbol{\mathcal{C}})$; (iv) $R_{\mathrm{eff}}/R \to Z^2/E[\hat{Z}^2 \mid \boldsymbol{\mathcal{C}}]$; (v) one-step RB: $\mathrm{Var}(o_j) \leq \mathrm{Var}(1\{j \in S\})$ conditionally; (vi) for fixed R, N → ∞ consistency and CLTs: cited.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:512

Seat A, R1 S2 · EQ-PIP — Statement. P1: $|E[\widehat{\mathrm{PIP}}_{rj}] - \mathrm{PIP}_j| \leq$ TV(law of the delivered terminal population, π); after locking (all particles at S_0) the bias is ≤ δ_1 and the L^1 error ≤ (L+4)$\bar{\rho}$ + u_π + δ_1 (LH, fixed p). Thm R (DELIV-POP) deferred.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:397

Seat B, R1 D14 · Proposition PIP-eq — statement. $\mathrm{PIP}^{\mathrm{eq}} \to E[\widehat{\mathrm{PIP}}_{1j} | \boldsymbol{\mathcal{C}}] \neq \mathrm{PIP}_j$ at fixed N; $E[\hat{\bar{Z}} \cdot \mathrm{PIP}^{\mathrm{eq}}] - Z \cdot \mathrm{PIP} = -(1-1/R)\mathrm{Cov}$; $|E[\widehat{\mathrm{PIP}}_{rj}] - \mathrm{PIP}_j| \leq$ TV(delivered population, π); splitting N_e into more islands is neutral for $\hat{Z}$ to first order (exact-refresh idealisation) but not for $\mathrm{PIP}^{\mathrm{eq}}$.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_D.md:438,461

Seat D, R1 S5 · Pooling rules (Prop P) and R_eff — Statement (line 438). Fixed N: $\hat{Z}$-weighted pool → π(h) a.s. as R′ → ∞; equal-weight → $E[\widehat{\mathrm{PIP}}_{1j} | \boldsymbol{\mathcal{C}}]$ (bias of order 1/N); $E[\hat{\bar{Z}} \cdot \mathrm{PIP}^{\mathrm{eq}}] - Z \cdot \mathrm{PIP} = -(1-1/R)\mathrm{Cov}$; fixed R: island estimates → π(h) as N → ∞ (cited); $R_{\mathrm{eff}}/R \to Z^2/E[\hat{Z}^2 | \boldsymbol{\mathcal{C}}]$; one-step RB variance reduction.
Seat D, R1 S7 · Equal-weight PIP (A4) — Statement (line 461). $|E[\widehat{\mathrm{PIP}}_{rj}] - \mathrm{PIP}_j| \leq$ TV(delivered population, π); after locking ≤ δ_1 (LH fixed p). Thm R deferred (DELIV-POP OPEN; general LH locking FALSE per A-REV7).

Seat A's clause (ii), the finite-R bound, moved to M2(b) and M1(f) (TH01.f). The locking clauses ("after locking …") rest on LH locking, which is "general LH locking FALSE per A-REV7" (OBG-LH-lock-general: XF ERROR). Only P1 is carried into the final card.
Chair-specified fallback (LP-REST), verbatim: "Prop P(i),(ii) (MA-PropP, 'PROVED (elementary)', author-only: checked in the HQ); the recording-marginal P1 comparison as checked (W2_PC1 B.S10); R_eff as the Kish identity (cited)". Stop rule: "Generic rule; general K57 accuracy stays FW" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-REST.md:90].
Dependencies. Uses M1 and M3 [NBH/02_theory.md:1200]. OB04(ii) has its final placement in S-POOL [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:71]. The N14(c) IND-T extension (OB14) is used by S-POOL [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1879]. Cited: CL02 (K47). Deferred: CJ07, CJ08, CJ09. Dissent on K57: DS06, Seat D [NBH/branch_snapshots/MEET-THM/MINUTES.md:86].

Plain reading

Given the frozen calibration, the primary ($\hat{Z}$-weighted) pooled estimate of a bounded summary converges almost surely to the exact posterior mean as the number of islands grows, at any fixed particle count (Seat A's clause (i) [NBH/branch_snapshots/MEET-THM/R1_A.md:223], = METH-AUDIT Prop P(ii) [NBH/branch_snapshots/METH-AUDIT/PROOFS.md:126-128]; MA-PropP: "PROVED (elementary), author-only"; for general h it rests on N1, a proof sketch). The equal-weight (secondary) PIP converges instead to the single-island mean, which differs from the posterior value "in general" [NBH/branch_snapshots/MEET-THM/R1_A.md:223]: equal weighting retains any single-island bias for every R (METH-AUDIT Prop P(i) [NBH/branch_snapshots/METH-AUDIT/PROOFS.md:122-124]). That bias is at most the TV distance between the law of one uniformly sampled recorded state and π (P1). R_{eff} is the Kish effective number of islands.

Not claimed / not proved

- No general fixed-budget accuracy for the equal-weight PIP (K57 = `CJ08`).
- No masked consistency of the equal-weight PIP or MPM (K58 = `CJ07`; DELIV-POP OPEN).
- No R-versus-N allocation result (K59 = `CJ09`).
- General LH locking is false (OBG-LH-lock-general XF ERROR).
- MA-PropP is author-only.
- MA-CorV is a PROOF SKETCH.
- ADRAFT-ThmA is Astra-only; cite Del Moral instead.

TH18 · S-LIM · Limitations as theorems (K79, K81, K100, K101); K99 qualitative remark

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): CEX-localmax(A2), CEX2, CEX-ESSneutral(A3), PropD-T03, SHARE-PB: COUNTEREXAMPLE (constructions); INP I5 PROVED [NBH/ID_REGISTRY.md:98]

Decision note. "Duplicates: PIP = E[# selected in group]/d; numeric limits only with their p qualification" [NBH/branch_snapshots/MEET-THM/MINUTES.md:65]. TLF, verbatim: "Duplicates clause (MINUTES; SYNTH_R2 §3): PIP = E[# selected in group]/d; the numeric limit 2/3 only for p = 2; with an orthogonal null it is 3/5; numeric limits only with their p qualification. The seats distribute clauses differently (A puts the weak-local-maximum limitation CEX-localmax in S-CERT; B has it as (b) here and K74 here; MINUTES places K74 in S-CERT); the duplicates clause is identical. C cites these." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1107]. Seat B's text is the fullest and is transcribed first.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1115-1117

Seat B (R2_B) S6 · Proposition LIM (K79, K74, K81, K100, K101; K99 remark) · supplement.
Final statement: (a) no transcript diagnostic certifies absence of an unqueried mode; (b) a certified weak local maximum can be far below the mode; (c) ESS = N throughout is compatible with genealogical collapse; (e) d exact duplicates: by symmetry PIP = E[number selected in the group]/d; the limit is 1/d only if exactly one is included asymptotically; under the ridge/BB target with two duplicates and fixed noise it is 2/3, group inclusion → 1; (f) MPM may exceed K; (g′) qualitative barrier: SHARE-PB's k-way contrast construction, as a remark. Repairs: (e) corrected (PC1 §18 COUNTEREXAMPLE conceded); (d) withdrawn (IMPLAUSIBLE); (g) withdrawn with K80 (FUTURE_WORK). Status: CEX-localmax(A2), CEX2, CEX-ESSneutral(A3), PropD-T03, SHARE-PB: COUNTEREXAMPLE (constructions); INP I5 PROVED. Rest unchanged from R1 §D D15.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1126

Seat A (R2_A) S9. S9 · Proposition LIM (K79, K81, K100, K101). (a), (b), (d) unchanged from R1 §D S3. (c) duplicates: d interchangeable columns have equal PIPs with PIP = E[number selected in the group]/d; the limit is 1/d only if exactly one is included asymptotically; under the ridge/BB target with two duplicates and fixed noise the limit is 2/3 (PC1:281); group inclusion → 1. Supplement.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1135

Seat D (R2_D) S11 row. S11 limitations — K79, K81, K99 (qualitative), K100 (corrected): Duplicates: PIP = E[#selected in group]/d (cluster 18). [status: unchanged]

Round-1 base statements (superseded wherever the Round-2 texts above repair or withdraw a clause).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:410

Seat B, R1 D15 · Proposition LIM — statement. (a) no diagnostic from the transcript certifies absence of an unqueried mode (changing its mass leaves the transcript unchanged); (b) a certified flip/swap local maximum can be far below the mode; (c) ESS = N throughout is compatible with genealogical collapse; (d) cross-block masked contrasts of order k ≥ 4 in a saturated cluster are unreachable at budget $W \cdot (2p)^{-k}$ while every diagnostic passes; (e) duplicates: PIP → 1/d, group inclusion → 1; (f) MPM may exceed K; (g) E2 fires w.p. ≈ $P(\chi^2_3 > 3/s^2)$ under a declared alternative.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_A.md:531

Seat A, R1 S3 · LIMITS — Statement. (a) E1c/E2: changing an unqueried state's mass leaves the transcript unchanged, so no run diagnostic certifies absence of unseen modes (oracle statement; positive certificates can still detect some failures); (b) flat weights do not prevent genealogical collapse (CEX-ESSneutral); (c) duplicates: equal PIPs, PIP → 1/d with group inclusion → 1, no estimator distinguishes duplicates; $|S_0| > K$ has posterior zero; (d) the MPM may exceed K; HPM ≠ MPM in general.

Superseded in Round 2:

- B's (d) was withdrawn (IMPLAUSIBLE); its quantitative form is CJ15.
- B's (g) was withdrawn with K80 (CJ17).
- "PIP → $1/d$" in both base texts is replaced by PIP = E[# selected in group]/d (PC1 §18 COUNTEREXAMPLE).

Chair-specified fallback (LP-REST), verbatim: "Each limitation restated with its filed construction: E1c/V12 for K79; duplicates with PIP = E[# selected in group]/d, and numeric limits only with their p qualification". Stop rule: "Generic rule; a limitation whose proof fails moves to FW. It is never kept as an assertion" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-REST.md:91].
Dependencies. Cited by M4 [NBH/02_theory.md:1201]. Related method-level limitations: LIM04 (no universal diagnostic), LIM08 (cap K). Deferred quantitative barrier: CJ15.

Plain reading

These are limitations, each backed by a filed construction (status: "COUNTEREXAMPLE (constructions); INP I5 PROVED"):

- No diagnostic computed from the run's transcript can certify that an unvisited mode is absent.
- A certified weak local maximum can be far below the mode.
- Effective sample size $= N$ at every stage is compatible with all particles descending from one ancestor.
- Exact duplicate columns share PIP $=$ E[# selected in group]$/d$, which need not tend to $1/d$ (numeric limits only with their p qualification: $2/3$ only for $p = 2$).
- The MPM may have more than K variables.

Not claimed / not proved

- The numeric duplicate limit $2/3$ holds "only for $p = 2$"; with an orthogonal null it is $3/5$. MINUTES/TLF make numeric limits valid "only with their p qualification" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1107]. The seat texts and the one-line REG-A description ("$2/3$ for two duplicates, fixed noise") omit this qualification [NBH/ID_REGISTRY.md:98]. Use the qualified form.
- K99 is a qualitative remark only. Its quantitative k-way barrier is CJ15.
- Flag power (K80) is CJ17.
- K74's placement: MINUTES places K74 in S-CERT (TH14). Seat B lists it here.

TH19 · S-REP · Run-card reporting corollary: $\hat{J}$ (B/C/D form) and $\hat{\varepsilon}_x$ with companions (K104/K11)

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): Reporting assembly: PROOF SKETCH, unchecked. N1: PROOF SKETCH. PC1c3: PROVED, with N1 as premise. W2P2 C16 and repaired W2P1 T4/T5 are statement/route checks, not a full proof upgrade [NBH/ID_REGISTRY.md:99]

Decision note. TLF, verbatim: "Final card is R3_C §S2 S-REP (R3-6, unanimous: A withdrew its TI form). **$\hat{\mathbf{u}}$ is D's global block-table envelope** $\hat{u} = \Sigma_b \max_{U \subseteq C_b} x_b(U)$, not a maximum over V (C's objection, R3_C §S1, adopted; a maximum over $V \subsetneq \Omega$ does not bound all fits). ... the earlier R2 wording 'maximal fit fraction on V' (SYNTH_R2 §2.5; D:182) is withdrawn. Growth rates for the informative regime: the two stacking configurations of MINUTES §2.3 §2.6. d is a default member of $\mathcal{H}$ (AMENDMENT_HQ_02). The accumulator is read-only (HQ Q4)." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1199] [NBH/branch_snapshots/MEET-THM/MINUTES.md:72]. The final card follows in full, split into three boxes for page breaking.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1207-1222

Final card (R3_C §S2 S-REP) — run-card reporting corollary, part 1/3.
IDs. S-REP: K104/K11 reporting corollary, using M1/K46/K52 and M2/K102.
Hypotheses. D0, EX-A and the MC contract. d is fixed from data, blocks and tables before final randomness. Let $D = \max_\Omega d - \min_\Omega d$, $W_r = \widehat{Z}_r/Z$, $M_2 = E(W_r^2 \mid \mathcal{C})$.
Implementation premises. Read-only initial and measured-endpoint discrepancy accumulators; no new draws, transitions or scores; no feedback. Existing values are read when available. Missing historical records yield NA. Integrated accumulator and extended-stream fidelity require audit.
Initial records. $A_r = N^{-1}\sum_{i=1}^{N} d(S_{ri}^{(0)})$, using each final island's initial iid start particles. Initial particles are not reweighted by final island evidence.
Endpoint records. $B_r = (4N)^{-1}\sum_{k=1}^{4}\sum_{i=1}^{N} d(S_{ri}^{(k)})$, at the four prescribed measured endpoints. These are plain discrepancies, not marginal RB PIP records.
Estimator. $\hat{J} = \dfrac{\sum_{r=1}^{R'} \widehat{Z}_r B_r}{\sum_{r=1}^{R'} \widehat{Z}_r} - \dfrac{1}{R'}\sum_{r=1}^{R'} A_r$. Preserve its signed value. **Evidence:** W2P2 §4, "§R5 · $\hat{J}$ definition", COMPLIANT; W2P1 B.R5(i), T1–T4.
Target identity. $J = KL(\pi\|\pi_0) + KL(\pi_0\|\pi) = E_\pi d - E_{\pi_0} d$. **Evidence:** W2P2 §4, "C16 · $\hat{J}$ theorem"; W1P1 §16.
Finite-R' statement. For $\epsilon > 0$, $P(|\hat{J} - J| > \epsilon \mid \mathcal{C}) \le \min\{1, 4(M_2-1)/R' + 16D^2M_2/(R'\epsilon^2) + D^2/(R'N\epsilon^2)\}$. **Evidence:** W2P2 §4, C16 theorem; W2P1 D.R5(i), T1–T3.
Limits and zero range. Fixed data/calibration: $\hat{J} \to J$ almost surely under independent stacking. If $D = 0$, both equal zero. On W, the adopted explicit stacking configurations yield consistency using $D \le D_* = O_P(1)$, $M_2 \le e^{D_*}$. **Evidence:** W2P2 §4 and §6, C16.
Bound disclosure. The bound can equal one at default $R' = 4$, as in W2P2's check; there is no default-budget accuracy or diagnostic-power promise. A large true J does not ensure a large observed $\hat{J}$.
Output form. Under GUARD-W, replace the failure event by $\mathcal{O} \cap \{|\hat{J} - J| > \epsilon\}$ with the same upper bound. **Evidence:** W1P2 cluster 3, D12; W2P2 §6, C16.
Numerical label. "Post hoc Monte Carlo estimate of Jeffreys divergence; not a certificate; floating-point error unvalidated." Report nats. The exact branch may report NA for this bridge diagnostic.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1223-1230

Final card (R3_C §S2 S-REP), part 2/3.
Correlation proxy. $\hat{\varepsilon}_\times = \max_{b\neq c,\ j\in C_b,\ k\in C_c} |X_j^\top X_k|/(n-1)$, empty maximum zero; field `cross_block_rmax`. **Evidence:** W2P2 §4, proxy definition, COMPLIANT.
Proxy label. "Descriptive maximum cross-block correlation proxy; not signal leakage or a confidence bound; floating-point error unvalidated."
Global companions. $\widehat{\kappa} = \min_b \lambda_{\min}(G_{C_b}/(n-1))$. If positive, $\widehat{\delta} = (K-1)\hat{\varepsilon}_\times/\widehat{\kappa}$. Let $\widehat{u} = \sum_b \max_{U\subseteq C_b} x_b(U)$, using already available block-score gains. This is a conservative global envelope, not a maximum over V.
Companion domain. Require $\widehat{\kappa} > 0$, $\widehat{\delta} < 1$, $\widehat{u} < 1$, $\widehat{\delta}\widehat{u} < (1-\widehat{\delta})(1-\widehat{u})$. Otherwise print derived $\hat{\eta}$ as **"n/a"**. **Evidence:** W2P1 D.R5(ii), T4/T5; B.R5(ii) repair; SYNTH_R2 §3.
Derived envelope. Put $\hat{r} = \widehat{\delta}\widehat{u}/[(1-\widehat{\delta})(1-\widehat{u})]$, $\hat{\eta} = K\log(1/(1-\widehat{\delta})) + \nu\log(1/(1-\hat{r}))$. On the domain, $\sup_{S\in\Omega} |d(S) - d_\sigma(S)| \le \hat{\eta}$. **Evidence:** W2P1 D.R5(ii), T4/T5.
Per-model norm clause. For nonempty S, the standardized cross-block relative norm obeys $\varepsilon_S \le (|S|-1)\hat{\varepsilon}_\times/\widehat{\kappa}$. Treat the empty model separately with norm zero. **Evidence:** W2P1 D.R5(ii), T4/T5.
Regularized proxy bounds. With G_D block diagonal, $E = G - G_D$, $A_D = G_D + I$, $H_D = (n+1)G_D + I$: $\delta_A \le \frac{(n-1)(p-1)\hat{\varepsilon}_\times}{1+(n-1)\widehat{\kappa}}$, $\delta_H \le \frac{(n^2-1)(p-1)\hat{\varepsilon}_\times}{1+(n^2-1)\widehat{\kappa}}$. **Evidence:** W2P2 §4, C16 proxy theorem.
Scope of derived bounds. M2 still requires its full matrix/residual domain. Restricted-set calculations remain restricted-set calculations. Neither a lower bound of order $n\hat{\varepsilon}_\times|\beta_j\beta_k|$ nor global rescaling from V survives W2P1.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1231-1242

Final card (R3_C §S2 S-REP), part 3/3.
Guarantee. The permitted reporting fields have precise mathematical targets and qualified error bounds without influencing the sampling procedure.
Goals. Objective (1); proposal §5.3 discrepancy reporting, §5.4 approximation and §13.1 bounded expectations; HQ Q4. No separate Song–Liang theorem-equivalence claim.
Status/self-check flag. Reporting assembly: **PROOF SKETCH, unchecked**. N1: **PROOF SKETCH**. PC1c3: **PROVED**, with N1 as premise. W2P2 C16 and repaired W2P1 T4/T5 are statement/route checks, not a full proof upgrade.
Fidelity. Reporting amendment authorized by Q4, to be recorded in AMENDMENT_HQ_02. Per-particle d values exist during execution but are not currently retained. Historical availability is not inferred. No machine-certified bound is claimed.
Novelty/citation. **MODEST/NEW-COMP** reporting consequence. Independent-system evidence weighting is established in Dai–Heng–Jacob–Whiteley, §4.2 (https://arxiv.org/html/2007.11936v3). The Jeffreys identity and norm inequalities are foundations; matching the prescribed initial and endpoint records is the contribution.
Proof route/key lemma. Apply M1 to globally range-scaled d; split the error between endpoint ratio and initial mean. Use Popoviciu, Chebyshev and a union bound without assuming independence between those two estimates. Proxy bounds use row sums and block eigenvalues.
Difficulty/Lean core. Routine mathematics; moderate accumulator audit. Lean: finite ratio bound, variance/union inequalities, range scaling, empty-set conventions and row-sum norm bounds.
Dependencies. M1; M2 for informative W rates; M3 for discrepancy definitions; S-KERNEL for exact initialization; S-OUT for delivered-event transfer.
Risk. Wrong endpoint timing; final-evidence weighting of initial particles; using V-range globally; invalid residual domain; interpreting a descriptive proxy as coefficient leakage or flag power.
Fallback/stop rule. Retain precisely labelled descriptive fields if quantitative proof closure fails. If records are unavailable, report NA. Stop after one failed proof/audit repair cycle; remove quantitative reporting claims. Failure inherited from N1 or M2 triggers their binding-goal escalation, not a silent waiver.
IC3 exhibit/check. W has $\widehat{\kappa} = 3/4$, bounded discrepancy/moments and actual SMC dispatch. W2P2 `w_family.py` checks its proxy bounds; `jhat_mc.py` checks the estimator on a coded-grid component replica. W2P1 T4/T5 checks the global-envelope version.
Numerical companion. None newly run. Proposed: at most ten seeds on enumerable correlated component designs; compare exact J, accumulated endpoint/start means and analytic proxy bounds. Replay identical streams with reporting enabled/disabled to test zero feedback. Report NA rather than reconstruct absent historical accumulators.

Charter (LP-REP). "None. MINUTES §2.5, wave 4: 'Nothing binding waits on them.' Failure inherited from N1 or M2 triggers their binding-goal escalation, not a silent waiver." The named premises are: "Read-only initial and measured-endpoint discrepancy accumulators; integrated accumulator and extended-stream fidelity require audit (S-REP card); AMENDMENT_HQ_02 definitions ($\hat{J}$, $\hat{\varepsilon}_\times$ with companions, d in $\mathcal{H}$, the two R′ growth rates)" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-REP.md:61-65]. The read-only d accumulator is a named premise. Per-particle d values "are not retained today", so historical fits report NA [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1884]. The reporting definitions are DF19 and DF20 (DF19, DF20); the HQ ruling allowing them is RL04 (RL04).
Dependencies. As in the card's own Dependencies row [NBH/02_theory.md:1202]. The binding upstream obligation is OB01 (N1). Cited: CL11 (Dai–Heng–Jacob–Whiteley §4.2).

Plain reading

Two new run-card fields are read off records the sampler already produces, with no feedback into sampling:

- $\hat{J}$ estimates the Jeffreys divergence J between start and posterior. Under D0, EX-A and the MC contract, given the frozen calibration, it has an explicit finite-R' tail bound (vacuous at the default $R' = 4$), converges a.s. for fixed data under independent stacking, and is consistent on the reporting family W under the adopted stacking rates. The assembly is a proof sketch resting on N1.
- $\hat{\varepsilon}_\times$ is the largest cross-block sample correlation. On a stated domain it yields a computable bound on how far the bridge potential d can be from its shared-variance part.

Both fields are labelled post-hoc and descriptive.

Not claimed / not proved

- No accuracy at the default $R' = 4$: the bound can equal one there.
- No diagnostic power: a large true J does not ensure a large observed $\hat{J}$.
- The proxy is not signal leakage and not a confidence bound.
- No machine-certified bound; floating point is unvalidated.
- Historical runs give NA.
- The whole card rests on `OB01` (N1, PROOF SKETCH). The assembly is a PROOF SKETCH, unchecked.

TH20 · S-LEAK · B's leakage clauses (F), (S) as a supplementary corollary to M2

APPROVED (F.B., 2026-11-07) Status (REG-A, verbatim): OWS-WS6(b): PROVED (SF: none; XF: none), PC1 B.D6(S) TRUE; the m lower bound: PROVED (by this seat, unchecked); (b6) OWS-WS53F: PROVED-UNDER (SF: none; XF: none); PC1 B.D6(F) TRUE / DESIGN-VERIFIED (D_B) [NBH/ID_REGISTRY.md:100]

Decision note. "R3-5, unanimous" [NBH/branch_snapshots/MEET-THM/MINUTES.md:71]. TLF, verbatim: "(F), (S) are non-perturbative / different scale; they are not part of HQ's (a)(b)(c); (b5)'s pathwise bound stays in M2 … R3_C: 'Preserve all inherited hypotheses; no positive uniform $q_2(S_0)$ conclusion' … Seats B and D state (S) differently … Only B states (F). See 'Chair's choices' item 4 (compatible; both ship)." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1145]. The chair's choice is transcribed first.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1344-1348

Chair's choice 4 (S-LEAK, (F) and (S)). Ship:
– (F) as B states it (`R2_B.md` §R3 (b));
– (S) as D's exact identity $1 + \chi^2(\pi \| \pi_0) = \pi(S_0)^2/(\pi_0(S_0) \cdot q_2(S_0))$, **plus** B's data-dependent bound $m \geq \pi_0(S_0) \cdot e^{-2 \cdot \mathrm{osc}\, d}$, **with** D's caveat that m has no design-uniform lower bound.
These are compatible. W2_PC1 B.R3(b)(5),(6) finds the m bound TRUE pointwise, and the q_2 route withdrawn. No uniform positive $q_2(S_0)$ conclusion is made (R3_C).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1162

Clause (F) — Seat B (R2_B), fragment (b6) of the §R3 card row (other clauses of the line not reproduced).
(b6) Favourable start under leakage: G, R, SE-blk(φ_0), EBM(M_n), CBL(η), $\eta < \eta_A \Rightarrow 1 - \pi_0(S_0) = O_P(2^B \sqrt{\log n} \cdot n^{-\kappa})$.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1180

Clause (S) — Seat D (R2_D), the non-perturbative layer.
Non-perturbative layer (any ε, every grid; Seat B's (S), PC1 TRUE-AS-WRITTEN): $\chi^2(q_{\varphi_t}\|q_{\varphi_{t-1}}) \le 1/m^2 - 1$ with $m = \min\{\pi_0(S_0), q_2(S_0)\}$, exact $1 + \chi^2(\pi\|\pi_0) = \pi(S_0)^2/(\pi_0(S_0)q_2(S_0))$; m has no design-uniform lower bound (q_2 route IMPLAUSIBLE uniformly, PC1).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1153

Clause (S), m-form — Seat B (R2_B), fragment of the (b5) clause (other clauses of the line not reproduced).
and $\max_t \chi^2(q_{\varphi_t}\|q_{\varphi_{t-1}}) \le 1/m^2 - 1, m := \min\{\pi_0(S_0), q_2(S_0)\} \ge \pi_0(S_0)e^{-2\cdot\mathrm{osc}\, d}$.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1171

Seat B: status fragments for (S), (F), m-bound.
OWS-WS6(b): PROVED (SF: none; XF: none), PC1 B.D6(S) TRUE; [...] the m lower bound: PROVED (by this seat, unchecked; $q_2(S_0)/\pi_0(S_0) = e^{2d(S_0)}/E_{\pi_0}e^{2d}$). [...] (b6) OWS-WS53F: PROVED-UNDER (SF: none; XF: none); PC1 B.D6(F) TRUE / DESIGN-VERIFIED (D_B).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1189

Seat A (R2_A) D4 row (A places B's (F),(S) in a leakage supplement; states no clause).
D4 | amend | One merged package (§R3): WS3/WS4 route, now VERIFIED cross-family (CHK:13–14), with C's pathwise $M_2 \le e^{\mathrm{osc}\, d}$ clause (PC2:125) and B's (F),(S) clauses as a leakage supplement. Concede A.C9(c) "hence within" (PC1:138), the exhibit (PC1:142), the (e) "iff" (PC1:141). B's (N) and q_2 lemma not adopted (PC1:145, 147). | A proof of B's q_2 tilt-tail lemma would add a non-perturbative clause.

Hypotheses of (F). Seat B's §R3 row names G, R, SE-blk(φ_0), EBM(M_n), CBL(η) and $\eta < \eta_A$ without defining them [NBH/branch_snapshots/MEET-THM/R2_B.md:216]. They are defined in Seat B's Round-1 card D6 (Theorem WSTAB-C), transcribed next. That card also carries the original forms of (F) and (S). Its clause (O) is now M2(c) material. Its clause (N) was not adopted (Seat A D4 above: "B's (N) and q_2 lemma not adopted").

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/R1_B.md:505

Seat B, R1 card D6 · Theorem WSTAB-C — statement (Round-1 base; superseded by the chair's choice above where they differ).
Blocks any partition (the v3 one included); SE-blk(φ_0): $\lambda_{\min}(R_{C_b}) \ge \varphi_0^2$ ∀b; G, R; cross-block leakage CBL(η): $\max_b \Lambda_b \le \eta\sigma^{*2}\log n$ with $\Lambda_b = \|(P_{C_b} - P_{U_b^*})\mu_{-b}\|^2$ (the energy other blocks' signal leaks into block b beyond its own truths), $\eta < \eta_A$; EBM(M_n): $\min_{b,j\in U_b^*}|\tilde{\theta}_{b,j}| \ge [\sqrt{M_n(\sigma^{*2}+2M)\log n} + \sqrt{\Lambda_b}]/(\varphi_0\sqrt{n-1})$, $M_n \to \infty$. Then: (F) $1 - \pi_0(S_0) = O_P(2^B\sqrt{\log n}\cdot n^{-\kappa})$ (favourable start). (S) For **every** grid $0 = \varphi_0 < \ldots < \varphi_T = 1$ and every t: $\chi^2(q_{\varphi_t}\|q_{\varphi_{t-1}}) \le 1/m^2 - 1, m := \min\{\pi_0(S_0), q_2(S_0)\}$, and $1 + \chi^2(\pi\|\pi_0) = \pi(S_0)^2/(\pi_0(S_0)q_2(S_0))$ exactly; with T = 1, $\mathrm{Var}_{\mathcal{C}}(\hat{Z}_r/Z) = \chi^2(\pi\|\pi_0)/N$. (O) Obstruction: under ORTH with $|N_0|$ truth-free null predictors and p = n−1, $\chi^2(\pi\|\pi_0) \to \infty$ for every $\bar{R}^2 > \frac{1}{2}$ while $\mathrm{TV}(\pi, \pi_0) \to 0$; conversely under masking $\mathrm{ESS}_\infty \le \pi_0(S_0)/\pi(S_0)^2 = p^{-c}n^{-c/2}e^{O_P(1)}$. (N) Necessity: if $\pi_0(S_0) \not\to 1$ then infinitely often coefficients cancel or leakage is large.

The tilted law $q_t \propto \gamma_0 e^{td}$, so that q_2 is the tilt at $t = 2$, is defined in DF07 (DF07).
Chair-specified fallback (IC7, close audit #8), verbatim: "the finite-law exact identity $1 + \chi^2(\pi\|\pi_0) = \pi(S_0)^2/(\pi_0(S_0)\cdot q_2(S_0))$, plus the data-dependent bound $m \ge \pi_0(S_0)\cdot e^{-2\cdot\mathrm{osc}\, d}$. No design-

uniform lower bound on m. (F) is dropped if its leakage route fails. Stop rule: the generic rule." [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-REP.md:96].

Dependencies. Uses M2 TH02, M3 TH03 and S-T TH09 [NBH/02_theory.md:1203].

Plain reading

These clauses hold for any ε and any grid, outside the perturbative §5.3 package:

- (S): the χ^2 divergence from start to posterior is an exact closed form in three probabilities of the true model. Every bridge increment is controlled by the smaller of the start and tilted masses of S_0, which is at least $\pi_0(S_0)e^{-2\,\mathrm{osc}\,d}$ for the given data.
- (F): under B's leakage conditions (G, R, SE-blk(φ_0), EBM(M_n), CBL(η), $\eta < \eta_A$), $1 - \pi_0(S_0) = O_P(2^B\sqrt{\log n}\,n^{-\kappa})$. This is asymptotic, in probability, and PROVED-UNDER with no SF or XF check.

Not claimed / not proved

- No design-uniform lower bound on m. The q_2 route is "IMPLAUSIBLE uniformly" and was withdrawn, and "No uniform positive $q_2(S_0)$ conclusion is made".
- (F) is PROVED-UNDER with no SF or XF check. Its conditions are DESIGN-VERIFIED only on D_B.
- The m lower bound is "PROVED (by this seat, unchecked)".
- (F) and (S) are not part of HQ's §5.3 (a)(b)(c).

5.3 Remarks TH21 (dissolved card) and TH22 (Thesis CD)

TH21 · S-STGI · K29/K30 — DISSOLVED (not a list item)

Status (REG-A, verbatim): DISSOLVED (MINUTES §2.2, corrected) [NBH/ID_REGISTRY.md:101]

Decision, verbatim (corrected MINUTES §2.2): "**Dissolved** (corrected 2026-11-04 after the registrar's cross-check, REGISTRY_PAPER1.md §7). K30 is stated once, as M7(d), labelled ideal two-state grid. K29 → FUTURE_WORK FW23. Its conditional implication under (LL) or the over-tilted q_w bound may be cited as an attributed lemma, not a numbered claim". On the vote: "A (R3_A:43, 'object: dissolve'), B (FW) and C (R3_C §S1, 'object to K29 as a numbered contribution') prevail. **Dissent:** D (R3_D, 'confirm K29 here'). The reason: (LL) is not a model/design condition, and no family exhibits it (W2PC1 A.S11, DESIGN-ASSERTED/GAP). v1 of these minutes misreported A's position" [NBH/branch_snapshots/MEET-THM/MINUTES.md:70].

Where the content went.

- K30 (the ideal two-state grid crossing, $\log_2 A_0 + O(1)$ stages) is clause (d) of M7, TH07.d. Chair's choice 2: "Stated once, as M7(d) in the final M7 card ... The other seats' wordings (C14, A S11, D STG′(c)) are references for the M7 HQ" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1338].
- K29's unconditional favourable-regime application is CJ23 (FW23).
- K29's conditional implication may be cited as an attributed lemma. Chair's choice 1: "B's STG(f) wording is the reference text. Dissent: D" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1337].

That reference text is reproduced below. It is *not* a list claim.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:879

Seat B (R2_B) STG card M7, clause (f) only (fragment; the rest of the row is M7 material): (f) [supplement] favourable O(1) stages under STG-F's over-tilted q_w bound or LL.

Round-3 objections, verbatim (what would bring K29 back).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:906-924

A (R3_A:43). S-STGI (K29, K30): **object: dissolve.** K30 → M7(d) labelled ideal grid (the chair's M7 row already lists it); K29 → FUTURE_WORK | K29 fails IC3: (LL) is not a model/design condition and no family exhibits it (W2PC1:187 DESIGN-ASSERTED, GAP) | A design family on which (LL) or the over-tilted q_w bound is verified; then K29 returns as a supplement clause.
B (R3_B:61). S-STGI (K29 conditional on (LL); K30 ideal): **object on K29** | (LL)/the over-tilted q_w bound is not a model/design condition and no family satisfying it has been exhibited (W2PC1 A.S11: DESIGN-ASSERTED, GAP). IC3 fails; a list item cannot be conditional on an unexhibited hypothesis. K30 is already M7(d); state it once. | An exhibited design family satisfying (LL), checked in one wave: then K29 returns as a supplement clause. Until then FUTURE_WORK.
C (R3_C:40). §2.2 S-STGI: Object to K29 as a numbered contribution | Its LL/over-tilted input lacks an exhibited model/design route: W2P1 A.S11 is DESIGN-ASSERTED/GAP. Cite the conditional implication; defer its unconditional favourable-regime application. Keep K30 as the ideal-grid result. | A checked IC3 family establishing the missing input.
D (R3_D:45). S-STGI (K29 conditional on (LL), K30 ideal): confirm K29 here; K30 also stated as M7's labelled clause (c) (R3-8) | — | none.

The other historical seat texts are kept "for reference only; they are not list claims" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:830-837]. Their spans:

- Seat C's C14 (K30) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:853-861];
- Seat A's S11 (K29 under (LL); K30) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:870];
- Seat D's STG′ [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:888];
- Seat D's S10 row [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:897].

Recorded dissents: DS03 (D, K29) and DS04 (C, K30 in the supplement).

Not claimed / not proved

- No favourable $O(1)$ stage law is claimed. The input it needs, (LL) or the over-tilted q_w bound, is not a model/design condition and no family exhibits it.
- No "as-coded" masked stage count (CJ06, Conjecture P OPEN).
- K30 is an ideal-grid comparison result, not a statement about the coded grid.

UNCERTAIN — propagation of the corrected vote

The final audit found that old S-STGI vote and placement text survived in TLF:831,833, in REGISTRY_PAPER1 and in charters M2/M6/M7 and LP-IDENT. The chair applied the corrections without another audit [NBH/branch_snapshots/MEET-THM/CLOSE_AUDIT.md:12]. The TLF snapshot carries the corrected heading [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:830]. The archive did not re-audit REGISTRY_PAPER1 [NBH/02_theory.md:2787]. Treat "K30 = M7(d); K29 = FW23" as final. Treat any older text in REGISTRY_PAPER1 that places S-STGI in LP-IDENT as superseded.

TH22 · Thesis CD (K96) · scoped remark after the main theorems

Status (REG-A, verbatim): remark, not a theorem [NBH/ID_REGISTRY.md:102]
Decision. "Thesis CD (K96): a scoped remark after the main theorems, unanimous since Round 2" [NBH/branch_snapshots/MEET-THM/MINUTES.md:54]. The thesis sentence adopted earlier at [pos 1044] was: "In split-and-merge, correlation breaks correctness. In local-to-global SMC, correlation costs only sequential depth and particles, never correctness." [NBH/03_results_and_conclusions.md:115]. Chair's choice 5, verbatim: "Use **Seat C's** scoped wording (R2_C.md §R6 agenda 6 table): 'Remark only; there is no retained non-corollary thesis sentence', with its fixed-data / asymptotic / not-claimed scopes. All seats conceded D2 to C's position. The other three wordings are kept above for the paper's authors." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1349].

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1278-1283

Seat C (R2_C, §R6 agenda 6) — the chosen wording.
K96 disposition. Remark only. There is no retained non-corollary thesis sentence.
Fixed data. For each admissible finite input and successful frozen calibration, the idealized start, kernels and unnormalized identities target the specified posterior. Normalized finite-run posterior estimates are not generally unbiased or exact. Independent stacking gives fixed-data consistency for the stated records.
Asymptotic accuracy. Exact-target selection uses C9's model/design conditions. Growing-dimensional summary accuracy is established by the proposed C1/C2 assembly on W with explicit stacking. Masked output/store accuracy uses LH and LS0 with output/retention premises. These are separate scopes.
Correlation and cost. Dependence affects the start discrepancy, available bounds and search difficulty. The results do not prove that correlation only increases cost, that all correlated designs admit polynomial accuracy budgets, or that the posterior itself remains statistically consistent under arbitrary correlation.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1260

Seat A (R2_A, agenda item 6). Item 6, thesis scope (CD). Fixed data: M1 identities, M2(a)–(f), M5, M3(b) (all finite inequalities). Asymptotic: M2(g), M6(b1-LH), S1, S2, the ideal-grid depth laws. Favourable vs masked: M1, M2(a), M3, M5 are regime-free; depth laws differ by regime (M7(d) vs S11); selection is proved only on masked LH (M6). The sentence not a corollary of other cards: none. CD is a **remark** ("correlation enters the guarantees only through (δ_1, osc d_σ, η_Ω, A_0), each computable from (X, y); it enters no identity and no limit"), placed after M8.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1269

Seat B (R2_B, agenda item 6). Item 6, thesis scope. Remark only (concede D2). Scope of the remark: fixed data for (a) identities and (c) particle cost ($M_2 \le e^{\operatorname{osc} d}$, $R' \asymp e^{\operatorname{osc} d}/\varepsilon^2$), both for every X; depth (b) is a fixed-data bound $T \le P + \lceil \operatorname{osc}_V d/0.3246 \rceil + 1$ plus the ideal-grid crossing law $\log_2 A_0 + O(1)$; selection (d) is asymptotic, favourable (b6 start, T03♯) or masked (HPM b1-LH) on separate sequences. Candidate non-corollary sentence "correlation increases depth and M_2" is false (non-monotone: R1 check, PC1 §11); the true non-monotone sentence is Proposition NEC's cancellation point. So: no theorem.

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1292

Seat D (R2_D, agenda item 6). Item 6, thesis theorem. Scope: fixed data for the identities and finite-R′ bounds (H′(a),(b)); asymptotic for depth and selection (STG′(c), Theorem 5(b1-LH), SL′), with favourable and masked families kept on separate sequences. There is no sentence of CD that is not a corollary of H′, STG′, WS′, Theorem 5 and SL′ (the candidate "$\varepsilon \mapsto$ (d, χ^2_t, T, M_φ) is continuous at 0 while $\varepsilon \mapsto P(\mathcal{O} \cap \{\hat{S} \neq S_0\})$ is bounded through the same quantities" is WS′(b) plus Theorem 5(0)). **Remark only**, placed as the organising paragraph of the main text. "Never correctness" = H′(a); "costs depth" = STG′ + WS′(b)(iii); "costs particles/islands" = H′(b) with $M_2 \geq 1/P_{\mathcal{C}}$(cross) disclosed and $M_2 \leq e^{\mathrm{osc}\, d}$ where (a) applies; not monotone in correlation, never conditional on output.

Card labels inside the A, B and D wordings (e.g. M2(g), M6(b1-LH), S11, H′, STG′, WS′, Theorem 5, SL′) are each seat's Round-2 labels, not the final M1–M8 lettering. The final lettering is in TH01–TH08.

Plain reading

There is no separate "thesis theorem". What paper 1 can say about correlation is the sum of the cards, each with its own scope:

- fixed-data identities (idealised start, kernels and unnormalised identities) hold for each admissible finite input and successful frozen calibration; normalised finite-run estimates are not generally unbiased or exact;
- accuracy and selection results are asymptotic, with separate scopes: exact-target selection under C9's (T03♯) model/design conditions, summary accuracy on W with explicit stacking, masked output/store accuracy on LH with LS0 and output/retention premises;
- dependence affects the start discrepancy, the available bounds and search difficulty (Seat C's wording); no monotone "correlation only increases cost" statement is proved.

Not claimed / not proved

- That correlation only increases cost. "correlation increases depth and M_2" is false: it is non-monotone.
- That all correlated designs admit polynomial accuracy budgets.
- That the posterior stays consistent under arbitrary correlation.
- That normalised finite-run estimates are unbiased or exact.

The thesis sentence of [pos 1044] is a slogan. It is not a theorem.

5.4 Proof obligations OB01–OB17 (= N1–N17)

Origin. The method meeting MEET-v31 handed seventeen unproved properties to the theorem meeting as obligations N1–N17 [pos 1280] [pos 1290]. Their texts are in THEOREM_REVISION_INPUTS.md §3 (INP). Each gives a statement, hypotheses, status, the decision it supports, what happens if it fails, and the INV matrix cell. INP's header rule: "**Every PROVED here is a candidate** (not reviewed, not adopted). No status is upgraded." and " 'Seat-proved, unchecked' = a seat gave an argument that no proof check (PC1/PC2) has covered" [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:7-10].

Naming. Item labels inside the INP texts (M2, M4, M7, M9–M14, A1, A3, C1, D1–D4, R3, R5, R7, V1–V13) are labels of the draft list TL = THM-SYNTH THEOREM_LIST.md, not of the final M1–M8 [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:12]. The final placement of each obligation is the "Final" column of COVERAGE_MATRIX §(d), quoted in each card below [NBH/branch_snapshots/MEET-THM/

COVERAGE_MATRIX.md:62-84]. N18 existed only under item-1 Outcome 1 and "was not opened". Item 1 closed as Outcome 2, so there are seventeen [NBH/02_theory.md:2563].

Audit obligations are premises, not future work. MINUTES §2.3, verbatim: "N2(b), N2(c), N2(e), N14(c), the N3 DP invariant, and the integrated streams/LS0/GUARD-W/execution-DAG audits are **named implementation premises in the charters** and are recorded in REGISTRY_PAPER1.md" [NBH/branch_snapshots/MEET-THM/MINUTES.md:90]. REGISTRY_PAPER1 §8 adds: "They are code-audit or execution facts, not model/design assumptions and not newly proved lemmas" [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1870]. Its name map, verbatim: "GUARD-W = N2(e) = C's A-GUARD. IND-T extension = N14(c) (IND-T for streams 10..9+R′). 'Integrated streams' = C's A-STREAM = N2(b) (persistent per-worker stream state) + N14(c). 'LS0' premise = N2(c) + OP-LS0 + the emitted centre (N3). 'Execution DAG' = C's A-DAG = D's trajectory inventory." [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1872].

Binding safeguards (MINUTES §2.7). "§13.1: if N1's general record identity fails, the narrower theorem ships and §13.1 escalates. §5.3: if both the D_* route and the correlated-family route fail, it escalates. Also: loss of the masked statement without an approved replacement escalates (C §S5)" [NBH/branch_snapshots/MEET-THM/MINUTES.md:89].

In each card, "Clauses that depend on it" lists the TH IDs (cards or clause rows) that the cited sources tie to the obligation. The ranking in §5.8 counts these lists.

OB01 · N1 · Measured-round island identity for general bounded h (Theorem H crux; V1(a2))

Status (REG-A, verbatim): PROOF SKETCH; PC1 claim 3 takes it as a premise. Binding §13.1: if it fails the goal is not met by withdrawal; HQ escalates to Felipe [NBH/ID_REGISTRY.md:145]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:127-134

N1. Measured-round island identity for general bounded h (Theorem H crux; V1(a2)).
– *Statement.* Conditional on a successful frozen $\mathcal{C}$, for each final island r and each h ∈ $\mathcal{H}$ with 0 ≤ h ≤ 1, measurable in (X, y, $\mathcal{C}$): $E[\hat{Z}_r \cdot H_r(h) \mid \mathcal{C}] = Z \cdot \pi(h)$, with $H_r(h)$ as in M9.
– *Hypotheses.* EX-A, EX-S, CAL, V5 (each φ = 1 round π-invariant), multinomial resampling, R′ declared before stream 10.
– *Route* (R2C :200–204): the finite-state resample–move recursion gives $E[\hat{Z}_r \eta_{r,T}(f)] = Z\pi(f)$; terminal rounds preserve π, so the identity holds at each measured endpoint and for their average. R3A (unchecked): the base case of the locked terminal-round lemma.
– *Status:* **PROOF SKETCH**; PC1 claim 3 takes it as a premise.
– *Supports:* M9, A1, C1(d) joint clause, D3(a) for general h.
– *If it fails:* the binding goal is not met by withdrawal: HQ escalates to Felipe for a method reconsideration or an explicit scope change (proposal §13.1, binding). Until that decision: $\hat{\mu}(h)$ carries no theorem; A1 shrinks to marginal PIPs (locked); M4 survives (Prop P(ii) for PIPs).
– *Cell:* row 8 × V (islandwise identity, extended); row 10 × "NONE: joint inclusion probabilities".

Final placement. "M1 (Lemma N1, final (b)); binding §13.1 safeguard." Seat C's note: "M1/L-N1; S-REP and S-IDENT consume it; failure escalates §13.1" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:68].

Clauses that depend on it.

- TH01.b, TH01.e, TH01.f, TH01.g, TH01.i: the R3_A edge reads "N1 → M1(b,e,f,g,i)" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:28].
- TH13 (S-IDENT) and TH19 (S-REP). TH19's status words include "N1: PROOF SKETCH".

Who discharges it. The M1 HQ, wave 1: "N1 (binding §13.1) is the longest pole and starts first" [NBH/branch_snapshots/MEET-THM/MINUTES.md:110]. M1 charter, verbatim [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:114-118]:

- first lemma: "N1 (measured-endpoint identity E[$\hat{Z}_r$ H_r(h) | $\mathcal{C}$] = Z·π(h)) via the terminal-round prefix identity; then (e) as its corollary on 1{S ≠ S_0}";

- check: “N1 proof to PC1 (Codex Sol); the Lean file to the registrar; a p ≤ 8 coded-grid replica of E[$\hat{Z}_r$ H_r(h)] (10 seeds) by PC2”;
- fallback: “marginal PIPs only (locked identity) for (e)–(i); one HQ wave on N1”;
- escalation: “N1 refuted or unclosed after one wave (→ §13.1 unmet, HQ)”.

Lean: T12(i.a) is partially formalised. “The resample–move recursion `hstep` is an explicit hypothesis: the probabilistic island model is not formalised” [NBH/03_results_and_conclusions.md:145].
Attempts recorded. Only the route sketch above (R2C, R3A, unchecked). The numerical checks (W1PC2:101, W2PC2 §0) “are statement-sanity only” and do not upgrade N1 [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1902].

OB02 · N2 · Integrated v3.1 core: Prop W, W-cal, LS0, stacking, GUARD-W

Status (REG-A, verbatim): Lemma S PROVED. (a) PROVED by code reading, executed on smoke + 3/3 designs. (b) PROVED by code reading, not executed. (c) not in the production core. (d) PROVED (PC1 claim 6, deterministic). (e) not audited [NBH/ID_REGISTRY.md:146]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:136-141

N2. Integrated v3.1 core: Prop W, W-cal, LS0, stacking, GUARD-W. Code-audit facts plus executed replays, not mathematical theorems (TL:119).
– *Statement.* For fixed (X, y, seed, binary), process workers, Lemma S merges: (a) final islands bitwise equal to serial (Prop W); (b) the 4 diagnostic islands, with persistent per-worker stream state, bitwise equal to serial (W-cal); (c) LS0 leaves island outputs, $\hat{Z}$, PIPs bitwise unchanged, gives $\mathcal{T}_{new} = top_c(\mathcal{T}_{old} \cup \mathcal{S}_{added})$, fires no E3; (d) R′ = 4k with the extra search leaves islands 1–4 unchanged and nests the post-search scored set; (e) **GUARD-W:** v3.1 worker, merge and LS0 code satisfy “any worker, LS0 or guard stop ⇒ no output” (an LS0 stop is an operational interruption; a completed capped LS0 returns `local_limit`) (new code-audit obligation, class (iv); S3 R3-5(ii)).
– *Status.* Lemma S PROVED. (a) PROVED by code reading, executed on smoke + 3/3 designs. (b) PROVED by code reading, **not executed**. (c) not in the production core. (d) PROVED (PC1 claim 6, deterministic). (e) **not audited** (API-AUDIT covered the serial v3 guard only).
– *Supports:* M2, M5, M6, M7, card G2 premise (i), “the 2,394 study fits’ saved sampler estimates are reusable as v3.1” (full v3.1 records pending: LS0 integration and flags, NA interface, certificates, primary rescoring).
– *If it fails:* (b) → LAW-PRESERVING, wall-clock claims capped at the D4 table, “bitwise” claims narrow; (e) → CEX-guard applies to v3.1 (N6); fixes are engineering; decisions survive.
– *Cell:* row 8 × V; row 7 × V; row 12 × “NONE: LS0 absent from production”; row 14 × V.

Final placement. “Named implementation premises in the charters (M1, M4, M5, M8; LP-AUDIT) and recorded in `REGISTRY_PAPER1.md` (MINUTES §2.3)” [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:69].
Clauses that depend on it (REGISTRY_PAPER1 §8 “Used by” column [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1876-1883]).

- N2(b) W-cal: S-OUT `TH15` (K83) and M8 `TH08`.
- N2(c) LS0 integration: M5 `TH05`, M4(c) `TH04.c` and M8.
- N2(e) GUARD-W: M1 `TH01.c` and `TH01.d`; M4(e) `TH04.e`; M5; S-OUT; S-REP `TH19` (output form).

Through N6(b), premise (i) of the certificate-card bound also rests on N2(e) (`OB06`).
Who discharges it. The code-audit wave (wave 3): M5, M4, M8 and LP-AUDIT. “Code-audit wave: N2, N3, LS0, GUARD-W and the DAG serve M5, M4 and M8 together” [NBH/branch_snapshots/MEET-THM/MINUTES.md:112]. Escalations, verbatim: “GUARD-W or IND-T extension audit fails → M1 (c),(d) re-

vert to potential-run statements and N2 reopens (R3_A:192); GUARD-W audit fails → M4(e) in potential-run form only (R3_B:214); N3 fails outright or the integrated LS0 differs from the specified LS0 → HQ decides (R3_D:225); the DAG audit finds an uncounted serial component → M8 (a)/(b) restated, not dropped (R3_D:174)" [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1886].

Attempts recorded.

- Prop W was executed bitwise on smoke and 3/3 designs.
- W-cal was argued by code reading only and has not been executed.
- API-AUDIT covered the serial v3 guard only.
- LS0 changed the HPM in 0/2,394 test-export fits. LS0 is not in the production core [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1882].
- The read-only d accumulator for S-REP is a further named premise of the same kind [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1884].

UNCERTAIN — N2(b) label

The TLF registrar note says the label N2(b) "is not defined in any R3 file; its definition is in `THEOREM_REVISION_INPUTS.md` (N2), which is not on this branch" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1329]. The archive resolves it: N2(b) = W-cal (the 4 diagnostic islands bitwise equal to serial), as defined above [NBH/02_theory.md:2788].

OB03 · N3 · LS0-based T38(b1-LH) about the shipped method; e4 tie-DP invariant

Status (REG-A, verbatim): (b1-LH) PROVED (candidate), XF VERIFIED (A-REV8) for "v3 + LS0"; bridge to code = N2(c), not done. DP invariant OPEN [NBH/ID_REGISTRY.md:147]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:143-149

N3. LS0-based T38(b1-LH) about the shipped method; e4 tie-DP invariant.
– *Statement.* For v3.1 with LS0, on LH exact Gram (p ≤ n − 1, c linked pairs, $g_{\max} \le B < 2g_{\min}$): $P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$, with no open statistical hypothesis. Sub-obligation: on truncation the tie-aware DP returns the prescribed smallest-size, lexicographically smallest maximiser.
– *Hypotheses.* EX-A, EX-S, CAL, SR, GUARD + GUARD-W, (P), (M1)–(M3) (verified on LH), **OP-LS0**, the **LS0 centre rule** (e4), $\ell \ge \max\{\Phi(S^\circ) - 1, 0\}$ (N11).
– *Status.* (b1-LH) PROVED (candidate), XF VERIFIED (A-REV8) for "v3 + LS0"; bridge to code = N2(c), not done. DP invariant **OPEN**. R3A and R3B sketch the same route (unchecked): greedy choice j = 1..p, keeping j iff some size-$k_{\min}$ maximiser consistent with earlier choices contains j, each test a max-plus feasibility DP over (block, remaining size) (`v3_start.h:179–200`); polynomial, no tie enumeration. Study: 0/2,400 truncated; all had one tie.
– *Supports:* M2, M12 e4.
– *If it fails:* headline falls back to (b2-LH) on STAB-LH without LS0; M2 survives (ADDITIVE; HPM unchanged 2,394/2,394) without theorem motivation.
– *Cell:* row 11 × A; row 12 × V/A.

Final placement. "Named premise (MINUTES §2.3) + K09 proposition C10 in S-KERNEL; not FW" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:70].

Clauses that depend on it. `TH05.b1LH` (status "N3 OPEN"), `TH04.f` (N3 carried as a hypothesis) and `TH12` (K09, C10) [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1880].

Who discharges it. The M5 HQ, wave 3. Its first lemma, verbatim: "the e4 tie-DP invariant (N3): the DP returns the prescribed smallest-size, lexicographically smallest start-mode maximiser (W1PC2 D9: TRUE on 400 random instances, DESIGN-ASSERTED). It is the only OPEN mathematical obligation in the chain. In parallel, the merged-LS0 replay (N2(c)) against LS-score's

centre rule." Fallback: "(b1-LH) becomes 'conditional on N3' after one unproductive HQ cycle; (0) + (a) + (b2-LH) always ship" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:101-104]. The finite proposition is Seat C's C10 in TH12: PROOF SKETCH, unchecked. Its route is "Optimal-completion invariant and prefix induction".
Attempts recorded. R3A and R3B sketched the greedy max-plus route (unchecked). W1PC2 D9 found it TRUE on 400 random instances (DESIGN-ASSERTED, not a proof).

OB04 · N4 · Finite-R accuracy of the $\hat{Z}$-weighted PIP and $\hat{\mu}(h)$; the constant M under masking (V11)

Status (REG-A, verbatim): (i) PROVED, given N1 for general h or the locked identity for PIPs; (ii) PROVED (candidate); (iii) Obligation: CONJECTURE, OPEN for c ≥ 2 [NBH/ID_REGISTRY.md:148]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:151-157

N4. Finite-R accuracy of the $\hat{Z}$-weighted PIP and $\hat{\mu}(h)$; the constant M under masking (V11).
– (i) PC1 claim 3: $P(|\hat{\mu} - \mu| > \varepsilon) \leq \min\{1, 4(M-1)/R + 4M/(R\varepsilon^2)\}$, $M = E[(\hat{Z}_r/Z)^2 \mid \boldsymbol{\mathcal{C}}]$. **PROVED**, given N1 for general h or the locked identity for PIPs.
– (ii) Prop P(iii) with M1 (PROVED, candidate): regime-free distance to $1\{j \in S_0\}$.
– (iii) **Obligation:** an upper bound on M on the production grid through a masked crossing (V11, "resolved grid ⇒ $O_P(\sqrt{T/N})$": CONJECTURE, OPEN for c ≥ 2). Known lower bound M ≳ $1/P_{\mathcal{C}}$(cross), PROOF SKETCH (TL A1(v)); do not cite F-MASKPIP M3 (XF ERROR).
– *Supports:* M4; M7's R′ guidance; R_{eff} reporting.
– *If (iii) fails:* M4 survives (R-consistency holds; equal-weight has none); the paper claims correctness, not efficiency.
– *Cells:* row 10 × A; row 9 × "NONE: two-sided relative accuracy of $\hat{Z}$ (masked)"; row 8 × "NONE: island variance vs correlation"; row 7 × A.

Final placement. "(i) M1(f); (ii) S-POOL; (iii) M2(b5) on the domain; masked precision = FW24 (separate from FW2)." The seats differed on (iii): "A: masked M → FW2. D: V11 → FW2. C: M1 finite-R′, M2 design-derived moments, masked V11 FW (separate)" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:71].
Clauses that depend on it. TH01.f (i), TH17 (ii) and TH02.b5 (iii, on the domain). The open masked part is CJ24.
Who discharges it. Part (i) is discharged once OB01 is. Part (iii) on the domain belongs to the M2 HQ. Masked (iii) is future work (CJ24; reason: "V11 OPEN").
Do not retry / do not cite. "do not cite F-MASKPIP M3 (XF ERROR)".

OB05 · N5 · Calibration contract

Status (REG-A, verbatim): (a) assembly of T12/T31(i) (PROVED-LOCKED, conditional on $\boldsymbol{\mathcal{C}}$), IND-T (PROVED, XF VERIFIED*, A-REV1 only), Lemma RO; (b) OPEN [NBH/ID_REGISTRY.md:149]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:159-163

N5. Calibration contract.
– *(a) Validity for any frozen $\mathcal{C}$:* every final-run identity (V1, N1, N6) holds conditional on any successful $\mathcal{C}$ on streams 0–9, final streams ≥ 10 independent. Status: assembly of T12/T31(i) (PROVED-LOCKED, conditional on $\mathcal{C}$), IND-T (PROVED, XF VERIFIED*, A-REV1 only), Lemma RO; IND-T for streams 10..9+R′ is a code-audit item. ESS 0.90 (pilot = grid) is a hypothesis.
– *(b) OPEN* (all NONE in INV row 7 and §5 item 9): (b1) rebuild-loop convergence within ≤ ℓ rebuilds; (b2) P($S_0 \in V$), "V contains the competing pair"; (b3) P(schedule_limit), P(grid_unconverged) → 0, incl. as-coded masked T_n (Conjecture P).
– *If (b) fails:* M8 **survives** (validity conditional on any frozen $\mathcal{C}$); only delivery and efficiency suffer. Each is stated OPEN beside the POST HOC tabulation of rebuilds, tier, T, convergence.
– *Cells:* row 7 × V (a); row 7 × "NONE" (b1–b3 stay empty); row 14 × "NONE: P(schedule_limit)"; row 5 × "NONE: realised T_n".

Final placement. "(a) M1(c); coded finite geometry and K33 count form M7 (D's S-IDENT placement recorded as dissent, MINUTES §2.3); (b) FW6" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:72].
Clauses that depend on it. TH01.c for (a) and TH07 (K33 count form) for the coded geometry. Part (b) is CJ06.
Who discharges it. Part (a): the M1 HQ, together with the IND-T extension audit (OB14(c)). Part (b) is future work: "Rebuild convergence is FW in either placement of K33" [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:14].
Not an inherent limitation. MEET-v31 lists "rebuild convergence, P($S_0 \in V$) and P(schedule_limit) → 0" and "the as-coded masked T_n law" as "Not inherent (OPEN; must not be presented as limitations)" [NBH/branch_snapshots/MEET-v31/MINUTES.md:96-99].

OB06 · N6 · Output-event coverage statements

Status (REG-A, verbatim): (a) PROVED (potential run; output form CORRECT WITH AMENDMENT). (b) PROVED modulo premise (i), whose GUARD-W part is the code-audit obligation N2(e) [NBH/ID_REGISTRY.md:150]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:165-173

N6. Output-event coverage statements.
– (a) $P(\mathcal{O} \cap \{\max(Z_{\mathcal{T}}, 0.05\hat{Z}) > Z\}) \le 0.05 \cdot P(\text{SMC} \cap \mathcal{A}) \le 0.05$; $\{L > Z\} = \{0.05\hat{Z} > Z\}$.
– (b) Card level: P($\mathcal{O}$ ∩ {some G2, L or R item false}) ≤ 0.05·P(SMC ∩ $\mathcal{A}$).
– (c) Every result is joint with $\mathcal{O}$, never conditional on it: "the probability that a run returns a card with a false claim is at most 5%".
– *Hypotheses* (PC1 claim 2): (i) Lemma RO + GUARD-W; (ii) successful $\mathcal{C}$ with $E[\bar{\hat{Z}} \mid \mathcal{C}] \le Z$; (iii) CAL, EX-S; (iv) $\hat{Z}$ = Z on the exact branch; plus EX-A and distinct store entries. Rigorous fp coverage also needs validated SC and VAL-FP (N16).
– *Status:* (a) **PROVED** (potential run; output form CORRECT WITH AMENDMENT). (b) **PROVED modulo premise (i)**, whose GUARD-W part is the code-audit obligation N2(e).
– *Supports:* M10, M11.
– *If premise (i) fails:* CEX-guard applies and the output-event statements fail for v3.1; fix is engineering.
– *Cells:* row 9 × V; row 14 × V.

Final placement. "M1(d), M4(e), S-OUT" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:73].
Clauses that depend on it. TH01.d, TH04.e and TH15.
Who discharges it. Nothing mathematical remains except premise (i). That premise is the GUARD-W audit N2(e) (OB02), owned by the M4 HQ and LP-AUDIT. The M4 charter's first lemma,

verbatim: "N13 retention under the coded LS0 rule … confirmed against the code …; then the GUARD-W stop-only coupling audit N2(e), on which (e) rests" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M4.md:100]. Rigorous floating-point coverage additionally needs OB16.

OB07 · N7 · Certificate card G1/G2/L/R (PC2 verdicts)

Status (REG-A, verbatim): G1 PROVED; G2 PROVED; L1 CORRECT WITH AMENDMENT; L2 PROVED; L3 amended form PROVED; R PROVED with amendments; Card bound (C6) PROVED modulo premise (i); C6′ PROVED. Open sub-obligation SC(η_F)/VAL-FP (N16) [NBH/ID_REGISTRY.md:151]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:175-182

N7. Certificate card G1/G2/L/R (PC2 verdicts; card as adopted in M10).
– *By item:* **G1** (C1(a) amended, excl. $\hat{S}$) PROVED, deterministic. **G2** (C1(b)) PROVED, output event, joint. **L1** (C3(a); empty maxima +∞; fp margin $2\eta_F$) CORRECT WITH AMENDMENT; on the finally returned $\hat{S}$. **L2** (C3(b)) PROVED. **L3** (C3(c), realised $\mathcal{S}$, $|\mathcal{T}| \geq 2$) amended form PROVED; premise holds under LS0. **R** (C4 + Lemma M) PROVED with amendments (drop |L| > K leaves; S ⊆ U, |S| ≤ K; SC covers Φ and M); C4(b) PROVED as written (isolating $\{\hat{S}\}$ is sharpness only; R3C withdrew). Card bound (C6) PROVED modulo premise (i). C6′ PROVED.
– Budget and per-record clauses: as M10.
– *FALSE or not a theorem:* C1(a) third clause; C1(b) "unscored mass" (COUNTEREXAMPLE `t4b`); the conditional C6 reading; "SC checkable by replay"; rigorous `global_certified` (η uses a *lower* condition-number estimate); the C4 cost remark.
– *Excluded from the output:* C2 (conditional on PW2), C5 (needs S_0; a study-level item).
– *Open sub-obligation:* SC(η_F) by validated arithmetic and VAL-FP(η, ε) at production size (N16); until then, labelled numerical.
– *Supports:* M10. *If SC is never validated:* the label stays (POST HOC).
– *Cells:* row 13 × F (per-fit certificates, **not** flag power); row 9 × F; row 12 × "NONE: meaning of `local_certified`" (L1, deterministic); row 11 × F.

Final placement. "S-CERT (K73 without region clause) + M4(e); N16 = FW16" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:74].
Clauses that depend on it. TH14 and TH04.e.
Who discharges it. LP-AUDIT (S-CERT) and the M4 HQ. The remaining sub-obligation is OB16.
Do not retry. The items listed as "FALSE or not a theorem" above are settled negatives. They include C1(b) "unscored mass" (COUNTEREXAMPLE `t4b`) and a rigorous `global_certified`.

OB08 · N8 · PART-pairing for the study generator g = 5 (2g ≤ B)

Status (REG-A, verbatim): OPEN; Prop Q PROOF SKETCH; W7-PART CE2 COUNTEREXAMPLE [NBH/ID_REGISTRY.md:152]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:184-188

N8. PART-pairing for the study generator g = 5 (2g ≤ B).
– *Statement (as far as sources go):* a partition law for complete linkage + size cut + next-fit packing with equal groups 2g ≤ B (study g = 5, B = 14): which groups are co-blocked, and does that remove masking (Prop Q)?
– *Status:* **OPEN**; Prop Q **PROOF SKETCH** (R2B row 6′); W7-PART CE2 COUNTEREXAMPLE (unequal groups not merged).
– *Supports:* M12 e1. *If it fails:* e1 survives (changing B or the cut is WORKHORSE); the study generator is disclosed as outside every selection theorem.
– *Cells:* row 3 × "NONE: partition law for the study's packing"; row 11 × "NONE: study generator".

Final placement. "S-BLOCKS (preservation, SIZE equality); FW25" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:75].
Clauses that depend on it. TH11 ships without it, since PART-pairing is "excluded". The open part is CJ25.
Who discharges it. No HQ: it is future work (CJ25). The study generator is disclosed as outside every selection theorem.
Do not retry. W7-PART CE2 is a COUNTEREXAMPLE: unequal groups are not merged. "SIZE does not determine pairings when two groups fit in one block" [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:33]. "the study generator's g = 5 partition theorem" is listed as not inherent: OPEN, and not to be presented as a limitation [NBH/branch_snapshots/MEET-v31/MINUTES.md:103].

OB09 · N9 · V5: kernel reversibility, self-adjointness, exact start sampler (closure gate)

Status (REG-A, verbatim): "by construction, consultant-checked, no registry row"; gate closed: N9-KERNEL "PROVED (candidate)" with exact numerical verification on production headers ([pos 1550]: 7 sub-claims; [pos 1593]: 8 sub-claims); cross-family VERIFIED WITH AMENDMENT (CHK_INHERITED) [NBH/ID_REGISTRY.md:153]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:190-195

N9. V5: kernel reversibility, self-adjointness, exact start sampler.
– *Statement:* each elementary move (Gibbs flip, uniform swap, neighbour swap, independence move with π_0 proposal) is γ_φ-reversible; a uniformly random order of reversible kernels is self-adjoint; the start sampler draws exactly from π_0 with exact $\bar{Z}_0$.
– *Status:* by construction, consultant-checked, **no registry row** (INV §5 item 12; TL:77); one-page route (TL V1(v)).
– *Closure gate (HQ, 2026-11-04):* N9 is a closure gate, not an ordinary obligation; it is being checked against the code (branch `theory/N9-KERNEL`). Open method questions: none, **conditional on N9** (running). If N9 fails, the method question reopens before the theorem meeting. Status words above unchanged.
– *Supports:* M1, V1, N1, rejection of C10. *If it fails:* V1 fails; the method question reopens before the theorem meeting. The binding goal (proposal §5.1) is not met by withdrawal: HQ escalates to Felipe for a method reconsideration or an explicit scope change.
– *Cells:* row 6 × V; row 4 × V.

Final placement. "S-KERNEL (cited N9 with CHK amendments)" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:76].
Clauses that depend on it. TH12 and TH01.a (status "N9K: PROVED (candidate), code-checked") [NBH/ID_REGISTRY.md:31].
Who discharges it. The gate is closed: N9-KERNEL was checked against production code and verified cross-family with amendment. What remains is the HQ-level registry row. REGISTRY_PAPER1 recommends "XF: VERIFIED WITH AMENDMENT (CHK); open the registry row" [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1898]. LP-FOUND cites N9 with the CHK amendments.

UNCERTAIN — 7 vs 8 sub-claims

[pos 1550] says 7 sub-claims and [pos 1593] says 8. MEET-v31 MINUTES §9 says seven. The archive does not reconcile them [NBH/02_theory.md:2786].

OB10 · N10 · V10: L versus t_mix

Status (REG-A, verbatim): NONE / OPEN [NBH/ID_REGISTRY.md:154]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:197-201

N10. V10: L versus t_{mix}.
– *Statement:* do $L = \max(4, \lceil \log_2(n+1) \rceil)$ terminal φ = 1 rounds give populations close to π?
– *Status:* **NONE / OPEN** (INV row 6); YWJ/ZYVR route blocked as stated (INV §2 D1). Validity (V1, N1) holds at any L; L affects accuracy and variance.
– *Supports:* M1. *If it fails:* theorems needing mixing carry L as a hypothesis.
– *Cell:* row 6 × "NONE: mixing / spectral gap".

Final placement. "FW4 (K24); S-IDENT minorisation/terminal-law TV ship" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:77].
Clauses that depend on it. None of TH01–TH20 has it as a premise. TH13 ships the minorisation/terminal-law bounds without it. It blocks CJ04 and is one of the two inputs that CJ02 needs ("needs V11 or N10, both OPEN").
Who discharges it. No HQ: future work (CJ04).
Do not retry as stated. "the YWJ/ZYVR route is blocked as stated" [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:12].

OB11 · N11 · ℓ versus the moves needed

Status (REG-A, verbatim): on LH with c ≤ s_0 fixed, "eventually true" (elementary, unchecked); growing s_0: OPEN [NBH/ID_REGISTRY.md:155]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:203-207

N11. ℓ versus the moves needed.
– *Statement:* $\ell = \lceil \log_2(n+p+1) \rceil \geq \max\{\Phi(S^\circ) - 1, 0\}$ (moves LS0 needs, LS-score), and the caps on the final local search and rebuilds are not binding w.p. → 1.
– *Status:* on LH with c ≤ s_0 fixed, "eventually true" (R2B row 7; elementary, unchecked); growing s_0: **OPEN**.
– *Supports:* M2. *If it fails:* R3/R5 carry ℓ as a hypothesis; growing support would need a bigger budget (method-induced).
– *Cells:* row 12 × C/A; row 7 × C (rebuild cap).

Final placement. "M5 hypothesis (+ S-CERT); growing support / cap nonbinding = FW19" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:78].
Clauses that depend on it. TH05, where ℓ is a hypothesis, and TH14. The open part is CJ19.
Who discharges it. The fixed-support LH part ("eventually true", unchecked) could be checked inside the M5 HQ. Growing s_0 is future work (CJ19).

OB12 · N12 · General (n, p) scope, per regime

Status (REG-A, verbatim): see the table in the source; p ≤ e^{Cn^α}: OPEN, not impossible [NBH/ID_REGISTRY.md:156]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:209-222

N12. General (n, p) scope, per regime (S2 P-9 theory scope; R2A–R2C §4).

Regime	What theory may claim	Cause of any restriction
Tiny n	No asymptotic claim. n ≤ 5 exact for **all p** (chair-checked). n = 2 extension (e3, proposed; current code rejects n < 3): PROOF SKETCH (R2C E1); its p-fold-tie output statement (M12) is R3B's, elementary, **seat-derived, unchecked**; duplicates kept (chair-checked), which is a code fact, not a proof of the n = 2 output. n = 1 → `degenerate_n`; zero predictors → `no_predictors`; n = 0 rejected.	None from the method.
Fixed p, n → ∞	Eventually exact dispatch (R2C §4.4, seat-proved, unchecked), so V2–V4 apply.	—
Growing p ≪ n	Polynomial-p target results under their assumptions; logarithmic L, ℓ may be short (N10, N11).	Method budgets (theorem hypotheses).
p ≈ n	Exact-Gram LH needs p ≤ n − 1; centred designs have rank ≤ n − 1.	Proof family's geometry, not the target.
Polynomial p ≫ n ($p \le n^A$)	T03/T03♯; T38(b1-LH-iid) XF GAP; study generator N8.	**Prior and proof technique** (INV §3b), not the sampler.
$p \le e^{Cn^\alpha}$	No theorem; whether our prior admits it is **OPEN**, not impossible.	—
Joint growth	Budgets polynomial; by-definition limits: fixed allocation, Θ(p) round; w4 boundary n + p ≥ 2^{31}.	Method-induced: K excludes s_0 > K; fixed B limits whole-group recovery to g ≤ B ("recovery as a single block", R3C); bounded stores; log ℓ and L; $N_0 \ge p$.

w1 and w3 affect proposal efficiency only; validity rests on the Hastings ratio.
– *Cells:* row 1 × A and row 1 × "NONE: growing s_0"; INV §3b "p up to e^{Cn^α}: LACK".

The REG-A status cell writes the exponent as "e^{Cn^α}"; it is rendered above in text form.
Final placement. "FW18; FW19; S-TARGET K76" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:79].
Clauses that depend on it. TH10: K76 eventual exact dispatch is "seat-proved, unchecked". The open parts are CJ18 and CJ19.
Who discharges it. LP-FOUND can check the K76 eventual-dispatch argument. Exponential p is future work (CJ18): "No completed prior-tail/contraction route; p ≤ n^A stays in scope" [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:26]. "subexponential-p theory under our prior" is listed as not inherent: OPEN [NBH/branch_snapshots/MEET-v31/MINUTES.md:102].

OB13 · N13 · Retention lemma

Status (REG-A, verbatim): PROVED (PC1 claim 5; R2C D7) [NBH/ID_REGISTRY.md:157]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:224-226

N13. Retention lemma. If $S_0 \in \mathcal{S}$ and $\delta_1 < ½$ then $S_0 \in \mathcal{T} = \mathrm{top}_{40p+1}(\mathcal{S})$, and the HPM and $\mathrm{MPM}^{\mathcal{T}}$ certificates are equivalent.
– *Status:* **PROVED** (PC1 claim 5; R2C D7); premises Lemma S and the inheritance merge (PC2 code facts). Unqualified retention stays false.
– *Supports:* A3 ∘ R5, card L3, M2's eviction wording. *Cell:* row 11 × A.

Final placement. "M4(c)" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:80]. Dependent clause: TH04.c.
Remaining work. Mathematically discharged, as a candidate. The M4 charter's first lemma re-

confirms it "against the code" under the coded LS0 rule, which depends on OB02(c). If the code's retention rule differs from union-then-top-c, "N13's premise fails → HQ" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M4.md:100-104]. Unqualified retention is false: do not attempt it.

OB14 · N14 · Stacking statements

Status (REG-A, verbatim): (a) PROVED (PC1 claim 6); (c) code audit, not done [NBH/ID_REGISTRY.md:158]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:228-234

N14. Stacking statements.
– (a) Cor W-R(b): post-search nesting, HPM(R′) ≥ HPM(4) with the extra search. PROVED (PC1 claim 6).
– (b) Every listed bound holds per run for any R′ declared before final randomness (PC1 claim-6 table).
– (c) IND-T for streams 10..9+R′: code audit, not done.
– (d) One certificate card per record (M7, M10).
– *Supports:* M7. *Without the extra search:* PC1's three uses (monotonicity, $D_n(R') \leq D_n(4)$, transfer of R = 4 measurements) go; no listed theorem affected.
– *Cell:* row 8 × V; row 8 × "NONE: choice of R vs N" (D4 PROOF SKETCH, v2).

Final placement. "S-OUT + M1(c); N14(c) = named audit premise (MINUTES §2.3)" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:81].

Clauses that depend on it.

- TH15 and TH01.c.
- N14(c) is "Used by" M1, S-OUT K84/K85/K86 and S-POOL TH17 [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1879].
- N14(c) is also named as a premise in the M5 TH05 and M8 TH08 charters [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:79] [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M8.md:61].

Who discharges it. LP-AUDIT and M1 run the IND-T extension audit (code). If it fails, "M1 (c),(d) revert to potential-run statements and N2 reopens".

UNCERTAIN — N14(c) label in the charters

The final audit found N14(c) "mislabeled as extra-search nesting" in M1:92, M5:79, M8:61 and LP-AUDIT:70. The correction: "Name IND-T extension under N14(c); name extra-search nesting under K84/N14(a)" [NBH/branch_snapshots/MEET-THM/CLOSE_AUDIT.md:14]. The charter snapshots read correctly now, for example [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:79]. Read N14(c) as the IND-T extension in every source.

OB15 · N15 · Work and depth at the implemented level

Status (REG-A, verbatim): Prop D PROVED (counting); calibration share seat-derived (R2A, R2B), not proof-checked [NBH/ID_REGISTRY.md:159]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:236-240

N15. Work and depth at the implemented level (D1(b) restated).
– *Statement:* depth and work for island-parallel v3.1: setup, (h) tables, calibration ($(1+4I_e)\cdot H_e$ work and $(1 + I_e) \cdot H_e$ depth per tier; ≤ ℓ rebuilds per tier, ≤ 2ℓ with escalation), final run $N_e(T + L + 4)m$ sequential step slots (not scorer operations or seconds; add score costs and finite-worker scheduling; R′ above the worker count runs in waves), LS0, extra search, population reductions.
– *Status:* Prop D PROVED (counting); calibration share seat-derived (R2A, R2B), not proof-checked.
– *Supports:* M5, M8, M13. *If it fails:* recount.
– *Cells:* row 8 × "NONE: work–depth theorem" (filled as counting; speed-up vs sequential MCMC stays empty, empirical); row 7 × C; row 5 × "NONE: depth vs correlation" (ideal grid via A_0 only).

Final placement. "M8; DAG audit named premise" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:82]. Dependent card: TH08.
Who discharges it. The M8 HQ, wave 3. First lemma, verbatim: "the trajectory inventory of the execution DAG (every executed pilot, diagnostic, rebuild and final trajectory with its own (N_i, T_i, L_i); every completed search), read from `v3_engine.h`, `v3_calib.h`, `v3_fit.R`; this is the W1PC1 §13 repair and the open audit (W2PC2:40)" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M8.md:79]. The execution-DAG audit status is "NOT REPAIRED / GAP" [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1883].
Do not claim. No comparator speed-up: "never state a comparator speed-up" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M8.md:82]. See also DR05 and CJ12.

OB16 · N16 · Floating-point contract

Status (REG-A, verbatim): OPEN. VAL-FP MEASURED at p ≤ 40 only [NBH/ID_REGISTRY.md:160]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:242-245

N16. Floating-point contract. Validated SC(η_F) for scores, Φ, M, $Z_{\mathcal{T}}^{-}$, and VAL-FP(η, ε) for $\hat{Z}$ at production size.
– *Status:* **OPEN.** VAL-FP MEASURED at p ≤ 40 only; η a documented heuristic (`v3_score.h:240`); PC2's exact-rational test found no violation in 60 cases (max |error|/η = 0.069): not a proof.
– *Supports:* M10 labels, G2 fp coverage, e2, e5. *If never done:* labels stay "numerical".
– *Cells:* row 14 × "NONE: floating-point error theorem"; row 9 × "VAL-FP MEASURED at p ≤ 40 only".

Final placement. "FW16; K108 in S-CERT" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:83].
Clauses that depend on it. No clause has it as a premise. It decides the "numerical" label on the certificate card. The M4 charter, verbatim: "N16 floating point (card is 'numerical' until SC + VAL-FP)" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M4.md:82]. That label sits on TH04. It is also the open sub-obligation of OB07. The future-work item is CJ16.
Do not retry. T33-errbound is a COUNTEREXAMPLE: "exact-arithmetic bounds do not validate rounded production computations" [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:24]. The 60-case exact-rational test is "not a proof".

OB17 · N17 · Analytic bounds on Z and π(Ŝ)

Status (REG-A, verbatim): PROVED (PC1 claim 1; PC2 C1(c)) [NBH/ID_REGISTRY.md:161]

Statement (verbatim from source) NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:247-249

N17. Analytic bounds on Z and π(Ŝ). $Z \le C_n(2b_0 + y^\top y/(n+1))^{-\nu}$; equivalently $\pi(\hat{S}) \ge e^{F(\hat{S})}(2b_0 + y^\top y/(n+1))^{\nu}/C_n$.
– *Status:* **PROVED** (PC1 claim 1; PC2 C1(c)). Its gap to Z is data-dependent: the ≈ 530-nat figure at n = 200 compares against m(∅), not against Z, and is not a universal looseness (n = 200, p = 1, y = x_1 gives log(U/Z) = 3.83; review check, SOL1 #27). Supports M14(ii).
– *Cell:* row 9 × "NONE … an upper bound on Z is impossible from a run": **INV's wording is literally false**; correct to what is proved (T31(c)'s non-implication plus this bound), not to a blanket "no useful upper bound".

Final placement. "M4(d) (final letter) and S-TARGET (K44)" [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:84]. Dependent clauses: TH04.d and TH10.
Remaining work. None mathematical (candidate PROVED). Do not state a universal looseness figure for it.

Notes carried with the obligations (INP, verbatim). "**Carried over unchanged (already in TL):** DELIV-POP (A4, OPEN); Conjecture P (in N5(b3)); Π(X) upper bound (R7); LH-iid XF GAP (R5); S4 weight stability (§5.3)." and "**Not obligations:** the HPM terminal frequency (descriptive, M9); an RB joint estimator (not shipped; as a v3.2 option it would need its own identity, R3A/R3B)." [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:255-257]. N18 ("Depth comparison between starts", OPEN, no route) existed only under item-1 Outcome 1 and was not opened [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:251-253].

5.5 Future work CJ01–CJ27 (= FW1–FW27)

Source and status. All 27 items come from MEET-THM/FUTURE_WORK.md, "Opened by the chair 2026-11-04; completed at close. Items here get no per-theorem HQ for paper 1." The header reads: "FW1 and FW2 are unchanged. FW3–FW27 were added by the clerk from MINUTES §2.3/§2.4, SYNTH_R2 §2.4 and R3_C §S4 'Required additions'; reasons are as the sources give them" [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:3-5]. Every CJ row of REG-A carries the same status, verbatim: FUTURE_WORK (deferred; reason in 02a §A5.1) [NBH/ID_REGISTRY.md:103-129]. The package was announced at [pos 1626].

How to read the table.

- "Item" and "Reason" are the verbatim cells of FUTURE_WORK.md; row FWk is at line $8 + k$.
- "Serves" (what the item would buy) is taken only from the coverage mapping of proposal goals, Song–Liang items and INV matrix rows [NBH/02_theory.md:2742-2770]. An empty cell (—) means the coverage mapping names no goal for it.

The FUTURE_WORK footer, verbatim: "Items recorded as 'outside the fixed method' (adaptive/overlapping partitions; backfit start) are dispositions in COVERAGE_MATRIX.md, not FUTURE_WORK. The audit obligations (N2(b),(c),(e), N14(c), the N3 DP invariant, the integrated streams/LS0/GUARD-W/DAG audits) are named implementation premises in HQ_CHARTERS/, not FUTURE_WORK (MINUTES §2.3)." [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:48].

ID (FW)	Item (verbatim)	Reason (verbatim)	Serves
CJ01 FW1	Frequentist coverage of the exact posterior's credible intervals	A target-theory question outside proposal §5.4, which concerns the computed vs the exact posterior (HQ ruling Q2, 2026-11-04) [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:9]	§5.4 credible-interval coverage (frequentist half); INV row 10

ID (FW)	Item (verbatim)	Reason (verbatim)	Serves
CJ02 FW2	TV/KL error bound for the final pooled estimate in the masked regime	Not binding (§5.4 lists "possible metrics"; HQ ruling Q3); needs V11 or N10, both OPEN [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:10]	§5.4 TV, KL, masked accuracy (not binding); row 10
CJ03 FW3	K13: start-mode consistency under orthogonal blocks and a σ^2-inflated beta-min margin	Needs TG4-type probability bounds not supplied [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:11]	—
CJ04 FW4	K24: polynomial mixing of the $\varphi = 1$ round (Yang–Wainwright–Jordan transfer); also useful general production mixing and masked population accuracy (N10)	N10 OPEN; the YWJ/ZYVR route is blocked as stated [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:12]	INV row 6 (moves); discharges OB10
CJ05 FW5	K23: Dirichlet/spectral comparison of the random-order round with the random-scan mixture	Its only consumer, K24, is deferred (A, B, D, R3-4). Seat C dissents: it would keep the finite squared-kernel Dirichlet comparison as an attributed lemma in S-KERNEL; MINUTES allows the S-KERNEL HQ to cite it without a theorem number [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:13]	row 6; optional citation CL22
CJ06 FW6	K31: as-coded masked stage count (realised T_n; Conjecture P OPEN); N5(b): rebuild convergence within ℓ rebuilds, candidate-pair coverage / P($S_0 \in V$), P(schedule_limit) and calibration-failure probabilities	Conjecture P OPEN; V's complement is never inspected (I9(d)); only $T \le T_e$ and the coverage identity are provable; successful-grid properties do not prove successful calibration. Rebuild convergence is FW in either placement of K33 [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:14]	rows 5, 7, 14; discharges OB05(b)
CJ07 FW7	K58: masked consistency of the equal-weight PIP/MPM	Reduces to population delivery; FMP-DELIVPOP is OPEN [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:15]	row 10
CJ08 FW8	K57, general fixed-budget equal-weight accuracy (only the recording-marginal bias comparison P1 ships in S-POOL)	A recording-marginal bias bound does not close estimator accuracy or DELIV-POP; needs a checked estimator-level theorem for the actual records and budgets [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:16]	row 10; dissent DS06 (D)
CJ09 FW9	K59: R vs N at fixed total particles (allocation)	PROOF SKETCH only [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:17]	row 8
CJ10 FW10	K71: (b1-LH-iid) on LH with iid final draws, also under general DELIV	(b1-LH-iid) GAP; T38-b1-LH-iidfail XF ERROR [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:18]	—
CJ11 FW11	K75: NP-completeness of the global-certificate decision problem for the fixed prior and penalty	Needs a score-specific reduction that has not been supplied [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:19]	—
CJ12 FW12	K92: separation from single-chain flip/swap MCMC (moat construction)	The moat construction and the accuracy-matched comparator are unproved; speed-up claims are empirical by decision [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:20]	Song–Liang §6 complexity (accuracy-matched comparison)
CJ13 FW13	K82: delivery, P(U) → 0 (also delivery/resource-failure laws, P(resource_limit))	No route; every output theorem is stated jointly with the output event in liminf form [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:21]	rows 11, 14
CJ14 FW14	K94: finite-n bound on the posterior-miss probability Π(X)	Envelope GAP [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:22]	row 1 (target)
CJ15 FW15	K99 quantitative: k-way barrier for cross-block masked contrasts	k-way barrier unproved (the qualitative remark ships in S-LIM) [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:23]	— (qualitative form in TH18)

ID (FW)	Item (verbatim)	Reason (verbatim)	Serves
CJ16 FW16	K107: validated arithmetic / rounding-error bound; production validated numerics (N16)	N16 OPEN; T33-errbound COUNTEREXAMPLE; exact-arithmetic bounds do not validate rounded production computations. K108 (deterministic perturbation lemma) stays in S-CERT [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:24]	§5.1 numerical fidelity; rows 2, 9, 14; discharges OB16
CJ17 FW17	K80: positive power of the E1/E2/E3 and low_empirical_ESS flags	Flags are heuristics by decision; no sampling law for the flags [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:25]	§4.4/§8.2 diagnostics; row 13
CJ18 FW18	N12 / exponential p: exponential-dimension contraction and selection (Song–Liang Cor 1)	No completed prior-tail/contraction route; p ≤ n^A stays in scope [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:26]	Song–Liang Cor 1; row 1; OB12
CJ19 FW19	Growing support and cap non-binding (N11/N12)	Fixed-support LH does not discharge these extensions [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:27]	row 1; OB11, OB12
CJ20 FW20	Hellinger / predictive contraction (Song–Liang Thm 1)	Not delivered; selection counterpart in S-T only [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:28]	Song–Liang Thm 1; row 1
CJ21 FW21	K97: Song–Liang analytic separation (null-exclusion/FDR)	The iteration-uniform null-BF/prior-odds step is NOT REPAIRED (W2PC1) and is an unproved hypothesis about SL's algorithm, not a model/design condition. LIT-PAR G4 is cited in the paper as labelled numerical evidence [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:29]	Song–Liang Cor 2; CL23
CJ22 FW22	Sharp ε-scale necessity statement for §5.3	All three seat forms are COUNTEREXAMPLEs; C's is a range obstruction for the D_* route only. The scale disclosure ships (not a theorem) [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:30]	§5.3 weight stability (binding)
CJ23 FW23	K29: favourable O(1) stage law, unconditional application	The (LL)/over-tilted q_w input is not a model/design condition and no family exhibits it (W2PC1:187 DESIGN-ASSERTED, GAP). Only the conditional implication is cited in S-STGI [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:31]	— (see TH21)
CJ24 FW24	Useful two-sided masked evidence precision (N4(iii)/V11)	Distinct from FW2's final-estimate TV/KL and from one-shot variance failure; V11 OPEN [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:32]	rows 8, 9; OB04(iii)
CJ25 FW25	Random/adaptive partition quality and study packing (N8; K71 as C pairs them), including the study generator (PART-pairing, g = 5, 2g ≤ B)	SIZE does not determine pairings when two groups fit in one block; PART-pairing OPEN; the study design is not the LH exhibit [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:33]	rows 3, 4, 11; OB08
CJ26 FW26	Finite-n Gibbs discovery (HPM residual)	Existing asymptotic or vacuous bounds do not supply useful finite-n discovery (ThmB-T24 vacuous at n = 200; no route) [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:34]	row 11
CJ27 FW27	Unbounded functionals: integrability extensions of M1 (§13.1)	M1 is stated for bounded h ∈ [0, 1]; unbounded functionals require integrability extensions (from C's coverage table, not its "Required additions" table) [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:35]	§13.1 (binding goal); row 10

For agents — future work: what would reopen an item

The sources name the missing input for some items. Only these are restated here:

- CJ23 (K29) "returns as a supplement clause" given "A design family on which (LL) or the over-tilted q_w bound is verified" (A), "checked in one wave" (B), or "A checked IC3 family establishing the missing input" (C) [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:906-918].
- CJ08 needs "A checked estimator-level theorem for the actual records and budgets" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1097].
- CJ02 needs V11 or N10.
- CJ21 is blocked by the iteration-uniform null-BF/prior-odds step, which is NOT REPAIRED. The archive records a recommendation for a scout HQ on it (decision P2) [NBH/05_open_questions_and_next_steps.md:67].

Items the method meeting listed as "Not inherent (OPEN; must not be presented as limitations)" must never be written up as limitations (§5.6).

5.6 Drops DR01–DR06 and inherent limitations LIM01–LIM09

Drops (candidates removed from the list; do not re-propose without new evidence)

The drops are recorded in MINUTES §2.4 and in FUTURE_WORK.md under "Dropped (MINUTES §2.4; not FUTURE_WORK)" [NBH/branch_snapshots/MEET-THM/MINUTES.md:96-102] [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:37-46]. K12 and K34 were dropped by all four seats in Round 3 [NBH/branch_snapshots/MEET-THM/MINUTES.md:81]. The "Candidate" column below quotes the registrar's description of the candidate [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1663-1668].

ID (K)	Status (REG-A, verbatim)	Candidate and evidence (REGISTRY_PAPER1 §4, verbatim fragments)
DR01 K12	DROPPED. Reason: A test oracle, not a guarantee	"Test oracle, not a guarantee (A, B, D drop; C would defer)"
DR02 K34	DROPPED. Reason: TI evidence is not in the method (the $\hat{J}$ half ships as reporting, S-REP)	"DROP as a theorem … TI evidence is not in the method; $\hat{J}$ half ships as reporting (S-REP). (A, B, D drop; C would defer)". The $\hat{J}$ half is TH19.
DR03 K66	DROPPED. Reason: PARTLY-FALSE (C3-PGD, CEX-localmax(A2))	"HP3 landscape PARTLY-FALSE (C3-PGD, CEX-localmax(A2))". C3-PGD (COUNTEREXAMPLE): "SE and beta-min do not imply progress-greedy dominance: a null that looks like a combination of two truths is added first."
DR04 K67	DROPPED. Reason: Relies on K66	"Relies on K66". No registry row for HP4. Its LH analogue is T38-b1-LH (in M5, TH05.b1LH).
DR05 K91	DROPPED. Reason: No accuracy-matched comparator; OPW2-cex	"Particles-vs-depth speed-up: no accuracy-matched comparator; OPW2-cex refutes the perfect-mixing formula". INP §5 row 8: speed-up vs sequential MCMC → "empty (empirical)".
DR06 K98	DROPPED. Reason: KNOWN-FALSE (SHARE-M(b), FD-NM)	"Naive merge KNOWN-FALSE (SHARE-M(b), FD-NM)".

REG-A status cells: [NBH/ID_REGISTRY.md:130-135].

UNCERTAIN — line numbers of the drops in MINUTES

FUTURE_WORK.md cites the drops as `MINUTES.md:94–99` [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:41-46]. The archive's conflict list says they "sit at lines 98–103" in the snapshot [NBH/02_theory.md:2789]. In the snapshot read for this document they are at lines 97–102, under the heading "Drops, with reasons:" at line 96 [NBH/branch_snapshots/MEET-THM/MINUTES.md:96-102]. The content is identical in all three. Only the line numbers drift.

Inherent limitations (method meeting MEET-v31, §5)

These nine items are the method meeting's list. The rule, verbatim: "Each is either PROVED impossible or true by definition (rule: SYNTH_R1 §3 as amended by the seats). Re-checked under HQ amendment 01: items that do not meet this rule were moved to the 'Not inherent' list below." [NBH/branch_snapshots/MEET-v31/MINUTES.md:84]. The positions are [pos 1128], [pos 1280] and [pos 1302]. The paper-level theorem form of several of them is TH18 (S-LIM).

LIM01 true by definition. "**Per-round sequential factor m = p + B + 2.** True by definition of a random-scan full-coordinate Gibbs round: each update conditions on the previous one. This is inherent to the chosen kernel, not to SMC or BVS. Removing it (C10) is WORKHORSE and breaks the Gibbs-path theorems (T38(b2), BG3, V5)." [NBH/branch_snapshots/MEET-v31/MINUTES.md:86]

LIM02 true by definition. "**Every worker needs scoring access to all predictors** (X or sufficient statistics, 8np bytes, local or shared/distributed at a communication cost). True by definition of a full-coordinate round. A private $8p^2$ Gram per worker is **not** inherent (C7)." [NBH/branch_snapshots/MEET-v31/MINUTES.md:87]

LIM03 T27(i), PROVED-LOCKED (conditional, in its design class). "**Masking of the product-of-blocks start built from block-local scores against the full y, under T27(i)'s hypotheses (H2, H3; verified on LH).** T27(i), PROVED-LOCKED (conditional, in its design class). This is a law for that construction and design class; it does not say that every product start is masked on every design. The bridge exists to correct it (SEP-B)." [NBH/branch_snapshots/MEET-v31/MINUTES.md:88]

LIM04 V12 and E1c PROVED; Chatterjee–Diaconis Thm 2.1. "**No diagnostic from the observed record (the run's own weights or islands) has a universal detection guarantee for never-visited modes.** V12 and E1c PROVED; Chatterjee–Diaconis Thm 2.1. This does not rule out alerts on particular runs: positive one-sided certificates (decision 10, e.g. $\hat{Z} < Z_{\mathcal{T}}$) can still detect some failures." [NBH/branch_snapshots/MEET-v31/MINUTES.md:89]

LIM05 T31(c), PROVED-LOCKED; PC2 C1(c), PROVED relative to it. "**The properties of $\hat{Z}$ behind the 95% bound imply no two-sided relative-error bound** (positive, finitely many values, mean Z, iid islands: some such law has $P(\hat{Z} < c_0 Z) \geq 1 - \eta$ for any $c_0 > 0$, $\eta \in (0, 1)$, so $\{Z \geq \alpha\hat{Z}\}$ places no upper limit on $Z/\hat{Z}$) (T31(c), PROVED-LOCKED; PC2 C1(c), PROVED relative to it). This is a non-implication from the estimator's law, not a data-specific impossibility; it excludes neither structural nor record-based bounds. A finite analytic bound exists (PC1 claim 1, PROVED): $Z \leq C_n(2b_0 + y^\top y/(n+1))^{-\nu}$. Its gap to Z is data-dependent; there is no universal looseness figure. The literal 'no run can upper-bound Z' was FALSE, and the blanket 'no useful upper bound on Z from the run's record alone' is withdrawn as an inherent limitation." [NBH/branch_snapshots/MEET-v31/MINUTES.md:90]

LIM06 NOT inherent (moved); seat-derived, unchecked. "*(Moved by HQ amendment 01.)* Markov sharpness ('$0.05\hat{Z}$ is sharp given only nonnegativity and the mean') is Seat C's argument, **seat-derived, unchecked** (not covered by PC1), so it is no longer listed as inherent; see the list below. Unchanged and PROVED: the assumption-free probabilistic lower bound $\max\{Z_{\mathcal{T}}, 0.05\hat{Z}\}$ keeps ≥ 95% coverage (PC1 claim 2). Sharper bounds need information beyond positivity and the mean (not necessarily a variance hypothesis)." [NBH/branch_snapshots/MEET-v31/MINUTES.md:91]

LIM07 true by definition of the fixed target (N200, T07). "**Posterior misses at n = 200** (565 strict competitor defeats of S_0 among the 2,394 completed fits, with six stops: at least 565 target-mode misses, not the exact count of global-posterior misses on all 2,400 data sets). True by definition of the fixed target: no search can make the HPM equal S_0 when $F(S_0) < F(\hat{S})$ (N200, T07). This is a fact about these data and this prior, not a universal error rate." [NBH/branch_snapshots/MEET-v31/MINUTES.md:92]

LIM08 true by definition (V13). "**The cap K excludes truths with $s_0 > K$,** by definition of the prior. **A fixed allocation cannot support unrestricted growth of (n, p),** by definition (V13)." [NBH/branch_snapshots/MEET-v31/MINUTES.md:93]

LIM09 true by definition. "**The restricted ESS on V cannot bound mass outside V** without further assumptions, by definition of a restricted diagnostic." [NBH/branch_snapshots/MEET-v31/MINUTES.md:94]

REG-A status cells: [NBH/ID_REGISTRY.md:136-144]. REG-A maps LIM01 = I1, LIM02 = I2, LIM03 = I3, LIM04 = I5, LIM05 = I7, LIM06 = I8, LIM07 = I6, LIM08 = I9(a),(e) and LIM09 = I9(d) in the INP I-table.

For agents — not inherent: do not present these as limitations

MEET-v31 MINUTES §5, verbatim: "**Not inherent (OPEN; must not be presented as limitations):**

- the severity of weight variance through a masked crossing on the production grid (V11);
- rebuild convergence, $P(S_0 \in V)$ and P(schedule_limit) → 0;
- the as-coded masked T_n law;
- DELIV-POP;
- weak-dependence stability (proposal §5.3; a binding goal, so the escalation rule of §9 applies);
- subexponential-p theory under our prior;
- the study generator's g = 5 partition theorem;
- the measured-round identity for general h (proposal §13.1; a binding goal, so the escalation rule of §9 applies);
- Markov sharpness of the assumption-free lower bound on Z (moved from item 6: seat-derived, unchecked);
- the blanket 'no useful upper bound on Z from the run's record alone' (withdrawn from item 5: 'useful' is undefined and the claim is not proved; only T31(c) and PC1 claim 1 stand).

These are theorem obligations in `THEOREM_REVISION_INPUTS.md`, each with the decision it supports and what happens if the obligation fails. Where a failure would leave a binding proposal goal unmet (general h, §13.1; weight stability, §5.3; N9, §5.1), the goal is not met by withdrawal: HQ escalates to Felipe for a method reconsideration or an explicit scope change (§9)." [NBH/branch_snapshots/MEET-v31/MINUTES.md:96-108]

Likewise the lemma packs: a clause whose proof fails "is never relabelled as a limitation" (IC7), and for S-LIM "a limitation whose proof fails moves to FW. It is never kept as an assertion" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-REST.md:91].

UNCERTAIN — which list of inherent limitations

Three counts coexist in the archive [NBH/02_theory.md:2783]:

- MEET-v31 MINUTES §5 has 9 numbered items, of which item 6 is a "moved" stub. LIM01–LIM09 follow this list.

- THEOREM_REVISION_INPUTS §4.A has I1–I7 and I9–I11 (I8 moved). This includes I4 (one-shot IS needs ≈ Q*), I10 (equal-weight bias) and I11 (sequential rebuilds), which are absent from the MINUTES nine.
- FOR_FELIPE_FINAL_METHOD §4 lists 7.

The archive registers the MINUTES nine because HQ also says "nine" at [pos 1280]. The I-table is at [NBH/branch_snapshots/MEET-v31/THEOREM_REVISION_INPUTS.md:261-286].

5.7 Cited lemmas CL01–CL23

These come from TLF Part III, "Cited lemmas (not numbered claims of paper 1)": "Collected from the final cards and Round 3 files. Each is cited in the main text or inside a card; none is a theorem number of paper 1 unless said so." [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1297-1299]. The REG-A status of each is "cited (not a numbered claim of paper 1); used in: …" [NBH/ID_REGISTRY.md:212-234]. Columns 2 and 3 are verbatim. The last column is the archive's own source pointer (paths relative to MEET-THM).

ID	Cited result (verbatim)	Used in	Archive pointer
CL01	N9 kernel facts: γ_φ-reversibility of each elementary move, q_φ-invariance and self-adjointness of the uniformly random-order round, exact draw_pi0 and $\bar{Z}_0$ (K08, K20, K21), with the CHK amendments (VERIFIED WITH AMENDMENT)	M1(a), S-KERNEL	R3_A:114,120; MINUTES:39,64
CL02	K47 (pooling, cited)	M1; S-POOL	MINUTES:39; R3_A:114
CL03	SMC identities and fixed-data limits: Del Moral 2004; Del Moral–Doucet–Jasra 2006; Whiteley–Lee–Heine 2016; Dai et al. 2022	M1(a), (f)	R3_A:127; R3_D:183
CL04	Markov lower bound: Grosse–Ghahramani–Adams 2015	M1(d), M4(e)	R3_A:127; R3_B:208
CL05	Greedy-schedule optimality: Štefankovič–Vempala–Vigoda 2009	M7(a), M2(b6)	R3_A:154; R3_B:179
CL06	K28: $\sqrt{c} \cdot T^*(c) \to \Lambda_{\text{len}}$ (Syed et al. 2024 Thm 3; Gelman–Meng 1998)	M7(e) (cited)	R3_A:147,154; MINUTES:45
CL07	Chatterjee–Diaconis 2018 Thms 1.1–1.3 (exp(KL) scale; with log-weight-tail qualifications)	M6, M2(c2)	R3_C:199; R3_B:179; R3_D:183
CL08	Cérou–Del Moral–Guyader 2011 Thms 1.4–1.5 (variance theorems; novelty of S-Z (i) to the citation checker)	M2, S-Z	R3_B:179; R2_B S4
CL09	Frazier et al. 2023, Assumption 6(iv) (generic χ^2 precedent; not re-fetched)	M2	R3_B:179
CL10	Papaspiliopoulos–Rossell 2017 §§3.1–3.2 (block computation precedent)	M3	R3_C:170
CL11	Dai–Heng–Jacob–Whiteley §4.2 (independent-system evidence weighting)	S-REP	R3_C:236
CL12	K60, K61: Barbieri–Berger 2004 HPM/MPM facts (cited lemmas)	M5, S-TARGET	R3_D:151,183; R2_D:46
CL13	Rossell 2022 Prop 1 (the $2E_0\delta_1$ term)	M5(0)	R3_D:151
CL14	Yang–Wainwright–Jordan 2016; Zhou–Yang–Vats–Roberts–Rosenthal 2022 (context next to M5; "with their priors LH is covered by rapid mixing; ours is for the paper's own target and start")	M5, M4	R3_D:151; R3_B:208
CL15	Hou–Wang–Atchadé arXiv:2407.20580 (never "first finite-n")	M5, M4	R3_D:151; R3_B:208
CL16	Brent 1974 (W/w + D scheduling)	M8(d)	R3_D:170,183
CL17	T22(iii) (K16) PROVED-LOCKED CONDITIONAL, with G, R, within-block SE, EXP′, VIS, NL_η, $\text{NL}_{\text{avg}}(\kappa)$ retained	M3 (supplementary clause)	R3_C:163
CL18	CHK-verified WS3, WS4 (M2(a), M3(e)); WS6(d), WS7 with amendments (M2(c))	M2, M3	R3_B:160,170–171; MINUTES:23

ID	Cited result (verbatim)	Used in	Archive pointer
CL19	T03♯ (S-T) target consistency, T01–T03 PROVED-LOCKED (external review queued)	M1(i), M3, M4(f), M5, M6	R2_C C9; MINUTES:60
CL20	Lemma RO (event containment), GUARD-W (stop-only coupling; audit open)	M1(d), M4(e), S-OUT	R3_B:194; R2_C C13
CL21	Furnival–Wilson 1974 (leaps-and-bounds superset bound, K05)	S-CERT	R0_DIFF K05 row (line 74)
CL22	K23 squared-kernel Dirichlet comparison, as an attributed lemma without theorem number (C's dissent; optional for the S-KERNEL HQ)	S-KERNEL	MINUTES:79; R3_C:59
CL23	LIT-PAR G4 (Song–Liang numerical comparison), cited as labelled numerical evidence; the analytic separation K97 is FUTURE_WORK	Coverage Cor 2	MINUTES:78,84

TLF table rows: [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1303-1325]. The local cross-references are: CL01 → TH12, OB09; CL08 → TH16; CL11 → TH19; CL12 → TH10; CL19 = TH09; CL20 → TH15; CL21 → TH14; CL22 = CJ05 (optional citation); CL23 → CJ21.

Not claimed / not proved — cited lemmas

- A cited lemma is not a numbered claim of paper 1. Its status is that of the cited source.
- CL09 is marked "not re-fetched".
- CL08's relation to S-Z (i) is unchecked: "Novelty of (i) vs Cérou–Del Moral–Guyader Thm 1.4 unchecked" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:1035].
- CL15 must never be described as "first finite-n".
- CL20 carries an open audit (GUARD-W, OB02(e)).
- CL17 is CONDITIONAL.

UNCERTAIN — archive pointers

The pointers in the last column are the archive's own. They refer to the MEET-THM seat files at the line numbers the archive records. Two of them had line drift in other places: MINUTES line numbers cited by FUTURE_WORK.md, and the registrar's MINUTES:87 for the audit-obligations row, which is at line 90 in this snapshot [NBH/branch_snapshots/MEET-THM/MINUTES.md:90]. This agent did not re-verify every CL pointer line.

5.8 Which obligations block the most card clauses

For agents — ranking of obligations by number of dependent card clauses

Method. This is a count, with no judgement added. For each obligation, the sources tie a list of TH IDs (cards or clause rows) to it. Those lists are the "Clauses that depend on it" rows in §5.4. They come from the COVERAGE_MATRIX "Final" column, the REGISTRY_PAPER1 §8 "Used by" column, the charters' named premises and the card status words. The table counts distinct TH IDs. It omits OB13 and OB17, whose status is PROVED (a candidate status), even though TH IDs depend on them (OB13: TH04.c; OB17: TH04.d, TH10). Ties keep the OB order.

#	ID	Open part (status words)	Count	Dependent TH IDs
1	OB02 (N2)	(b) not executed; (c) not in production core; (e) GUARD-W not audited	8	TH01.c, TH01.d, TH04.c, TH04.e, TH05, TH08, TH15, TH19
2	OB01 (N1)	PROOF SKETCH (binding §13.1)	7	TH01.b, TH01.e, TH01.f, TH01.g, TH01.i, TH13, TH19
3	OB14 (N14)	(c) code audit, not done	5	TH01.c, TH05, TH08, TH15, TH17
4	OB03 (N3)	DP invariant OPEN; bridge to code not done	3	TH04.f, TH05.b1LH, TH12
5	OB04 (N4)	(iii) CONJECTURE, OPEN for $c \geq 2$	3	TH01.f, TH02.b5, TH17
6	OB06 (N6)	(b) modulo premise (i) = N2(e)	3	TH01.d, TH04.e, TH15
7	OB05 (N5)	(b) OPEN	2	TH01.c, TH07
8	OB07 (N7)	sub-obligation SC/VAL-FP (N16)	2	TH04.e, TH14
9	OB09 (N9)	PROVED (candidate), VERIFIED WITH AMENDMENT; registry row to open	2	TH01.a, TH12
10	OB11 (N11)	growing s_0 OPEN; LH part unchecked	2	TH05, TH14
11	OB08 (N8)	OPEN	1	TH11 (ships without it)
12	OB12 (N12)	K76 seat-proved, unchecked; exp. p OPEN	1	TH10
13	OB15 (N15)	calibration share not proof-checked; DAG audit GAP	1	TH08
14	OB16 (N16)	OPEN	1	TH04 (“numerical” label only)
15	OB10 (N10)	NONE / OPEN	0	— (blocks CJ04, CJ02)

Chains the sources state.

- OB06(b) reduces to N2(e) of OB02.
- OB04(i) and TH19 reduce to OB01.
- OB07’s open part is OB16.
- The M5 charter calls N3 “the only OPEN mathematical obligation in the chain” of M5 [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:101].

Kinds of obligation. OB01, OB03 (DP invariant) and OB04(iii) are mathematical. OB02, OB14(c), OB15 (DAG) and OB16 are code-audit or validated-numerics items. The theory loop can only prepare the latter, for example by writing the exact audit statement; executing them is engineering [NBH/branch_snapshots/MEET-THM/REGISTRY_PAPER1.md:1870].

6 Work done: the campaign, reversals and lessons

This section is the memory of the project. It records how the theory reached its present state, which claims were reversed or refuted and why, which routes are known dead ends, how results were checked, and how the work sits against the literature. Every reversal below was paid for with real compute and reviewer time; the loop must not pay for it again. Statuses of campaign-history IDs (three-digit IDs, “Part B” of the archive registry) are copied verbatim into the registry and rendered in Appendix B; where a status is quoted here it is inside `\status{}` and copied from [NBH/ID_REGISTRY.md].

For agents — how to use this section

Before you attack any target, (1) find it in the reversal entries (§6.3) and in the do-not-retry table (§6.4); (2) check whether a campaign-era result already covers part of it (its three-digit ID, Appendix B); (3) note which check level its inputs carry (§6.5). A route listed as refuted may be revisited only with a new idea that addresses the recorded reason for failure, and you must say so explicitly in your output. The old repository documents (§6.2) are *not* current; never quote their statuses or targets as the present state.

6.1 Timeline of the theory work

The theory went through five stages: a pre-campaign architecture run and proof waves on the old repository (October 2026); the campaign thread "New BVS Headquarters" (2026-11-02 to 04: cloud waves W1–W8, Astra reviews, API runs, literature phase); the method meeting MEET-v31; the theorem meeting MEET-THM; and the transfer of the archive into this repository, followed by Felipe's approval of the theorem list on 2026-11-07. Positions [pos N] refer to the thread transcript `NBH/transcript/`.

When (UTC)	Stage	What happened (sources)
2026-10-19 → 10-20	Pre-campaign architecture run	Fable 5.1-led council, review group, judge (Astra + two Sol) and lead's rulings produced 36 conjectures T01–T36 (22 core, 10 secondary, 4 optional), `THEORY.tex`, `THEOREM_MAP.md`, `DROPPED.md`; "RUN COMPLETE 2026-10-20" [Docs/Theory/STATUS.md:6-17].
Oct 2026	Pre-campaign proof waves	Three proof waves locked 21 entries in `proofs/LOCKED_PROOFS.md`; review depth differs by wave (wave 3 same-family only, external review queued); none adopted [Docs/THEORY.md:13-21].
2026-10-28	Frozen paper targets	HQ and Felipe froze the JRSS-B list (Thm 1 = T03, Prop 1 = T07, Thm 2 = T22+T27, Thm 3 = T37, Thm 4 = T21, Thm 5 = T38, Thm 6 = T26, Prop 2 = T12) and a Biometrika cut of 5; new conjectures T37–T39 added [Docs/THEORY_PRIORITIES.md:1-40]. State at campaign start: 39 items, 21 locked entries, none adopted [NBH/02b_theory_campaign_history.md:65].
11-02 14:40 → 15:54	Phase 0 ([pos 0–180])	Thread becomes the "New HQ"; GH-PACK transfer lands on `main` as `c4ef0dc`; HQ reads the transferred theory: "So the paper hinges on T37 → T38" [pos 156]; literal T37 already failing on data [pos 178]; green light [pos 180] [NBH/01_overview_and_timeline.md:13-38].
11-02 15:54 → 20:36	Phase 1 ([pos 180–800]): cloud waves W1–W8	Wave 1 (5 Opus routines: W1-T37R, W1-T24G, W1-T38E, W1-T03R, W1-EXP), Waves 2–8 (verifications, p ≫ n family, PGD, T38 assembly, SHARE, repairs W6-FIXREV1/2, W7-SEP/PART/SEC, W8-SYNTH), plus subscription teams (F-*, O-*, A-REV1–4, L-LEAN, S-ELS, F-REFEREE) [NBH/01_overview_and_timeline.md:48-65]. Literal T37, H24 and the sharper T02 cutoff fall; T03# proved; "tempering is needed" refuted by W7-SEP [NBH/01_overview_and_timeline.md:70-89].
11-02 21:08 → 22:36	Phase 2 ([pos 800–960])	Astra re-reviews A-REV5–9; E-SPINE opens PR #1 (spine correction, unmerged); T38-RECORD statement of record v3 and registry v2 (431 results, 26 conflicts, LOCKABLE 48 → 55); API credit set up; API wave (PXNGAP, N200, MASKZPIP, RESID, FAVIID, AUDIT, T26BUD, POS, VSTAB); LH-LOCK dies on the 64K output limit [NBH/01_overview_and_timeline.md:115-131].
11-02 22:36 → 11-03 16:34	Phase 3 ([pos 960–1110])	API results; literature phase LIT-SMC/MCMC/STAT/VENUE, then LIT-PAR; Felipe sets Biometrika + Lean [pos 1015] and re-aims the project at posterior accuracy and parallelism [pos 1020], [pos 1030]; thesis sentence [pos 1044]; METH-AUDIT, THM-INV, THM-SYNTH; method decision v3.1 [pos 1107] [NBH/01_overview_and_timeline.md:137-160].

When (UTC)	Stage	What happened (sources)
11-03 16:34 → 11-04 00:01	Phase 4 ([pos 1110–1300]): method meeting MEET-v31	Three seats (Fable CP, Fable CW, Astra), chair Opus; decisions MD01–MD13; block start kept (MD01); 49 settings audited (DF25); 9 inherent limitations, 17 obligations [NBH/01_overview_and_timeline.md:164-194].
11-04 00:01 → 15:16	Phase 5 ([pos 1300–1676]): Round 0, Sol trial, N9 gate, theorem meeting MEET-THM, ABL-CONF	Fresh-eyes Fable and raw Astra (Round 0); Sol 1/Sol 2 reviews (73 findings, none changed a method decision); N9 kernel gate passes; theorem meeting closes at 05:27 with 8 main + 12 supplement cards, 27 future-work items, 6 drops [pos 1626]; HQ approves with one change (RL06) [pos 1639]; ABL-CONF verdicts [pos 1669] [NBH/01_overview_and_timeline.md:198-223].
11-04 15:10	Thread end	Felipe: "I will take a look at the theorems and give you feedback / questions later" [pos 1673]; the last three runs fail because the working directory had moved [NBH/01_overview_and_timeline.md:396-398].
2026-11-04	Archive	Text-only archive NBH/ written (transcript, 84 branch snapshots, registry) [NBH/README.md:12-16].
2026-11-07	Approval	Felipe approves the paper-1 theorem list (M1–M8 = TH01–TH08, supplement cards TH09–TH20) as the current targets APPROVED (F.B., 2026-11-07) [Theory_work/State_of_Project/sources/approval_2026-11-07.md:3-5]; recorded as decision DEC01 (§7.1).

How the theorem count evolved. 8 main + about 11 lemmas + about 5 remarks (frozen Oct 28) → "6–7 main results and 12–15 in the supplement" [pos 957] → "About 22 main theorems in total. Around 14 would go in paper 1" [pos 1348] → final 8 main + 12 supplement cards [pos 1626], [pos 1650] [NBH/03b_reasoning_and_traces.md:125-127]. The draft list entering the theorem meeting had 24 entries, not 19 as earlier summaries said [NBH/03b_reasoning_and_traces.md:366].

6.2 The old repository theory documents and what supersedes them

The repository still carries the pre-campaign theory documents. They were written by the previous HQ ("SMC_BVS Headquarters") and its GH-PACK crew, and they describe the state *before* the campaign. Docs/THEORY_PRIORITIES.md says so itself: "The proof statuses in this repo are unchanged since 2026-11-02. Any results of the theory campaign running elsewhere (D77) are not reflected here" [Docs/THEORY_PRIORITIES.md:3]; the old HQ stopped theory work under D77 ("the theory campaign is being executed elsewhere") [Docs/HQ/DECISIONS.md:231], [Docs/PLAN.md:3]. Where the archive and an older repo document disagree, the archive's final-state files win.

Old document	What it is / what is stale in it	Superseded by
Docs/THEORY.md	2026-11-02 status draft of T01–T39 with the pre-campaign vocabulary PROVED-LOCKED / CONDITIONAL / PARTLY PROVED / CONJECTURED / DROPPED [Docs/THEORY.md:1-21]. T37, T38, T39 appear as conjectures; the campaign refuted literal T37 and restated T38 (§6.3).	§4, §5 (cards TH01–TH22); NBH/02_theory.md; campaign history NBH/02b_theory_campaign_history.md.

Old document	What it is / what is stale in it	Superseded by
`Docs/THEORY_PRIORITIES.md`	Frozen 2026-10-28 JRSS-B targets (Thm 1–6, Prop 1–2), Biometrika cut of 5, pursuit order "T37 first" [Docs/THEORY_PRIORITIES.md:27-40]. Its story "consistent because the truth gets scored" [Docs/THEORY_PRIORITIES.md:25] was replaced by posterior accuracy + parallelism (reversal H.7). Its n = 200 disclosure "beta-min fails by about 100×" [Docs/THEORY_PRIORITIES.md:66] is retired (**RF105**).	Approved list M1–M8 + supplement APPROVED (F.B., 2026-11-07); venue Biometrika + Lean [pos 1015]; objectives [NBH/START_HERE.md:9-14].
`Docs/PLAN.md`	Old HQ plan (2026-11-03): phases P4–P9, proof pursuit order T37 → T02/T03 → T22+T27 → T38 → (T24, T39) → bookkeeping; open decisions 1–17 [Docs/PLAN.md:92-145].	Per-theorem HQ waves and charters (§7.5); pending decisions DEC01–DEC14.
`Docs/Theory/theorems/`, `one_pagers/`, `THEOREM_MAP.md`, `THEORY.tex`	Pre-campaign conjecture files T01–T36 and one-pagers T01–T39 [README.md:31-34]. Shared definitions in `theorems/00_COMMON.md` (G, R, R1, SE, BM, EX-A, EX-S, CAL, $F(S)$, ...) are still the definitions used by the campaign results [NBH/notation_and_conventions.md:447].	Notation: Appendix A; statements: the cards.
`Docs/Theory/proofs/LOCKED_PROOFS.md`	Register of the 21 locked entries (PROVED-LOCKED); "Locked is not adopted" [Docs/THEORY.md:21]. Locked statuses are inherited verbatim into the cards (e.g. T12, T31(i) in M1).	Card statuses (§4). Adoption remains open (DEC12).
`Docs/Theory/DROPPED.md`, `RULINGS.md`	Candidates dropped during the pre-campaign run with reasons LF/UR/TR/UA/DM [Docs/Theory/DROPPED.md:1-3]; lead's rulings of 2026-10-20 on the judge's changes [Docs/Theory/RULINGS.md:1-3]. Archived as `RF106–RF118` and `RF120` (status "DROPPED (LF)" etc.).	Still valid as history: do not resurrect a dropped item without addressing its recorded reason.
`Docs/Theory/FAILED_ATTEMPTS.md`	Operational log of the pre-campaign council (truncated agent replies, stalls, a DNS outage), not mathematics [Docs/Theory/FAILED_ATTEMPTS.md:1-8].	Infrastructure rules (§7.6).
`Docs/HQ/DECISIONS.md`	Old HQ decision log D1–D94 for the *empirical* New Campaign [Docs/HQ/DECISIONS.md:1]; the empirical campaign completed at D94 [Docs/HQ/DECISIONS.md:294].	Still the record for the empirical study (§3); not a theory source.

DIVERGENCE — old docs vs final state: PR #1 not merged

The spine correction (every "the bridge/sampler is needed" → "the run must move off its start") was written by E-SPINE as PR #1 against `Docs/PLAN.md`, `Docs/THEORY.md`, `Docs/THEORY_PRIORITIES.md`, a new `Docs/Theory/AMENDMENTS_20261102.md` and the T27/T38/T39 one-pagers [NBH/04_code_and_files_index.md:138]. "It isn't merged" [pos 850]; the archive (its information cutoff) does not record its later fate [NBH/05_open_questions_and_next_steps.md:29]. Observed on 2026-11-07: still OPEN and unmerged [Theory_work/State_of_Project/sources/PR1_status_2026-11-07.json:1]; the merge-or-close decision is pending (DEC09). Hence the files on `main` still carry the pre-correction wording. Decision DEC09.

6.3 Reversals and refutations

Each entry: the claim as it was held, why it failed (counterexample or proof), what replaced it, and the IDs and sources. Entries H.1–H.8 are the ones the lead listed as mandatory; H.9–H.21 are the other major reversals found in the sources. The decision-log, dead-end and HQ-correction catalogues they come from are [NBH/03b_reasoning_and_traces.md:264-374]; the full per-result history is in `NBH/02b_theory_campaign_history.md`.

H.1 · Literal T37 (truth dominance ⇒ greedy repair) is false → Theorem A

Claim. T37 (CJ117), filed 2026-10-28 as the new primary route for the paper's Thm 3: finite-n truth dominance TD(d, m) (CJ129) implies that the production local search repairs from any near-truth centre [NBH/03b_reasoning_and_traces.md:268-269].

Failed. RF101 (literal T37): REFUTED: TH198 (Prop1-TD) PROVED, SF VERIFIED (W3-VT37R), XF VERIFIED (A-REV2); d = 0 counterexample RF101 (CEX1) VERIFIED (W3-VT37R, reproduced by A-REV2) [NBH/ID_REGISTRY.md:1003]. TH198 (Prop1-TD) shows TD(d, m) is self-contradictory for $d \geq 1$; at $d = 0$ it fails without positivity. On the data, 1 of 2,394 fits was favourable and local search repaired 0 [NBH/03b_reasoning_and_traces.md:270]. HQ's first reading ("T37 as written proves the wrong mechanism" [pos 178]) was itself refined: the original is false even at $d = 0$ [NBH/03b_reasoning_and_traces.md:344]. Same family: RF165.

Replacement. TH199 (Theorem A, refined T37(i)): greedy repair holds exactly under TD′ with positivity (final form M5(a), TH05.a: from a centre $C \in \mathrm{Lat}(S^\circ)$ under TD'_∞, with $\ell \geq \tau(C)$ and a completed final scan) (PROVED (W6-FIXREV2 candidate); SF VERIFIED (W3-VT37R, 720 designs 0 violations); XF VERIFIED* (A-REV2 A1-A4 and POS-repaired A5; original A5(a) ERROR); A-REV5 core VERIFIED, literal cap-free lattice wording ERROR); TH201 (Theorem B (T37), union bound); TH204 (Theorem W, PGD) [NBH/03b_reasoning_and_traces.md:272]. Theorem A's abstract core (Lemma 0, A1–A4, concrete lattice forms) is sorry-free in Lean (`T37Greedy.lean`, `T37Lattice.lean`); the capped-scan certification interface is not yet formalised (§4.6) [NBH/03_results_and_conclusions.md:144] and is a core of M5 (TH05).

Side refutations. Theorem W in region form does not cover the masked regime: RF176, exact tie on masked LH with $c \geq 2$, TH217 (conflict C15 [NBH/05_open_questions_and_next_steps.md:152]); Lemma 0(b) "every unrepaired centre is a posterior miss" RF167 (conflict C1; true on the study data, open in general as CJ320; failures partitioned by TH200); the two-way failure dichotomy RF168; "TD$'_P$ iff (L) and all D_S" fails with exact ties RF169 (conflict C2) [NBH/03b_reasoning_and_traces.md:321], [NBH/05_open_questions_and_next_steps.md:138-139].

H.2 · H24 (uniform exit) is false in correlated designs → Theorem B (restated T24)

Claim. T24 terminal repair with input H24 (CJ120): a uniform one-round truth-scoring floor a_n and exit bound η_n with $(L + 4)\eta_n \to 0$; this was the input of T21 route (e); constant $1/31!$ [NBH/03b_reasoning_and_traces.md:274-275].

Failed. RF102 (H24): REFUTED as a general asymptotic input (CE144 (CE1) COUNTEREXAMPLE, SF VERIFIED* W2-VT24G, XF VERIFIED* A-REV1; broad asymptotic wording GAP, narrow explicit sequence VERIFIED A-REV5) [NBH/ID_REGISTRY.md:1004]. Counterexample CE144 (CE1): exit probability ≥ 0.39 in every correlated scenario [NBH/03b_reasoning_and_traces.md:276]; explicit sequence CE145 ($p = \lceil n^{1/3} \rceil$). The factorial constant $c_n = 1/(s_0 + d + B + 2)!$ was retired as useless (RF183); RF182 records the retirement of H24 in the pre-campaign register.

Replacement. TH218 (Theorem B, restated T24; polynomial constant $\sim 1/(B + 3)$): PROVED (candidate); SF VERIFIED* (W2-VT24G, after Lemma F′ repair); XF VERIFIED after F′ repair (A-REV1; statistical application GAP); XF VERIFIED finite feasible uninterrupted bound (A-REV5); vacuous at n=200 [NBH/ID_REGISTRY.md:366]; T21 route (e′) TH239. Finite-n Gibbs discovery stays future work CJ26 ("ThmB-T24 vacuous at $n = 200$") [NBH/05_open_questions_and_next_steps.md:72].

Side errors (Astra). Prop. U (TH227, conflict C3; repaired as Prop. U′), Lemma S attainment (LM174, conflict C4), E3(ii) without $q < 1$ (RF188, conflict C5), the limiting constant $0.52 \to 0.51$ (RF134, conflict C6) [NBH/05_open_questions_and_next_steps.md:140-143].

H.3 · A sharper T02 size cutoff is impossible → T03♯

Claim. Replacing T02(i)'s geometric tail by the exact BB$(1,p)$ tail would cut the size cutoff to $m_n = s_0 + O(s_0 \log n / \log p)$ under G and R [NBH/03b_reasoning_and_traces.md:280-282].

Failed. **RF103**: REFUTED under G and R alone: CE111 (PropD-T03) COUNTEREXAMPLE, SF VERIFIED (W2-VT03R row 11), XF VERIFIED (A-REV2); TH126 (E3-T03) barrier VERIFIED* [NBH/ID_REGISTRY.md:1005]. **CE111**: duplicated columns satisfy G and R but force $m_n \geq \frac{1}{2}\log_2 n - O(1)$ [NBH/ID_REGISTRY.md:1207].

Replacement. **TH128** (T03♯) with the exact BB$(1,p)$ tail and weaker eigenvalue conditions [NBH/03b_reasoning_and_traces.md:283]; final card TH09 (S-T). **Onset correction:** the wave-1 disclosure "thresholds start near $n = 1.2 \cdot 10^5$" was wrong (**RF123**); the proof's thresholds start near $n \approx 2 \cdot 10^6$ [NBH/03b_reasoning_and_traces.md:346]. See FR07.

H.4 · "Tempering is needed" is false → "moving off the start is needed"

Claim. The paper's conceptual spine (from the old HQ story): in the masked regime start-only discovery fails, so the tempering bridge/sampler is needed. W7-SEP was set up to prove it [NBH/03b_reasoning_and_traces.md:296-297].

Failed. **RF104**: REFUTED on LH by proof (TH388 (SEP-GU) Cor. GU-LH: COUNTEREXAMPLE to the task's claim (a); numerically 386/386 replicates); SF: none (A-REV7 pending then), XF: TH388 (SEP-GU) VERIFIED* (A-REV7), GAP in A-REV9 review [NBH/ID_REGISTRY.md:1006]. On the paper's own masked family LH, one greedy search from the start mode reaches S_0 with probability tending to one (Cor. GU-LH of **TH388**, Prop GU: exact Gram, isolated relevant set, SE on it, fixed signals; PROVED (candidate; SF: none; XF: VERIFIED* (A-REV7) as greedy best-neighbour theorem, GAP as standalone no-LS0 growing-p theorem; A-REV9: GAP in that review, Lemma M not re-audited) [NBH/ID_REGISTRY.md:536]; **RF262**); duplicate row **RF264**.

Replacement. Felipe: "Make all necessary changes then" [pos 798]; E-SPINE rewrote every "needed" as "the run must move off its start, and v3's moves do that"; edits went to PR #1, not to `main`; T38/T39 conjecture files were not edited because of the Oct 28 freeze hashes [NBH/03b_reasoning_and_traces.md:96-101]. **PR #1 is unmerged**: "It isn't merged" at [pos 850], and still OPEN and unmerged when queried on 2026-11-07 [Theory_work/State_of_Project/sources/PR1_status_2026-11-07.json:1] (DEC09). Final theory: TH06 (M6 · Theorem SEP + Lemma KL: start-only obstruction at the LH threshold, one-shot correction needs polynomially many draws; in the card the one-shot part is a second-moment (L^2) obstruction, not a high-probability error lower bound, and the final assembly is a proof sketch), built on **TH380**, **TH382** (SEP A/B) [NBH/03b_reasoning_and_traces.md:300].

Honest scope. "masking on LH is a property of the start; no published BVS example has MCMC provably slow while SMC succeeds" (**TH401**, **CJ290**) [NBH/03b_reasoning_and_traces.md:300]. The masked-start theorems "repair a problem created by the start. They don't show the method beats MCMC" [NBH/03_results_and_conclusions.md:323].

H.5 · Pooled $\hat{Z}$ × pooled PIP is biased → islandwise identities → $\hat{Z}$-weighted PIP

Claim. The pooled evidence times the pooled PIP is an unbiased island-combination estimator of $Z \cdot \mathrm{PIP}_j$ [NBH/03b_reasoning_and_traces.md:285-286].

Failed. **RF226**: REFUTED (Prop 2(c)); SF reproduced to 1e-58; XF VERIFIED (A-REV4) [NBH/ID_REGISTRY.md:1128]. Exact bias by **TH298** (Prop2(c)): PROVED; SF VERIFIED/CORRECT (W5-LOCKBOOK, reproduced to 1e-58); XF VERIFIED (A-REV4, Owls 50-digit reproduction -0.1739741778 and -0.1317163950; A-REV6 VERIFIED) [NBH/ID_REGISTRY.md:446]; numerical instance **CE182**.

Replacement. Islandwise form **TH297** (Prop2 (a,b,d,e)), then the $\hat{Z}$-weighted PIP as primary (MD08), because only it provably converges as islands are added (**TH397**, Prop P) [NBH/03b_reasoning_and_traces.md:174-176], [NBH/03b_reasoning_and_traces.md:288].

Related. "$E[\hat{Z} \mid \text{completed}] = Z$" is false (RF221, CEX-guard; also RF227); replaced by output-intersected statements $P(\mathcal{O}, \cdot)$ via Lemma RO [NBH/ID_REGISTRY.md:1123].

H.6 · The shared (common plug-in) noise-variance start is rejected

Claim. F-DRAFT flagged a common plug-in σ^2 for the block tables as a possible cure for the $n = 200$ dilution masking [pos 616]; O-MIX arm C and F-SIGMA developed it [NBH/03b_reasoning_and_traces.md:83-85], [NBH/03b_reasoning_and_traces.md:311].

Failed. RF150: rejected (E-SPINE 2.2, W8-SYNTH 4.5); refuted by O-DILUTE 5, F-SIGMA Thm S6, API-RESID Step 0 on 2400 real fits (0/2400 favourable) [NBH/ID_REGISTRY.md:1052]. "Two agents independently found it floods the start in study-like designs" [pos 796]; oracle σ^2 gives median $\log_{10} \pi_0(S_0) = -690$ [NBH/03b_reasoning_and_traces.md:85]. F-SIGMA: "the exact cure for dilution under block orthogonality, but catastrophic in random designs" (TH186, TH188: "vacuous as an improvement" over v3) [NBH/01_overview_and_timeline.md:84]; ranking ceiling for every common σ^2 TH176; O-MIX TH183 (conflict C23, "decision: rejected").

Replacement. The block start with its own σ^2-coupling decomposition, TH03 (M3) [NBH/03b_reasoning_and_traces.md:311]. Round 0 found independently that "orthogonal blocks still interact through the shared noise variance σ^2" [NBH/01_overview_and_timeline.md:213]; this coupling is a term of M2(a) and M3, not a reason to share σ^2 in the start.

UNCERTAIN — was the rejection ruled by Felipe?

HQ recommended "Common-σ^2 start: reject" [pos 796] and the PR #1 amendments record it; "Whether Felipe explicitly ruled is UNCERTAIN" [NBH/03b_reasoning_and_traces.md:84]. Recorded as DEC11.

H.7 · From HPM consistency to posterior accuracy and parallelism [pos 1020–1044]

Claim. The pre-campaign story: "What SMC-BVS reports is consistent because the truth gets **scored**, not because the sampler converges" [Docs/THEORY_PRIORITIES.md:25]; the paper hinged on T37 → T38 (HPM consistency) [pos 156].

Reversed. Felipe [pos 1020]: "precision of the BVS part is not the main target neither is consistency or any theoretical property as there are already good BVS algorithms that provide accuracy and have theoretical guarantees. [...] primary target is precision in terms of recovering the true posterior and secondary BVS accuracy"; the motivation is parallel computing (Song 2015); [pos 1030] "retire MCMC for BVS" [NBH/03b_reasoning_and_traces.md:141-145]. HQ: "The literature readers were assessing against the wrong target" [pos 1025].

Replacement. Objectives in order: (1) accuracy in recovering the true posterior, (2) parallel scalability over particles and islands, not over variables [pos 1113], (3) selection accuracy [NBH/START_HERE.md:9-13]; theory re-ranked (accuracy of $\hat{Z}$, PIPs, joint inclusion > cross-block cost > parallel cost > selection accuracy) [pos 1028]; thesis [pos 1044]: "In split-and-merge, correlation breaks correctness. In local-to-global SMC, correlation costs only sequential depth and particles, never correctness." The "never correctness" half was recorded as known and the "costs only depth" half as open, per THM-INV; in the final list the thesis is only the scoped remark TH22, whose fixed-data scope covers the unnormalised identities given a frozen calibration [pos 1107] [NBH/03b_reasoning_and_traces.md:144]. HPM consistency survives as M5 (TH05), objective 3.

H.8 · One-shot weight stability (the draft's §5.3 mechanism) is false → bridge form (M2)

Claim. Felipe's first draft, §5.3 Theorem 1 (CJ251): weight stability of the merged start under weak cross-block dependence [NBH/03b_reasoning_and_traces.md:290-291].

Failed. **RF256**: REFUTED for v3 tables: d=d_sigma>=0 under exact BO; chi2 can diverge at zero cross-block correlation (WS6(d)); F-DRAFT: false for v3 tables [NBH/ID_REGISTRY.md:1158]. The discrepancy is driven by per-block σ^2 estimation (dilution), not by cross-block correlation [pos 616]; F-DRAFT's Lemma W2 was an ERROR with an explicit counterexample (A-REV4) [NBH/03b_reasoning_and_traces.md:292]. For the shipped method the literal one-shot form is refuted (OWS-WS6(d), OWS-WS7, "VERIFIED WITH AMENDMENT" by the cross-family inherited check) [NBH/03b_reasoning_and_traces.md:294], [NBH/03_results_and_conclusions.md:20].

Replacement. **TH362** (Theorem W under H-ORTH), **TH350** (Theorem WS3; exact BO structure of d), the exact dilution form **LM212**, and the final bridge-form package TH02 (M2): "Weight stability: start-to-posterior distance controlled by cross-block correlation; bridge stable as correlation grows; the literal one-shot form proved false" [pos 1650] (HQ's plain words; the card's bounds hold inside the matrix domain H-DOM, the general D_* route is a proof sketch, and useful constants need $\varepsilon \lesssim n^{-1/2}/p$ on W). HQ ruling RL01 accepts the bridge form as the faithful reading of §5.3. A related HQ correction: "§5.3 is empty" was wrong (O-WSTAB had results under an orthogonal idealisation) [NBH/03b_reasoning_and_traces.md:367].

H.9 · Weak-signal masking → dilution masking; the $n = 200$ disclosure numbers

Claim. W1-EXP explained the $n = 200$ masking by weak signals [NBH/03b_reasoning_and_traces.md:317].

Failed. **RF136**: ERROR (W3-VEXP row 27 'evidence against'; W3-PXN Prop W); retired in favour of dilution [NBH/ID_REGISTRY.md:1038]; also **RF151**.

Replacement. Dilution masking **TH158** (PXN-W), the exact identity $-\log \pi_0(S_0) = V + C - \log A$ (**TH172**, Cor. E1; final form TH03.d) and the finite-n formula **TH177** (Prop. D′, PROOF SKETCH) [NBH/03_results_and_conclusions.md:268-270]. Disclosure numbers corrected: "beta-min fails by about 100×" → at least 43× (RF105: CORRECTED (number retired; not a theorem). Treat as a disclosure fact: beta-min fails by at least 43x (E-SPINE 2.3)); single "favourable from $n \approx$ …" thresholds withdrawn (**RF160**: withdrawn: sources disagree (O-DILUTE 590-730; W3-VEXP 800-1,600; W5-FAVIID 1,050-3,200); omit or attribute (W8-SYNTH 4.13; E-SPINE 2.3; T38_STATEMENT D6 open)) [NBH/ID_REGISTRY.md:1007], [NBH/ID_REGISTRY.md:1062].

H.10 · General LH locking is false; "no masked theorem without LS0" is retired

Claim. BG3's premises hold on the whole LH family (LH-LOCK task, Fable) [NBH/03b_reasoning_and_traces.md:302-303].

Failed. **RF208**: ERROR (A-REV7): C2 fails e.g. g=4, rho=0.85, psi=0.5; reproduced by Claude (API-VSTAB) [NBH/ID_REGISTRY.md:1110]. The LH-LOCK run also died on the 64K output limit and was not relaunched (its goal was refuted) [NBH/03b_reasoning_and_traces.md:250].

Replacement. Conditional fixed-p implication **TH261**; fixed-p corollary on (CERT) **TH276**; no-LS0 subfamily theorem **TH275**; B2-LH under STAB-LH **TH272** (PROVED-UNDER STAB-LH (author, W5-B2ENV); XF VERIFIED* conditional theorem (A-REV7, eta_n corrected); theorem-specific parts cross-checked by Claude (API-VSTAB)) [NBH/ID_REGISTRY.md:420]; final: TH05.b2LH labelled no-LS0 variant of M5 [NBH/03b_reasoning_and_traces.md:306]. "Without LS0 no masked-regime consistency theorem" was retired (**RF212**, **RF218**, **RF280**). STAB-LH is a genuine restriction (fails at $(\rho, \psi) = (0.85, 0.5)$ for every g tested) [NBH/02b_theory_campaign_history.md:7934].

H.11 · Naive merge recovers the posterior: false

Claim. Merging block fits (SHARE-M(b), F-DRAFT Prop NM) recovers the posterior / gives vanishing crossing temperature.

Failed. **RF203** (SHARE-M(b)): XF ERROR by A-REV3, retained by A-REV6 (quadratic-expansion remainder of order n; crossing temperature fails under BM-C) [NBH/ID_REGISTRY.md:1105]; **RF265** (FD-NM) ERROR (A-REV4); the consolidated claim **RF266** (K98): KNOWN-FALSE (SHARE-M(b) XF ERROR A-REV3/A-REV6; FD-NM XF ERROR A-REV4; ADRAFT-PropM COUNTEREXAMPLE); DROP [NBH/ID_REGISTRY.md:1168] → drop DR06.

Replacement. The surviving naive-merge counterexample **CE208** (Prop M, Astra) [NBH/03b_reasoning_and_traces.md:325]; the assembly **TH254** (SHARE-B2W4) carries PROVED given inputs (W4-SHARE); SF VERIFIED (W5-B2ENV, L+5 rounds note); XF VERIFIED* (A-REV3: conditional inequality VERIFIED, exit-loss line ERROR) [NBH/ID_REGISTRY.md:402] (conflict C17); M6 (TH06) states why the merge alone fails.

H.12 · Rapid-mixing theorems "apply to LH": qualified

Claim. API-POS: the rapid-mixing theorems (Yang–Wainwright–Jordan, Zhou–Yang–Vats–Roberts–Rosenthal) apply to the linked-hub family [pos 978].

Failed. LIT-MCMC [pos 1009]: their hypotheses fail for our target in 12 of 12 study-like data sets by a factor $\geq p^{0.87}$ (TH401, Prop OB) [NBH/03b_reasoning_and_traces.md:319]. **RF274**: QUALIFIED, UNCERTAIN: API-POS headline 2 itself says the theorems apply "with their own priors" to LH-iid and headline 3 that "for the paper's own target, neither theorem applies as stated"; LIT-MCMC refutes only a sufficient condition (R > M), not fast mixing; THM-INV D1 and THM-SYNTH REPORT §4.1: say both; replaced by Prop OB (TH401) and the positioning paragraph (TH404) [NBH/ID_REGISTRY.md:1176]; LIT-VENUE's "poly-time recovery on LH also follows" contradicted (**RF275**).

Lesson. "Say which object a literature theorem is about: a sufficient condition failing is not slow mixing" [NBH/02b_theory_campaign_history.md:17502]. The spectral-gap question for the $\varphi = 1$ round stays open (CJ04, CJ05); positioning in **TH404**.

H.13 · Parallelism and depth claims that were withdrawn

- "Methods with guarantees are sequential" (**RF268**): REFUTED as a general statement (LIT-PAR §1 item 2: "FALSE in the work/depth sense. Do not print it."); replaced by CJ282 S-GAP (three-part form); THM-SYNTH REPORT §3.3 "Do not print"; counterexample candidate CE212 [NBH/ID_REGISTRY.md:1170]. The three-part gap claim **CJ282** survives "TRUE WITH QUALIFICATIONS".
- "Depth = number of stages T" (**RF269**): a round is $p + B + 2$ sequential steps; replaced by island-level depth accounting in Prop D (M8, TH08): the final card gives per-island depth $N_e(T + L + 4)m$ and inventories calibration trajectories separately [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:407]; the archive registry's formula $(I + 2)(T + L + 4)(p + B + 2)$ [NBH/ID_REGISTRY.md:1171] is an earlier form (see the depth factor in §2.7).
- "Depth free of N, H, R" (**RF271**): struck for v3.1; paper 1 claims island-level depth only (MD06) [NBH/ID_REGISTRY.md:1173].
- "More islands always helps" (**RF273**): false for the equal-weight PIP under a split of a fixed budget; replaced by stacking at fixed N_e and the $\hat{Z}$-weighted primary PIP [NBH/ID_REGISTRY.md:1175]. Post-search HPM monotone in R' without the extra search: false (**RF272**).
- METH-AUDIT's "bitwise on 3/3 designs" (**RF270**): an overclaim at report time (1/1 + smoke), re-established by HQ re-runs as 3/3 [NBH/ID_REGISTRY.md:1172], [NBH/03b_reasoning_and_traces.md:354].
- Rejected method changes with reasons ([pos 1107]): "Splitting a fixed particle budget into many small islands. It provably makes the equal-weight PIP worse."; "Shortening calibration. It weakens three existing theorems."; "A new move kernel to cut the sequential cost per particle [...] would break the move-path theorems." [NBH/03b_reasoning_and_traces.md:157].

H.14 · Evidence and output glosses corrected

- "$\hat{Z}$ can only fail downward" (**RF233**): PROVED false (PC1 claim 4, theorem-meeting phase); formal tail statement untouched; also owned by theme 12 (THM-SYNTH) [NBH/ID_REGISTRY.md:1135]. Replacement wording (PC1 claim 4), to be used if the paper discusses the masked regime: "$\hat{Z}$ underestimates Z with high probability; the complementary rare event carries compensating overestimates of the same total mean; Markov caps them, $P(\hat{Z} > 20Z) \leq 0.05$; the certificate $\hat{Z} < Z_\mathcal{T}$ detects underestimation only." [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:173].
- "No useful upper bound on Z" as an inherent limitation: Astra's analytic bound $Z \leq C_n(2b_0 + y^\top y/(n+1))^{-\nu}$ (OB17) showed it is not inherent; then narrowed by HQ: "What's actually proved is narrower than the documents said" [pos 1512]; see LIM05, LIM06 [NBH/03b_reasoning_and_traces.md:361].
- "PIP Monte Carlo SD ≤ 0.01" is not a guarantee (resampling induces dependence); "$\Theta(p^2)$ work" holds only when $n = O(p)$; the local-search scan bound is wrong when $K > p/2$; the MPM threshold is $\frac{1}{2}$, not $\frac{1}{3}$ [NBH/03b_reasoning_and_traces.md:359-365].

H.15 · The Song–Liang analytic separation was not obtained

Claim. Split-and-merge (Song–Liang 2015) loses the masked true predictor whenever the random split separates the linked pair; LIT-PAR G4 sketch [NBH/ID_REGISTRY.md:1178].

Failed. **RF276**: CORRECTED by THM-SYNTH REPORT §1.6(a) / Theorem SL(e) (checked numerically, scripts/check_lemmas.py; CE214) and THM-INV D10; replaced by Theorem SL (TH417, PROOF SKETCH); final: K97 -> FUTURE_WORK FW21 (null-exclusion/FDR step NOT REPAIRED, W2PC1). LIT-PAR's 5/318 used a singleton masked variable ($g = 1$) (**CE214**); the draft clause "$\mathrm{cov}(x_t, y) = 0$ on isolation" had a counterexample (**RF267**); the crux of **TH417** (PIP → 0 under the SL priors) was "NOT REPAIRED (W2PC1)" [NBH/02b_theory_campaign_history.md:17495]. "The meeting couldn't get a sound proof that their split-and-merge loses true predictors under correlation" [pos 1650].

Current state. Future work CJ21 (FW21, K97); numerical counterexample **CE213** (20/20 for the exact posterior vs 5/318 for split-and-merge, LH-like design, $p = 16$, whose masked variable is a singleton, i.e. group size 1, not the LH family; CE214) cited as labelled evidence CL23 [NBH/03_results_and_conclusions.md:263]. Scout HQ pending: DEC02, frontier FR06.

H.16 · "Stage count continuous in ε" → sandwich + local constancy

The chair's Q5 asked for a stage count continuous in the cross-block correlation ε. PC1 found $T = 3$ between the baseline counts $\{2, 4\}$ at $\varepsilon = 10^{-5}$ [pos 1621]; HQ accepted the stage-count content only as "sandwich + local constancy at non-critical thresholds + continuous upper envelope", with the integer count stated to jump [NBH/03b_reasoning_and_traces.md:236-237], [NBH/03b_reasoning_and_traces.md:327]. Ruling RL05; clauses TH02.b6, TH02.b8 and TH07.

H.17 · The statement of Theorem 5 (T38) and the LS0 decision

- T38(b)-R (**TH241**) was superseded by T38-(0) (**TH281**) after A-REV1 (conflicts C8, C11); the implications (b1′) **TH285** and (b2) **TH286** stand as implications; the statement of record is **TH290** (T38-v3, "nobody reviewed v3 itself") [NBH/05_open_questions_and_next_steps.md:145-148], [NBH/ID_REGISTRY.md:438].
- The iid-failure clause of (b1-LH) for $c \geq 2$ (**RF217**) is an ERROR (A-REV5) / GAP (A-REV8), replaced by a growth condition; LS0 consistency unaffected; the iid case is future work CJ10 [NBH/ID_REGISTRY.md:1119]. The main masked clause (b1-LH) with LS0 is **TH287** (conflict C24).

- LS0 was adopted after the 0-LS0 data check (0 of 2,394 HPMs changed) [NBH/03b_reasoning_and_traces.md:68-75]; the headline choice "LS0 vs no-LS0" (**CJ215**) was resolved in the final list by TH05.b1LH (with LS0) and TH05.b2LH (labelled variant without LS0). LS0 existed "only in a test export" (**CJ213**): still OPEN as an implementation premise (ENG02).
- Stale CONJECTURE labels on T32(i) (**CJ112**) and the cost corollary **TH289** versus PROVED in W5-LOCKBOOK (same family only) (conflict C18; **RF219**); same staleness for **TH386** (C19) [NBH/05_open_questions_and_next_steps.md:155-156].

H.18 · Backfit/residualised start: not adopted; its rationale withdrawn

CJ162: candidate, unproven (API-RESID: 'unproven candidate'; no theorem, no SMC evidence); rejected for paper 1, recorded as future work (MEET-v31 MINUTES item 1) [NBH/ID_REGISTRY.md:832]. Final reasons: "your ground rules; switching to backfit would mean a full re-run; no exact backfit sampler with a known normaliser exists, so its $\hat{Z}$ would not be unbiased; there is no theorem for it" [pos 1480]. The earlier argument "the backfit start could gain at most 6 of 2,400 fits" "doesn't follow" (**RF164**) [NBH/03b_reasoning_and_traces.md:89-94].

H.19 · The start ablation: a vacuous pre-registered rule → ABL-CONF

ABL-START's rule (inconclusive when evidences differ by > 1 nat) made every design inconclusive because the prior start's evidence was poor; official verdict Outcome 2 ("keep, no speed claim", **CJ166**); HQ: "That flaw in the rule is mine to own as HQ" [pos 1280] [NBH/03b_reasoning_and_traces.md:161-166]. Replaced by ABL-CONF with a 75-minute cap and non-delivery as a cost failure: Claim 2 (prior start under-estimates log evidence) LICENSED (pre-registered rule: holds on S1, S4, S5; median log Z error -13.4/-10.8/-5.7 nats); XF none (**CJ165**); Claim 1 (block start cheaper) INCONCLUSIVE (pre-registered rule: S1 inconclusive, S4 holds_nondeliv, S5 holds); XF none (**CJ164**) [NBH/ID_REGISTRY.md:834-835]. "Below the block start on 12 of 12" was replaced by the certified 10 of 12 [NBH/03b_reasoning_and_traces.md:228].

H.20 · p ≫ n: the GAP that every Astra reviewer flagged

The LH-iid p ≫ n form of Thm 5(b1) (**TH160**) carries PROVED (candidate); SF VERIFIED (W5-VMASK); XF GAP (A-REV3, A-REV6, A-REV8) [NBH/ID_REGISTRY.md:308]; PXN-P6 (**TH153**) is XF GAP (conflict C16). API-PXNGAP (**TH161**) answers each objection with a specific lemma but has PROVED (candidate, API-PXNGAP) given T22(i),(ii), T03, CONC, W7 Thm B; no SF/XF review in the archive [NBH/ID_REGISTRY.md:309]. Its own report names two obstructions: the family needs $g \leq B_n < 2g$, so the study's groups of 5 are excluded; and super-polynomial p is blocked because T03 is locked only for $p \leq n^A$ [NBH/branch_snapshots/API-PXNGAP/REPORT.md:17-20]. Open as FR01, FR02.

H.21 · Pre-campaign drops

The pre-campaign run dropped candidates as likely false, unworkable, trivial or with unreasonable assumptions (archived as RF106–RF120), e.g. T19 Tier A uniform coupling (hypotheses refuted on every dataset), T5(iii) additive band (false for slowly growing p), T11(ii) "no ordering constant for $a = 1$" (counterexample), T22(iv) existence of usable upper bounds (P8 counterexample) [Docs/Theory/DROPPED.md:8-18]. Pre-campaign failed experiments (BFDR, expected-F1, GFM, block-MAP, …) carry "do not retry as is" (RF119) [NBH/ID_REGISTRY.md:1021].

For agents — HQ claims that were later corrected

The archive lists 34 claims made or relayed by HQ that were later corrected (R01–R34), including billing (routines bill the plan), the $c = 1$ accuracy correction (SMC-BVS is behind SuSiE/SSL/horseshoe at $c = 1$, best at $c = 0.5$), the N9 sub-claim count, and the draft-list count

[NBH/03b_reasoning_and_traces.md:339-374]. Lesson recorded there: headlines of the form "$N/3$" may overclaim; statuses written by an author for its own work are same-family or author-only unless a cross-family review is named [NBH/02b_theory_campaign_history.md:17502].

6.4 Do-not-retry table

Routes that are refuted, vacuous or blocked, with the recorded reason. "Retry" means: re-attempt the same route without a new idea addressing the reason.

Route	Why it fails (as recorded)	IDs / source
Literal truth dominance TD(d, m) as a premise for greedy repair	Self-contradictory for $d \geq 1$ on any design; false at $d = 0$ without positivity; literal TD never holds in the study design	RF101, CJ129 [NBH/03b_reasoning_and_traces.md:270]
Theorem W region form as the general sufficient condition in the masked regime	Exact tie on masked LH with $c \geq 2$; PGD(η) fails for every $\eta > 0$; use the lattice form	RF176, TH217 [NBH/05_open_questions_and_next_steps.md:152]
H24 / uniform exit $(L + 4)\eta_n \to 0$	False in correlated designs (exit ≥ 0.39)	RF102, CE144 [NBH/03b_reasoning_and_traces.md:276]
Factorial constants $1/(s_0 + d + B + 2)!$	Useless; replaced by polynomial constants	RF183 [NBH/ID_REGISTRY.md:1085]
Sharper size cutoff from the BB$(1, p)$ tail under G and R alone	Duplicated-column counterexample	RF103, CE111 [NBH/03b_reasoning_and_traces.md:282]
Pooled $\hat{Z}$ × pooled PIP; conditioning on completion	Biased ($-\frac{3}{4}$ Cov); $E[\hat{Z} \mid \text{completed}] \neq Z$; state output-intersected events instead	RF226, RF221 [NBH/03b_reasoning_and_traces.md:287]
Weak-signal explanation of $n = 200$ masking	Evidence against; the mechanism is σ^2-dilution	RF136 [NBH/03b_reasoning_and_traces.md:317]
Common plug-in σ^2 start (cure for dilution)	Floods the start in study-like designs; catastrophic in random designs	RF150 [NBH/03b_reasoning_and_traces.md:311]
Mixture of block starts (B-merge)	"does nothing for dilution"; not in v3.1	**TH181** [NBH/03b_reasoning_and_traces.md:315]
Backfit/residualised start for paper 1	No exact sampler with a known normaliser (so $\hat{Z}$ not unbiased); no theorem; full re-run	CJ162 [NBH/03b_reasoning_and_traces.md:92]
The draft's one-shot weight-stability mechanism for v3 tables	$d = d_\sigma \geq 0$ under exact BO; χ^2 can diverge at zero cross-block correlation	RF256, CJ251 [NBH/ID_REGISTRY.md:1158]
Proving "tempering is needed" (start-only fails in the masked regime)	One greedy search from the start mode suffices on LH	RF104, RF262 [NBH/03b_reasoning_and_traces.md:298]
General LH locking (BG3 premises on all of LH)	C2 fails, e.g. $g = 4, \rho = 0.85, \psi = 0.5$	RF208 [NBH/03b_reasoning_and_traces.md:304]
Naive merge / SHARE-M(b) / vanishing crossing temperature	Quadratic remainder of order n; crossing temperature false under BM-C	RF203, RF266 [NBH/02b_theory_campaign_history.md:7931]

Route	Why it fails (as recorded)	IDs / source
Set analogue of T22(iv); "start mass + bridge mass + exit control ⇒ delivery"	Both false (C0, C1 counterexamples; extinction 0.48/0.995 at $N = 10^5/10^6$)	**RF198**, **RF199** [NBH/02b_theory_campaign_history.md:7927]
Lattice-indicator potentials to certify delivery	Cannot certify delivery at any budget; the right potential is the indicator of a closed region	**TH273** [NBH/02b_theory_campaign_history.md:7935]
Expecting any local method (LS0, route (b2), T39) to succeed on posterior-barrier designs	Barrier designs are a ceiling for every local method; keep R/SE-type hypotheses	**CE169**, **TH252** [NBH/02b_theory_campaign_history.md:7936]
Rapid-mixing theorems (YWJ, ZYVR) as a certificate for our target	Their sufficient condition $R > M$ fails 12/12 by $\geq p^{0.87}$; a failed sufficient condition is not slow mixing	RF274, TH401 [NBH/02b_theory_campaign_history.md:17492]
Printing "methods with guarantees are sequential", "first parallel exact BVS", "depth = T"	False in the work/depth sense; wordings never to print	RF268, RF269 [NBH/02b_theory_campaign_history.md:17486]
Parallel-within-particle moves (to cut depth)	A workhorse kernel change that breaks the Gibbs-path theorems; rejected	RF269 [NBH/02b_theory_campaign_history.md:17487]
Song–Liang separation via "$\mathrm{cov}(x_t, y) = 0$ on isolation" or the numerical design with a singleton masked variable (group size 1, not the g-prior's g)	Counterexample to the clause; the 5/318 used group size 1; null-BF/FDR step not repaired	RF267, CE214, TH417 [NBH/02b_theory_campaign_history.md:17495]
Union bound over Hamming balls for $\Pi(X)$	Failed route; the $\Pi(X)$ upper bound stays OPEN	**RF249**, **CJ230** [NBH/ID_REGISTRY.md:1151]
A-priori ε of VAL-FP at study budgets	Vacuous; measure by replay	**RF250** [NBH/ID_REGISTRY.md:1152]
Making the (D_S) margin non-vacuous at $n = 200$ by a worst-case union over the family	The $(2A + 1)\log n$ penalty is intrinsic to the worst-case union; a path-wise bound would be needed	[NBH/branch_snapshots/API-PXNGAP/REPORT.md:128-131]
Covering the study's groups of 5 with the LH family	With $2g \leq B$ next-fit packs two groups per block; by PXN Prop. Q (a sketch) the linked pair is co-blocked and masking disappears	[NBH/branch_snapshots/API-PXNGAP/REPORT.md:125-127]
Extending the PXNGAP route to superpolynomial p	T03 is locked for $p \leq n^A$ only, so the posterior half is unavailable	[NBH/branch_snapshots/API-PXNGAP/REPORT.md:123-124]
Literal "stage count continuous in ε"	The count is an integer step function ($T = 3$ between $\{2, 4\}$)	RL05 [NBH/03b_reasoning_and_traces.md:327]
Pre-registered rules with relative evidence gates	The ABL-START rule was vacuous; use absolute gates and caps	CJ166 [NBH/03b_reasoning_and_traces.md:323]

6.5 Verification infrastructure and what each check level means

Status words of authors. The campaign protocol allowed only: "PROVED (a complete proof you checked line by line), PROOF SKETCH (the crux is identified and the steps are plausible, but not every step is verified), COUNTEREXAMPLE (with a reproducible script or explicit construction), CONJECTURE, and OPEN. Never upgrade a status. A proof here is a **candidate**: it is not reviewed and not adopted. Review happens later, by other model families" [NBH/branch_snapshots/campaign/PROTOCOL.md:19]. In the theorem meeting a seat's own argument is "PROVED (by this seat, unchecked)" until a check is filed [NBH/branch_snapshots/MEET-THM/DOSSIER.md:104]. Pre-campaign "PROVED-LOCKED" results are same-family only with "external review queued" [NBH/03_results_and_conclusions.md:23]. Definitions of every word: Appendix D.

Check levels. **SF** = same-family check (Claude checking Claude); **XF** = cross-family check (Astra A-REV1–9; PC1 = gpt-6.1-sol on Seats A, B, D; PC2 = Opus 5.5 on Seat C, which was Astra; CHK = PC1 on inherited proofs) [NBH/03_results_and_conclusions.md:7]. Statement checks (TRUE-AS-WRITTEN, COUNTEREXAMPLE, NEEDS-HYPOTHESIS, AMBIGUOUS; design and route verdicts) are *not* proofs: "Your verdicts are about statement sanity; a TRUE-AS-WRITTEN is not a proof" [NBH/branch_snapshots/MEET-THM/CHECK_W1_BRIEF.md:17-23]. **Lean** certifies finite abstract cores only. **Adoption/locking** is Felipe's decision and nothing has been adopted [NBH/05_open_questions_and_next_steps.md:251]. External human review of the proofs has not happened [NBH/03_results_and_conclusions.md:340].

Astra cross-family reviews (2026-11-02). [NBH/03b_reasoning_and_traces.md:384-394]

Review	Scope and headline (from the file)	Main IDs affected
A-REV1	Theorem B (restated T24), T38(b)-R, D2, E3, Lemma S, Prop U, IND-T, CE1; the finite Gibbs-path bound "survives independent review after the F′ repair"; errors in Prop U, Lemma S attainment, E3(ii), Lemma U's limit	TH218, TH241, RF102
A-REV2	T03♯, refined T37 (Theorem A), EXP′: package GAP with T03♯ conditional VERIFIED; constant 0.52 → 0.51; exact-ties edge case	TH128, TH199, RF101
A-REV3	W3-PXN (p ≫ n), W4-SHARE, Prop M: "GAP as written"; M(b) ERROR	TH153, TH254, RF203
A-REV4	W2-T26, Lemma RO, F-DRAFT, A-DRAFT: "Overall: GAP"; F-DRAFT "two explicit false formulas"; pooled bias VERIFIED	TH298, RF226, RF265, CE208
A-REV5	Re-review of W6 repairs: principal repairs sound within their restrictions; package not VERIFIED	downstream E3, R2, PXN; T38 iid growth
A-REV6	Repairs of A-REV3/4 findings; does W7-PART close p ≫ n: "Only the restricted recovery premise; not the full result"	**TH253**, RF203, RF265
A-REV7	No-LS0 masked routes: B2-LH VERIFIED conditional on STAB-LH; ERROR: unconditional LH locking; "DOM is void when $c = 1$" ERROR	TH272, TH261, RF208, TH388
A-REV8	LS0 masked route: conditional consistency of Thm 5(b1) VERIFIED, "a theorem about v3 plus LS0"; p ≫ n interface flagged	TH290, TH160, TH153
A-REV9	Certificates, masked PIP/evidence variance, dilution, separation, partition, stages, weight stability: restricted affirmative verdicts; certificate card repaired	TH172, TH362, TH380, TH382

HQ's summary: Astra "is the most valuable check in the campaign" [pos 1238]; it caught errors after Claude passed them; weaknesses: "Many of its GAP verdicts concern conventions or wording", and one self-review [NBH/03b_reasoning_and_traces.md:396]. Standing caveat: repairs of Claude results found wrong by Astra "have not been re-checked by Astra" unless covered by A-REV5–9 [NBH/05_open_questions_and_next_steps.md:165].

Sol (GPT-6.1) checks. Method meeting: Sol 1 checked 60 claims (45 hold, 19 of them with corrections; 7 false; 6 gaps; 2 uncheckable); Sol 2 audit 13 findings (4 decision-level); dispositions in `MEET-v31/AMENDMENT_HQ_01.md`. Theorem meeting: W1_PC1 (Sol) COUNTEREXAMPLE 11, NEEDS-HYPOTHESIS 25, AMBIGUOUS 12, TRUE-AS-WRITTEN 53; W2_PC1 TRUE-AS-WRITTEN 70, COUNTEREXAMPLE 9, NEEDS-HYPOTHESIS 8, AMBIGUOUS 4; PC2 (Opus) on Seat C: W1_PC2 54 of 56 clauses TRUE-AS-WRITTEN (2 AMBIGUOUS), W2_PC2 29 of 29; "Two rounds of proof checks by a different model family found 20 counterexamples in draft statements, and all were repaired or removed" [pos 1650]; CHK_INHERITED: OWS-WS3/WS4 VERIFIED, WS6(d), WS7, N9 VERIFIED WITH AMENDMENT; close audit "PASS WITH CORRECTIONS" [NBH/03b_reasoning_and_traces.md:400-408].

Registry and LOCKABLE. LOCKABLE_v2 rule (strict): "author status **PROVED** (not PROVED-UNDER) **and** same-family verdict VERIFIED (or VERIFIED after repair) **and** cross-family verdict VERIFIED (or VERIFIED with qualification)"; counts: 55 strict candidates, 3 repaired by W6 but not re-reviewed, 43 same-family-only, 43 cross-family-only, 6 counterexamples confirmed by both, 5 conditional implications verified by both; "This file lists candidates; it does not adopt, lock or upgrade anything" [NBH/branch_snapshots/T38-RECORD/LOCKABLE_v2.md:3-5]. REGISTRY_v2: 431 named results (PROVED 256, PROVED-UNDER 51, PROOF SKETCH 42, COUNTEREXAMPLE 34, CONJECTURE 29, OPEN 9, REFEREE 10; XF-VERIFIED 118, XF-ADVERSE 46, SF-VERIFIED only 53, UNVERIFIED 192) [NBH/03_results_and_conclusions.md:136]. The registrar index at the theorem meeting: 166 cited IDs, 51 checked only by their author, 9 same-family only, 4 Astra only [NBH/01_overview_and_timeline.md:214].

Lean library. `L-LEAN`: `lake build` green, 0 errors, 0 `sorry`, 101 declarations; `#print axioms` shows only `propext`, `Classical.choice`, `Quot.sound` for the 56 audited declarations. Sorry-free targets: T21 (C1) (**TH418**, **TH420**, **TH422**), T07 (**TH423**; **TH106**), T11(iii) (**TH109**), T37(i)/Theorem A (TH199). T12(i.a) is partial: "The resample–move recursion `hstep` is an explicit hypothesis". Fidelity checks (Fable, Astra round 2) GREEN for statement fidelity only. "The Lean files are abstract finite versions. They do not certify M1–M8" [NBH/03_results_and_conclusions.md:143-147]. No main card is Lean-certified as a whole [NBH/03_results_and_conclusions.md:22].

Headline campaign-era results behind the cards. Statuses as abridged by the source [NBH/03_results_and_conclusions.md:119-131]:

ID (native)	Status (exact words, abridged by the registry)
TH115 (T21 composition)	PROVED-LOCKED (C1 and routes (a)-(f), wave 2) + CONDITIONAL on open hypotheses: H24 (route (e)) and the route-specific occurrence / exposure conditions; not adopted
TH422 (t21_C1; Lean)	Lean: sorry-free (lake build green, 0 sorry); Fable 5.1 GREEN; Astra r1 GREEN (statement fidelity only)
TH334 (Theorem N200)	PROVED (elementary, every n); author-only (no SF, no XF); Pi(X) upper bound OPEN
TH106 (T07)	PROVED-LOCKED (all parts, wave 3); not adopted
TH109 (T11; T11(iii))	PARTLY PROVED: (iii) PROVED-LOCKED (wave 1); (i), (ii), (iv)-(vii) not locked (CONJECTURED)
TH199 (Theorem A)	PROVED (W6-FIXREV2 candidate); SF VERIFIED (W3-VT37R, 720 designs 0 violations); XF VERIFIED* (A-REV2 A1-A4 and POS-repaired A5; original A5(a) ERROR)
LM189 (LS-score)	PROVED; SF VERIFIED (W5-VMASK row 20); XF VERIFIED* (A-REV8 …; move count 'at most Phi(C0)-1' ERROR at C0=S0, repaired to max{Phi(C0)-1,0})
LM204 (Lemma M)	PROVED; SF VERIFIED (O-VCERT: 0 violations in 476,160 exhaustive pairs); XF VERIFIED in exact arithmetic (A-REV9); LOCKABLE_v2
TH394 (Prop W, concurrent islands)	PROVED (code reading; executed check …); author-only; executed on smoke + 3/3 designs
TH395 (Prop W-cal)	PROVED (code reading only; not executed); still not executed at MEET-v31 close

6.6 Literature positioning, novelty and venue

The literature phase (LIT-SMC, LIT-MCMC, LIT-STAT, LIT-VENUE, then LIT-PAR) ran on the API credit [pos 985–1009], [pos 1072]. Its positions, as the sources state them [NBH/03_results_and_conclusions.md:313-323]:

Work	Position of the project
Rossell 2022	T03♯ is "a modest extension"; "The bound $P(S^\dagger \neq S_0) \leq 2E[1-\pi(S_0)]$ is his Proposition 1 word for word, so Thm 4 and Thm 5 must credit him" (CL13; **RF130**); ours is the cap K of order n under BB$(1,p)$.
Štefankovič–Vempala–Vigoda 2009	The greedy ESS grid as the shortest valid grid (O-STAGES Theorem G(i)) "is already in the literature"; cite as known (CL05); the G(ii) sandwich is a routine v3-specific extension.
Whiteley–Lee–Heine 2016; Dai–Heng–Jacob–Whiteley 2022	Frozen-run unbiasedness, $\hat{Z}$-weighted pooling of independent islands and freeze-the-adaptive-run advice are standard and must be cited (CL03, CL11); the non-standard items are the negatives (pooled × pooled is not an identity; $E[\hat{Z} \mid \text{completed}] \neq Z$).
Song–Liang 2015 (split-and-merge)	Model for "how much theory to get out of a method" [pos 1150]; their results are about selection, not the posterior; Condition 5 / Cor 2 violated under cancelling correlated partners; the analytic separation is not proved (CJ21); the paper "compares, and does not claim to exceed, Thm 1, Thm 2 and Thm 4"; 5/318 is labelled evidence (CL23).
Yang–Wainwright–Jordan 2016; Zhou–Yang–Vats–Roberts–Rosenthal 2022	Rapid mixing under their priors covers LH-like designs; their condition $R > M$ fails for our target at study scale (12/12, factor $\geq p^{0.87}$; TH401); "No published BVS example has MCMC provably slow while tempering or SMC succeeds" (open, CJ290); the masked-start theorems "repair a problem created by the start. They don't show the method beats MCMC" (CL14).
Hou–Wang–Atchadé (arXiv:2407.20580)	Never claim "first finite-n" (CL15).
Lindsten et al. 2017 (D&C-SMC)	"closest conceptual antecedent"; "The general local-to-global idea already exists" [pos 616]; pitch "when and why a scored-HPM SMC succeeds", not "local-to-global correction".
Schäfer–Chopin 2013; Papaspiliopoulos–Rossell 2017; Syed et al. 2024	Ancestor of the method (no theorems); exact enumeration of correlation-clustered blocks (CL10); dense-schedule barrier analogue (CL06).
Barbieri–Berger 2004	Cited lemmas K60, K61 ("HPM/MPM facts") used by M5 and S-TARGET (CL12) [NBH/02_theory.md:2085]. LIT-STAT: the HPM is generally *not* the optimal predictive model; say "Bayes rule for 0–1 support loss (exact recovery)" and note that under predictive loss the MPM can be optimal; the thresholding-at-$\frac{1}{2}$ proof behind T07(ii)(b) is standard and cited [NBH/branch_snapshots/LIT-STAT/REPORT.md:137-140].

What is genuinely ours (LIT-VENUE, LIT-SMC). (i) the exact, algorithm-agnostic finite-n split of the reported-model error into a posterior-miss term and a search-miss term measurable per run (Theorem N200 / T21 composition); (ii) the masked-start lower bound paired with a repair theorem; (iii) consistency of an SMC-BVS output at all ("No paper in the sample proves anything about the output of an SMC sampler for BVS"); (iv) run certificates; (v) Lean-checked finite cores. Polynomial-time recovery on LH already follows for YWJ/ZYVR priors: "T38(c), 'polynomial cost', is not a selling point on its own" [NBH/03_results_and_conclusions.md:325]. LIT-PAR: "The gap is real" (no parallel BVS method with a theorem combines an exact model posterior with unbiased $\hat{Z}$, communication-free parallel computation and a proved depth bound under correlation), with the qualification never to print "methods with guarantees are sequential" (CJ282, RF268) [NBH/03_results_and_conclusions.md:264].

Venue. Before the literature phase HQ judged "JRSS-B is realistic, probably after one major revision" [pos 933]. LIT-VENUE: "The package realistically supports Biometrika, the 5-result cut, only if the ref-

eree's major point is met: Thm 5(b) for the algorithm as run, with the transition shown at growing n [...] JRSS-B is a stretch. AoS is not realistic"; unmet expectations: real-data analyses, head-to-head comparisons (PARNI, MAdaSub, TGS, SuSiE/SSL), simulations at growing n, a main theorem not tied to LH, external human review [NBH/03_results_and_conclusions.md:330]. Felipe's decision [pos 1015]: "Let's try to build something rigorous enough to make it to Biometrika. Down to Lean certificates. Then we will analyze what more challenging results can we go to." Paper 1 = Biometrika, paper 2 = trees [NBH/03_results_and_conclusions.md:331].

UNCERTAIN — venue view shifted

HQ's view moved from "JRSS-B realistic" [pos 933] to "JRSS-B a stretch, Biometrika conditional" [pos 1009]; the final state carries the later one, with Biometrika as the target by Felipe's instruction [NBH/03_results_and_conclusions.md:332].

Maturity and confidence (the sources' own words). At [pos 929]: "about two-thirds to three-quarters of what a strong JRSS-B submission needs". After the meetings the list is complete as a list, but "Proof maturity is lower than list completeness" [NBH/03_results_and_conclusions.md:305-308]. High confidence: the T21 composition / N200 inequality and its Lean core, the 2,400-fit decomposition, LS0 neutrality, N9. Medium: the masked-regime theorems on a constructed family; p ≫ n only via API-PXNGAP. Low or open: N1, GUARD-W, N3, floating point, masked two-sided $\hat{Z}$, finite-n $\Pi(X)$ bound, delivery, exponential p [NBH/03_results_and_conclusions.md:336-338]. The hostile-referee simulation F-REFEREE recommended "major revision" (8 major points; registry rows **CJ303–CJ306**) [NBH/03b_reasoning_and_traces.md:417].

7 Open problems, pending decisions and plan

This section says what is open and what comes next. It has six parts: Felipe's decisions, closed and pending (§7.1); the theory frontier, as precise open problems FR01–FR14 (§7.2); engineering and audit items ENG01–ENG20 that gate claims (§7.3); every recorded conflict or uncertain item (§7.4); the planned next phase (§7.5); and the infrastructure rules for the new loop (§7.6). The obligations OB01–OB17, future work CJ01–CJ27, drops DR01–DR06 and limitations LIM01–LIM09 are defined in §5; they are referenced here and not restated.

7.1 Decisions: closed and pending

At the end of the source thread HQ had approved the method (v3.1) and the theorem package, and "Felipe had not yet approved the theorem list" [NBH/05_open_questions_and_next_steps.md:15]. That decision is now closed (DEC01). The others in the archive's list [NBH/05_open_questions_and_next_steps.md:17-30] remain open unless a source closes them. "Status" gives the state as this document records it; the archive's own status words are in the sources cited.

ID	Decision (content, recommendation, sources)	Status
DEC01	**Approve the paper-1 theorem list** (8 main cards TH01–TH08, supplement TH09–TH20); was P1 [NBH/05_open_questions_and_next_steps.md:19], [pos 1650]. Approved by Felipe on 2026-11-07 as the current targets [Theory_work/State_of_Project/sources/approval_2026-11-07.md:3-5]; recorded in this document's changelog (Appendix E). Approval changes no proof status.	DECIDED 2026-11-07 APPROVED (F.B., 2026-11-07)
DEC02	**Song–Liang separation scout HQ.** "I recommend one small scout HQ on the separation alongside wave 1. If it succeeds it becomes a ninth main theorem; nothing else waits on it" [pos 1650]; = CJ21 (FW21, K97) + CL23 [NBH/05_open_questions_and_next_steps.md:20]. Frontier FR06.	OPEN

ID	Decision (content, recommendation, sources)	Status
DEC03	**Prop D (M8, TH08) in the main text or the supplement.** HQ recommends main "for parallel scalability. The alternative leaves 7 main theorems" [pos 1650]; Seat D's alternative: main = 7 [NBH/05_open_questions_and_next_steps.md:21], [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M8.md:57].	OPEN (see box)
DEC04	**HQ's reading of proposal §5.3 (bridge form).** RL01, RL05: "§5.3 is met by the bridge form. One caveat: the example family gives useful constants only for very weak cross-block correlation, of order $\varepsilon \lesssim n^{-1/2}/p$"; sharp ε-scale necessity is CJ22; "Felipe may overrule" [NBH/05_open_questions_and_next_steps.md:22].	OPEN (HQ ruling stands unless overruled)
DEC05	**HQ's reading of §5.4.** RL02: "coverage" = distance from the exact posterior's credible quantities, not frequentist coverage (CJ01); RL03: TV/KL of the final estimate under masking is future work (CJ02), with at least one masked accuracy statement shipped (TH04.f) [NBH/05_open_questions_and_next_steps.md:23].	OPEN (HQ ruling stands unless overruled)
DEC06	**The stage count as a step function.** "The stage count is a step function of correlation strength, held within two-sided bounds" (TH02.b6, TH02.b8, TH07; RL05) [NBH/05_open_questions_and_next_steps.md:24].	OPEN (for Felipe to see)
DEC07	**Items HQ decided on its own authority.** (i) Reporting-only run-card numbers $\hat{J}$ and $\hat{\varepsilon}_\times$ (DF19, DF20; RL04; MEET-v31/AMENDMENT_HQ_02.md); (ii) growth rate (ii) $R'_n = 4\lceil (n+p)^6/4 \rceil$ kept only inside a theorem (RL06, "Felipe may overrule"); (iii) the overnight kill fixed by systemd units [NBH/05_open_questions_and_next_steps.md:25].	OPEN (Felipe may overrule)
DEC08	**Registry updates for the registry owner**: MEET-THM/REGISTRY_PAPER1.md §9, "five recommended registry updates" [NBH/05_open_questions_and_next_steps.md:26].	OPEN
DEC09	**PR #1** (spine correction, branch theory/E-SPINE): "It isn't merged" [pos 850] [NBH/05_open_questions_and_next_steps.md:29]; on 2026-11-07 GitHub reports it OPEN, not merged, last updated 2026-11-02 [Theory_work/State_of_Project/sources/PR1_status_2026-11-07.json:1]. The repo's Docs/ still carry the pre-correction wording (§6.2). Decision pending: merge or close.	OPEN (PR open, unmerged on 2026-11-07)
DEC10	**LH-SCALE go-ahead** (the simulated referee's one compute request; ENG18): "Needs your OK" [pos 824] [NBH/05_open_questions_and_next_steps.md:29], [NBH/05_open_questions_and_next_steps.md:216].	OPEN (never approved or run)
DEC11	**Explicit ruling on the common-σ^2 start rejection.** HQ recommended "Common-σ^2 start: reject" [pos 796]; "Whether Felipe explicitly ruled is UNCERTAIN" [NBH/03b_reasoning_and_traces.md:84] (reversal H.6).	RECOMMENDED BY HQ; FELIPE'S RULING NOT RECORDED
DEC12	**Adoption / locking of results.** "Nothing has been adopted or locked by Felipe; adoption is his decision" [NBH/START_HERE.md:44]; the LOCKABLE_v2 list "lists candidates; it does not adopt, lock or upgrade anything" [NBH/branch_snapshots/T38-RECORD/LOCKABLE_v2.md:3]; the old plan's "Adopt the 21 locked proof entries" was "never answered" [Docs/PLAN.md:119].	OPEN
DEC13	**Sol as a second referee / fourth seat.** The sources record no plan beyond HQ's offer "If you want to try Sol, I'd add it as a fourth seat or as a proof checker, not as a replacement for Astra" [pos 1302] [NBH/05_open_questions_and_next_steps.md:224].	OPEN
DEC14	**Residualised (backfit) start as an optional method-side investment.** "not decided"; HQ said it "attacks the method's main weakness at its source" [pos 937]; it remains an unproven candidate (CJ162); MEET-v31 item 1 rejected it for paper 1 [NBH/05_open_questions_and_next_steps.md:221].	OPEN (not decided)
DEC15	**Island-level depth only for paper 1; per-particle streams = v3.2** (MD06); one of the four method-process decisions "All approved" [pos 1304] [NBH/05_open_questions_and_next_steps.md:27].	DECIDED 2026-11-04 (date conflict, box below)

ID	Decision (content, recommendation, sources)	Status
DEC16	**Novelty check of "report the best model ever visited" against Hans, Dobra & West (2007) shotgun stochastic search**, carried forward by the old HQ: "Matters for the T21/T38 novelty claims" [Docs/PLAN.md:136]. No resolution was found in the archive's narrative documents.	OPEN (not addressed in the archive)

UNCERTAIN — does the 2026-11-07 approval settle the placement of M8 (DEC03)?

The approval record gives its scope as the main cards M1–M8 (TH01–TH08) and the supplement cards (TH09–TH20) [Theory_work/State_of_Project/sources/approval_2026-11-07.md:5], i.e. the list as HQ presented it with Prop D as M8. HQ asked a *separate* question, "Prop D (M8) in the main text or the supplement" [pos 1650], and the approval record states that this placement question is not covered [Theory_work/State_of_Project/sources/approval_2026-11-07.md:6]. No source records an answer to it. Until Felipe says otherwise, treat M8 as listed among the main cards and DEC03 as open.

UNCERTAIN — date of the island-level depth decision (U14)

METHOD_v3.1_FINAL.md §3 says 2026-11-03; in the transcript it was proposed at [pos 1263] and [pos 1302] and approved at [pos 1304] on 2026-11-04T00:05Z [NBH/05_open_questions_and_next_steps.md:123].

HQ rulings and recorded dissents. HQ ruled on the chair's questions Q1–Q5 under Felipe's delegation ("you have full autonomy wrt the specs of the meeting" [pos 1616]); Felipe may overrule them (DEC04, DEC05, DEC07) [NBH/03b_reasoning_and_traces.md:230-242]. Seats' disagreements with placements are recorded as dissents. Statuses verbatim [NBH/ID_REGISTRY.md:187-192], [NBH/ID_REGISTRY.md:206-211].

ID	Native	Content (registry)	Status
RL01	HQ ruling Q1 (§5.3)	Bridge form accepted with three conditions: (a) discrepancy half, (b) bridge-form stability, (c) refutation of the one-shot form (OWS-WS6(d), WS7) after a cross-family check	ACCEPTED ("Q1 (§5.3): ACCEPTED, with three conditions"); Q1(c) met by CHK_INHERITED (VERIFIED WITH AMENDMENT)
RL02	HQ ruling Q2 (§5.4 coverage)	"Coverage error" means $\lvert\pi(C(\hat{\pi})) - \text{nominal}\rvert$ plus MC error; frequentist coverage → CJ01	ACCEPTED
RL03	HQ ruling Q3 (§5.4 TV/KL under masking)	Obligation with fallback; the list must contain at least one masked-regime accuracy statement (→ TH04(f)); masked final-estimate TV/KL → CJ02	NOT binding
RL04	HQ ruling Q4 ($\hat{f}$ and leakage proxy)	Reporting allowed: from existing quantities, no random numbers, no feedback, precisely specified	REPORTING, allowed
RL05	HQ ruling Q5 (stage count continuous in ε)	Sandwich + local constancy + continuous envelope; real-valued parts continuity/Lipschitz; T is a step function ($T = 3$ between $\{2, 4\}$ at $\varepsilon = 10^{-5}$)	ACCEPTED, with two conditions
RL06	HQ approval amendment (growth rate (ii))	Package approved; $R_n' = 4\lceil(n+p)^6/4\rceil$ kept only as a sufficient rate inside the theorem, not a method configuration ($\approx 10^{17}$ islands at study sizes)	adopted by HQ; Felipe may overrule
DS01	Dissent K33 (Seat D)	D would place K33 in S-IDENT; A, B, C: M7	recorded dissent
DS02	Dissent K23 (Seat C)	C would keep the squared-kernel Dirichlet comparison as an attributed lemma in S-KERNEL; A, B, D: FUTURE_WORK (CJ05)	recorded dissent
DS03	Dissent K29 / S-STGI (Seat D)	D: "confirm K29 here"; A, B, C prevail: S-STGI dissolved (TH21)	recorded dissent

ID	Native	Content (registry)	Status
DS04	Dissent K30 (Seat C)	C: K30 in the supplement as S-STGI/C14; A, B, D: M7(d)	recorded dissent
DS05	Dissent K70 (Seat C)	C: K70 as a supplementary attributed variant; A, B, D: labelled no-LS0 variant clause of M5	recorded dissent
DS06	Dissent K57 (Seat D)	D: general K57 in the supplement; B, C: only P1 in S-POOL, general accuracy → FW (CJ08)	recorded dissent

Standing safeguard (P10). "if a theorem behind one of your mandatory proposal goals (§5.1–5.4, §13.1) fails, it comes back to you. It is never silently dropped"; currently live for OB01 (N1, binding §13.1) [NBH/05_open_questions_and_next_steps.md:28]. Standing instructions of Felipe that govern the next phase: "extract until the last drop of information the method has into provable results. No rock can remain unturned" [pos 1150]; method stays composite-start/SMC/HPM unless "irrefutable reasons" [pos 1155]; any (n, p) with no arbitrary choices [pos 1214]; Biometrika first, "Down to Lean certificates" [pos 1015] [NBH/05_open_questions_and_next_steps.md:32-37]. The old plan's other open items (Docs/PLAN.md §4, items 3–10, 12–17) concern the empirical campaign, benchmarks, the public repository and operations [Docs/PLAN.md:116-145]; they are outside the theory loop's scope.

7.2 The theory frontier

Each frontier item states the gap, why it matters (which objective or card), what was tried and why it failed, candidate routes *named in the sources*, and what "done" means according to the sources. Unless a charter says more, "done" follows the plan's general standard: each per-theorem HQ "loops to a refereed proof plus a Lean certificate" [NBH/START_HERE.md:57], with a cross-family check on every claim, every self-proved claim labelled "unchecked", and Lean coverage per the cards' "Lean core" rows [NBH/05_open_questions_and_next_steps.md:209-212]; the card ships in the narrowest form its filed checks support, and a clause whose proof or cross-family check fails after one repair cycle moves to FUTURE_WORK.md with the reason, never relabelled as a limitation (IC7) [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-FOUND.md:112]. Obligations and future-work items are referenced by ID; their exact statuses are in §5.

FR01 · p ≫ n: a cross-family-checked masked consistency theorem, and beyond polynomial p

Gap. No p ≫ n version of the masked-regime HPM result has passed a cross-family review. The W5-VMASK form TH160 is XF GAP (A-REV3, A-REV6, A-REV8). The repair candidate API-PXNGAP (TH161) is "PROVED (candidate, API-PXNGAP) given T22(i),(ii), T03, CONC, W7 Thm B; no SF/XF review in the archive" [NBH/ID_REGISTRY.md:308-309]; it is campaign-era and outside the final list [NBH/03_results_and_conclusions.md:337]. Beyond polynomial p: superpolynomial p ($\log p = o(n)$) is "CONJECTURE, recorded in (O2)" [NBH/branch_snapshots/API-PXNGAP/REPORT.md:123-124]; exponential p is CJ18 / OB12 ("$p \le e^{Cn^{\alpha}}$: OPEN, not impossible") [NBH/05_open_questions_and_next_steps.md:98].
Why it matters. The first of the main open edges named by the archive [NBH/START_HERE.md:47]; M5's masked clauses (b1-LH), (b2-LH) (TH05) are stated on LH exact Gram with $p \le n - 1$ [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:292]; the target card S-T (TH09) keeps the scope "$p \le n^A$ from prior and technique; $e^{Cn^{\alpha}}$ OPEN" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:478]; the study runs at $p \in \{500, 1000\}$, $n = 200$.
Tried. A-REV3's objections and API-PXNGAP's lemma-by-lemma answers (Lemma Tr-J, Lemma E, Lemma T, Lemma LS, Theorem R) [NBH/branch_snapshots/API-PXNGAP/REPORT.md:25-40]. Routes abandoned and why: keeping PXN's Lemma Tr; citing W1-EXP; citing Theorem A(A3) for the lattice route; superpolynomial p (T03 locked only for $p \le n^A$); groups of 5 (FR02); non-vacuous (D_S) at $n = 200$ [NBH/branch_snapshots/API-PXNGAP/REPORT.md:112-131].
Candidate routes (sources). The report's own list of items needing an independent check: Lemma Tr-J(c) constants; Lemma E (E2), (E3); Lemma Λ(a) Haar-span step; Theorem R (= W7 Theorem B); the operational closure OP-LS0 + GUARD; Lean-readiness of Lemma LS and (C)

[NBH/branch_snapshots/API-PXNGAP/REPORT.md:133-144]. Lemma E "could replace the W1-EXP citations in W3/W4/W5" [NBH/branch_snapshots/API-PXNGAP/REPORT.md:152-154].
Done when. The candidate's own card: "Confidence it is provable at reasonable cost: high — it is written; what remains is cross-family review" [NBH/branch_snapshots/API-PXNGAP/REPORT.md:164]. A cross-family VERIFIED on PROOF §§1–10 with the six items above, then the generic standard. Whether and where it enters the list is a loop output: "Future demotions/reformulations/additions are outputs of the theory loop" [Theory_work/State_of_Project/AUTHORING.md:32].

FR02 · Covering the study's own design (groups of 5; dilution masking)

Gap. The masked-regime theorems are proved on the linked-hub family LH with $g \le B_n < 2g$; "with the study's groups of 5 (2g ≤ B) the partition packs ≥ 2 groups per block and LH masking is not available, so the theorem does not describe the study's own generator" [NBH/branch_snapshots/API-PXNGAP/REPORT.md:17-20]. PART-pairing for the study generator is OB08 / CJ25 ("OPEN; Prop Q PROOF SKETCH; W7-PART CE2 COUNTEREXAMPLE") [NBH/05_open_questions_and_next_steps.md:94].
Why it matters. The $n = 200$ masking in the study is σ^2-dilution, not LH linkage (reversal H.9); the referee simulation and LIT-VENUE ask for theory that describes the algorithm as run and "a main theorem not tied to the constructed LH family" [NBH/05_open_questions_and_next_steps.md:217].
Tried. Covering groups of 5 on LH: impossible (next-fit packs two groups per block, W7 CE2; by PXN Prop. Q (a sketch) the linked pair is co-blocked and masking disappears) [NBH/branch_snapshots/API-PXNGAP/REPORT.md:125-127]. The common-σ^2 cure for dilution: refuted (RF150).
Candidate route (sources). "The right object for the study regime is a different family (dilution masking, PXN §14, CONJECTURE)" [NBH/branch_snapshots/API-PXNGAP/REPORT.md:126-127]; the dilution identity TH172 and the finite-n formula TH177 (PROOF SKETCH) are the existing inputs (M3, TH03).
Done when. CJ25 is future work: "Items here get no per-theorem HQ for paper 1" [NBH/05_open_questions_and_next_steps.md:43]. No acceptance criterion beyond the generic standard is stated in the sources.

FR03 · From explicit exhibits (LH, W) to classes of designs

Gap. The correlated-design guarantees are exhibited on explicit families: LH for M5's masked clause, M6 and M4(f); Seat C's family W (within-group correlation 1/4, $\varepsilon_n = 1/(np) > 0$, two $n^{-1/4}$ signals, blocks = groups for $n \ge 9^6$) for M1(i) and M2 [NBH/02_theory.md:765]. M2's useful constants need $\varepsilon \lesssim n^{-1/2}/p$ on the exhibited family; at fixed ε, D_* grows ($534 \to 6.45 \cdot 10^4$ on W at $\varepsilon = .01$) [NBH/02_theory.md:823].
Why it matters. LIT-VENUE's unmet expectation of a main theorem not tied to LH [NBH/03_results_and_conclusions.md:330]; the F-REFEREE points on stylised designs [NBH/03b_reasoning_and_traces.md:417].
Tried and failed. General LH locking (RF208); the region form of Theorem W as a general sufficient condition (RF176); STAB-LH is a genuine restriction (fails at $(\rho, \psi) = (0.85, 0.5)$; $c = 2$ partly open, **CJ204**) [NBH/02b_theory_campaign_history.md:7934]; posterior-barrier designs are a ceiling for every local method (CE169).
Candidate routes (sources). M2's charter: if the general D_* matrix route stalls, "prove on W directly" (fallback), and "§5.3 escalates" only if both the D_* route and the correlated-family route fail [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M2.md:83], [NBH/02_theory.md:835]. The general D_* decomposition (M2(a1)) is Seat C's PROOF SKETCH and is the charter's first lemma [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M2.md:117].
Done when. For M2: per charter, cross-family proofs of (a1) and (b2)–(b7), ≤ 20 seeds on W [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M2.md:119-121]. A theorem for a class of designs beyond the exhibits is not chartered for paper 1.

FR04 · N1: the measured-round island identity for general bounded h (binding §13.1)

Gap. OB01: "PROOF SKETCH; PC1 claim 3 takes it as a premise. Binding §13.1: if it fails the goal is not met by withdrawal; HQ escalates to Felipe" [NBH/05_open_questions_and_next_steps.md:87]. The NEW-SYNTH Theorem H (**TH412**): "crux not verified: island identity $E[W_r \hat{h}_r] = \pi(h)$ for the measured-round average" [NBH/ID_REGISTRY.md:560].

Why it matters. M1 (TH01) clauses (b, e, f, g, i) rest on N1 ("Core: N1 → M1(b,e,f,g,i)") [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:28]; objective 1 (posterior accuracy for every reported summary, joint inclusions); proposal §13.1 is binding.

Route (charter). "First lemma: N1 (measured-endpoint identity $E[\hat{Z}_r H_r(h) \mid \mathcal{C}] = Z \cdot \pi(h)$) via the terminal-round prefix identity; then (e) as its corollary on $1\{S \neq S_0\}$"; Lean target T12Core `unbiased_of_step` with `hstep` for multinomial resampling [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:114-115]. The card's route text locates the identity: "N1: finite-state resample–move recursion; the state before each measured record is Zπ-weighted because the prefix of a uniformly random order is state-independent (terminal-round prefix identity, N9 §3)" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:119]. The N9 text it refers to (the RB-PIP case): "**PIP identities** (RB marginal PIP, primary estimator): need only that every step is γ_1-invariant and that the order is independent of the state. Then the state just before each Gibbs step j is π-distributed, and E[o_j] = π(j ∈ S). Both hold." [NBH/branch_snapshots/N9-KERNEL/REPORT.md:47-49]; full proofs of the moves and the round in `NBH/branch_snapshots/N9-KERNEL/PROOF.md` §1–§2. The proof text behind it: "*Invariance is weaker and is what the estimators use.* Each step is γ_φ-invariant (reversible ⇒ invariant), so every realised order is γ_φ-invariant, and so is the round. The state immediately before any Gibbs step j of a round started from γ_φ is γ_φ-distributed, whatever the order (the order is independent of the state); this is what the RB identity E[o_j] = π_φ(j ∈ S) needs." [NBH/branch_snapshots/N9-KERNEL/PROOF.md:241-244]; state independence of the order: "the m−1 uniforms are independent of P's state (they are read before any step of the round, at a stopping time of the stream), and Fisher–Yates maps the (m−1)-tuple of choices bijectively onto the m! permutations [...]. So σ is uniform on S_m and independent of the state." [NBH/branch_snapshots/N9-KERNEL/PROOF.md:218-220]. The route sketch from the method meeting (Seat C): "The proof route is concrete. Conditional on calibration, the finite-state resample–move recursion gives $E[\hat{Z}_r \eta_{r,T}(f)] = Z\pi(f)$. Terminal rounds preserve π, so this identity extends to each measured endpoint and their average. This is a **PROOF SKETCH** pending the implementation-level recursion check." [NBH/branch_snapshots/MEET-v31/R2_C.md:200-204]; Seat A, explicitly as its own reading: "(a2) for the plain average is the base case of the same terminal-round unbiasedness that the RB identity rests on ($E[\hat{Z}_r \eta_r(f)] = Z\pi(f)$ for a bounded f at a φ = 1 round), so N1 should be a short extension of the locked lemma rather than new theory; its status stays PROOF SKETCH until checked" [NBH/branch_snapshots/MEET-v31/R3_A.md:38]. Lead's reading, comparing the two cited statements: N9's text covers the per-step RB marginal record; N1 (M1(b), [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:111]) asks for $E[\hat{Z}_r H_r(h) \mid \mathcal{C}] = Z \cdot \pi(h)$ with H_r the plain measured-endpoint average of a general bounded h.

Done when. Cross-family check of the N1 proof by PC1 (Codex Sol); the Lean file to the registrar; a $p \leq 8$ coded-grid replica of $E[\hat{Z}_r H_r(h)]$ (10 seeds) by PC2. Fallback: marginal PIPs only (locked identity) for (e)–(i); one HQ wave on N1. Escalation: "N1 refuted or unclosed after one wave (→ §13.1 unmet, HQ)" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:116-118]. Wave 1; "the longest pole and starts first" [NBH/05_open_questions_and_next_steps.md:203].

FR05 · N3: the e4 tie-DP invariant

Gap. OB03: "(b1-LH) PROVED (candidate), XF VERIFIED (A-REV8) for 'v3 + LS0'; bridge to code = N2(c), not done. DP invariant OPEN" [NBH/05_open_questions_and_next_steps.md:89].

Why it matters. M5 (TH05), objective 3: the e4 invariant is "the only OPEN mathematical obligation in the chain" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:101].

Route (charter). "the DP returns the prescribed smallest-size, lexicographically smallest start-mode maximiser (W1PC2 D9: TRUE on 400 random instances, DESIGN-ASSERTED)"; in parallel the merged-LS0 replay (ENG02) [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:101]; engineering companion ENG10.
Done when. Cross-family check of the assembly; Thm5-LH (SAME-FAMILY-ONLY) and T38-(a) (XF none) to a cross-family check; ≤ 20-seed LH replica with LS0. Fallback: "(b1-LH) becomes 'conditional on N3' after one unproductive HQ cycle; (0) + (a) + (b2-LH) always ship". Escalation: a counterexample to T38-b1-LH on LH, the integrated LS0 differing from the specified one, or N3 failing outright [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M5.md:103-105]. Wave 3.

FR06 · The Song–Liang separation (analytic)

Gap. CJ21 (FW21, K97): "Iteration-uniform null-BF/prior-odds step NOT REPAIRED (W2PC1); LIT-PAR G4 cited as numerical evidence; scout HQ recommended (P2)" [NBH/05_open_questions_and_next_steps.md:67]. Only numerical evidence exists: exact posterior 20/20 vs split-and-merge 5/318 on an LH-like masked design with $p = 16$ whose masked variable is a singleton (group size 1, not the LH family; CE213, CE214, CL23) [NBH/03_results_and_conclusions.md:263].
Why it matters. The thesis [pos 1044] contrasts split-and-merge with local-to-global SMC; "our side, validity under any correlation, is proved (M1)" [pos 1650] (HQ's words; in the card, validity is clause M1(a) [cited], under the named implementation premises, with the clause statuses of TH01.a, and the general-h identity N1 is a proof sketch). If it succeeds, "it becomes a ninth main theorem" (DEC02).
Tried. THM-SYNTH Theorem SL (TH417, PROOF SKETCH; crux "PIP → 0 under the SL priors" NOT REPAIRED); LIT-PAR's analytic sketch corrected (RF276); the 5/318 used group size 1, i.e. a singleton masked variable (CE214; this "g" is a group size, not the g-prior's $g = n$); the clause "$\mathrm{cov}(x_t, y) = 0$ on isolation" has a counterexample, replaced by population cancellation $x_t^\top \mu = 0$ (RF267); correct citation is Song–Liang Thm 4 / Cor 2 + eq. (19) + Condition 5 [NBH/02b_theory_campaign_history.md:17495].
Done when. A scout HQ is only recommended; its charter is not written. The paper otherwise "compares, and does not claim to exceed, Thm 1, Thm 2 and Thm 4" of Song–Liang [NBH/03_results_and_conclusions.md:316].

FR07 · Finite-n versus asymptotic onset ($n \approx 2 \cdot 10^6$ versus $n = 200$)

Gap. The statistical theorems are asymptotic; T03♯'s proof thresholds start near $n \approx 2 \cdot 10^6$ (RF123 corrected the earlier $1.2 \cdot 10^5$); beta-min fails by at least 43× at the study's signals (RF105) [NBH/START_HERE.md:51]. Theorem N200 (TH334) gives the finite-n account at the level of $\Pi(X)$ predictions ("a lower bound plus a measured term"), 558 predicted vs 571 observed wrong reports [NBH/03_results_and_conclusions.md:339], but its $\Pi(X)$ upper bound is OPEN (CJ14, CJ230: "OPEN (Envelope GAP)"). Finite-n Gibbs discovery is CJ26 ("ThmB-T24 vacuous at $n = 200$; no route").
Why it matters. F-REFEREE's first major point: asymptotic theorems against $n = 200$ simulations [NBH/03_results_and_conclusions.md:308]; the theory "can explain the mechanism, not the $n = 200$ numbers" [NBH/03_results_and_conclusions.md:339].
Tried. The union bound over Hamming balls for $\Pi(X)$ failed (RF249); the worst-case (D_S) margin is negative at $n = 200$ [NBH/branch_snapshots/API-PXNGAP/REPORT.md:56-58].
Candidate routes (sources). "a path-wise bound using O(1) prior differences on the lattice would be the natural finite-n statement for T37" [NBH/branch_snapshots/API-PXNGAP/REPORT.md:155-156]; the growing-n simulation LH-SCALE (ENG18) to show the predicted transition [NBH/05_open_questions_and_next_steps.md:216].
Done when. CJ14, CJ26 are future work (no paper-1 HQ). LH-SCALE awaits DEC10.

FR08 · Forward-KL control of the terminal law

Gap. On W, the logarithmic number of terminal rounds gives terminal marginal TV and reverse-KL convergence in probability; "Forward-KL convergence additionally needs control of $\log(1/\pi_{\min})$, which is not supplied" [NBH/02_theory.md:1592]. The C15-KL clause's fallback: "No forward-KL convergence rate without controlling $\pi_{\min}$; no silent smoothing" [NBH/02_theory.md:1558].
Why it matters. Proposal §5.4 lists KL among the accuracy measures; the KL calculus sits in the S-IDENT card TH13 (wave 2, LP-IDENT) and M2(a2) [NBH/02_theory.md:1522].
Done when. Not chartered as a separate item; under RL03 the masked final-estimate TV/KL is future work CJ02. Any forward-KL statement needs a bound on $\log(1/\pi_{\min})$, per the sources.

FR09 · The §5.3 scale: useful constants only for $\varepsilon \lesssim n^{-1/2}/p$

Starting material. The M2 charter's first lemma: "C's D_* decomposition (a1) via the restriction-monotonicity lemma (PC2 Appendix A steps 2–4), written as a filed proof; in parallel the pilot-progress lemma behind (b6′)" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M2.md:117]; Seat C's bounds defining b_X, C_σ, D_* [NBH/branch_snapshots/MEET-THM/R2_C.md:65] and its discrepancy half [NBH/branch_snapshots/MEET-THM/R2_C.md:261]; the PC2 check files are in NBH/branch_snapshots/MEET-THM/evidence/W2_PC2/.
Gap. "the §5.3 constants, useful only for $\varepsilon \lesssim n^{-1/2}/p$" [NBH/START_HERE.md:49]. The sharp ε-scale necessity is CJ22: "All three seat forms COUNTEREXAMPLE; scale disclosure ships" [NBH/05_open_questions_and_next_steps.md:68].
Why it matters. Binding goal §5.3 (M2, TH02; RL01); HQ's caveat to Felipe [pos 1650] (DEC04).
Done when. The scale is disclosed, not proved sharp; M2 ships with the disclosure [NBH/02_theory.md:823]. A sharp necessity theorem is future work.

FR10 · Masked-regime two-sided accuracy of $\hat{Z}$ and PIPs

Gap. Masked $\hat{Z}$ and PIP statements are one-sided. OB04(iii) "Obligation: CONJECTURE, OPEN for $c \geq 2$" = CJ24 (useful two-sided masked evidence precision; V11 OPEN); masked consistency of the equal-weight PIP/MPM CJ07; TV/KL of the final pooled estimate under masking CJ02 [NBH/05_open_questions_and_next_steps.md:48], [NBH/05_open_questions_and_next_steps.md:53], [NBH/05_open_questions_and_next_steps.md:70], [NBH/05_open_questions_and_next_steps.md:90].
Why it matters. Objective 1; the F-REFEREE point "evidence and PIPs mean nothing in the masked regime" [NBH/03b_reasoning_and_traces.md:417]; listed as "Low or open" confidence [NBH/03_results_and_conclusions.md:338]. Paper 1 ships one masked accuracy statement (TH04.f) per RL03.
Done when. Future work for paper 1 (no HQ).

FR11 · Delivery, calibration and mixing

Gap. Delivery $P(U) \to 0$ and resource-failure laws (CJ13: "No route; output theorems stated jointly with $\mathcal{O}$ in liminf form"); the calibration contract's open half OB05(b) = CJ06 (as-coded masked stage count, Conjecture P OPEN); polynomial mixing of the $\varphi = 1$ round and the L terminal rounds versus mixing time (CJ04, OB10: "NONE / OPEN"); the random-order versus random-scan Dirichlet comparison (CJ05, dissent DS02) [NBH/05_open_questions_and_next_steps.md:50-52], [NBH/05_open_questions_and_next_steps.md:59], [NBH/05_open_questions_and_next_steps.md:96].
Why it matters. Every output theorem is stated jointly with the output event because delivery is not proved; "Low or open" confidence for delivery [NBH/03_results_and_conclusions.md:338].
Tried. The YWJ/ZYVR transfer route "blocked as stated" (CJ04); "start mass + bridge mass + exit control ⇒ delivery" false (RF199).
Done when. Future work for paper 1.

FR12 · No MCMC-slow / SMC-fast separation

Gap. "No published BVS example has MCMC provably slow while tempering or SMC succeeds" (CJ290, OPEN as a literature claim) [NBH/03_results_and_conclusions.md:323]; separation from single-chain flip/swap MCMC (CJ12: "Moat construction and accuracy-matched comparator unproved"); DR05 dropped the comparison for lack of an accuracy-matched comparator [NBH/05_open_questions_and_next_steps.md:58], [NBH/05_open_questions_and_next_steps.md:75].
Why it matters. Felipe's goal "to retire MCMC for BVS" [pos 1030]; the honest scope: the masked-start theorems "don't show the method beats MCMC" (reversal H.4).
Done when. Future work; no charter.

FR13 · The "costs only depth" half of the thesis

Gap. Of the thesis "correlation costs only sequential depth and particles, never correctness", "the 'never correctness' half is known; 'costs only depth' half open per THM-INV" [NBH/03b_reasoning_and_traces.md:144]. The stage laws (M7, TH07; HQ's plain words: "correlation costs about $\log_2$(masking depth) sequential stages", which in the card is clause (d) on the ideal exact-frontier two-state grid only, while the coded grid has $T \le T_e$ and the count bound of (c)) and Prop D (M8, TH08, given the execution DAG as specified, audit open) address it at island level; the as-coded masked stage count is CJ06; the favourable $O(1)$ stage law unconditional is CJ23 (S-STGI dissolved, TH21; dissent DS03); Prop D's calibration share is "seat-derived, unchecked" (OB15) [NBH/05_open_questions_and_next_steps.md:69], [NBH/05_open_questions_and_next_steps.md:101].
Why it matters. Objective 2 (parallel scalability); the remark TH22 records that "there is no retained non-corollary thesis sentence" [NBH/03_results_and_conclusions.md:115].
Done when. M7 (wave 2) and M8 (wave 3) per charters; M8: cross-family code read reproducing (a)–(c) and the timing tabulation on ≤ 20 saved fits (ENG09); escalation if depth grows with R' at $w \ge R'$ [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M8.md:81-83].

FR14 · Lean coverage of the main cards

Gap. "No main card is Lean-certified as a whole"; every main card has a "Lean core" still to be written [NBH/03_results_and_conclusions.md:22]. T12(i.a) is partial: the resample–move recursion `hstep` is an explicit hypothesis [NBH/03_results_and_conclusions.md:145]. Floating-point: the label "numerical implementation of an exact-arithmetic certificate" stays until OB16 / CJ16 ("N16 OPEN; T33-errbound COUNTEREXAMPLE") is closed (ENG05).
Why it matters. Venue target "Down to Lean certificates" [pos 1015].
Route (sources). "T12 `hstep` discharge for multinomial resampling is the first named target (`L-LEAN/T12_PLAN.md`), keep `stmtdiff.py` frozen-statement check; Astra/Fable fidelity checks per file" [NBH/05_open_questions_and_next_steps.md:212]; per-card Lean targets are in the charters (e.g. M1, M2, M5, M7, M8).
Done when. Per card: the charter's Lean target compiles with 0 `sorry` and passes cross-family statement-fidelity checks.

Priority of the frontier problems (derived)

The sources do not rank FR01–FR14 directly. The order below is *derived by the lead* from two source facts only: which proposal goals are binding (§13.1 and §5.3) [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:16], [NBH/branch_snapshots/MEET-THM/COVERAGE_MATRIX.md:23], and the chartered wave order [NBH/branch_snapshots/MEET-THM/MINUTES.md:110-113]. Problems outside the waves are scope extensions or scouts; their order among themselves is not set by any source.

Priority	Problems	Basis in the sources
1	FR04 (N1)	Binding §13.1; wave 1: "N1 (binding §13.1) is the longest pole and starts first" [NBH/branch_snapshots/MEET-THM/MINUTES.md:110].
1 (same wave)	M3 START assembly (TH03); LP-FOUND = S-TARGET, S-BLOCKS, S-KERNEL (TH10–TH12)	Also wave 1 ("Foundations") [NBH/branch_snapshots/MEET-THM/MINUTES.md:110]; no FR card because they are assemblies of card clauses, not open frontier problems. M3 first lemma: "Match production $\gamma_0, \bar{Z}_0$ to L-DIL; then audit LH's normalized reference mass and joint sequential ratio premises" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M3.md:93]; LP-FOUND: "(none specified by the seats for the pack; each card's route is in its seat card)" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-FOUND.md:111].
2	FR09 (§5.3 scale)	Binding §5.3; wave 2 (M2 with M7, M6, LP-IDENT) [NBH/branch_snapshots/MEET-THM/MINUTES.md:111].
3	FR08, FR13	Wave-2 cards (S-IDENT KL calculus; M7 stage laws), and wave 3 for M8 [NBH/branch_snapshots/MEET-THM/MINUTES.md:111-112].
4	FR05 (N3)	Wave 3, the code-audit wave: "N2, N3, LS0, GUARD-W and the DAG serve M5, M4 and M8 together. M4(f) needs M5" [NBH/branch_snapshots/MEET-THM/MINUTES.md:112].
de-ferred	FR10, FR11	Future work for paper 1 (no HQ) [NBH/branch_snapshots/MEET-THM/FUTURE_WORK.md:3]; they touch wave-3 cards (M4, M5) but are not chartered.
5	FR14 (Lean)	Runs inside every wave: each HQ "loops all the way to Lean" (§7.5).
scope	FR01, FR02, FR03, FR07	Not on the chartered waves. They decide whether the theorems cover the study's own regime ($p \gg n$, groups of 5, $n = 200$); no source orders them.
scout	FR06, FR12	Recommended scout alongside wave 1 (DEC02: "nothing else waits on it" [pos 1650]); FR12 is explicitly not claimed.

7.3 Engineering and audit items that gate claims

"None of these can change a method decision [...] Each one gates only the claim named" [NBH/05_open_questions_and_next_steps.md:171]. Items ENG01–ENG17 are the archive's E1–E17; ENG18–ENG20 are the simulation and review items of its next-step list.

ID	Item	Gates	Status
ENG01	W-cal: bitwise replay of `v3_calibrate` with concurrent diagnostic islands on S1 r001, S4 r001, S5 r002 (about 1 h on Owls); Prop W-cal TH395 [NBH/05_open_questions_and_next_steps.md:175]	"bitwise" for calibration; any wall-clock claim	OPEN (not executed)
ENG02	LS0 merge into the production core plus merged-LS0 replay on the 3 designs (HPM, stores, flags, `ls0_improved`, a deterministic eviction/tie example); CJ213, **CJ178**; LS0 cost data **CJ228**, outcome neutrality **CJ226** [NBH/05_open_questions_and_next_steps.md:176]	"LS0 is in the shipped core"; T38(b1-LH) as a theorem about the shipped method (OB02(c), OB03)	OPEN
ENG03	Stop-handling / GUARD-W audit: "any worker stop, LS0 stop or guard stop ⇒ no output"; forced-stop table after the `log_Z` NA fix [NBH/05_open_questions_and_next_steps.md:177]	premise (i) of G2 and every $P\{\mathcal{O}, \cdot\}$ statement (OB02(e), OB06(b))	OPEN
ENG04	IND-T for streams 10..9 + R' [NBH/05_open_questions_and_next_steps.md:178]	stacking theorems; frozen-$\mathcal{C}$ contract (OB14(c))	OPEN
ENG05	VAL-FP at production size: long-double replay on 2 study designs at $p = 500$ (measured only at $p \leq 40$); validated arithmetic [NBH/05_open_questions_and_next_steps.md:179]	removal of the "numerical implementation" label (OB16, CJ16)	OPEN

ID	Item	Gates	Status
ENG06	Preflight fixes (e6) and allocation audit of concurrent peak memory and stop behaviour [NBH/05_open_questions_and_next_steps.md:180]	`resource_limit` estimates	OPEN
ENG07	Checked arithmetic (e7): overflow boundary tests [NBH/05_open_questions_and_next_steps.md:181]	support at huge (n, p)	OPEN
ENG08	Full $\hat{Z}$-weighted PIP rescoring of all 2,394 fits on NOTS/laptop bundles (`pip_rescore.R`), with R_{eff} per fit; only 3 designs done (max ΔPIP $1.29 \cdot 10^{-3}$) [NBH/05_open_questions_and_next_steps.md:182]	primary PIP and $\text{MPM}_{\hat{Z}}$ for the 2,400 fits (MD08)	OPEN
ENG09	Calibration timing tabulation from ≤ 20 saved fits (Owls ≤ 8 cores) [NBH/05_open_questions_and_next_steps.md:183]	cost wording; the 0.46/0.67/1.69 vs 0.45/0.62/1.76 formula	OPEN
ENG10	e4 tie-DP invariant tests (N3) and $n = 2$ boundary tests [NBH/05_open_questions_and_next_steps.md:184]	LS0 under truncation; $n = 2$ support (FR05)	OPEN
ENG11	Certificate card on saved fits; C5 screening event over the 2,400 fits (CJ238); per-dataset matching of O-SWAP checks [NBH/05_open_questions_and_next_steps.md:185]	certificate statements on study fits	OPEN
ENG12	Per-particle random streams (C6): implement and law-check iff paper 1 claims particle-parallel depth or shows a fit with $p > 2{,}500$ [NBH/05_open_questions_and_next_steps.md:186]	particle-parallel depth (v3.2)	NOT TRIGGERED
ENG13	$X^\top X$-free scorer (C7), the declared v3.2 path [NBH/05_open_questions_and_next_steps.md:187]	scalability beyond $p \approx 2{,}500$ (LIM02)	v3.2 PATH
ENG14	Joint summaries (pair inclusion, model size): "need a small new run, because those particles weren't saved" [NBH/05_open_questions_and_next_steps.md:188]	joint-h numerics (TH01.g)	OPEN
ENG15	AMENDMENT_HQ_02 reporting fields ($\hat{J}$, $\hat{\varepsilon}_\times$, d in $\mathcal{H}$, read-only accumulator) [NBH/05_open_questions_and_next_steps.md:189]	TH19 (S-REP)	NOT CODED: "Reporting fields 1–3 are specified, not yet coded" [NBH/branch_snapshots/MEET-v31/AMENDMENT_HQ_02.md:15]; the open-questions file lists no code status
ENG16	Data caveat: Owls regeneration ≠ NOTS bits (`data_hash_check.csv` "5/7 match=FALSE rows") [NBH/05_open_questions_and_next_steps.md:190]	claims mixing Owls and NOTS records	UNCERTAIN (interpretation not recorded)
ENG17	Reproducible record for ABL-CONF fits; NOTS reorganisation (`projects/BVS`) by the old HQ [NBH/05_open_questions_and_next_steps.md:191], [NBH/05_open_questions_and_next_steps.md:235]	all empirical claims	IN PROGRESS at [pos 1673]
ENG18	LH-SCALE: production code on LH at growing n; "3 × 4 × 2 × 100 fits; $n \in \{400, 1600, 6400, 25600\}$; $g = 9$; $\rho \in \{0.6, 0.85\}$; $\psi \in \{0.3, 0.5\}$; $s_0 = 2c + 4$" (about 2,400 fits); report $P(\hat{S} = S_0)$, discovering mechanism, $\pi_0(S_0)$ vs the T27 rate, stage counts [NBH/05_open_questions_and_next_steps.md:216]	F-REFEREE's transition at growing n (FR07)	OPEN (awaits DEC10; needs NOTS)

ID	Item	Gates	Status
ENG19	Real data and head-to-head comparisons (PARNI, MAdaSub, TGS, SuSiE, SSL, Song–Liang numerics); no $\hat{Z}$/PIP head-to-head against other Bayesian methods exists [NBH/05_open_questions_and_next_steps.md:217]	LIT-VENUE expectations	OPEN
ENG20	Cross-family re-check of Claude repairs not covered by A-REV5–9 [NBH/05_open_questions_and_next_steps.md:165], [NBH/05_open_questions_and_next_steps.md:224]	every repaired campaign result used by a card	OPEN

7.4 Recorded conflicts and uncertain items

All items below are recorded in the archive as unresolved conflicts between sources ("both sides recorded, none closed by a new review unless stated") [NBH/05_open_questions_and_next_steps.md:107-165]. They are reproduced so the loop does not treat either side as settled.

Final-state items U1–U14 (from the final-state theory file)

UNCERTAIN — U1 · growth rate (ii)

MINUTES, AMENDMENT_HQ_02_DEFINITIONS §B and NOTATION say "two optional rates"; HQ adopted only (i) as a method setting, (ii) "only as a sufficient rate inside the theorem". How the final M1 states (ii) is not recorded; a task text paraphrases it as "$R' \geq c(n+p)^6$" [NBH/05_open_questions_and_next_steps.md:110]. See RL06.

UNCERTAIN — U2 · size of the draft list

"19 items" [pos 1280], [pos 1302], [pos 1201] vs "24 entries, not 19" [pos 1492] (MINUTES §1 uses 24) [NBH/05_open_questions_and_next_steps.md:111].

UNCERTAIN — U3 · number of inherent limitations

MINUTES §5: 9; THEOREM_REVISION_INPUTS §4.A: I1–I7, I9–I11; FOR_FELIPE_FINAL_METHOD §4: 7. Registered: the MINUTES nine (LIM01–LIM09) [NBH/05_open_questions_and_next_steps.md:112].

UNCERTAIN — U4 · M1 K-ID header

The M1 K-ID header differs between the card and MINUTES §2.1; the registry follows MINUTES [NBH/05_open_questions_and_next_steps.md:113].

UNCERTAIN — U5 · M5 δ_1

δ_1 is defined as an expectation and then used pointwise; integration note 4 (CLOSE_AUDIT #7) fixes the pointwise reading [NBH/05_open_questions_and_next_steps.md:114].

UNCERTAIN — U6 · N9 sub-claims

7 sub-claims ([pos 1550]; MEET-v31 MINUTES §9) vs 8 ([pos 1593]) [NBH/05_open_questions_and_next_steps.md:115].

UNCERTAIN — U7 · S-STGI / K29 / K30 propagation

Treat "K30 = M7(d); K29 = FW23 (CJ23)" as final; `REGISTRY_PAPER1.md` not re-audited [NBH/05_open_questions_and_next_steps.md:116].

UNCERTAIN — U8 · label N2(b)

N2(b) (W-cal) is defined in THEOREM_REVISION_INPUTS (on MEET-v31), not in the R3 files [NBH/05_open_questions_and_next_steps.md:117].

UNCERTAIN — U9 · FUTURE_WORK line numbers

Line-number drift in FUTURE_WORK citations (content identical) [NBH/05_open_questions_and_next_steps.md:118].

UNCERTAIN — U10 · supplement-card order

The order differs between MINUTES §2.2 and the TLF header table; IDs follow the TLF order [NBH/05_open_questions_and_next_steps.md:119].

UNCERTAIN — U11 · who Seat D was

"Seat D" is "fresh-eyes Fable" in FOR_FELIPE and [pos 1584]; NOTATION lists the seats as A (Fable 5.1), B (Fable 5.1), C (gpt-6-astra), D (Fable 5.1) [NBH/05_open_questions_and_next_steps.md:120].

UNCERTAIN — U12 · approvals

METH31's header "Not yet approved by HQ or Felipe" predates the approvals; at the archive's end Felipe's approval of the theorem list was not recorded [NBH/05_open_questions_and_next_steps.md:121]. *Resolved for the theorem list* by the 2026-11-07 approval (DEC01); Felipe's explicit sign-off of the plain-language final method is still not recorded [NBH/01_overview_and_timeline.md:399].

UNCERTAIN — U13 · names of the checkers

A task text calls the check waves "Sol" and "Opus"; the files call them PC1 (gpt-6.1-sol on Seats A, B, D) and PC2 (Opus 5.5 on Seat C); CHK = PC1 [NBH/05_open_questions_and_next_steps.md:122].

UNCERTAIN — U14 · date of the island-depth decision

See the box under DEC15 [NBH/05_open_questions_and_next_steps.md:123].

Empirical and bookkeeping conflicts

UNCERTAIN — concurrency "32 islands within 2e-4"

Holds only for S1 (max ΔPIP $2 \cdot 10^{-4}$); S4 $2.3 \cdot 10^{-3}$, S5 $7 \cdot 10^{-4}$; log Z gaps 0.027, 0.012, 0.47 nats [NBH/05_open_questions_and_next_steps.md:126].

UNCERTAIN — favourable count of the unmodified v3 start

0/2,400 (API-RESID) vs 1/2,394 (O-DILUTE §1) [NBH/05_open_questions_and_next_steps.md:127].

UNCERTAIN — thresholds and numerical inconsistencies

"Favourable from n": 590–730 (O-DILUTE), 800–1,600 (W3-VEXP), 1,050–3,200 (W5-FAVIID) (RF160); start-mode misses "6–12 of 14" vs "holds 2.5–6 of 14" vs "7 of 14 masked"; beta-min shortfall $\geq$ 43× vs "about 100×" (old); SE columns $\leq$ 114 vs 215–229; stopped fits 6 vs 4 (O-SWAP); LS0 cost "about $\ell \cdot p$" vs $1 + (\ell + 1) \max_k [p + k(p - k)]$ ($\approx 1.7 \cdot 10^5$ at $p = 1{,}000$); terminal segment $L + 4$ vs $L + 5$ (resolved: both right for different boundaries). The source recommends S-ELS for counts, O-DILUTE for start masses, W6-FIXREV1 for LS0 cost, and to drop or attribute the "favourable from $n \approx \ldots$" threshold [NBH/05_open_questions_and_next_steps.md:128].

UNCERTAIN — credits

Cloud credits $245 vs $250; $21 (LEDGER) vs $19 [pos 1673]; API spend by run vs the implied $79 [NBH/05_open_questions_and_next_steps.md:129].

UNCERTAIN — Lean count wording

"101 declarations" vs "101 theorems/results"; 56 audited for axioms [NBH/05_open_questions_and_next_steps.md:130].

UNCERTAIN — conflict counts

26 (REGISTRY_v2) vs 37 (MEET-THM registrar index); they refer to different registries and are not reconciled [NBH/05_open_questions_and_next_steps.md:131], [NBH/03_results_and_conclusions.md:137].

UNCERTAIN — SSL numbers at $c = 1$

HQ's SSL numbers (569/582) vs SSL-RS (supplement) 593/592 [NBH/05_open_questions_and_next_steps.md:132].

UNCERTAIN — venue view

Shifted from [pos 933] to [pos 1009] (§6.6) [NBH/05_open_questions_and_next_steps.md:133].

UNCERTAIN — `T11Miss.lean` and TH06.a

Whether `T11Miss.lean` is reused for TH06.a is not stated [NBH/05_open_questions_and_next_steps.md:134].

UNCERTAIN — PR #1 and LH-SCALE

Resolution not recorded (DEC09, DEC10) [NBH/05_open_questions_and_next_steps.md:135].

Registry conflicts C1–C26 (campaign registry REGISTRY_v2)

UNCERTAIN — C1 · Lemma 0(b), R2 dichotomy

RF167, RF168: W1-T37R PROVED vs W3-VT37R / A-REV2 ERROR; W6-FIXREV2 replaces them by Lemma 0(b′) and a three-way Prop. F (same-family, unreviewed) [NBH/05_open_questions_and_next_steps.md:138].

UNCERTAIN — C2 · TD'_P iff DS

RF169: SF VERIFIED vs XF ERROR (ties); W6-FIXREV2 Lemma T + Lemma NT [NBH/05_open_questions_and_next_steps.md:139].

UNCERTAIN — C3 · Prop U

TH227: SF VERIFIED (bookkeeping) vs XF ERROR; Theorem B unaffected [NBH/05_open_questions_and_next_steps.md:140].

UNCERTAIN — C4 · Lemma S attainment

LM174: "tight" claim VERIFIED vs XF ERROR on the full domain; the inequality itself VERIFIED [NBH/05_open_questions_and_next_steps.md:141].

UNCERTAIN — C5 · E3(ii)

RF188: SF VERIFIED vs XF ERROR (omits $q < 1$) [NBH/05_open_questions_and_next_steps.md:142].

UNCERTAIN — C6 · EXP constant

RF134: $0.52/c_*$ vs actual $0.51/c_*$; finite bound unaffected [NBH/05_open_questions_and_next_steps.md:143].

UNCERTAIN — C7 · Thm 4 "iff"

RF135: the "iff" gloss is an ERROR; VEXP's added claim not justified [NBH/05_open_questions_and_next_steps.md:144].

UNCERTAIN — C8 · T38b-R (whole)

TH241: SF VERIFIED whole vs A-REV1 GAP (superseded by W6-FIXREV1, see C11) [NBH/05_open_questions_and_next_steps.md:145].

UNCERTAIN — C9 · ORTH-joint, T26-PropC

CJ210, **TH325**: PROOF SKETCH (weaker) recorded [NBH/05_open_questions_and_next_steps.md:146].

UNCERTAIN — C10 · Thm2-BO label mismatch

The author's PROOF SKETCH kept [NBH/05_open_questions_and_next_steps.md:147].

UNCERTAIN — C11 · T38-(0) scope

TH281 / T38b-R(whole): A-REV1 reviewed W1-T38E, not W4-T38FULL; W6-FIXREV1 supersedes [NBH/05_open_questions_and_next_steps.md:148].

UNCERTAIN — C12 · Lemma 0(a,c) wording

LM144: wording dispute on "initial centre is the truth" [NBH/05_open_questions_and_next_steps.md:149].

UNCERTAIN — C13 · version of W1-T37R reviewed

Reviewed at d492f22 (unfinished REPORT) vs 3bcc429; PROOF.md identical [NBH/05_open_questions_and_next_steps.md:150].

UNCERTAIN — C14 · F-REFINE legend

F-REFINE "[P]" is same-family only [NBH/05_open_questions_and_next_steps.md:151].

UNCERTAIN — C15 · Theorem W vs VPGD-PropT

TH204 vs TH217: Theorem W (region form) fails with an exact tie on masked LH ($c \geq 2$); the lattice form W_{Lat} holds [NBH/05_open_questions_and_next_steps.md:152].

UNCERTAIN — C16 · PXN-P6

TH153: W5-VMASK verified vs A-REV3 GAP (A-REV3 did not read W5-VMASK) [NBH/05_open_questions_and_next_steps.md:153].

UNCERTAIN — C17 · SHARE-M(b), SHARE-B2W4

RF203, TH254: PROOF SKETCH / upgrade vs A-REV3 ERROR [NBH/05_open_questions_and_next_steps.md:154].

UNCERTAIN — C18 · Cost-T38, T32(i)

TH289, CJ112: stale CONJECTURE labels vs W5-LOCKBOOK PROVED (same family only) [NBH/05_open_questions_and_next_steps.md:155].

UNCERTAIN — C19 · SEP-E

TH386: same staleness [NBH/05_open_questions_and_next_steps.md:156].

UNCERTAIN — C20 · no-LS0 masked candidates on LH

TH285, TH286, **TH263**, **TH267**, **TH269**: none XF reviewed at the time (later A-REV7) [NBH/05_open_questions_and_next_steps.md:157].

UNCERTAIN — C21 · O-CERT §4

O-VCERT "§4 empty" vs a later illustration (not reviewed) [NBH/05_open_questions_and_next_steps.md:158].

UNCERTAIN — C22 · STG-Wp

TH367: O-STAGES Cor. W′ rests on (LL) vs O-WSTAB COUNTEREXAMPLE on the divergent side [NBH/05_open_questions_and_next_steps.md:159].

UNCERTAIN — C23 · OMIX-C1

TH183: common plug-in σ^2 vs O-DILUTE/F-SIGMA; rejected (E-SPINE §2.2) [NBH/05_open_questions_and_next_steps.md:160].

UNCERTAIN — C24 · T38-b1-LH iid failure

TH287, RF217: A-REV5 ERROR vs A-REV8 GAP on "$Q_f \pi_0(S_0) \to 0$ when $c \geq 2$" [NBH/05_open_questions_and_next_steps.md:161].

UNCERTAIN — C25 · SEP-GU

TH388: A-REV7 VERIFIED vs A-REV9 GAP (import not re-audited) [NBH/05_open_questions_and_next_steps.md:162].

UNCERTAIN — C26 · LH-LAT sharpness

TH293, **RF144**: m_0 "sharp" false on 32/94 heterogeneous settings; exact margin formula R5 [NBH/05_open_questions_and_next_steps.md:163].

UNCERTAIN — further conflict lists

W8-SYNTH §4 (15 conflict groups, 02b P.4); the retired/refuted checklist (02b P.2); UNCERTAIN markers by theme authors (02b P.5, e.g. BASIN used for two objects; "D18" low-p finding not in the snapshots; P8 counterexample identity; label-collision proposals) [NBH/05_open_questions_and_next_steps.md:164]. METH-AUDIT's "1/1 vs 3/3" residue is still listed as a source conflict (RF270).

7.5 The planned next phase: per-theorem HQs

After the approval of the list (DEC01, now given), the plan is one HQ per main theorem plus five lemma packs, in waves of 3–4 "because subscriptions cannot run 14 HQs at once" [pos 1348] [NBH/05_open_questions_and_next_steps.md:198-209]. Charters: NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/ (M1–M8, LP-FOUND, LP-IDENT, LP-AUDIT, LP-REST, LP-REP); each gives wave, covered K-IDs, dependencies, Lean core, risks, numerical companion ($\leq$ 20 seeds) and fallback.

Wave	HQs (cards)	Reason (MINUTES §2.5) and first lemma (charter)
1	M1 V/H (TH01); M3 START (TH03); LP-FOUND = S-TARGET TH10 + S-BLOCKS TH11 + S-KERNEL TH12 (+ S-T TH09 cited)	"Foundations. N1 (binding §13.1) is the longest pole and starts first" [NBH/05_open_questions_and_next_steps.md:203]. M1: N1 (FR04) [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:114]. M3: "Match production $\gamma_0, \bar{Z}_0$ to L-DIL; then audit LH's normalized reference mass and joint sequential ratio premises" [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M3.md:93]. LP-FOUND: none specified; §5.1 "exact correction" rests on S-KERNEL [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/LP-FOUND.md:85].
2	M2 WS (TH02, binding §5.3); M7 STG (TH07); M6 SEP + KL (TH06); LP-IDENT = S-IDENT (TH13)	"The §5.3 package and its inputs (d_σ from M3, the count from M7, the lower bound from M6)" [NBH/05_open_questions_and_next_steps.md:204]. M2: D_* decomposition (a1) and the pilot-progress lemma [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M2.md:117]; M7: Lemma RS, then the coverage identity [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M7.md:83]; M6: binary coarsening plus the singleton second-moment term [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M6.md:77].
3	M5 HPM (TH05); M4 $Z\mathcal{T}$ + masked statement (TH04); M8 Prop D (TH08); LP-AUDIT = S-CERT (TH14) + S-OUT (TH15)	"Code-audit wave: N2, N3, LS0, GUARD-W and the DAG serve M5, M4 and M8 together. M4(f) needs M5" [NBH/05_open_questions_and_next_steps.md:205]. M5: N3 (FR05); M4: N13 retention under the coded LS0 rule, then the GUARD-W audit [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M4.md:100]; M8: trajectory inventory of the execution DAG [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M8.md:79].
4	LP-REST = S-Z (TH16) + S-POOL (TH17) + S-LIM (TH18); LP-REP = S-REP (TH19) + S-LEAK (TH20)	"Consequences and reporting corollaries. Nothing binding waits on them" [NBH/05_open_questions_and_next_steps.md:206].

Process rules carried over: each HQ "loops all the way to Lean"; cross-family check on every claim; batch math claims into one proof check; label every self-proved claim "unchecked"; freeze the agenda before dispatch; cap seat output [NBH/05_open_questions_and_next_steps.md:209]. Binding-goal safeguards: §13.1 (M1) and §5.3 (M2) escalate to Felipe if their routes fail; "an orthogonal-only substitute does not close" §5.3 [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M2.md:83]; named implementation premises (N2(b), N2(c), N2(e), N14(c), the N3 DP invariant, the streams/LS0/GUARD-W/execution-DAG audits) are recorded in the charters [NBH/branch_snapshots/MEET-THM/HQ_CHARTERS/M1.md:99]. The scout HQ on the Song–Liang separation would run alongside wave 1 if approved (DEC02) [NBH/05_open_questions_and_next_steps.md:210].

7.6 Infrastructure rules for the new loop

The archive's rules, carried over [NBH/05_open_questions_and_next_steps.md:242-251]:

1. **Server restarts cancel delegated agents.** Long jobs must not depend on the agent session.
2. **systemd-scope kill rule.** "T3 runs each agent in its own system container, and when the agent's session ends it kills everything that agent started; launching jobs 'detached' [`setsid`] doesn't escape that." Launch long jobs as independent `systemd-run --user` units (Owls `POLICY.md`).
3. **Owls limits.** All compute through `owls_run`; low priority, single-threaded BLAS, 2 cores per job, 8 cores total; "Big work goes to NOTS, one ssh connection at a time".
4. **Billing.** Routines bill the weekly plan, only `claude --cloud` sessions bill the cloud credit; the promotional API credit covers the Agent SDK, not the Claude Code CLI.
5. **64K output-token failure rule.** "tell agents to write long proofs in several files/commits, never one giant response".

6. **Thread launches** must use a branch that exists where the thread starts.
7. **Briefs.** "N/3"-type headlines may overclaim; wait-for-run agents exit early; stale branch heads can mislead reviews.
8. **Authority.** HQ has full authority for method and theorem decisions [pos 1061] but never adopts or locks results; the burden of proof is on any method change ("investment rule"); Felipe's preference "use CW instead of CP".

Credits as of [pos 1673]: $19 cloud (expired 2026-11-05) and $121 of the API credit (expiry 2026-11-17); a new $200 API credit was announced for 2026-11-15 [NBH/05_open_questions_and_next_steps.md:228-232]. Data and scripts of the empirical study were being moved to NOTS `projects/BVS` by the old HQ [NBH/05_open_questions_and_next_steps.md:235].

For agents — suggested first moves for the loop (ordered by the sources' own priorities)

1. Start wave 1 as chartered: **M1** with N1 first (FR04; "the longest pole and starts first"), **M3**, and **LP-FOUND** [NBH/05_open_questions_and_next_steps.md:203]. Read the charter files first.
2. Put to Felipe the still-open decisions that change wave 1's scope or run alongside it: the Song–Liang scout (DEC02), then DEC03–DEC07 (HQ readings he may overrule) [NBH/START_HERE.md:57].
3. Before wave 3, the engineering premises its theorems rest on: LS0 in the production core with replay (ENG02), GUARD-W audit (ENG03), IND-T (ENG04), N3 tests (ENG10) [NBH/05_open_questions_and_next_steps.md:205].
4. Outstanding cross-family re-checks of Claude repairs (ENG20) and of API-PXNGAP (FR01) are recorded as owed [NBH/05_open_questions_and_next_steps.md:224], [NBH/branch_snapshots/API-PXNGAP/REPORT.md:164].
5. Never retry a route of §6.4 without addressing its recorded reason; keep every status word verbatim.

A Notation and conventions

Sources: the archive's notation file [NBH/notation_and_conventions.md:1-7], which embeds the registrar's table `MEET-THM/NOTATION.md` verbatim [NBH/notation_and_conventions.md:48-121], and the shared definitions of the final theorem list [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:52-88]. **Caveat** (archive editor): "the final cards in THEOREM_LIST_FINAL (02a TH01–TH20) are copied verbatim from the seats' Round-3 cards and keep the seats' own symbols [...]; the registrar's canonical renamings [...] govern the HQ charters and future writing but are not applied inside the verbatim cards" [NBH/notation_and_conventions.md:17]. Where the registrar flagged a clash it is marked [C] below; "canonical" means the registrar's proposal, "adopted for the HQ charters unless an HQ objects" (no objection is recorded). Positions [pos N] are transcript positions.

A.1 Definition IDs DF01–DF25

Each row: archive ID (defined here), registry name, exact registry status, where this document gives the object, and the registry row. Statuses are copied verbatim from `NBH/ID_REGISTRY.md`.

ID	name	status (exact)	given in
DF01	Model, likelihood, g-prior, σ^2 prior	unchanged in v3.1 ("Felipe's model")	§2.2 [NBH/ID_REGISTRY.md:162]
DF02	Size prior BB(1,p) and cap K	unchanged	§2.2 [NBH/ID_REGISTRY.md:163]
DF03	Score m(S), evidence Z, posterior π	unchanged	§2.2 [NBH/ID_REGISTRY.md:164]
DF04	Preprocessing / standardisation	rule 1 changed by e3 (n ≥ 2 proposed)	§2.1 [NBH/ID_REGISTRY.md:165]
DF05	Blocks and tables	unchanged; table stage parallel (#4)	§2.4 [NBH/ID_REGISTRY.md:166]
DF06	Block-product start π_0	kept (item 1 Outcome 2: no depth/speed claim)	§2.4 [NBH/ID_REGISTRY.md:167]
DF07	Discrepancy d and tempering bridge γ_φ	unchanged	§2.4, §2.5 [NBH/ID_REGISTRY.md:168]
DF08	Kernel moves and round	unchanged; validity N9 (OB09, TH12)	§2.5 [NBH/ID_REGISTRY.md:169]
DF09	Resampling, pilot, ESS 0.90	unchanged; ESS class (c)	§2.6 [NBH/ID_REGISTRY.md:170]
DF10	Candidate set V, restricted ESS, grid, rebuild loop	unchanged; contract #18	§2.6 [NBH/ID_REGISTRY.md:171]
DF11	Calibration record $\mathcal{C}$, tiers, final runs	contract #18; streams extended #7	§2.6, §2.7 [NBH/ID_REGISTRY.md:172]
DF12	Islands and stacking	new in v3.1 (#7)	§2.7 [NBH/ID_REGISTRY.md:173]
DF13	Evidence estimator $\hat{Z}$_r, pooled $\hat{Z}$, Markov bound	unchanged; card adds max{Z_$\mathcal{T}$, 0.05$\hat{Z}$}	§2.8 [NBH/ID_REGISTRY.md:174]
DF14	PIP rules, MPM, R_eff	swap of primary/secondary (#11)	§2.8 [NBH/ID_REGISTRY.md:175]
DF15	HPM decision and local search	unchanged; scans parallel (#3)	§2.8 [NBH/ID_REGISTRY.md:176]
DF16	LS0	new (#8–#10)	§2.8 [NBH/ID_REGISTRY.md:177]
DF17	Reported store $\mathcal{T}$, Z_$\mathcal{T}$ and the certificate card	new (#13, #14; §2.1)	§2.8.5 [NBH/ID_REGISTRY.md:178]
DF18	General-h accumulator $\mathcal{H}$	new (#12)	§2.8.4 [NBH/ID_REGISTRY.md:179]
DF19	$\hat{J}$ (J_hat)	reporting amendment	§2.8.4 [NBH/ID_REGISTRY.md:180]
DF20	$\hat{\varepsilon}$_× (cross_block_rmax) with $\hat{\kappa}$, $\hat{\delta}$, $\hat{u}$, $\hat{r}$, $\hat{\eta}$	reporting amendment	§2.8.4 [NBH/ID_REGISTRY.md:181]
DF21	Output convention and event $\mathcal{O}$, GUARD-W	new wording (#6, #16)	§2.8, §2.7 [NBH/ID_REGISTRY.md:182]

ID	name	status (exact)	given in
DF22	Shared implementation premises	premises (not model conditions)	§A.3 [NBH/ID_REGISTRY.md:183]
DF23	Design families G, LH, W, ORTH	families used for IC3	§A.3 [NBH/ID_REGISTRY.md:184]
DF24	H-DOM and constants a, δ_A, δ_H, r, C_σ, b_X, D_*	definition	§A.3 [NBH/ID_REGISTRY.md:185]
DF25	Settings table (49 rows), classes (a)–(e)	METH31 §5 (ground rule 9)	§2.9 [NBH/ID_REGISTRY.md:186]

A.2 Symbols

"N-tab" = the registrar's table as embedded in `notation_and_conventions.md` (line cited). "M" = `METHODOLOGY_v3.md`; "M31" = `METHOD_v3.1_FINAL.md`; "TLF" = `THEOREM_LIST_FINAL.md`. The last column records a change of convention during the campaign ("—" = none recorded).

symbol	meaning	defined in	changed when
Data, model, target			
n, p	sample size; number of retained (non-constant) predictors	M [Docs/Methodology/METHODOLOGY_v3.md:64]	input domain $n \geq 3 \to n \geq 2$ proposed (e3) [pos 1251] [NBH/notation_and_conventions.md:400]
X, y; s_y	standardised design and response (max-abs, centre, SD with $n-1$); $s_y^2 = \|Cy\|^2/(n-1)$	M [Docs/Methodology/METHODOLOGY_v3.md:85-87]	—
S, $k = \|S\|$, S_b	model (index set), its size, $S \cap C_b$	M [Docs/Methodology/METHODOLOGY_v3.md:97]	—
K; Ω; k^*	cap $\min(p, \lfloor n/2 \rfloor)$; model space $\{\|S\| \leq K\}$; $k^* = \min(K, \lfloor p/2 \rfloor)$	N-tab [NBH/notation_and_conventions.md:96-97]	scan size $p + K(p-K) \to p + k^*(p-k^*)$ [pos 1480] [NBH/notation_and_conventions.md:395]. [C] K_n (Song–Liang), K-IDs
G_S, z_S, H_S, Q_S	$X_S^\top X_S$; $X_S^\top y$; $(n+1)G_S + I$; $y^\top y - n z_S^\top H_S^{-1} z_S$	M [Docs/Methodology/METHODOLOGY_v3.md:97-105]	[C] G also model assumption, card G1/G2, Theorem G; Q also draw counts [NBH/notation_and_conventions.md:102]
a_0, b_0; ν; $\bar{Q}$	IG hyperparameters 0.01; $\nu = a_0 + n/2$; $\bar{Q} = 2b_0 + y^\top y$	N-tab [NBH/notation_and_conventions.md:101] [NBH/notation_and_conventions.md:142]	"negligible" claim withdrawn [NBH/notation_and_conventions.md:396]
C_n, c_n	normalising constant of m (M); cards' common constant c_n	M [Docs/Methodology/METHODOLOGY_v3.md:107]; TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:65]	—
$m(S)$, $\mathrm{BF}(S)$	marginal likelihood; $m(S)/m(\emptyset)$	M [Docs/Methodology/METHODOLOGY_v3.md:106] [Docs/Methodology/METHODOLOGY_v3.md:176]	—
$w(k)$, W_K, $\varpi(S)$	BB$(1,p)$ weight $\frac{1}{2}/\binom{2p-1}{k}$; normaliser; capped prior $w(k)1\{k \leq K\}/W_K$	M [Docs/Methodology/METHODOLOGY_v3.md:103]	[C] W also island weight, subset, exhibit [NBH/notation_and_conventions.md:76]

symbol	meaning	defined in	changed when
$F(S)$	score $\log \varpi(S) + \log m(S)$; $e^{F(S)} = \varpi(S)m(S)$	N-tab [NBH/notation_and_conventions.md:91]	[C] Frobenius F_S (A:153, D:120), canonical Fro_S; M's V9 uses $F(S) = -(n/2)\log \bar{\sigma}^2(S) - (\lvert S\rvert/2)\log n$ [Docs/Methodology/METHODOLOGY_v3.md:372]
π, Z	posterior $\varpi m/Z$; evidence $\sum_\Omega \varpi m$	N-tab [NBH/notation_and_conventions.md:53] [NBH/notation_and_conventions.md:108]	[V] $\pi_\mathcal{T}, \pi^\mathcal{T}, \hat{\pi}$; canonical $\pi_\mathcal{T}$
$S_0, s_0; \delta_1$	true support and size; $\delta_1 := 1 - \pi(S_0)$	N-tab [NBH/notation_and_conventions.md:98] [NBH/notation_and_conventions.md:65]	[C] S° reference vs launch centre
θ^*, β	true coefficients; generic coefficient	N-tab [NBH/notation_and_conventions.md:112]	—
Blocks, start, bridge			
B, B^*	block cap $\min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, B^*)$, $B^* = 16$	N-tab [NBH/notation_and_conventions.md:95]	B^* "implementation limit" → (c) budget (e1) [pos 1226] [NBH/notation_and_conventions.md:402]
$C_1..C_J, J$	blocks (packing order); number of blocks	M [Docs/Methodology/METHODOLOGY_v3.md:168-169]	[C] J also Jeffreys; canonical block count N_{blk} [NBH/notation_and_conventions.md:73]
$r_b(U), e_b(j)$	block table $m(U)/m(\emptyset)$; size profile $\sum_{\lvert U\rvert=j} r_b(U)$	M [Docs/Methodology/METHODOLOGY_v3.md:170-175]	—
$\gamma_0, \pi_0, \bar{Z}_0$	start weight $\varpi \prod_b r_b(S \cap C_b)$; start law; its exact normaliser	N-tab [NBH/notation_and_conventions.md:54]	—
$\tilde{S}_0$	start mode $\arg\max \gamma_0$ (size, then lex ties)	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:150]	—
d; $\gamma_\varphi, q_\varphi$ (q_t)	discrepancy $\log \mathrm{BF} - \sum_b \log r_b$; bridge $\gamma_0 e^{\varphi d}$; normalised $\gamma_\varphi/\bar{Z}_\varphi$	N-tab [NBH/notation_and_conventions.md:55-56]	[C] d duplicates count, cell count; canonical q_φ, drop π_φ
$\varphi, \varphi_t, T, T_e, T_0$	bridge index; grid points; realised stage count; ceiling at tier e; $\max(1000, \ell^2)$	N-tab [NBH/notation_and_conventions.md:84-85] [NBH/notation_and_conventions.md:107]	[C] $T^\sigma/T_{\mathrm{bd}}, T^*, T_n$
$J, \hat{J}, A_r, B_r$	Jeffreys $\mathrm{KL}(\pi\Vert\pi_0) + \mathrm{KL}(\pi_0\Vert\pi)$; run-card estimate; initial and endpoint d means	N-tab [NBH/notation_and_conventions.md:73-74] [NBH/notation_and_conventions.md:114]	Seat A's trapezoid form withdrawn [pos 1626] [NBH/notation_and_conventions.md:408]
d_σ, d_X	σ^2-coupling discrepancy; cross-block part $d - d_\sigma$	N-tab [NBH/notation_and_conventions.md:57-58]	[C] σ noise sd; baseline superscript "bd"
$a, \delta_A, \delta_H, r, C_\sigma, b_X, D_*$	H-DOM constants of M2	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:148]	[C] many; canonical r_* for C's r [NBH/notation_and_conventions.md:66]
$x_b, \bar{x}_S$	block fit fractions $n z_{S_b}^\top H_{S_b}^{-1} z_{S_b}/\bar{Q}, \sum_b x_b$	N-tab [NBH/notation_and_conventions.md:111]	—
$\hat{\varepsilon}_\times, \hat{\kappa}, \hat{\delta}, \hat{u}, \hat{r}, \hat{\eta}$	cross-block proxy and companions	N-tab [NBH/notation_and_conventions.md:69-72] [NBH/notation_and_conventions.md:117]	$\hat{\delta}$: $K \to K-1$; $\hat{u}$: max over V → table envelope [NBH/notation_and_conventions.md:406-407]

symbol	meaning	defined in	changed when
Λ_{len}	thermodynamic length $\int_0^1 \text{sd}_{q_\varphi}(d)\, d\varphi \le \sqrt{J}$	N-tab [NBH/notation_and_conventions.md:100]	renamed from conflated Λ [NBH/notation_and_conventions.md:409]
Sampler, calibration, islands			
r_n, $N(j)$	neighbour threshold $\sqrt{2\log p/n}$; neighbour set	M [Docs/Methodology/METHODOLOGY_v3.md:181-183]	scale wording w1 (not $\sqrt{\log n/n}$) [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:192]
m; o_j	round length $p + B + 2$; recorded Gibbs conditional	M [Docs/Methodology/METHODOLOGY_v3.md:187-192]	—
ℓ, L	local-search/rebuild cap $\lceil \log_2(n+p+1) \rceil$; burn-in rounds $\max(4, \lceil \log_2(n+1) \rceil)$	N-tab [NBH/notation_and_conventions.md:82-83]	[C] card items L1–L3; ℓ_0 log-odds levels
N_0, N_e, H_0, H_e	final/calibration populations at tier e	N-tab [NBH/notation_and_conventions.md:80-81] [NBH/notation_and_conventions.md:113]	2,500 floor = budget, not MC guarantee [NBH/notation_and_conventions.md:398]
e (tier)	$e \in \{0, 1\}$, factor 4	M [Docs/Methodology/METHODOLOGY_v3.md:73]	[C] Euler's e, items e1–e8
V, F_V, ρ_V, $\bar{\rho}$, c_ρ	candidate set; restricted log-normaliser; restricted ESS; threshold 0.9; $\sqrt{\log(10/9)}$	N-tab [NBH/notation_and_conventions.md:88-89] [NBH/notation_and_conventions.md:103]	ESS 0.90 class (c) with B's caveat [NBH/notation_and_conventions.md:403]
$\mathcal{C}$	frozen calibration record (grid, tier, N, L, budgets) before stream 10	N-tab [NBH/notation_and_conventions.md:109]	—
R, R', R'_n	default 4 islands; declared islands $4k$; growth rates	N-tab [NBH/notation_and_conventions.md:78-79]	R'_n rates adopted [pos 1626] [pos 1639] [NBH/notation_and_conventions.md:405]
c_4	centre of islands 1–4 (extra search under stacking)	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:142]	—
$\hat{Z}_r$, $\hat{Z}$	island evidence, "unbiased for Z given $\mathcal{C}$" (clause TH01.a, under its named implementation premises); pooled $(1/R')\sum_r \hat{Z}_r$	N-tab [NBH/notation_and_conventions.md:75]	[V] $\bar{\hat{Z}}$ dropped
W_r, M_2, R_{eff}	$\hat{Z}_r/Z$; $E[W_r^2 \mid \mathcal{C}]$; $(\sum W_r)^2 / \sum W_r^2$	N-tab [NBH/notation_and_conventions.md:76-77]; M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:158]	—
$S_{ri}^{(k)}$	particle i of island r at measured round k ($k = 0$ initial)	N-tab [NBH/notation_and_conventions.md:115]	—
Outputs and certificates			
$\widehat{\text{PIP}}_j$, $\text{MPM}_{\hat{Z}}$, MPM_{eq}	primary ($\hat{Z}$-weighted) and secondary PIPs; MPMs at $\frac{1}{2}$	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:158]	primary/secondary swapped in v3.1 [pos 1251] [NBH/notation_and_conventions.md:390]
$\hat{S}$; $\mathcal{S}$, $\mathcal{S}^+$	returned HPM; scored set, post-search scored set	N-tab [NBH/notation_and_conventions.md:110]	—
$\mathcal{T}$, $Z_{\mathcal{T}}$	deduplicated reported top store ($40p + 1$); $\sum_{\mathcal{T}} e^F$	N-tab [NBH/notation_and_conventions.md:90-91]	after LS0 $\mathcal{T}_{\text{new}} = \text{top}_c(\cdot)$ [NBH/notation_and_conventions.md:392]
$\eta(S)$, η_F	scorer's heuristic error expression (64ε …)	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:337]	[C] discrepancy bound η_S, $\hat{\eta}$; canonical e_F [NBH/notation_and_conventions.md:67]

symbol	meaning	defined in	changed when
g_{add}, g_{del}, g_{swap}, g_{sub}, ρ_{loc}	card gaps; local posterior share	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:234-240]	[C] g prior scale, group size [NBH/notation_and_conventions.md:104]
$\mathcal{H}$, h, $H_r(h)$, $\hat{\mu}(h)$	declared list; test function; measured-round island average; pooled estimate	N-tab [NBH/notation_and_conventions.md:87] [NBH/notation_and_conventions.md:119]	d added to default $\mathcal{H}$ [pos 1626]
$\mathcal{O}$, $\mathcal{U}$, D_n	output event; $\mathcal{O}^c$; $P(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+\})$ (overbar = potential run)	N-tab [NBH/notation_and_conventions.md:92]; TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:86]	[C] U, U_n no-output event in seats
U_m, U_Z	analytic bound $m(S) \le C_n(2b_0 + y^\top y/(n+1))^{-\nu}$; envelope	N-tab [NBH/notation_and_conventions.md:93] [NBH/notation_and_conventions.md:146]	"≈ 530 nats loose" withdrawn [pos 1512]
$\Phi(S)$	$\lvert S \triangle S_0 \rvert$ lattice distance	N-tab [NBH/notation_and_conventions.md:99]	—
Local search, lattice, LH variants (symbols used in M3, M5; definitions quoted from the cited source)			
S°	"S° appears in the start-mass bound and as the launch centre"; "S° is a reference model in $\pi_0(S^\circ) \ge (1-q)^{s+1}/(1+e)^s$ […] and the launch/start-mode centre in Lat(S°), Φ(S°) […]. The sources do not say whether the two coincide."	N-tab [NBH/notation_and_conventions.md:98]	[C] registrar: "S° only if the card states which object"; local-search centre written $\tilde{S}_0$
$M(S)$ (greedy map)	"best(S) := the $\prec$-least element of $\arg\max_{T \in N(S)} F(T)$"; "M(S) := best(S) if F(best(S)) > F(S), and M(S) := ⊥ otherwise"	W6-FIXREV2 [NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md:93-94]	—
TD′$_\infty$, TD′$_P$; TD′	"(L): M(S_0) = ⊥." "TD′$_\infty(\mathcal{R})$: (L), and for every $S \in \mathcal{R} \setminus \{S_0\}$: M(S) ≠ ⊥ and $M(S) \in \mathcal{R}$." "TD′$_P(\mathcal{R})$: (L), and for every $S \in \mathcal{R} \setminus \{S_0\}$: M(S) ≠ ⊥ and $M(S) \in \mathcal{P}(S)$", with "$\mathcal{P}(S) := \{T \in N(S) : \Phi(T) < \Phi(S)\}$". TD′ without subscript is defined only implicitly: the notation file lists the family "TD(d,m), TD′, TD′_P, TD′_∞" [NBH/notation_and_conventions.md:452]; first use in the final list: "(a) TD′ lattice descent (TheoremA)" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:299]	W6-FIXREV2 [NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md:97] [NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md:101-103]	literal TD(d,m) unsatisfiable (Prop 1-TD) [NBH/notation_and_conventions.md:452]
Lat(S)	"$R_{a,d} := \{U \cup P : U \subseteq S_0, \lvert S_0 \setminus U \rvert \le a, P \subseteq S_0^c, \lvert P \rvert \le d, \lvert U \cup P \rvert \le K\}$; Lat($S$) := $\{U' \cup P' : U \subseteq U' \subseteq S_0, P' \subseteq P\}$ for $S = U \cup P$"	W6-FIXREV2 [NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md:98-99]	—
C (centre); $\Phi(C)$, $\tau(C)$	C is defined only implicitly, as the centre the local search starts from; first use: "local search from C ∈ Lat(S°) with $\ell \ge \tau$(C) and a completed final scan reaches and certifies S_0 within Φ(C) moves" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:292]. Φ as in the row $\Phi(S)$. τ: "S_0 is reached after exactly τ(S) = m(S) + e(S) − σ(S) moves, where σ(S) is the number of extra→truth swaps used; max{m(S), e(S)} ≤ τ(S) ≤ Φ(S)", with "m(S) = $\lvert S_0 \setminus S \rvert$, e(S) = $\lvert S \setminus S_0 \rvert$"	W6-FIXREV2 [NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md:151-152] [NBH/branch_snapshots/W6-FIXREV2/STATEMENTS.md:97]	the W6 text writes the centre as S_c; R2_D: "$\ell \ge \tau(C) = \Phi(S^\circ) - 1 = c - 1$ fixed on LH" [NBH/branch_snapshots/MEET-THM/R2_D.md:194]

symbol	meaning	defined in	changed when
STAB-LH	"(STAB-LH), a finite population condition on the fixed LH parameters. $m_{\text{STAB}} := \min\{\Delta(V-t)-\Delta(V),\ \Delta(V-t+l)-\Delta(V) : S^\circ \subseteq V \subseteq \mathcal{SU},\ t \in S^\circ,\ l \in \mathcal{SU}\setminus V\} > 0$" ($\mathcal{SU}$ = the signal units; Δ = population residual variance). M5's (b2-LH) uses the instance "STAB-LH (g = 9, c = 2, ρ = 0.4, ψ = 0.2)" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:292]	02b [NBH/02b_theory_campaign_history.md:7852]	fails e.g. at $(\rho,\psi) = (0.85, 0.5)$ [NBH/02b_theory_campaign_history.md:7852]
L-DIL	Seat C's dependency-graph node name, defined only by its node-list entry "L-DIL: factorization" and the edge "S-TARGET → L-DIL; L-DIL → M3(a–d) | Start normalization and dilution." (used in the M3 charter's first lemma, "Match production $\gamma_0, \bar{Z}_0$ to L-DIL")	R3_C [NBH/branch_snapshots/MEET-THM/R3_C.md:250] [NBH/branch_snapshots/MEET-THM/R3_C.md:259]	—

A.3 Assumption, premise and label names

name	meaning (source words, abridged where marked)	defined in
D0	"$n \geq 3, p \geq 1$; nonconstant columns and response, centered and standardized" with the shipped target	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:64]
G (sampling model)	fixed raw design; $y^r = \alpha\mathbf{1} + X^r\beta + \epsilon, \epsilon \sim N(0,\sigma^2 I)$, fixed $\sigma^2 > 0$; standardise afterwards (DF23)	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:68]
R1, R	fixed-design triangular array, fixed $s_0 \geq 1$, $p \leq n^A$, $s_0 \leq K$ eventually; G; $\|\mu\|^2 \leq Mn$; truth Gram eigenvalue $\geq \phi_0^2$	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:448]
$m_n^\sharp$, j_0	tail cutoff $\min\{m \geq s_0 : R_m \leq n^{-(s_0+2)/2}\}$; $j_0 = \lceil (2A+1)s_0 \rceil + 1$	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:449-450]
$\text{SE}_\lambda(j_0)$, $\text{SE}_w(m_n^\sharp; g_0)$	anchored normalized / unnormalized minimum-eigenvalue conditions on $S_0 \cup J$	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:450]
BM-∞, BM(c)	beta-min $\theta_{\min}^{*2} n\phi_0^2/(\sigma^2 \log n) \to \infty$; or fixed $c > c_0(s_0, A)$	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:451]
LH	$p = 9\lfloor n^{1/3}/9 \rfloor$, groups of nine, within 0.4, two cross-group links 0.2, coefficients $(1, -0.2)$, $\sigma^2 = 1$; exact Gram $X^\top X = (n-1)\Sigma$	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:69-70]
LH-iid	the $p \gg n$ random-design LH family of W3-PXN (P1–P8); "(b1-LH-iid)" means different objects in FW10/K71 and in 02b	[NBH/02b_theory_campaign_history.md:4469] [NBH/notation_and_conventions.md:450]
LH-LAT	lattice margin condition on LH ($m_0 = 0.0325 > 0$ on the exhibited design)	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:304]
W	reporting exhibit: groups of nine, within $1/4$, cross $\varepsilon_n = 1/(np)$ (also $n^{-1/2}/p$), two coefficients $n^{-1/4}$; $D_* = O_P(1)$	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:71]
ORTH	$X^\top X = (n-1)I, p \leq n-1$	[NBH/notation_and_conventions.md:164]
SIZE	$g_{\max} \leq B < 2g_{\min}$; holds on LH for $n \geq 9^6$ with $g = 9$	[NBH/notation_and_conventions.md:434]
H-DOM	$a < 1, \delta_A < 1, \delta_H < 1, r < 1$ (matrix domain of M2; DF24)	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:148]
EX-A	"Exact mathematical scores, arithmetic, sums and comparisons. This is an idealisation, not a validated floating-point property."	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:72]

name	meaning (source words, abridged where marked)	defined in
MC contract; Output contract	computational premises of the final run; delivery transfer "by intersection with $\mathcal{O}$, never by conditioning on delivery"	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:73-74]
EX-S (H-U), CAL, IND-T, GUARD-W, SR, OP-LS0, e4, complete scans	see the verbatim box below (DF22)	TLF [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:86]
Lemma RO; CEX-guard	stop-only guard: $P(B \mid \mathcal{C}) \le \alpha \Rightarrow P(\mathcal{O} \cap B \mid \mathcal{C}) \le \alpha$, not $P(B \mid \mathcal{O}, \mathcal{C}) \le \alpha$; "GUARD-W: not audited"	[NBH/notation_and_conventions.md:433]
VAL-FP, SC	floating-point validation of $\hat{Z}$ at production size; validated score arithmetic	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:216] [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:402]
W-cal; Prop W	bitwise replay of concurrent calibration; bitwise equivalence of concurrent final islands	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:398] [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:131]
PART-pairing	partition theorem for the study generator $g = 5$ ($2g \le B$); obligation N8 (OB08)	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:180]
obligation Nk	N1–N17 = OB01–OB17; written "obligation N1"	[NBH/notation_and_conventions.md:435]
K-IDs	K01–K108 meeting candidate IDs, written "K-ID K46"	[NBH/notation_and_conventions.md:436]
classes (a)–(e)	modelling convention; function of (n, p) or X; declared budget; numerical tolerance; arbitrary	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:288-292]
ADDITIVE, POST HOC, LAW-PRESERVING, WORKHORSE	change classes of METH-AUDIT	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:16-20]
items e1–e8, w1–w7	item-9 fixes and wording fixes	M31 [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:175-198]
objectives (1)–(3)	accuracy in recovering the true posterior; parallel scalability; BVS selection accuracy (ground rule 6)	[NBH/notation_and_conventions.md:11-13]

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:86

Shared implementation premises (named once, used by both cards): **EX-A** exact scores and arithmetic; **EX-S** fresh independent ideal uniforms (H-U); **CAL** a successful frozen calibration record $\mathcal{C}$ on streams 0–9; **IND-T** final streams 10..9+R′ independent of $\mathcal{C}$ and of each other, R′ = 4k declared before stream 10; **GUARD-W** stop-only guard (operational interruption ⇒ no output); **SR** the method's stop/return rule as in the T38 hypotheses; **OP-LS0** LS0 integrated as specified (METHOD §2 #8); **e4** emitted LS0 centre = the prescribed smallest-size, lexicographically smallest start-mode maximiser (DP invariant N3 OPEN); **complete scans** no operational interruption of any search. $\mathcal{O}$ is the output event; statements are $P\{\mathcal{O} \cap \cdot\}$; $D_n := P(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+\})$ in potential-run notation (the overbar is the potential run, not set complement; W2PC1:122).

Statement (verbatim from source) NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:73-74

MC contract: Exact iid start draws, multinomial resampling, actual invariant move rounds, independent ideal uniforms; successful full calibration $\mathcal{C}$ frozen before final randomness; independent final islands; common N; $R' = 4k$ declared in advance. These are computational premises. **Output contract:** Potential uninterrupted final run first. GUARD-W identifies delivered statistical records with potential records on $\mathcal{O}$. Its integrated audit remains open. Bounds transfer by intersection

with $\mathcal{O}$, never by conditioning on delivery.

UNCERTAIN — the D0 data domain vs v3.1 e3

TLF's D0 says "$n \geq 3, p \geq 1$" [NBH/branch_snapshots/MEET-THM/THEOREM_LIST_FINAL.md:64]; v3.1 e3 says "**n = 2 is admitted**" as "a proposed extension: current preprocessing still rejects n < 3" [NBH/branch_snapshots/MEET-v31/METHOD_v3.1_FINAL.md:182]. The theory cards use $n \geq 3$; the code rejects $n < 3$.

A.4 Conventions that changed, and name collisions

The archive lists 35 changed conventions C1–C35 with old → new, source and position [NBH/notation_and_conventions.md:386-422]. Those that touch the method or its reporting: top-store MPM threshold $\frac{1}{3} \to \frac{1}{2}$ (C1); "$\hat{Z}$ can only fail downward" withdrawn (C2); primary PIP equal-weight → $\hat{Z}$-weighted (C3); "no useful upper bound on Z" narrowed to T31(c) (C4); stores nested/bitwise under LS0 struck (C5); "depth free of N, H, R" struck (C6); C6 trigger widened to the union (C7); local-search scan size (C8); IG negligibility (C9); N_0 cost $\Theta(p \cdot \max(n, p))$ (C10); 2,500 floor a budget (C11); Requirement (1) and the output convention (C12); input domain $n \geq 2$ proposed (C13); `global_certified` heuristic (C14); B^* a (c) budget (C15); ESS 0.90 class (c) (C16); growth-rate edits allowed (C17); R'_n rates (C18); $\hat{\delta}$ (C19); $\hat{u}$ (C20); $\hat{f}$ form (C21); Λ_{len} (C22); N9 gate passed (C30); binding-goal escalation (C31); item 1 Outcome 2 (C33).

Known collisions ("Always resolve through `ID_REGISTRY.md`"): "Lemma S" (shift lemma / ESS-share lemma / HPM order-independence lemma), "Lemma U", "Prop M", "Theorem SEP" vs "Lemma SEP", "BASIN", "Lemma R", "Theorem B", "Theorem A", "E1/E2" (W1-T03R lemmas vs API-MASKZPIP theorems, and also the calibration flags E1–E3), "IND", "C1", "T05" [NBH/notation_and_conventions.md:457-473].

B Registry of stable IDs

Generated from `registry/*.tsv` by `scripts/registry_to_tex.py`. The TSV files are the machine-readable source of truth for status; this appendix is a rendering of them.

B.1 Main cards and their clauses (68)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
TH01	M1 · Theorem V/H (validity of v3.1 and accuracy of reported bounded functionals) ; K46, K32, K86, K35, K36, K50, K51, K52, K48, K62	Validity of v3.1 (island identities, frozen-$\mathcal{C}$ contract, output event) and accuracy of reported bounded summaries (islandwise bound, finite-R′ bound, declared classes, decision transfer, growth-rate asymptotics)	T12, T31(i): PROVED-LOCKED; Prop2(a,b,d,e): PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: GAP (A-REV4)); N9K: PROVED (candidate) , code-checked; CHK_INHERITED: VERIFIED WITH AMENDMENT; N1: PROOF SKETCH; A1new: PROOF SKETCH (NEW-SYNTH); FMP-M1: PROVED (SF: none; XF: VERIFIED (A-REV9)); PC1c3: PROVED (PC1 claim 3), 'E[W_r ĥ_r] = π(h) is still a premise'; MA-PropP: PROVED (elementary), author-only; IND-T: PROVED (SF: none; XF: VERIFIED* (A-REV1)); LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK) ; XF: VERIFIED* (A-REV6)); CEX-guard: COUNTEREXAMPLE; N6(a): PROVED (potential run; output form CORRECT WITH AMENDMENT); T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)); (h), (i): PROVED (by this seat, unchecked).	any X (a)-(h), under the named implementation premises; (i) : T03♯ hyps (incl. p<= n^A) + osc d<=D_*=O_P(1) + stacking R′_n, asymptotic, jointly with output (exhibited on W)	TH12;TH15; TH10;TH02; TH09;OB01
TH02	M2 · Theorem WS, the §5.3 package; K102, K10, K11, K25, K53; K103/K104; K27	§5.3 package: (a) discrepancy between start and posterior controlled by cross-block terms plus σ^2-coupling; (b) bridge-form stability incl. stage-count sandwich and step-function statement; (c) one-shot reading is false (ONE-SHOT)	OWS-WS3, OWS-WS4: PROVED (SF: none; XF: none) + CHK "VERIFIED (PC1, cross-family, line by line)". OWS-WS6d, OWS-WS7: PROVED (SF: none; XF: none) + CHK "VERIFIED WITH AMENDMENT". OWS-WS1: PROVED, AUTHOR-ONLY. FD-W1: PROVED, XF VERIFIED (A-REV4). AREV9-OWSTAB: "VERIFIED under H-ORTH exactly as stated" (not used beyond H-ORTH). Seat C's D_* decomposition and actual-grid moment: PROOF SKETCH (NEW-COMP); W1PC2/ W2PC2 TRUE-AS-WRITTEN (statement sanity, not a proof). (b2) –(b4), (b6) assemblies: PROVED (by Seats A/B/D, unchecked), W2PC1 TRUE-AS-WRITTEN. C8 pilot extension: PROOF SKETCH, unchecked (W2PC2 PLAUSIBLE) ; T26BUD-G: PROVED, deterministic, AUTHOR-ONLY. (b7): PROVED (by this seat, unchecked) . T03: PROVED-LOCKED; T27(i): PROVED-LOCKED, CONDITIONAL. OWS-WS6e: CONJECTURE; OWS-WS8d: OPEN (neither used).	any X on H-DOM, method's own partition (fixed data); W exhibit; (c1) ORTH asymptotic; (c2) masked (LH)	TH03;TH07; TH01;TH12; TH09;TH06; TH11
TH03	M3 · Theorem START; K15, K17, K18, K10 (K16 supplementary)	Deterministic start mass, exact factorization, dilution, σ^2-coupling decomposition, and the masking law on LH	Status: deterministic: T22(i)(ii)(iv): **PROVED-LOCKED**. DIL-E: **PROVED; SF: none; XF: VERIFIED (A-REV9)**. DIL-E1: **PROVED; SF: none; XF: VERIFIED* (A-REV9)**. Status: statistical: T27(i): **PROVED-LOCKED, CONDITIONAL**. Thm3/4-EXP: **PROVED; SF: VERIFIED (W3-VEXP); XF: VERIFIED (A-REV2)**. Thm5-LH: **PROVED; SF: VERIFIED (W3-VEXP); XF: none; SAME-FAMILY-ONLY**. Status: coupling/ assembly: OWS-WS3: **PROVED; SF: none; XF: none; AUTHOR-ONLY** in the inherited ledger; CHK separately records **VERIFIED (PC1, cross-family, line by line)**. Final assembly: **PROOF SKETCH, unchecked**.	(a)-(d): any admissible X (D0, EX-A) ; (e): block orthogonality; (f): LH under G	TH10;TH11; TH09

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
TH04	M4 · Theorem Z$\mathcal{T}$ and the certificate card, with the masked-regime statement; K42, K43, K45, K44; K36, K78; K95 cited	Deterministic two-sided brackets from the reported store, retention at $\delta_1 < ½$, analytic envelope, output-event certificate, and the masked-regime statement (HQ Q3)	MZ-P3: ”PROVED (P3 top store, deterministic); P3′ conditional on T03 + discovery”; ”candidate: not reviewed, not adopted”; AUTHOR-ONLY. N13: PROVED (PC1 claim 5). PC1c5: (a) PROVED, ½ sharp. N17: PROVED (PC1 claim 1; PC2 C1(c)). PC1c2: PROVED for the potential run; N6(a) output form CORRECT WITH AMENDMENT; N6(b) PROVED modulo premise (i) (GUARD-W not audited). LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)). CEX-guard: COUNTEREXAMPLE (XF VERIFIED*). OCERT-C1: XF ERROR as quantified (O-VCERT R1 wording used). T38-b1-LH: PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8)). Thm5-LH: PROVED (SF: VERIFIED (W3-VEXP); XF: none), SAME-FAMILY-ONLY. T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)). (f) assembly: PROOF SKETCH (seat-assembled; W2PC1/W2PC2 TRUE-AS-WRITTEN as an assembly).	any X (a)-(e) ,(g); (f) LH (exact Gram) under M5 (b1-LH) hyps	TH05;TH09; TH15;TH10; TH14;TH01; TH11;OB03
TH05	M5 · Theorem HPM; K72, K69, K65, K68; K63 hyp.; K70 variant; K82 remark	Composition bound for the reported HPM at every n; Theorem A / LS-score; reported HPM with LS0 on LH (b1-LH); labelled variants (a-fav), (b2-LH)	Thm4-C1 ”PROVED (SF: VERIFIED* (W5-LOCKBOOK); XF: none)”; T38-(0) ”PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))”; N200 ”PROVED (elementary, every n); Π(X) upper bound OPEN” (author-only); TheoremA ”PROVED (SF: VERIFIED (W3-VT37R); XF: VERIFIED* (A-REV2))”; LS-score ”PROVED (SF: VERIFIED (W5-VMASK); XF: VERIFIED* (A-REV8))”; T38-b1-LH ”PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8))” (registry ⚑ CONFLICT on the iid sub-claim, excluded) ; Thm-b1 ”PROVED-UNDER (SF: VERIFIED (W5-VMASK); XF: VERIFIED (A-REV8))”; T38-(a) ”PROVED-UNDER (SF: VERIFIED* (inputs); XF: none)”; B2E-B2LH ”PROVED-UNDER (SF: none; XF: VERIFIED* (A-REV7))”; STAB-cert(A7) ”REFEREE (Astra-authored) ”; PART-ThmA ”PROVED (XF: VERIFIED* (A-REV9))”; N3 ”OPEN”; Thm5-LH (input via M3) ”PROVED (SF: VERIFIED (W3-VEXP); XF: none)” SAME-FAMILY-ONLY. Assembly: PROOF SKETCH (by this seat, unchecked). **Fidelity:** v3.1 with LS0 (METHOD #8) for (0), (a), (b1-LH); any frozen grid; pooling not involved (scored-set maximum); (b2-LH) is v3 without LS0, labelled; (a-fav) is an iid-draw variant, labelled.	any X, every n (0); TD′_ ∞ on Lat(S°) with ℓ >= τ(C) and a completed final scan (a); LH (exact Gram, p<=n-1) with LS0, T03♯ hyps on LH, SIZE, premises incl. e4 (N3 OPEN), joint with output, no delivery claim (b1-LH); STAB-LH without LS0 (b2-LH) ; (F) with iid π0 draws (a-fav)	TH09;TH03; TH11;TH14; TH15;OB03
TH06	M6 · Theorem SEP + Lemma KL; K19, K106, K105	Start-only discovery obstruction (LH threshold p^2n), singleton second-moment bound for one-shot evidence, and Lemma KL	SEP-A/SEP-B: **PROVED; SF: none; XF: VERIFIED (A-REV9)**. KL: **PROOF SKETCH (NEW-SYNTH) **, **AUTHOR-ONLY** in IDX. Statement checks remain TRUE-AS-WRITTEN; they do not upgrade KL. Final assembly: **PROOF SKETCH, unchecked**.	finite Ω (a) ,(c),KL; LH under G (D0, EX-A) for (b) and the LH consequences	TH03;TH09; TH10

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
TH07	M7 · Theorem STG; K26, K25, K27, K30, K33; K28 cited	Stage laws of the coded grid: monotonicity/ termination, restricted-ESS coverage identity, interval inequality and stage-count bound, ideal-grid crossing $\log_2 A_0 + O(1)$	STG-G, STG-L0, STG-L1: PROVED (SF: none; XF: none); T26BUD-G: PROVED, deterministic (AUTHOR-ONLY, W2PC2:155); STG-M: PROVED (SF: none; XF: VERI-FIED* (A-REV9)), explicit deci-mal constants GAP (W2PC2 C14) ; STG-L: PROVED (SF: none; XF: none); ConjP: OPEN; STG-cexW2: COUNTEREXAMPLE; OWS-WS7(d): PROVED (SF: none; XF: none) + CHK VERIFIED WITH AMEND-MENT; (b), (c): D/E proposals, TRUE-AS-WRITTEN (W1PC1: 192–193); (c′): PROOF SKETCH (Seat C, unchecked), W2PC2 TRUE-AS-WRITTEN; STG-F: PROVED (SF: none; XF: none), cited only (K29 FW).	any X, coded grid (a)-(c′) ; ideal two-state grid (d); ideal fine grid (e, cited)	TH10;TH03; TH01;TH13
TH08	M8 · Proposition D; K87, K88, K89, K06, K64; K90 remark	Island-level work, depth and width ac-counting (final run, calibration, searches) , polynomial work and Brent scheduling, ceilings; no speed claim	MA-PropD ”PROVED (counting, given the code’s structure)” AUTHOR-ONLY; T32(i) ”PROVED (SF: VERIFIED (W7-SEC); XF: none) ”, displayed rate GAP (A-REV5), reg-istry ⚑ CONFLICT marker carried; Cost-T38 ”PROVED-UNDER (SF: VERIFIED (W3-VT38E); XF: GAP (A-REV5))”; calibration share ”seat-derived, unchecked” (N15); assem-bly PROOF SKETCH (by this seat, unchecked). **Fidelity:** island level, v3.1 as shipped; ”depth free of N, H, R” struck (MEET-v31 deci-sion 6); K88’s particle-parallel crit-ical path is v3.2 and not claimed.	any admissi-ble input; (d) p<=n^A	TH07;TH14; TH15;TH01; TH05
TH01.a	M1(a) Validity [cited]	Each elementary move γ_φ-reversible; random-order round q_φ-invariant and self-adjoint; draw_pi0 and $\tilde{Z}_0$ exact; $E[\hat{Z}_r\|\mathcal{C}] = Z$; $E[\hat{Z}_r \hat{PIP}_j\|\mathcal{C}] = Z \cdot PIP_j$	T12, T31(i): PROVED-LOCKED; Prop2(a,b,d,e): PROVED (SF: VERI-FIED (W5-LOCKBOOK); XF: GAP (A-REV4)); N9K: PROVED (candidate) , code-checked; CHK_INHERITED: VERIFIED WITH AMENDMENT	any X	TH01;TH12; TH10
TH01.b	M1(b) Lemma N1	$E[\hat{Z}_r H_r(h)\|\mathcal{C}] = Z \cdot \pi(h)$ with H_r the plain measured-endpoint average (obligation OB01)	N1: PROOF SKETCH	any X	TH01
TH01.c	M1(c) Contract	(a),(b) hold condition-ally on any successful frozen $\mathcal{C}$; any R′ de-clared before stream 10 keeps them; adap-tive R′ or conditioning on completion need not	CEX-guard: COUNTEREXAMPLE; W1PC1:127 AMBIGUOUS → ”need not” (W2PC1:92 REPAIRED)	any X	TH01;TH15
TH01.d	M1(d) Output event	$P(\mathcal{O} \cap \{0.05\hat{Z} > Z\}) \le 0.05 \cdot P(SMC \cap \mathcal{A}) \le 0.05$, never conditional on $\mathcal{O}$	N6(a): PROVED (potential run; out-put form CORRECT WITH AMEND-MENT); LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)); GUARD-W audit open	any X	TH01;TH15
TH01.e	M1(e) Islandwise bound	Every regime and frozen grid: $E[W_r \cdot \lvert H_r(h) - h(S_0)\rvert \mid \mathcal{C}] \le 2\delta_1$, $\delta_1 = 1 - \pi(S_0)$ [given N1 (PROOF SKETCH) ; otherwise marginal PIPs only]	PC1c3: PROVED (PC1 claim 3), ’E[W_r ĥ_r] = π(h) is still a premise’ (card status row); verdict rows W1PC1:118,120; W2PC1:93 TRUE	any X	TH01;OB01

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TH01.f	M1(f) Finite-R′ bound	P(|μ̂(h) − π(h)| > ε | $\mathcal{C}$) ≤ min{1, 4(M₂−1)/R′ + 4M₂/(R′ε²)}; second tail bound; vacuous at R′ = 4 [given N1 (PROOF SKETCH); otherwise marginal PIPs only]	same as TH01.e (PC1c3: PROVED, N1 a premise); fixed data μ̂(h) → π(h) a.s. [cited]	any X	TH01;OB01
TH01.g	M1(g) Declared classes	Joint inclusions h_A on ≤ 2pB declared pairs (plain averages); interval probabilities h_I for a finite declared list, union bound; posterior-content error only, no quantile claim (HQ Q2)	verdicts W1PC1:263 (interval claim narrowed); same status block as TH01.e	any X	TH01;OB01
TH01.h	M1(h) Decision transfer	Exact PIPs ≥ δ from ½ and uniform error < δ ⇒ reported MPM = exact MPM	(h), (i): PROVED (by this seat, unchecked)	any X	TH01
TH01.i	M1(i) Asymptotic with R′_n	With R′_n = 4⌈log₂(n+p+1)⌉: μ̂(h) → h(S₀) in probability, jointly with output; liminf P($\mathcal{O}$ ∩ correct) = 1 − limsup P(U_n)	(h), (i): PROVED (by this seat, unchecked); T03♯: PROVED (SF: VERIFIED (W2-VT03R); XF: VERIFIED (A-REV2)); second rate: see U1	fixed design (T03♯ hyps) + osc d<=D_*=O_P(1) + stacking R′_n; exhibited on W	TH01;TH09;TH02.b5
TH02.a1	M2(a1) Decomposition	Under H-DOM, d = d_σ + d_X with 0 ≤ d_σ ≤ C_σ, |d_X| ≤ b_X; osc_Ω d ≤ D_*	Seat C's D_* decomposition: PROOF SKETCH (NEW-COMP); W1PC2/W2PC2 TRUE-AS-WRITTEN (statement sanity, not a proof)	any X on H-DOM	TH02;TH03
TH02.a2	M2(a2) Divergences	KL(π‖π₀), KL(π₀‖π), J ≤ D_*; TV(π,π₀) ≤ tanh(osc d/2); 1 + χ²(π‖π₀) ≤ e^{osc d}	as a1; W1PC1 §16 KL/TV rows TRUE-AS-WRITTEN	any X on H-DOM	TH02;TH02.a1
TH02.a3	M2(a3) Block-orthogonal case and per-model η-form	E = 0: d = d_σ ≥ 0; per-model |d_X(S)| ≤ η_S	OWS-WS3, OWS-WS4: PROVED (SF: none; XF: none) + CHK "VERIFIED (PC1, cross-family, line by line)"; FD-W1: PROVED, XF VERIFIED (A-REV4)	any X (E=0 or η-form domain)	TH02;TH03.e
TH02.b1	M2(b1) Per-stage χ²	χ²(q_{u+Δ}‖q_u) ≤ e^{ΔD_*} − 1	W2PC2 C2(b)·increments TRUE-AS-WRITTEN (statement sanity)	any X on H-DOM	TH02;TH02.a1
TH02.b2	M2(b2) Four-normaliser transfer	multiplicative e^{±4νb_X} bounds on the ratio of 1+χ² quantities (same data vs block-diagonal baseline q^σ); on nonempty W ⊆ Ω the same with e^{±4νη_W}, plus a TV bound	(b2)–(b4), (b6) assemblies: PROVED (by Seats A/B/D, unchecked), W2PC1 TRUE-AS-WRITTEN	same data, fixed partition; b_X from the H-DOM matrix domain; η_W per-model envelope on W	TH02;TH02.a1
TH02.b3	M2(b3) First stage	1 + N_e Var(Ẑ_stage1/…) = 1 + χ²(q_{φ₁}‖π₀) exactly; within e^{±4φ₁η} of block-orthogonal value	as b2	any X, fixed partition	TH02;TH02.b2
TH02.b4	M2(b4) Doeblin constants	M_φ ≤ e^{2φη}M^σ_φ; minorisation constant ≥ e^{−2φη}/M^σ_φ	as b2	any X, fixed partition	TH02;TH02.b2
TH02.b5	M2(b5) Pathwise island moment	W_r ∈ [e^{−osc d}, e^{osc d}], E[W_r|$\mathcal{C}$] = 1, M₂ ≤ e^{D_*}	Seat C's D_* decomposition and actual-grid moment: PROOF SKETCH (NEW-COMP); W1PC2/W2PC2 TRUE-AS-WRITTEN (statement sanity, not a proof)	any X on H-DOM, successful coded grid	TH02;TH01.a;TH02.a1

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TH02.b6	M2(b6) Stage count: the sandwich	T_bd(.9e^{−4η}) ≤ T ≤ T_bd(.9e^{4η}) on exact-frontier greedy ESS-.9 grids	(b2)–(b4), (b6) assemblies: PROVED (by Seats A/B/D, unchecked), W2PC1 TRUE-AS-WRITTEN	exact-frontier greedy ESS-.9 grids on a fixed common nonempty V with common pilot targets, same-data block-diagonal baseline d_σ on the fixed partition, and .9e^{4η} < 1; coded grid adds the 2^-60 bi-section and 1e-12 snap/validation qualifications	TH02;TH07
TH02.b6p	M2(b6′) Local constancy, integer bound, envelope (b6′)	T = T_bd(.9) for small ε off critical thresholds; T ≤ 2⌈D_*/c_ρ⌉ + 2; checked fallback T ≤ P + ⌈D_*/c_ρ⌉ + 1	C8 pilot extension: PROOF SKETCH, unchecked (W2PC2 PLAUSIBLE); T26BUD-G: PROVED, deterministic, AUTHOR-ONLY	coded grid under pilot premises	TH02;TH07.cp
TH02.b7	M2(b7) Q5 condition (1): continuity/ Lipschitz in ε	On E = εE$_1$: δ_A(ε) = c_Aε, δ_H(ε) = c_Hε; b_X, D_*, per-stage χ^2 bounds continuous, b_X Lipschitz with constant L(ε$_1$)	PROVED (by this seat, unchecked), elementary calculus on the checked formulas	fixed-partition family E=εE1	TH02;TH02.a1
TH02.b8	M2(b8) Q5 condition (2): step function statement	”T(ε) is a step function of ε. It can jump at critical thresholds: PC1 exhibits an actual design (n = 40, p = 2, B = 1) with one-sided baseline counts {2, 4} and T = 3 at ε = 10^{-5}.”	stated verbatim in the theorem (card row (b8)); required by HQ Q5 condition (2); no proof status needed	PC1 design n=40,p=2,B=1	TH02
TH02.c1	M2(c1) Zero-dependence obstruction (WS6(d) with CHK amendment)	Under G, ORTH, fixed S$_0$, null count m → ∞ …: D_α(π‖π$_0$) → ∞ in probability; χ^2(π‖π$_0$) → ∞; TV → 0 in bounded-block setting	OWS-WS6d: PROVED (SF: none; XF: none) + CHK ”VERIFIED WITH AMENDMENT”	ORTH, asymptotic	TH02;TH09
TH02.c2	M2(c2) Masked obstruction (WS7 with CHK amendments)	If π$_0$(S$_0$) = p^{−c}n^{−c/2}e^{O_P(1)} and π(S$_0$) → 1: ESS_∞ → 0, KL → ∞, TV → 1	OWS-WS7: PROVED (SF: none; XF: none) + CHK ”VERIFIED WITH AMENDMENT”; T27(i): PROVED-LOCKED, CONDITIONAL	masked (instantiated on LH), asymptotic	TH02;TH03.f; TH06.KL
TH03.a	M3(a) deterministic mass	If q < 1, π$_0$(S°) ≥ (1−q)^{s+1}/(1+e)^s; ratio formula for feasible S$_0$	T22(i)(ii)(iv): PROVED-LOCKED	any X (D0, EX-A)	TH03;TH10
TH03.b	M3(b) bridge mass	inf_{0≤t≤1} q_t(S$_0$) ≥ min{π$_0$(S$_0$), π(S$_0$)}	T22(iv): PROVED-LOCKED	any X (D0, EX-A)	TH03;TH10
TH03.c	M3(c) exact factorization	π$_0$(S) = A(|S|) ∏_b q_b(S_b)	Final assembly: PROOF SKETCH, unchecked	any X (production tables)	TH03;TH10; TH11
TH03.d	M3(d) dilution	−log π$_0$(S$_0$) = V + C − log A(|S$_0$|)	DIL-E: PROVED; SF: none; XF: VERIFIED (A-REV9). DIL-E1: PROVED; SF: none; XF: VERIFIED* (A-REV9)	any X (production tables)	TH03;TH03.c
TH03.e	M3(e) shared variance	Under block orthogonality d_X = 0 and 0 ≤ d_σ ≤ νΣ_{b<c}x_bx_c/(1−x)	OWS-WS3: PROVED; SF: none; XF: none; AUTHOR-ONLY in the inherited ledger; CHK separately records VERIFIED (PC1, cross-family, line by line)	block orthogonality	TH03

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TH03.f	M3(f) masked law	Under G on LH, log $\pi_0(S_0)$ + 2 log p + log n = O_P(1) whereas $\pi(S_0) \to 1$	T27(i): PROVED-LOCKED, CONDITIONAL. Thm3/4-EXP: PROVED; SF: VERIFIED (W3-VEXP); XF: VERIFIED (A-REV2). Thm5-LH: PROVED; SF: VERIFIED (W3-VEXP); XF: none; SAME-FAMILY-ONLY	LH under G	TH03;TH09; TH11
TH03.K16	K16 placement (supplementary conditional lemma)	Cite T22(iii), favourable rate regime, with G, R, within-block SE, EXP′, VIS, NL_η, NL_ avg(κ) retained	T22(iii), PROVED-LOCKED CONDITIONAL	favourable regime (G, R, within-block SE, EXP′, VIS, NL_η, NL_ avg(κ))	TH03
TH04.a	M4(a) Deterministic identities	$Z_{\mathcal{T}} \le Z$; $TV(\pi_{\mathcal{T}}, \pi) = 1 - Z_{\mathcal{T}}/Z$; $KL(\pi_{\mathcal{T}} \| \pi) = \log(Z/Z_{\mathcal{T}})$; $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$	MZ-P3: ”PROVED (P3 top store, deterministic); P3′ conditional on T03 + discovery”; ”candidate: not reviewed, not adopted”; AUTHOR-ONLY; W1PC1 §3 TRUE-AS-WRITTEN	any X (finite Ω)	TH04
TH04.b	M4(b) Brackets when the truth is stored	If $S_0 \in \mathcal{T}$: $1 - \delta_1 \le Z_{\mathcal{T}}/Z \le 1$; $TV \le \delta_1$; $\lvert PIP^{\mathcal{T}}_j - PIP_j \rvert \le \delta_1$; $\lvert \pi_{\mathcal{T}}(h) - \pi(h) \rvert \le 2\delta_1 \lVert h \rVert_\infty$	same as (a); W1PC1 §3 B.D3(b) COUNTEREXAMPLE repaired with $\lVert h \rVert_\infty$	any X (S0 in store)	TH04
TH04.c	M4(c) Retention (N13)	If $S_0 \in \mathcal{S}$ and $\delta_1 < ½$ then $S_0 \in \mathcal{T}$ and $MPM^{\mathcal{T}} = S_0$; ½ is sharp; unqualified retention is false	N13: PROVED (PC1 claim 5)	any X	TH04;OB13
TH04.d	M4(d) Analytic envelope (N17)	$m(S) \le U_m := C_n(2b_0 + y^T y/(n+1))^{-\nu}$; $Z \le Z_{\mathcal{T}} + U_m[1 - \Sigma_{\mathcal{T}}\, \varpi]$	N17: PROVED (PC1 claim 1; PC2 C1(c))	any X	TH04;TH10; OB17
TH04.e	M4(e) Output-event certificate	$P(\mathcal{O} \cap \{\max(Z_{\mathcal{T}}, 0.05\hat{Z}) > Z\}) \le 0.05 \cdot P(SMC \cap \mathcal{A}) \le 0.05$; card level ≤ 0.05; never conditional on completion	PC1c2: PROVED for the potential run; N6(a) output form CORRECT WITH AMENDMENT; N6(b) PROVED modulo premise (i) (GUARD-W not audited); LemmaRO: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV6)); CEX-guard: COUNTEREXAMPLE (XF VERIFIED*)	any X	TH04;TH15; TH01.a
TH04.f	M4(f) Masked-regime statement (HQ Q3)	On LH under M5’s hypotheses: $P(\mathcal{O} \cap \{TV(\pi_{\mathcal{T}}, \pi) > \varepsilon\}) \to 0$; $P(\mathcal{O} \cap \{MPM^{\mathcal{T}} \ne S_0\}) \to 0$; surely $\hat{Z} < Z_{\mathcal{T}} \Rightarrow \hat{Z} < Z$. Reading: vanishing P(output and error), not conditional success	(f) assembly: PROOF SKETCH (seat-assembled; W2PC1/W2PC2 TRUE-AS-WRITTEN as an assembly); T38-b1-LH: PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8)); Thm5-LH: PROVED (SF: VERIFIED (W3-VEXP) ; XF: none), SAME-FAMILY-ONLY	LH (exact Gram, n>= 9^6) under M5 (b1-LH) hyps	TH04;TH05. b1LH;TH09; TH04.c;TH11
TH04.g	M4(g) Definitional facts	For a proper store $KL(\pi \| \pi_{\mathcal{T}}) = \infty$; positive-mass state absent from records makes $KL(\pi \| \hat{\pi}) = \infty$; masked final-estimate TV/KL bound is FW2 (CJ02)	definitional	any X	TH04
TH05.0	M5(0) Composition at every n, any X	$P(\mathcal{O} \cap \{\hat{S}^+ \ne S_0\}) \le 2E_0\delta_1 + D_n$; if $S_0 \in \mathcal{S}^+$ and $1 - \pi(S_0) < ½$ then $\hat{S}^+ = S_0$ (see integration note 4 / U5)	T38-(0) ”PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))”; N200 ”PROVED (elementary, every n); Π(X) upper bound OPEN” (author-only)	any X, every n	TH05;TH15; TH09
TH05.a	M5(a) Theorem A / LS-score	Under TD'_∞ on Lat(S°), local search from $C \in Lat(S°)$ with $\ell \ge \tau(C)$ and a completed final scan reaches and certifies S_0 within $\Phi(C)$ moves; LS0 scores (not certifies) S_0 within $\max\{\Phi(S°) - 1, 0\}$ moves	TheoremA ”PROVED (SF: VERIFIED (W3-VT37R); XF: VERIFIED* (A-REV2))”; LS-score ”PROVED (SF: VERIFIED (W5-VMASK); XF: VERIFIED* (A-REV8))”	TD'_∞ on Lat(S°), ℓ >= τ(C), completed final scan (deterministic, EX-A)	TH05;TH14

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TH05.b1LH	M5(b1-LH) HPM with LS0 on LH	$P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$; $\liminf P(\mathcal{O} \cap \{\hat{S}^+ = S_0\}) = 1 - \limsup P(U_n)$	T38-b1-LH "PROVED (SF: VERIFIED* (W5-VMASK); XF: VERIFIED* (A-REV8))" (registry ⚑ CONFLICT on the iid sub-claim, excluded) ; Thm-b1 "PROVED-UNDER (SF: VERIFIED (W5-VMASK); XF: VERIFIED (A-REV8))"; N3 "OPEN"	LH (exact Gram, p<=n-1) with LS0; T03♯ hyps on LH; SIZE; premises incl. e4 (N3 OPEN); asymptotic, joint with output, no delivery claim	TH05;TH03; TH09;TH11; TH05.a;TH04.c;OB03
TH05.afav	M5(a-fav) labelled variant	Under (F) with iid π_0 draws, $D_n \to 0$	T38-(a) "PROVED-UNDER (SF: VERIFIED* (inputs); XF: none)"; labelled variant	(F) with iid π0 draws (labelled variant)	TH05;TH09
TH05.b2LH	M5(b2-LH) labelled variant (K70)	Without LS0 on STAB-LH ($g = 9$, $c = 2$, $\rho = 0.4$, $\psi = 0.2$), any frozen grid, same conclusion	B2E-B2LH "PROVED-UNDER (SF: none; XF: VERIFIED* (A-REV7))"; STAB-cert(A7) "REFEREE (Astra-authored)"	STAB-LH without LS0 (labelled variant)	TH05;TH03; TH09
TH06.a	M6(a) discovery	$P(S_0$ absent from all Q iid draws from $\pi_0) = (1-a)^Q \le e^{-Qa}$	SEP-A/SEP-B: PROVED; SF: none; XF: VERIFIED (A-REV9)	finite Ω, iid π0 draws	TH06
TH06.b	M6(b) LH threshold	$Q^*_n = p^2 n$: $Q_n/Q^*_n \to 0 \Rightarrow$ miss probability $\to 1$; $\to \infty \Rightarrow \to 0$	SEP-A/SEP-B: PROVED; SF: none; XF: VERIFIED (A-REV9)	LH under G (D0, EX-A), asymptotic	TH06;TH03.f
TH06.c	M6(c) singleton second moment	$1 + \chi^2(\pi\|\|\pi_0) \ge b^2/a$; evidence estimator variance identity	SEP: PROVED; XF VERIFIED (A-REV9); CHK §2(a),(b)	finite Ω; LH consequence	TH06;TH03.f
TH06.KL	Lemma KL	$KL(\pi\|\|\pi_0) \ge b \log(1/a) - \log 2$	KL: PROOF SKETCH (NEW-SYNTH) , AUTHOR-ONLY in IDX. Statement checks remain TRUE-AS-WRITTEN; they do not upgrade KL	finite Ω; LH consequence	TH06;TH03.f
TH07.a	M7(a) Monotonicity and construction	ρ_V monotone; bisection accepts only intervals with $\rho_V \ge 0.9$; build_grid terminates; $T \le T_e$ on success; greedy optimality among refinements of the retained pilot targets	STG-G, STG-L0, STG-L1: PROVED (SF: none; XF: none); W1PC1:190 TRUE; exact/minimal W1PC1:191 COUNTEREXAMPLE repaired	any X, coded grid	TH07;TH10
TH07.b	M7(b) Coverage identity	$\rho_{full}(u,v) = \rho_V(u,v) \cdot a_u\, a_w/a_v^2$, $w = 2v-u$; $a_u, a_w \ge 1-\eta \Rightarrow \rho_{full} \ge 0.9(1-\eta)^2$	(b), (c): D/E proposals, TRUE-AS-WRITTEN (W1PC1:192–193)	any X, any nonempty V	TH07;TH13
TH07.c	M7(c) Interval inequality and count	$T \le P + \lceil 2/h_V \rceil + 1 = P + \lceil D_V/c_\rho \rceil + 1$ with resolution $\le h_V/2$; uniform full-space grid existence	W1PC1:194 COUNTEREXAMPLE repaired to P + …; (b), (c) TRUE-AS-WRITTEN	any X, coded grid with resolution premise	TH07;TH07.b
TH07.cp	M7(c′) Pilot-count clause (cited from M2(b))	$P \le \lceil 2/h \rceil + 1$, hence $T \le 2\lceil D_*/c_\rho \rceil + 2$; $D_* = 0 \Rightarrow T = 1$	(c′): PROOF SKETCH (Seat C, unchecked), W2PC2 TRUE-AS-WRITTEN	coded grid; global range bound D_* and 2^-60 <= h/2; successful-grid (no-ceiling/ snap) qualifications of M7(c)	TH07;TH02.a1

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TH07.d	M7(d) Ideal grid (K30), labelled, not the coded grid	Dominance crossing of gap A_0 costs $\log_2 A_0$ + O(1) stages; restricted ESS alone forces only $\log_2(9.5\ell_0)$ stages	STG-M: PROVED (SF: none; XF: VERIFIED* (A-REV9)), explicit decimal constants GAP (W2PC2 C14); OWS-WS7(d): PROVED (SF: none; XF: none) + CHK VERIFIED WITH AMENDMENT	ideal grid (labelled, not coded) : exact-frontier two-state ESS-0. 9 grid, initial log-odds $-A_0$, terminal log-odds bounded away from 0 on the positive side	TH07;TH03
TH07.e	M7(e) [cited]	$\sqrt{c}\cdot T^*(c) \to \Lambda_len := \int_0^1$ sd_{q_φ}(d)dφ ≤ √J as c → 0	KNOWN—CITE Syed et al. 2024 Thm 3, Gelman–Meng 1998	ideal fine grid, c->0	TH07;TH13
TH07.f	M7(f) as-coded masked T_n	Conjecture P, OPEN → FUTURE_WORK (K31) ; shipped guarantee T ≤ T_e; favourable O(1) (K29) → FUTURE_ WORK	ConjP: OPEN; STG-cexW2: COUN-TEREXAMPLE; STG-F: PROVED (SF: none; XF: none), cited only	n/a	TH07
TH08.a	M8(a) Final run	Work = R′·N_e·(T + L + 4)·m slots; island depth = N_e·(T + L + 4)·m in ⌈R′/w⌉ waves; width R′	MA-PropD ”PROVED (counting, given the code’s structure)” AUTHOR-ONLY	any admissi-ble input	TH08;TH07
TH08.b	M8(b) Calibration	Work = Σ_i N_i(T_i + L_ i + 4)m over executed trajectories; per tier ≤ (1 + 4I_e)H_e trajecto-ries in work	”calibration share seat-derived, unchecked” (N15)	any admissi-ble input	TH08;TH07
TH08.c	M8(c) Searches	Each local search scores ≤ (ℓ + 1)[p + k*(p − k*)] models	T32(i) ”PROVED (SF: VERIFIED (W7-SEC); XF: none)”, displayed rate GAP (A-REV5)	any admissi-ble input	TH08;TH14
TH08.d	M8(d) Polynomial work and scheduling	For p ≤ n^A, under the budgets above and any permitted polynomial R′, total work polynomial in n; on w workers time ≤ W/w + D (Brent)	Cost-T38 ”PROVED-UNDER (SF: VERIFIED (W3-VT38E); XF: GAP (A-REV5))”	p<=n^A; shipped budgets; any permitted polynomial R′	TH08;TH08.a; TH08.b;TH08. c
TH08.e	M8(e) Ceilings	Θ(p) slots per round and $\Theta(p^2)$ Gram mem-ory are not removed by islands or particles	assembly PROOF SKETCH (by this seat, unchecked)	any admissi-ble input	TH08
TH21	S-STGI · K29/K30 (DISSOLVED)	Former supplement card; K30 is M7(d) , K29 → FUTURE_ WORK FW23 (CJ23); conditional implication citable as attributed lemma	DISSOLVED (MINUTES §2.2, cor-rected)	n/a	
TH22	Thesis CD (K96)	Scoped remark after the main theorems; ”there is no retained non-corollary the-sis sentence” (Seat C wording)	remark, not a theorem	n/a	TH01;TH02; TH03;TH04; TH05;TH06; TH07;TH08

B.2 Supplement cards (12)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
TH09	S-T · Theorem 1, target consistency T03♯ (K93, K04)	Posterior concentration $1 - \pi(S_0) = O_P(\sqrt{(\log n/n)})$ under R1, G, R, SE_λ, SE_w, BM; exact HPM and MPM consistent; large-signal inclusion without beta-min	T03♯: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2) . T02(i♯), T02(ii♯): PROVED; SF/ XF VERIFIED. T01–T03: PROVED-LOCKED (wave 3; external review queued), SAME-FAMILY-ONLY	fixed design (T03♯ hyps: R1 fixed s0, p<=n^A; G; R; SE_λ; SE_ w; BM)	
TH10	S-TARGET · Propriety, invariance, envelopes, exact branch, odds certificate (K01–K05, K76, K95)	Shipped target proper (incl. rank-deficient); Q_S ≥ Y/(n+1); analytic envelope; affine invariance and permutation equivariance of the target; exact branch; odds certificate $A \le \eta < 1 \Rightarrow HPM = MPM = S_0$	N17: PROVED. T01: PROVED-LOCKED (wave 3; external review queued); SAME-FAMILY-ONLY. Remaining assembly: PROOF SKETCH, unchecked. N12 eventual dispatch: seat-proved, unchecked; T26-exact-branch-paren: XF ERROR (corrected)	any X, fixed data, n>=3	TH09
TH11	S-BLOCKS · Separation ⇒ groups kept; SIZE ⇒ equality (K07, K14, K06)	Complete linkage keeps strictly separated groups; with B < 2g_ min output blocks equal groups; $\lceil p/B \rceil \le J \le 2\lceil p/B \rceil$; Σ_b 2^{|C_ b|} ≤ p2^B/B; capped graph ≤ pB edges, no isolated vertex for p > 1	PART-ThmA: PROVED; SF none; XF VERIFIED*. Counting/graph assembly: PROOF SKETCH, unchecked. PART-pairing remains OPEN, excluded. PART-CE2: COUNTEREXAMPLE	strict separation and every group of size <= B (preservation) ; + B < 2 g_ min (SIZE) for equality	
TH12	S-KERNEL · N9 kernel facts, graph (p ≥ 2), start sampler, minorisation (K20, K21, K08, K09, K22, K103)	Move reversibility under N9's sufficient conditions, exact start sampler, minorisation Q_φ(S,·) ≥ π_φ/M_φ, tie-aware start-mode DP (K09)	N9K: PROVED (candidate), N9-KERNEL §§0–3 VERIFIED WITH AMENDMENT (PC1, cross-family) ; T09: PROVED-LOCKED (wave 1); N3: OPEN; K22, K23, K103: PROVED (by this seat, unchecked), PC1 §19 TRUE-AS-WRITTEN. C10 (K09): PROOF SKETCH, unchecked	any X	OB09
TH13	S-IDENT · Restricted-ESS coverage identity, TI/KL/TV calculus, C15-CI clause (K25 lemma, K104, K55, K103)	Terminal-law TV ≤ q^L, KL bounds, empirical TV/content for predeclared cells, and the named posterior-content credible-set clause C15-CI	C15-TV, C15-KL, C15-CI: PROOF SKETCH, unchecked (assembly); Independence-sampler foundation: KNOWN—CITE; S9: PROVED (by this seat, unchecked); OWS-WS1: PROVED (SF: none; XF: none)	fixed data; W for convergence	TH12;TH01; TH02;TH07; OB01
TH14	S-CERT · Local search, certificate semantics, B&B soundness, perturbation lemma (K64, K73, K74, K05, K108)	'local_certified' = weak local maximum (ties allowed); ≤ (ℓ+1) scans; card R exact on completion; perturbation lemma $\tilde{Z}/Z \in e^{\pm\varepsilon}$, TV ≤ tanh ε	N7 L1: CORRECT WITH AMENDMENT; OCERT-LemmaM: PROVED (SF: VERIFIED (O-VCERT); XF: VERIFIED (A-REV9)); OCERT-C4: PROVED (XF: VERIFIED*)	exact arithmetic	
TH15	S-OUT · Output-event validity, worker equivalence, stacking (K77, K78, K83–K86, K63)	Under stop-only GUARD-W a potential-run event with $P(B\|\mathcal{C}) \le \alpha$ gives $P(\mathcal{O} \cap B\|\mathcal{C}) \le \alpha$; not $P(B\|\mathcal{O},\mathcal{C}) \le \alpha$	LemmaRO: PROVED; SF VERIFIED; XF VERIFIED*. CEX-guard: COUNTEREXAMPLE; XF VERIFIED*. N6 potential-run: PROVED; output form: CORRECT WITH AMENDMENT. GUARD-W: not audited	potential run; GUARD-W premise	OB02;OB14
TH16	S-Z · Evidence variance, masked tails (K37–K41, K53)	$E[\hat{Z}_r\|\mathcal{C}] = Z$; exact variance identity; masked lower tail in exception-probability form; $Var(\hat{Z}_r/Z\|\mathcal{C}) \le (T+1)e^{2D}/N$ and $Var(W_r\|\mathcal{C}) \le e^{2\cdot osc d} - 1$ (both ship)	OPW2-ThmA: PROVED (SF: none; XF: VERIFIED (A-REV9)); MZ-E1E2: PROVED (E1, E2), candidate: not reviewed, not adopted; FMP-Na: PROVED (SF: none; XF: GAP (A-REV9)); PC1c4: PROVED; T26-ThmA: PROVED (SF: VERIFIED (W5-LOCKBOOK); XF: VERIFIED* (A-REV4)); T26 overall CONJECTURE	fixed data (given frozen C); masked tail in exception form	TH02;TH03; TH04;TH10
TH17	S-POOL · Pooling rules, recording-marginal equal-weight bias comparison P1, R_eff (K49, K50, K54, K56, K57-P1)	Pooling rules; P1: bias of the equal-weight PIP bounded by TV of the uniformly sampled one-record marginal; R_eff	MA-PropP: PROVED (elementary), author-only; Prop2(c): PROVED (XF VERIFIED); MZ-P1P2: P1 PROVED; MA-CorV: PROOF SKETCH; ADRAFT-ThmA Astra-only, cite Del Moral; OBG-LH-lock-general: XF ERROR	fixed data (given frozen C)	TH01;TH03; OB14

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
TH18	S-LIM · Limitations as theorems (K79, K81, K100, K101) ; K99 qualitative remark	No transcript diagnostic certifies an unqueried mode; certified weak local max can be far below the mode; ESS = N compatible with genealogical collapse; duplicates PIP = E[#selected in group]/d (numeric limit 2/3 only for p = 2 with two duplicates and fixed noise; 3/5 with an orthogonal null); MPM may exceed K	CEX-localmax(A2), CEX2, CEX-ESSneutral(A3), PropD-T03, SHARE-PB: COUNTEREXAMPLE (constructions); INP I5 PROVED	any X (constructions)	
TH19	S-REP · Run-card reporting corollary: $\hat{J}$ (B/C/D form) and $\hat{\varepsilon}_{\times}$ with companions (K104/K11)	$\hat{J}$ estimator with finite-R′ tail bound and zero-range convention; $\hat{\varepsilon}_{\times}$ = cross_block_rmax with $\hat{\kappa}$, $\hat{\delta}$, $\hat{u}$, $\hat{r}$, $\hat{\eta}$ and regularized proxy bounds; labels	Reporting assembly: PROOF SKETCH, unchecked. N1: PROOF SKETCH. PC1c3: PROVED, with N1 as premise. W2P2 C16 and repaired W2P1 T4/T5 are statement/route checks, not a full proof upgrade	fixed data; W for consistency	TH01;TH02; TH03;TH12; TH15;OB01
TH20	S-LEAK · B's leakage clauses (F), (S) as a supplementary corollary to M2	(S) exact identity $1 + \chi^2(\pi\|\pi_0) = \pi(S_0)^2/(\pi_0(S_0)q_2(S_0))$ and $m \geq \pi_0(S_0)e^{-2\cdot osc}$ with no design-uniform lower bound; (F) favourable start under leakage $1 - \pi_0(S_0) = O_P(2^B\sqrt{\log n}\cdot n^{-\kappa})$	OWS-WS6(b): PROVED (SF: none; XF: none), PC1 B.D6(S) TRUE; the m lower bound: PROVED (by this seat, unchecked); (b6) OWS-WS53F: PROVED-UNDER (SF: none; XF: none); PC1 B.D6(F) TRUE / DESIGN-VERIFIED (D_B)	(S) any eps, every grid; (F) G,R,SE-blk,EBM,CBL	TH02;TH03; TH09

B.3 Proof obligations (17)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
OB01	N1 · Measured-round island identity for general bounded h (Theorem H crux; V1(a2))	$E[\hat{Z}_r \cdot H_r(h)\|\mathcal{C}] = Z\cdot\pi(h)$ for $h \in \mathcal{H}$, $0 \leq h \leq 1$, measurable in $(X, y, \mathcal{C})$	PROOF SKETCH; PC1 claim 3 takes it as a premise. Binding §13.1: if it fails the goal is not met by withdrawal; HQ escalates to Felipe	n/a	
OB02	N2 · Integrated v3. 1 core: Prop W, W-cal, LS0, stacking, GUARD-W	Code-audit facts plus executed replays (a) final islands bitwise equal to serial, (b) diagnostic islands bitwise, (c) LS0 leaves islands unchanged, $\mathcal{T}_{new}$ = top_c(…), (d) R′ = 4k with extra search nests the scored set, (e) GUARD-W	Lemma S PROVED. (a) PROVED by code reading, executed on smoke + 3/3 designs. (b) PROVED by code reading, not executed. (c) not in the production core. (d) PROVED (PC1 claim 6, deterministic). (e) not audited	n/a	
OB03	N3 · LS0-based T38(b1-LH) about the shipped method; e4 tie-DP invariant	$P(\mathcal{O} \cap \{\hat{S}^+ \neq S_0\}) \to 0$ on LH for v3.1 with LS0; sub-obligation: tie-aware DP returns the prescribed smallest-size, lexicographically smallest maximiser	(b1-LH) PROVED (candidate), XF VERIFIED (A-REV8) for "v3 + LS0"; bridge to code = N2(c), not done. DP invariant OPEN	n/a	OB02;OB11
OB04	N4 · Finite-R accuracy of the $\hat{Z}$-weighted PIP and $\hat{\mu}(h)$; the constant M under masking (V11)	(i) $P(\|\hat{\mu} - \mu\| > \varepsilon) \leq \min\{1, 4(M-1)/R + 4M/(R\varepsilon^2)\}$; (ii) regime-free distance to $1\{j \in S_0\}$; (iii) upper bound on M through a masked crossing	(i) PROVED, given N1 for general h or the locked identity for PIPs; (ii) PROVED (candidate); (iii) Obligation: CONJECTURE, OPEN for $c \geq 2$	n/a	OB01

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
OB05	N5 · Calibration contract	(a) validity for any frozen $\mathcal{C}$; (b) OPEN: rebuild convergence, P(S_0 ∈ V), P(schedule_limit), P(grid_unconverged), as-coded masked T_n (Conjecture P)	(a) assembly of T12/T31(i) (PROVED-LOCKED, conditional on $\mathcal{C}$), IND-T (PROVED, XF VERIFIED*, A-REV1 only), Lemma RO; (b) OPEN	n/a	OB14
OB06	N6 · Output-event coverage statements	(a) P($\mathcal{O}$ ∩ {max(Z_$\mathcal{T}$, 0.05$\hat{Z}$) > Z}) ≤ 0.05·P(SMC ∩ $\mathcal{A}$) ≤ 0.05; (b) card level; (c) joint with $\mathcal{O}$, never conditional	(a) PROVED (potential run; output form CORRECT WITH AMENDMENT). (b) PROVED modulo premise (i), whose GUARD-W part is the code-audit obligation N2(e)	n/a	OB02;OB16
OB07	N7 · Certificate card G1/G2/L/R (PC2 verdicts)	Per-item semantics of the card	G1 PROVED; G2 PROVED; L1 CORRECT WITH AMENDMENT; L2 PROVED; L3 amended form PROVED; R PROVED with amendments; Card bound (C6) PROVED modulo premise (i); C6′ PROVED. Open sub-obligation SC(η_F)/VAL-FP (N16)	n/a	OB16
OB08	N8 · PART-pairing for the study generator g = 5 (2g ≤ B)	Partition law for complete linkage + size cut + next-fit packing with equal groups 2g ≤ B	OPEN; Prop Q PROOF SKETCH; W7-PART CE2 COUNTEREXAMPLE	n/a	
OB09	N9 · V5: kernel reversibility, self-adjointness, exact start sampler (closure gate)	Each elementary move γ_φ-reversible; uniformly random order self-adjoint; start sampler exact with exact $\tilde{Z}_0$	”by construction, consultant-checked, no registry row”; gate closed: N9-KERNEL ”PROVED (candidate)” with exact numerical verification on production headers ([pos 1550]: 7 sub-claims; [pos 1593]: 8 sub-claims); cross-family VERIFIED WITH AMENDMENT (CHK_INHERITED)	n/a	
OB10	N10 · V10: L versus t_mix	Do L = max(4, ⌈$\log_2$(n+1)⌉) terminal φ = 1 rounds give populations close to π?	NONE / OPEN	n/a	
OB11	N11 · ℓ versus the moves needed	ℓ = ⌈$\log_2$(n+p+1)⌉ ≥ max{Φ(S°) − 1, 0}; caps not binding w.p. → 1	on LH with c ≤ s_0 fixed, ”eventually true” (elementary, unchecked); growing s_0: OPEN	n/a	
OB12	N12 · General (n, p) scope, per regime	Per-regime table of what theory may claim	see the table in the source; p ≤ e^{Cn^α}: OPEN, not impossible	n/a	
OB13	N13 · Retention lemma	If S_0 ∈ $\mathcal{S}$ and δ_1 < ½ then S_0 ∈ $\mathcal{T}$ = top_{40p+1}($\mathcal{S}$), HPM and MPM^$\mathcal{T}$ certificates equivalent	PROVED (PC1 claim 5; R2C D7)	n/a	OB02
OB14	N14 · Stacking statements	(a) post-search nesting HPM(R′) ≥ HPM(4); (b) every bound holds per run for any R′ declared before final randomness; (c) IND-T for streams 10..9+R′; (d) one card per record	(a) PROVED (PC1 claim 6); (c) code audit, not done	n/a	
OB15	N15 · Work and depth at the implemented level	Island-parallel depth/work account (see TH08)	Prop D PROVED (counting); calibration share seat-derived (R2A, R2B), not proof-checked	n/a	
OB16	N16 · Floating-point contract	Validated SC(η_F) for scores, Φ, M, Z_$\mathcal{T}^-$, and VAL-FP(η, ε) for $\hat{Z}$ at production size	OPEN. VAL-FP MEASURED at p ≤ 40 only	n/a	
OB17	N17 · Analytic bounds on Z and π($\hat{S}$)	Z ≤ C_n($2b_0 + y^T y/(n+1)$)^{−ν}	PROVED (PC1 claim 1; PC2 C1(c))	n/a	

B.4 Future work, drops and inherent limitations (42)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CJ01	FW1	Frequentist coverage of the exact posterior's credible intervals	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ02	FW2	TV/KL error bound for the final pooled estimate in the masked regime	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB10
CJ03	FW3	K13: start-mode consistency under orthogonal blocks and a σ^2-inflated beta-min margin	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ04	FW4	K24: polynomial mixing of the $\varphi = 1$ round (Yang–Wainwright–Jordan transfer); also useful general production mixing and masked population accuracy (N10)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB10
CJ05	FW5	K23: Dirichlet/spectral comparison of the random-order round with the random-scan mixture	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ06	FW6	K31: as-coded masked stage count (realised T_n; Conjecture P OPEN); N5(b): rebuild convergence within ℓ rebuilds, candidate-pair coverage / P($S_0 \in$ V), P(schedule_limit) and calibration-failure probabilities	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB05
CJ07	FW7	K58: masked consistency of the equal-weight PIP/MPM	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ08	FW8	K57, general fixed-budget equal-weight accuracy (only the recording-marginal bias comparison P1 ships in S-POOL)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ09	FW9	K59: R vs N at fixed total particles (allocation)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ10	FW10	K71: (b1-LH-iid) on LH with iid final draws, also under general DELIV	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ11	FW11	K75: NP-completeness of the global-certificate decision problem for the fixed prior and penalty	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ12	FW12	K92: separation from single-chain flip/swap MCMC (moat construction)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ13	FW13	K82: delivery, P(U) $\to$ 0 (also delivery/ resource-failure laws, P(resource_limit))	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ14	FW14	K94: finite-n bound on the posterior-miss probability $\Pi(X)$	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ15	FW15	K99 quantitative: k-way barrier for cross-block masked contrasts	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CJ16	FW16	K107: validated arithmetic / rounding-error bound; production validated numerics (N16)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB16
CJ17	FW17	K80: positive power of the E1/E2/E3 and low_empirical_ESS flags	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ18	FW18	N12 / exponential p: exponential-dimension contraction and selection (Song–Liang Cor 1)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB12
CJ19	FW19	Growing support and cap non-binding (N11/N12)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB11;OB12
CJ20	FW20	Hellinger / predictive contraction (Song–Liang Thm 1)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ21	FW21	K97: Song–Liang analytic separation (null-exclusion/FDR)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ22	FW22	Sharp ε-scale necessity statement for §5.3	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ23	FW23	K29: favourable O(1) stage law, unconditional application	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	TH21
CJ24	FW24	Useful two-sided masked evidence precision (N4(iii)/V11)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB04
CJ25	FW25	Random/adaptive partition quality and study packing (N8; K71 as C pairs them), including the study generator (PART-pairing, g = 5, 2g ≤ B)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	OB08
CJ26	FW26	Finite-n Gibbs discovery (HPM residual)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
CJ27	FW27	Unbounded functionals: integrability extensions of M1 (§13.1)	FUTURE_WORK (deferred; reason in 02a §A5.1)	n/a	
DR01	K12	Dropped candidate K12	DROPPED. Reason: A test oracle, not a guarantee	n/a	
DR02	K34	Dropped candidate K34	DROPPED. Reason: TI evidence is not in the method (the ∫ half ships as reporting, S-REP)	n/a	
DR03	K66	Dropped candidate K66	DROPPED. Reason: PARTLY-FALSE (C3-PGD, CEX-localmax(A2))	n/a	
DR04	K67	Dropped candidate K67	DROPPED. Reason: Relies on K66	n/a	
DR05	K91	Dropped candidate K91	DROPPED. Reason: No accuracy-matched comparator; OPW2-cex	n/a	
DR06	K98	Dropped candidate K98	DROPPED. Reason: KNOWN-FALSE (SHARE-M(b), FD-NM)	n/a	
LIM01	Inherent limitation 1: Per-round sequential factor m = p + B + 2	True by definition of a random-scan full-coordinate Gibbs round; inherent to the chosen kernel, not to SMC or BVS (I1)	true by definition	n/a	
LIM02	Inherent limitation 2: Every worker needs scoring access to all predictors	X or sufficient statistics, 8np bytes; a private $8p^2$ Gram per worker is not inherent (C7) (I2)	true by definition	n/a	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
LIM03	Inherent limitation 3: Masking of the product-of-blocks start built from block-local scores against the full y, under T27(i)'s hypotheses (H2, H3; verified on LH)	Law for that construction and design class only (I3)	T27(i), PROVED-LOCKED (conditional, in its design class)	n/a	
LIM04	Inherent limitation 4: No diagnostic from the observed record has a universal detection guarantee for never-visited modes	Positive one-sided certificates (e.g. $\hat{Z} < Z_{\mathcal{T}}$) can still detect some failures (I5)	V12 and E1c PROVED; Chatterjee–Diaconis Thm 2.1	n/a	
LIM05	Inherent limitation 5: The properties of $\hat{Z}$ behind the 95% bound imply no two-sided relative-error bound	Non-implication from the estimator's law, not data-specific; analytic bound Z ≤ C_n(2b$_0$ + $y^T y/(n+1))^{-\nu}$ exists (I7)	T31(c), PROVED-LOCKED; PC2 C1(c), PROVED relative to it	n/a	
LIM06	Inherent limitation 6: (Moved by HQ amendment 01) Markov sharpness of $0.05\hat{Z}$	No longer listed as inherent: seat-derived, unchecked (I8)	NOT inherent (moved); seat-derived, unchecked	n/a	
LIM07	Inherent limitation 7: Posterior misses at n = 200	565 strict competitor defeats of S_0 among the 2,394 completed fits (six stops): at least 565 target-mode misses (I6)	true by definition of the fixed target (N200, T07)	n/a	
LIM08	Inherent limitation 8: The cap K excludes truths with $s_0 > K$; a fixed allocation cannot support unrestricted growth of (n, p)	(I9(a),(e))	true by definition (V13)	n/a	
LIM09	Inherent limitation 9: The restricted ESS on V cannot bound mass outside V without further assumptions	(I9(d))	true by definition	n/a	

B.5 Definitions, rulings, method decisions, dissents, cited lemmas (73)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
MD01	MEET-v31 decision 1	Block (local-to-global) start kept; backfit start rejected for paper 1	decided (MEET-v31 MINUTES §2) ; closed in: R1 (all three); R2 rule; R3-6 procedure; result §3	n/a	
MD02	MEET-v31 decision 2	HPM is the only decision rule; MPMs descriptive; no HPM-vs-MPM flag	decided (MEET-v31 MINUTES §2); closed in: R1 (all three)	n/a	
MD03	MEET-v31 decision 3	LS0 in the shipped core after the final local search; own provenance and ls0_ improved flag; does not fire E3	decided (MEET-v31 MINUTES §2); closed in: R2 D3c	n/a	
MD04	MEET-v31 decision 4	Parallel execution layer over islands, scans and tables; process workers; GUARD-W; NA fields	decided (MEET-v31 MINUTES §2); closed in: R1–R2 (D3e, D3f)	n/a	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
MD05	MEET-v31 decision 5	Optional stacking R-prime = 4k at N_e declared before stream 10; extra search from c4; baseline and expanded records	decided (MEET-v31 MINUTES §2); closed in: R2 D3a, D3b	n/a	
MD06	MEET-v31 decision 6	Depth and cost stated at island level; C6/C7 declared v3.2 path; C6 union trigger does not fire	decided (MEET-v31 MINUTES §2) ; closed in: R2 D3d, D4; C6 trigger: HQ (AMENDMENT_HQ_01)	n/a	
MD07	MEET-v31 decision 7	Calibration unchanged with a frozen-C contract; shortening (C9) rejected	decided (MEET-v31 MINUTES §2); closed in: R1–R2 (all three)	n/a	
MD08	MEET-v31 decision 8	Zhat-weighted PIP primary, equal-weight secondary, both MPMs and R_eff reported (POST HOC)	decided (MEET-v31 MINUTES §2); closed in: R2 D5 (all three)	n/a	
MD09	MEET-v31 decision 9	Flags E1-E3, ESS and grid flags kept as calibration triggers and labelled heuristics; no PSIS/Rhat; card never feeds back	decided (MEET-v31 MINUTES §2); closed in: R1 (all three)	n/a	
MD10	MEET-v31 decision 10	POST HOC certificate card (PC2 minimal card G1, G2, L1-L3, optional R) with one declared budget and numerical label	decided (MEET-v31 MINUTES §2); closed in: R2 D2; R3-1	n/a	
MD11	MEET-v31 decision 11	General-h estimator H_r(h) with Zhat-weighted and equal-weight pools; default H; plain averages for joint h	decided (MEET-v31 MINUTES §2); closed in: R2 D7; R3-3	n/a	
MD12	MEET-v31 decision 12	Output convention: estimates on O are functions of (X, y, seed, binary); guarantees as P{O, .}	decided (MEET-v31 MINUTES §2); closed in: R2 item 9 e8	n/a	
MD13	MEET-v31 decision 13	Item 9 generality in (n, p): no class (e) setting; fixes e1-e8 and wordings w1-w7 adopted	decided (MEET-v31 MINUTES §2); closed in: R2–R3-4	n/a	
DF01	Model, likelihood, g-prior, σ^2 prior	Model: Gaussian likelihood (no intercept, centred y), g-prior g = n with unit ridge, sigma^2 ~ IG(0.01, 0.01)	unchanged in v3.1 (”Felipe’s model”)	any X	
DF02	Size prior BB(1,p) and cap K	Capped BB(1,p) size prior w(k) = 1/(2 C(2p-1,k)), normaliser W_K, cap K = min(p, floor(n/2))	unchanged	any X	
DF03	Score m(S), evidence Z, posterior π	Marginal likelihood m(S), score F(S) = log varpi(S) + log m(S), posterior pi = varpi m / Z, evidence Z	unchanged	any X	DF01;DF02
DF04	Preprocessing / standardisation	Preprocessing: input checks, exact-constant column drop, stable standardisation (max-abs, centre, SD with n-1), duplicates kept and reported	rule 1 changed by e3 (n ≥ 2 proposed)	any X	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
DF05	Blocks and tables	Blocks (complete linkage on 1-|r|, size-only cut at B, next-fit packing) and exact block tables r_b(U) = m(U)/m(empty)	unchanged; table stage parallel (#4)	any X	DF03
DF06	Block-product start π_0	Block-product start pi_0 = gamma_0 / Zbar_0 with exact convolution normaliser and exact iid sampler draw_pi0	kept (item 1 Outcome 2: no depth/speed claim)	any X	DF02;DF05
DF07	Discrepancy d and tempering bridge γ_φ	Discrepancy d = log BF - sum_b log r_b and tempering bridge gamma_phi = gamma_0 exp(phi d), phi in [0,1]	unchanged	any X	DF06
DF08	Kernel moves and round	Move round of m = p + B + 2 labelled steps (p Gibbs, 1 uniform swap, B neighbour swaps, 1 independence proposal) in a fresh uniformly random order	unchanged; validity N9 (OB09, TH12)	any X	DF07
DF09	Resampling, pilot, ESS 0.90	Multinomial resampling every stage; adaptive pilot with ESS 0.90 H_e, 60 halvings, floor 1/T_e	unchanged; ESS class (c)	any X	DF07
DF10	Candidate set V, restricted ESS, grid, rebuild loop	Candidate set V, restricted log-normaliser F_V, restricted ESS rho_V, candidate grid construction, rebuild loop of at most ell iterations	unchanged; contract #18	any X	DF09
DF11	Calibration record $\mathcal{C}$, tiers, final runs	Calibration record C (grid, tier, budgets, R-prime, inherited leaders) frozen before stream 10; tiers e in {0, 1}; final streams 10..9+ R-prime	contract #18; streams extended #7	any X	DF10
DF12	Islands and stacking	Final islands: R = 4 default; optional stacking R-prime = 4k at unchanged N_e declared before stream 10; extra search from c4; baseline and expanded records	new in v3.1 (#7)	any X	DF11
DF13	Evidence estimator $\hat{Z}$_r, pooled $\hat{Z}$, Markov bound	Island evidence estimate Zhat_r (unbiased given C under the named implementation premises, TH01.a), pooled Zhat = mean of Zhat_r, Markov lower bound 0.05 Zhat (card: max(Z_T, 0.05 Zhat))	unchanged; card adds max{Z_$\mathcal{T}$, 0.05$\hat{Z}$}	any X	DF06;DF07; DF11
DF14	PIP rules, MPM, R_eff	PIPs: Zhat-weighted pooled RB-PIP primary, equal-weight secondary; MPM_Z and MPM_eq at 1/2; R_eff	swap of primary/secondary (#11)	any X	DF13
DF15	HPM decision and local search	HPM = strict maximum of F over every scored model (ties: smaller |S|, then lexicographic); local search of at most ell moves plus one scan	unchanged; scans parallel (#3)	any X	DF03

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
DF16	LS0	LS0: one local search from the exact start mode after the final search; reads no uniforms; provenance ls0; flag ls0_improved; does not fire E3	new (#8–#10)	any X	DF06;DF15
DF17	Reported store $\mathcal{T}$, Z_$\mathcal{T}$ and the certificate card	Reported store T (top 40p+1 by F), Z_T = sum over T of exp(F) , and the POST HOC certificate card G1, G2, L1-L3, optional R	new (#13, #14; §2.1)	any X	DF15
DF18	General-h accumulator $\boldsymbol{\mathcal{H}}$	General-h accumulator: declared list H of bounded h, H_r(h) after each measured round, Zhat-weighted and equal-weight pools	new (#12)	any X	DF12;DF13
DF19	$\hat{\jmath}$ (J_hat)	Jhat: post hoc Jeffreys-divergence estimate from initial and measured-endpoint means of d	reporting amendment	any X	DF07;DF13
DF20	$\hat{\varepsilon}$_× (cross_block_rmax) with $\hat{\kappa}$, $\hat{\delta}$, $\hat{u}$, $\hat{r}$, $\hat{\eta}$	Cross-block correlation proxy epsilon-hat (cross_block_rmax) with companions kappa-hat, delta-hat, u-hat, r-hat, eta-hat	reporting amendment	any X	DF05
DF21	Output convention and event $\mathcal{O}$, GUARD-W	Output convention: estimates on the output event O are functions of (X, y, seed, binary) ; statements as P{O, . }; GUARD-W stop-only guard; NA fields	new wording (#6, #16)	any X	
DF22	Shared implementation premises	Shared implementation premises EX-A, EX-S/H-U, CAL, IND-T, GUARD-W, SR, OP-LS0, e4, complete scans, MC contract	premises (not model conditions)	n/a	
DF23	Design families G, LH, W, ORTH	Design families: sampling model G, LH (exact-Gram masked family), W (reporting exhibit), ORTH	families used for IC3	n/a	
DF24	H-DOM and constants a, δ_A, δ_H, r, C_σ, b_X, D_*	H-DOM matrix domain of M2 and constants a, delta_A, delta_H, r, C_sigma, b_X, D_star	definition	n/a	
DF25	Settings table (49 rows), classes (a)–(e)	Settings table of v3.1 (49 rows) with classes (a)-(e); no class (e) in the method law	METH31 §5 (ground rule 9)	any X	
RL01	HQ ruling Q1 (§5.3)	Bridge form ACCEPTED with three conditions: (a) discrepancy half, (b) bridge-form stability, (c) refutation of the one-shot form (OWS-WS6(d), WS7) after a cross-family check	ACCEPTED ("Q1 (§5.3): ACCEPTED, with three conditions"); Q1(c) met by CHK_INHERITED (VERIFIED WITH AMENDMENT)	n/a	
RL02	HQ ruling Q2 (§5.4 coverage)	Posterior-content / MC reading: "Coverage error" means |π(C($\hat{\pi}$)) − nominal| plus MC error; frequentist coverage → FUTURE_WORK (CJ01)	ACCEPTED	n/a	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
RL03	HQ ruling Q3 (§5.4 TV/KL under masking)	NOT binding; obligation with fallback; list must contain at least one masked-regime accuracy statement paper 1 proves (→ TH04(f)); masked final-estimate TV/KL → CJ02	NOT binding	n/a	
RL04	HQ ruling Q4 ($\hat{J}$ and leakage proxy)	REPORTING allowed: from existing quantities, no random numbers, no feedback, precisely specified	REPORTING, allowed	n/a	
RL05	HQ ruling Q5 (stage count continuous in ε)	Stage-count content of Q1(b) = sandwich + local constancy + continuous envelope; real-valued parts must be continuity/Lipschitz statements; theorem and paper must say T is a step function (T = 3 between {2, 4} at ε = 10…	ACCEPTED, with two conditions	n/a	
RL06	HQ approval amendment (growth rate (ii))	Package approved; $R'_n = 4\lceil (n+p)^6/4 \rceil$ kept only as a sufficient rate inside the theorem (M1 simultaneous control), not as a method configuration (≈ 10^{17} islands at study sizes)	adopted by HQ; Felipe may overrule	n/a	
DS01	Dissent K33 (Seat D)	D would place K33 (calibration deterministic termination/count form) in S-IDENT; A, B, C: M7; rebuild convergence N5(b) → FW either way	recorded dissent	n/a	
DS02	Dissent K23 (Seat C)	C would keep the finite squared-kernel Dirichlet comparison as an attributed lemma in S-KERNEL (no theorem number); A, B, D: FUTURE_WORK (CJ05)	recorded dissent	n/a	
DS03	Dissent K29 / S-STGI (Seat D)	D: "confirm K29 here"; A, B, C prevail: S-STGI dissolved	recorded dissent	n/a	
DS04	Dissent K30 (Seat C)	C: K30 in the supplement as S-STGI / C14; A, B, D: M7(d)	recorded dissent	n/a	
DS05	Dissent K70 (Seat C)	C: K70 as a supplementary attributed variant; A, B, D: labelled no-LS0 variant clause of M5	recorded dissent	n/a	
DS06	Dissent K57 (Seat D)	D: general K57 in the supplement; B, C (and A:39,108): only P1 in S-POOL, general accuracy → FW (CJ08)	recorded dissent	n/a	
CL01	Cited lemma/result (TLF Part III)	N9 kernel facts: γ_{φ}-reversibility of each elementary move, q_{φ}-invariance and self-adjointness of the uniformly random-order round, exact draw_pi0 and $\tilde{Z}_0$ (K08, K20, K21) , with the CHK amendments (VERIFIED WITH AMENDMENT)	cited (not a numbered claim of paper 1); used in: M1(a), S-KERNEL	n/a	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CL02	Cited lemma/result (TLF Part III)	K47 (pooling, cited)	cited (not a numbered claim of paper 1); used in: M1; S-POOL	n/a	
CL03	Cited lemma/result (TLF Part III)	SMC identities and fixed-data limits: Del Moral 2004; Del Moral–Doucet–Jasra 2006; Whiteley–Lee–Heine 2016; Dai et al. 2022	cited (not a numbered claim of paper 1); used in: M1(a), (f)	n/a	
CL04	Cited lemma/result (TLF Part III)	Markov lower bound: Grosse–Ghahramani–Adams 2015	cited (not a numbered claim of paper 1); used in: M1(d), M4(e)	n/a	
CL05	Cited lemma/result (TLF Part III)	Greedy-schedule optimality: Štefankovič–Vempala–Vigoda 2009	cited (not a numbered claim of paper 1); used in: M7(a), M2(b6)	n/a	
CL06	Cited lemma/result (TLF Part III)	K28: $\sqrt{c}\cdot T^*(c) \to \Lambda$_len (Syed et al. 2024 Thm 3; Gelman–Meng 1998)	cited (not a numbered claim of paper 1); used in: M7(e) (cited)	n/a	
CL07	Cited lemma/result (TLF Part III)	Chatterjee–Diaconis 2018 Thms 1.1–1.3 (exp(KL) scale; with log-weight-tail qualifications)	cited (not a numbered claim of paper 1); used in: M6, M2(c2)	n/a	
CL08	Cited lemma/result (TLF Part III)	Cérou–Del Moral–Guyader 2011 Thms 1.4–1.5 (variance theorems; novelty of S-Z (i) to the citation checker)	cited (not a numbered claim of paper 1); used in: M2, S-Z	n/a	
CL09	Cited lemma/result (TLF Part III)	Frazier et al. 2023, Assumption 6(iv) (generic χ^2 precedent; not re-fetched)	cited (not a numbered claim of paper 1); used in: M2	n/a	
CL10	Cited lemma/result (TLF Part III)	Papaspiliopoulos–Rossell 2017 §§3.1–3.2 (block computation precedent)	cited (not a numbered claim of paper 1); used in: M3	n/a	
CL11	Cited lemma/result (TLF Part III)	Dai–Heng–Jacob–Whiteley §4.2 (independent-system evidence weighting)	cited (not a numbered claim of paper 1); used in: S-REP	n/a	
CL12	Cited lemma/result (TLF Part III)	K60, K61: Barbieri–Berger 2004 HPM/MPM facts (cited lemmas)	cited (not a numbered claim of paper 1); used in: M5, S-TARGET	n/a	
CL13	Cited lemma/result (TLF Part III)	Rossell 2022 Prop 1 (the $2E_0\delta_1$ term)	cited (not a numbered claim of paper 1); used in: M5(0)	n/a	
CL14	Cited lemma/result (TLF Part III)	Yang–Wainwright–Jordan 2016; Zhou–Yang–Vats–Roberts–Rosenthal 2022 (context next to M5; "with their priors LH is covered by rapid mixing; ours is for the paper's own target and start")	cited (not a numbered claim of paper 1); used in: M5, M4	n/a	
CL15	Cited lemma/result (TLF Part III)	Hou–Wang–Atchadé arXiv:2407.20580 (never "first finite-n")	cited (not a numbered claim of paper 1); used in: M5, M4	n/a	
CL16	Cited lemma/result (TLF Part III)	Brent 1974 (W/w + D scheduling)	cited (not a numbered claim of paper 1); used in: M8(d)	n/a	
CL17	Cited lemma/result (TLF Part III)	T22(iii) (K16) PROVED-LOCKED CONDITIONAL, with G, R, within-block SE, EXP′, VIS, NL_η, NL_avg(κ) retained	cited (not a numbered claim of paper 1); used in: M3 (supplementary clause)	n/a	
CL18	Cited lemma/result (TLF Part III)	CHK-verified WS3, WS4 (M2(a), M3(e)) ; WS6(d), WS7 with amendments (M2(c))	cited (not a numbered claim of paper 1); used in: M2, M3	n/a	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CL19	Cited lemma/result (TLF Part III)	T03♯ (S-T) target consistency, T01–T03 PROVED-LOCKED (external review queued)	cited (not a numbered claim of paper 1); used in: M1(i), M3, M4(f), M5, M6	n/a	
CL20	Cited lemma/result (TLF Part III)	Lemma RO (event containment), GUARD-W (stop-only coupling; audit open)	cited (not a numbered claim of paper 1); used in: M1(d), M4(e), S-OUT	n/a	
CL21	Cited lemma/result (TLF Part III)	Furnival–Wilson 1974 (leaps-and-bounds superset bound, K05)	cited (not a numbered claim of paper 1); used in: S-CERT	n/a	
CL22	Cited lemma/result (TLF Part III)	K23 squared-kernel Dirichlet comparison, as an attributed lemma without theorem number (C's dissent; optional for the S-KERNEL HQ)	cited (not a numbered claim of paper 1); used in: S-KERNEL	n/a	
CL23	Cited lemma/result (TLF Part III)	LIT-PAR G4 (Song–Liang numerical comparison), cited as labelled numerical evidence; the analytic separation K97 is FUTURE_WORK	cited (not a numbered claim of paper 1); used in: Coverage Cor 2	n/a	

B.6 Campaign-history results, refutations and counterexamples (165)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CE111	Prop D	n>=3, p>=2 (e.g. p= n), X^r=[x,...,x] with p identical columns, beta*=(beta1,0,...,0) fixed beta1 != 0: G and R hold (s0=1) but on E_ n (P0(E_n)->1) Pi(/S/ >m/y) >= 2^(-(m+4)) exp{-m(m-1)/(4(p-m))} for 0<=m<=K-1, so any cu…	COUNTEREXAMPLE (reproducible) (W1-T03R); SF VERIFIED (W2-VT03R row 11; constants re-derived, 0.25157 >= 1/4); XF VERIFIED (A-REV2: G and R only, violates even size-two SE)		
CE144	CE1	At boundary state (S0 minus j) union P a look-alike exits with prob >= (1-o(1))/(B+d+4); MC exit 0.62-0.98 in S2-S6 at n=200, exact lower bound >=0.39	COUNTEREXAMPLE (numerical, reproducible; asymptotic PROOF SKETCH); SF CONFIRMED (W2-VT24G, wording GAP); XF VERIFIED* finite obstruction (A-REV1)		
CE145	CE1-narrow	Exact-Gram sequence, one truth, one proxy at correlation r, p_n=ceil(n^{1/3}): (L+4)eta_ n -> infinity for every fixed d	COUNTEREXAMPLE (PROVED from Lemma E and Lemma U(C5) , W6-FIXREV1); referee-authored in A-REV1; XF VERIFIED as stated candidate (A-REV5); not universal		
CE169	SHARE-PB	BVS design with lambda_min(R_{S0}) = 0.004, pi(S0)=0.94-0. 97, start mode {12} ⊂ S0 (pi0≈0.98): delivery to R_0 in 200/200 islands, LT fails (median log Zhat/Z = -67), S0 scored only through a rare gate in 82% (n= 4000) a…	COUNTEREXAMPLE (finite n, reproducible; W4-SHARE; no verifier assigned); defeats route (b2), LS0 and T39 alike		

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CE182	T12(iv.b) component instance	n=20,p=2,T=1,N=1,M=4,L'=L=5,j=1: E[P-bar_1]-PIP_1=-0.173974177850, pooled relative bias -0.131716, D=8.14676240634, PIP_1=0.990618012243	reproduced by W2-BOOK (50 digits), W5-LOCKBOOK (60 digits), A-REV4 (Owls)		
CE208	Prop M	p=2, X'X=(n-1)[[1,rho],[rho,1]], y=b x1+eps: local PIPs ->1, product/union put mass ->1 on {1,2}, full posterior ->1 on {1}, TV ->1; table rho=0.8 b=sigma*=1	COUNTEREXAMPLE, complete proof (author A-DRAFT = Astra; used by A-REV4 as valid repair; no campaign verification)		
CE213	SaM on an LH-like masked design	LH-like design (LIT-PAR construction, not the exact W1-EXP family): p = 16; groups {0..3}, {5,6,7}, {8..11}, {12..15} equicorrelated rho = 0.7; x4 a singleton linked only to hub x0 with corr psi = 0.5; truth theta_0 = b…	COUNTEREXAMPLE (numerical, reproducible) to "SaM recovers S0" on an LH-like masked design when the random split separates the linked pair [LIT-PAR §3.1 G4]; analytic part PROOF SKETCH, corrected by RF276/TH417; author-only; cited at MEET-THM as labelled numerical evidence		
CE214	isolation probability of a masked hub under a uniform equipartition (THM-SYNTH…	Exactly P(j_b isolated from g given columns) = C(P_n-1-g, s_n-1)/C(P_n-1, s_n-1); the union bound is vacuous at small K_n: at K_n = 2, P(isolated) = 0.53 for g = 1 (LIT-PAR design, P = 16) but 0.05 for g = 4, so at fixe…	numerical check (THM-SYNTH scripts/check_lemmas.py, seed 20261103); part of Theorem SL (TH417, PROOF SKETCH); shows LIT-PAR's 5/318 used g = 1		
CJ112	T32	Branch-specific arithmetic and storage counts (old label T23-C)	CONJECTURED (not proved)		
CJ117	T37	Finite-n truth dominance and deterministic greedy repair (new conjecture, added 2026-10-28)	CONJECTURED (not proved); NEW, added 2026-10-28 to fill the finite-n repair hole		
CJ120	H24	Hypothesis of T21 route (e): uniform one-round truth-scoring floor a_n and exit bound eta_n on the near-truth region R_d, with a_n >= c_n/2 and (L+4)eta_n -> 0	RETIRED (REGISTRY_v2 D): its asymptotic input (L+4)eta_n -> 0 is false in correlated designs (CE1, LemmaE); replaced by Theorem B / B-sharp (route e'). See RF102 for the refutation and CE row CE144 (CE1) (Theme 5).		
CJ129	TD	TD(d,m): for S in R_d not containing S0 some missing true variable has addition gain >= m + every other move; for S strictly containing S0 some extra has deletion gain >= m + every other move; at S0 every move has gain…	REFUTED as written: 'false for d >= 1 on any design (Prop1-TD, V1-TDfails); false at d = 0 without positivity (CEX1)'; replaced by Theorem A + POS and Theorem W. See RF101; CE: RF101 (CEX1) (Theme 4).		
CJ162	Backfit start	v3 tables scored on partial residuals y - X_{-b} beta_hat_{-b} (backfit): a favourable, usable start (3 sweeps 1888/2400, fixed point 2203/2400 favourable on real fits)	candidate, unproven (API-RESID: 'unproven candidate'; no theorem, no SMC evidence); rejected for paper 1, recorded as future work (MEET-v31 MINUTES item 1)		

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CJ164	ABL-CONF Claim 1	at equal accuracy the block start (A) costs less than singleton factors (B)	INCONCLUSIVE (pre-registered rule: S1 inconclusive, S4 holds_ nondeliv, S5 holds); XF none		
CJ165	ABL-CONF Claim 2	at equal declared budgets the prior start (C) underestimates the evidence and is less accurate than the block start (A)	LICENSED (pre-registered rule: holds on S1, S4, S5; median log Z error -13.4/-10.8/-5.7 nats); XF none		
CJ166	ABL-START Outcome 1 depth claim	block start (A) needs no more stages/ rebuilds than singleton (B) or prior (C) starts on >=9/12 designs with accuracy not worse	not licensed: Outcome 2 (keep, reframe; no depth or speed claim) under the pre-registered rule; v1 and corrected v2 script agree; arm C certified low on 10/12 designs		
CJ178	LS0-4	In-engine LS0 run with emit_centre= true and HPM tie rule untested; post hoc union equals integrated run unverified	OPEN implementation check (O-LS0)		
CJ204	B2E-open47	(b2) on LH when m_ STAB < 0 (e.g. rho=0. 85, psi=0.5; configuration B)	OPEN (W5-B2ENV; two plausible repairs; runs numerically unaffected)		
CJ210	ORTH-joint	Joint family for (a) with (P): X'X=(n-1)I, p<=n-1, fixed theta, so Theorem 1, Theorem 2(iii), Theorem 5(a) and Theorem 6 hold jointly	PROOF SKETCH (W2-T26 Prop. C); W4-T38FULL 'new composition checked line by line here' (a6 PROVED-UNDER the verified a3, no verifier); A-REV4 and A-REV6 GAP; registry ORTH-joint conflict		
CJ213	N2(c)	LS0 is executed with its centre emitted and scans completed on every counted fit (OP-LS0); production core has no LS0; an in-engine integrated test fit (merged-LS0 replay) is still to run	OPEN as an implementation premise (T38-RECORD II.7 obligations; MEET-v31/METHOD_v3. 1_FINAL.md line 48: 'not yet in the production core, and the integrated replay is pending'; final cards N2(c) pending); O-LS0 data: 0/2,394 HPM changes in the post hoc export		
CJ215	D2 (T38-RECORD Part IV)	Decision: headline of Theorem 5(b) is (b1-LH) with LS0 or a no-LS0 LH theorem ((b2-LH) on STAB-LH, (b1'-LH) for c=1)	open at v3 (condition met: A-REV7 verified a route; API-VSTAB later verified the STAB certificate [pos 978]); v3 states both and does not choose; final M5 card: (b1-LH) with LS0 as main clause, (b2-LH) as labelled no-LS0 variant (K70)		
CJ226	LS0-1	[data fact] LS0 post hoc on all 2,400 regenerated fits: best LS0 model never beats the production HPM (0/ 2,394); recovery unchanged (1,119/1,200 at c=1, 710/1,200 at c= 0.5); job B reproduced job A exactly	data (O-LS0-WRAP aggregation); Owls regeneration != NOTS bits (F(S0) agrees to 5e-8; every margin >= 0.002)		
CJ228	LS0-3	[data fact] LS0 median 0.17 s (p=500), 0.37 s (p=1000), about 5e4 and 1.1e5 scores, under 0.1% of a fit	data		
CJ230	Upper bound on Pi(X)	Finite-n upper bound on the posterior-miss probability Pi(X)=P0(S-dagger != S0 / X) at n= 200 (e.g. P0(L/X)+r_n(X))	OPEN (Envelope GAP); remainder Pi-P0(L) measured <= 1.1 points per cell, not bounded		

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CJ238	Bookkeeping: C5 screening event over 2,400 fits	Tabulate {S0 in U(X,y, S-hat)} over the 2,400 fits; match O-SWAP realised check with SMC outputs dataset by dataset; replay eta_F in long double/mpfr	OPEN bookkeeping		
CJ251	Section 5.3 Theorem 1	Draft target: divergence between product of block posteriors and posterior controlled by cross-block covariance, so importance-weight variance bounded	proposed by the draft; class (b)+(d); mechanism false for v3 tables (RF256); replaced by TH350, TH362, TH363; final card M2 Theorem WS		
CJ282	S-GAP	Among published BVS methods none simultaneously (a) targets the exact model posterior with an unbiased evidence estimate, (b) parallelises in communication-free width (islands), (c) has a proved sequential-depth (stage-…	CONJECTURE (about the literature; evidence from §2) [LIT-PAR REPORT §4.1]; headline verdict ”TRUE WITH QUALIFICATIONS”; confidence medium-high (”a literature claim can always be overturned by a missed paper; 2023-2026 coverage was by search, not exhaustive”); the literal sentence ”methods with guarantees are sequential” is FALSE (RF268); the planned Astra novelty cross-check is not found in the sources		
CJ290	LIT-MCMC L2	No published BVS family on which standard BVS MCMC provably mixes slowly while a tempering or SMC method provably succeeds; the only proved BVS slow-mixing example is YWJ App. A.2 (null truth, saturated-model trap), cur…	OPEN (as a literature claim); medium confidence [LIT-MCMC §2.1 L2]; search not exhaustive; consistent with the W7-SEP result ”moving off the start is needed, not tempering”		
CJ303	F-REFEREE M1 (JRSS-B referee simulation, major point)	All theory is asymptotic, thresholds at n of about 2e6, simulations at n = 200 [REFEREE_REPORT §1]; referee recommendation: ”major revision”; the one change that most raises the tier: make Theorem 5(b) a theorem about t…	referee objection; state at T38-RECORD cut: open (R2 LH-SCALE, R9 first-n table not started); S-ELS decomposition and O-LS0 data exist [registry H]		
CJ306	F-REFEREE M4 (JRSS-B referee simulation, major point)	The favourable regime is where the sampler is unnecessary; what does the bridge provably buy (separation; masked Zhat/PIP statements) [REFEREE_REPORT §1]; referee recommendation: ”major revision”; the one change that mo…	referee objection; state at T38-RECORD cut: in progress, with a twist: W7-SEP supports ”moving off the start is needed”, not ”tempering is needed” (E-SPINE); F-MASKPIP M1-M3 [registry H]		
CJ320	Lemma 0(b) and E_LS repair	Lean covers Lemma 0 (a),(c) only; Lemma 0(b) (informal ERROR per A-REV2) and its three-way-partition repair are not in Lean	OPEN		
LM144	Lemma 0(a),(c)	F strictly increases along the search path; certified stop at S0 gives S-hat=S0 up to ties without pi(S0)>1/2	PROVED; SF VERIFIED (W3-VT37R); XF VERIFIED* (A-REV2, A-REV5)		

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
LM174	LemmaS-ineq	Weights with ESS >= eH: any subset of fraction f has weighted share >= max{0, f - sqrt(f(1-f)(1/e - 1))}; attained only for 1-e < f < 1	inequality PROVED (W1-T38E) , SF VERIFIED (W3-VT38E), XF VERIFIED (A-REV1); 'attained by two-level weights' on the full domain ERROR (A-REV1); Lemma S' PROVED (W6-FIXREV1)		
LM189	LS-score	If D_S holds on Lat(C0) ∩ {Phi >= 2}, l >= Phi(C0)-1, and C0 != S0 or the centre is emitted, the production local search from C0 scores S0 (certification not needed); move count max{Phi(C0)-1,0}	PROVED; SF VERIFIED (W5-VMASK row 20); XF VERIFIED* (A-REV8: scoring VERIFIED; move count 'at most Phi(C0)-1' ERROR at C0=S0, repaired to max{Phi(C0)-1,0})		
LM204	Lemma M	As S grows Q_S de- creases and D_S in- creases (and w non- increasing on {0..K}); Q_S = y'y/(n+1) + n/(n+1) y'(I+(n+1)M_S)^- 1 y; D_S = det(I+(n+1) M_S)/det(I+M_S)	PROVED; SF VERIFIED (O-VCERT: 0 violations in 476,160 exhaus- tive pairs); XF VERIFIED in exact arithmetic (A-REV9); LOCKABLE_ v2		
LM212	Lemma W2 (O- WSTAB)	Under BO LR=nu[g(y)-g((1-kappa)y)] and kappa G<=LR<= kappa G/(1-y), kappa the realised dilution fraction	PROVED (author); SF verified (CHECK_SEP_W, F5 exposition); replaces F-DRAFT W2 (ERROR, A- REV4)		
RF101	literal T37	Literal T37(i): un- der TD(d, 0) with strict inequalities the production local search walks mono- tonically from any centre in R_d to S0 in exactly /S symmetric- difference S0/ moves; TD(d,m) itself is self- contradictory for...	REFUTED: TH198 (Prop1-TD) PROVED, SF VERIFIED (W3-VT37R) , XF VERIFIED (A-REV2); d = 0 counterexample RF101 (CEX1) VERIFIED (W3-VT37R, reproduced by A-REV2)		
RF102	H24	H24: uniform in R_d, K1(x,{S0}) >= a_n and one-round first-exit probability <= eta_n, with a_n >= c_n/2 and (L+4) eta_n -> 0; the vanishing-exit input is false in correlated designs	REFUTED as a general asymp- totic input (CE144 (CE1) COUN- TEREXAMPLE, SF VERIFIED* W2- VT24G, XF VERIFIED* A-REV1; broad asymptotic wording GAP, narrow explicit sequence VERIFIED A-REV5)		
RF103	T02 sharp size cutoff	Replacing T02(i)'s geometric tail 2^-m by the exact BB(1,p) tail would cut the cutoff m_n from about 200 to s0 + O(s0 log n / log p) under G and R; false without SE (duplicated columns)	REFUTED under G and R alone: CE111 (PropD-T03) COUNTEREX- AMPLE, SF VERIFIED (W2-VT03R row 11), XF VERIFIED (A-REV2); TH126 (E3-T03) barrier VERIFIED*		
RF104	tempering is needed	Claim that start-only discovery fails in the masked regime so the tempering bridge / sampler is needed; false on the paper's own masked family LH: one greedy search from the start mode suffices (Prop. GU)	REFUTED on LH by proof (TH388 (SEP-GU) Cor. GU-LH: COUNTEREX- AMPLE to the task's claim (a); numerically 386/386 replicates); SF: none (A-REV7 pending then), XF: TH388 (SEP-GU) VERIFIED* (A- REV7), GAP in A-REV9 review		

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RF105	beta-min about 100x	Pre-campaign disclosure figure: the explicit BM-C constant fails by about 100x in squared signal at n = 200; corrected to at least 43x once T03-sharp replaced M*	CORRECTED (number retired; not a theorem). Treat as a disclosure fact: beta-min fails by at least 43x (E-SPINE 2.3)		
RF123	W1-T03R disclosure: the proof deterministic thresholds start near n = 1.2e5	W1-T03R REPORT headline 4/section 3/ section 5: the proof's own deterministic thresholds start near n ~ 1.2*10^5 and other thresholds bind earlier. False: q_n <= 1/2 first holds at n about 2.1e6 (p=n^1.17) and 2.5e6 (p=n…	ERROR (SF: W2-VT03R verdict row 29); ERROR confirmed XF (A-REV2 section 3.5: q=5.5e22 and 7.9e28 at n=130,328); registry T03-1.2e5 [APPLIED]		
RF130	Claim that T03-sharp(iii) P(S_dagger != S0) <= 2E0[1-pi(S0)] is a new contribut…	LIT-STAT: this inequality is Rossell (2022, Bayesian Analysis) Proposition 1 verbatim (also for the MPM); T21 and Thm 5(0) inherit the term; it must be credited to Rossell.	Literature verdict KNOWN (LIT-STAT, single reader; Rossell read via arXiv:1806.04071); HQ acted on it [pos 1142]		
RF134	EXP-const	Displayed limit kappa_ EXP -> 0.52/c_* = 4. 16(M+2 sigma^2)/ sigma^2; actual limit is 0.51/c_* = 4.08(M+2 sigma^2)/sigma^2 (0. 52/c_*+o(1) valid only as a looser eventual bound); replaced by kappa_n decreasing to 0.51/c_E (…	ERROR (A-REV1, A-REV2 5.1); SF W3-VEXP VERIFIED (missed it); repair PROVED (W6-FIXREV1/2, A-REV5 VERIFIED); registry conflict C6		
RF135	Thm4-iff	'masking persists iff eta < 1/2.02, a sufficient condition' (self-contradictory) and VEXP's 'p^-c alone forces pi_0(S0)->0 whatever eta'; only sufficiency holds	ERROR (A-REV2 5.3; W3-VEXP R4 partial); registry conflict C7		
RF136	WeakSignal	W1's guess that the n= 200 start masking is weak-signal masking, CL(L) with bounded L on weak truths, with pi(S0) also small; replaced by dilution	ERROR (W3-VEXP row 27 'evidence against'; W3-PXN Prop W); retired in favour of dilution		
RF144	LH-LAT-sharp	Prop LH-LAT states m0 is sharp; true only for homogeneous pairs (108/108), not 32/94 heterogeneous settings (down to x0.002)	GAP (overstatement, W5-VMASK R5); conflict C26; A-REV8: valid not sharp		
RF150	Common plug-in sigma^2 is the right tool for the n = 200 dilution masking (O-MI…	O-MIX recommendation of (C) for the dilution mechanism	rejected (E-SPINE 2.2, W8-SYNTH 4.5); refuted by O-DILUTE 5, F-SIGMA Thm S6, API-RESID Step 0 on 2400 real fits (0/2400 favourable)		
RF151	W1-EXP weak-signal masking explains n = 200	CL(L) with bounded L for a few weak truths, c = number of weak truths, pi0 concentrates on S0 minus them; pi(S0) also small	ERROR / retired (W3-VEXP row 27, W3-PXN Prop W: 'evidence against'); replaced by dilution masking; XF none		

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RF160	Favourable from n ~ 800-1,600 (and other single thresholds)	single 'favourable from n ~' figure for the start	withdrawn: sources disagree (O-DILUTE 590-730; W3-VEXP 800-1,600; W5-FAVIID 1,050-3,200); omit or attribute (W8-SYNTH 4.13; E-SPINE 2.3; T38_STATEMENT D6 open)		
RF164	Headline gain capped at 6/2,400 as rationale against a start change	argument that no start change can improve study accuracy beyond 6 fits (API-RESID, R1_A)	rationale withdrawn (MEET-v31 MINUTES: a changed start does not nest scored sets); data statistic stands (565 posterior misses, 6 resource stops)		
RF165	literal T37	Literal truth dominance TD(d,m) with clauses (i) exact /S Delta S0/ moves, (ii) union bound, (iii) TD(d, 0) w.p.->1	REFUTED (Prop 1 for d>=1; CEX-1 for d=0); pre-campaign CONJECTURED (not proved)		
RF167	Lemma0(b)	On every posterior miss F(S0)<=F(S-hat) no local search delivers S0; so 0/541 repairs is forced by monotonicity	ERROR (SF: inference ERROR, remark GAP; XF ERROR A-REV2) ; registry CONFLICT C1; true on study data (S-ELS)		
RF168	R2-dichotomy	Every failed fit is E_LS^c or E_LS with terminal score>=F(S0)	ERROR (XF A-REV2: omits inferior local maximum and capped searches); SF none (authored by verifier)		
RF169	TDP-iff-DS	TD'_P(R) iff (L) and strict one-state events D_S hold for all S	ERROR with exact ties (XF A-REV2) ; SF VERIFIED (W3-VT37R) in conflict; registry CONFLICT C2		
RF176	Theorem W region form covers masked regime	Region-form Theorem W (a>=2) as general sufficient condition for masked LH	COUNTEREXAMPLE (W5-VPGD: eta=0, 20/20; D_S fails 200/200); registry CONFLICT C15		
RF182	H24	T24 terminal-repair inputs: K1(x,{S0})>=a_n and one-round exit<= eta_n uniformly on R_d with (L+4)eta_n->0	Retired; CE1 COUNTEREXAMPLE (numerical, SF CONFIRMED) ; CE1-narrow COUNTEREXAMPLE PROVED (XF VERIFIED A-REV5); H24 as written trivially satisfiable (a_n=0, eta_n=1) per W2-VT24G; register keeps 'not proved' [UNCERTAIN wording]		
RF183	c_n = 1/(s0+d+B+2)!	T24's entrance constant is 1/31! at s0=14 and useless	Retired as useless (FABLE_REVIEW; confirmed W1-T24G)		
RF188	E3(ii)	Start mass >= (1-q)^{s+1}/(1+e)^s used without the locked hypothesis q<1	ERROR (A-REV1; SF VERIFIED incl. (ii) by W3-VT38E) [CONFLICT C5]; repaired E3' (A-REV5 VERIFIED); F-REFINE Card 3(i) repeated it (A-REV5 ERROR)		
RF198	set analogue of T22(iv)	Bridge mass of a set is never below the smaller endpoint mass	REFUTED (C0); SF VERIFIED		
RF199	start mass + bridge mass + exit control imply delivery	Start mass on R_d, uniform bridge mass and exit control imply a near-truth particle at the terminal rounds	REFUTED (C1, reproducible; SF VERIFIED* after kernel repair)		
RF203	SHARE-M(b)	Under G,R,SE,BM-inf every T in R_d missing a truth has an addition with odds >= 1 for all phi >= phi_n = lambda_n/(lambda_n+g_n) -> 0 (no posterior barrier near the truth)	PROOF SKETCH (W4-SHARE); SF argument correct, upgraded on LH (W5-B2ENV); XF ERROR (A-REV3, retained A-REV6): quadratic expansion remainder of order n; vanishing crossing temperature false under BM-C; GAP no deletion/traversal bound		
RF208	OBG-LH-lock-general	Strict gaps C1-C3 (LH locking) hold for all LH parameters with /psi/ < rho	ERROR (A-REV7): C2 fails e.g. g=4, rho=0.85, psi=0.5; reproduced by Claude (API-VSTAB)		
RF212	Without LS0 no masked-regime consistency theorem	Without LS0 the paper has no masked-regime consistency theorem	retired (A-REV7 VERIFIED B2-LH under STAB-LH and D2-LH; API-VSTAB VERIFIED the certificate); unrestricted LH still not claimed		

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RF217	T38-b1-LH-iidfail	(b1-LH) in v1/v2: 'Q_ f pi0(S0) -> 0 when c >= 2 and p -> infinity, or c = 1 and p/(sqrt(n) log^2 n) -> infinity'	ERROR (A-REV5, counterexample p ~ log log n) / GAP (A-REV8 G1) ; recorded at the weaker verdict (ERROR); replaced by the growth condition n^{1-c/2} p^{-c} log^2 n -> 0 (c=2: p/log n -> infinity); LS0 consistency unaffected; FUTURE_ WORK FW10		
RF218	Without LS0 the paper has no masked-regime consistency theorem whose premises a…	W4-T38FULL II.9: 'So without LS0, the paper has no masked-regime consistency theorem whose premises are all proved on an explicit family.' (also v2 Not claimed, F-REFINE)	retired: no-LS0 candidates on LH now exist ((b1'-LH) A-REV7 VERIFIED qualitative; (b2-LH) A-REV7 VERIFIED conditional + Astra STAB certificate, API-VSTAB verified); replaced by 'candidates on LH, pending cross-family review' then by (b1'-LH), (b2-LH) in I.C		
RF219	T32(i) is a CONJECTURE (stale status in W6-FIXREV1 k1 and W7-SEP I11)	W6-FIXREV1 dependency k1: 'T32(i) ceiling: CONJECTURE'; W7-SEP input I11	retired as stale: T32(i) PROVED by W5-LOCKBOOK, re-counted by W7-SEC (SF); XF A-REV6 'supported by loop accounting', A-REV9 deferred (registry conflict C18)		
RF221	E[Zhat / completed] = Z	Identities in expectation (E[Zhat]=Z, E[Zhat PIPhat]=Z PIP) hold conditionally on output / completion	REFUTED (CEX-guard, exact stylised counterexample; W5-LOCKBOOK reproduced it and also under the engine-faithful cost rule; A-REV4 VERIFIED qualitatively) ; replaced by output-intersected statements P(O, .) via Lemma RO; also theme 8 (Prop 2)		
RF226	pooled Z-hat x pooled PIP unbiased	Claim that the pooled evidence times the pooled primary PIP is unbiased for Z PIP_j; exact bias is -(3/4)Cov(PIP_1j,Z_1)	REFUTED (Prop 2(c)); SF reproduced to 1e-58; XF VERIFIED (A-REV4)		
RF227	E[Z-hat/completed]= Z	Claim that conditioning on the output-available event preserves E[Z-hat]=Z	COUNTEREXAMPLE (CEX-guard, exact enumeration); SF reproduced (also engine-faithful costs); XF VERIFIED* (A-REV4, stylised)		
RF233	Z-hat can only fail downward in the masked regime	Plain-language claim that in the masked regime Z-hat errs only downward	PROVED false (PC1 claim 4, theorem-meeting phase); formal tail statement untouched; also owned by theme 12 (THM-SYNTH)		
RF249	Union bound over Hamming balls bounds Pi(X)	Distance-2 superset union bound ~0.07 at p=500, c=1, as large as the whole one-add rate	Failed route (API-N200); see CE195; CJ230 remains OPEN		
RF250	A-priori epsilon of VAL-FP certifiable at study budgets	Union bound over 1e8-1e11 final-run decisions: epsilon from 2e-3 to »1	Vacuous (W7-SEC); measure by replay		
RF256	Draft section 5.3 premise for v3 tables	Divergence controlled by cross-block covariance so importance-weight variance stays bounded	REFUTED for v3 tables: d=d_ sigma>=0 under exact BO; chi2 can diverge at zero cross-block correlation (WS6(d)); F-DRAFT: false for v3 tables		
RF262	task clause (a)	Claim in W7-SEP task: Q iid draws + production local search (and LS0 alone) score S0 w. p. -> 0 for every polynomial budget unless basin contains S~0	COUNTEREXAMPLE (by proof, Cor GU-LH; 386/386 replicates; SF: none; XF: GU VERIFIED* A-REV7); replaced by TH384, TH388, TH389		
RF264	tempering is needed	Spine claim: the tempering bridge/ sampler is needed in the masked regime (That is exactly why the sampler exists)	REFUTED on LH by W7-SEP Prop GU (candidate-based); replaced by 'moving off the start is needed' (E-SPINE PR #1; T38 v3 Remark 6; decision D4)		

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RF265	FD-NM	F-DRAFT Proposition NM (naive merge fails, rigorous): general proxy-split family with O_P(1) log BF remainder, start mode, naive union and O(1) ideal grid stages	ERROR (A-REV4; scope GAP); F-DRAFT status PROVED modulo T03/EXP'/Cor. P1; repairs: A-DRAFT Prop M (CE208), NM-R (TH393, pending Claude); not applied to F-DRAFT text		
RF266	K98	Proposal claim that naively merging local inclusion probabilities of correlated predictors recovers the posterior	KNOWN-FALSE (SHARE-M(b) XF ERROR A-REV3/A-REV6; FD-NM XF ERROR A-REV4; ADRAFT-PropM COUNTEREXAMPLE); DROP		
RF267	K97 draft clause cov(x_t,y)=0 on isolation	Draft clause that isolation makes the observed response exactly orthogonal to the hub (exact orthogonal null column, cov(x_t,y) =0)	COUNTEREXAMPLE (W1 PC1; conceded R2); replaced by population cancellation x_t^T mu = 0		
RF268	methods with guarantees are sequential	The sentence "methods that come with guarantees are sequential, and parallel-friendly methods with guarantees break under correlated designs" (HQ framing of Felipe's belief) and the wordings "no analogue speed-up in the…	REFUTED as a general statement (LIT-PAR §1 item 2: "FALSE in the work/depth sense. Do not print it. "); replaced by CJ282 S-GAP (three-part form); THM-SYNTH REPORT §3.3 "Do not print"; counterexample candidate CE212		
RF269	depth = T stages	HQ framing: "Sequential depth: the only sequential part is the stage count, and O-STAGES already bounds it" i.e. depth = T; in score evaluations a round is p+B+2 sequential elementary steps per particle, so depth is T L…	REFUTED / withdrawn (LIT-PAR §1 item 4 "Self-critique"; METH-AUDIT §7 "Depth = T stages without the round factor fails"); replaced by Prop D depth (I+2)(T+L+4)(p+B+2) at island level (TH399, M8); factor costs about 2-3 ms per round at p = 500 (METH-AUDIT §6)		
RF270	METH-AUDIT REPORT s1.2: "bitwise on the smoke problem and on 3/3 study designs…	METH-AUDIT REPORT §1 item 2 states that the bitwise reproduction holds "on the smoke problem and on 3/3 study designs (§6)" and C8 cites "measured in §6 (R = 32 x 312) "; the worker exited while waiting on runs: only smo…	RETRACTED as written (LEDGER 16:50 UTC: "its 3/3 headline is an overclaim"; THM-SYNTH task file: "treat METH-AUDIT's 3/3 as 1/1 + smoke"; MEET-v31 R1_B: "METH-AUDIT 3/3 is not supported by committed files"); empirically re-established by HQ re-runs: S4_r001 (pos 1175) and S5_r002 (pos 1241) bitwise PASS, so the executed check is smoke + 3/3 (concurrency log) ; UNCERTAIN residue: REGISTRY_ INDEX §5.B B1 and REGISTRY_ PAPER1 still list "1/1 vs 3/3" as a source CONFLICT		

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RF271	depth free of N, H and R for v3.1 (Prop D clause	Prop D / Theorem D(c) / Theorem CD(b) : "D does not depend on N, H or R" for the critical path with one worker per particle per island; false for v3.1 as implemented, whose particle rounds are serial within an island (de…	STRUCK for v3.1 (MEET-v31 decision 6: "Depth and cost are stated at the implemented (island) level. Free of N, H, R is removed. Calibration is counted."; METHOD_v3. 1_FINAL: "Depth free of N, H, R is not claimed for v3.1"); Felipe decided (2026-11-03) that paper 1 claims island-level depth only; C6 is implemented and law-checked iff paper 1 claims particle-parallel depth or demonstrates any fit with p > 2,500 (neither fires); the as-recorded METH-AUDIT statement itself is conditional on width R N so not false as a particle-parallel statement		
RF272	post-search HPM monotone in the island count R prime without the extra local se…	Reading of Cor W-R: since the pre-local-search scored set and HPM are nested in R prime, the reported (post-local-search) HPM is monotone in R prime; the deterministic implication "pre-search sets nested => post-search…	FALSE (PC1 claim 6, Claude Opus 5.5 same-family proof check for the MEET-v31 chair; COUNTEREXAMPLE CE220); METH-AUDIT PROOFS.md already carried the caveat; replaced by Cor W-R with the extra local search from c4 (TH396); no listed theorem uses post-search monotonicity		
RF273	more islands always helps	"More islands always helps": false for the equal-weight (primary) PIP under a split of a fixed total, because its limit is the single-island mean E[PIPhat_ 1j/C] whose bias is set by N per island (Prop P(i)); Zhat is neu…	REJECTED (METH-AUDIT §4 C8 WORKHORSE in effect, §7 "More islands always helps fails"); replaced by stacking R prime = 4k at fixed N_ e (TH396, MEET-v31 item 5) and the Zhat-weighted primary PIP (MEET-v31 item 8)		
RF274	rapid-mixing theorems (YWJ, ZYVR, Chang-Zhou) apply to / certify the paper targ…	Ledger gloss of API-POS: "Rapid-mixing theorems (YWJ, ZYVR) apply to LH"; HQ correction at pos 1009: "API-POS said the rapid-mixing theorems apply to the linked-hub family. LIT-MCMC finds they don't, at least for our ow…	QUALIFIED, UNCERTAIN: API-POS headline 2 itself says the theorems apply "with their own priors" to LH-iid and headline 3 that "for the paper's own target, neither theorem applies as stated"; LIT-MCMC refutes only a sufficient condition (R > M), not fast mixing; THM-INV D1 and THM-SYNTH REPORT §4.1: say both; replaced by Prop OB (TH401) and the positioning paragraph (TH404)		
RF275	LIT-VENUE s1.3: poly-time recovery on LH also follows from Chang-Zhou R > M "si…	LIT-VENUE headline item 3: polynomial-time recovery of S0 on LH "also follows from Chang-Zhou's unimodality condition R > M, since on LH S0 is the unique local mode"; unimodality gives R > 1, not R > M, and R > M fails…	CONTRADICTED by LIT-MCMC exact enumeration (CE216) and THM-INV D1; "LIT-MCMC's computation contradicts this: unimodality (R > 1) does not give R > M"		
RF276	LIT-PAR G4 analytic sketch: Song-Liang loses the masked variable whenever the s…	LIT-PAR G4: SaM recovers S0 in 5/318 splits that separate the linked pair "(probability ~ 1/ 2 for two halves, -> (K-1)/K for K groups)" and the analytic sketch via "Song-Liang Thm 5" read from slides; corrections: the f…	CORRECTED by THM-SYNTH REPORT §1.6(a) / Theorem SL(e) (checked numerically, scripts/ check_lemmas.py; CE214) and THM-INV D10; replaced by Theorem SL (TH417, PROOF SKETCH); final: K97 -> FUTURE_WORK FW21 (null-exclusion/FDR step NOT REPAIRED, W2PC1)		

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RF280	F-REFINE: without LS0 the paper has no masked-regime consistency theorem	F-REFINE ARCHITECTURE §4: "So without LS0 the paper has no masked-regime consistency theorem whose premises are all proved on an explicit family" (also W4-T38FULL II. 9, W6-FIXREV1 "Not claimed")	SUPERSEDED: no-LS0 candidates on LH now exist (W5-B2ENV, W5-VMASK R1, O-BGIBBS BG3, F-SHARE F1/F2, W7-SEP); registry: "candidates on LH, pending cross-family review (A-REV7)"; A-REV7 VERIFIED B2-LH under STAB-LH; F-REFINE amended in place by E-SPINE (2026-11-02)		
TH106	T07	Size-only ranking ceiling and Bayes decisions (old label A6 = T5-C)	PROVED-LOCKED (all parts, wave 3); not adopted (adoption is Felipe's decision) [review record: wave 3: Opus 5.5 lead; two Opus 5.5 xhigh (same-family Claude); statement checker where noted; external OpenAI review queued (no cross-family review done); not adopted]		
TH109	T11	Unified scoring and discovery ledger (old label L-LED / T14-D)	PARTLY PROVED: (iii) PROVED-LOCKED (wave 1); (i), (ii), (iv)-(vii) not locked (CONJECTURED); not adopted (adoption is Felipe's decision) [review record: wave 1: Opus 5.5 lead; Astra #1 (Stage A); Astra #2 + Opus 5.5 xhigh (Stage B) ; Fable 5.1 high + Astra xhigh (Stage C): correct; same-family and cross-family (Astra); not adopted]		
TH115	T21	Composition theorem for the reported HPM (old label T14-H)	PROVED-LOCKED (C1 and routes (a)-(f), wave 2) + CONDITIONAL on open hypotheses: H24 (route (e)) and the route-specific occurrence / exposure conditions; not adopted (adoption is Felipe's decision) [review record: wave 2: Astra lead; two Sol 6.1 high; Astra xhigh closing review; Claude Opus audit: correct; conditional theorem; not adopted]		
TH128	Theorem 1	Posterior target consistency (statement of record W6-FIXREV2 section 1.2): under R1, G, R, SE_lambda(j0), SE_w(m#_n; g0) and BM-inf or BM(c>c0): 1-pi(S0)=O_P(sqrt(log n/ n)); PIP/MPM/global-HPM consistency with P0(S_dagg…	PROVED (candidate) (W1-T03R) ; SF VERIFIED (W2-VT03R, no GAP/ERROR in the mathematics) ; XF VERIFIED under exactly its hypotheses (A-REV2); VERIFIED core (A-REV5, onset gloss ERROR) ; LIT-STAT: modest extension of Rossell 2022, clause (iii) known; final card: PROVED; SF VERIFIED (W2-VT03R); XF VERIFIED (A-REV2)		
TH153	PXN-P6	On LH-iid w.p. -> 1: (L) S0 locally optimal and (D) D_S on Lat(S°)\{S0}, so the production local search from S° returns S0 certified in c moves; LS0 scores and certifies S0	PROVED (W3-PXN); SF VERIFIED (W5-VMASK rows 14-15); XF GAP (A-REV3, A-REV6, A-REV8; conflict C16: Lambda(a) uses Tr(c) beyond range)		
TH158	PXN-W	Block-orthogonal iid design: start keep-j log-odds LO_0(j) = -log((2p-s0)/s0) - (1/ 2)log(n+1) + V_0(j) + O_P(1) with V_0 = (n/ 2) theta_j^2 Var(x_j/X_ T)/v_T; under fixed s0 + BM-infinity dilution vanishes	PROOF SKETCH (W3-PXN, asymptotic part); numerics in REPORT sec 5; no review		

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
TH160	Thm 5(b1-LH-iid) (W5-VMASK form)	LH-iid (c linked pairs, g<=B<2g, n^a<=p<= n^A) + EX-A, EX-S, CAL, SR, GUARD/OP-LS0 + LS0 with emitted centre: P(O ∩ {S+hat != S0}) -> 0 and liminf P(O ∩ {S+hat = S0}) >= 1 - limsup P(U)	PROVED (candidate); SF VERIFIED (W5-VMASK); XF GAP (A-REV3, A-REV6, A-REV8)		
TH161	API-PXNGAP theorem	LH-iid with (SIZE), 18<=p<=n^A, p->inf, EX-A/EX-S/CAL/SR/ GUARD/OP-LS0: (R) P(blocks != groups) <= (2/(sqrt(pi)(1-rho))) p^2 sqrt(n) e^{-(n-4) I*}; (S)(E)(T)(P) and (C) P(O ∩ {S+hat != S0}) <= 2E0 delta1 + D_n -> 0	PROVED (candidate, API-PXNGAP) given T22(i),(ii), T03, CONC, W7 Thm B; no SF/XF review in the archive		
TH172	Corollary E1	-log pi0(S0)=V+C-log A (visibility + competition - size) via softplus(-lambda_b), log(1+eta_ b)	PROVED (author); SF none; XF VERIFIED* (A-REV9: production BF tables, r_b(empty)=1; GAP for arbitrary positive tables without normalisation)		
TH176	Proposition S	pi0(S0)<=1/prod_i(1+ [M_(i)-(i-1)]_+) for v3 tables and any common sigma^2	PROVED (author); SF none; XF GAP (A-REV9: replacement assignments not injective)		
TH177	Proposition D’	lambda_j ~ chi2_1(n rho_j^2)/2 - log(n)/2 - log((2p-s0-1)/(s0+1)), rho_j^2=(1-kappa_ j)x_j<=omega_j saturating in c; assembled formula for log pi0(S0)	PROOF SKETCH (author); numerically checked; SF none; XF none		
TH181	Theorem B3	mixture of v3 partition and linked-merged partition: pi0mix(S0) >=1/2-O_P(2^{2g}n^{-1/2}); T38(b) on LH without LS0 by route (a); cost log 2 nats	PROVED-UNDER (verified W1-EXP Thm 6, locked T03, T11(iii), T21, Prop B1) (author only); LH-iid PROOF SKETCH; XF none		
TH183	Theorem C1 (O-MIX)	under BO: pi0^C(S0) >=pi(S0)exp(-nu psi(u(S0))); plug-in with / Q_S’-Q_S0/=o(sqrt n) gives pi0^C(S0)>=pi(S0)(1-o_P(1))	PROVED (author only; BO, deterministic); registry CONFLICT C23 (as a method recommendation); decision: rejected (E-SPINE 2.2); XF none		
TH186	Theorem S2	with a=sigma^2/ sigma-hat^2: concentration iff a<1+1/(2A) ; rates n^{-1/2}, sqrt(log p)n^{-1/2}, p^{a-1}n^{-1/2}; flooding above threshold	PROVED-UNDER (BO) (author: ’PROVED under BO’); heavy-tail step to be re-checked; XF none		
TH188	Theorem S4	under S2 with a_+ <1+1/(2A) and T03 hypotheses: Q_f pi0^C(S0)->inf, P(S_hat != S0) ->0 without LS0 (BO)	PROVED-UNDER (Theorem S2, locked T03, T11(iii), T21) (author only); ’vacuous as an improvement’ over v3 in fixed-beta regime		
TH198	Prop 1	Literal TD(d,m) of T37 cannot hold for d>=1 (m=0 and F(S0+l)=F(S0) forced); P0(TD(d,0))=0 under G	PROVED; SF VERIFIED (W3-VT37R, 720 designs 0/720); XF VERIFIED (A-REV2, A-REV5)		
TH199	Theorem A	Production greedy search from S_c in R_{a, d} returns S0 certified iff l>=tau(S_c) under TD’_P (A1-A4); A5 literal TD needs POS; TD’_P implies TD’_inf	PROVED (W6-FIXREV2 candidate) ; SF VERIFIED (W3-VT37R, 720 designs 0 violations); XF VERIFIED* (A-REV2 A1-A4 and POS-repaired A5; original A5(a) ERROR); A-REV5 core VERIFIED, literal cap-free lattice wording ERROR		

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TH200	Proposition F	Every fit with S-hat != S0 lies in exactly one class by F_pre, F_post vs F(S0): I (E_LS^c), II (search miss), III (explored competitor)	PROVED (immediate, W6-FIXREV2) ; XF VERIFIED (A-REV5)		
TH201	Theorem B (T37)	Failure probability of the search <= P(S_c not in C)+P0(NA^c)+P0(TL^c)+sum over Lat(C) of P0(D_S^c) for a data-independent class C	PROVED (inclusions, union bound) ; PROOF SKETCH (single events) ; SF VERIFIED (W3-VT37R); XF VERIFIED (A-REV2; selected centre needs covering class); cap-count correction pending (A-REV5)		
TH204	Theorem W	PGD(eta)+SE_lambda(d+1)+BM-inf+(H1/H2/ H3) imply P0(TD'_P(R_{a,d}(D)))->1 and search from any centre in region returns S0 certified w.p.->1	PROVED (candidate); SF VERIFIED (W5-VPGD, no ERROR, minor GAPs repaired R1-R4); XF none (A-REV5: implication only); region form fails on masked LH		
TH217	Prop T	If a present truth u is worthless at S+j then rho0>=rho+ and PGD(eta) fails for all eta on the region; D_S fails w. p.->1	PROVED (W5-VPGD); no second check; registry CONFLICT C15 with F-REFINE description		
TH218	Theorem B	Restated T24: P_x(S0 not scored within H terminal rounds) <= P(Bin(H,c_*)<v)+H*eta <= v(1-c_*)^floor(H/v)+ H*eta, exact polynomial constants (c2 >= 1/(B+ 3)-eps_del), no factorial	PROVED (candidate); SF VERIFIED* (W2-VT24G, after Lemma F' repair) ; XF VERIFIED after F' repair (A-REV1; statistical application GAP); XF VERIFIED finite feasible uninterrupted bound (A-REV5); vacuous at n=200		
TH227	Prop. U'	Bad(W) = leaves family or raises V or removes a truth, for every label; extra-for-extra swap neutral iff it stays in family; independence step charged by beta_I(W)	Original Prop. U: PROVED (candidate), SF VERIFIED as book-keeping (W2-VT24G), XF ERROR (A-REV1) [CONFLICT C3]; repaired Prop. U': PROVED (W6-FIXREV1, same-family); XF VERIFIED (A-REV5); Theorem B unaffected		
TH239	T21-e'	Theorem B (via beta_ H) replaces H24 in T21 route (e); multi-particle form (e-M)	PROVED (candidate) as an implication; SF VERIFIED* (W2-VT24G with three edits; W5-LOCKBOOK CONSISTENT); XF none for W2-BOOK Thm 4(e')		
TH241	T38b-R(whole)	Composition of the discovery term into liminf P(U^c, Shat+ = S0) >= 1 - limsup P(U) via routes (b1) LS0+ TD', (b1') pilot D2, (b2) DELIV+REPAIR	PROVED given named inputs (W1-T38E); SF VERIFIED (W3-VT38E) ; XF GAP as supplied, composition and (b1) VERIFIED, (b1'),(b2) GAP, repaired formulation VERIFIED (A-REV1); superseded by T38-(0) (W6-FIXREV1 v2/T38-RECORD v3)		
TH252	SHARE-X	P(some particle ever in U) <= N pi0(U) + N sum_k a_k per island (U = set entered from outside with round-entry probability a_k) ; posterior barrier is a gate	PROVED (W4-SHARE) main inequality; SF VERIFIED* (W5-B2ENV: Moreover and Use clauses GAP); no XF row		
TH253	SHARE-M(a)	Deterministic core: if lambda_min(R_ {TUM})>0 and M=S0\T nonempty then max_j delta_{T,j}^2 >= (n-1) lambda^2 //theta_M/ /_2^4///theta_M//_ 1^2 >= (n-1) lambda^2 theta_min^2	PROVED (W4-SHARE); SF VERIFIED (W5-B2ENV; 20,000 instances); XF VERIFIED (A-REV3)		

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TH254	SHARE-B2W4	Route (b2) assembly: P(S0 not in S/C) <= [eps_E + exp{-N pi0(h0)(1-e^{-c})}]^4 + eps_rep given (TRACK), (OSC), confinement and REPAIR on W	PROVED given inputs (W4-SHARE) ; SF VERIFIED (W5-B2ENV, L+ 5 rounds note); XF VERIFIED* (A-REV3: conditional inequality VERIFIED, exit-loss line ERROR)		
TH261	OBG-LH-lock	Premises of BG3 on LH exact Gram with p fixed under strict gaps C1-C3 (and block gaps C1',C2') : pi0(U^c) <= e^{-c3 n}, eps_U <= m e^{-c3 n}, beta_1 >= (1-o(1))/m!	PROOF SKETCH (author only); XF VERIFIED* as a conditional fixed-p implication from strict C1-C3 and block keep margins (A-REV7, Astra-authored completion); C1', C2' VERIFIED/PROVED on LH; unrestricted locking ERROR		
TH263	OBG-T38b-BG3	P(O ∩ {Shat != S0}) -> 0 for v3 as run (no LS0) on LH exact Gram with p fixed, every fixed c (BG3 + LH-lock + T38(0))	PROOF SKETCH (composite carries the weakest input's word); XF VERIFIED* (A-REV7: BG3 + conditional LH-lock for fixed p, component lemma not the headline)		
TH267	FSH-T38b2	Gibbs-path route with delivery by closure: F2 + scoring on W(S°) (Section 4) + T38(0), every fixed c, no LS0	PROOF SKETCH (composite; no separate status word; author P(true) 0.85, P(provable) 0.5); XF GAP (A-REV7: not to be cited as a growing-p no-LS0 guarantee)		
TH269	B2E-D2LH	LH exact Gram, c=1, p/ sqrt(n)->infinity: P(S0 not in S_cal) = O(sqrt(n)/p) + o(1) -> 0, so the original method (no LS0) is consistent	PROVED-UNDER (author, given D2, E-T1, E-L4, E-T5d); XF VERIFIED for qualitative consistency, GAP for the stated unconditional O(sqrt n/ p) rate (A-REV7)		
TH272	B2E-B2LH	LH exact Gram, every c, under STAB-LH, any frozen grid, no LS0: P(S0 not in S/C) <= N pi0(N^c) + N T_e a_n + (1-c0+r_n)^N -> 0, hence liminf P(U^c, Shat = S0) >= 1 - limsup P(U) for v3 as run	PROVED-UNDER STAB-LH (author, W5-B2ENV); XF VERIFIED* conditional theorem (A-REV7, eta_n corrected); theorem-specific parts cross-checked by Claude (API-VSTAB)		
TH273	B2E-latpot	Lattice-indicator potentials have Gamma + sum eta >= log(NT) - O(1) while D5 needs <= log(N pi0(h0)/T) - omega(1): cannot certify delivery at any budget; on LH the right potential 1_N gives Gamma = 0	PROOF SKETCH (author only; numerically confirmed)		
TH275	B2LH-subfamily(A7)	Unchanged v3, no LS0, on LH exact Gram satisfying (CERT) (explicit member g=9, c=2,rho=0.4,psi=0.2, beta=1,sigma^2=1,p_ n=9 floor(n/18)): P(O ∩ {Shat != S0}) -> 0	PROVED (candidate) given cited inputs (API-VSTAB); B2-LH XF VERIFIED* (A-REV7); certificate and subfamily Astra-authored, Claude VERIFIED; IND-T flagged without Claude verifier		
TH276	fixed-p component corollary on (CERT)	On (CERT) the strict gaps C1-C3 of LH-lock hold at every fixed p, so the fixed-p BG3/LH-lock bound holds with no further hypothesis	PROVED (API-VSTAB, one run)		

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TH281	T38-(0)	For every n (both variants, strict convention) : P(O and S-hat+ != S0) <= 2E0[delta1] + D_n, D_n := P(O-bar and S0 not in S-bar+); if D_n -> 0 then P(O and S-hat+ != S0) -> 0 and P(O, S-hat+ = S0) - (1-P(U)) -> 0 (joint…	PROVED; SF VERIFIED (W3-VT38E) ; XF VERIFIED (A-REV1 s7.4; A-REV5 output/execution interface; A-REV8 'W6 versus A-REV1 s7') ; Lean t21_C1 sorry-free, fidelity GREEN; registry T38-(0) 'PROVED (SF: VERIFIED (W3-VT38E); XF: VERIFIED* (A-REV5))'		
TH285	T38-(b1')	No method change: D2-infinity rate margins (fixed b0>1/10, /D_n/ <=k_D, Dep_n<=1-c0, P(NB'^c)->0) imply the potential first stage of the tier-0 pilot scores S0, S-hat+ = S-hat	PROVED (implication) via D2-infinity; SF VERIFIED (W3-VT38E, finite D2); XF VERIFIED (A-REV1 finite D2 and D2-inf; A-REV5 rate margins); masked NB' instance on LH c=1 is part (b1'-LH); registry T38-b1' 'PROVED-UNDER ... GAP' for the statistical instance		
TH286	T38-(b2)	No method change: DELIV(X_n) (assumed) and REP(X_n) with a data-measurable eligible set X_n of repair-certificate states, H' = L+5 rounds at phi=1, capped beta_H; D_n -> 0	PROVED (implication); XF VERIFIED as an interface (A-REV1 s7.2; A-REV5); DELIV(X_n) OPEN in general; D3 surrogate for DELIV GAP (A-REV5)		
TH287	T38-b1-LH	Linked-hub family LH in exact Gram (p<=n-1) : (P), (M1), (M2) with fixed margin m0>0, (M3) hold; pi0(S0)=p^{-c}n^{-c/2}e^{O_P(1) }; with LS0 P(O and S-hat+ != S0) -> 0 with no statistical hypothesis left open	PROVED (candidate); SF VERIFIED (W5-VMASK; W3-VEXP); XF VERIFIED (A-REV8 'exact-Gram b1-LH consistency VERIFIED ... with LS0; c=1 needs only its initial scan'); iid-failure clause ERROR (A-REV5) / GAP (A-REV8), repaired by V18		
TH289	T38-(c)	Score evaluations <= T32(i)'s ceiling plus 1 + (l+1) max_{k<= K}[p + k(p-k)] for LS0, polynomial in n under the v3 budgets; displayed order O((n+p)(p+B)log^2(n+ p) + p^2 log^2(n+p)) carries an XF GAP	PROVED-UNDER T32(i); T32(i) PROVED (W5-LOCKBOOK), SF recount W7-SEC; XF A-REV6 'supported by loop accounting' (no verdict word), A-REV9 deferred; displayed log^2 rate GAP (A-REV6) ; LS0 term PROVED and VERIFIED (A-REV1, A-REV5); registry Cost-T38 'PROVED-UNDER (SF: VERIFIED (W3-VT38E); XF: GAP (A-REV5))'		
TH290	T38-v3	Statement of record v3 of Theorem 5: (0), (0_H), (a), (b1), (b1'), (b2), (b1-LH), (b1'-LH), (b2-LH), (b1-LH-iid), (c), with change log V1-V21, Part III conflicts, Part IV decisions D1-D6; no new mathematics	PROVED-UNDER (registry 'T38-v3') ; nobody reviewed v3 itself; every clause carries its source status; all PROVED are candidates		
TH293	LH-LAT	On LH (exact Gram) , POP-LAT(S°, m0) holds with fixed m0 : = min{(1-kappa_max) R_min, min_i beta_i^2(1-psi_i^2)^2, min_h theta_h^2} > 0; valid lower bound, sharp for homogeneous pairs, exact margin m0-sharp (W5-VMASK R5)	PROVED; SF VERIFIED (W5-VMASK rows 23, 23"; 'm0 is sharp' withdrawn as GAP overstatement); XF VERIFIED* (A-REV8: VERIFIED for fixed exact-Gram LH parameters; m0 a valid lower bound, not universal sharp margin)		

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TH297	Prop2(a,b,d,e)	On the potential run given frozen C: E[Z_ r]=Z, E[Z_r PIP_rj]=Z PIP_j, pooled island product unbiased, P(Z>=0.05 Z-hat)>= 0.95, E log Z-hat<= log Z; output-level P(reported and 0.05 Z-hat>Z)<=0.05 P(A) with no conditioning...	PROVED (candidate; corollary of locked T12, T31(i), Lemma RO) ; SF VERIFIED/CORRECT (W5-LOCKBOOK); XF GAP (A-REV4: fixed-data probability convention B3, exact-branch extension B4); A-REV6 identity VERIFIED, convention GAP, W5 full-output expectation ERROR; repair [PENDING], A-REV6 R2 candidate		
TH298	Prop2(c)	E[Z-hat P-bar_j/C]=Z PIP_j - (3/4)Cov(PIP_ 1j,Z_1/C) (factor (I-1)/ I for I islands); E[P-bar_ j/C]-PIP_j=-Cov/Z; both nonzero in the T12(iv. b) instance (<-0.17 and <-0.1275 Z)	PROVED; SF VERIFIED/CORRECT (W5-LOCKBOOK, reproduced to 1e-58); XF VERIFIED (A-REV4, Owls 50-digit reproduction -0. 1739741778 and -0.1317163950; A-REV6 VERIFIED)		
TH325	T26-PropC	On XtX=(n-1)I_p, p<= min(n-1,n^A), fixed s0 and nonzero theta: G,R,SE,BM-C,VIS,NL_ eta(eta=0),NL_avg(kappa<=1/2) and strict EXP-prime hold jointly (so Theorem B applies) ; numerics max/R/ 0. 11-0.16	PROOF SKETCH; SF PROOF SKETCH kept (W5-LOCKBOOK); XF GAP (A-REV4, A-REV6); conflict C9 with W4-T38FULL "no open statistical input on ORTH"		
TH334	Theorem N200	P(U or S-hat != S0) = P(U) + P(O, S-hat after S0) + P(O, S0 after S-hat) <= P(U) + Pi(X) + P(O, S0 not scored), Pi(X)=P0(S-dagger != S0 / X); matching lower bound minus P(O, S-dagger not scored); implies T21 C1 term by...	PROVED (elementary, every n); author-only (no SF, no XF); Pi(X) upper bound OPEN		
TH350	Theorem WS3	Under exact block orthogonality d=d_ sigma>=0 is pure noise-variance sharing, zero iff at most one block active; with known sigma^2 d=0 and pi0=pi	PROVED; XF VERIFIED (A-REV4 F-DRAFT W1; A-REV6 exact sample-BO identity); CHK_INHERITED VERIFIED (PC1, cross-family, line by line)		
TH362	Theorem W (ORTH)	Under ORTH, fixed s0, theta*, p log p=o(n): both ends concentrate, KL,TV->0, chi2 = p^{r+ -eps}/sqrt(n) if p^r sqrt(log p)/sqrt(n)->0, log(1+chi2)>=e^{(L_n-C)/r} and <=C e^{L_n/ r}(log p)^{5(1+r)/(2r)} on divergent side; t...	PROVED (author); SF verified except W(d) upper half restricted (CHECK_SEP_W: F1 gap, F3 constant; SEP-prime and W(g) not reviewed); XF A-REV9 VERIFIED under H-ORTH exactly as stated for W(a)-(f) including the stronger upper-side premise; W(g) not covered		
TH367	O-STAGES Corollary W prime	Given layer law (LL): /T*-ceil(log2(1+1/g_ n))/<=1 and T*<=ceil(log2(1+2AR2))+1=O(1) for p<=n^A	PROVED-UNDER (LL; PROOF SKETCH hypothesis); registry conflict C22 (LL may need restating for 2Ar>1); SF none; XF none		
TH380	SEP(A)	Q iid pi0 draws score S0 w.p. -> 0 or 1 as Q/ Q* -> 0 or inf, sharp Q*=p^c n^{c/2} on LH exact Gram; polynomial budget n^alpha fails iff alpha<c(a+1/2) (p~n^a)	PROVED (author W7-SEP, candidate; SF: none; XF: VERIFIED (A-REV9) as an implication from imported start-mass and concentration premises)		

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TH382	SEP(B)	1+chi2(pi//pi0) >= pi(S0)^2/pi0(S0) = Q* e^{-Mstat}(1+o_P(1)) ; KL lower bound; relative variance (1/pi0(S0)-1)/Q; estimator of pi(S0) equals 0 w.p. -> 1 below Q*	PROVED (author W7-SEP, candidate; SF: none; XF: VERIFIED (A-REV9) as implications from imported premises)		
TH386	SEP(E)	p~n^a: iid/IS exponent c(a+1/2) sharp; LS0 or draw+search <=2a; v3 with LS0 <= a+max(1, a); iid vs v3 separated iff c>=2; no super-polynomial gap in R1	PROVED-UNDER (registry; v3 row conditional on T32(i) listed as CONJECTURE, stale; SF: none; XF: none); registry CONFLICT C19		
TH388	Prop GU	Exact Gram, isolated relevant set R, SE on R, fixed signals: production local search from every C subset of R reaches, scores and certifies S0 within tau_GU moves w.p. -> 1; no PGD needed	PROVED (candidate; SF: none; XF: VERIFIED* (A-REV7) as greedy best-neighbour theorem, GAP as standalone no-LS0 growing-p theorem; A-REV9: GAP in that review, Lemma M not re-audited); registry CONFLICT C25		
TH394	Proposition W	Final islands run as concurrent workers (own ScoreCache and Ledger, stream sigma_r = 9+r), merged by Lemma S before the final local search, reproduce the serial run bitwise: Zhat_r, PIP_r, ESS, phi, increments, move cou…	PROVED (code reading; executed check in REPORT §6) [PROOFS.md heading]; candidate, not reviewed, author-only (MEET-THM IDX 92) ; executed bitwise by the author on the smoke problem and design p0500_S1_r001 only, then p0500_S4_r001 (pos 1175) and p0500_S5_r002 (pos 1241) by HQ re-run = 3/3 (MEET-v31 evidence/2026-11-03_concurrency_runs.md); source CONFLICT 1/1 vs 3/3 recorded in MEET-THM REGISTRY_INDEX §5. B B1; no cross-family check of the code reading		
TH395	Proposition W-cal	The 4 diagnostic islands of cpp_cal_diag run as persistent workers (own stream state across iterations, own cache and ledger) , stores and HPM merged into the calibration ledger before the local search and before V is bu…	PROVED (code reading only; not executed) [PROOFS.md heading]; author-only; still not executed at MEET-v31 close (METHOD_v3.1_FINAL: "(b) has not been executed yet")		
TH397	Proposition P	Islands conditionally iid given the calibration: (i) equal-weight PIP converges a.s. to E[PIPhat_1j / C], keeping the single-island bias b_j(N); (ii) Zhat-weighted PIP is consistent as R -> infinity (SLLN); (iii) P(max_…	PROVED (elementary) [PROOFS.md heading]; author-only (MEET-THM IDX 70, FLAG AUTHOR-ONLY) ; input M1 XF VERIFIED (A-REV9) ; (i)-(ii) known (Verge et al. 2015; Whiteley-Lee-Heine 2016 s5.1; Dai et al. 2022 (4.8)) per LIT-PAR/THM-SYNTH; PC1 consulted only its quoted statement; practical scope caveat: E[W^2] can be exponentially large in the masked regime		
TH401	Proposition OB	For the paper target on M(s) (s0 < s <= K) , gamma_e = S0 + {e}: (i) varpi(S0)/varpi(gamma_e) = w(k-1)/w(k) = (2p-k)/k; (ii) pi(S0)/pi(gamma_e) = [(2p-k)/k] sqrt(n+1) exp(-T_e/2)(1+o_P(1)); (iii) if max_e T_e >= (2-eps)…	(i) PROVED; (ii)-(iii) PROOF SKETCH [LIT-MCMC REPORT §2.2]; "the step the best neighbour of gamma_e* is S0 is used only as an upper bound ... not verified"; evidence: ob_check 12/12 (CE215), rz_lh (CE216); author-only		

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TH404	Statement R3	(i) under the YWJ-prior with a fixed cap, YWJ Thm 2/Cor 2 and ZYVR Thm 1 apply to LH-iid (p >= n, p <= n^A) : recovery of S0 and mixing from any sparse start in O(alpha p n log p) scores (YWJ) or O(n) informed iterations…	(i) PROOF SKETCH; (ii) PROVED for the definitional failures Y0, Y8, Z2 and PROOF SKETCH for Z4(1a) ; (iii) via Prop U PROOF SKETCH, YWJ-type spectral-gap proof for the paper prior a CONJECTURE (CJ297); author-only; qualified by LIT-MCMC: R > M fails on LH at n = 200, 1600 and in 12/12 study-like data sets (RF274)		
TH412	Theorem H	h: Omega -> [0,1], hhat_r the island plain (or Rao-Blackwellised) estimate: (a) E[W_r / hhat_r - h(S0)/ / C] <= delta_1 for the plain average (<= 2 delta_1 + c_h(rho_bar+u_pi) for the RB version); (b) P(/ hhat^zw - h(S0)/…	PROOF SKETCH (NEW-SYNTH) [THM-SYNTH REPORT §2]; crux not verified: island identity E[W_ r hhat_r] = pi(h) for the measured-round average; PC1 claim 3 (same-family Claude check): finite-R bound P(/muhat - mu/ > eps) <= min{1, 4(M-1)/R + 4M/(R eps^2) } PROVED (add W_r >= 0), the identity ”is still a premise”; (d) R_n condition uncheckable without an upper bound on E[W^2/C_n]; final MEET-THM: N1 (general-h identity) PROOF SKETCH		
TH417	Theorem SL	LH exact Gram with c linked pairs, Song-Liang procedure (uniform equipartition into K_n subsets of size s_n, Stage-1 PIP threshold q, SL priors) ; E_n = some masked hub isolated in its subset from its partner and its own…	COUNTEREXAMPLE (numerical, LIT-PAR G4) + PROOF SKETCH (NEW-SYNTH); crux (b) PIP -> 0 under the SL priors ”NOT REPAIRED (W2PC1)” at MEET-THM; LH-iid v2; final: FUTURE_WORK FW21, LIT-PAR G4 cited as labelled numerical evidence		
TH418	hpm_eq_truth	If post F A S0 > 1/2, S0 in scored set (subset of A) and Shat maximises F over it, then Shat = S0 (any tie rule)	Lean: sorry-free (lake build green, 0 sorry); Fable 5.1 GREEN; Astra r1 GREEN (statement fidelity only)		
TH420	t21_decomposition’	P(U or Shat != S0) <= P(U) + P(S0 not in scored) + P(U^c, S0 in scored, pi(S0) <= 1/2)	Lean: sorry-free (lake build green, 0 sorry); Fable 5.1 GREEN; Astra r1 GREEN (statement fidelity only)		
TH422	t21_C1	P.real(U or Shat != S0) <= P.real(U) + 2*integral(1 - pi(S0)) + P.real(U^c, S0 not in scored); abstract probability space, measurable scores	Lean: sorry-free (lake build green, 0 sorry); Fable 5.1 GREEN; Astra r1 GREEN (statement fidelity only); informal TH278 (Thm4-C1) PROVED (SF: VERIFIED* (W5-LOCKBOOK) ; XF: none); T21 pre-campaign PROVED-LOCKED, CONDITIONAL		
TH423	ceiling	If S0 is the least argmax over Omega of (positive size-only prior) x m then s0 <= K and S0 is the least argmax of m in its size class	Lean: sorry-free (lake build green, 0 sorry); Fable 5.1 GREEN; Astra r1 GREEN (statement fidelity only); informal Prop1-T07 PROVED (SF: VERIFIED* (W5-LOCKBOOK); XF: VERIFIED* (A-REV4))		
CJ222	ELS-1; E_LS ledger data fact; final local search 0 moves	[data fact] Final local search made 0 moves in 2,394/2,394 completed PROD-H fits; S0 scored in every completed fit (store F(S0) equals offline F(S0) to 1e-12); E_LS empty (0/2,394) (S-ELS reading; the repo reports leave this unidentified, unreconciled conflict EMP12)	data from production ledgers (S-ELS); consistent with O-LS0		

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
CJ223	ELS-2; classification of the 2,400 fits	[data fact] 1,829 exact recoveries, 565 strict posterior misses (F(S0)<F(S-hat); smallest margin 0.0049 nats; 23 within 0.1), 0 search misses, 0 ties, 6 resource-limit stops (all S5)	data (S-ELS); O-SWAP ”4 no-output fits” superseded		
CJ247	W8-SYNTH 4.13 numerical inconsistencies	[data fact] Table of quantities with conflicting values across reports (start mode 6-12/2.5-6/7 of 14; favourable-n 590-730/800-1,600/1,050-3,200; beta-min 43x vs 100x; stopped fits 6 vs 4; LS0 cost; L+4 vs L+5)	Open reconciliation; recommendation: use S-ELS for counts		
TH116	T22; T16 (old label) ; aggregate start bounds; conditional concentration; T22(i)(ii)(iv); T22(iii) ; JRSS-B Thm 2; THEORY.md row T22	Aggregate start bounds and conditional concentration (old label T16)	PARTLY PROVED: (i), (ii), (iv) PROVED-LOCKED (wave 1; (iv-c) premises 1 - pi(S0) -> 0 from T03 and 1 - pi0(S0) -> 0 from (iii)); (iii) PROVED-LOCKED (wave 2) + CONDITIONAL on EXP’, VIS, NL_eta, NL_avg(kappa); (v) CONJECTURED (not proved); not adopted (adoption is Felipe’s decision) [review record: wave 1: Opus 5.5 lead; Astra #1 (Stage A); Astra #2 + Opus 5.5 xhigh (Stage B); Fable 5.1 high + Astra xhigh (Stage C): correct; same-family and cross-family (Astra); not adopted]		
TH119	T27; T20-C (old label); sequential cancellation; iid-discovery budget; T27(i); masking law; THEORY.md row T27	Sequential cancellation and the iid-discovery budget (old label T20-C)	PARTLY PROVED: (i) PROVED-LOCKED (wave 2) + CONDITIONAL on H2, H3; (ii)-(iv) not locked (CONJECTURED); not adopted (adoption is Felipe’s decision) [review record: wave 2: Astra lead; two Sol 6.1 high; Astra xhigh closing review; Claude Opus audit: correct; conditional theorem; not adopted]		
TH110	T12; T8; frozen-run evidence; recorded RB identities; JRSS-B Prop 2; THEORY.md row T12	Frozen-run evidence and recorded-RB identities (old label T8)	PROVED-LOCKED (all parts (i)-(iv) , wave 1); not adopted (adoption is Felipe’s decision) [review record: wave 1: Opus 5.5 lead; Astra #1 (Stage A); Astra #2 + Opus 5.5 xhigh (Stage B); Fable 5.1 high + Astra xhigh (Stage C): correct; same-family and cross-family (Astra); not adopted]		
TH123	T03; A2; T3; locked T03; PROOF_v2; H03 (= T03(i) conclusion)	Pre-campaign (locked) selection consistency: under G, R, full SE, BM-C (M*) or BM-inf and H-T02, 1-pi(S0)=O_P(r_n); PIP/MPM/HPM consistency; screening without beta-min; scale invariance.	PROVED-LOCKED (wave 3; statement checker approved; two review rounds, both reviewers verified; external OpenAI review queued) ; MEET-THM flag SAME-FAMILY-ONLY; numerical check 1958/1958; statement unchanged; conditional on H-T02 which T02(i) closes		
TH283	T38-(a); Theorem 5 (a); T38 (a) ; favourable regime; K68 (a-fav)	Under (F): 1 - pi0(S0) -> 0 in probability, D_n -> 0 for both variants (HPM convention adds D_n^{sl} <= P(SMC and A^c), uncontrolled); inputs locked T11(iii) , locked T22(iii) with EXP’, W1-EXP Thms 2, 6	PROVED-UNDER (F); SF VERIFIED* (inputs, W3-VEXP); ORTH joint family PROOF SKETCH; XF: none for the assembly (inputs A-REV2, A-REV5 VERIFIED for fixed designs)		

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
TH284	T38-(b1); (b1); Theorem b1; Thm-b1; LS0 route; T38b-R(b1); Theorem 5(b1)	With LS0: assume G, (P), OP-LS0, deterministic S°=U°∪P° with /P°/<=d and Phi(S°)<=phi-bar, (M1) ELIG, (M2) POP-LAT(S°, m_n), (M3); then S∼0=S° and LS0 scores S0 within max{Phi(S°)-1, 0} moves, D_n -> 0 and P(O and S-hat+ != S0) -> 0	PROVED-UNDER (M1)-(M3); SF VERIFIED (W5-VMASK row 22); XF VERIFIED (A-REV8 'Thm 5(b1) conditional consistency VERIFIED'; A-REV5 'b1 direct route VERIFIED')		
TH330	Corollary C5; Cor C5; OCERT-C5; posterior-miss certificate	On {S0 in U}: S-hat != S0 and C4(a) imply F(S0) <= F(S-hat), so the miss is a posterior miss; needs strict inequality (or the HPM total order) and /S0/ <= K	PROVED; SF VERIFIED* (O-VCERT tie caveat); XF ERROR on tie-insensitive prose, strict-score event inclusion VERIFIED (A-REV9)		
CJ224	ELS-3; the 541 unrepaired centres forced by Lemma 0	[data fact] All 541 fits with S_c in R_2 minus {S0} have F(S0)<F(S_c) (median gap -1.33, range -9.05..-0.0049); S0 a one-move neighbour in 434; W3-VT37R sampled F(S0)>F(S_c) not reproducible	data; settles Lemma0(b) on the study data only		

B.7 Empirical results (27)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
EMP01	Low-p HPM exactness; 0 search misses	SMC HPM = exact HPM 2,400/2,400 primary and 12,000/12,000 repeats; HPM = truth 92.3%; 2,216 hits, 0 search misses, 184 posterior misses	Terra tie-outs PASS, Sol ACCEPT (D75, D94)	study design n=200, p in {20,30}, S1-S6 (groups of 5)	
EMP02	Low-p evidence accuracy	log Zhat error bias -0.0002, RMSE 0.012; 0.05 Zhat lower bound covers Z in 14,400/14,400; mean relative error -0.00011 (SE 0.00010)	Terra tie-outs PASS, Sol ACCEPT (D75, D94)	study design n=200, p in {20,30}, S1-S6 (groups of 5)	
EMP03	Low-p PIP and MPM accuracy	mean per-fit max |PIP-exact| 0.0005 (RB), 0.0020 (plain); MPM = exact MPM 99.9% (RB), 99.8% (plain)	Terra tie-outs PASS, Sol ACCEPT (D75, D94)	study design n=200, p in {20,30}, S1-S6 (groups of 5)	
EMP04	Low-p seed stability	all 6 seeds give the same HPM in 2,400/2,400 cases; sd of log Zhat across seeds has mean 0.011	Terra tie-outs PASS, Sol ACCEPT (D75, D94)	study design n=200, p in {20,30}, S1-S6 (groups of 5)	
EMP05	Low-p recovery vs benchmarks	pooled /2,400: SSL 2,251; SMC-BVS HPM (= exact HPM) 2,216; SuSiE 2,183; aLasso 2,066; MCP 2,061; ABESS 2,051; horseshoe 2,028; lasso 1,852; EN 1,692	Terra tie-outs PASS, Sol ACCEPT (D75, D94)	study design n=200, p in {20,30}, S1-S6 (groups of 5)	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
EMP06	High-p reproduction	v2.2.1 rerun reproduces the 2026-10-23 fits on 11 fields in 2,400/2,400, max |diff| 0	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP07	High-p exact recovery	HPM exact recovery / 600: 559, 560 (c=1) and 370, 340 (c=0.5) at p=500, 1000	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP08	High-p RB vs plain MPM	MPM(RB) = MPM(plain) in 2,391/2, 394 completed fits; max |RB PIP - plain PIP| 0.0067; internal agreement only (D22)	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP09	High-p rankings vs 8 benchmarks	c=0.5: first or tied-first in 11/12 cells, +67 (p500) and +85 (p1000) over SSL; c=1: 4th of 9 behind SuSiE, SSL, horseshoe	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP10	High-p stops	2,394 completed + 6 resource_limit (all S5) counted as misses; P5 retry not run (D62); at most 6 recoveries (0.25 points) at stake	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP11	High-p operational miss decomposition	1,829 hits; 565 completed misses each with a scored competitor strictly above S0 (min gap 0.0049 nats); 0 detected search misses; 6 no output	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP12	Was S0 scored in the 565 completed misses? (conflict)	repo reports: not identified (0 to 565; registry in HLL mode); S-ELS: S0 in the top store in all 2,394 completed fits; no source reconciles them	UNCERTAIN (repo: not identified, 0 to 565; S-ELS: S0 in the top store in all 2,394 completed fits; unreconciled). Repo half: Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87); S-ELS half: no tie-out	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	EMP11
EMP13	Shapes of misses	high p: 207 of 565 completed misses are equal-size; c=1 misses 70/77 supersets; low p: 94 superset, 78 same-size, 10 subset, 2 mixed of 184	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {20,30,500, 1000}, S1-S6 (groups of 5)	
EMP14	Final local search never moved (high p)	0 improving moves in 2,394 completed fits; the 541 misses with centre in R2 minus S0 all have F(centre) >= F(S0)	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP15	Masked start	high p: Q_f pi0(S0) < 1 in 2,393/2,394; cell median log10 pi0(S0) -12.4 to -25.6; low p: pi0(S0) spans 38 orders and SMC matches the exact HPM rate in every start-mass class	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {20,30,500, 1000}, S1-S6 (groups of 5)	
EMP16	Truth dominance fails generically	low p: literal TD fails 2,400/2,400 (d=1,2); toward form holds 63. 0% (d=1), every such case a hit; high p: TD(0,0) holds strictly in 0/ 196 audited cases	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {20,30,500, 1000}, S1-S6 (groups of 5)	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
EMP17	First discovery in the pilot	high p: reported HPM first scored in the pilot in all 2,394 completed fits (hits: Gibbs 1,724, nswap 104, uswap 1) ; low p: truth always first scored in the pilot	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {20,30,500, 1000}, S1-S6 (groups of 5)	
EMP18	Cost scaling	p 500 to 1000 medians: x2^1.07 fresh scores, x2^1.5 distinct models, x2^1.4 runtime; low p median 2.7 s per fit, flat	Terra tie-outs PASS, Sol ACCEPT WITH FIXES ×2, narrow F3 check ACCEPT (D78, D87)	study design n=200, p in {20,30,500, 1000}, S1-S6 (groups of 5)	
EMP19	O-LS0 post hoc LS0	LS0 best model never beats the production HPM (0/2,394); recovery unchanged 1,119/1,200 and 710/1,200; median cost 0.17 s / 0.37 s	data (O-LS0-WRAP aggregation); Owls regeneration != NOTS bits (F(S0) agrees to 5e-8; every margin >= 0.002)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP20	API-N200 prediction	predicted wrong 558.1 vs observed 571 over 2,400 fits; RMS z 1.08 over 24 cells; exact checks at n=40, p=12	PROVED (elementary, every n); author-only (no SF, no XF); Pi(X) upper bound OPEN	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	EMP11
EMP21	Common sigma^2 and residualised start pilots	common sigma-hat^2 plug-ins: 0/2,400 favourable; residualised tables 1,888/2,400 (3 sweeps) and 2,203/2,400 (fixed point) favourable	candidate, unproven (API-RESID: 'unproven candidate'; no theorem, no SMC evidence); rejected for paper 1, recorded as future work (MEET-v31 MINUTES item 1)	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP22	Concurrency bitwise checks	R=4 concurrent final islands bitwise identical to serial on 3/3 designs (358->189 s, 672->352 s, 413->219 s); Prop W-cal not executed	PROVED (code reading; executed check in REPORT §6) [PROOFS.md heading]; candidate, not reviewed, author-only (MEET-THM IDX 92) ; executed bitwise by the author on the smoke problem and design p0500_S1_r001 only, then p0500_S4_r001 (pos 1175) and p0500_S5_r002 (pos 1241) by HQ re-run = 3/3 (MEET-v31 evidence/2026-11-03_concurrency_runs.md); source CONFLICT 1/1 vs 3/3 recorded in MEET-THM REGISTRY_INDEX §5. B B1; no cross-family check of the code reading	3 study designs at p=500	
EMP23	ABL-START and ABL-CONF start ablations	prior start underestimates the evidence (median log Z error -13.4/-10.8/-5.7 nats; LICENSED); block-start cost claim INCONCLUSIVE; no depth or speed claim	LICENSED (pre-registered rule: holds on S1, S4, S5; median log Z error -13.4/-10.8/-5.7 nats); XF none	3 designs (S1, S4, S5 at p=500, r001, c=1) x 10 seeds	
EMP24	Calibration cost formula check	calibration/final ratio predicted 0.46/0.67/1.69 vs observed 0.45/0.62/1.76 on 3 designs	seat-derived, unchecked	3 study designs at p=500	
EMP25	LIT-PAR SaM counterexample numerics	LH-like design p=16: exact posterior recovers S0 in 20/20 datasets; SaM in 5/318 splits that separate the linked pair	COUNTEREXAMPLE (numerical, reproducible) to "SaM recovers S0" on an LH-like masked design when the random split separates the linked pair [LIT-PAR §3.1 G4]; analytic part PROOF SKETCH, corrected by RF276/TH417; author-only; cited at MEET-THM as labelled numerical evidence	LH-like design p=16 (not the exact W1-EXP family; masked variable a singleton, group size 1; CE214)	

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
EMP26	Dilution numbers	on 480 regenerated study fits V carries 83% of -log pi0(S0), C 11%, size term 6%; about 7 of 14 truths invisible at n=200	PROOF SKETCH (author); numerically checked; SF none; XF none	study design n=200, p in {500, 1000}, S1-S6 (groups of 5)	
EMP27	O-CERT exhaustive Lemma M check	0 violations in 476,160 exhaustive (S, S+j) pairs; 0 violations over 285,712 boxes on 48 designs	PROVED; SF VERIFIED (O-VCERT: 0 violations in 476,160 exhaustive pairs); XF VERIFIED in exact arithmetic (A-REV9); LOCKABLE_v2	48 small exact designs	

B.8 Pending decisions, engineering items and the frontier (50)

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
DEC01	P1 approve paper-1 theorem list	Approve the paper-1 list: 8 main cards TH01-TH08 and supplement cards TH09-TH20	DECIDED 2026-11-07	n/a	TH01; TH09
DEC02	P2 Song-Liang separation scout HQ	Run one small scout HQ on the Song-Liang separation alongside wave 1; success makes a ninth main theorem	OPEN	n/a	CJ21; CL23
DEC03	P3 Prop D (M8) main vs supplement	Place Prop D (TH08) in the main text (HQ recommendation) or the supplement (main = 7)	OPEN	n/a	TH08
DEC04	P4 reading of proposal §5.3	HQ reading: §5.3 met by the bridge form; useful constants only for eps <~ n^(-1/2)/p (RL01, RL05; CJ22)	OPEN (HQ ruling stands unless overruled)	n/a	RL01; RL05
DEC05	P5 reading of proposal §5.4	HQ reading: coverage = distance from exact-posterior credible quantities (RL02); masked TV/KL future work (RL03)	OPEN (HQ ruling stands unless overruled)	n/a	RL02; RL03
DEC06	P6 stage count as step function	The stage count is a step function of correlation strength held within two-sided bounds (TH02.b6, TH02.b8, TH07)	OPEN (for Felipe to see)	n/a	RL05; TH07
DEC07	P7 HQ-authority items	Reporting-only J-hat and eps-hat-x (DF19, DF20, RL04); growth rate (ii) theorem-side only (RL06); systemd fix	OPEN (Felipe may overrule)	n/a	RL04; RL06; DF19; DF20
DEC08	P8 registry updates	Five recommended registry updates in MEET-THM/REGISTRY_PAPER1.md §9 for the registry owner	OPEN	n/a	
DEC09	PR #1 (spine correction)	Merge or not PR #1 (theory/E-SPINE): replace ”tempering is needed” by ”moving off the start is needed” in Docs/	OPEN (PR open, unmerged on 2026-11-07)	n/a	RF104; RF264
DEC10	LH-SCALE go-ahead	Approve the referee-requested LH-SCALE simulation on NOTS (about 2,400 fits, n 400-25,600)	OPEN (never approved or run)	n/a	ENG18

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
DEC11	common-sigma^2 rejection ruling	Explicit ruling by Felipe on HQ recommendation to reject the common plug-in sigma^2 start	RECOMMENDED BY HQ; FELIPE'S RULING NOT RECORDED	n/a	RF150
DEC12	adoption / locking of results	Adopt or lock results (LOCKABLE_v2 candidates; the 21 pre-campaign locked entries)	OPEN	n/a	
DEC13	Sol as second referee	Add Sol as a fourth seat or proof checker (no plan recorded)	OPEN	n/a	
DEC14	residualised start investment	Optional method-side investment in the residualised (backfit) start; rejected for paper 1	OPEN (not decided)	n/a	CJ162
DEC15	island-level depth only (MD06)	Paper 1 claims island-level depth only; per-particle streams are v3.2	DECIDED 2026-11-04	n/a	MD06
DEC16	novelty check vs shotgun stochastic search	Check "report the best model ever visited" against Hans, Dobra & West (2007) for the T21/T38 novelty claims	OPEN (not addressed in the archive)	n/a	TH115
ENG01	E1 W-cal replay	Bitwise replay of v3_calibrate with concurrent diagnostic islands on 3 designs	OPEN (not executed)	n/a	TH395
ENG02	E2 LS0 merge into production core	Merge LS0 into the production core; merged-LS0 replay on 3 designs	OPEN	n/a	CJ213; CJ178; CJ226; CJ228; OB02; OB03
ENG03	E3 GUARD-W audit	Stop-handling audit: any worker, LS0 or guard stop implies no output	OPEN	n/a	OB02; OB06
ENG04	E4 IND-T for streams 10..9+R'	Independence of final streams from calibration streams for every declared R'	OPEN	n/a	OB14
ENG05	E5 VAL-FP at production size	Long-double replay at p = 500; validated arithmetic	OPEN	n/a	OB16; CJ16
ENG06	E6 preflight and allocation audit	Preflight fixes and concurrent peak-memory and stop audit	OPEN	n/a	
ENG07	E7 checked arithmetic	Overflow boundary tests (integer, stream-id, allocation)	OPEN	n/a	
ENG08	E8 full Z-hat-weighted PIP rescoring	Rescore all 2,394 fits with the Z-hat-weighted PIP (3 designs done)	OPEN	n/a	MD08
ENG09	E9 calibration timing tabulation	Tabulate calibration/final timings on <= 20 saved fits	OPEN	n/a	OB15; TH08
ENG10	E10 tie-DP invariant tests	e4 tie-DP invariant (N3) and n = 2 boundary tests	OPEN	n/a	OB03
ENG11	E11 certificate card on saved fits	Certificate card on saved fits; C5 screening event over 2,400 fits	OPEN	n/a	CJ238; OB07
ENG12	E12 per-particle streams trigger	Implement per-particle streams iff particle-parallel depth claimed or p > 2,500 shown	NOT TRIGGERED	n/a	MD06
ENG13	E13 Gram-free scorer	X^T X-free scorer, declared v3.2 path	v3.2 PATH	n/a	LIM02

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
ENG14	E14 joint-summaries run	Small new run for pair inclusion and model size summaries	OPEN	n/a	TH01
ENG15	E15 AMENDMENT_ HQ_02 fields	Implementation of the reporting fields of AMENDMENT_HQ_02	NOT CODED (AMENDMENT_HQ_ 02.md:15: 'Reporting fields 1–3 are specified, not yet coded'); NBH/05 lists no code status	n/a	TH19; RL04
ENG16	E16 Owls vs NOTS data caveat	Interpret data_hash_ check.csv (Owls regen- eration vs NOTS bits)	UNCERTAIN (interpretation not recorded)	n/a	CJ226
ENG17	E17 reproducible record / NOTS reor- ganisation	Reproducible record of ABL-CONF fits; NOTS projects/BVS reorganisation	IN PROGRESS at pos 1673	n/a	
ENG18	LH-SCALE simulation	Production code on LH at n in {400,1600,6400, 25600}, about 2,400 fits	OPEN (awaits DEC10; needs NOTS)	n/a	DEC10
ENG19	real data and head- to-head comparisons	Real-data analyses; comparisons with PARNI, MAdaSub, TGS, SuSiE, SSL, Song-Liang	OPEN	n/a	
ENG20	XF re-check of Claude repairs	Cross-family re-check of Claude repairs not covered by A-REV5-9	OPEN	n/a	
FR01	p » n masked con- sistency, beyond polynomial p	XF-reviewed p » n masked HPM consis- tency (API-PXNGAP candidate unreviewed) ; superpolynomial and exponential p	OPEN		TH160; TH161; CJ18; OB12
FR02	covering the study's own design	Groups of 5 (2g <= B) excluded from LH; dilution-masking fam- ily (PXN §14) conjec- ture	OPEN		OB08; CJ25; TH172; TH177
FR03	from exhibits (LH, W) to classes of designs	Correlated-design guarantees shown only on explicit families LH and W	OPEN		RF208; RF176; CJ204
FR04	N1 general-h island identity	Measured-round island identity for general bounded h (binding §13.1)	OPEN		OB01; TH412
FR05	N3 tie-DP invariant	e4 tie-DP invariant, the only open mathemati- cal obligation in the M5 chain	OPEN		OB03; ENG10
FR06	Song-Liang analytic separation	Analytic proof that split-and-merge loses true predictors under correlation; numerical 20/20 vs 5/318 only	OPEN		CJ21; TH417; CE213; CL23
FR07	finite-n vs asymptotic onset	Onset n ~ 2e6 vs study n = 200; Pi(X) upper bound; finite-n Gibbs discovery	OPEN		TH334; CJ14; CJ230; CJ26; RF123
FR08	forward-KL control	Forward-KL conver- gence of the terminal law needs control of log(1/pi_min), not supplied	OPEN		CJ02
FR09	§5.3 epsilon scale	Useful constants only for eps <~ n^(-1/2)/p; sharp necessity open	OPEN		CJ22
FR10	masked two-sided Z- hat and PIP accuracy	Two-sided masked evidence and PIP pre- cision; masked TV/ KL	OPEN		OB04; CJ24; CJ07; CJ02

ID	Native name	Short description	Status (verbatim)	Regime	Depends on
FR11	delivery, calibration and mixing	Delivery P(U) -> 0; calibration contract (b) ; mixing of the phi = 1 round	OPEN		CJ13; CJ06; OB05; CJ04; OB10; CJ05
FR12	MCMC-slow / SMC-fast separation	No published or proved BVS separation from MCMC	OPEN		CJ290; CJ12
FR13	costs-only-depth half of the thesis	The thesis half "correlation costs only sequential depth and particles" is open	OPEN		CJ06; CJ23; OB15
FR14	Lean coverage of the main cards	No main card Lean-certified; T12 hstep first target; floating-point contract	OPEN		OB16; CJ16; ENG05

C Files, code and data index

Paths are relative to the repository root (`FbStatsQuant/Bayesian-Variable-Selection`, commit `21ebc28`); `NBH/` = `Theory_work/Old_Projects/New_BVS_Headquarters/`. Datasets, fits and binaries are not in git; their locations are listed in §C.4. The archive's own complete index is `NBH/04_code_and_files_index.md` (every branch, commit, thread, worktree, machine path and script) [NBH/04_code_and_files_index.md:3].

C.1 The archive of the theory campaign (`NBH/`)

File	Purpose
`NBH/START_HERE.md`	One-page state of the theory, reading order, "which file for what" [NBH/START_HERE.md:59-84]
`NBH/README.md`	What the archive is, source thread IDs, conventions, integrity checks, file index [NBH/README.md:1-123]
`NBH/01_overview_and_timeline.md`	Arc of the thread by phase, dated timeline, Felipe's standing rules, final state
`NBH/02_theory.md`	Final state (Part A): v3.1 setup, cards M1–M8 and supplements verbatim, obligations, future work, drops, rulings, coverage
`NBH/02b_theory_campaign_history.md`	Campaign history (Part B, 2.7 MB; grep it): every result, refutation and counterexample (IDs $\geq$ 101), 14 themes, dead ends, conflicts
`NBH/03_results_and_conclusions.md`	Final theoretical and empirical results with numbers; conclusions, novelty, venue, confidence
`NBH/03b_reasoning_and_traces.md`	Decision log (DEC01–DEC47 of the archive), dead ends B01–B19, HQ corrections R01–R34, reviewer findings, trace index
`NBH/04_code_and_files_index.md`	Every branch, commit, PR, thread, worktree, machine path, script, result file and billing fact
`NBH/05_open_questions_and_next_steps.md`	Pending decisions, future work, obligations, conflicts, engineering TODOs, next steps, infrastructure rules
`NBH/ID_REGISTRY.md`	Every archive ID (Part A final state; Part B campaign history) with native names and exact status; alias index
`NBH/notation_and_conventions.md`	Symbols, priors, the v3.1 settings table and (n, p) rules, convention changes, name collisions
`NBH/inputs/Local_to_Global_draft.txt`	Felipe's first proposal draft (sets the mandatory goals §5.1–5.4, §13.1) [NBH/README.md:33]
`NBH/transcript/INDEX.md`, `part_*.md`, `traces/`	Verbatim thread by position (`grep -n "BEGIN pos N "`); HQ reasoning, delegate briefs and file changes [NBH/README.md:38]

File	Purpose
NBH/branch_snapshots/	Verbatim .md/.tex/.lean of all 84 theory/* branches and main@c4ef0dc/Docs: the authoritative mathematical texts; SNAPSHOT_MANIFEST.txt (ref, path, sha256); NONTEXT_INDEX.tsv (non-text files by git blob) [NBH/README.md:34]

C.2 Key branch snapshots (NBH/branch_snapshots/<BRANCH>/)

File	Purpose
MEET-THM/THEOREM_LIST_FINAL.md	Final cards M1–M8, supplements, cited lemmas (TLF)
MEET-THM/FOR_FELIPE_THEOREM_LIST.md	Plain-language theorem list presented for approval
MEET-THM/REGISTRY_PAPER1.md	K-number registry of paper 1; §9 recommended registry updates (DEC08)
MEET-THM/MINUTES.md, CLOSE_AUDIT.md, COVERAGE_MATRIX.md, FUTURE_WORK.md	Theorem-meeting minutes (waves §2.5, dissents), Sol close audit, coverage of goals, FW1–FW27 with reasons
MEET-THM/HQ_CHARTERS/	Charters of the per-theorem HQs (M1–M8) and lemma packs (LP-*)
MEET-THM/evidence/	HQ rulings Q1–Q5 (2026-11-04_HQ_RULINGS_Q1-Q4.md, ..._HQ_RULING_Q5.md); check waves W1_PC1/PC2, W2_PC1/PC2; CHK_INHERITED
MEET-THM/DOSSIER.md, CHECK_W1_BRIEF.md	Inclusion criteria IC1–IC7 and the card format; verdict vocabulary of the check waves
MEET-v31/METHOD_v3.1_FINAL.md, FOR_FELIPE_FINAL_METHOD.md	Method spec v3.1 and its plain-language version
MEET-v31/AMENDMENT_HQ_01.md, AMENDMENT_HQ_02.md	Sol-finding dispositions; reporting fields and growth rates (RL06)
T38-RECORD/T38_STATEMENT.md, REGISTRY_v2.md, LOCKABLE_v2.md	Statement of record of Theorem 5 (v3), the 431-result campaign registry (conflicts C1–C26), the 55 lockable candidates
L-LEAN/ (SmcBvs/, STATUS.md, T12_PLAN.md, fidelity/)	Lean 4 library (101 declarations, 0 sorry); plan for T12 hstep
A-REV1/ … A-REV9/ (REVIEW.md)	Astra cross-family reviews
API-PXNGAP/REPORT.md, PROOF.md	p ≫ n repair candidate (TH161, FR01)
API-N200/REPORT.md, PROOF.md	Theorem N200, 558 vs 571 (TH334)
O-DILUTE/REPORT.md, THEORY.md	Dilution identity and numbers (TH172, TH177)
S-ELS/REPORT.md, O-LS0/REPORT.md	Data facts on the 2,400 fits; LS0 data check (0/2,394)
W7-SEP/, LIT-PAR/REPORT.md, LIT-*/REPORT.md	Separation and Prop GU; literature verdicts
F-REFEREE/REFEREE_REPORT.md, RESPONSE_PLAN.md	Hostile JRSS-B referee simulation, response plan (LH-SCALE)
ABL-CONF/REPORT.md, PREREG.md	Confirmatory start test (CJ164, CJ165)
N9-KERNEL/	Kernel validity gate (OB09, TH12)
E-SPINE/	PR #1 contents (spine correction, Docs/Theory/AMENDMENTS_20261102.md)
campaign/LEDGER.md, campaign/PROTOCOL.md	Campaign ledger (times, costs) and protocol (status words)
W8-SYNTH/SYNTHESIS.md	Synthesis of 2026-11-02 evening (conflict groups §4)

C.3 Repository: results, code, old docs

Path	Purpose
Results/2026-10-28_New_Campaign/L_20261102/ (HEADLINE_L.md, EVIDENCE_L.md)	Low-p results (exact posterior by enumeration) [README.md:83]
Results/2026-10-28_New_Campaign/H2_20261102/ (HEADLINE_H.md, EVIDENCE_H.md)	High-p results (2,400 fits) [README.md:84]
Results/SMC_BVS_v3_production/	2026-10-23 production fits, reproduced exactly by v2.2.1 [README.md:88]
Docs/HQ/STATUS.md	Results on one screen [README.md:86]
Scripts/Campaigns/2026-10-28_New_Campaign/	Frozen New Campaign code (code/core/v2_2_1/: SMC-BVS v2.2.1), bundles, analysis scripts, Sol reviews; verify_manifests.sh [README.md:51-77]
Scripts/SMC_BVS/	v2.1 method code (2026-10-23 production) [README.md:41]
Docs/Methodology/	METHODOLOGY_v3 (the v3 method spec) [README.md:35]
Docs/THEORY.md, THEORY_PRIORITIES.md, PLAN.md, Docs/Theory/	Pre-campaign theory documents; superseded (§6.2)
Docs/HQ/DECISIONS.md	Old HQ decision log D1–D94 (empirical campaign)
Docs/Records/DATA_INDEX.md	Where the bulk data lives [Docs/Records/DATA_INDEX.md:1-3]
Papers/	Reference papers (e.g. Papers/Priors_and_decision_rules/)
Theory_work/State_of_Project/	This document: sections/, registry/*.tsv (machine-readable registry), scripts/ (check.py, build.sh), AUTHORING.md

C.4 Data locations, machines and branches

Location	What is there
Felipe's PC, OneDrive "NOTS Data/New_Campaign/"	Mirror of every New Campaign run (inputs, results, logs), sha256-verified [Docs/Records/DATA_INDEX.md:9]
Felipe's PC, Documents/NOTS_BVS	Earlier campaigns: the 2026-10-23 v2.1 production fits, benchmark study, profiling [Docs/Records/DATA_INDEX.md:11]
NOTS /scratch/fb31/smcbvs_v3_deploy_20261022/claude/prod_*/results/	Per-island pip_r, log_Z_r of the 2,394 completed fits (needed for ENG08) [NBH/04_code_and_files_index.md:88]
NOTS /home/fb31/projects/BVS/	Code/scripts; target of the NOTS reorganisation by the old HQ (ENG17) [NBH/04_code_and_files_index.md:87], [NBH/05_open_questions_and_next_steps.md:235]
Owls /home/fb31/Bayesian-Variable-Selection	Current repo clone with all local theory/* branches and T3 worktrees [NBH/04_code_and_files_index.md:21]
Owls /home/fb31/ls0_work, ablconf_work, abl_work, api_workers, hq_reviews, fresh_eyes	O-LS0 run on the 2,400 regenerated datasets; ABL-CONF and ABL-START fits; Agent SDK workers; Sol reviews; fresh-eyes Round 0 [NBH/04_code_and_files_index.md:26-32]
Owls ~/.config/bvs-hq/	HQ private config, launchers, Owls guard rails (secrets never in the archive) [NBH/04_code_and_files_index.md:23-24]
GitHub origin/theory/* (84 branches incl. campaign)	All campaign work; snapshotted in NBH/branch_snapshots/ [NBH/04_code_and_files_index.md:150]
PR #1, https://github.com/FbStatsQuant/Bayesian-Variable-Selection/pull/1	Head theory/E-SPINE (tip cec1680), the spine correction; unmerged (DEC09) [NBH/04_code_and_files_index.md:138]; still OPEN on 2026-11-07 [Theory_work/State_of_Project/sources/PR1_status_2026-11-07.json:1]
main	c4ef0dc (GH-PACK transfer, 2026-11-02), d46ebed (post-campaign refresh, 2026-11-03) [NBH/04_code_and_files_index.md:147-148]; then 620d8b7 ("Transfer", 2026-11-04: archive branch snapshots), 79c483c (README, 2026-11-05), 21ebc28 (theory archive docs added, 2026-11-07; the frozen source commit of this document) [Theory_work/State_of_Project/sources/git_log_main_2026-11-07.txt:1-3]

D Glossary of status words and review vocabulary

Meanings are as the sources define them; the source of each definition is cited. Status words are always copied verbatim and never upgraded. Mathematical symbols and design families are defined in Appendix A.

D.1 Status words (proof status of a result)

Word	Meaning (source)
PROVED	"a complete proof you checked line by line" (campaign protocol). Every campaign PROVED is a *candidate* [NBH/02b_theory_campaign_history.md:46].
PROVED (candidate)	"A proof here is a candidate: it is not reviewed and not adopted. Review happens later, by other model families" [NBH/branch_snapshots/campaign/PROTOCOL.md:19]; "candidate" means AI-proved and not adopted [NBH/branch_snapshots/THM-SYNTH/THEOREM_LIST.md:3-4].
PROVED (by this seat, unchecked)	Theorem-meeting status of a seat's own argument until a check is filed [NBH/branch_snapshots/MEET-THM/DOSSIER.md:104]. A checker who reads it line by line may write only "argument checked line by line by PCk: no error found" [NBH/branch_snapshots/MEET-THM/CHECK_W1_BRIEF.md:23].
PROVED-LOCKED	A row of `Docs/Theory/proofs/LOCKED_PROOFS.md`: statement and proof frozen by hash, always "not adopted" [NBH/02b_theory_campaign_history.md:49]; pre-campaign, same-family only with "external review queued" [NBH/03_results_and_conclusions.md:23].
PROVED-UNDER	"PROVED given/under named inputs" (REGISTRY_v2 author-status class) [NBH/02b_theory_campaign_history.md:52]. Excluded from LOCKABLE.
CONDITIONAL	A locked theorem whose hypotheses (H24, PW2, EXP′, VIS, NL_η, NL_{avg}, H2, H3, BASIN, COVER, exposure/occurrence conditions) are assumed, not proved [NBH/02b_theory_campaign_history.md:50].
PARTLY PROVED	Only named parts are locked; the others are CONJECTURED [NBH/02b_theory_campaign_history.md:51], [Docs/THEORY.md:15].
PROOF SKETCH	"the crux is identified and the steps are plausible, but not every step is verified" [NBH/02b_theory_campaign_history.md:46]. Not proved.
CONJECTURE / CONJECTURED (not proved)	Not proved; pre-campaign items not in the register and T37–T39 [Docs/THEORY.md:16].
OPEN	One of the five status words the protocol allows; it gives no further definition [NBH/branch_snapshots/campaign/PROTOCOL.md:19].
COUNTEREXAMPLE	"with a reproducible script or explicit construction" [NBH/02b_theory_campaign_history.md:46]; "(numerical, reproducible)" marks a computed one.
REFUTED, RETIRED, CORRECTED, RETRACTED, STRUCK, QUALIFIED, SUPERSEDED, withdrawn	Registry Part B dispositions of claims that are false, obsolete, numerically corrected, withdrawn by their author, removed from a statement, or weakened; the exact word and its reason are in each row, e.g. RF101, RF105, RF270, RF271, RF274 [NBH/ID_REGISTRY.md:1003-1007].
KNOWN-FALSE	A claim on the meetings' list of known-false results (e.g. SHARE-M(b), FD-NM, general LH locking), which a card may not rely on without addressing the refutation [NBH/branch_snapshots/MEET-THM/CHECK_W1_BRIEF.md:21]; example RF266.
DROPPED (LF / UR / TR / UA / DM)	Pre-campaign drops: LF likely false; UR unprovable or unworkable route; TR trivial or vacuous; UA unreasonable, insufficient or unverifiable assumptions; DM demoted/merged [Docs/Theory/DROPPED.md:3]. In the final list, "drop" = DR01–DR06.
REFEREE	A repair or counterexample supplied by a cross-family referee without a PROVED label [NBH/02b_theory_campaign_history.md:53].
[APPLIED] / [PENDING]	A repair a campaign run has written / not yet written [NBH/02b_theory_campaign_history.md:55].

Word	Meaning (source)
not adopted; adoption; locking	"Locked" = files frozen and audited; "Adopted" = Felipe accepted the result into the paper's theory; nothing is adopted [Docs/THEORY.md:21]; adoption and locking are Felipe's decision (DEC12).
APPROVED (F.B., 2026-11-07)	This document's mark for the paper-1 theorem list approved by Felipe as the current targets; it changes no proof status (DEC01).

D.2 Check levels and review verdicts

Term	Meaning (source)
SF (same-family)	"a Claude run (Opus or Fable) checked a Claude run" [NBH/02b_theory_campaign_history.md:54].
XF (cross-family)	A review by a different model family: Astra A-REV1–9; PC1 = gpt-6.1-sol on Seats A, B, D; PC2 = Opus 5.5 on Seat C (Astra); CHK = PC1 on inherited proofs [NBH/03_results_and_conclusions.md:7].
VERIFIED, VERIFIED*, GAP, ERROR	Verdict tags of the SF/XF reviews; VERIFIED* = "verified after a repair or with a stated qualification". "A re-review never raises an earlier adverse code on the original text; repaired texts carry the re-review verdict" [NBH/02b_theory_campaign_history.md:54].
VERIFIED WITH AMENDMENT; CORRECT WITH AMENDMENT	The check accepted the proof or output form subject to amendments; for N9-KERNEL §§0–3 under CHK_INHERITED the source records "(amendments adopted)" [NBH/02_theory.md:727]. Whether an amendment was adopted is stated per item, not implied by the verdict.
SAME-FAMILY-ONLY	Checked only within the Claude family (e.g. PROVED-LOCKED items; Thm5-LH) [NBH/03_results_and_conclusions.md:23], [NBH/03_results_and_conclusions.md:337].
AUTHOR-ONLY	Checked by nobody but its author (registrar flag; 51 of 166 cited IDs at the theorem meeting) [NBH/01_overview_and_timeline.md:214].
TRUE-AS-WRITTEN / COUNTEREXAMPLE / NEEDS-HYPOTHESIS / AMBIGUOUS	Check-wave verdict on a clause: no counterexample found (say how you searched) / explicit construction + script / a missing hypothesis named / the readings and which is true [NBH/branch_snapshots/MEET-THM/CHECK_W1_BRIEF.md:17]. "a TRUE-AS-WRITTEN is not a proof" [NBH/branch_snapshots/MEET-THM/CHECK_W1_BRIEF.md:23].
DESIGN-VERIFIED / DESIGN-ASSERTED / DESIGN-FAILS / NONE-GIVEN	Non-vacuity verdict: a concrete (for correlation theorems: genuinely correlated) design checked to satisfy every hypothesis / named but not verified / violates a hypothesis / none given [NBH/branch_snapshots/MEET-THM/CHECK_W1_BRIEF.md:18].
PLAUSIBLE / GAP / IMPLAUSIBLE (route)	Route verdict: key lemma right and steps fit / a named step missing / why not [NBH/branch_snapshots/MEET-THM/CHECK_W1_BRIEF.md:19].
W1PC1, W1PC2, W2PC1, W2PC2	Verdict rows of check wave 1 or 2 by checker PC1 (Sol) or PC2 (Opus) [NBH/02_theory.md:727]; files in MEET-THM/evidence/W1_PC1/ etc. W2PC1 also tallied repairs: "repaired 57, repaired differently 16, new 15, not repaired 3" [NBH/03b_reasoning_and_traces.md:404].
CHK, CHK_INHERITED	PC1's line-by-line check of inherited proofs (OWS-WS3/4 VERIFIED; WS6(d), WS7, N9 VERIFIED WITH AMENDMENT) [NBH/03b_reasoning_and_traces.md:407].
A-REV1 … A-REV9	The nine Astra cross-family reviews of 2026-11-02 (scopes in §6.5) [NBH/03b_reasoning_and_traces.md:384-394].
PASS WITH CORRECTIONS	Verdict of the Sol close audit of the theorem package; "the chair applied all of them" [NBH/03b_reasoning_and_traces.md:408].
LOCKABLE (LOCKABLE_v2)	Strict rule: author status PROVED (not PROVED-UNDER) and SF VERIFIED (or after repair) and XF VERIFIED (or with qualification); the list "does not adopt, lock or upgrade anything" [NBH/branch_snapshots/T38-RECORD/LOCKABLE_v2.md:3].

Term	Meaning (source)
Lean: sorry-free; `lake build` green; fidelity GREEN/RED	Formal proof with no `sorry`; the build passes; a reviewer judged the Lean statement faithful to the frozen informal statement ("statement fidelity only") or not; `stmtdiff.py` checks frozen statements [NBH/03_results_and_conclusions.md:143-146].
NEW / NEW-COMP / KNOWN—CITE / MODEST	Novelty tags of the theorem-meeting cards, "with checked citations" [NBH/branch_snapshots/MEET-THM/DOSSIER.md:73]. KNOWN—CITE results become cited lemmas (IC4).
NEW-SYNTH	A new statement proposed by THM-SYNTH, "PROOF SKETCH at most", "a proposal for an HQ, not a result" [NBH/branch_snapshots/THM-SYNTH/THEOREM_LIST.md:5-6].
IC1–IC7	Inclusion criteria of the theorem meeting: statement, route, non-vacuous (exhibited design), not merely known, serves the goals, Lean core named, fallback [NBH/branch_snapshots/MEET-THM/DOSSIER.md:60-69].
LICENSED / INCONCLUSIVE / holds_nondeliv	Outcomes of the pre-registered ABL-CONF rule (CJ164, CJ165); holds_nondeliv = the comparison arm failed to deliver within the cap [NBH/03b_reasoning_and_traces.md:260].
ACCEPTED; NOT binding; REPORTING, allowed	Statuses of HQ rulings RL01–RL05 [NBH/ID_REGISTRY.md:187-191].
recorded dissent	A seat's disagreement with a placement, kept in the minutes (DS01–DS06) [NBH/ID_REGISTRY.md:206-211].
UNCERTAIN	Marks a conflict between sources or something they do not settle [NBH/05_open_questions_and_next_steps.md:7]; rendered here as `uncertain` boxes.
DECIDED / OPEN / NOT TRIGGERED / IN PROGRESS	This document's status words for its own DEC, ENG, FR items (§7).

D.3 People, models, accounts and roles

Term	Meaning (source)
Felipe (F.B.)	Project lead; sets objectives and rules, approves theorem targets [NBH/01_overview_and_timeline.md:9].
HQ	The orchestrating assistant of the source thread (Claude Opus 5.5 on Claude Personal, "New HQ") [NBH/01_overview_and_timeline.md:9]; the "old HQ" ("SMC_BVS Headquarters") ran the empirical campaign and wrote the old `Docs/`. Per-theorem HQs are the planned next-phase leads (§7.5).
Astra	GPT-6 Astra (`gpt-6-astra`), OpenAI, on Codex; the cross-family reviewer of the campaign [NBH/03b_reasoning_and_traces.md:380].
Sol	GPT-6.1 Sol (`gpt-6.1-sol`), OpenAI, on Codex; Sol 1 (proof check), Sol 2 (audit), PC1 checker and close auditor [NBH/03b_reasoning_and_traces.md:380].
Terra	A Codex worker used by the old HQ for numerical tie-outs of the empirical analyses ("Terra tie-out PASS") [Docs/HQ/DECISIONS.md:45], [Docs/HQ/DECISIONS.md:76].
Fable 5.1, Opus 5.5, Sonnet, Haiku	Claude models: Fable the strongest, Opus the default worker; Sonnet/Haiku for clerical work, "out of every mathematical judgment" [NBH/01_overview_and_timeline.md:9], [NBH/03b_reasoning_and_traces.md:205].
CP, CW	Claude Personal (Ultra) and Claude Work subscriptions [NBH/01_overview_and_timeline.md:9].
Seats A–D; chair	Theorem-meeting seats: A (Fable 5.1), B (Fable 5.1), C (gpt-6-astra), D (Fable 5.1; "fresh-eyes Fable" in some files, U11); chair Opus 5.5 (CW) [NBH/05_open_questions_and_next_steps.md:120], [NBH/03b_reasoning_and_traces.md:207].
PC1, PC2	Proof checkers of the theorem meeting: PC1 = gpt-6.1-sol on Seats A, B, D; PC2 = Opus 5.5 on Seat C [NBH/05_open_questions_and_next_steps.md:122].
Round 0; fresh eyes	Isolated fresh Fable and raw Astra planning round with a method-only packet [NBH/03b_reasoning_and_traces.md:209-212].

Term	Meaning (source)
Registrar; clerk	Sonnet support roles (registry index, citation checks, charter drafting) [NBH/03b_reasoning_and_traces.md:205].

D.4 Identifiers and abbreviations

Term	Meaning (source)
Two-digit IDs (TH01–TH22, OB, CJ01–CJ27, DR, LIM, DF, RL, MD, DS, CL)	Final-state archive IDs (registry Part A) [NBH/README.md:39].
Three-digit IDs (TH101+, LM101+, CJ101+, RF101+, CE101+)	Campaign-history IDs (registry Part B): results, lemmas, conjectures/open problems, refuted claims, counterexamples [NBH/README.md:39].
DEC, ENG, FR	This document's own IDs: pending decisions, engineering items, frontier problems (§7). EMP = empirical facts (§3).
M1–M8; S-*; LP-*	Main cards (= TH01–TH08); supplement cards (S-T, S-TARGET, …); lemma packs of the per-theorem HQs [NBH/05_open_questions_and_next_steps.md:208].
K-numbers (K01–K108)	The theorem meeting's candidate IDs (MEET-THM/R0_DIFF.md), written "K-ID K46"; not archive IDs [NBH/notation_and_conventions.md:436].
FW1–FW27; N1–N17; E1–E17	Future-work items = CJ01–CJ27; obligations = OB01–OB17 [NBH/05_open_questions_and_next_steps.md:81]; engineering TODOs = ENG01–ENG17.
T01–T39	Pre-campaign theorem/conjecture numbers (Docs/Theory/) [Docs/THEORY.md:3].
Q1–Q5; P1–P12; U1–U14; C1–C26	The chair's questions to HQ (rulings RL01–RL05); the archive's pending items; final-state uncertain items; campaign-registry conflicts (§7.4).
H.1–H.21	This document's reversal entries (§6.3).
02a, 02b, REG-A, REG-B, TLF, METH31	NBH/02_theory.md; NBH/02b_theory_campaign_history.md; parts A/B of NBH/ID_REGISTRY.md; MEET-THM/THEOREM_LIST_FINAL.md; MEET-v31/METHOD_v3.1_FINAL.md [NBH/README.md:40-44].
[pos N]	Position *N* in the source-thread transcript (grep -n "BEGIN pos N " in NBH/transcript/part_*.md) [NBH/README.md:38].
W1–W8; F-*, O-*, A-*, S-*, API-*, LIT-*, L-LEAN, MEET-*	Campaign waves and the branch families (Fable, Opus, Astra, Sonnet synthesis, API-credit, literature, Lean, meetings); each is a branch theory/<ID> snapshotted in NBH/branch_snapshots/ [NBH/01_overview_and_timeline.md:48-65].
Routines; claude --cloud; API credit; Agent SDK	Routines bill the weekly plan; --cloud sessions bill the cloud credit; the promotional API credit covers the Agent SDK, not the CLI [NBH/05_open_questions_and_next_steps.md:245-246].

E Changelog

v1.0 (2026-11-07) First unified version, compiled from the New_BVS_Headquarters archive (commit 21ebc28) and the repository results. Theorem list approved by F.B. on 2026-11-07 [Theory_work/State_of_Project/sources/approval_2026-11-07.md:3-5]. Post-commit facts added by the lead, each with its own source file in sources/: the approval; PR #1 still open on 2026-11-07; the main commit log. Council review rounds: review/rounds/. Council result: passed in round 5 of 5 (Sol gpt-6.1-sol 8.9, Opus 5.5 8.9; all hard gates passed; question bank 30/30 for both); rounds 1–4 each failed gate G4 on one summary line that claimed more than its source, all corrected (summary in review/rounds/R5_reviews.md). Known cosmetic issue: about 78 overfull lines.